Linear Algebra

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Vectors

Exercise 0.1 (Store a data point as one object)**.** A house has a floor area of 120 square meters, 3 bedrooms, and an age of 15 years.

1. Write these three numbers as a single vertical list $\underset{\sim}{x}$, in the order given, and call its entries x_1, x_2 and x_3. What is x_2?
2. Does the list $(3, 120, 15)$ describe the same house?

Solution

Solution 0.1.

1. Stacking the three numbers in order gives

$$\underset{\sim}{x} = \begin{bmatrix} 120 \\ 3 \\ 15 \end{bmatrix},$$

   so $x_2 = 3$, the number of bedrooms.

2. No. The list $(3, 120, 15)$ would describe a house with 3 square meters of floor area and 120 bedrooms. An entry's position says what it measures, so the order of the entries matters.

Definition 0.1 (Column vector)**.** A **column vector** [1] of length p is an ordered list of p real numbers, written vertically as a $p \times 1$ array:

$$\underset{\sim}{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix}$$

Remark (Vectors are columns by default)

Remark 0.1 (Vectors are columns by default). In these notes, as in most statistics textbooks, a vector is a column vector unless stated otherwise. A vector written inline as a list in parentheses is still a column vector: for example, $\underset{\sim}{x} = (2, -1, 5)$ means

$$\underset{\sim}{x} = \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}.$$

Remark (Why the vector mark sits under the symbol)

Remark 0.2 (Why the vector mark sits under the symbol). In these notes, the mark that shows a symbol is a vector is a tilde under the letter, as in $\underset{\sim}{x}$. Many books instead use a bold letter or an arrow over the letter. We chose a mark underneath because bold letters are hard to write by hand on a board, and because the space above a letter is already used by the hat that marks an estimated value, as in $\widehat{\beta}$. The notation choices page gives the full reasons and an example of an estimated vector$\widetilde{\text{r}}$.

Exercise 0.2 (Reading one entry of a column vector of temperatures)**.** A week of daily readings is often stored as a column vector, with the reading for day i in entry i. The noon temperatures on the Celsius scale on the first four days of a week are

$$\underset{\sim}{x} = \begin{bmatrix} 12 \\ 15 \\ 9 \\ 11 \end{bmatrix}.$$

Find x_3.

Hint

In Definition 0.1, x_i is the entry in position i, counting from the top.

Solution

Solution 0.2. $x_3 = 9$, the third entry from the top, so the noon temperature on day 3 was 9 on the Celsius scale.

Definition 0.2 (The set $\mathbb{R}^p$)**.** For a positive integer p, $\mathbb{R}^p$ is the set of all column vectors of length p whose entries are real numbers.

Remark (Reading $\underset{\sim}{x} \in \mathbb{R}^p$)

Remark 0.3 (Reading $\underset{\sim}{x} \in \mathbb{R}^p$). Writing $\underset{\sim}{x} \in \mathbb{R}^p$ says, in one symbol, that $\underset{\sim}{x}$ is a column vector with p real entries. For example, $(1, -2, 0.5) \in \mathbb{R}^3$. The vector $(1, 2)$ is not in $\mathbb{R}^3$, because it has only 2 entries; it is in $\mathbb{R}^2$. In Exercise 0.1, $\underset{\sim}{x} \in \mathbb{R}^3$.

Exercise 0.3 (Testing whether a vector with entries $\sqrt{2}$ and π lies in $\mathbb{R}^3$)**.** Entries such as $\sqrt{2}$ and π have no exact decimal form, but they are still real numbers. Let

$$\underset{\sim}{v} = \begin{bmatrix} \sqrt{2} \\ \pi \\ -1 \end{bmatrix}.$$

Is $\underset{\sim}{v}$ in $\mathbb{R}^3$?

Hint

Definition 0.2 has two conditions: the number of entries of the column vector, and the kind of number in each entry.

Solution

Solution 0.3. Yes. $\underset{\sim}{v}$ is a column vector with 3 entries, and each of $\sqrt{2}$, π and -1 is a real number, so $\underset{\sim}{v} \in \mathbb{R}^3$ by Definition 0.2.

Definition 0.3 (Scalar)**.** In linear algebra, a **scalar** [2] is a single real number, as opposed to a vector or a matrix, which holds several numbers.

Example 0.1 (Scalars and a vector)**.** The numbers 5 and -0.2 are scalars. The column vector $(5, -0.2)$ is not a scalar: it has 2 entries, so it is in $\mathbb{R}^2$ (Definition 0.2), and each of its 2 entries is a scalar.

Exercise 0.4 (Picking out the scalars among class summaries)**.** A summary of a class of 30 students can report single numbers or whole lists of numbers. Consider these three quantities:

- the height of the tallest student in the class,
- the list of every student's height,
- the number of students in the class.

Which of these three quantities are scalars?

Hint

By Definition 0.3, a scalar is one real number. Count the numbers in each quantity.

Solution

Solution 0.4. The tallest student's height and the number of students are scalars: each is one real number. The list of heights holds 30 numbers, so it is a column vector in $\mathbb{R}^{30}$, not a scalar.

Definition 0.4 (Real-valued function, scalar-valued function)**.** A function $f : A \to \mathbb{R}$, whose values are numbers, is **real-valued** (also called **scalar-valued**, since its values are scalars).

Definition 0.5 (Vector-valued function)**.** A function $f : A \to \mathbb{R}^m$, whose values are vectors (Definition 0.2), is **vector-valued**.

Example 0.2 (A real-valued and a vector-valued function)**.** On $\mathbb{R}^2$, $f(\underset{\sim}{x}) = x_1 + x_2$ is real-valued and $g(\underset{\sim}{x}) = \begin{bmatrix} x_1 + x_2 \\ x_1 - x_2 \end{bmatrix}$ is vector-valued:

$$f\left(\begin{bmatrix} 3 \\ 4 \end{bmatrix}\right) = 3 + 4 = 7, \qquad g\left(\begin{bmatrix} 3 \\ 4 \end{bmatrix}\right) = \begin{bmatrix} 3 + 4 \\ 3 - 4 \end{bmatrix} = \begin{bmatrix} 7 \\ -1 \end{bmatrix}$$

Exercise 0.5 (Testing whether a price prediction is real-valued)**.** A model predicts a house's price from three of its features, stored in $\underset{\sim}{x} \in \mathbb{R}^3$, with the function

$$f(\underset{\sim}{x}) = 2x_1 + 5x_2 - x_3.$$

Is f real-valued?

Hint

Evaluate f at one point, and count the numbers in the result.

Solution

Solution 0.5. Yes. For every $\underset{\sim}{x} \in \mathbb{R}^3$, $2x_1 + 5x_2 - x_3$ is one real number, so $f : \mathbb{R}^3 \to \mathbb{R}$ is real-valued. For example, $f\left(\begin{bmatrix}1\\1\\1\end{bmatrix}\right) = 2 + 5 - 1 = 6$.

Exercise 0.6 (Evaluating a vector-valued function at one input)**.** A vector-valued function returns several numbers for one input. Let $g : \mathbb{R}^2 \to \mathbb{R}^3$ be

$$g(\underset{\sim}{x}) = \begin{bmatrix} x_1 + x_2 \\ x_1 x_2 \\ x_1 - x_2 \end{bmatrix}.$$

Find $g\left(\begin{bmatrix}2\\3\end{bmatrix}\right)$.

Hint

Substitute $x_1 = 2$ and $x_2 = 3$ into each entry, as in Example 0.2.

Solution

Solution 0.6. Substituting $x_1 = 2$ and $x_2 = 3$,

$$g\left(\begin{bmatrix}2\\3\end{bmatrix}\right) = \begin{bmatrix} 2+3 \\ 2 \cdot 3 \\ 2-3 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ -1 \end{bmatrix},$$

a column vector in $\mathbb{R}^3$.

Definition 0.6 (Transpose)**.** The **transpose** of a column vector $\underset{\sim}{x}$ is the row vector with the same sequence of entries, written horizontally:

$$\underset{\sim}{x}^{\top} \equiv \underset{\sim}{x}' \equiv [x_1,\ x_2,\ \ldots,\ x_p]$$

Example 0.3 (Transposing a column vector)**.** The transpose of the column vector with entries 2, −1, 5 is the row vector with the same entries:

$$\begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}^{\top} = [2,\ -1,\ 5].$$

Exercise 0.7 (Writing one observation as a row vector)**.** A data table stores each observation on its own line, so an observation held as a column vector is written in the table as its transpose. Let

$$\underset{\sim}{x} = \begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix}.$$

Find $\underset{\sim}{x}^{\top}$.

Hint

By Definition 0.6, keep the entries in the same order and write them horizontally.

Solution

Solution 0.7. By Definition 0.6,

$$\underset{\sim}{x}^{\top} = [4,\ -1,\ 0].$$

Definition 0.7 (Vector addition)**.** The **sum** of two column vectors $\underset{\sim}{x}$ and y of the same length p is the column vector $\underset{\sim}{x} + \underset{\sim}{y}$ of length p obtained by adding entry by entry:

$$(\underset{\sim}{x} + \underset{\sim}{y})_i \overset{\text{def}}{=} x_i + y_i, \quad i = 1, \ldots, p$$

Example 0.4 (Adding two vectors)**.**

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1+4 \\ 2+5 \\ 3+6 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}$$

Exercise 0.8 (Adding two months of sales for three items)**.** A shop records how many units of each of three items it sells in a month, as a column vector in $\mathbb{R}^3$. In two months it sells

$$\underset{\sim}{x} = \begin{bmatrix} 12 \\ 5 \\ 8 \end{bmatrix} \quad \text{and} \quad \underset{\sim}{y} = \begin{bmatrix} 10 \\ 7 \\ 3 \end{bmatrix}$$

units.
Find $\underset{\sim}{x} + \underset{\sim}{y}$, the units of each item sold over both months.

Hint

Use Definition 0.7, as in Example 0.4.

Solution

Solution 0.8. By Definition 0.7,

$$\underset{\sim}{x} + \underset{\sim}{y} = \begin{bmatrix} 12+10 \\ 5+7 \\ 8+3 \end{bmatrix} = \begin{bmatrix} 22 \\ 12 \\ 11 \end{bmatrix}.$$

Definition 0.8 (Dot product)**.** For any two real-valued vectors $\underset{\sim}{x} = (x_1, \ldots, x_p)$ and $\underset{\sim}{y} = (y_1, \ldots, y_p)$ of the same length p, the **dot product** of $\underset{\sim}{x}$ and $\underset{\sim}{y}$ is:

$$\underset{\sim}{x} \cdot \underset{\sim}{y} = \underset{\sim}{x}^\top \underset{\sim}{y} \overset{\text{def}}{=} \sum_{i=1}^{p} x_i y_i \tag{1}$$

See also the definitions in:

- Dobson and Barnett [3], Section 1.3 (equation 1.1, page 7)
- Kaplan [4], chapter on vectors
- wikipedia

The dot product has a different generalization for two matrices; see wikipedia for more.

Example 0.5 (A dot product)**.** For $\underset{\sim}{x} = (1, 2, 3)$ and $\underset{\sim}{y} = (4, 5, 6)$:

$$\begin{aligned} \underset{\sim}{x} \cdot \underset{\sim}{y} &= 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 && \text{(definition of the dot product)} \\ &= 4 + 10 + 18 && \text{(multiply)} \\ &= 32 && \text{(add)} \end{aligned}$$

Definition 0.9 (Linear combination)**.** A **linear combination** of the numbers $a_1, \ldots, a_k$ with coefficients $c_1, \ldots, c_k$ is the sum

$$c_1 a_1 + \cdots + c_k a_k = \sum_{i=1}^{k} c_i a_i$$

A linear combination of vectors $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k$ of the same length is defined entry by entry: entry j of $\sum_{i=1}^{k} c_i \underset{\sim}{v}_i$ is the linear combination of the jth entries of $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k$ with the same coefficients.

Definition 0.10 (Coefficients of a linear combination)**.** The numbers $c_1, \ldots, c_k$ in a linear combination (Definition 0.9) $\sum_{i=1}^{k} c_i a_i$ are its **coefficients**.

Example 0.6 (Coefficients of a linear combination of numbers)**.** The linear combination of the numbers 4 and 5 with coefficients 2 and 3 is $2 \cdot 4 + 3 \cdot 5 = 8 + 15 = 23$.

Definition 0.11 (Weighted sum)**.** A **weighted sum** of the numbers $a_1, \ldots, a_k$ is a linear combination of them, with coefficients $w_1, \ldots, w_k$:

$$w_1 a_1 + \cdots + w_k a_k = \sum_{i=1}^{k} w_i a_i$$

Multiply each a_i by its coefficient w_i, then add.

Definition 0.12 (Weight)**.** The coefficients $w_1, \ldots, w_k$ of a weighted sum $\sum_{i=1}^{k} w_i a_i$ are called its **weights**. The weight w_i says how much a_i counts.

Example 0.7 (The total price of a shopping basket)**.** A basket holds 3 items that cost \$2 each and 2 items that cost \$5 each. The total price is the weighted sum of the prices 2 and 5 with weights 3 and 2, the count of items at each price:

$$\begin{aligned} 3 \cdot 2 + 2 \cdot 5 &= 6 + 10 && \text{(multiply each price by its weight)} \\ &= 16 && \text{(add)} \end{aligned}$$

Definition 0.13 (Weighted average, weighted mean)**.** A **weighted average** (or **weighted mean**) of the numbers $a_1, \ldots, a_k$ is a weighted sum of them whose weights $w_1, \ldots, w_k$ are non-negative and add to 1:

$$\sum_{i=1}^{k} w_i a_i \quad \text{with} \quad w_i \geq 0 \text{ for each } i \quad \text{and} \quad \sum_{i=1}^{k} w_i = 1$$

An a_i with a larger weight counts for more.

Example 0.8 (A course grade)**.** A course grade counts the midterm for 1/4 and the final for 3/4. A student scores 70 on the midterm and 90 on the final. The weights 1/4 and 3/4 are non-negative and add to 1, so the grade is a weighted average of 70 and 90:

$$\begin{aligned} \frac{1}{4} \cdot 70 + \frac{3}{4} \cdot 90 &= 17.5 + 67.5 && \text{(multiply each score by its weight)} \\ &= 85 && \text{(add)} \end{aligned}$$

The grade 85 lies between 70 and 90, and nearer to 90, which has the larger weight.

Example 0.9 (A linear combination of two vectors)**.** The linear combination of $(1, 0)^\top$ and $(0, 1)^\top$ with coefficients 2 and 3 is $(2 \cdot 1 + 3 \cdot 0,\ 2 \cdot 0 + 3 \cdot 1)^\top = (2, 3)^\top$.

Remark (The dot product is a linear combination)

Remark 0.4 (The dot product is a linear combination). The dot product $\underset{\sim}{x} \cdot \underset{\sim}{y}$ is a linear combination (Definition 0.9) of the entries of y, with coefficients $x_1, \ldots, x_p$. In Example 0.5, $\underset{\sim}{x} \cdot \underset{\sim}{y} = 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6$ is the linear combination of 4, 5, and 6 with coefficients 1, 2, and 3.
The dot product is also the standard *inner product* on $\mathbb{R}^p$; "inner product" is the general notion (Definition 0.5, later in these notes), of which the dot product is one example.

Theorem 0.1 (Dot product is symmetric)**.** *The dot product is symmetric:*

$$\underset{\sim}{x} \cdot \underset{\sim}{y} = \underset{\sim}{y} \cdot \underset{\sim}{x}$$

Proof

Proof.

$$\begin{aligned} \underset{\sim}{x} \cdot \underset{\sim}{y} &= \sum_{i=1}^{p} x_i y_i && \text{(definition of the dot product)} \\ &= \sum_{i=1}^{p} y_i x_i && \text{(products of numbers are symmetric)} \\ &= \underset{\sim}{y} \cdot \underset{\sim}{x} && \text{(definition of the dot product)} \end{aligned}$$

□

Special vectors

Definition 0.14 (Zero vector)**.** The **zero vector** $\underset{\sim}{0}$ of length p has all entries equal to zero:

$$\underset{\sim}{0} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Remark (The zero vector is the additive identity)

Remark 0.5 (The zero vector is the additive identity). Adding the zero vector to a vector leaves it unchanged (Definition 0.7): $\underset{\sim}{x} + \underset{\sim}{0} = \underset{\sim}{x}$ for any vector $\underset{\sim}{x}$ of the same length, so $\underset{\sim}{0}$ is the identity element for vector addition. For example,

$$\begin{aligned} (2, -1) + (0, 0) &= (2 + 0, -1 + 0) \\ &= (2, -1). \end{aligned}$$

Exercise 0.9 (Picking the zero vector with three entries from three candidates)**.** A zero vector has a number of entries as well as zero entries, so $\underset{\sim}{0} \in \mathbb{R}^2$ and $\underset{\sim}{0} \in \mathbb{R}^3$ are different vectors. Consider three candidates:

$$\underset{\sim}{a} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \underset{\sim}{b} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \qquad \underset{\sim}{c} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

Which of the three candidates is the zero vector with 3 entries?

Hint

Definition 0.14 has two conditions: 3 entries, and 0 in every entry.

Solution

Solution 0.9. Only $\underset{\sim}{b}$. $\underset{\sim}{a}$ has only 2 entries, so it is the zero vector in $\mathbb{R}^2$, and $\underset{\sim}{c}$ has a nonzero second entry.

Definition 0.15 (Mean, average)**.** The **mean** [5] (or **average**) of n numbers $y_1, \ldots, y_n$ is their sum divided by how many there are:

$$\bar{y} \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^{n} y_i$$

Example 0.10 (The mean of three numbers)**.** For $y_1 = 2$, $y_2 = -1$ and $y_3 = 5$:

$$\begin{aligned} \bar{y} &= \frac{1}{3}(2 + (-1) + 5) && \text{(definition of the mean, with } n = 3) \\ &= \frac{1}{3} \cdot 6 && \text{(add)} \\ &= 2 && \text{(multiply)} \end{aligned}$$

Remark (The sample mean estimates the population mean)

Remark 0.6 (The sample mean estimates the population mean). When the numbers are observed data values, $\bar{y}$ is also called the sample mean, an estimator of the population mean.

Exercise 0.10 (Finding the mean age of four siblings)**.** Four siblings are 3, 7, 10 and 12 years old.
Find the mean of their ages.

Hint

Use Definition 0.15 with $n = 4$, as in Example 0.10.

Solution

Solution 0.10. By Definition 0.15 with $n = 4$,

$$\bar{y} = \frac{1}{4}(3 + 7 + 10 + 12) = \frac{32}{4} = 8,$$

so the mean age is 8 years.

Definition 0.16 (Ones vector)**.** The **ones vector** $\underset{\sim}{1}$ of length p has all entries equal to one:

$$\underset{\sim}{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

Exercise 0.11 (Adding one to every entry with the ones vector)**.** Adding the ones vector to a vector raises every entry by 1, for example to shift a list of counts that start at 0 so that they start at 1. Let

$$\underset{\sim}{x} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}.$$

Find $\underset{\sim}{x} + \underset{\sim}{1}$, where $\underset{\sim}{1}$ has 3 entries.

Hint

Write out $\underset{\sim}{1}$ with Definition 0.16, then use Definition 0.7.

Solution

Solution 0.11. By Definition 0.16, $\underset{\sim}{1} = (1, 1, 1)^\top$, so by Definition 0.7

$$\underset{\sim}{x} + \underset{\sim}{1} = \begin{bmatrix} 2+1 \\ 0+1 \\ -1+1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}.$$

Definition 0.17 (Sum of the elements of a vector)**.** The **sum of the elements** of a vector $\underset{\sim}{x} \in \mathbb{R}^p$ is the sum of its entries:

$$s_{\underset{\sim}{x}} \overset{\text{def}}{=} \sum_{i=1}^{p} x_i.$$

In words, $s_{\underset{\sim}{x}}$ is the entries of $\underset{\sim}{x}$ added together.

Example 0.11 (The sum of the elements of a vector with three entries)**.** For $\underset{\sim}{x} = (2, -1, 5)$,

$$\begin{aligned} s_{\underset{\sim}{x}} &= 2 + (-1) + 5 && \text{(definition of the sum of the elements, with } p = 3\text{)} \\ &= 6. && \text{(add)} \end{aligned}$$

Exercise 0.12 (Totalling the units sold across three items)**.** A shop sells 22, 12 and 11 units of three items over two months, stored as

$$\underset{\sim}{x} = \begin{bmatrix} 22 \\ 12 \\ 11 \end{bmatrix}.$$

Find $s_{\underset{\sim}{x}}$, the sum of the elements of $\underset{\sim}{x}$.

Hint

Use Definition 0.17 with $p = 3$, as in Example 0.11.

Solution

Solution 0.12. By Definition 0.17,

$$s_{\underset{\sim}{x}} = 22 + 12 + 11 = 45,$$

so the shop sold 45 units in all.

Theorem 0.2 (The ones vector sums the entries)**.** *For every $\underset{\sim}{x} \in \mathbb{R}^p$, the sum of the elements of $\underset{\sim}{x}$ is the dot product of the ones vector with $\underset{\sim}{x}$:*

$$s_{\underset{\sim}{x}} = \underset{\sim}{1} \cdot \underset{\sim}{x}.$$

Proof

Proof.

$$\begin{aligned} \underset{\sim}{1} \cdot \underset{\sim}{x} &= \sum_{i=1}^{p} 1 \cdot x_i && \text{(Definition 0.8, with every entry of } \underset{\sim}{1} \text{ equal to 1)} \\ &= \sum_{i=1}^{p} x_i && \text{(multiply each entry by 1)} \\ &= s_{\underset{\sim}{x}}. && \text{(Definition 0.17)} \end{aligned}$$

□

Example 0.12 (The ones vector applied to a vector with three entries)**.** For $\underset{\sim}{x} = (2, -1, 5)$ and the ones vector $\underset{\sim}{1}$ with three entries,

$$\begin{aligned} \underset{\sim}{1} \cdot \underset{\sim}{x} &= 1 \cdot 2 + 1 \cdot (-1) + 1 \cdot 5 && \text{(Definition 0.8)} \\ &= 6, && \text{(multiply and add)} \end{aligned}$$

which is $s_{\underset{\sim}{x}}$, as in Example 0.11. Dividing by $p = 3$ gives the mean of the entries, $6/3 = 2$.

Exercise 0.13 (Summing entries with a dot product against the ones vector)**.** Software that computes dot products quickly can total the entries of a vector through Theorem 0.2. Let

$$\underset{\sim}{x} = \begin{bmatrix} 4 \\ -2 \\ 7 \\ 1 \end{bmatrix},$$

and let $\underset{\sim}{1}$ be the ones vector with 4 entries.
Find $\underset{\sim}{1} \cdot \underset{\sim}{x}$.

Hint

By Theorem 0.2, $\underset{\sim}{1} \cdot \underset{\sim}{x}$ is the sum of the elements of $\underset{\sim}{x}$.

Solution

Solution 0.13. By Theorem 0.2,

$$\underset{\sim}{1} \cdot \underset{\sim}{x} = s_{\underset{\sim}{x}} = 4 + (-2) + 7 + 1 = 10.$$

Definition 0.18 (Indicator vector, standard basis vector)**.** The j-th **indicator vector** (or **standard basis vector** [6]) $\underset{\sim}{e}_j$ of length p has a 1 in position j and 0s elsewhere:

$$(\underset{\sim}{e}_j)_i = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \qquad \underset{\sim}{e}_j = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \leftarrow \text{position } j$$

Remark (Indicator vectors build every vector)

Remark 0.7 (Indicator vectors build every vector). For $p = 3$, the indicator vectors are $\underset{\sim}{e}_1 = (1, 0, 0)$, $\underset{\sim}{e}_2 = (0, 1, 0)$, and $\underset{\sim}{e}_3 = (0, 0, 1)$. Every vector of length 3 is a weighted sum $c_1\underset{\sim}{e}_1 + c_2\underset{\sim}{e}_2 + c_3\underset{\sim}{e}_3$ of them, with weights equal to its entries; for example, $(2, -1, 5) = 2\underset{\sim}{e}_1 - 1\underset{\sim}{e}_2 + 5\underset{\sim}{e}_3$. No other weights work: entry j of $c_1\underset{\sim}{e}_1 + c_2\underset{\sim}{e}_2 + c_3\underset{\sim}{e}_3$ is c_j, so each weight c_j must equal entry j of the vector.

Exercise 0.14 (Encoding the second of four categories as an indicator vector)**.** One-hot encoding stores a value that falls in one of p categories as the indicator vector of its category. A survey answer falls in the second of 4 categories.
Find $\underset{\sim}{e}_2$ with 4 entries, the one-hot encoding of that answer.

Hint

By Definition 0.18, $\underset{\sim}{e}_2$ has a 1 in position 2 and 0s elsewhere.

Solution

Solution 0.14. By Definition 0.18 with $j = 2$ and $p = 4$,

$$\underset{\sim}{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}.$$

Theorem 0.3 (Indicator vectors select entries)**.** *For any vector* $\underset{\sim}{x}$ *of length* p *and any* $j \in \{1, \ldots, p\}$*:*

$$\underset{\sim}{e}_j \cdot \underset{\sim}{x} = x_j$$

Proof

Proof. Writing the dot product componentwise:

$$
\begin{aligned}
\underset{\sim}{e}_j \cdot \underset{\sim}{x} &= \sum_{i=1}^{p} (\underset{\sim}{e}_j)_i \, x_i && \text{(definition of the dot product)} \\
&= \sum_{i=1}^{p} \begin{cases} 1 \cdot x_i & \text{if } i = j \\ 0 \cdot x_i & \text{if } i \neq j \end{cases} && \text{(definition of } \underset{\sim}{e}_j\text{)} \\
&= x_j && \text{(only the } i = j \text{ term is nonzero)}
\end{aligned}
$$

□

Exercise 0.15 (Selecting one entry with an indicator vector)**.** Theorem 0.3 turns reading one entry of a vector into a dot product. Let

$$
\underset{\sim}{x} = \begin{bmatrix} 5 \\ -1 \\ 8 \\ 2 \end{bmatrix}.
$$

Find $\underset{\sim}{e}_3 \cdot \underset{\sim}{x}$, where $\underset{\sim}{e}_3$ is the indicator vector with 4 entries.

Hint

Use Theorem 0.3 with $j = 3$.

Solution

Solution 0.15. By Theorem 0.3 with $j = 3$, $\underset{\sim}{e}_3 \cdot \underset{\sim}{x} = x_3 = 8$. Computing the dot product directly gives the same answer:

$$
\underset{\sim}{e}_3 \cdot \underset{\sim}{x} = 0 \cdot 5 + 0 \cdot (-1) + 1 \cdot 8 + 0 \cdot 2 = 8.
$$

Theorem 0.4 (A vector is a linear combination of indicator vectors)**.** *Every vector $\underset{\sim}{x} \in \mathbb{R}^p$ is the linear combination (Definition 0.9) of the indicator vectors $\underset{\sim}{e}_1, \ldots, \underset{\sim}{e}_p$ (Definition 0.18) with coefficients $x_1, \ldots, x_p$:*

$$
\underset{\sim}{x} = \sum_{j=1}^{p} x_j \underset{\sim}{e}_j
$$

Proof

Proof. Compare the two sides entry by entry. For each $i \in \{1, \ldots, p\}$:

$$
\begin{aligned}
\left(\sum_{j=1}^{p} x_j \underset{\sim}{e}_j\right)_i &= \sum_{j=1}^{p} x_j (\underset{\sim}{e}_j)_i && \text{(Definition 0.9, entry by entry)} \\
&= \sum_{j=1}^{p} \begin{cases} x_j \cdot 1 & \text{if } j = i \\ x_j \cdot 0 & \text{if } j \neq i \end{cases} && \text{(Definition 0.18)} \\
&= x_i \cdot 1 + \sum_{j \neq i} x_j \cdot 0 && \text{(split off the } j = i \text{ term)} \\
&= x_i + 0 && \text{(times 1 changes nothing; times 0 gives 0)} \\
&= x_i && \text{(adding 0 changes nothing)}
\end{aligned}
$$

□

Example 0.13 (Expanding a vector in indicator vectors)**.**

$$
\begin{aligned}
\begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix} &= 4 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + (-1) \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\
&= 4\underset{\sim}{e}_1 - \underset{\sim}{e}_2 + 2\underset{\sim}{e}_3
\end{aligned}
$$

Exercise 0.16 (Writing a vector as a linear combination of indicator vectors)**.** By Theorem 0.4, every vector in $\mathbb{R}^3$ is a linear combination of the indicator vectors $\underset{\sim}{e}_1$, $\underset{\sim}{e}_2$ and $\underset{\sim}{e}_3$. Let

$$
\underset{\sim}{x} = \begin{bmatrix} -3 \\ 0 \\ 6 \end{bmatrix}.
$$

Find the numbers c_1, c_2 and c_3 with $\underset{\sim}{x} = c_1\underset{\sim}{e}_1 + c_2\underset{\sim}{e}_2 + c_3\underset{\sim}{e}_3$.

Hint

Use Theorem 0.4, as in Example 0.13.

Solution

Solution 0.16. By Theorem 0.4, the numbers c_j are the entries of $\underset{\sim}{x}$: $c_1 = -3$, $c_2 = 0$ and $c_3 = 6$. As a check,

$$-3\begin{bmatrix}1\\0\\0\end{bmatrix}+0\begin{bmatrix}0\\1\\0\end{bmatrix}+6\begin{bmatrix}0\\0\\1\end{bmatrix}=\begin{bmatrix}-3\\0\\6\end{bmatrix}.$$

Orthogonality

Definition 0.19 (Orthogonal vectors)**.** Two vectors $\underset{\sim}{x}$ and $\underset{\sim}{y}$ of the same length are **orthogonal** [7] (written $\underset{\sim}{x} \perp \underset{\sim}{y}$) if their dot product (2) is zero:

$$\underset{\sim}{x} \perp \underset{\sim}{y} \iff \underset{\sim}{x} \cdot \underset{\sim}{y} = 0$$

Remark (Orthogonal means perpendicular)

Remark 0.8 (Orthogonal means perpendicular). Orthogonality extends the geometric idea of perpendicular lines to vectors with any number of entries. In the plane, $(1, 2)$ and $(-2, 1)$ are perpendicular, and their dot product is $1 \cdot (-2) + 2 \cdot 1 = 0$. The same test works in $\mathbb{R}^3$, where pictures are harder to draw:

$$\begin{aligned}(1, 1, 0) \cdot (1, -1, 5) &= 1 - 1 + 0\\ &= 0,\end{aligned}$$

so $(1, 1, 0) \perp (1, -1, 5)$.

Example 0.14 (Vectors that are not orthogonal)**.**

$$\begin{aligned}(1, 2) \cdot (1, 1) &= 1 \cdot 1 + 2 \cdot 1\\ &= 3\\ &\neq 0,\end{aligned}$$

so $(1, 2)$ and $(1, 1)$ are not orthogonal.

Exercise 0.17 (Choosing c to make two vectors orthogonal)**.** Principal component analysis describes data along orthogonal vectors, so testing for orthogonality comes up often. For $c \in \mathbb{R}$, let

$$\underset{\sim}{x} = \begin{bmatrix}2\\c\\1\end{bmatrix}, \qquad \underset{\sim}{y} = \begin{bmatrix}1\\3\\-5\end{bmatrix}.$$

Find the value of c for which $\underset{\sim}{x} \perp \underset{\sim}{y}$.

Hint

By Definition 0.19, solve $\underset{\sim}{x} \cdot \underset{\sim}{y} = 0$ for c, using 2.

Solution

Solution 0.17. $c = 1$. By Definition 0.19, $\underset{\sim}{x} \perp \underset{\sim}{y}$ exactly when

$$\begin{aligned} 0 &= \underset{\sim}{x} \cdot \underset{\sim}{y} && \text{(Definition 0.19)} \\ &= 2 \cdot 1 + c \cdot 3 + 1 \cdot (-5) && \text{(Definition 0.8)} \\ &= 3c - 3, && \text{(multiply and collect terms)} \end{aligned}$$

which holds only for $c = 1$.

Definition 0.20 (Euclidean norm (length, L_2 norm))**.** The **Euclidean norm** [8] (or **length**, also called the L_2 **norm** and written $\|\underset{\sim}{x}\|_2$) of a vector $\underset{\sim}{x}$ of length p is

$$\|\underset{\sim}{x}\| \stackrel{\text{def}}{=} \sqrt{\underset{\sim}{x} \cdot \underset{\sim}{x}} = \sqrt{\sum_{i=1}^{p} x_i^2} \tag{2}$$

where $\sqrt{\cdot}$ is the square root.

Example 0.15 (The length of a vector)**.** For $\underset{\sim}{x} = (3, 4)$:

$$\begin{aligned} \|\underset{\sim}{x}\| &= \sqrt{3^2 + 4^2} && \text{(definition of the norm)} \\ &= \sqrt{25} && \text{(square and add)} \\ &= 5 \end{aligned}$$

The vector $(0.6, 0.8)$ has norm $\sqrt{0.36 + 0.64} = 1$.

Exercise 0.18 (Finding the norm of a vector with four entries)**.** The Euclidean norm works the same way in $\mathbb{R}^4$ as in the plane, although there is no picture to draw. Let $\underset{\sim}{x}$ be the vector

$$\underset{\sim}{x} = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 4 \end{bmatrix}.$$

Find $\|\underset{\sim}{x}\|$.

Hint

Use Definition 0.20 with $p = 4$, as in Example 0.15.

Solution

Solution 0.18. By Definition 0.20,

$$\|\underset{\sim}{x}\| = \sqrt{1^2 + 2^2 + 2^2 + 4^2} = \sqrt{25} = 5.$$

Definition 0.21 (Unit vector)**.** A **unit vector** [9] is a vector whose Euclidean norm (Definition 0.20) is 1. A unit vector is said to have *unit length*.

Example 0.16 (A unit vector and a vector that is not one)**.** For $\underset{\sim}{x} = (0.6, 0.8)$:

$$\begin{aligned} \|\underset{\sim}{x}\| &= \sqrt{0.6^2 + 0.8^2} && \text{(definition of the norm)} \\ &= \sqrt{0.36 + 0.64} && \text{(square)} \\ &= \sqrt{1} && \text{(add)} \\ &= 1 && \text{(take the square root)} \end{aligned}$$

so $(0.6, 0.8)$ is a unit vector. The vector $(1, 1)$ is not:

$$\begin{aligned} \|(1,1)\| &= \sqrt{1^2 + 1^2} \\ &= \sqrt{2} \\ &\neq 1. \end{aligned}$$

Exercise 0.19 (How far apart are two points?)**.** Let $\underset{\sim}{x} = (1, 2)$ and $\underset{\sim}{y} = (4, 6)$.

1. Compute the vector $\underset{\sim}{x} - \underset{\sim}{y}$.
2. Compute the length of $\underset{\sim}{x} - \underset{\sim}{y}$, using the Euclidean norm.
3. Is the result the same if you compute the length of $\underset{\sim}{y} - \underset{\sim}{x}$ instead?

Solution

Solution 0.19.

1.
$$\begin{aligned} \underset{\sim}{x} - \underset{\sim}{y} &= (1 - 4, 2 - 6) \\ &= (-3, -4). \end{aligned}$$

2.
$$\begin{aligned}\|\underset{\sim}{x}-\underset{\sim}{y}\| &= \sqrt{(-3)^2+(-4)^2}\\ &= \sqrt{25}\\ &= 5.\end{aligned}$$

3. $\underset{\sim}{y}-\underset{\sim}{x}=(3,4)$, and
$$\begin{aligned}\|\underset{\sim}{y}-\underset{\sim}{x}\| &= \sqrt{3^2+4^2}\\ &= 5.\end{aligned}$$

Yes, the result is the same.

Definition 0.22 (Euclidean distance)**.** The **Euclidean distance** [10] between two vectors $\underset{\sim}{x}$ and $\underset{\sim}{y}$ of length p is the Euclidean norm of their difference:
$$d(\underset{\sim}{x},\underset{\sim}{y}) \overset{\text{def}}{=} \|\underset{\sim}{x}-\underset{\sim}{y}\| = \sqrt{\sum_{i=1}^{p}(x_i-y_i)^2} \tag{3}$$

Remark (Distance from the origin)

Remark 0.9 (Distance from the origin). The distance from the origin $\underset{\sim}{0}$ to a vector $\underset{\sim}{x}$ is the norm of $\underset{\sim}{x}$:
$$\begin{aligned}d(\underset{\sim}{x},\underset{\sim}{0}) &= \|\underset{\sim}{x}-\underset{\sim}{0}\|\\ &= \|\underset{\sim}{x}\|\end{aligned}$$
[11, p. 39].
For example, the distance from $\underset{\sim}{0}$ to $(3,4)$ is $\|(3,4)\|=5$, as in Example 0.15.

Exercise 0.20 (Finding the distance between two customers' feature vectors)**.** A nearest-neighbour method compares two customers through the Euclidean distance between their feature vectors. Two customers have feature vectors
$$\underset{\sim}{x}=\begin{bmatrix}1\\0\\3\end{bmatrix}, \qquad \underset{\sim}{y}=\begin{bmatrix}3\\1\\1\end{bmatrix}.$$
Find $d(\underset{\sim}{x},\underset{\sim}{y})$.

Hint

Use Definition 0.22 with $p = 3$.

Solution

Solution 0.20. By Definition 0.22,

$$\begin{aligned} d(\underset{\sim}{x}, \underset{\sim}{y}) &= \sqrt{(1-3)^2 + (0-1)^2 + (3-1)^2} && \text{(Definition 0.22)} \\ &= \sqrt{4+1+4} && \text{(subtract and square)} \\ &= 3. && \text{(add and take the square root)} \end{aligned}$$

Definition 0.23 (Orthonormal vectors)**.** A set of vectors $\{\underset{\sim}{x}_1, \underset{\sim}{x}_2, \ldots, \underset{\sim}{x}_k\}$ is **orthonormal** if the vectors are mutually orthogonal (Definition 0.19) and each has norm 1 (Definition 0.20), that is, unit length:

$$\underset{\sim}{x}_i \cdot \underset{\sim}{x}_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Example 0.17 (Indicator vectors are orthonormal)**.** The indicator vectors $\underset{\sim}{e}_1, \underset{\sim}{e}_2, \ldots, \underset{\sim}{e}_p$ (Definition 0.18) form an orthonormal set. By Theorem 0.3, $\underset{\sim}{e}_i \cdot \underset{\sim}{e}_j$ is entry i of $\underset{\sim}{e}_j$, which is 1 if $i = j$ and 0 if $i \neq j$. For $p = 2$:

$$\begin{aligned} \underset{\sim}{e}_1 \cdot \underset{\sim}{e}_1 &= 1 \cdot 1 + 0 \cdot 0 \\ &= 1, \end{aligned}$$

$$\begin{aligned} \underset{\sim}{e}_2 \cdot \underset{\sim}{e}_2 &= 0 \cdot 0 + 1 \cdot 1 \\ &= 1, \end{aligned}$$

and

$$\begin{aligned} \underset{\sim}{e}_1 \cdot \underset{\sim}{e}_2 &= 1 \cdot 0 + 0 \cdot 1 \\ &= 0. \end{aligned}$$

Example 0.18 (Two ways to fail to be orthonormal)**.**

- $(1, 1)$ and $(1, -1)$ are orthogonal, since $1 \cdot 1 + 1 \cdot (-1) = 0$, but not orthonormal: $(1, 1) \cdot (1, 1) = 2 \neq 1$.
- $(1, 0)$ and $(0.6, 0.8)$ both have unit length, since $(1, 0) \cdot (1, 0) = 1$ and

$$(0.6, 0.8) \cdot (0.6, 0.8) = 0.36 + 0.64$$
$$= 1,$$

but they are not orthonormal: $(1, 0) \cdot (0.6, 0.8) = 0.6 \neq 0$.

Exercise 0.21 (Compute an inner product and a norm)**.** Let

$$\underset{\sim}{a} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}, \qquad \underset{\sim}{b} = \begin{bmatrix} 0 \\ 4 \\ -2 \end{bmatrix}$$

Compute $\underset{\sim}{a}^\top \underset{\sim}{b}$ and $\|\underset{\sim}{a}\|_2$.

Solution

Solution 0.21. Multiply entry by entry and add:

$$\begin{aligned} \underset{\sim}{a} \cdot \underset{\sim}{b} &= (3)(0) + (-1)(4) + (2)(-2) \\ &= 0 - 4 - 4 \\ &= -8 \end{aligned}$$

For the norm, take the inner product of $\underset{\sim}{a}$ with itself first:

$$\begin{aligned} \underset{\sim}{a} \cdot \underset{\sim}{a} &= 3^2 + (-1)^2 + 2^2 \\ &= 9 + 1 + 4 \\ &= 14 \end{aligned}$$

so

$$\|\underset{\sim}{a}\|_2 = \sqrt{14} \approx 3.742$$

The inner product came out negative while the norm cannot: a norm is a square root of a sum of squares, so it is never below zero.
Python has both operations built in, and R builds them from `sum`.

```python
import numpy as np

a = np.array([3.0, -1.0, 2.0])
b = np.array([0.0, 4.0, -2.0])

inner = float(a @ b)
norm_a = round(float(np.linalg.norm(a)), 3)

print(inner, norm_a)
#> -8.0 3.742
```

```r
a <- c(3, -1, 2)
b <- c(0, 4, -2)

print(c(sum(a * b), round(sqrt(sum(a^2)), 3)))
#> [1] -8.000  3.742
```

Both print -8 and 3.742.

Video lecture

Hutchinson's Linear Function Basics (24 min) covers dot products and the Euclidean norm, and goes on to hyperplanes (Definition 0.27) and level sets [12]. The login for the video site is posted on Canvas.

Matrices

Definition 0.1 (Matrix)**.** A **matrix** [13] of dimensions $m \times n$ is a rectangular array of $m \cdot n$ numbers, arranged in m rows and n columns:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Remark (Matrix entries, and vectors as matrices)

Remark 0.1 (Matrix entries, and vectors as matrices). The entry in row i and column j of $\mathbf{A}$ is written a_{ij} or $(\mathbf{A})_{ij}$. For example, for the 2×3 matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix},$$

$a_{12} = 2$ and $a_{23} = 6$.
A column vector of length p is a $p \times 1$ matrix, and a row vector of length p is a $1 \times p$ matrix. For example, the column vector $(7, 8)$ is a 2×1 matrix, and the row vector $[7,\ 8]$ is a 1×2 matrix.

Definition 0.2 (Equal matrices)**.** Two matrices $\mathbf{A}$ and $\mathbf{B}$ are **equal**, written $\mathbf{A} = \mathbf{B}$, if both are $m \times n$ and $a_{ij} = b_{ij}$ for every $i = 1, \dots, m$ and $j = 1, \dots, n$.

Example 0.1 (Equal and unequal matrices)**.**

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \neq \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix} \neq \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

The second pair differs in one entry. The third pair lists the same four numbers in the same order, but one matrix is 1×4 and the other 2×2, so the two are not equal.

Exercise 0.1 (Solving for an unknown entry)**.** Two matrices are equal only when every pair of matching entries agrees, so one matrix equation stands for several equations between numbers. Let

$$\begin{bmatrix} 2 & x+1 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 0 & 5 \end{bmatrix}.$$

Find x.

Hint

Both matrices are 2×2, so by Definition 0.2 each entry on the left is the same number as the entry in the same position on the right.

Solution

Solution 0.1. Both matrices are 2×2, so they are equal when their entries in each position agree. Three positions already agree, and the entries in position $(1, 2)$ give

$$x + 1 = 4 \quad \text{(Definition 0.2)}$$
$$x = 3. \quad \text{(subtract 1 from both sides)}$$

Definition 0.3 (Row of a matrix)**.** The ith **row** of an $m \times n$ matrix $\mathbf{A}$ is the $1 \times n$ row vector of the entries in row i:

$$\mathbf{A}_{i,:} = \begin{bmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{bmatrix}.$$

Definition 0.4 (Column of a matrix)**.** The jth **column** of an $m \times n$ matrix $\mathbf{A}$ is the column vector of the entries in column j:

$$\mathbf{A}_{:,j} = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}.$$

Example 0.2 (Rows and columns of a matrix)**.** For $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, the rows are $\mathbf{A}_{1,:} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ and $\mathbf{A}_{2,:} = \begin{bmatrix} 4 & 5 & 6 \end{bmatrix}$, and the columns are $\mathbf{A}_{:,1} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$, $\mathbf{A}_{:,2} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$ and $\mathbf{A}_{:,3} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$.

Exercise 0.2 (Reading off a column of a matrix)**.** When each row of a matrix records one observation, each column holds one measured quantity for every observation, so reading off a column extracts that quantity's values. Let

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 3 & 4 \end{bmatrix}.$$

Find $\mathbf{A}_{:,2}$, the second column of $\mathbf{A}$.

Hint

The jth column of an $m \times n$ matrix lists the entries $a_{1j}, \ldots, a_{mj}$ from top to bottom, as in Example 0.2.

Solution

Solution 0.2. $\mathbf{A}$ has $m = 2$ rows, so its second column holds the two entries a_{12} and a_{22}:

$$\mathbf{A}_{:,2} = \begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

Exercise 0.3 (Counting the rows and columns of a matrix)**.** Data with one row per observation and one column per measured quantity is stored as a matrix, so the matrix's number of rows is the number of observations and its number of columns is the number of quantities. Let

$$\mathbf{C} = \begin{bmatrix} 4 & 0 \\ 1 & -3 \\ 2 & 7 \end{bmatrix}.$$

Find m and n such that $\mathbf{C}$ is an $m \times n$ matrix.

Hint

In Definition 0.1, m counts the rows and n counts the columns.

Solution

Solution 0.3. $\mathbf{C}$ has three rows, $\begin{bmatrix} 4 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 & -3 \end{bmatrix}$ and $\begin{bmatrix} 2 & 7 \end{bmatrix}$, and each row has two entries, so $\mathbf{C}$ has two columns. So $m = 3$ and $n = 2$: $\mathbf{C}$ is a 3×2 matrix, with $3 \cdot 2 = 6$ entries.

Exercise 0.4 (Reading off a row of a matrix)**.** When each row of a matrix records one observation, reading off a row extracts every measured quantity for that one observation. Let

$$\mathbf{D} = \begin{bmatrix} 1 & -4 & 2 \\ 0 & 6 & 3 \\ 5 & 5 & -1 \end{bmatrix}.$$

Find $\mathbf{D}_{3,:}$, the third row of $\mathbf{D}$.

Hint

A row of a matrix lists the entries $a_{i1}, \ldots, a_{in}$ from left to right, as in Example 0.2.

Solution

Solution 0.4. Write d_{ij} for the entry of $\mathbf{D}$ in row i and column j. $\mathbf{D}$ has $n = 3$ columns, so its third row holds the three entries d_{31}, d_{32} and d_{33}:

$$\mathbf{D}_{3,:} = \begin{bmatrix} d_{31} & d_{32} & d_{33} \end{bmatrix} = \begin{bmatrix} 5 & 5 & -1 \end{bmatrix}.$$

Theorem 0.1 (A matrix is a column of its rows)**.** *An $m \times n$ matrix $\mathbf{A}$ is the column vector whose entries are its rows:*

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{1,:} \\ \vdots \\ \mathbf{A}_{m,:} \end{bmatrix}.$$

Exercise 0.5 (Stacking rows into a matrix)**.** Theorem 0.1 says that a matrix is determined by its rows, so data recorded one observation at a time can be assembled into a matrix by stacking those records. Let $\mathbf{M}$ be the 3×2 matrix with

$$\mathbf{M}_{1,:} = \begin{bmatrix} 1 & 2 \end{bmatrix}, \qquad \mathbf{M}_{2,:} = \begin{bmatrix} 3 & -1 \end{bmatrix}, \qquad \mathbf{M}_{3,:} = \begin{bmatrix} 0 & 4 \end{bmatrix}.$$

What is $\mathbf{M}_{:,2}$, the second column of $\mathbf{M}$?

Hint

Stack the rows from top to bottom, as in Theorem 0.1, then read down the second column.

Solution

Solution 0.5. Stacking the three rows from top to bottom gives

$$\begin{aligned} \mathbf{M} &= \begin{bmatrix} \mathbf{M}_{1,:} \\ \mathbf{M}_{2,:} \\ \mathbf{M}_{3,:} \end{bmatrix} && \text{(Theorem 0.1)} \\ &= \begin{bmatrix} 1 & 2 \\ 3 & -1 \\ 0 & 4 \end{bmatrix}, && \text{(substitute the rows)} \end{aligned}$$

so the second column of $\mathbf{M}$ lists the second entry of each row:

$$\mathbf{M}_{:,2} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}.$$

Theorem 0.2 (A matrix is a row of its columns)**.** *An $m \times n$ matrix $\mathbf{A}$ is the row vector whose entries are its columns:*

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{:,1} & \cdots & \mathbf{A}_{:,n} \end{bmatrix}.$$

Exercise 0.6 (Placing columns side by side)**.** Theorem 0.2 says that a matrix is determined by its columns, so measurements recorded one quantity at a time can be assembled into a matrix by placing those lists side by side. Let $\mathbf{N}$ be the 2×3 matrix with

$$\mathbf{N}_{:,1} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \qquad \mathbf{N}_{:,2} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}, \qquad \mathbf{N}_{:,3} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}.$$

What is $\mathbf{N}_{2,:}$, the second row of $\mathbf{N}$?

Hint

Place the columns from left to right, as in Theorem 0.2, then read across the second row.

Solution

Solution 0.6. Placing the three columns from left to right gives

$$\begin{aligned} \mathbf{N} &= \begin{bmatrix} \mathbf{N}_{:,1} & \mathbf{N}_{:,2} & \mathbf{N}_{:,3} \end{bmatrix} && \text{(Theorem 0.2)} \\ &= \begin{bmatrix} 2 & 0 & 4 \\ -1 & 3 & 1 \end{bmatrix}, && \text{(substitute the columns)} \end{aligned}$$

so the second row of $\mathbf{N}$ lists the second entry of each column:

$$\mathbf{N}_{2,:} = \begin{bmatrix} -1 & 3 & 1 \end{bmatrix}.$$

Definition 0.5 (Tensor)**.** A **tensor** [14] of order $k \geq 1$ is an array of numbers with k indices,

$$\mathcal{A} = \left(a_{i_1 i_2 \cdots i_k}\right), \quad i_1 = 1, \ldots, n_1, \ \ldots, \ i_k = 1, \ldots, n_k,$$

and its dimensions are $n_1 \times n_2 \times \cdots \times n_k$.

Example 0.3 (Vectors, matrices and images as tensors)**.** A vector of length p is a tensor of order 1, and an $m \times n$ matrix is a tensor of order 2. A 28×28 grayscale image is a matrix: the entry a_{ij} is the brightness of the pixel in row i and column j. A 28×28 color image stores three numbers per pixel (red, green, blue), so it is an order-3 tensor with dimensions $28 \times 28 \times 3$, and a_{ijc} is the value of color channel c at the pixel in row i and column j. It has $28 \cdot 28 \cdot 3 = 2352$ entries.

Exercise 0.7 (Counting the entries of a batch of photos)**.** Software for neural networks often stores a batch of color photos as one tensor, with one index for the photo, two for the pixel's position, and one for the color channel, and the software stores one number per entry, so the count of entries determines how much memory the batch needs. Let $\mathcal{B}$ be the order-4 tensor for a batch of 10 color photos of 32×32 pixels, with $n_1 = 10$, $n_2 = 32$, $n_3 = 32$ and $n_4 = 3$.
How many entries does $\mathcal{B}$ have?

Hint

Each index i_r in Definition 0.5 takes n_r values, and each choice of all four indices picks one entry, as in Example 0.3.

Solution

Solution 0.7. An entry of $\mathcal{B}$ is picked by choosing i_1 from 10 values, i_2 and i_3 from 32 values each, and i_4 from 3 values, so the number of entries is

$$\begin{aligned} n_1 \cdot n_2 \cdot n_3 \cdot n_4 &= 10 \cdot 32 \cdot 32 \cdot 3 && \text{(substitute the index ranges)} \\ &= 30720. && \text{(multiply)} \end{aligned}$$

Matrix transpose

Definition 0.6 (Matrix transpose)**.** The **transpose** [15] of an $m \times n$ matrix $\mathbf{A}$ is the $n \times m$ matrix $\mathbf{A}^\top$ obtained by swapping the rows and columns of $\mathbf{A}$:

$$(\mathbf{A}^\top)_{ij} = a_{ji}$$

Example 0.4 (Transposing a 2×3 matrix)**.** For $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$,

$$\mathbf{A}^\top = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix},$$

a 3×2 matrix. For instance, entry $(1, 2)$ of the transpose is $a_{21} = 4$.

Exercise 0.8 (Transposing a 3×2 matrix)**.** The transpose turns the rows of a matrix into columns, just as the transpose of a vector turns a column into a row. Let

$$\mathbf{B} = \begin{bmatrix} 1 & -2 \\ 0 & 3 \\ 4 & 5 \end{bmatrix}.$$

Compute $\mathbf{B}^\top$.

Hint

Entry (i, j) of the transpose is entry (j, i) of the original matrix, as in Example 0.4.

Solution

Solution 0.8. $\mathbf{B}$ is 3×2, so its transpose is 2×3. The ith row of $\mathbf{B}^\top$ is the ith column of $\mathbf{B}$:

$$\mathbf{B}^\top = \begin{bmatrix} 1 & 0 & 4 \\ -2 & 3 & 5 \end{bmatrix}.$$

For instance, entry $(2, 3)$ of $\mathbf{B}^\top$ is $b_{32} = 5$.

Matrix addition

Definition 0.7 (Zero matrix)**.** The $m \times n$ **zero matrix** [16] $\mathbf{0}_{m \times n}$ (or $\mathbf{0}$ when dimensions are clear from context) has all entries equal to zero:

$$\mathbf{0}_{m \times n} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}$$

Example 0.5 (Zero matrices of two shapes)**.**

$$\mathbf{0}_{2\times 3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \mathbf{0}_{3\times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Both have all entries 0, but one is 2×3 and the other 3×2, so they are different matrices.

Exercise 0.9 (Whether a matrix of zeros is a given zero matrix)**.** A zero matrix carries its shape in its subscript, because, as Example 0.5 shows, zero matrices of different shapes are different matrices. Let

$$\mathbf{Z} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Is $\mathbf{Z} = \mathbf{0}_{2\times 3}$?

Hint

Compare the shapes of the two matrices, using Definition 0.2.

Solution

Solution 0.9. No. $\mathbf{Z}$ has three rows and two columns, so it is 3×2. But $\mathbf{0}_{2\times 3}$ is 2×3. Two matrices of different shapes are not equal, even when every entry of both is 0. Instead, $\mathbf{Z}$ is the zero matrix $\mathbf{0}_{3\times 2}$.

Definition 0.8 (Matrix addition)**.** Two matrices $\mathbf{A}$ and $\mathbf{B}$ of the same dimensions $m\times n$ can be added element-wise; their **matrix sum** [17] is:

$$(\mathbf{A}+\mathbf{B})_{ij} = a_{ij} + b_{ij}$$

Example 0.6 (Adding matrices, and a sum that is not defined)**.**

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1+0 & 2+1 \\ 3+1 & 4+0 \end{bmatrix} \quad \text{(add entry by entry)}$$

$$= \begin{bmatrix} 1 & 3 \\ 4 & 4 \end{bmatrix}. \quad \text{(add)}$$

A 2×2 matrix and a 2×3 matrix cannot be added: they do not have the same dimensions, so entry $(1,3)$ of the sum would have no first term.

Exercise 0.10 (Adding two 2×2 matrices)**.** A matrix sum combines two tables of the same shape entry by entry, such as two months of counts recorded in the same rows and columns. Let

$$\mathbf{A} = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 3 & 2 \\ -2 & 5 \end{bmatrix}.$$

Compute $\mathbf{A} + \mathbf{B}$.

Hint

Add the entries in the same position, as in Example 0.6.

Solution

Solution 0.10. Both matrices are 2×2, so the matrix sum is defined:

$$\begin{aligned} \mathbf{A} + \mathbf{B} &= \begin{bmatrix} 1+3 & -1+2 \\ 2+(-2) & 0+5 \end{bmatrix} && \text{(add entry by entry)} \\ &= \begin{bmatrix} 4 & 1 \\ 0 & 5 \end{bmatrix}. && \text{(add)} \end{aligned}$$

Theorem 0.3 (Matrix addition is commutative)**.** *For $m \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$, matrix addition is commutative:*

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$$

Exercise 0.11 (Proving that matrix addition is commutative)**.** Theorem 0.3 lets two matrices be added in either order, just as two numbers can be (sums are symmetric). Prove Theorem 0.3.

Hint

Two $m \times n$ matrices are equal when their (i, j) entries agree for every i and j. Express the (i, j) entry of each side with Definition 0.8.

Solution

Solution 0.11. Both sides are $m \times n$ matrices, so they are equal when their (i, j) entries agree for every i and j. For each i and j,

$$\begin{aligned}(\mathbf{A}+\mathbf{B})_{ij} &= a_{ij}+b_{ij} && \text{(Definition 0.8)}\\ &= b_{ij}+a_{ij} && \text{(sums of numbers are symmetric)}\\ &= (\mathbf{B}+\mathbf{A})_{ij}, && \text{(Definition 0.8)}\end{aligned}$$

where the middle step uses the symmetry of sums of numbers. So $\mathbf{A}+\mathbf{B}=\mathbf{B}+\mathbf{A}$.

Theorem 0.4 (Matrix addition is associative)**.** *For $m\times n$ matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$, matrix addition is associative:*

$$(\mathbf{A}+\mathbf{B})+\mathbf{C}=\mathbf{A}+(\mathbf{B}+\mathbf{C})$$

Exercise 0.12 (Proving that matrix addition is associative)**.** Theorem 0.4 lets three matrices be added without parentheses, just as three numbers can be (sums are associative). Prove Theorem 0.4.

Hint

Two $m\times n$ matrices are equal when their (i,j) entries agree for every i and j. Apply Definition 0.8 twice on each side.

Solution

Solution 0.12. Both sides are $m\times n$ matrices, so they are equal when their (i,j) entries agree for every i and j. For each i and j,

$$\begin{aligned}((\mathbf{A}+\mathbf{B})+\mathbf{C})_{ij} &= (\mathbf{A}+\mathbf{B})_{ij}+c_{ij} && \text{(Definition 0.8)}\\ &= (a_{ij}+b_{ij})+c_{ij} && \text{(Definition 0.8)}\\ &= a_{ij}+(b_{ij}+c_{ij}) && \text{(sums of numbers are associative)}\\ &= a_{ij}+(\mathbf{B}+\mathbf{C})_{ij} && \text{(Definition 0.8)}\\ &= (\mathbf{A}+(\mathbf{B}+\mathbf{C}))_{ij}, && \text{(Definition 0.8)}\end{aligned}$$

where the middle step uses the associativity of sums of numbers. So $(\mathbf{A}+\mathbf{B})+\mathbf{C}=\mathbf{A}+(\mathbf{B}+\mathbf{C})$.

Theorem 0.5 (Zero matrix is the additive identity)**.** *For an $m\times n$ matrix $\mathbf{A}$:*

$$\mathbf{A}+\mathbf{0}_{m\times n}=\mathbf{A}$$

Exercise 0.13 (Proving that the zero matrix is the additive identity)**.** Theorem 0.5 says that adding the zero matrix leaves a matrix unchanged, just as adding the number 0 leaves a number unchanged.
Prove Theorem 0.5.

Hint

Express the (i, j) entry of $\mathbf{A} + \mathbf{0}_{m\times n}$ with Definition 0.8, and use that every entry of $\mathbf{0}_{m\times n}$ is 0.

Solution

Solution 0.13. Both sides are $m \times n$ matrices, so they are equal when their (i, j) entries agree for every i and j. For each i and j,

$$\begin{aligned}(\mathbf{A} + \mathbf{0}_{m\times n})_{ij} &= a_{ij} + (\mathbf{0}_{m\times n})_{ij} && \text{(Definition 0.8)}\\ &= a_{ij} + 0 && \text{(Definition 0.7)}\\ &= a_{ij}. && \text{(adding 0 to a number leaves it unchanged)}\end{aligned}$$

So $\mathbf{A} + \mathbf{0}_{m\times n} = \mathbf{A}$.

Theorem 0.6 (Additive inverse)**.** *For any $m\times n$ matrix $\mathbf{A}$, the $m\times n$ matrix $-\mathbf{A}$ (defined by $(-\mathbf{A})_{ij} = -a_{ij}$) satisfies:*

$$\mathbf{A} + (-\mathbf{A}) = \mathbf{0}_{m\times n}$$

Exercise 0.14 (Proving that a matrix plus its negation is zero)**.** Theorem 0.6 says that adding $-\mathbf{A}$ to a matrix $\mathbf{A}$ gives the zero matrix, just as adding $-a$ to a number a gives 0. Here $-\mathbf{A}$, the negation of $\mathbf{A}$, negates every entry of $\mathbf{A}$.
Prove Theorem 0.6.

Hint

Express the (i, j) entry of $\mathbf{A} + (-\mathbf{A})$ with Definition 0.8, and use $(-\mathbf{A})_{ij} = -a_{ij}$ from Theorem 0.6.

Solution

Solution 0.14. Both sides are $m \times n$ matrices, so they are equal when their (i, j) entries agree for every i and j. For each i and j,

$$
\begin{aligned}
(\mathbf{A}+(-\mathbf{A}))_{ij} &= a_{ij}+(-\mathbf{A})_{ij} && \text{(Definition 0.8)}\\
&= a_{ij}+(-a_{ij}) && \text{(definition of } -\mathbf{A} \text{ in Theorem 0.6)}\\
&= 0 && \text{(a number plus its negative is 0)}\\
&= (\mathbf{0}_{m\times n})_{ij}. && \text{(Definition 0.7)}
\end{aligned}
$$

So $\mathbf{A}+(-\mathbf{A})=\mathbf{0}_{m\times n}$.

Scalar multiplication

Definition 0.9 (Scalar multiplication)**.** The **scalar multiple** [18] of a matrix $\mathbf{A}$ by a scalar c is:

$$(c\mathbf{A})_{ij} = c\cdot a_{ij}$$

Example 0.7 (A scalar multiple)**.**

$$
\begin{aligned}
3\begin{bmatrix}1 & -2\\ 0 & 4\end{bmatrix} &= \begin{bmatrix}3\cdot 1 & 3\cdot(-2)\\ 3\cdot 0 & 3\cdot 4\end{bmatrix}\\
&= \begin{bmatrix}3 & -6\\ 0 & 12\end{bmatrix}.
\end{aligned}
$$

Exercise 0.15 (Multiplying a matrix by a negative factor)**.** A scalar multiple rescales every entry of a matrix by the same factor, as when a table of distances is converted from meters to centimeters. A negative factor flips the sign of every entry, and a factor between -1 and 1 shrinks every entry toward 0. Let

$$\mathbf{C} = \begin{bmatrix}4 & -2\\ 0 & 6\end{bmatrix}.$$

Compute $-\frac{1}{2}\mathbf{C}$.

Hint

Multiply each entry by $-\frac{1}{2}$, as in Example 0.7.

Solution

Solution 0.15.

$$-\frac{1}{2}\mathbf{C} = \begin{bmatrix} -\frac{1}{2}\cdot 4 & -\frac{1}{2}\cdot(-2) \\ -\frac{1}{2}\cdot 0 & -\frac{1}{2}\cdot 6 \end{bmatrix} \quad \text{(Definition 0.9)}$$

$$= \begin{bmatrix} -2 & 1 \\ 0 & -3 \end{bmatrix}. \quad \text{(multiply)}$$

Matrix multiplication

Definition 0.10 (Matrix multiplication)**.** The **product** [19] of an $m \times k$ matrix $\mathbf{A}$ and a $k \times n$ matrix $\mathbf{B}$ is the $m \times n$ matrix $\mathbf{C} = \mathbf{AB}$ with entries:

$$c_{ij} = \sum_{s=1}^{k} a_{is}\, b_{sj}$$

Remark (Matrix multiplication is not commutative)

Remark 0.2 (Matrix multiplication is not commutative). When $\mathbf{AB}$ and $\mathbf{BA}$ are both defined, they are usually different. For example, with

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix},$$

$$\mathbf{AB} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$$

$$\mathbf{BA} = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix},$$

so $\mathbf{AB} \neq \mathbf{BA}$.

Example 0.8 (Dot product as matrix multiplication)**.** The dot product of two column vectors $\underset{\sim}{x}$ and $\underset{\sim}{\beta}$ can be written as a matrix product of the row vector $\underset{\sim}{x}^{\top}$ with the column vector $\underset{\sim}{\beta}$:

$$
\begin{aligned}
\underset{\sim}{x} \cdot \underset{\sim}{\beta} &= \underset{\sim}{x}^{\top} \underset{\sim}{\beta} \\
&= [x_1,\ x_2,\ \ldots,\ x_p] \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix} \\
&= x_1\beta_1 + x_2\beta_2 + \cdots + x_p\beta_p
\end{aligned}
$$

Exercise 0.16 (Computing the product of a 2×3 and a 3×2 matrix)**.** The product of two matrices is defined only when the first has as many columns as the second has rows. Each entry of the product combines one row of the first factor with one column of the second. Let

$$
\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 4 \end{bmatrix}.
$$

Find $\mathbf{AB}$.

Hint

$\mathbf{A}$ is 2×3, and $\mathbf{B}$ is 3×2. By 4, $\mathbf{AB}$ is therefore 2×2. Its entry c_{ij} is $a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j}$.

Solution

Solution 0.16. $\mathbf{A}$ has three columns, and $\mathbf{B}$ has three rows. So the product is defined and is 2×2. Its entries are

$$
\begin{aligned}
\mathbf{AB} &= \begin{bmatrix} 1\cdot 2 + 0\cdot 0 + 2\cdot 1 & 1\cdot 1 + 0\cdot(-1) + 2\cdot 4 \\ (-1)\cdot 2 + 3\cdot 0 + 1\cdot 1 & (-1)\cdot 1 + 3\cdot(-1) + 1\cdot 4 \end{bmatrix} && \text{(Definition 0.10)} \\
&= \begin{bmatrix} 4 & 9 \\ -1 & 0 \end{bmatrix}. && \text{(multiply and add)}
\end{aligned}
$$

Theorem 0.7 (Matrix multiplication is associative)**.** *For an $m \times k$ matrix $\mathbf{A}$, a $k \times l$ matrix $\mathbf{B}$, and an $l \times n$ matrix $\mathbf{C}$:*

$$
\underbrace{(\mathbf{AB})\mathbf{C}}_{m\times n} = \underbrace{\mathbf{A}(\mathbf{BC})}_{m\times n}
$$

Proof

Proof. Entry (i, j) of each side, from 4:

$$\begin{aligned}
[(\mathbf{AB})\mathbf{C}]_{ij} &= \sum_{t=1}^{l} (\mathbf{AB})_{it}\, c_{tj} && \text{(definition of } (\mathbf{AB})\mathbf{C}) \\
&= \sum_{t=1}^{l} \left(\sum_{s=1}^{k} a_{is}\, b_{st} \right) c_{tj} && \text{(definition of } \mathbf{AB}) \\
&= \sum_{s=1}^{k} a_{is} \left(\sum_{t=1}^{l} b_{st}\, c_{tj} \right) && \text{(distribute and swap the finite sums)} \\
&= \sum_{s=1}^{k} a_{is}\, (\mathbf{BC})_{sj} && \text{(definition of } \mathbf{BC}) \\
&= [\mathbf{A}(\mathbf{BC})]_{ij} && \text{(definition of } \mathbf{A}(\mathbf{BC}))
\end{aligned}$$

□

Exercise 0.17 (Grouping a product of three matrices)**.** By Theorem 0.7, a product of three matrices can be grouped either way, so $\mathbf{AB}\underset{\sim}{x}$ can be computed as $\mathbf{A}(\mathbf{B}\underset{\sim}{x})$, without ever forming $\mathbf{AB}$. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}, \qquad \underset{\sim}{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

where $\underset{\sim}{x}$ is a 2×1 matrix.
Find $\mathbf{AB}\underset{\sim}{x}$ by computing $\mathbf{B}\underset{\sim}{x}$ first.

Hint

By 4, $\mathbf{B}\underset{\sim}{x}$ is a 2×1 matrix; multiply $\mathbf{A}$ by it.

Solution

Solution 0.17. First,

$$\begin{aligned}
\mathbf{B}\underset{\sim}{x} &= \begin{bmatrix} 2 \cdot 1 + 0 \cdot 2 \\ 1 \cdot 1 + 1 \cdot 2 \end{bmatrix} && \text{(Definition 0.10)} \\
&= \begin{bmatrix} 2 \\ 3 \end{bmatrix}, && \text{(multiply and add)}
\end{aligned}$$

so

$$\begin{aligned}
\mathbf{AB}\underset{\sim}{x} &= \mathbf{A}(\mathbf{B}\underset{\sim}{x}) && \text{(Theorem 0.7)} \\
&= \begin{bmatrix} 1\cdot 2 + 1\cdot 3 \\ 0\cdot 2 + 1\cdot 3 \end{bmatrix} && \text{(Definition 0.10)} \\
&= \begin{bmatrix} 5 \\ 3 \end{bmatrix}. && \text{(multiply and add)}
\end{aligned}$$

Grouping the other way gives the same answer: $\mathbf{AB} = \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}$, and $(\mathbf{AB})\underset{\sim}{x} = \begin{bmatrix} 3+2 \\ 1+2 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$.

Theorem 0.8 (Matrix multiplication is distributive over addition)**.** *Matrix multiplication distributes over addition from the left and from the right. For an $m \times k$ matrix $\mathbf{A}$ and $k \times n$ matrices $\mathbf{B}$ and $\mathbf{C}$:*

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$$

For $m \times k$ matrices $\mathbf{A}$ and $\mathbf{B}$ and a $k \times n$ matrix $\mathbf{C}$:

$$(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}$$

Exercise 0.18 (Proving that matrix multiplication distributes from the left)**.** Theorem 0.8 lets a product with a matrix sum be expanded term by term, just as multiplication of numbers distributes over addition. Let $\mathbf{A}$ be an $m \times k$ matrix, and let $\mathbf{B}$ and $\mathbf{C}$ be $k \times n$ matrices.
Prove that $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$.

Hint

Express the (i, j) entry of the left side with 4 and Definition 0.8, then split the result with the rule for a sum of sums.

Solution

Solution 0.18. Both sides are $m \times n$ matrices, so they are equal when their (i, j) entries agree for every i and j. For each i and j,

$$
\begin{aligned}
[\mathbf{A}(\mathbf{B}+\mathbf{C})]_{ij} &= \sum_{s=1}^{k} a_{is}(\mathbf{B}+\mathbf{C})_{sj} && \text{(Definition 0.10)} \\
&= \sum_{s=1}^{k} a_{is}(b_{sj}+c_{sj}) && \text{(Definition 0.8)} \\
&= \sum_{s=1}^{k} (a_{is}b_{sj}+a_{is}c_{sj}) && \text{(multiplication of numbers distributes over addition)} \\
&= \sum_{s=1}^{k} a_{is}b_{sj} + \sum_{s=1}^{k} a_{is}c_{sj} && \text{(a sum of sums splits)} \\
&= (\mathbf{AB})_{ij} + (\mathbf{AC})_{ij} && \text{(Definition 0.10)} \\
&= (\mathbf{AB}+\mathbf{AC})_{ij}. && \text{(Definition 0.8)}
\end{aligned}
$$

The third step uses that multiplication of numbers distributes over addition, and the fourth uses the rule for a sum of sums. So $\mathbf{A}(\mathbf{B}+\mathbf{C}) = \mathbf{AB}+\mathbf{AC}$.

Theorem 0.9 (Product of two sums of matrices)**.** *For $m \times k$ matrices $\mathbf{A}$ and $\mathbf{B}$ and $k \times n$ matrices $\mathbf{C}$ and $\mathbf{D}$,*

$$(\mathbf{A}+\mathbf{B})(\mathbf{C}+\mathbf{D}) = \mathbf{AC}+\mathbf{AD}+\mathbf{BC}+\mathbf{BD}.$$

Each product keeps its factors in the order they appear on the left, because matrix multiplication is not commutative. So $\mathbf{AD}$ and $\mathbf{BC}$ are not collected into $2\mathbf{AD}$.

Proof

Proof.

$$
\begin{aligned}
(\mathbf{A}+\mathbf{B})(\mathbf{C}+\mathbf{D}) &= (\mathbf{A}+\mathbf{B})\mathbf{C} + (\mathbf{A}+\mathbf{B})\mathbf{D} && \text{(Theorem 0.8, from the left)} \\
&= \mathbf{AC}+\mathbf{BC}+\mathbf{AD}+\mathbf{BD} && \text{(Theorem 0.8, from the right, twice)} \\
&= \mathbf{AC}+\mathbf{AD}+\mathbf{BC}+\mathbf{BD} && \text{(matrix addition is commutative)}
\end{aligned}
$$

□

Example 0.9 (Why the middle terms do not combine)**.** Take $\mathbf{A} = \mathbf{C} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $\mathbf{B} = \mathbf{D} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$. Then $\mathbf{A}+\mathbf{B} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, so

$$(\mathbf{A}+\mathbf{B})^2 = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix},$$

$$\mathbf{A}^2+\mathbf{AB}+\mathbf{BA}+\mathbf{B}^2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}.$$

Here $\mathbf{AB} \neq \mathbf{BA}$, so the middle terms do not combine into $2\mathbf{AB}$. The squared norm of a sum of vectors, $\|\underset{\sim}{x}+\underset{\sim}{y}\|^2$, is a case where they do combine (Theorem 0.2), because $\underset{\sim}{x}\cdot\underset{\sim}{y} = \underset{\sim}{y}\cdot\underset{\sim}{x}$. Matrices that commute, such as $\mathbf{A}$ and $\mathbf{A}^2$, combine the same way.

Exercise 0.19 (Comparing $(\mathbf{A}+\mathbf{B})(\mathbf{A}+\mathbf{B})$ with the rule for numbers)**.** For numbers, $(a+b)^2 = a^2+2ab+b^2$. Theorem 0.9 expands a product of two matrix sums into four terms instead, because $\mathbf{AB}$ and $\mathbf{BA}$ can differ. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Is $(\mathbf{A}+\mathbf{B})(\mathbf{A}+\mathbf{B})$ equal to $\mathbf{AA}+2\mathbf{AB}+\mathbf{BB}$?

Hint

Compute $\mathbf{A}+\mathbf{B}$ and its product with itself, then compute $\mathbf{AA}$, $\mathbf{AB}$ and $\mathbf{BB}$ with 4.

Solution

Solution 0.19. No. On the left, $\mathbf{A}+\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, so

$$(\mathbf{A}+\mathbf{B})(\mathbf{A}+\mathbf{B}) = \begin{bmatrix} 1\cdot 1+1\cdot 0 & 1\cdot 1+1\cdot 0 \\ 0\cdot 1+0\cdot 0 & 0\cdot 1+0\cdot 0 \end{bmatrix} \quad \text{(Definition 0.10)}$$

$$= \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}. \quad \text{(multiply and add)}$$

On the right, the same rule gives $\mathbf{AA} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\mathbf{AB} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and $\mathbf{BB} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, so

$$\mathbf{AA}+2\mathbf{AB}+\mathbf{BB} = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}.$$

The entries in position $(1,2)$ differ, $1 \neq 2$, so the two matrices are not equal. The rule for numbers fails because $\mathbf{BA} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ differs from $\mathbf{AB}$: Theorem 0.9 gives $\mathbf{AA}+\mathbf{AB}+$

$\mathbf{BA} + \mathbf{BB} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, which does match the left side.

Exercise 0.20 (Which of these products are defined?)**.** Let $\mathbf{A}$ be 4×3, let $\mathbf{B}$ be 3×2, and let $\underset{\sim}{x} \in \mathbb{R}^3$. For each expression, say whether it is defined, and if so give the shape of the result:

1. $\mathbf{AB}$
2. $\mathbf{BA}$
3. $\mathbf{A}\underset{\sim}{x}$
4. $\underset{\sim}{x}^\top \underset{\sim}{x}$
5. $\underset{\sim}{x}\underset{\sim}{x}^\top$

Solution

Solution 0.20. A vector in $\mathbb{R}^3$ is a 3×1 matrix, so the same rule settles every case.

1. $\mathbf{AB}$: $(4 \times 3)(3 \times 2)$. The inner pair is 3 and 3, so it is defined, and the result is 4×2.
2. $\mathbf{BA}$: $(3 \times 2)(4 \times 3)$. The inner pair is 2 and 4, which do not match, so it is **not defined**. Matrix multiplication is not commutative, and this is the blunt form of that: swapping the order can leave an expression that means nothing.
3. $\mathbf{A}\underset{\sim}{x}$: $(4 \times 3)(3 \times 1)$, defined, result 4×1 — a vector in $\mathbb{R}^4$.
4. $\underset{\sim}{x}^\top \underset{\sim}{x}$: $(1 \times 3)(3 \times 1)$, defined, result 1×1 — a single number. The product is the inner product of Equation 1.
5. $\underset{\sim}{x}\underset{\sim}{x}^\top$: $(3 \times 1)(1 \times 3)$, defined, result 3×3 — a matrix.

The last two use the same two vectors and differ only in order, and they return objects of different kinds. Reading a transpose as decoration rather than as a shape change is the commonest way to lose track of an expression.

Square and identity matrices

Definition 0.11 (Square matrix)**.** A matrix is **square** [20] if it has the same number of rows as columns.

Example 0.10 (Square and not square)**.** $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is square (2 rows, 2 columns); $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ is not (2 rows, 3 columns).

Exercise 0.21 (Whether a data matrix times its transpose is square)**.** Many statistical methods start from an $n \times p$ data matrix $\mathbf{X}$, with one row per observation, and form the product $\mathbf{X}^\top\mathbf{X}$ of its transpose with itself. Whether that product is square matters, because some operations, such as taking the inverse of a matrix, apply only to square matrices.
Is $\mathbf{X}^\top\mathbf{X}$ square?

Hint

Find the shape of $\mathbf{X}^\top$ from Definition 0.6, then the shape of the product from 4.

Solution

Solution 0.21. Yes. The transpose $\mathbf{X}^\top$ is $p \times n$, so $\mathbf{X}^\top$ has n columns. $\mathbf{X}$ has n rows, so the product $\mathbf{X}^\top\mathbf{X}$ is defined and is $p \times p$. It has p rows and p columns, so it is square for every n and p.

Definition 0.12 (Main diagonal)**.** The **main diagonal** [21] of an $m \times n$ matrix $\mathbf{A}$ is the list of its entries $a_{11}, a_{22}, \ldots, a_{kk}$, where k is the smaller of m and n.

Definition 0.13 (Diagonal entry)**.** The entries on the main diagonal (Definition 0.12) of a matrix $\mathbf{A}$ are the **diagonal entries** of $\mathbf{A}$.

Definition 0.14 (Off-diagonal entry)**.** In an $m \times n$ matrix $\mathbf{A}$ with main diagonal (Definition 0.12), every entry a_{ij} with $i \neq j$ is an **off-diagonal entry**.

Example 0.11 (Diagonal entries of a square and a non-square matrix)**.**

- For $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, the diagonal entries are $a_{11} = 1$ and $a_{22} = 4$, and the off-diagonal entries are $a_{12} = 2$ and $a_{21} = 3$.
- For $\mathbf{B} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, $m = 2$ and $n = 3$, so $k = 2$, and the diagonal entries are $b_{11} = 1$ and $b_{22} = 5$; the other four entries, 2, 3, 4 and 6, are off-diagonal.

Exercise 0.22 (The main diagonal of a tall matrix)**.** The main diagonal of a matrix that is not square stops when it runs out of rows or columns, whichever comes first. A tall matrix, with more rows than columns, runs out of columns first. Let

$$\mathbf{E} = \begin{bmatrix} 7 & -2 \\ 1 & 3 \\ 0 & 5 \\ 4 & -6 \end{bmatrix}.$$

Find the main diagonal of $\mathbf{E}$.

Hint

Find m, n and k in Definition 0.12.

Solution

Solution 0.22. Write e_{ij} for the entry of $\mathbf{E}$ in row i and column j. $\mathbf{E}$ is 4×2, so $m = 4$, $n = 2$ and $k = 2$. The main diagonal of $\mathbf{E}$ is therefore the list e_{11}, e_{22}, that is, $7, 3$. The list stops at e_{22} because $\mathbf{E}$ has no third column. It therefore has no entry e_{33}.

Exercise 0.23 (Adding the diagonal entries of a square matrix)**.** The sum of the diagonal entries of a square matrix summarizes that matrix in one number, called its trace. Let

$$\mathbf{F} = \begin{bmatrix} 2 & 5 & -1 \\ 0 & -3 & 8 \\ 4 & 1 & 6 \end{bmatrix}.$$

Find the sum of the diagonal entries of $\mathbf{F}$.

Hint

Find the main diagonal of $\mathbf{F}$ first.

Solution

Solution 0.23. Write f_{ij} for the entry of $\mathbf{F}$ in row i and column j. $\mathbf{F}$ is 3×3, so its diagonal entries are $f_{11} = 2$, $f_{22} = -3$ and $f_{33} = 6$, and adding them gives

$$f_{11} + f_{22} + f_{33} = 2 + (-3) + 6 = 5.$$

Exercise 0.24 (Counting the off-diagonal entries of a square matrix)**.** In a table of pairwise comparisons among several items, the off-diagonal entries are the comparisons between two different items, so counting them counts the ordered pairs of different items. Let $\mathbf{A}$ be a 5×5 matrix, so $\mathbf{A}$ is square.
How many off-diagonal entries does $\mathbf{A}$ have?

Hint

Count all the entries, then subtract the diagonal entries.

Solution

Solution 0.24. $\mathbf{A}$ has $5{\cdot}5 = 25$ entries. Its diagonal entries are the 5 entries a_{ii}, $i = 1, \ldots, 5$, and every other entry has $i \neq j$, so $\mathbf{A}$ has $25 - 5 = 20$ off-diagonal entries.

Definition 0.15 (Identity matrix)**.** The $p \times p$ **identity matrix** [22] $\mathbf{I}_p$ (or $\mathbf{I}$ when the size is clear from context) has ones on the main diagonal and zeros elsewhere:

$$(\mathbf{I}_p)_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \qquad \mathbf{I}_p = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

Example 0.12 (The 3×3 identity)**.**

$$\mathbf{I}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

so, for instance, $(\mathbf{I}_3)_{22} = 1$ and $(\mathbf{I}_3)_{23} = 0$.

Exercise 0.25 (Whether a matrix with the identity pattern is an identity matrix)**.** An identity matrix has ones on its main diagonal and zeros elsewhere. The matrix below has the same pattern of ones and zeros. Let

$$\mathbf{J} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Is $\mathbf{J}$ an identity matrix?

Hint

Compare the shape of $\mathbf{J}$ with the shape of $\mathbf{I}_p$ in Definition 0.15.

Solution

Solution 0.25. No. Every identity matrix $\mathbf{I}_p$ is $p \times p$, so it is square. $\mathbf{J}$ is 2×3, so it is not square. Therefore $\mathbf{J}$ is not an identity matrix, even though its main diagonal holds

ones and its other entries are zeros.

Remark (The identity matrix and the Kronecker delta)

Remark 0.3 (The identity matrix and the Kronecker delta). The entries of the identity matrix are given by the Kronecker delta: $(\mathbf{I}_p)_{ij} = \delta_{ij}$.

Theorem 0.10 (Identity matrix is a multiplicative identity)**.** *For any $m \times p$ matrix $\mathbf{A}$:*

$$\mathbf{A}\,\mathbf{I}_p = \mathbf{A}$$

$$\mathbf{I}_m\,\mathbf{A} = \mathbf{A}$$

Exercise 0.26 (Multiplying by an identity matrix)**.** Theorem 0.10 says that an identity matrix of the right size leaves a matrix unchanged under matrix multiplication, just as multiplying a number by 1 leaves it unchanged. Let

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 3 & 4 \end{bmatrix}.$$

Compute $\mathbf{I}_2\mathbf{A}$ entry by entry, from the definition of matrix multiplication.

Hint

Entry (i, j) of $\mathbf{I}_2\mathbf{A}$ is the dot product of the ith row of $\mathbf{I}_2$ with the jth column of $\mathbf{A}$, as in Example 0.8.

Solution

Solution 0.26. $\mathbf{I}_2$ is 2×2 and $\mathbf{A}$ is 2×3, so the product is defined and is 2×3. The first row of $\mathbf{I}_2$ is $\begin{bmatrix} 1 & 0 \end{bmatrix}$ and the second is $\begin{bmatrix} 0 & 1 \end{bmatrix}$, so by 4

$$\begin{aligned} \mathbf{I}_2\mathbf{A} &= \begin{bmatrix} 1\cdot 2 + 0\cdot 5 & 1\cdot 0 + 0\cdot 3 & 1\cdot(-1) + 0\cdot 4 \\ 0\cdot 2 + 1\cdot 5 & 0\cdot 0 + 1\cdot 3 & 0\cdot(-1) + 1\cdot 4 \end{bmatrix} && \text{(Definition 0.10)} \\ &= \begin{bmatrix} 2 & 0 & -1 \\ 5 & 3 & 4 \end{bmatrix} && \text{(multiply and add)} \\ &= \mathbf{A}. \end{aligned}$$

Each row of $\mathbf{I}_2$ has a single 1, so entry (i, j) of the product picks out entry (i, j) of $\mathbf{A}$, as Theorem 0.10 says.

Matrix-vector multiplication

Definition 0.16 (Matrix-vector multiplication)**.** The **matrix-vector product** of an $m \times p$ matrix $\mathbf{A}$ and a $p \times 1$ column vector $\underset{\sim}{x}$ is the $m \times 1$ column vector $\mathbf{A}\underset{\sim}{x}$ with entries:

$$(\mathbf{A}\underset{\sim}{x})_i = \sum_{j=1}^{p} a_{ij}\, x_j$$

Remark (Each entry is a dot product)

Remark 0.4 (Each entry is a dot product)*.* Matrix-vector multiplication extends the dot product (2): entry i of $\mathbf{A}\underset{\sim}{x}$ is the dot product of row i of $\mathbf{A}$ with $\underset{\sim}{x}$. For example,

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 2 \cdot (-1) \\ 3 \cdot 1 + 4 \cdot (-1) \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}.$$

Exercise 0.27 (A matrix-vector product with more columns than rows)**.** A matrix-vector product $\mathbf{A}\underset{\sim}{x}$ computes one weighted sum of the entries of $\underset{\sim}{x}$ for each row of $\mathbf{A}$, so one multiplication computes several weighted sums at once. The matrix $\mathbf{A}$ below has more columns than rows, so $\mathbf{A}\underset{\sim}{x}$ is shorter than $\underset{\sim}{x}$. Let

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 3 & 4 \end{bmatrix}, \qquad \underset{\sim}{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Compute $\mathbf{A}\underset{\sim}{x}$.

Hint

Entry i of $\mathbf{A}\underset{\sim}{x}$ is the dot product of the ith row of $\mathbf{A}$ with $\underset{\sim}{x}$, as in Remark 0.4.

Solution

Solution 0.27. $\mathbf{A}$ is 2×3 and $\underset{\sim}{x}$ is 3×1, so the matrix-vector product is defined and is 2×1:

$$\begin{aligned}
\mathbf{A}\underset{\sim}{x} &= \begin{bmatrix} 2\cdot 1 + 0\cdot 2 + (-1)\cdot 3 \\ 5\cdot 1 + 3\cdot 2 + 4\cdot 3 \end{bmatrix} && \text{(Definition 0.16)} \\
&= \begin{bmatrix} 2 + 0 - 3 \\ 5 + 6 + 12 \end{bmatrix} && \text{(multiply)} \\
&= \begin{bmatrix} -1 \\ 23 \end{bmatrix}. && \text{(add)}
\end{aligned}$$

Definition 0.17 (Linear system, system of linear equations)**.** A **linear system** [23] (or **system of linear equations**) with m equations in n unknowns is an equation $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ in which the $m \times n$ matrix $\mathbf{A}$ and the vector $\underset{\sim}{b} \in \mathbb{R}^m$ are given. A **solution** of the system is a vector $\underset{\sim}{x} \in \mathbb{R}^n$ for which $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ holds.

Example 0.13 (Two equations in two unknowns)**.** By Definition 0.16, row i of $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ is the equation $a_{i1}x_1 + \cdots + a_{in}x_n = b_i$. So the two equations $x_1 + 2x_2 = 5$ and $3x_1 + 4x_2 = 6$ form the linear system $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \underset{\sim}{x} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$. The vector $\underset{\sim}{x} = (-4, 4.5)$ is a solution:

$$\begin{aligned}
\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} -4 \\ 4.5 \end{bmatrix} &= \begin{bmatrix} 1\cdot(-4) + 2\cdot 4.5 \\ 3\cdot(-4) + 4\cdot 4.5 \end{bmatrix} && \text{(Definition 0.16)} \\
&= \begin{bmatrix} -4 + 9 \\ -12 + 18 \end{bmatrix} && \text{(multiply)} \\
&= \begin{bmatrix} 5 \\ 6 \end{bmatrix} && \text{(add)}
\end{aligned}$$

The vector $(1, 2)$ is not a solution: its first equation holds, since $1 + 2 \cdot 2 = 5$, but its second gives $3 \cdot 1 + 4 \cdot 2 = 11 \neq 6$.

Exercise 0.28 (Checking whether a vector solves a linear system)**.** Checking whether a vector solves a linear system takes only one matrix-vector product, which is much less work than solving the system. Let

$$\mathbf{A} = \begin{bmatrix} 3 & 1 \\ 1 & -2 \end{bmatrix}, \qquad \underset{\sim}{b} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}, \qquad \underset{\sim}{x} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Is $\underset{\sim}{x}$ a solution of $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$?

Hint

Compute $\mathbf{A}\underset{\sim}{x}$ with Definition 0.16. Then compare it with $\underset{\sim}{b}$, as in Example 0.13.

Solution

Solution 0.28. Yes. The matrix-vector product is

$$\begin{aligned} \mathbf{A}\underset{\sim}{x} &= \begin{bmatrix} 3\cdot 2 + 1\cdot(-1) \\ 1\cdot 2 + (-2)\cdot(-1) \end{bmatrix} && \text{(Definition 0.16)} \\ &= \begin{bmatrix} 5 \\ 4 \end{bmatrix}, && \text{(multiply and add)} \end{aligned}$$

which matches $\underset{\sim}{b}$ entry by entry, so $\underset{\sim}{x}$ is a solution of the system.

Linear and affine maps

Definition 0.18 (Additive function)**.** A function $f : \mathbb{R}^p \to \mathbb{R}^m$ is **additive** [24] if it preserves vector addition: for all $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^p$,

$$f(\underset{\sim}{x} + \underset{\sim}{y}) = f(\underset{\sim}{x}) + f(\underset{\sim}{y})$$

Example 0.14 (An additive function, and one that is not)**.**

- $f(x) = 2x$ on $\mathbb{R}$ is additive. For $x, y \in \mathbb{R}$:

$$\begin{aligned} f(x+y) &= 2(x+y) \\ &= 2x + 2y \\ &= f(x) + f(y) \end{aligned}$$

- $g(x) = x + 1$ on $\mathbb{R}$ is not additive, since $g(0+0) = 1$ but $g(0) + g(0) = 2$.

Exercise 0.29 (Testing whether $x_1 x_2$ is additive)**.** An additive function can be evaluated at $\underset{\sim}{x} + \underset{\sim}{y}$ by evaluating it at $\underset{\sim}{x}$ and at $\underset{\sim}{y}$ separately, so additivity is worth checking before splitting a computation that way. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be $f(\underset{\sim}{x}) = x_1 x_2$.
Is f additive?

Hint

Try $\underset{\sim}{x} = (1, 0)$ and $\underset{\sim}{y} = (0, 1)$ in Definition 0.18.

Solution

Solution 0.29. No. For $\underset{\sim}{x} = (1, 0)$ and $\underset{\sim}{y} = (0, 1)$, $\underset{\sim}{x} + \underset{\sim}{y} = (1, 1)$, so

$$f(\underset{\sim}{x} + \underset{\sim}{y}) = 1 \cdot 1 = 1, \qquad f(\underset{\sim}{x}) + f(\underset{\sim}{y}) = 1 \cdot 0 + 0 \cdot 1 = 0.$$

These differ, so f does not satisfy Definition 0.18 for this pair of inputs, and f is not additive.

Definition 0.19 (Homogeneous function)**.** A function $f : \mathbb{R}^p \to \mathbb{R}^m$ is **homogeneous** [24] if it preserves scalar multiplication: for all $\underset{\sim}{x} \in \mathbb{R}^p$ and $c \in \mathbb{R}$,

$$f(c\underset{\sim}{x}) = cf(\underset{\sim}{x})$$

Example 0.15 (Homogeneous functions, and one that is not)**.**

- $f(x) = 2x$ on $\mathbb{R}$ is homogeneous. For $x, c \in \mathbb{R}$:

$$\begin{aligned} f(cx) &= 2cx \\ &= c(2x) \\ &= cf(x) \end{aligned}$$

- $h(x) = x^2$ on $\mathbb{R}$ is not homogeneous, since $h(2 \cdot 1) = 4$ but $2\,h(1) = 2$.
- $k(\underset{\sim}{x}) = (x_1^3 + x_2^3)^{1/3}$ on $\mathbb{R}^2$ is homogeneous but not additive. For $\underset{\sim}{x} \in \mathbb{R}^2$ and $c \in \mathbb{R}$:

$$\begin{aligned} k(c\underset{\sim}{x}) &= (c^3x_1^3 + c^3x_2^3)^{1/3} \\ &= (c^3(x_1^3 + x_2^3))^{1/3} \\ &= c(x_1^3 + x_2^3)^{1/3} \\ &= c\,k(\underset{\sim}{x}) \end{aligned}$$

  but $k((1, 0) + (0, 1)) = k((1, 1)) = 2^{1/3}$, while $k((1, 0)) + k((0, 1)) = 2$.

Exercise 0.30 (Testing whether the absolute value is homogeneous)**.** A homogeneous function turns a rescaled input into the same rescaling of its output, including a rescaling by a negative number. Let $f : \mathbb{R} \to \mathbb{R}$ be the absolute value, $f(x) = |x|$.
Is f homogeneous?

Hint

Try a negative c in Definition 0.19.

Solution

Solution 0.30. No. Take $c = -1$ and $x = 1$. Then

$$f(cx) = |-1| = 1, \qquad cf(x) = -1 \cdot |1| = -1.$$

These differ, so f is not homogeneous. It does satisfy $f(cx) = cf(x)$ for every $c \geq 0$, since then $|cx| = c|x|$.

Definition 0.20 (Linear map, linear transformation, linear function)**.** A function $f : \mathbb{R}^p \to \mathbb{R}^m$ is a **linear map** [24] (also called a **linear transformation** or **linear function**) if it is both additive and homogeneous.

Example 0.16 (A linear map, and two functions that are not)**.**

- $f(\underset{\sim}{x}) = (2x_1,\ x_1 + x_2)$ on $\mathbb{R}^2$ is linear. For $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^2$ and $c \in \mathbb{R}$:

$$\begin{aligned} f(\underset{\sim}{x} + \underset{\sim}{y}) &= (2(x_1 + y_1),\ (x_1 + y_1) + (x_2 + y_2)) \\ &= f(\underset{\sim}{x}) + f(\underset{\sim}{y}) \end{aligned}$$

  and

$$\begin{aligned} f(c\underset{\sim}{x}) &= (2cx_1,\ cx_1 + cx_2) \\ &= cf(\underset{\sim}{x}). \end{aligned}$$

- $g(x) = x+1$ on $\mathbb{R}$ is not linear: additivity fails, since $g(0+0) = 1$ but $g(0)+g(0) = 2$.
- $h(x) = x^2$ on $\mathbb{R}$ is not linear: $h(2 \cdot 1) = 4$ but $2\,h(1) = 2$, so $h(cx) \neq c\,h(x)$ for $c = 2$, $x = 1$.

Exercise 0.31 (Testing whether a difference of scaled measurements is a linear map)**.** A score that scales two measurements and takes their difference, such as $3x_1 - x_2$, is a common building block of a model. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be $f(\underset{\sim}{x}) = 3x_1 - x_2$.
Is f a linear map?

Hint

Check the two conditions in Definition 0.20 one at a time, for any $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^2$ and $c \in \mathbb{R}$.

Solution

Solution 0.31. Yes. For any $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^2$, f is additive:

$$\begin{aligned} f(\underset{\sim}{x} + \underset{\sim}{y}) &= 3(x_1 + y_1) - (x_2 + y_2) && \text{(Definition 0.7)} \\ &= (3x_1 - x_2) + (3y_1 - y_2) && \text{(rearranging the terms)} \\ &= f(\underset{\sim}{x}) + f(\underset{\sim}{y}). \end{aligned}$$

For any $\underset{\sim}{x} \in \mathbb{R}^2$ and $c \in \mathbb{R}$, f is homogeneous:

$$\begin{aligned} f(c\underset{\sim}{x}) &= 3cx_1 - cx_2 && \text{(Definition 0.9)} \\ &= c(3x_1 - x_2) && \text{(factoring out } c\text{)} \\ &= cf(\underset{\sim}{x}). \end{aligned}$$

Since f is both, it is a linear map.

Definition 0.21 (Linear operator)**.** A **linear operator** on $\mathbb{R}^p$ is a linear map (Definition 0.20) from $\mathbb{R}^p$ to itself, that is, a linear map $f : \mathbb{R}^p \to \mathbb{R}^p$ whose input and output have the same length. Some authors use "operator" more loosely, for any linear map.

Example 0.17 (A linear operator, and a linear map that is not one)**.** The map $f(\underset{\sim}{x}) = (2x_1,\ x_1 + x_2)$ from Example 0.16 sends $\mathbb{R}^2$ to $\mathbb{R}^2$, so it is a linear operator on $\mathbb{R}^2$. The linear map $g(\underset{\sim}{x}) = x_1 + x_2$ sends $\mathbb{R}^2$ to $\mathbb{R}$, so it is linear but not an operator: its outputs have length 1, not 2.

Exercise 0.32 (Testing whether a projection is a linear operator)**.** Dropping the last coordinate of a point in $\mathbb{R}^3$ projects it onto a plane, as a shadow does. Let $f : \mathbb{R}^3 \to \mathbb{R}^2$ be $f(\underset{\sim}{x}) = (x_1, x_2)$. This f is a linear map.
Is f a linear operator on $\mathbb{R}^3$?

Hint

Write f as $f : \mathbb{R}^p \to \mathbb{R}^m$, and compare p with m as in Definition 0.21.

Solution

Solution 0.32. No. A linear operator on $\mathbb{R}^3$ sends $\mathbb{R}^3$ to $\mathbb{R}^3$. The outputs of f lie in $\mathbb{R}^2$, not $\mathbb{R}^3$, so f is a linear map but not a linear operator.

Theorem 0.11 (A linear map sends zero to zero)**.** *If $f : \mathbb{R}^p \to \mathbb{R}^m$ is a linear map (Definition 0.20), then $f(\underset{\sim}{0}) = \underset{\sim}{0}$.*

Proof

Proof. On the left, $\underset{\sim}{0}$ is the zero vector of length p; on the right, it is the zero vector of length m.

$$\begin{aligned} f(\underset{\sim}{0}) &= f(0 \cdot \underset{\sim}{0}) && \text{(every entry of } 0 \cdot \underset{\sim}{0} \text{ is } 0 \cdot 0 = 0\text{; Definition 0.9)} \\ &= 0 \cdot f(\underset{\sim}{0}) && \text{(homogeneity, with } c = 0 \text{ and } \underset{\sim}{x} = \underset{\sim}{0}\text{)} \\ &= \underset{\sim}{0} && \text{(every entry of } 0 \cdot f(\underset{\sim}{0}) \text{ is 0; Definition 0.9)} \end{aligned}$$

□

Example 0.18 (Ruling out linearity at zero)**.** In Example 0.16, $g(x) = x + 1$ has $g(0) = 1 \neq 0$, so by Theorem 0.11 it is not linear, without checking additivity or homogeneity. Sending $\underset{\sim}{0}$ to $\underset{\sim}{0}$ is necessary but not sufficient: $h(x) = x^2$ in the same example has $h(0) = 0$ and is still not linear.

Exercise 0.33 (Ruling out a linear map from its value at zero)**.** Theorem 0.11 gives a quick test: a function that does not send $\underset{\sim}{0}$ to $\underset{\sim}{0}$ cannot be a linear map. Let $g : \mathbb{R}^2 \to \mathbb{R}^2$ be

$$g(\underset{\sim}{x}) = (x_1 + x_2,\ x_2 - 1).$$

Is g a linear map?

Hint

Compute $g(\underset{\sim}{0})$.

Solution

Solution 0.33. No.

$$g(\underset{\sim}{0}) = (0 + 0,\ 0 - 1) = (0, -1) \neq \underset{\sim}{0}.$$

Every linear map sends $\underset{\sim}{0}$ to $\underset{\sim}{0}$ by Theorem 0.11, so g is not one.

Theorem 0.12 (Multiplying by a matrix is a linear map)**.** *For any $m \times p$ matrix $\mathbf{A}$, the function $f : \mathbb{R}^p \to \mathbb{R}^m$ defined by $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x}$ is a linear map (Definition 0.20).*

Proof

Proof. Both conditions are checked one entry at a time. For each $i \in \{1, \dots, m\}$, additivity:

$$\begin{aligned}
(\mathbf{A}(\underset{\sim}{x}+\underset{\sim}{y}))_i &= \sum_{j=1}^{p} a_{ij}(\underset{\sim}{x}+\underset{\sim}{y})_j && \text{(Definition 0.16)}\\
&= \sum_{j=1}^{p} a_{ij}(x_j+y_j) && \text{(Definition 0.7)}\\
&= \sum_{j=1}^{p} (a_{ij}x_j + a_{ij}y_j) && \text{(multiplication distributes over addition)}\\
&= \sum_{j=1}^{p} a_{ij}x_j + \sum_{j=1}^{p} a_{ij}y_j && \text{(a sum of sums splits)}\\
&= (\mathbf{A}\underset{\sim}{x})_i + (\mathbf{A}\underset{\sim}{y})_i && \text{(Definition 0.16)}\\
&= (\mathbf{A}\underset{\sim}{x} + \mathbf{A}\underset{\sim}{y})_i && \text{(Definition 0.7)}
\end{aligned}$$

and homogeneity:

$$\begin{aligned}
(\mathbf{A}(c\underset{\sim}{x}))_i &= \sum_{j=1}^{p} a_{ij}(c\underset{\sim}{x})_j && \text{(Definition 0.16)}\\
&= \sum_{j=1}^{p} a_{ij}\, c\, x_j && \text{(Definition 0.9)}\\
&= \sum_{j=1}^{p} c\, a_{ij}x_j && \text{(products are symmetric)}\\
&= c\sum_{j=1}^{p} a_{ij}x_j && \text{(a constant factor comes out of a sum)}\\
&= c\,(\mathbf{A}\underset{\sim}{x})_i && \text{(Definition 0.16)}\\
&= (c\,\mathbf{A}\underset{\sim}{x})_i && \text{(Definition 0.9)}
\end{aligned}$$

The four algebra rules cited here are on the algebra page: distributivity, a sum of sums, symmetry of products, and a constant factor. □

Exercise 0.34 (Recognizing a linear map as multiplication by a matrix)**.** Theorem 0.12 saves checking additivity and homogeneity by hand: a function that multiplies its input by a fixed matrix is a linear map. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be

$$f(\underset{\sim}{x}) = (x_1 + 2x_2,\ 3x_1 - x_2).$$

Is f a linear map?

Hint

Find a 2×2 matrix $\mathbf{A}$ with $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x}$, using Definition 0.16.

Solution

Solution 0.34. Yes. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}.$$

By Definition 0.16,

$$\mathbf{A}\underset{\sim}{x} = \begin{bmatrix} 1 \cdot x_1 + 2 \cdot x_2 \\ 3 \cdot x_1 + (-1) \cdot x_2 \end{bmatrix} = \begin{bmatrix} x_1 + 2x_2 \\ 3x_1 - x_2 \end{bmatrix} = f(\underset{\sim}{x}).$$

So f multiplies its input by $\mathbf{A}$, and f is a linear map by Theorem 0.12.

Theorem 0.13 (A linear map preserves linear combinations)**.** *If $f : \mathbb{R}^p \to \mathbb{R}^m$ is a linear map (Definition 0.20), then for any vectors $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k \in \mathbb{R}^p$ and any coefficients $c_1, \ldots, c_k \in \mathbb{R}$:*

$$f\left(\sum_{j=1}^{k} c_j \underset{\sim}{v}_j\right) = \sum_{j=1}^{k} c_j f(\underset{\sim}{v}_j)$$

Proof

Proof. By induction on k. For $k = 1$, the claim $f(c_1\underset{\sim}{v}_1) = c_1 f(\underset{\sim}{v}_1)$ is homogeneity. Suppose the claim holds for k vectors. Then for $k + 1$:

$$
\begin{aligned}
f\left(\sum_{j=1}^{k+1} c_j \underset{\sim}{v}_j\right) &= f\left(\sum_{j=1}^{k} c_j \underset{\sim}{v}_j + c_{k+1}\underset{\sim}{v}_{k+1}\right) && \text{(split off the last term)} \\
&= f\left(\sum_{j=1}^{k} c_j \underset{\sim}{v}_j\right) + f(c_{k+1}\underset{\sim}{v}_{k+1}) && \text{(additivity)} \\
&= \sum_{j=1}^{k} c_j f(\underset{\sim}{v}_j) + f(c_{k+1}\underset{\sim}{v}_{k+1}) && \text{(induction hypothesis)} \\
&= \sum_{j=1}^{k} c_j f(\underset{\sim}{v}_j) + c_{k+1} f(\underset{\sim}{v}_{k+1}) && \text{(homogeneity)} \\
&= \sum_{j=1}^{k+1} c_j f(\underset{\sim}{v}_j) && \text{(join the last term back on)}
\end{aligned}
$$

□

Exercise 0.35 (Evaluating a linear map on a linear combination)**.** A linear map is determined on a linear combination of vectors by its values on those vectors alone. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear map with

$$
f(\underset{\sim}{v}_1) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad f(\underset{\sim}{v}_2) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.
$$

Find $f(3\underset{\sim}{v}_1 - 2\underset{\sim}{v}_2)$.

Hint

Apply Theorem 0.13 with $c_1 = 3$ and $c_2 = -2$.

Solution

Solution 0.35.

$$
\begin{aligned}
f(3\underset{\sim}{v}_1 - 2\underset{\sim}{v}_2) &= 3f(\underset{\sim}{v}_1) - 2f(\underset{\sim}{v}_2) && \text{(Theorem 0.13)} \\
&= 3\begin{bmatrix} 1 \\ 2 \end{bmatrix} - 2\begin{bmatrix} 0 \\ -1 \end{bmatrix} && \text{(the given values)} \\
&= \begin{bmatrix} 3 \\ 6 \end{bmatrix} - \begin{bmatrix} 0 \\ -2 \end{bmatrix} && \text{(Definition 0.9)} \\
&= \begin{bmatrix} 3 \\ 8 \end{bmatrix} && \text{(subtracting entry by entry)}
\end{aligned}
$$

Theorem 0.14 (Every linear map is multiplication by a matrix)**.** *If $f : \mathbb{R}^p \to \mathbb{R}^m$ is a linear map (Definition 0.20), then there is exactly one $m \times p$ matrix $\mathbf{A}$ such that*

$$f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x} \quad \text{for all } \underset{\sim}{x} \in \mathbb{R}^p$$

Column j of $\mathbf{A}$ is $f(\underset{\sim}{e}_j)$, the value of f at the jth indicator vector (Definition 0.18); that is, $a_{ij} = \big(f(\underset{\sim}{e}_j)\big)_i$.

Proof

Proof. **Existence.** Let $\mathbf{A}$ be the $m \times p$ matrix with $a_{ij} = \big(f(\underset{\sim}{e}_j)\big)_i$. For any $\underset{\sim}{x} \in \mathbb{R}^p$,

$$\begin{aligned} f(\underset{\sim}{x}) &= f\left(\sum_{j=1}^{p} x_j \underset{\sim}{e}_j\right) && \text{(Theorem 0.4)} \\ &= \sum_{j=1}^{p} x_j f(\underset{\sim}{e}_j) && \text{(Theorem 0.13)} \end{aligned}$$

and entry i of that sum is

$$\begin{aligned} \left(\sum_{j=1}^{p} x_j f(\underset{\sim}{e}_j)\right)_i &= \sum_{j=1}^{p} x_j \big(f(\underset{\sim}{e}_j)\big)_i && \text{(Definition 0.9, entry by entry)} \\ &= \sum_{j=1}^{p} x_j a_{ij} && \text{(definition of } \mathbf{A}\text{)} \\ &= \sum_{j=1}^{p} a_{ij} x_j && \text{(products are symmetric)} \\ &= (\mathbf{A}\underset{\sim}{x})_i && \text{(Definition 0.16)} \end{aligned}$$

Uniqueness. Suppose $\mathbf{B}$ is any $m \times p$ matrix with $f(\underset{\sim}{x}) = \mathbf{B}\underset{\sim}{x}$ for all $\underset{\sim}{x}$. Taking $\underset{\sim}{x} = \underset{\sim}{e}_j$, for each i:

$$\begin{aligned} \big(f(\underset{\sim}{e}_j)\big)_i &= (\mathbf{B}\underset{\sim}{e}_j)_i && \text{(assumption on } \mathbf{B}\text{)} \\ &= \sum_{k=1}^{p} b_{ik} (\underset{\sim}{e}_j)_k && \text{(Definition 0.16)} \\ &= b_{ij} \cdot 1 + \sum_{k \neq j} b_{ik} \cdot 0 && \text{(Definition 0.18, splitting off the } k = j \text{ term)} \\ &= b_{ij} + 0 && \text{(times 1 changes nothing; times 0 gives 0)} \\ &= b_{ij} && \text{(adding 0 changes nothing)} \end{aligned}$$

So

$$
\begin{aligned}
b_{ij} &= \left(f(\underset{\sim}{e}_j)\right)_i \\
&= a_{ij}
\end{aligned}
$$

for every i and j, and $\mathbf{B} = \mathbf{A}$. □

Definition 0.22 (Matrix of a linear map)**.** The **matrix of** a linear map $f : \mathbb{R}^p \to \mathbb{R}^m$ is the unique $m \times p$ matrix $\mathbf{A}$ with $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x}$ for all $\underset{\sim}{x}$ (Theorem 0.14). Its column j is $f(\underset{\sim}{e}_j)$.

Example 0.19 (Finding the matrix of a linear map)**.** For $f(\underset{\sim}{x}) = (2x_1,\ x_1 + x_2)$ from Example 0.16, the values of f at the indicator vectors are

$$
f(\underset{\sim}{e}_1) = f\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = \begin{bmatrix}2\cdot 1\\1+0\end{bmatrix} = \begin{bmatrix}2\\1\end{bmatrix}, \qquad f(\underset{\sim}{e}_2) = f\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = \begin{bmatrix}2\cdot 0\\0+1\end{bmatrix} = \begin{bmatrix}0\\1\end{bmatrix}
$$

and these are the columns of the matrix of f (Definition 0.22): $\mathbf{A} = \begin{bmatrix}2 & 0\\1 & 1\end{bmatrix}$.

Exercise 0.36 (Evaluating a linear map from its values on the indicator vectors)**.** By Theorem 0.14, the values of a linear map at the indicator vectors determine it everywhere. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear map with

$$
f(\underset{\sim}{e}_1) = \begin{bmatrix}1\\3\end{bmatrix}, \qquad f(\underset{\sim}{e}_2) = \begin{bmatrix}-2\\0\end{bmatrix}.
$$

Find $f\left(\begin{bmatrix}4\\1\end{bmatrix}\right)$.

Hint

Build the matrix of f from the two given values, then multiply.

Solution

Solution 0.36. By Theorem 0.14, $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x}$, where column j of $\mathbf{A}$ is $f(\underset{\sim}{e}_j)$:

$$
\mathbf{A} = \begin{bmatrix}1 & -2\\3 & 0\end{bmatrix}.
$$

So

$$
f\left(\begin{bmatrix}4\\1\end{bmatrix}\right) = \begin{bmatrix}1\cdot 4 + (-2)\cdot 1\\3\cdot 4 + 0\cdot 1\end{bmatrix} = \begin{bmatrix}2\\12\end{bmatrix}.
$$

Exercise 0.37 (Finding the matrix of a linear map into $\mathbb{R}^3$)**.** The matrix of a linear map turns the map into a table of numbers that software can store and multiply. Let $f : \mathbb{R}^2 \to \mathbb{R}^3$ be the linear map

$$f(\underset{\sim}{x}) = (x_2,\ x_1,\ x_1 - x_2).$$

Find the matrix of f.

Hint

Column j of the matrix is $f(\underset{\sim}{e}_j)$.

Solution

Solution 0.37. The values of f at the indicator vectors are

$$f(\underset{\sim}{e}_1) = f\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = \begin{bmatrix}0\\1\\1\end{bmatrix}, \qquad f(\underset{\sim}{e}_2) = f\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = \begin{bmatrix}1\\0\\-1\end{bmatrix}.$$

These values are the columns of the matrix of f:

$$\mathbf{A} = \begin{bmatrix}0 & 1\\1 & 0\\1 & -1\end{bmatrix}.$$

It is 3×2, since f sends $\mathbb{R}^2$ to $\mathbb{R}^3$.

Corollary 0.1 (Linear functions from vectors to numbers)**.** *A function $f : \mathbb{R}^p \to \mathbb{R}$ is linear (Definition 0.20) if and only if there is a vector $\underset{\sim}{w} \in \mathbb{R}^p$ such that*

$$f(\underset{\sim}{x}) = \underset{\sim}{w} \cdot \underset{\sim}{x} \quad \textit{for all } \underset{\sim}{x} \in \mathbb{R}^p$$

and then $w_j = f(\underset{\sim}{e}_j)$ for each j.

Proof

Proof. **If f is linear.** Theorem 0.14 with $m = 1$ gives a $1 \times p$ matrix $\mathbf{A}$ with $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x}$ and $a_{1j} = f(\underset{\sim}{e}_j)$. Let $\underset{\sim}{w}$ be the vector with $w_j = a_{1j}$, so that $\mathbf{A} = \underset{\sim}{w}^\top$. Then

$$\begin{aligned} f(\underset{\sim}{x}) &= \mathbf{A}\underset{\sim}{x} && \text{(Theorem 0.14)}\\ &= \underset{\sim}{w} \cdot \underset{\sim}{x} && (\mathbf{A} = \underset{\sim}{w}^\top) \end{aligned}$$

and

$$\begin{aligned} w_j &= a_{1j} \\ &= f(\underset{\sim}{e}_j). \end{aligned}$$

If $f(\underset{\sim}{x}) = \underset{\sim}{w}^\top \underset{\sim}{x}$. Here $\underset{\sim}{w}^\top$ is a $1 \times p$ matrix, so f is linear by Theorem 0.12. □

Example 0.20 (A linear function from $\mathbb{R}^2$ to $\mathbb{R}$)**.** $f(\underset{\sim}{x}) = 3x_1 + 5x_2$ is $\underset{\sim}{w} \cdot \underset{\sim}{x}$ with $\underset{\sim}{w} = (3, 5)^\top$, so it is linear by Corollary 0.1. The entries of $\underset{\sim}{w}$ are the values at the indicator vectors:

$$\begin{aligned} f(\underset{\sim}{e}_1) &= 3 \cdot 1 + 5 \cdot 0 \\ &= 3 \end{aligned}$$

and

$$\begin{aligned} f(\underset{\sim}{e}_2) &= 3 \cdot 0 + 5 \cdot 1 \\ &= 5. \end{aligned}$$

$g(\underset{\sim}{x}) = 3x_1 + 5x_2 + 1$ is not linear, because $g(\underset{\sim}{0}) = 1 \neq 0$ (Theorem 0.11).

Exercise 0.38 (Finding the weight vector of a linear score)**.** A linear score of p measurements is a dot product of the measurements with a vector of weights, by Corollary 0.1. Let $f : \mathbb{R}^3 \to \mathbb{R}$ be a linear map with

$$f(\underset{\sim}{e}_1) = 2, \qquad f(\underset{\sim}{e}_2) = 0, \qquad f(\underset{\sim}{e}_3) = -1.$$

Find the vector $\underset{\sim}{w}$ with $f(\underset{\sim}{x}) = \underset{\sim}{w} \cdot \underset{\sim}{x}$ for every $\underset{\sim}{x} \in \mathbb{R}^3$.

Hint

Corollary 0.1 gives each entry w_j of $\underset{\sim}{w}$.

Solution

Solution 0.38. By Corollary 0.1, $w_j = f(\underset{\sim}{e}_j)$ for each j, so

$$\underset{\sim}{w} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}.$$

Then $f(\underset{\sim}{x}) = 2x_1 - x_3$ for every $\underset{\sim}{x} \in \mathbb{R}^3$.

Videos

Two chapters of the *Essence of linear algebra* series by 3Blue1Brown show the results above geometrically:

- Linear transformations and matrices (Chapter 3) shows why a linear map is fixed by where it sends the indicator vectors, which become the columns of its matrix (Definition 0.22).
- Dot products and duality (Chapter 9) shows why every linear map from $\mathbb{R}^p$ to $\mathbb{R}$ is a dot product with a fixed vector (Corollary 0.1).

Definition 0.23 (Translation)**.** For a fixed vector $\underset{\sim}{b} \in \mathbb{R}^m$, the **translation** [25] by $\underset{\sim}{b}$ is the function $t_{\underset{\sim}{b}} : \mathbb{R}^m \to \mathbb{R}^m$ defined by

$$t_{\underset{\sim}{b}}(\underset{\sim}{y}) = \underset{\sim}{y} + \underset{\sim}{b}$$

Example 0.21 (Translating two points)**.** The translation by $\underset{\sim}{b} = (1, -2)^\top$ sends

$$t_{\underset{\sim}{b}}\left(\begin{bmatrix}0\\0\end{bmatrix}\right) = \begin{bmatrix}0+1\\0+(-2)\end{bmatrix} = \begin{bmatrix}1\\-2\end{bmatrix}, \qquad t_{\underset{\sim}{b}}\left(\begin{bmatrix}3\\4\end{bmatrix}\right) = \begin{bmatrix}3+1\\4+(-2)\end{bmatrix} = \begin{bmatrix}4\\2\end{bmatrix}$$

Both points move one unit right and two units down.

Remark (A translation moves every point by the same vector)

Remark 0.5 (A translation moves every point by the same vector)*.* A translation by $\underset{\sim}{b}$ moves every point by the same vector $\underset{\sim}{b}$.

Exercise 0.39 (Finding the translation that moves one point to another)**.** A translation moves every point by the same vector, so where it sends one point fixes the whole translation. A translation $t_{\underset{\sim}{b}}$ of $\mathbb{R}^2$ sends $(2, 3)^\top$ to $(5, 1)^\top$.
Find $\underset{\sim}{b}$.

Hint

Solve $(2, 3)^\top + \underset{\sim}{b} = (5, 1)^\top$ entry by entry.

Solution

Solution 0.39. By Definition 0.23, $t_{\underset{\sim}{b}}\left((2, 3)^\top\right) = (2, 3)^\top + \underset{\sim}{b}$. Matching this vector with $(5, 1)^\top$ entry by entry gives

$$\underset{\sim}{b} = \begin{bmatrix} 5-2 \\ 1-3 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}.$$

Definition 0.24.

Affine map, affine transformation, affine function
A function $f : \mathbb{R}^p \to \mathbb{R}^m$ is an **affine map** (also called an **affine transformation** or **affine function**) if it is a matrix product followed by a translation (Definition 0.23) (a composition, with the matrix product as the inner function): there are an $m \times p$ matrix $\mathbf{A}$ and a vector $\underset{\sim}{b} \in \mathbb{R}^m$ such that

$$\begin{aligned} f(\underset{\sim}{x}) &= t_{\underset{\sim}{b}}(\mathbf{A}\underset{\sim}{x}) \\ &= \mathbf{A}\underset{\sim}{x} + \underset{\sim}{b} \quad \text{for all } \underset{\sim}{x} \in \mathbb{R}^p \end{aligned}$$

Exercise 0.40 (Testing whether a rescale-and-shift is an affine map)**.** Rescaling a measurement and then shifting it, as when converting units with a different zero point, is the typical affine map. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be

$$f(\underset{\sim}{x}) = (2x_1 + 1,\ x_2 - 3).$$

Is f an affine map?

Hint

Look for a 2×2 matrix $\mathbf{A}$ and a vector $\underset{\sim}{b}$ with $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x} + \underset{\sim}{b}$.

Solution

Solution 0.40. Yes. Let

$$\mathbf{A} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \underset{\sim}{b} = \begin{bmatrix} 1 \\ -3 \end{bmatrix}.$$

By Definition 0.16, $\mathbf{A}\underset{\sim}{x} = (2x_1, x_2)^\top$, so

$$\mathbf{A}\underset{\sim}{x} + \underset{\sim}{b} = \begin{bmatrix} 2x_1 + 1 \\ x_2 - 3 \end{bmatrix} = f(\underset{\sim}{x})$$

for every $\underset{\sim}{x} \in \mathbb{R}^2$. So f is an affine map.

Definition 0.25 (Offset of an affine map, intercept, bias)**.** In Definition 0.24, the vector $\underset{\sim}{b}$ is called the **offset** of f (statistics also calls it the **intercept**, and machine learning the **bias**).

Remark (Real-valued affine maps)

Remark 0.6 (Real-valued affine maps)*.* For a real-valued function (Definition 0.4, $m = 1$), $\mathbf{A}$ is a single row, written $\underset{\sim}{w}^\top$, and $\underset{\sim}{b}$ is a single number b, so an affine function is a linear function $\underset{\sim}{w} \cdot \underset{\sim}{x}$ (Corollary 0.1) plus an offset:

$$f(\underset{\sim}{x}) = \underset{\sim}{w} \cdot \underset{\sim}{x} + b, \qquad \underset{\sim}{w} \in \mathbb{R}^p, \quad b \in \mathbb{R} \tag{4}$$

Exercise 0.41 (Reading off the offset of an affine map)**.** In a fitted model, the offset is the part of the output that does not depend on the input. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be the affine map

$$f(\underset{\sim}{x}) = (x_1 - x_2 + 4,\ 3x_2 - 1).$$

Find the offset of f.

Hint

Write $f(\underset{\sim}{x})$ as $\mathbf{A}\underset{\sim}{x} + \underset{\sim}{b}$ as in Definition 0.24.

Solution

Solution 0.41.

$$f(\underset{\sim}{x}) = \begin{bmatrix} 1 & -1 \\ 0 & 3 \end{bmatrix} \underset{\sim}{x} + \begin{bmatrix} 4 \\ -1 \end{bmatrix},$$

so the offset of f is $\underset{\sim}{b} = (4, -1)^\top$. Each entry of $\underset{\sim}{b}$ is the term with no x_j in the matching entry of $f(\underset{\sim}{x})$.

Definition 0.26 (Augmented vector)**.** The **augmented vector** of $\underset{\sim}{x} \in \mathbb{R}^p$ is the vector $\underset{\sim}{x}^{\text{aug}} \in \mathbb{R}^{p+1}$ formed by placing a 1 above the entries of $\underset{\sim}{x}$:

$$\begin{aligned} \underset{\sim}{x}^{\text{aug}} &= \begin{bmatrix} 1 \\ \underset{\sim}{x} \end{bmatrix} \\ &= (1, x_1, \dots, x_p)^\top \end{aligned}$$

Example 0.22 (An affine function as a dot product with an augmented vector)**.** For $\underset{\sim}{x} = (2, 5)^\top$, the augmented vector is $\underset{\sim}{x}^{\text{aug}} = (1, 2, 5)^\top$. With $\underset{\sim}{w} = (3, -1)^\top$ and $b = 4$, the affine function $\underset{\sim}{w} \cdot \underset{\sim}{x} + b$ (Equation 4) equals the dot product of $(b, w_1, w_2)^\top = (4, 3, -1)^\top$ with $\underset{\sim}{x}^{\text{aug}}$:

$$\begin{aligned} \underset{\sim}{w} \cdot \underset{\sim}{x} + b &= 3 \cdot 2 + (-1) \cdot 5 + 4 \\ &= 5, \\ (4, 3, -1)^\top \cdot \underset{\sim}{x}^{\text{aug}} &= 4 \cdot 1 + 3 \cdot 2 + (-1) \cdot 5 \\ &= 5 \end{aligned}$$

Exercise 0.42 (Absorbing an offset into a dot product with the augmented vector)**.** The augmented vector lets software treat an affine score as a dot product, with the offset as one more weight. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be

$$f(\underset{\sim}{x}) = 2x_1 - x_2 + 5.$$

Find the vector $\underset{\sim}{v} \in \mathbb{R}^3$ with $f(\underset{\sim}{x}) = \underset{\sim}{v} \cdot \underset{\sim}{x}^{\text{aug}}$ for every $\underset{\sim}{x} \in \mathbb{R}^2$.

Hint

Follow Example 0.22. The first entry of $\underset{\sim}{x}^{\text{aug}}$ is 1.

Solution

Solution 0.42. Let $\underset{\sim}{v} = (5, 2, -1)^\top$. Since $\underset{\sim}{x}^{\text{aug}} = (1, x_1, x_2)^\top$ by Definition 0.26,

$$\underset{\sim}{v} \cdot \underset{\sim}{x}^{\text{aug}} = 5 \cdot 1 + 2x_1 + (-1)x_2 = f(\underset{\sim}{x}).$$

The first entry of $\underset{\sim}{v}$ is the offset 5, and the other two are the numbers multiplying x_1 and x_2.

Theorem 0.15 (An affine map sends zero to its offset)**.** *An affine map $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x} + \underset{\sim}{b}$ (Definition 0.24) satisfies $f(\underset{\sim}{0}) = \underset{\sim}{b}$.*

Proof

Proof. First, $\mathbf{A}\underset{\sim}{0} = \underset{\sim}{0}$: for each i,

$$
\begin{aligned}
(\mathbf{A}\underset{\sim}{0})_i &= \sum_{j=1}^{p} a_{ij} \cdot 0 && \text{(Definition 0.16)} \\
&= \sum_{j=1}^{p} 0 && \text{(any number times 0 is 0)} \\
&= 0 && \text{(a sum of zeros is 0)}
\end{aligned}
$$

Then

$$
\begin{aligned}
f(\underset{\sim}{0}) &= \mathbf{A}\underset{\sim}{0} + \underset{\sim}{b} && \text{(Definition 0.24)} \\
&= \underset{\sim}{0} + \underset{\sim}{b} && (\mathbf{A}\underset{\sim}{0} = \underset{\sim}{0}) \\
&= \underset{\sim}{b} && \text{(adding } \underset{\sim}{0} \text{ changes no entry; Definition 0.7)}
\end{aligned}
$$

□

Exercise 0.43 (Recovering a column of an affine map's matrix from two values)**.** Theorem 0.15 says that the value of an affine map at $\underset{\sim}{0}$ is its offset, so subtracting that value isolates the matrix part. Let $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x} + \underset{\sim}{b}$ be an affine map from $\mathbb{R}^2$ to $\mathbb{R}^2$ with

$$
f(\underset{\sim}{0}) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 4 \\ 2 \end{bmatrix}.
$$

Find the first column of $\mathbf{A}$.

Hint

$\mathbf{A}\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is the first column of $\mathbf{A}$.

Solution

Solution 0.43. By Theorem 0.15, $\underset{\sim}{b} = f(\underset{\sim}{0}) = (1, 2)^\top$. So

$$
\begin{aligned}
\mathbf{A}\begin{bmatrix} 1 \\ 0 \end{bmatrix} &= f\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) - \underset{\sim}{b} && \text{(Definition 0.24)} \\
&= \begin{bmatrix} 4-1 \\ 2-2 \end{bmatrix} && \text{(the given values)} \\
&= \begin{bmatrix} 3 \\ 0 \end{bmatrix}.
\end{aligned}
$$

By Definition 0.16, $\mathbf{A}(1, 0)^\top$ is the first column of $\mathbf{A}$, which is therefore $(3, 0)^\top$.

Theorem 0.16 (An affine map is linear exactly when its offset is zero)**.** *An affine map* $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x} + \underset{\sim}{b}$ *(Definition 0.24) is linear (Definition 0.20) if and only if* $\underset{\sim}{b} = \underset{\sim}{0}$.

Proof

Proof. If $\underset{\sim}{b} = \underset{\sim}{0}$, then

$$\begin{aligned} f(\underset{\sim}{x}) &= \mathbf{A}\underset{\sim}{x} + \underset{\sim}{0} \\ &= \mathbf{A}\underset{\sim}{x}, \end{aligned}$$

which is linear by Theorem 0.12. Conversely, if f is linear, then $f(\underset{\sim}{0}) = \underset{\sim}{0}$ by Theorem 0.11, and $f(\underset{\sim}{0}) = \underset{\sim}{b}$ by Theorem 0.15, so $\underset{\sim}{b} = \underset{\sim}{0}$. □

Example 0.23 (Two affine maps with the same matrix)**.** Let $\mathbf{A} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$, the matrix of the linear map in Example 0.16. $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x}$ has offset $\underset{\sim}{0}$, so it is linear. $g(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x} + \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ has offset $\begin{bmatrix} 1 \\ 0 \end{bmatrix} \neq \underset{\sim}{0}$, so by Theorem 0.16 it is not linear: indeed $g(\underset{\sim}{0}) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, not $\underset{\sim}{0}$.

Exercise 0.44 (Choosing a constant that makes an affine map linear)**.** By Theorem 0.16, an affine map is a linear map exactly when its offset is $\underset{\sim}{0}$. For $c \in \mathbb{R}$, let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be

$$f(\underset{\sim}{x}) = (x_1 - x_2 + c - 2,\ 3x_1 + c^2 - 4).$$

For which values of c is f linear?

Hint

Find the offset of f as a function of c.

Solution

Solution 0.44. Only for $c = 2$. The map is

$$f(\underset{\sim}{x}) = \begin{bmatrix} 1 & -1 \\ 3 & 0 \end{bmatrix} \underset{\sim}{x} + \begin{bmatrix} c - 2 \\ c^2 - 4 \end{bmatrix},$$

so its offset is $(c - 2,\ c^2 - 4)^\top$. By Theorem 0.16, f is linear exactly when both entries are 0. The first entry is 0 only for $c = 2$, and then the second entry is $2^2 - 4 = 0$ too.

Corollary 0.2 (Every linear map is affine)**.** *Every linear map $f : \mathbb{R}^p \to \mathbb{R}^m$ (Definition 0.20) is an affine map (Definition 0.24) with offset $\underset{\sim}{0}$.*

Proof

Proof. If f is linear, Theorem 0.14 gives a matrix $\mathbf{A}$ with

$$\begin{aligned} f(\underset{\sim}{x}) &= \mathbf{A}\underset{\sim}{x} \\ &= \mathbf{A}\underset{\sim}{x} + \underset{\sim}{0}, \end{aligned}$$

which is Definition 0.24 with $\underset{\sim}{b} = \underset{\sim}{0}$. □

Exercise 0.45 (Testing whether a linear score is an affine map)**.** Some results hold only for affine maps, so it helps to check whether a given score qualifies. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be $f(\underset{\sim}{x}) = x_1 - 4x_2$, a linear map.
Is f an affine map?

Hint

Use Corollary 0.2.

Solution

Solution 0.45. Yes. By Corollary 0.2, every linear map is an affine map with offset $\underset{\sim}{0}$. Here

$$f(\underset{\sim}{x}) = \begin{bmatrix} 1 & -4 \end{bmatrix} \underset{\sim}{x} + 0,$$

with the 1×2 matrix $\begin{bmatrix} 1 & -4 \end{bmatrix}$ and offset 0.

Theorem 0.17 (An affine map determines its matrix and offset)**.** *If $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x} + \underset{\sim}{b}$ and $f(\underset{\sim}{x}) = \mathbf{A}'\underset{\sim}{x} + \underset{\sim}{b}'$ for all $\underset{\sim}{x} \in \mathbb{R}^p$ (Definition 0.24), then $\mathbf{A}' = \mathbf{A}$ and $\underset{\sim}{b}' = \underset{\sim}{b}$.*

Proof

Proof. By Theorem 0.15, applied to each formula,

$$\begin{aligned} \underset{\sim}{b} &= f(\underset{\sim}{0}) \\ &= \underset{\sim}{b}'. \end{aligned}$$

Then

$$\begin{aligned}\mathbf{A}\underset{\sim}{x} &= f(\underset{\sim}{x}) - \underset{\sim}{b} \\ &= \mathbf{A}'\underset{\sim}{x}\end{aligned}$$

for all $\underset{\sim}{x}$, so $\mathbf{A}$ and $\mathbf{A}'$ are both matrices (Definition 0.22) of the linear map $\underset{\sim}{x} \mapsto f(\underset{\sim}{x}) - \underset{\sim}{b}$ (linear by Theorem 0.12), and the uniqueness in Theorem 0.14 gives $\mathbf{A}' = \mathbf{A}$. □

Exercise 0.46 (Matching the offsets of two formulas for one affine map)**.** Theorem 0.17 says that an affine map has only one matrix and one offset, so two ways of writing the same map must agree term by term. For $c \in \mathbb{R}$, suppose that

$$\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \underset{\sim}{x} + \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \underset{\sim}{x} + \begin{bmatrix} c \\ 1 \end{bmatrix} \quad \text{for every } \underset{\sim}{x} \in \mathbb{R}^2.$$

Find c.

Hint

Compare the offsets of the two sides.

Solution

Solution 0.46. $c = 3$. Both sides describe the same affine map. By Theorem 0.17, the two sides have the same offset, so $(3, 1)^\top = (c, 1)^\top$ and $c = 3$. Setting $\underset{\sim}{x} = \underset{\sim}{0}$ gives the same answer directly.

Remark ("Linear" in this site, and in machine learning and statistics)

Remark 0.7 ("Linear" in this site, and in machine learning and statistics). This site uses "linear" only in the sense of Definition 0.20, so by Theorem 0.16 a function with a nonzero offset is affine but not linear. For functions of one variable, this is the same split as affine $ax + b$ versus linear ax on the algebra page. (The algebra page writes these as $mx + b$ and mx; here m already names the number of outputs.)

Machine learning and statistics often call any function of the form Equation 4 "linear", as in a *linear model* in statistics or a *linear layer* in a neural network, and call the offset b the bias (machine learning) or the intercept (statistics), as Definition 0.24 notes. A model with an intercept is affine in $\underset{\sim}{x}$ but linear in its coefficients $(b, \underset{\sim}{w})$, and that is the sense of "linear" in "linear model": it stays linear in the coefficients even when the inputs are transformed: $w_1 x + w_2 x^2 + b$ is not affine in x, but it is linear in (b, w_1, w_2). Separately, the affine function $\underset{\sim}{w} \cdot \underset{\sim}{x} + b$ is a linear function of the augmented vector $\underset{\sim}{x}^{\text{aug}}$ (Definition 0.26), with b as the coefficient on its first entry.

Further reading

- Zhou [26], sections 6 ("Linear functions and operators"; see Definition 0.21) and 7 ("Affine functions"), proves that every linear function is a matrix-vector product by the same route as Theorem 0.14, and that every affine function has the form $\mathbf{A}\underset{\sim}{x} + \underset{\sim}{b}$. It defines "affine" differently, by $f(\alpha\underset{\sim}{x} + \beta y) = \alpha f(\underset{\sim}{x}) + \beta f(y)$ whenever $\alpha + \beta = 1$, and shows that this condition holds exactly for the maps $\mathbf{A}\underset{\sim}{x} + \underset{\sim}{b}$, so the two definitions agree.
- Boyd and Vandenberghe [27], chapter 2 ("Linear functions"), covers the vector-to-number case: linear functions as inner products, affine functions, the first-order Taylor approximation as an affine function, and the regression model. Section 8.1 ("Linear and affine functions") covers the vector-to-vector case. A free PDF is on the book's website.

Definition 0.27 (Hyperplane)**.** A **hyperplane** [28] in $\mathbb{R}^p$ is the set of points $\underset{\sim}{x} \in \mathbb{R}^p$ satisfying

$$\underset{\sim}{w} \cdot \underset{\sim}{x} + b = 0$$

for some vector $\underset{\sim}{w} \in \mathbb{R}^p \setminus \{\underset{\sim}{0}\}$ and number $b \in \mathbb{R}$. The number b is the offset of the hyperplane.

Definition 0.28 (Normal vector of a hyperplane)**.** In Definition 0.27, the vector $\underset{\sim}{w}$ is a **normal vector** [29] of the hyperplane.

Example 0.24 (A line in $\mathbb{R}^2$, and why the normal vector must be nonzero)**.**

- With $\underset{\sim}{w} = (1, 1)$ and $b = -1$, the hyperplane in $\mathbb{R}^2$ is the line $x_1 + x_2 - 1 = 0$. The point $(1, 0)$ is on it, since $1+0-1 = 0$; the point $(1, 1)$ is not, since $1+1-1 = 1 > 0$, so it lies in the half-space where $\underset{\sim}{w} \cdot \underset{\sim}{x} + b > 0$.
- With $\underset{\sim}{w} = \underset{\sim}{0}$, the equation reads $b = 0$: if $b \neq 0$ no point satisfies it, and if $b = 0$ every point does, so the set is empty or all of $\mathbb{R}^2$, not a line. That degenerate case is why the definition requires $\underset{\sim}{w} \neq \underset{\sim}{0}$.

Exercise 0.47 (Testing whether a vertical line is a hyperplane)**.** A vertical line in $\mathbb{R}^2$ is parallel to the x_2-axis. It cannot be written as $x_2 = mx_1 + k$ for any numbers m and k. Let L be the set of points $\underset{\sim}{x} \in \mathbb{R}^2$ with $x_1 = 3$.
Is L a hyperplane in $\mathbb{R}^2$?

Hint

Look for $\underset{\sim}{w} \neq \underset{\sim}{0}$ and b in Definition 0.27.

Solution

Solution 0.47. Yes. Take $\underset{\sim}{w} = (1, 0)^\top \neq \underset{\sim}{0}$ and $b = -3$. Then

$$\underset{\sim}{w} \cdot \underset{\sim}{x} + b = x_1 - 3,$$

which is 0 exactly when $x_1 = 3$. So L is the hyperplane with these $\underset{\sim}{w}$ and b.

Exercise 0.48 (Finding a normal vector of a plane in $\mathbb{R}^3$)**.** In $\mathbb{R}^3$ a hyperplane is a plane, and a normal vector points straight out of it. Let H be the hyperplane of points $\underset{\sim}{x} \in \mathbb{R}^3$ with

$$2x_1 - x_2 + 4x_3 = 7.$$

Find a normal vector of H.

Hint

Rewrite the equation in the form of Definition 0.27.

Solution

Solution 0.48. Subtracting 7 from both sides gives

$$2x_1 - x_2 + 4x_3 - 7 = 0,$$

which is $\underset{\sim}{w} \cdot \underset{\sim}{x} + b = 0$ with $\underset{\sim}{w} = (2, -1, 4)^\top$ and $b = -7$. So $\underset{\sim}{w} = (2, -1, 4)^\top$ is a normal vector of H. Any nonzero multiple of it is one too, since multiplying the equation by a nonzero number leaves H unchanged.

The course website's version of this page has an interactive plot here.

Exercise 0.49 (Write a rule in matrix form)**.** A clinic scores a patient's risk from three measurements.

symbol	meaning	units
x_1	age	years
x_2	systolic blood pressure	mmHg
x_3	body mass index	$\mathrm{kg/m^2}$

The rule in use is

$$\text{score} = 0.4\,x_1 + 0.2\,x_2 + 1.5\,x_3 - 30$$

Write this rule in the form of Equation 4, naming $\underset{\sim}{w}$, b and p explicitly.

Solution

Solution 0.49. Read the coefficients straight off the rule. There are three inputs, so $p = 3$:

$$\underset{\sim}{w} = \begin{bmatrix} 0.4 \\ 0.2 \\ 1.5 \end{bmatrix}, \qquad b = -30$$

and the rule is $f(\underset{\sim}{x}) = \underset{\sim}{w}^\top \underset{\sim}{x} + b$.

Exercise 0.50 (Score a patient)**.** Give the score from the rule in Exercise 0.49 for a patient aged 50 with blood pressure 130 and BMI 28.

Solution

Solution 0.50. For the patient,

$$\underset{\sim}{x} = \begin{bmatrix} 50 \\ 130 \\ 28 \end{bmatrix}$$

$$\begin{aligned} \underset{\sim}{w}^\top \underset{\sim}{x} &= (0.4)(50) + (0.2)(130) + (1.5)(28) \\ &= 20 + 26 + 42 \\ &= 88 \end{aligned}$$

$$\begin{aligned} f(\underset{\sim}{x}) &= 88 - 30 \\ &= 58 \end{aligned}$$

Two things are worth noticing. The coefficients are not comparable as they stand. The weight on BMI, 1.5, is 7.5 times the weight on blood pressure, 0.2, but the contribution of BMI, 42, is less than twice the contribution of blood pressure, 26, because the BMI value, 28, is much smaller than the blood pressure value, 130. So the weight on a feature says nothing on its own about how much that feature contributes to the score. And $b = -30$ changes the score: without it, the score would be 88 instead of 58.

Exercise 0.51 (Which of these are affine in $\underset{\sim}{w}$?)**.** Each of the five expressions is a function of the weights $\underset{\sim}{w} \in \mathbb{R}^p$, with the data $\underset{\sim}{x}$ and y held fixed. Say which ones are affine (Definition 0.24), that is, have the form Equation 4 — an inner product plus an offset — and which of those are also linear (Definition 0.20).

1. $\underset{\sim}{w}^\top \underset{\sim}{x}$
2. $\underset{\sim}{w}^\top \underset{\sim}{x} - y$
3. $(\underset{\sim}{w}^\top \underset{\sim}{x} - y)^2$
4. $\|\underset{\sim}{w}\|_2^2$
5. $w_1 x_1 + 3$

Solution

Solution 0.51.

1. **Affine, and linear.** It is Equation 4 with $b = 0$, which is a linear map by Theorem 0.12: the matrix $\underset{\sim}{x}^\top$ times $\underset{\sim}{w}$.
2. **Affine.** Still an inner product plus an offset; here the offset is $b = -y$, which is a constant because y is held fixed. It is linear only when $y = 0$: otherwise $\underset{\sim}{w} = \underset{\sim}{0}$ gives $-y \neq 0$, and a linear map sends $\underset{\sim}{0}$ to 0 (Theorem 0.11).
3. **Not affine.** Squaring is not an inner product plus an offset. This expression is the squared error of one observation, and its curvature in $\underset{\sim}{w}$ is exactly why fitting a model is an optimization problem rather than a lookup.
4. **Not affine.** By Equation 2 this is $\underset{\sim}{w}^\top \underset{\sim}{w}$, which has $\underset{\sim}{w}$ in both slots at once. Hold either slot fixed and it is linear in the other; as a function of the single variable $\underset{\sim}{w}$ it is quadratic.
5. **Affine, not linear.** It is Equation 4 with coefficient vector $\underset{\sim}{a} = (x_1, 0, \ldots, 0)^\top$ and $b = 3$; at $\underset{\sim}{w} = \underset{\sim}{0}$ it equals $3 \neq 0$, so it is not linear (Theorem 0.11). The data $\underset{\sim}{x}$ is untouched — $\underset{\sim}{a}$ is a new vector built from its first entry.

Items 1, 2 and 5 are affine and items 3 and 4 are not, which is the split that matters: a *model* can be affine in its weights while the *quantity we minimize* is not. Squared error and the norm penalty are both curved in $\underset{\sim}{w}$, and that curvature is what makes fitting an optimization problem at all: there is always a downhill direction to follow. Curvature alone does not promise a *single* answer. The squared error of one observation is completely flat along a whole line of weight vectors, every one of which fits that observation exactly, and pinning down one of them is part of what the norm penalty is added for in week 3.

Transposes of sums and products

Theorem 0.18 (Transpose of a sum)**.** *For $m \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$:*

$$(\mathbf{A}+\mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top$$

In particular, for column vectors $\underset{\sim}{x}$ and $\underset{\sim}{y}$ of the same length:

$$(\underset{\sim}{x}+\underset{\sim}{y})^\top = \underset{\sim}{x}^\top + \underset{\sim}{y}^\top$$

Proof

Proof. Entry (i, j) of each side:

$$\begin{aligned}
\left[(\mathbf{A}+\mathbf{B})^\top\right]_{ij} &= (\mathbf{A}+\mathbf{B})_{ji} && \text{(definition of the transpose)}\\
&= a_{ji} + b_{ji} && \text{(definition of matrix addition)}\\
&= (\mathbf{A}^\top)_{ij} + (\mathbf{B}^\top)_{ij} && \text{(definition of the transpose)}\\
&= \left[\mathbf{A}^\top + \mathbf{B}^\top\right]_{ij} && \text{(definition of matrix addition)}
\end{aligned}$$

□

Exercise 0.52 (Transposing a sum of two matrices)**.** Theorem 0.18 lets a transpose be taken before or after adding, whichever is less work. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 & 1 & 1 \\ 2 & 2 & -3 \end{bmatrix}.$$

Find $(\mathbf{A}+\mathbf{B})^\top$.

Hint

Add first with Definition 0.8, then take the transpose.

Solution

Solution 0.52. Adding entry by entry,

$$\mathbf{A}+\mathbf{B} = \begin{bmatrix} 1 & 3 & 1 \\ 5 & 1 & 1 \end{bmatrix},$$

so

$$(\mathbf{A}+\mathbf{B})^\top = \begin{bmatrix} 1 & 5 \\ 3 & 1 \\ 1 & 1 \end{bmatrix}.$$

By Theorem 0.18, adding $\mathbf{A}^\top = \begin{bmatrix} 1 & 3 \\ 2 & -1 \\ 0 & 4 \end{bmatrix}$ and $\mathbf{B}^\top = \begin{bmatrix} 0 & 2 \\ 1 & 2 \\ 1 & -3 \end{bmatrix}$ gives the same matrix.

Theorem 0.19 (Transpose of a product)**.** *For an $m \times k$ matrix $\mathbf{A}$ and a $k \times n$ matrix $\mathbf{B}$:*

$$\underbrace{(\mathbf{AB})^\top}_{n\times m} = \underbrace{\mathbf{B}^\top}_{n\times k}\ \underbrace{\mathbf{A}^\top}_{k\times m}$$

Proof

Proof. Entry (i, j) of each side:

$$\begin{aligned}
\left[(\mathbf{AB})^\top\right]_{ij} &= (\mathbf{AB})_{ji} && \text{(definition of the transpose)} \\
&= \sum_{s=1}^{k} a_{js}\, b_{si} && \text{(definition of matrix multiplication)} \\
&= \sum_{s=1}^{k} (\mathbf{B}^\top)_{is}\, (\mathbf{A}^\top)_{sj} && \text{(definition of the transpose; reorder each product)} \\
&= \left[\mathbf{B}^\top \mathbf{A}^\top\right]_{ij} && \text{(definition of matrix multiplication)}
\end{aligned}$$

□

Remark (The order of the factors reverses)

Remark 0.8 (The order of the factors reverses). Transposing a product reverses the order of the factors. Keeping the original order usually gives an undefined product or a different matrix. For example, with $\mathbf{A} = [1,\ 2]$ (1×2) and $\mathbf{B} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ (2×1),

$$\begin{aligned}
\mathbf{AB} &= 1 \cdot 3 + 2 \cdot 4 \\
&= 11,
\end{aligned}$$

so $(\mathbf{AB})^\top = 11$, and

$$\mathbf{B}^\top \mathbf{A}^\top = [3,\ 4] \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 11, \qquad \mathbf{A}^\top \mathbf{B}^\top = \begin{bmatrix} 1 \\ 2 \end{bmatrix} [3,\ 4] = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}.$$

If instead $\mathbf{A}$ is 2×3 and $\mathbf{B}$ is 3×1, then $\mathbf{B}^\top \mathbf{A}^\top$ is a $(1 \times 3)(3 \times 2)$ product, but $\mathbf{A}^\top \mathbf{B}^\top$ is a $(3 \times 2)(1 \times 3)$ product, which is undefined.

Exercise 0.53 (The order of the factors in a transposed square product)**.** Remark 0.8 shows that, for factors that are not square, keeping the original order of the factors in the transpose of a product can give an undefined product. For two square matrices of the same size, both $\mathbf{A}^\top\mathbf{B}^\top$ and $\mathbf{B}^\top\mathbf{A}^\top$ are defined, so the shapes alone cannot settle which order gives $(\mathbf{AB})^\top$. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

Does $\mathbf{A}^\top\mathbf{B}^\top$ equal $(\mathbf{AB})^\top$ for these $\mathbf{A}$ and $\mathbf{B}$?

Hint

Compute $\mathbf{AB}$ with 4 and take its transpose, then compute $\mathbf{A}^\top\mathbf{B}^\top$ and compare the two entry by entry.

Solution

Solution 0.53. No: the two matrices differ, so keeping the order of the factors gives the wrong transpose.
The product $\mathbf{AB}$ and its transpose are

$$\begin{aligned} \mathbf{AB} &= \begin{bmatrix} 1\cdot 1 + 2\cdot 1 & 1\cdot 0 + 2\cdot 1 \\ 0\cdot 1 + 1\cdot 1 & 0\cdot 0 + 1\cdot 1 \end{bmatrix} && \text{(Definition 0.10)} \\ &= \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}, && \text{(multiply and add)} \end{aligned}$$

so $(\mathbf{AB})^\top = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$.

The transposes are $\mathbf{A}^\top = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ and $\mathbf{B}^\top = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, so

$$\begin{aligned} \mathbf{A}^\top\mathbf{B}^\top &= \begin{bmatrix} 1\cdot 1 + 0\cdot 0 & 1\cdot 1 + 0\cdot 1 \\ 2\cdot 1 + 1\cdot 0 & 2\cdot 1 + 1\cdot 1 \end{bmatrix} && \text{(Definition 0.10)} \\ &= \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}. && \text{(multiply and add)} \end{aligned}$$

Entry $(1, 1)$ is 3 in $(\mathbf{AB})^\top$ and 1 in $\mathbf{A}^\top\mathbf{B}^\top$. In the reversed order, as Theorem 0.19 says,

$\mathbf{B}^\top\mathbf{A}^\top = \begin{bmatrix} 1\cdot 1+1\cdot 2 & 1\cdot 0+1\cdot 1 \\ 0\cdot 1+1\cdot 2 & 0\cdot 0+1\cdot 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$, which matches $(\mathbf{AB})^\top$.

Theorem 0.20 (A 1×1 product equals its own transpose)**.** *If $\underset{\sim}{u}$ and $\underset{\sim}{v}$ are vectors of length p and $\mathbf{M}$ is a $p\times p$ matrix, then $\underset{\sim}{u}^\top\mathbf{M}\underset{\sim}{v}$ is a number, and*

$$\underset{\sim}{u}^\top\mathbf{M}\underset{\sim}{v} = \underset{\sim}{v}^\top\mathbf{M}^\top\underset{\sim}{u}.$$

In particular, $\underset{\sim}{u}^\top\underset{\sim}{v} = \underset{\sim}{v}^\top\underset{\sim}{u}$.

Proof

Proof. The product $\underset{\sim}{u}^\top\mathbf{M}\underset{\sim}{v}$ is a $(1\times p)(p\times p)(p\times 1)$ product, so it is a 1×1 matrix. The transpose of a 1×1 matrix is itself, so

$$\begin{aligned} \underset{\sim}{u}^\top\mathbf{M}\underset{\sim}{v} &= \left(\underset{\sim}{u}^\top\mathbf{M}\underset{\sim}{v}\right)^\top && \text{(a } 1\times 1 \text{ matrix equals its transpose)} \\ &= \underset{\sim}{v}^\top\mathbf{M}^\top\left(\underset{\sim}{u}^\top\right)^\top && \text{(transpose of a product, twice)} \\ &= \underset{\sim}{v}^\top\mathbf{M}^\top\underset{\sim}{u} && \text{(transposing twice changes nothing)} \end{aligned}$$

The transpose step is Theorem 0.19. Taking $\mathbf{M}=\mathbf{I}$ gives the last statement. □

Example 0.25 (Two ways to write the same number)**.** Take $\underset{\sim}{u}=(1,2)$, $\underset{\sim}{v}=(3,-1)$ and $\mathbf{M}=\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$. Then $\mathbf{M}\underset{\sim}{v}=(1,-3)$, so

$$\underset{\sim}{u}^\top\mathbf{M}\underset{\sim}{v} = 1\cdot 1+2\cdot(-3) = -5.$$

Also $\mathbf{M}^\top=\begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$ and $\mathbf{M}^\top\underset{\sim}{u}=(1,8)$, so

$$\underset{\sim}{v}^\top\mathbf{M}^\top\underset{\sim}{u} = 3\cdot 1+(-1)\cdot 8 = -5.$$

The two products are equal, as the theorem says. In least squares, this is why the two cross terms of $(\mathbf{X}\underset{\sim}{\beta}-\underset{\sim}{y})^\top(\mathbf{X}\underset{\sim}{\beta}-\underset{\sim}{y})$ are equal and can be combined.

Exercise 0.54 (Evaluating a product of a vector, a matrix and a vector)**.** Theorem 0.20 turns $\underset{\sim}{v}^\top\mathbf{M}^\top\underset{\sim}{u}$ back into $\underset{\sim}{u}^\top\mathbf{M}\underset{\sim}{v}$, which can be the easier product of a vector, a matrix and a vector to compute. Let

$$\underset{\sim}{u}=\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad \underset{\sim}{v}=\begin{bmatrix} 2 \\ 3 \end{bmatrix}, \qquad \mathbf{M}=\begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}.$$

Find $\underset{\sim}{v}^\top\mathbf{M}^\top\underset{\sim}{u}$.

Hint

By Theorem 0.20, it is enough to compute $\underset{\sim}{u}^\top \mathbf{M} \underset{\sim}{v}$.

Solution

Solution 0.54. By Theorem 0.20, $\underset{\sim}{v}^\top \mathbf{M}^\top \underset{\sim}{u} = \underset{\sim}{u}^\top \mathbf{M} \underset{\sim}{v}$. By Definition 0.16,

$$\mathbf{M} \underset{\sim}{v} = \begin{bmatrix} 0 \cdot 2 + 1 \cdot 3 \\ 2 \cdot 2 + 0 \cdot 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix},$$

so

$$\underset{\sim}{u}^\top \mathbf{M} \underset{\sim}{v} = 1 \cdot 3 + (-1) \cdot 4 = -1.$$

The answer is -1.

Rank

Definition 0.1 (Linearly independent vectors)**.** Vectors $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k$ of the same length p are **linearly independent** [30] if the only numbers $c_1, \ldots, c_k$ with

$$c_1 \underset{\sim}{v}_1 + \cdots + c_k \underset{\sim}{v}_k = \underset{\sim}{0}$$

are

$$\begin{aligned} c_1 &= \cdots \\ &= c_k \\ &= 0. \end{aligned}$$

Vectors that are not linearly independent are **linearly dependent**.

Example 0.1 (Independent and dependent pairs of vectors)**.**

- $\underset{\sim}{v}_1 = (1, 0)$ and $\underset{\sim}{v}_2 = (1, 1)$ are linearly independent: $c_1 \underset{\sim}{v}_1 + c_2 \underset{\sim}{v}_2 = (c_1 + c_2, c_2)$, which is $\underset{\sim}{0}$ only if $c_2 = 0$ and then $c_1 = 0$.
- $\underset{\sim}{v}_1 = (1, 2)$ and $\underset{\sim}{v}_2 = (2, 4)$ are not: $2\underset{\sim}{v}_1 - \underset{\sim}{v}_2 = \underset{\sim}{0}$.

Exercise 0.1 (Testing whether two features and their total are linearly independent)**.** A table of data that stores two features and also their total holds redundant information, which the test below detects. Let

$$\underset{\sim}{v}_1 = \begin{bmatrix}1\\2\\0\end{bmatrix}, \qquad \underset{\sim}{v}_2 = \begin{bmatrix}0\\1\\1\end{bmatrix}, \qquad \underset{\sim}{v}_3 = \begin{bmatrix}1\\3\\1\end{bmatrix}.$$

Are $\underset{\sim}{v}_1$, $\underset{\sim}{v}_2$ and $\underset{\sim}{v}_3$ linearly independent?

Hint

Compare $\underset{\sim}{v}_3$ with $\underset{\sim}{v}_1 + \underset{\sim}{v}_2$.

Solution

Solution 0.1. No. $\underset{\sim}{v}_1 + \underset{\sim}{v}_2 = (1,3,1)^\top = \underset{\sim}{v}_3$, so

$$1 \cdot \underset{\sim}{v}_1 + 1 \cdot \underset{\sim}{v}_2 + (-1) \cdot \underset{\sim}{v}_3 = \underset{\sim}{0},$$

with numbers $c_1 = 1$, $c_2 = 1$ and $c_3 = -1$ that are not all 0. By Definition 0.1, the three vectors are linearly dependent.

Definition 0.2 (Rank)**.** The **rank** [31] of a matrix $\mathbf{A}$, written rank($\mathbf{A}$), is the largest number of columns of $\mathbf{A}$ that are linearly independent (Definition 0.1).

Example 0.2 (The rank of two 3×2 matrices)**.** The matrix $\begin{bmatrix}1 & 1\\1 & 2\\1 & 3\end{bmatrix}$ has rank 2: if $c_1(1,1,1) + c_2(1,2,3) = \underset{\sim}{0}$, then subtracting the first entry from the second gives $c_2 = 0$, and then $c_1 = 0$.
The matrix $\begin{bmatrix}1 & 2\\1 & 2\\1 & 2\end{bmatrix}$ has rank 1: its second column is twice its first, so the two columns are not linearly independent, but the first column on its own is.

Definition 0.3 (Full column rank)**.** An $n \times p$ matrix has **full column rank** if its rank (Definition 0.2) is p, so that all of its columns are linearly independent.

Remark (Full column rank needs at least as many rows as columns)

Remark 0.1 (Full column rank needs at least as many rows as columns). In Example 0.2, the first matrix has full column rank: its rank is 2, and it has 2 columns. The second does not, because its rank is 1.
An $n \times p$ matrix can have full column rank only when $p \leq n$, because more than n vectors

of length n are never linearly independent. For example, the 2×3 matrix $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ has 3 columns of length 2, and $1 \cdot (1,0) + 1 \cdot (0,1) - 1 \cdot (1,1) = \underset{\sim}{0}$, with coefficients that are not all zero. So its 3 columns are not linearly independent, and its rank is less than 3.

Exercise 0.2 (Finding the rank of a 3×3 matrix)**.** The rank of a data matrix counts how many of its columns carry new information. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}.$$

Find $\operatorname{rank}(\mathbf{A})$.

Hint

Add the first two columns of $\mathbf{A}$ and compare them with the third. Then check whether the first two are linearly independent.

Solution

Solution 0.2. $\operatorname{rank}(\mathbf{A}) = 2$. The third column of $\mathbf{A}$ adds up the first two: $(1,0,2)^\top + (0,1,1)^\top = (1,1,3)^\top$, so the three are not linearly independent. The first two are: if $c_1(1,0,2)^\top + c_2(0,1,1)^\top = \underset{\sim}{0}$, then the first entry gives $c_1 = 0$, and the second gives $c_2 = 0$. By Definition 0.2, the rank is 2.

Exercise 0.3 (Testing whether an all-ones column and one feature have full column rank)**.** Regression software often puts an all-ones column in the input matrix, beside the features. With one feature, measured at 2, 4 and 6, the input matrix is

$$\mathbf{X} = \begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 6 \end{bmatrix}.$$

Does $\mathbf{X}$ have full column rank?

Hint

By Definition 0.3, check whether the two columns of $\mathbf{X}$ are linearly independent.

Solution

Solution 0.3. Yes. Suppose $c_1(1,1,1)^\top + c_2(2,4,6)^\top = \underset{\sim}{0}$. The first two entries give

$$c_1 + 2c_2 = 0 \quad \text{and} \quad c_1 + 4c_2 = 0.$$

Subtracting the first equation from the second gives $2c_2 = 0$, so $c_2 = 0$, and then $c_1 = 0$. So the two columns of $\mathbf{X}$ are linearly independent, the rank of $\mathbf{X}$ is 2, and $\mathbf{X}$ has full column rank.

Subspaces

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This section is adapted from Zhou [32], used under the MIT License. The license text is:

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Definition 0.1 (Subspace, linear subspace)**.** A set $\mathcal{S}$ of vectors in $\mathbb{R}^p$ is a **subspace** [33] of $\mathbb{R}^p$ if it is not empty and it is closed under vector addition and scalar multiplication:

1. if $\underset{\sim}{u} \in \mathcal{S}$ and $\underset{\sim}{v} \in \mathcal{S}$, then $\underset{\sim}{u} + \underset{\sim}{v} \in \mathcal{S}$;
2. if $\underset{\sim}{u} \in \mathcal{S}$ and $\underset{\sim}{c} \in \mathbb{R}$, then $\underset{\sim}{c}\underset{\sim}{u} \in \mathcal{S}$.

A subspace is also called a **linear subspace**.

Example 0.1 (A line through the origin is a subspace)**.** Let $\mathcal{S} = \{c\,(1,2) : c \in \mathbb{R}\}$, the line in $\mathbb{R}^2$ through $(0,0)$ and $(1,2)$.

- **Addition:** $a\,(1,2) + b\,(1,2) = (a+b)\,(1,2)$, which is in $\mathcal{S}$.
- **Scalar multiplication:** $c\,(a\,(1,2)) = (ca)\,(1,2)$, which is in $\mathcal{S}$.

So $\mathcal{S}$ is a subspace of $\mathbb{R}^2$. Two other subspaces of $\mathbb{R}^p$ are the set $\{\underset{\sim}{0}\}$ that holds only the zero vector, and $\mathbb{R}^p$ itself.

Example 0.2 (A line that misses the origin is not a subspace)**.** Let $\mathcal{T} = \{(x,1) : x \in \mathbb{R}\}$, the horizontal line in $\mathbb{R}^2$ at height 1. The vectors $(0,1)$ and $(2,1)$ are in $\mathcal{T}$, but their sum $(2,2)$ is not, because its second entry is 2, not 1. So $\mathcal{T}$ is not closed under addition, and it is not a subspace.

Exercise 0.1 (Testing whether vectors with a nonnegative second entry are a subspace)**.** A constraint such as "this entry is at least 0" cuts out a set of vectors, and it is worth knowing whether that set is a subspace. Let

$$\mathcal{H} = \{(x,y) \in \mathbb{R}^2 : y \geq 0\},$$

the vectors whose second entry is nonnegative.
Is $\mathcal{H}$ a subspace of $\mathbb{R}^2$?

Hint

Check condition 2 of Definition 0.1 with $c = -1$.

Solution

Solution 0.1. No. The vector $(0,1)^\top$ is in $\mathcal{H}$, but $(-1)\cdot(0,1)^\top = (0,-1)^\top$ is not, because its second entry is negative. So $\mathcal{H}$ fails condition 2 of Definition 0.1, and it is not a subspace. $\mathcal{H}$ does satisfy condition 1, which shows that one condition alone is not enough.

Theorem 0.1 (Every subspace contains the zero vector)**.** *If $\mathcal{S}$ is a subspace of $\mathbb{R}^p$ (Definition 0.1), then $\underset{\sim}{0} \in \mathcal{S}$.*

Proof

Proof. A subspace is not empty, so it contains some vector $\underset{\sim}{u}$. Closure under scalar multiplication with $c = 0$ puts $0 \cdot \underset{\sim}{u}$ in $\mathcal{S}$, and $0 \cdot \underset{\sim}{u} = \underset{\sim}{0}$. □

Example 0.3 (Using the zero vector to rule out a subspace)**.** The plane $\{(x, y, z) : x + y + z = 1\}$ in $\mathbb{R}^3$ does not contain $\underset{\sim}{0} = (0, 0, 0)$, because $0 + 0 + 0 = 0 \neq 1$. So, by Theorem 0.1, it is not a subspace. The line $\mathcal{T}$ in Example 0.2 fails the same test: $(0, 0)$ is not in $\mathcal{T}$, because its second entry is not 1.
Containing $\underset{\sim}{0}$ is necessary but not sufficient. The union of the two coordinate axes in $\mathbb{R}^2$, $\{(x, 0) : x \in \mathbb{R}\} \cup \{(0, y) : y \in \mathbb{R}\}$, contains $\underset{\sim}{0}$, but $(1, 0) + (0, 1) = (1, 1)$ lies on neither axis, so the union is not closed under addition and is not a subspace.

Exercise 0.2 (Testing whether $p_1 + p_2 = 1$ defines a subspace)**.** A coin toss can land heads or tails, and the probabilities p_1 and p_2 of those two results satisfy $p_1 + p_2 = 1$. That condition defines the set

$$\mathcal{P} = \{(p_1, p_2) \in \mathbb{R}^2 : p_1 + p_2 = 1\}.$$

Is $\mathcal{P}$ a subspace of $\mathbb{R}^2$?

Hint

Use Theorem 0.1.

Solution

Solution 0.2. No. The zero vector $(0, 0)^\top$ is not in $\mathcal{P}$, because $0 + 0 = 0 \neq 1$. By Theorem 0.1, every subspace contains $\underset{\sim}{0}$, so $\mathcal{P}$ is not a subspace.

Definition 0.2 (Span)**.** The **span** [34] of vectors $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k \in \mathbb{R}^p$ is the set of all their linear combinations:

$$\operatorname{span}\{\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k\} \stackrel{\text{def}}{=} \{c_1 \underset{\sim}{v}_1 + \cdots + c_k \underset{\sim}{v}_k : c_1, \ldots, c_k \in \mathbb{R}\}.$$

The vectors $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k$ **span** a set $\mathcal{S}$ if their span equals $\mathcal{S}$.
By convention, the span of the empty list ($k = 0$) is $\{\underset{\sim}{0}\}$, and the empty list counts as linearly independent (Definition 0.1), since it has no coefficients that could fail to be zero.

Example 0.4 (Two spans in $\mathbb{R}^3$)**.**

- $\operatorname{span}\{(1,0,0),(0,1,0)\}$ is the set of vectors $(c_1, c_2, 0)$, the plane $z = 0$ in $\mathbb{R}^3$.
- $\operatorname{span}\{(1,2,3),(2,4,6)\}$ is only the line $\{c\,(1,2,3) : c \in \mathbb{R}\}$, because $(2,4,6) = 2\,(1,2,3)$, so $c_1(1,2,3) + c_2(2,4,6) = (c_1 + 2c_2)\,(1,2,3)$. Adding a vector to a list does not always enlarge its span.

Exercise 0.3 (Testing whether a vector lies in the span of two others)**.** Whether a target can be written as a linear combination of a few given vectors is a question about a span. Let

$$\underset{\sim}{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad \underset{\sim}{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \qquad \underset{\sim}{w} = \begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix}.$$

Is the vector $\underset{\sim}{w}$ in $\operatorname{span}\{\underset{\sim}{v}_1, \underset{\sim}{v}_2\}$?

Hint

Look for numbers c_1 and c_2 with $c_1\underset{\sim}{v}_1 + c_2\underset{\sim}{v}_2 = \underset{\sim}{w}$, one entry at a time.

Solution

Solution 0.3. Yes. Entry by entry, $c_1\underset{\sim}{v}_1 + c_2\underset{\sim}{v}_2 = (c_1,\ c_1 + c_2,\ c_2)^\top$. Matching $\underset{\sim}{w}$, the first entry gives $c_1 = 3$ and the third gives $c_2 = 2$. Then the second entry is $3 + 2 = 5$, which matches too, so $\underset{\sim}{w} = 3\underset{\sim}{v}_1 + 2\underset{\sim}{v}_2$ is in the span.

Theorem 0.2 (A span is a subspace)**.** *For any vectors $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k \in \mathbb{R}^p$, $\operatorname{span}\{\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k\}$ (Definition 0.2) is a subspace of $\mathbb{R}^p$ (Definition 0.1).*

Proof

Proof. The span is not empty: it contains $0 \cdot \underset{\sim}{v}_1 + \cdots + 0 \cdot \underset{\sim}{v}_k = \underset{\sim}{0}$.
Addition. Take two vectors in the span, $\underset{\sim}{u} = a_1\underset{\sim}{v}_1 + \cdots + a_k\underset{\sim}{v}_k$ and $\underset{\sim}{w} = b_1\underset{\sim}{v}_1 + \cdots + b_k\underset{\sim}{v}_k$. Then

$$\begin{aligned} \underset{\sim}{u} + \underset{\sim}{w} &= (a_1\underset{\sim}{v}_1 + \cdots + a_k\underset{\sim}{v}_k) + (b_1\underset{\sim}{v}_1 + \cdots + b_k\underset{\sim}{v}_k) && \text{(substitute the two combinations)} \\ &= (a_1 + b_1)\,\underset{\sim}{v}_1 + \cdots + (a_k + b_k)\,\underset{\sim}{v}_k && \text{(regroup, and distribute each } \underset{\sim}{v}_i\text{)} \end{aligned}$$

which is a linear combination of $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k$, so it is in the span.
Scalar multiplication. For $c \in \mathbb{R}$,

$$
\begin{aligned}
c\,\underset{\sim}{u} &= c\,(a_1\underset{\sim}{v}_1 + \cdots + a_k\underset{\sim}{v}_k) && \text{(substitute the combination)}\\
&= (ca_1)\,\underset{\sim}{v}_1 + \cdots + (ca_k)\,\underset{\sim}{v}_k && \text{(distribute } c \text{ over the sum)}
\end{aligned}
$$

which is again in the span. □

Example 0.5 (The line in Example 0.1 is a span)**.** The line $\mathcal{S} = \{c\,(1,2) : c \in \mathbb{R}\}$ in Example 0.1 is span$\{(1,2)\}$, so Theorem 0.2 gives a second proof that it is a subspace. Likewise, the plane $z = 0$ in Example 0.4 is a subspace of $\mathbb{R}^3$, because it is span$\{(1,0,0),(0,1,0)\}$.

Exercise 0.4 (Recognizing a set of vectors as a span)**.** Describing a set by free parameters often shows that it is a span. Let

$$\mathcal{S} = \{(a+b,\ a-b,\ 2a) : a, b \in \mathbb{R}\}.$$

Is $\mathcal{S}$ a subspace of $\mathbb{R}^3$?

Hint

Write $(a+b,\ a-b,\ 2a)$ as a times one vector plus b times another, then use Theorem 0.2.

Solution

Solution 0.4. Yes. For every $a, b \in \mathbb{R}$,

$$\begin{bmatrix} a+b \\ a-b \\ 2a \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix},$$

so $\mathcal{S} = \text{span}\left\{(1,1,2)^\top, (1,-1,0)^\top\right\}$. By Theorem 0.2, $\mathcal{S}$ is a subspace of $\mathbb{R}^3$.

Theorem 0.3 (A matrix-vector product combines the columns)**.** *If* $\mathbf{A}$ *is an* $m \times n$ *matrix with columns* $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_n$ *and* $\underset{\sim}{x} \in \mathbb{R}^n$*, then*

$$\mathbf{A}\underset{\sim}{x} = x_1\underset{\sim}{a}_1 + \cdots + x_n\underset{\sim}{a}_n.$$

Proof

Proof. Compare entry i of the two sides, for each $i = 1, \ldots, m$. Entry i of $\underset{\sim}{a}_j$ is a_{ij}, so

$$
\begin{aligned}
(\mathbf{A}\underset{\sim}{x})_i &= a_{i1}x_1 + \cdots + a_{in}x_n && \text{(Definition 0.16)} \\
&= x_1 a_{i1} + \cdots + x_n a_{in} && \text{(multiplication of numbers is commutative)} \\
&= (x_1 \underset{\sim}{a}_1 + \cdots + x_n \underset{\sim}{a}_n)_i && \text{(entry } i \text{ of each } x_j \underset{\sim}{a}_j \text{ is } x_j a_{ij}\text{)}
\end{aligned}
$$

□

Example 0.6 (A product as a combination of columns)**.**

$$
\begin{aligned}
\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} &= 1 \cdot \begin{bmatrix} 1 \\ 3 \end{bmatrix} + (-1) \cdot \begin{bmatrix} 2 \\ 4 \end{bmatrix} && \text{(Theorem 0.3)} \\
&= \begin{bmatrix} -1 \\ -1 \end{bmatrix}, && \text{(arithmetic)}
\end{aligned}
$$

the same answer as computing each entry as a row-times-vector dot product (Remark 0.4).

Exercise 0.5 (Computing a matrix-vector product as a combination of columns)**.** A linear model's predictions for every observation at once are a matrix-vector product $\mathbf{A}\underset{\sim}{x}$, and Theorem 0.3 writes it as a combination of the columns of $\mathbf{A}$. Let

$$
\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 3 \\ 1 & -1 \end{bmatrix}, \qquad \underset{\sim}{x} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.
$$

Find $\mathbf{A}\underset{\sim}{x}$ with Theorem 0.3.

Hint

Use Theorem 0.3 with $x_1 = 3$ and $x_2 = 2$.

Solution

Solution 0.5. By Theorem 0.3,

$$
\mathbf{A}\underset{\sim}{x} = 3 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 6+2 \\ 0+6 \\ 3-2 \end{bmatrix} = \begin{bmatrix} 8 \\ 6 \\ 1 \end{bmatrix}.
$$

Theorem 0.4 (A matrix maps a combination to the same combination of images)**.** *If* $\mathbf{A}$ *is an* $m \times n$ *matrix,* $\underset{\sim}{z}_1, \ldots, \underset{\sim}{z}_k \in \mathbb{R}^n$ *and* $c_1, \ldots, c_k \in \mathbb{R}$*, then*

$$
\mathbf{A}\left(c_1 \underset{\sim}{z}_1 + \cdots + c_k \underset{\sim}{z}_k\right) = c_1 \, \mathbf{A}\underset{\sim}{z}_1 + \cdots + c_k \, \mathbf{A}\underset{\sim}{z}_k.
$$

Proof

Proof. The function $\underset{\sim}{x} \mapsto \mathbf{A}\underset{\sim}{x}$ is a linear map by Theorem 0.12, and a linear map preserves linear combinations by Theorem 0.13. Applying that theorem with $f(\underset{\sim}{x}) = \mathbf{A}\underset{\sim}{x}$ and $\underset{\sim}{v}_i = \underset{\sim}{z}_i$ gives the identity. □

Example 0.7 (Applying a matrix to a combination)**.** Let $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $\underset{\sim}{z}_1 = (1, 0)$ and $\underset{\sim}{z}_2 = (0, 1)$. Then $\mathbf{A}\underset{\sim}{z}_1 = (1, 3)$ and $\mathbf{A}\underset{\sim}{z}_2 = (2, 4)$, and for the combination $2\underset{\sim}{z}_1 - \underset{\sim}{z}_2 = (2, -1)$:

$$\begin{aligned} \mathbf{A}\,(2, -1) &= (1 \cdot 2 + 2 \cdot (-1),\ 3 \cdot 2 + 4 \cdot (-1)) && \text{(Definition 0.16)} \\ &= (0, 2) && \text{(arithmetic)} \\ &= 2\,(1, 3) - (2, 4), && \text{(arithmetic)} \end{aligned}$$

which is $2\,\mathbf{A}\underset{\sim}{z}_1 - \mathbf{A}\underset{\sim}{z}_2$, as Theorem 0.4 says.

Exercise 0.6 (Applying a matrix to a combination without knowing the matrix)**.** Theorem 0.4 lets a matrix-vector product of a combination be found from the matrix-vector product of each part, even when the matrix itself is not known. Suppose that a matrix $\mathbf{A}$ satisfies

$$\mathbf{A}\underset{\sim}{z}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad \mathbf{A}\underset{\sim}{z}_2 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Find $\mathbf{A}\,(3\underset{\sim}{z}_1 + 2\underset{\sim}{z}_2)$.

Hint

Use Theorem 0.4 with $c_1 = 3$ and $c_2 = 2$.

Solution

Solution 0.6. By Theorem 0.4,

$$\mathbf{A}\,(3\underset{\sim}{z}_1 + 2\underset{\sim}{z}_2) = 3\begin{bmatrix} 1 \\ 2 \end{bmatrix} + 2\begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

Theorem 0.5 (More than p vectors in $\mathbb{R}^p$ are linearly dependent)**.** *If $k > p$, then any k vectors $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k \in \mathbb{R}^p$ are not linearly independent (Definition 0.1): some numbers $c_1, \ldots, c_k$, not all zero, satisfy $c_1\underset{\sim}{v}_1 + \cdots + c_k\underset{\sim}{v}_k = \underset{\sim}{0}_p$.*

Proof

Proof. Induction on p.
Base case, $p = 1$. Each $\underset{\sim}{v}_i$ is a single number v_i, and $k \geq 2$. If $v_1 = 0$, take $c_1 = 1$ and every other $c_i = 0$. Otherwise take $c_1 = v_2$, $c_2 = -v_1$ and every other $c_i = 0$; then $c_2 \neq 0$, and $v_2 v_1 - v_1 v_2 = 0$.
Inductive step. Let $p \geq 2$, assume the theorem holds for vectors in $\mathbb{R}^{p-1}$, and take $k > p$ vectors in $\mathbb{R}^p$. Write $v_{i,p}$ for the last entry of $\underset{\sim}{v}_i$.
Case 1: every $v_{i,p} = 0$. Deleting the last entry gives $k > p-1$ vectors in $\mathbb{R}^{p-1}$, which by the inductive hypothesis have a combination with coefficients not all zero that equals $\underset{\sim}{0}_{p-1}$. The same coefficients combine the original vectors to $\underset{\sim}{0}_p$, since every last entry is 0.
Case 2: some $v_{i,p} \neq 0$. Renumber the vectors so that $v_{k,p} \neq 0$, and for $i = 1, \ldots, k-1$ let

$$\underset{\sim}{w}_i \stackrel{\text{def}}{=} \underset{\sim}{v}_i - \frac{v_{i,p}}{v_{k,p}} \underset{\sim}{v}_k.$$

The last entry of $\underset{\sim}{w}_i$ is $v_{i,p} - \frac{v_{i,p}}{v_{k,p}} v_{k,p} = 0$. Deleting that zero last entry gives $k-1 > p-1$ vectors in $\mathbb{R}^{p-1}$, so by the inductive hypothesis there are numbers $c_1, \ldots, c_{k-1}$, not all zero, whose combination of the shortened vectors is $\underset{\sim}{0}_{p-1}$. Putting the zero last entries back gives $c_1 \underset{\sim}{w}_1 + \cdots + c_{k-1} \underset{\sim}{w}_{k-1} = \underset{\sim}{0}_p$. Then

$$\begin{aligned}
\underset{\sim}{0}_p &= c_1 \underset{\sim}{w}_1 + \cdots + c_{k-1} \underset{\sim}{w}_{k-1} && \text{(choice of } c_1, \ldots, c_{k-1}\text{)} \\
&= \sum_{i=1}^{k-1} c_i \left(\underset{\sim}{v}_i - \frac{v_{i,p}}{v_{k,p}} \underset{\sim}{v}_k \right) && \text{(definition of } \underset{\sim}{w}_i\text{)} \\
&= \sum_{i=1}^{k-1} c_i \underset{\sim}{v}_i - \sum_{i=1}^{k-1} \frac{c_i v_{i,p}}{v_{k,p}} \underset{\sim}{v}_k && \text{(distribute each } c_i\text{, and distribute the sum over the difference} \\
&= c_1 \underset{\sim}{v}_1 + \cdots + c_{k-1} \underset{\sim}{v}_{k-1} - \left(\sum_{i=1}^{k-1} \frac{c_i v_{i,p}}{v_{k,p}} \right) \underset{\sim}{v}_k, && \text{(factor } \underset{\sim}{v}_k \text{ out of the second sum)}
\end{aligned}$$

a combination of $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k$ equal to $\underset{\sim}{0}_p$ whose first $k-1$ coefficients are not all zero. □

Example 0.8 (Three vectors in $\mathbb{R}^2$)**.** Theorem 0.5 says the three vectors $(1,0)$, $(0,1)$ and $(1,1)$ in $\mathbb{R}^2$ are linearly dependent, and indeed $1 \cdot (1,0) + 1 \cdot (0,1) - 1 \cdot (1,1) = (0,0)$. The theorem says nothing about $k \leq p$ vectors: $(1,0)$ and $(0,1)$ are linearly independent, while $(1,2)$ and $(2,4)$ are not (Example 0.1).

Exercise 0.7 (Testing four vectors in $\mathbb{R}^3$ for linear independence)**.** A table of data with more features than observations has more columns than each column has entries, and

Theorem 0.5 settles such cases without any arithmetic. Let

$$\underset{\sim}{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \underset{\sim}{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \quad \underset{\sim}{v}_3 = \begin{bmatrix} 7 \\ 8 \\ 10 \end{bmatrix}, \quad \underset{\sim}{v}_4 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

Are these four vectors in $\mathbb{R}^3$ linearly independent?

Hint

Compare the number of vectors with $p = 3$ in Theorem 0.5.

Solution

Solution 0.7. No. There are $k = 4$ vectors in $\mathbb{R}^3$, and $4 > 3$, so by Theorem 0.5 they are linearly dependent.

Definition 0.3 (Basis)**.** A **basis** [35] of a subspace $\mathcal{S}$ (Definition 0.1) is a list of vectors that are linearly independent (Definition 0.1) and span $\mathcal{S}$ (Definition 0.2).

Example 0.9 (Two bases of $\mathbb{R}^2$)**.**

- $(1, 0)$ and $(0, 1)$ form a basis of $\mathbb{R}^2$: they are linearly independent, and any $(x, y) \in \mathbb{R}^2$ equals $x\,(1, 0) + y\,(0, 1)$.
- $(1, 0)$ and $(1, 1)$ also form a basis of $\mathbb{R}^2$: Example 0.1 shows they are linearly independent, and any (x, y) equals $(x - y)\,(1, 0) + y\,(1, 1)$, because $(x - y) + y = x$ in the first entry and $0 + y = y$ in the second.

Example 0.10 (Lists that are not bases of $\mathbb{R}^2$)**.**

- $(1, 0)$ and $(2, 0)$ are not a basis of $\mathbb{R}^2$. They are not linearly independent, since $2\,(1, 0) - (2, 0) = \underset{\sim}{0}$, and they do not span $\mathbb{R}^2$: every combination $c_1(1, 0) + c_2(2, 0)$ has second entry $\widetilde{0}$, so $(0, 1)$ is not in their span.
- $(1, 0)$, $(0, 1)$ and $(1, 1)$ span $\mathbb{R}^2$, but they are not a basis: $(1, 0) + (0, 1) - (1, 1) = \underset{\sim}{0}$, so they are not linearly independent.

Exercise 0.8 (Testing whether $(1, 1)$ and $(1, -1)$ form a basis of $\mathbb{R}^2$)**.** Describing every vector in $\mathbb{R}^2$ through $(1, 1)$ and $(1, -1)$ instead of the axes works only if those two vectors form a basis. Let

$$\underset{\sim}{a}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \underset{\sim}{a}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Is $\underset{\sim}{a}_1, \underset{\sim}{a}_2$ a basis of $\mathbb{R}^2$?

Hint

Check both conditions of Definition 0.3. For spanning, look for c_1 and c_2 with $c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2 = (x, y)^\top$.

Solution

Solution 0.8. Yes. They span $\mathbb{R}^2$: for every $(x, y)^\top \in \mathbb{R}^2$,

$$\frac{x+y}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{x-y}{2}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

They are linearly independent: if $c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2 = \underset{\sim}{0}$, then $c_1 + c_2 = 0$ and $c_1 - c_2 = 0$, and adding the two equations gives $c_1 = 0$ and then $c_2 = 0$. By Definition 0.3, $\underset{\sim}{a}_1, \underset{\sim}{a}_2$ is a basis of $\mathbb{R}^2$.

Theorem 0.6 (Coefficients in a basis are unique)**.** *If $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_k$ is a basis of a subspace $\mathcal{S}$ (Definition 0.3), then every $\underset{\sim}{x} \in \mathcal{S}$ is a linear combination $\underset{\sim}{x} = c_1\underset{\sim}{a}_1 + \cdots + c_k\underset{\sim}{a}_k$ for exactly one list of numbers $c_1, \ldots, c_k$.*

Proof

Proof. At least one such list exists, because the basis spans $\mathcal{S}$ (Definition 0.2). Suppose two lists both work:

$$\begin{aligned} c_1\underset{\sim}{a}_1 + \cdots + c_k\underset{\sim}{a}_k &= \underset{\sim}{x} \\ &= d_1\underset{\sim}{a}_1 + \cdots + d_k\underset{\sim}{a}_k. \end{aligned}$$

Subtracting the right side from the left side gives

$$\begin{aligned} \underset{\sim}{0} &= (c_1\underset{\sim}{a}_1 + \cdots + c_k\underset{\sim}{a}_k) - (d_1\underset{\sim}{a}_1 + \cdots + d_k\underset{\sim}{a}_k) && \text{(both combinations equal } \underset{\sim}{x}\text{)} \\ &= (c_1 - d_1)\,\underset{\sim}{a}_1 + \cdots + (c_k - d_k)\,\underset{\sim}{a}_k && \text{(regroup, and distribute each } \underset{\sim}{a}_i\text{)} \end{aligned}$$

Because $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_k$ are linearly independent (Definition 0.1), every coefficient $c_i - d_i$ is 0, so $c_i = d_i$ for each i. □

Example 0.11 (Coefficients of $(3,5)$ in two bases)**.** In the basis $(1,0),(0,1)$ of Example 0.9, $(3,5) = 3\,(1,0) + 5\,(0,1)$, and no other coefficients work. In the basis $(1,0),(1,1)$, the formula in Example 0.9 gives

$$\begin{aligned}(3,5) &= (3-5)\,(1,0) + 5\,(1,1)\\ &= -2\,(1,0) + 5\,(1,1).\end{aligned}$$

The same vector has different coefficients in different bases, but within one basis its coefficients are unique.
The list $(1,0),(0,1),(1,1)$ in Example 0.10 spans $\mathbb{R}^2$ but is not linearly independent, and the coefficients of $(3,5)$ are not unique:

$$\begin{aligned}(3,5) &= 3\,(1,0) + 5\,(0,1) + 0\,(1,1)\\ &= 0\,(1,0) + 2\,(0,1) + 3\,(1,1).\end{aligned}$$

Exercise 0.9 (Counting the ways to write a vector in a basis)**.** The vectors $\underset{\sim}{a}_1 = (1,1)^\top$ and $\underset{\sim}{a}_2 = (1,-1)^\top$ form a basis of $\mathbb{R}^2$, and

$$\begin{bmatrix}5\\1\end{bmatrix} = 3\underset{\sim}{a}_1 + 2\underset{\sim}{a}_2.$$

How many lists of numbers c_1, c_2 satisfy $(5,1)^\top = c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2$?

Hint

Use Theorem 0.6.

Solution

Solution 0.9. Exactly one, namely $c_1 = 3$ and $c_2 = 2$. By Theorem 0.6, a vector in the span of a basis has only one list of numbers, and $3, 2$ is a list that works.

Definition 0.4 (Coordinates in a basis)**.** Let $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_k$ be a basis of a subspace $\mathcal{S}$ (Definition 0.3), and let $\underset{\sim}{x} \in \mathcal{S}$. The **coordinates** [36] of $\underset{\sim}{x}$ in this basis are the numbers $c_1, \ldots, c_k$ with $\underset{\sim}{x} = c_1\underset{\sim}{a}_1 + \cdots + c_k\underset{\sim}{a}_k$. By Theorem 0.6 there is exactly one such list of numbers.

Definition 0.5 (Coordinate vector)**.** With $\underset{\sim}{x}$ and its coordinates $c_1, \ldots, c_k$ as in Definition 0.4, $(c_1, \ldots, c_k) \in \mathbb{R}^k$ is the **coordinate vector** of $\underset{\sim}{x}$.

Example 0.12 (Coordinates of $(4,1)$ in two bases of $\mathbb{R}^2$)**.** In the basis $(1,0),(1,1)$ of Example 0.9, the coordinates of $(4,1)$ are 3 and 1:

$$\begin{aligned} 3\,(1,0)+1\,(1,1) &= (3+1,\ 0+1) && \text{(Definition 0.9, entry by entry)} \\ &= (4,1). && \text{(add)} \end{aligned}$$

In the basis $(1,0),(0,1)$, its coordinates are its entries, 4 and 1.

Exercise 0.10 (Finding coordinates in a basis other than the indicator vectors)**.** The coordinates of a vector depend on the basis, and they match its entries only in the basis of indicator vectors. The vectors $\underset{\sim}{a}_1 = (2,0)^\top$ and $\underset{\sim}{a}_2 = (1,1)^\top$ form a basis of $\mathbb{R}^2$. Let

$$\underset{\sim}{x} = \begin{bmatrix} 7 \\ 3 \end{bmatrix}.$$

Find the coordinates of $\underset{\sim}{x}$ in the basis $\underset{\sim}{a}_1, \underset{\sim}{a}_2$.

Hint

Write $c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2$ entry by entry, and start from the second entry.

Solution

Solution 0.10. The coordinates of $\underset{\sim}{x}$ are 2 and 3. Entry by entry, $c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2 = (2c_1+c_2,\ c_2)^\top$. The second entry gives $c_2 = 3$, and then the first gives $2c_1 + 3 = 7$, so $c_1 = 2$. As a check, $2(2,0)^\top + 3(1,1)^\top = (7,3)^\top$.

Exercise 0.11 (Describing a point on a plane in $\mathbb{R}^3$ with two numbers)**.** A point on a plane through the origin in $\mathbb{R}^3$ needs only two numbers, its coordinate vector in a basis of the plane. The vectors $\underset{\sim}{a}_1 = (1,1,0)^\top$ and $\underset{\sim}{a}_2 = (0,1,1)^\top$ form a basis of a plane $\mathcal{S}$, which contains

$$\underset{\sim}{x} = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}.$$

Find the coordinate vector of $\underset{\sim}{x}$ in the basis $\underset{\sim}{a}_1, \underset{\sim}{a}_2$.

Hint

Write $c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2$ entry by entry, then use Definition 0.5.

Solution

Solution 0.11. Entry by entry, $c_1\underset{\sim}{a}_1 + c_2\underset{\sim}{a}_2 = (c_1,\ c_1 + c_2,\ c_2)^\top$. Matching $\underset{\sim}{x}$ gives $c_1 = 2$ and $c_2 = 3$, and the second entry checks: $2 + 3 = 5$. By Definition 0.5, the coordinate vector of $\underset{\sim}{x}$ is $(2, 3)^\top \in \mathbb{R}^2$.

Theorem 0.7 (All bases of a subspace have the same number of vectors)**.** *If $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_k$ and $\underset{\sim}{b}_1, \ldots, \underset{\sim}{b}_l$ are both bases of a subspace $\mathcal{S}$ of $\mathbb{R}^p$ (Definition 0.3), then $k = l$.*

Proof

Proof. Let $\mathbf{A}$ be the $p \times k$ matrix with columns $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_k$, and $\mathbf{B}$ the $p \times l$ matrix with columns $\underset{\sim}{b}_1, \ldots, \underset{\sim}{b}_l$.

Write $\mathbf{A}$ in terms of $\mathbf{B}$. Each $\underset{\sim}{a}_j$ is in $\mathcal{S}$, and the $\underset{\sim}{b}$'s span $\mathcal{S}$ (Definition 0.2), so $\underset{\sim}{a}_j = \mathbf{B}\underset{\sim}{c}_j$ for some $\underset{\sim}{c}_j \in \mathbb{R}^l$ (Definition 0.16). Let $\mathbf{C}$ be the $l \times k$ matrix with columns $\underset{\sim}{c}_1, \ldots, \underset{\sim}{c}_k$. Then $\underset{p\times k}{\mathbf{A}} = \underset{p\times l}{\mathbf{B}}\ \underset{l\times k}{\mathbf{C}}$, because column j of $\mathbf{BC}$ is $\mathbf{B}\underset{\sim}{c}_j$.

The columns of $\mathbf{C}$ are linearly independent. Suppose $\mathbf{C}\underset{\sim}{x} = \underset{\sim}{0}$ for some $\underset{\sim}{x} \in \mathbb{R}^k$. Then

$$\begin{aligned}
\mathbf{A}\underset{\sim}{x} &= (\mathbf{BC})\,\underset{\sim}{x} && \text{(substitute } \mathbf{A} = \mathbf{BC}\text{)}\\
&= \mathbf{B}\,(\mathbf{C}\underset{\sim}{x}) && \text{(Theorem 0.7)}\\
&= \mathbf{B}\,\underset{\sim}{0} && \text{(substitute } \mathbf{C}\underset{\sim}{x} = \underset{\sim}{0}\text{)}\\
&= \underset{\sim}{0} && \text{(a matrix times the zero vector is } \underset{\sim}{0}\text{)}
\end{aligned}$$

$\mathbf{A}\underset{\sim}{x}$ is the combination $x_1\underset{\sim}{a}_1 + \cdots + x_k\underset{\sim}{a}_k$ (Theorem 0.3), and the $\underset{\sim}{a}$'s are linearly independent (Definition 0.1), so $\underset{\sim}{x} = \underset{\sim}{0}$. So the only combination of the columns of $\mathbf{C}$ that equals $\underset{\sim}{0}$ has all coefficients 0.

Count. $\mathbf{C}$ has k linearly independent columns of length l. More than l vectors in $\mathbb{R}^l$ are never linearly independent (Theorem 0.5), so $k \le l$. Exchanging the roles of the two bases gives $l \le k$, so $k = l$. □

Example 0.13 (Both bases of $\mathbb{R}^2$ in Example 0.9 have two vectors)**.** The bases $(1, 0), (0, 1)$ and $(1, 0), (1, 1)$ of $\mathbb{R}^2$ in Example 0.9 both have 2 vectors, as Theorem 0.7 requires. So no list of 3 vectors is a basis of $\mathbb{R}^2$, which agrees with Example 0.10: the list $(1, 0), (0, 1), (1, 1)$ spans $\mathbb{R}^2$ but is not linearly independent.

Definition 0.6 (Dimension)**.** The **dimension** of a subspace $\mathcal{S}$, written $\dim(\mathcal{S})$, is the number of vectors in any basis of $\mathcal{S}$ (Definition 0.3). Theorem 0.7 makes this number well defined. By convention, the empty list is the basis of $\{\underset{\sim}{0}\}$, so $\dim(\{\underset{\sim}{0}\}) = 0$.

Example 0.14 (The dimensions of some subspaces)**.**

- $\dim(\{\underset{\sim}{0}\}) = 0$, by the convention in Definition 0.6.
- $\dim(\mathbb{R}^2) = 2$, by Example 0.9. More generally, $\dim(\mathbb{R}^p) = p$: the indicator vectors $\underset{\sim}{e}_1, \ldots, \underset{\sim}{e}_p$ (Definition 0.18) are linearly independent, since $c_1\underset{\sim}{e}_1 + \cdots + c_p\underset{\sim}{e}_p = (c_1, \ldots, c_p)$, which is $\underset{\sim}{0}_p$ only if every $c_i = 0$; and they span $\mathbb{R}^p$, since any $\underset{\sim}{x} = x_1\underset{\sim}{e}_1 + \cdots + x_p\underset{\sim}{e}_p$.
- The line in Example 0.1 has dimension 1: the single vector $(1, 2)$ is linearly independent and spans the line.
- The plane $z = 0$ in Example 0.4 has dimension 2, with basis $(1, 0, 0), (0, 1, 0)$.

Exercise 0.12 (Testing whether two vectors can form a basis of $\mathbb{R}^3$)**.** A list of two features cannot describe every point in $\mathbb{R}^3$, and Theorem 0.7 says why in terms of bases. The indicator vectors $\underset{\sim}{e}_1$, $\underset{\sim}{e}_2$ and $\underset{\sim}{e}_3$ form a basis of $\mathbb{R}^3$.
Can a list of 2 vectors be a basis of $\mathbb{R}^3$?

Hint

Use Theorem 0.7 with the basis $\underset{\sim}{e}_1, \underset{\sim}{e}_2, \underset{\sim}{e}_3$.

Solution

Solution 0.12. No. $\underset{\sim}{e}_1, \underset{\sim}{e}_2, \underset{\sim}{e}_3$ is a basis of $\mathbb{R}^3$ with 3 vectors, so by Theorem 0.7 every basis of $\mathbb{R}^3$ has 3 vectors, not 2.

Exercise 0.13 (Finding the dimension of a span that includes a multiple)**.** The dimension of a span can be smaller than the number of vectors that span it, for example when one vector is a multiple of another. Let

$$\mathcal{S} = \operatorname{span}\left\{(1, 2, 0)^\top,\ (2, 4, 0)^\top,\ (0, 0, 1)^\top\right\}.$$

Find $\dim(\mathcal{S})$.

Hint

Look for a vector in the list that is a multiple of another, then check that the rest are linearly independent.

Solution

Solution 0.13. $\dim(\mathcal{S}) = 2$. Since $(2, 4, 0)^\top = 2(1, 2, 0)^\top$, dropping $(2, 4, 0)^\top$ leaves the span unchanged, so $(1, 2, 0)^\top$ and $(0, 0, 1)^\top$ span $\mathcal{S}$. They are linearly independent: if

$c_1(1,2,0)^\top + c_2(0,0,1)^\top = \underset{\sim}{0}$, the first entry gives $c_1 = 0$ and the third gives $c_2 = 0$. So they form a basis of $\mathcal{S}$ with 2 vectors, and by Definition 0.6 the dimension is 2.

Theorem 0.8 (Adding a vector outside the span keeps a list independent)**.** *If $\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k$ are linearly independent (Definition 0.1) and $\underset{\sim}{w}$ is not in* $\operatorname{span}\{\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k\}$ *(Definition 0.2), then $\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k, \underset{\sim}{w}$ are linearly independent.*

Proof

Proof. Suppose $c_1 \underset{\sim}{v}_1 + \cdots + c_k \underset{\sim}{v}_k + d\,\underset{\sim}{w} = \underset{\sim}{0}$.
The coefficient d is 0. If $d \neq 0$, then

$$\begin{aligned} \underset{\sim}{w} &= \frac{1}{d}\,(d\,\underset{\sim}{w}) && (d \neq 0) \\ &= \frac{1}{d}\,(-c_1 \underset{\sim}{v}_1 - \cdots - c_k \underset{\sim}{v}_k) && \text{(solve the supposed equation for } d\,\underset{\sim}{w}) \\ &= \left(-\frac{c_1}{d}\right) \underset{\sim}{v}_1 + \cdots + \left(-\frac{c_k}{d}\right) \underset{\sim}{v}_k, && \text{(distribute } \tfrac{1}{d}) \end{aligned}$$

which puts $\underset{\sim}{w}$ in $\operatorname{span}\{\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k\}$, contrary to the assumption.
The other coefficients are 0. With $d = 0$, the equation reads $c_1 \underset{\sim}{v}_1 + \cdots + c_k \underset{\sim}{v}_k = \underset{\sim}{0}$, so every $c_i = 0$, because $\underset{\sim}{v}_1, \dots, \underset{\sim}{v}_k$ are linearly independent. □

Example 0.15 (Adding $(0,0,1)$ to two independent vectors in $\mathbb{R}^3$)**.** The vectors $(1,0,0)$ and $(0,1,0)$ are linearly independent, and their span is the plane $z = 0$ (Example 0.4). The vector $(0,0,1)$ has third entry 1, so it is not in that plane, and by Theorem 0.8 the three vectors $(1,0,0)$, $(0,1,0)$, $(0,0,1)$ are linearly independent. Adding $(1,1,0)$ instead would not work: $(1,1,0)$ is in the span, and $(1,0,0) + (0,1,0) - (1,1,0) = \underset{\sim}{0}$.

Exercise 0.14 (Adding a third feature to an independent pair)**.** Adding a new feature keeps a list of features linearly independent when the new one is not a combination of the old ones, which is what Theorem 0.8 says. Let

$$\underset{\sim}{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \qquad \underset{\sim}{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad \underset{\sim}{w} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Are $\underset{\sim}{v}_1$, $\underset{\sim}{v}_2$ and $\underset{\sim}{w}$ linearly independent?

Hint

Check whether $\underset{\sim}{w}$ is in $\operatorname{span}\{\underset{\sim}{v}_1, \underset{\sim}{v}_2\}$, then use Theorem 0.8.

Solution

Solution 0.14. Yes. $\underset{\sim}{v}_1$ and $\underset{\sim}{v}_2$ are linearly independent: $c_1\underset{\sim}{v}_1 + c_2\underset{\sim}{v}_2 = (c_1, c_2, 0)^\top$ is $\underset{\sim}{0}$ only for $c_1 = c_2 = 0$. Every vector in their span has third entry 0, and $\underset{\sim}{w}$ has third entry 1, so $\underset{\sim}{w}$ is not in the span. By Theorem 0.8, $\underset{\sim}{v}_1$, $\underset{\sim}{v}_2$ and $\underset{\sim}{w}$ are linearly independent.

Theorem 0.9 (An independent list extends to a basis)**.** *If $\mathcal{S}$ is a subspace of $\mathbb{R}^p$ (Definition 0.1) and $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k \in \mathcal{S}$ are linearly independent (Definition 0.1), then adding vectors of $\mathcal{S}$ to the list gives a basis of $\mathcal{S}$ (Definition 0.3). Starting from the empty list shows that every subspace has a basis.*

Proof

Proof. Repeat the following step. If the current list spans $\mathcal{S}$, stop: it is linearly independent and spans $\mathcal{S}$, so it is a basis. Otherwise, some $\underset{\sim}{w} \in \mathcal{S}$ is not in the span of the current list; add $\underset{\sim}{w}$ to the list, which stays linearly independent by Theorem 0.8.
Every list produced is a linearly independent list of vectors in $\mathbb{R}^p$, so it has at most p vectors (Theorem 0.5). Each step adds one vector, so the process stops after at most $p - k$ steps, and when it stops the list is a basis of $\mathcal{S}$.
For the empty list, which is linearly independent and spans $\{\underset{\sim}{0}\}$ by the conventions in Definition 0.2, the first step asks whether $\mathcal{S} = \{\underset{\sim}{0}\}$: if so, the empty list is already a basis; if not, the process adds vectors as described. □

Example 0.16 (Extending $(1, 1, 0)$ to a basis of $\mathbb{R}^3$)**.** Start with the single vector $(1, 1, 0)$.

1. $(1, 0, 0)$ is not in $\text{span}\{(1, 1, 0)\}$, because every multiple $c\,(1, 1, 0)$ has equal first and second entries. Add it.
2. $(0, 0, 1)$ is not in $\text{span}\{(1, 1, 0), (1, 0, 0)\}$, because every combination of those two vectors has third entry 0. Add it.
3. The list $(1, 1, 0), (1, 0, 0), (0, 0, 1)$ spans $\mathbb{R}^3$: $(x, y, z) = y\,(1, 1, 0) + (x - y)\,(1, 0, 0) + z\,(0, 0, 1)$. Stop.

The result is a basis of $\mathbb{R}^3$ that contains the starting vector.

Theorem 0.10 (An independent list in a subspace has at most dim vectors)**.** *Let $\mathcal{S}$ be a subspace of $\mathbb{R}^p$ with $\dim(\mathcal{S}) = d$ (Definition 0.6).*

1. *Any linearly independent list of vectors in $\mathcal{S}$ has at most d vectors.*
2. *If $\mathcal{T}$ is a subspace with $\mathcal{T} \subseteq \mathcal{S}$, then $\dim(\mathcal{T}) \leq \dim(\mathcal{S})$.*

Proof

Proof. **Part 1.** If $d = 0$, then $\mathcal{S} = \{\underset{\sim}{0}\}$, so every vector in a nonempty list from $\mathcal{S}$ is $\underset{\sim}{0}$, and $1 \cdot \underset{\sim}{0} = \underset{\sim}{0}$ shows the list is not linearly independent; only the empty list, with $0 = d$ vectors, is. Now let $d \geq 1$, and let $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_d$ be a basis of $\mathcal{S}$, and let $\underset{\sim}{u}_1, \ldots, \underset{\sim}{u}_k \in \mathcal{S}$ be linearly independent. Each $\underset{\sim}{u}_j$ is in $\mathcal{S}$, so $\underset{\sim}{u}_j = c_{1j}\underset{\sim}{a}_1 + \cdots + c_{dj}\underset{\sim}{a}_d$ for some coordinate vector $\underset{\sim}{c}_j = (c_{1j}, \ldots, c_{dj}) \in \mathbb{R}^d$. Suppose $k > d$. Then $\underset{\sim}{c}_1, \ldots, \underset{\sim}{c}_k$ are $k > d$ vectors in $\mathbb{R}^d$, so some $x_1, \ldots, x_k$, not all zero, satisfy $x_1\underset{\sim}{c}_1 + \cdots + x_k\underset{\sim}{c}_k = \underset{\sim}{0}_d$ (Theorem 0.5); entry i of that equation says $\sum_{j=1}^{k} x_j c_{ij} = 0$. Then

$$\begin{aligned}
\sum_{j=1}^{k} x_j \underset{\sim}{u}_j &= \sum_{j=1}^{k} x_j \sum_{i=1}^{d} c_{ij}\underset{\sim}{a}_i && \text{(substitute each } \underset{\sim}{u}_j\text{)} \\
&= \sum_{j=1}^{k}\sum_{i=1}^{d} x_j c_{ij}\,\underset{\sim}{a}_i && \text{(distribute each } x_j \text{ over the inner sum)} \\
&= \sum_{i=1}^{d}\sum_{j=1}^{k} x_j c_{ij}\,\underset{\sim}{a}_i && \text{(swap the order of the finite sums)} \\
&= \sum_{i=1}^{d}\left(\sum_{j=1}^{k} x_j c_{ij}\right)\underset{\sim}{a}_i && \text{(factor } \underset{\sim}{a}_i \text{ out of the inner sum)} \\
&= \sum_{i=1}^{d} 0 \cdot \underset{\sim}{a}_i && \text{(entry } i \text{ of } \textstyle\sum_j x_j\underset{\sim}{c}_j = \underset{\sim}{0}_d\text{)} \\
&= \underset{\sim}{0}_p, && \text{(a zero multiple of a vector is } \underset{\sim}{0}\text{)}
\end{aligned}$$

a combination of $\underset{\sim}{u}_1, \ldots, \underset{\sim}{u}_k$ with coefficients not all zero that equals $\underset{\sim}{0}_p$. That contradicts their linear independence, so $k \leq d$.

Part 2. $\mathcal{T}$ has a basis (Theorem 0.9) with $\dim(\mathcal{T})$ vectors. Those vectors are linearly independent and lie in $\mathcal{S}$, so by Part 1 there are at most $\dim(\mathcal{S})$ of them. □

Example 0.17 (Subspaces of a plane have dimension at most two)**.** The plane $\mathcal{S} = \{(x, y, 0) : x, y \in \mathbb{R}\}$ in $\mathbb{R}^3$ has dimension 2 (Example 0.14). By Theorem 0.10, no three vectors with third entry 0 are linearly independent; for instance, $(1, 0, 0) + (0, 1, 0) - (1, 1, 0) = \underset{\sim}{0}$. The line $\mathcal{T} = \text{span}\{(1, 1, 0)\}$ lies in $\mathcal{S}$, and

$$\begin{aligned}
\dim(\mathcal{T}) &= 1 \\
&\leq 2 \\
&= \dim(\mathcal{S}).
\end{aligned}$$

Theorem 0.11 (A subspace inside another of the same dimension equals it)**.** *If $\mathcal{T}$ and $\mathcal{S}$ are subspaces of $\mathbb{R}^p$ with $\mathcal{T} \subseteq \mathcal{S}$ and $\dim(\mathcal{T}) = \dim(\mathcal{S})$ (Definition 0.6), then $\mathcal{T} = \mathcal{S}$.*

Proof

Proof. Let $d = \dim(\mathcal{S})$, and let $\underset{\sim}{t}_1, \dots, \underset{\sim}{t}_d$ be a basis of $\mathcal{T}$ (Theorem 0.9); it has d vectors because $\dim(\mathcal{T}) = d$. (When $d = 0$ it is the empty list, whose span is $\{\underset{\sim}{0}\}$ by the convention in Definition 0.2, and the argument is unchanged.) Suppose some $\underset{\sim}{s} \in \mathcal{S}$ is not in $\operatorname{span}\{\underset{\sim}{t}_1, \dots, \underset{\sim}{t}_d\} = \mathcal{T}$. Then $\underset{\sim}{t}_1, \dots, \underset{\sim}{t}_d, \underset{\sim}{s}$ are linearly independent (Theorem 0.8), and they are $d + 1$ vectors of $\mathcal{S}$, which contradicts part 1 of Theorem 0.10. So every vector of $\mathcal{S}$ is in $\mathcal{T}$, that is, $\mathcal{S} \subseteq \mathcal{T}$. Together with $\mathcal{T} \subseteq \mathcal{S}$, this inclusion gives $\mathcal{T} = \mathcal{S}$. □

Example 0.18 (Two independent vectors in a plane span it)**.** Let $\mathcal{S} = \{(x, y, 0) : x, y \in \mathbb{R}\}$, the plane $z = 0$, which has dimension 2 (Example 0.14), and let $\mathcal{T} = \operatorname{span}\{(1, 0, 0), (1, 1, 0)\}$. Both vectors have third entry 0, so $\mathcal{T} \subseteq \mathcal{S}$. They are linearly independent: $c_1(1, 0, 0) + c_2(1, 1, 0) = (c_1 + c_2, c_2, 0)$, which is $\underset{\sim}{0}$ only if $c_2 = 0$ and then $c_1 = 0$. So $\dim(\mathcal{T}) = 2$, and Theorem 0.11 gives $\mathcal{T} = \mathcal{S}$ without solving for any coefficients. As a check, $(x, y, 0) = (x - y)\,(1, 0, 0) + y\,(1, 1, 0)$.

Example 0.19 (Equal dimensions without containment)**.** The lines $\operatorname{span}\{(1, 0, 0)\}$ and $\operatorname{span}\{(0, 1, 0)\}$ in $\mathbb{R}^3$ both have dimension 1, but they are not equal: $(1, 0, 0)$ is in the first and not the second. Theorem 0.11 does not apply, because neither line contains the other.

Column space and null space

Definition 0.1 (Column space, range, image)**.** The **column space** [37] of an $m \times n$ matrix $\mathbf{A}$ is

$$\mathcal{C}(\mathbf{A}) \stackrel{\text{def}}{=} \{\mathbf{A}\underset{\sim}{x} : \underset{\sim}{x} \in \mathbb{R}^n\},$$

the set of all vectors $\mathbf{A}\underset{\sim}{x}$ in $\mathbb{R}^m$. It is also called the **range** or **image** of $\mathbf{A}$.

Definition 0.2 (Row space)**.** The **row space** of an $m \times n$ matrix $\mathbf{A}$ is $\mathcal{C}(\mathbf{A}^\top)$, a set of vectors in $\mathbb{R}^n$.

Example 0.1 (The column space of a rank-one matrix)**.** Let

$$\mathbf{A} = \begin{bmatrix} 1 & -2 & -2 \\ 3 & -6 & -6 \end{bmatrix}.$$

For any $\underset{\sim}{x} \in \mathbb{R}^3$,

$$\begin{aligned}\mathbf{A}\underset{\sim}{x} &= (x_1 - 2x_2 - 2x_3,\ 3(x_1 - 2x_2 - 2x_3)) \\ &= (x_1 - 2x_2 - 2x_3)(1, 3)\end{aligned}$$

(Definition 0.16). The number $x_1 - 2x_2 - 2x_3$ can be any real number (take $x_2 = 0$ and $x_3 = 0$), so $\mathcal{C}(\mathbf{A}) = \{c(1, 3) : c \in \mathbb{R}\}$, a line in $\mathbb{R}^2$. The vector $(1, 0)$ is not in $\mathcal{C}(\mathbf{A})$: $c(1, 3)$ has second entry $3c$, which is 0 only if $c = 0$, and then the first entry is 0, not 1.

Theorem 0.1 (The column space is the span of the columns)**.** *If* $\mathbf{A}$ *is an* $m \times n$ *matrix with columns* $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_n$*, then*

$$\mathcal{C}(\mathbf{A}) = \text{span}\{\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_n\},$$

so $\mathcal{C}(\mathbf{A})$ *is a subspace of* $\mathbb{R}^m$*, and the row space* $\mathcal{C}(\mathbf{A}^\top)$ *is a subspace of* $\mathbb{R}^n$*.*

Proof

Proof. By Theorem 0.3, $\mathbf{A}\underset{\sim}{x} = x_1\underset{\sim}{a}_1 + \cdots + x_n\underset{\sim}{a}_n$, and as $\underset{\sim}{x}$ ranges over $\mathbb{R}^n$, the coefficients $x_1, \ldots, x_n$ range over all lists of n numbers. So the set of all $\mathbf{A}\underset{\sim}{x}$ (Definition 0.1) is the set of all linear combinations of the columns, which is their span (Definition 0.2). A span is a subspace (Theorem 0.2). Applying the same argument to $\mathbf{A}^\top$, whose columns are the rows of $\mathbf{A}$, shows the row space is a subspace of $\mathbb{R}^n$. □

Example 0.2 (Reading off the column space and row space from the columns and rows)**.** For $\mathbf{A}$ in Example 0.1, the columns are $(1, 3)$, $(-2, -6) = -2(1, 3)$ and $(-2, -6) = -2(1, 3)$, so by Theorem 0.1 $\mathcal{C}(\mathbf{A}) = \text{span}\{(1, 3)\}$, the same line found in Example 0.1. The rows are $(1, -2, -2)$ and $(3, -6, -6) = 3(1, -2, -2)$, so the row space is $\text{span}\{(1, -2, -2)\}$, a line in $\mathbb{R}^3$.

Definition 0.3 (Null space, kernel)**.** The **null space** of an $m \times n$ matrix $\mathbf{A}$ is

$$\mathcal{N}(\mathbf{A}) \stackrel{\text{def}}{=} \{\underset{\sim}{x} \in \mathbb{R}^n : \mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m\},$$

the set of solutions of the linear system $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$. It is also called the **kernel** of $\mathbf{A}$.

Definition 0.4 (Left null space)**.** The **left null space** of an $m \times n$ matrix $\mathbf{A}$ is $\mathcal{N}(\mathbf{A}^\top)$, a set of vectors in $\mathbb{R}^m$.

Example 0.3 (The null space and left null space of a rank-one matrix)**.** For $\mathbf{A}$ in Example 0.1, $\mathbf{A}\underset{\sim}{x} = (x_1 - 2x_2 - 2x_3,\ 3\,(x_1 - 2x_2 - 2x_3))$, so $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_2$ exactly when $x_1 = 2x_2 + 2x_3$. Writing

$$\begin{aligned}\underset{\sim}{x} &= (2x_2 + 2x_3, x_2, x_3) \\ &= x_2\,(2,1,0) + x_3\,(2,0,1)\end{aligned}$$

shows $\mathcal{N}(\mathbf{A}) = \operatorname{span}\{(2,1,0),(2,0,1)\}$, a plane in $\mathbb{R}^3$.
For the left null space, $\mathbf{A}^\top y = (y_1 + 3y_2,\ -2\,(y_1 + 3y_2),\ -2\,(y_1 + 3y_2))$, which is $\underset{\sim}{0}_3$ exactly when $y_1 = -3y_2$, so $\mathcal{N}(\mathbf{A}^\top) = \operatorname{span}\{(-3,1)\}$, a line in $\mathbb{R}^2$.

Exercise 0.1 (Testing whether a vector is in a column space)**.** A vector $\underset{\sim}{b}$ is in the column space of a matrix $\mathbf{A}$ exactly when some $\underset{\sim}{x}$ gives $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}, \qquad \underset{\sim}{b} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}.$$

Decide whether $\underset{\sim}{b}$ is in $\mathcal{C}(\mathbf{A})$.

Hint

Try $\underset{\sim}{x} = (x_1, 0)$ and solve for x_1.

Solution

Solution 0.1. Take $\underset{\sim}{x} = (2,0)$. Then

$$\mathbf{A}\underset{\sim}{x} = \begin{bmatrix} 1\cdot 2 + 2\cdot 0 \\ 3\cdot 2 + 6\cdot 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \end{bmatrix} = \underset{\sim}{b},$$

so $\underset{\sim}{b}$ is in $\mathcal{C}(\mathbf{A})$.

Exercise 0.2 (Testing whether a vector is in a row space)**.** The row space of a matrix $\mathbf{A}$ is the column space of its transpose. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$$

and let $\underset{\sim}{v}$ be the column vector $(1,2)$.
Decide whether $\underset{\sim}{v}$ is in the row space $\mathcal{C}(\mathbf{A}^\top)$.

Hint

Write out $\mathbf{A}^\top$ and look for $\underset{\sim}{u}$ with $\mathbf{A}^\top\underset{\sim}{u} = \underset{\sim}{v}$.

Solution

Solution 0.2. Here

$$\mathbf{A}^\top = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}.$$

Take $\underset{\sim}{u} = (1, 0)$. Then

$$\mathbf{A}^\top \underset{\sim}{u} = \begin{bmatrix} 1 \cdot 1 + 3 \cdot 0 \\ 2 \cdot 1 + 6 \cdot 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \underset{\sim}{v},$$

so $\underset{\sim}{v}$ is in the row space.

Exercise 0.3 (Showing that a column space is all of $\mathbb{R}^2$)**.** Theorem 0.1 describes a column space as the span of the columns of $\mathbf{A}$. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Show that $\mathcal{C}(\mathbf{A}) = \mathbb{R}^2$.

Hint

Two of the columns of $\mathbf{A}$ are the indicator vectors $\underset{\sim}{e}_1$ and $\underset{\sim}{e}_2$.

Solution

Solution 0.3. By Theorem 0.1, $\mathcal{C}(\mathbf{A}) = \text{span}\{(1, 0), (0, 1), (1, 1)\}$. Every $(b_1, b_2) \in \mathbb{R}^2$ is $b_1\,(1, 0) + b_2\,(0, 1) + 0\,(1, 1)$, so it is in that span. The column space also lies inside $\mathbb{R}^2$, so $\mathcal{C}(\mathbf{A}) = \mathbb{R}^2$.

Exercise 0.4 (Finding a null space)**.** The null space of a matrix $\mathbf{A}$ holds every $\underset{\sim}{x}$ with $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}$. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Find $\mathcal{N}(\mathbf{A})$ as a span.

Hint

Write out the two entries of $\mathbf{A}\underset{\sim}{x}$ and make each 0.

Solution

Solution 0.4. For $\underset{\sim}{x} = (x_1, x_2, x_3)$,

$$\mathbf{A}\underset{\sim}{x} = \begin{bmatrix} x_1 + 2x_2 \\ x_3 \end{bmatrix}.$$

This vector is $\underset{\sim}{0}_2$ exactly when $x_3 = 0$ and $x_1 = -2x_2$. So $\underset{\sim}{x} = (-2x_2, x_2, 0) = x_2\,(-2, 1, 0)$, and

$$\mathcal{N}(\mathbf{A}) = \text{span}\{(-2, 1, 0)\}.$$

Exercise 0.5 (Testing whether a vector is in a left null space)**.** The left null space of a matrix $\mathbf{A}$ is the null space of its transpose. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

and let y be the column vector $(2, -1)$.
Decide whether $\underset{\sim}{y}$ is in $\mathcal{N}(\mathbf{A}^\top)$.

Hint

Compute $\mathbf{A}^\top \underset{\sim}{y}$.

Solution

Solution 0.5. Here

$$\mathbf{A}^\top \underset{\sim}{y} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 - 2 \\ 4 - 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

so $\underset{\sim}{y}$ is in $\mathcal{N}(\mathbf{A}^\top)$. Here $\mathbf{A}$ equals its own transpose, so $\underset{\sim}{y}$ is in $\mathcal{N}(\mathbf{A})$ as well.

Theorem 0.2 (A null space is a subspace)**.** *For any $m \times n$ matrix $\mathbf{A}$, $\mathcal{N}(\mathbf{A})$ (Definition 0.3) is a subspace of $\mathbb{R}^n$ (Definition 0.1).*

Proof

Proof. $\mathbf{A}\underset{\sim}{0}_n = \underset{\sim}{0}_m$, so $\underset{\sim}{0}_n \in \mathcal{N}(\mathbf{A})$, and the null space is not empty. If $\underset{\sim}{u}, \underset{\sim}{w} \in \mathcal{N}(\mathbf{A})$ and $c \in \mathbb{R}$, then

$$\begin{aligned}
\mathbf{A}\,(\underset{\sim}{u}+\underset{\sim}{w}) &= \mathbf{A}\underset{\sim}{u}+\mathbf{A}\underset{\sim}{w} && \text{(Theorem 0.8)}\\
&= \underset{\sim}{0}_m+\underset{\sim}{0}_m && (\underset{\sim}{u}\text{ and }\underset{\sim}{w}\text{ are in the null space})\\
&= \underset{\sim}{0}_m && (\text{adding }\underset{\sim}{0}_m\text{ changes nothing})
\end{aligned}$$

and, for each entry $i = 1, \dots, m$,

$$\begin{aligned}
(\mathbf{A}\,(c\underset{\sim}{u}))_i &= a_{i1}(cu_1)+\cdots+a_{in}(cu_n) && \text{(Definition 0.16)}\\
&= c\,(a_{i1}u_1+\cdots+a_{in}u_n) && (\text{factor } c \text{ out of each term})\\
&= c\,(\mathbf{A}\underset{\sim}{u})_i && \text{(Definition 0.16)}\\
&= c\cdot 0 && (\underset{\sim}{u}\text{ is in the null space})\\
&= 0, && \text{(arithmetic)}
\end{aligned}$$

so $\mathbf{A}\,(c\underset{\sim}{u}) = \underset{\sim}{0}_m$. Therefore $\underset{\sim}{u}+\underset{\sim}{w}$ and $c\underset{\sim}{u}$ are in $\mathcal{N}(\mathbf{A})$. □

Example 0.4 (The solutions of $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ with $\underset{\sim}{b} \neq \underset{\sim}{0}$ are not a subspace)**.** In Example 0.3, $(2,1,0)$ and $(2,0,1)$ are in $\mathcal{N}(\mathbf{A})$, and so is their sum $(4,1,1)$, since $4-2\cdot 1-2\cdot 1 = 0$. By contrast, the solutions of $\mathbf{A}\underset{\sim}{x} = (1,3)$ for the same $\mathbf{A}$ are the vectors with $x_1-2x_2-2x_3 = 1$. That set does not contain $\underset{\sim}{0}_3$, so by Theorem 0.1 it is not a subspace.

Exercise 0.6 (Showing that the solutions of two equations are a subspace)**.** Theorem 0.2 turns a question about a subspace into a question about a null space. Let $\mathcal{S}$ hold the solutions $\underset{\sim}{x} \in \mathbb{R}^3$ of the two equations $x_1+x_2-x_3 = 0$ and $x_2 = 2x_3$.
Show that $\mathcal{S}$ is a subspace of $\mathbb{R}^3$.

Hint

Write both equations as $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_2$ for one 2×3 matrix $\mathbf{A}$.

Solution

Solution 0.6. Move every term to the left side: $x_1+x_2-x_3 = 0$ and $x_2-2x_3 = 0$. With

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -2 \end{bmatrix},$$

the two equations say $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_2$, so $\mathcal{S} = \mathcal{N}(\mathbf{A})$. By Theorem 0.2, $\mathcal{S}$ is a subspace of $\mathbb{R}^3$.

Theorem 0.3 (Independent columns exactly when the null space is trivial)**.** *For an $m\times n$ matrix $\mathbf{A}$, the following are equivalent:*

1. *The columns of $\mathbf{A}$ are linearly independent (Definition 0.1).*

2. $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$ *only for* $\underset{\sim}{x} = \underset{\sim}{0}_n$.
3. $\mathcal{N}(\mathbf{A}) = \{\underset{\sim}{0}_n\}$ *(Definition 0.3).*

So $\mathbf{A}$ *has full column rank (Definition 0.3) exactly when its null space contains only the zero vector.*

Proof

Proof. Write $\underset{\sim}{a}_1, \dots, \underset{\sim}{a}_n$ for the columns of $\mathbf{A}$. For $\underset{\sim}{x} = (x_1, \dots, x_n)$, the matrix-vector product is the combination of the columns with weights $x_1, \dots, x_n$:

$$\mathbf{A}\underset{\sim}{x} = x_1\underset{\sim}{a}_1 + \cdots + x_n\underset{\sim}{a}_n.$$

So $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$ says exactly that $x_1\underset{\sim}{a}_1 + \cdots + x_n\underset{\sim}{a}_n = \underset{\sim}{0}_m$. By Definition 0.1, the columns are linearly independent exactly when the only weights with this property are $x_1 = \cdots = x_n = 0$. That is statement 2, so statements 1 and 2 are equivalent. Statement 2 says that the only vector in $\mathcal{N}(\mathbf{A})$ is $\underset{\sim}{0}_n$, which is statement 3. The last sentence is the definition of full column rank (Definition 0.3). □

Example 0.5 (Checking independence through the null space)**.** For $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$,

$$\mathbf{A}\underset{\sim}{x} = \begin{bmatrix} x_1 + x_2 \\ x_1 + 2x_2 \\ x_1 + 3x_2 \end{bmatrix}.$$

This is $\underset{\sim}{0}_3$ only if $x_2 = 0$ (subtract the first entry from the second), and then $x_1 = 0$. So $\mathcal{N}(\mathbf{A}) = \{\underset{\sim}{0}_2\}$ and the two columns are linearly independent.
For $\mathbf{B} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix}$, $\mathbf{B}(2, -1) = \underset{\sim}{0}_3$, so $\mathcal{N}(\mathbf{B})$ contains a nonzero vector and the columns of $\mathbf{B}$ are linearly dependent.

Exercise 0.7 (Testing independent columns through the null space)**.** Theorem 0.3 turns a question about independence into one about solving $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}$. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 1 \end{bmatrix}.$$

Decide whether the columns of $\mathbf{A}$ are linearly independent.

Hint

Solve $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_3$, starting with the third entry.

Solution

Solution 0.7. For $\underset{\sim}{x} = (x_1, x_2)$,

$$\mathbf{A}\underset{\sim}{x} = \begin{bmatrix} x_1 + 2x_2 \\ 2x_1 + 4x_2 \\ x_2 \end{bmatrix}.$$

The third entry gives $x_2 = 0$, and then the first gives $x_1 = 0$. So $\mathcal{N}(\mathbf{A}) = \{\underset{\sim}{0}_2\}$, and by Theorem 0.3 the columns of $\mathbf{A}$ are linearly independent.

Theorem 0.4 (The row space and null space share only the zero vector)**.** *For any $m \times n$ matrix $\mathbf{A}$,*

$$\mathcal{C}(\mathbf{A}^\top) \cap \mathcal{N}(\mathbf{A}) = \{\underset{\sim}{0}_n\}.$$

Proof

Proof. Both sets are subspaces, by Theorem 0.1 and Theorem 0.2, so both contain $\underset{\sim}{0}_n$ (Theorem 0.1).
Now take any $\underset{\sim}{x}$ in both sets. Being in the row space means $\underset{\sim}{x} = \mathbf{A}^\top \underset{\sim}{u}$ for some $\underset{\sim}{u} \in \mathbb{R}^m$ (Definition 0.1), and being in the null space means $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$ (Definition 0.3). Then

$$\begin{aligned}
\underset{\sim}{x}^\top \underset{\sim}{x} &= \left(\mathbf{A}^\top \underset{\sim}{u}\right)^\top \underset{\sim}{x} && \text{(substitute } \underset{\sim}{x} = \mathbf{A}^\top \underset{\sim}{u} \text{ in the first factor)} \\
&= \left(\underset{\sim}{u}^\top \left(\mathbf{A}^\top\right)^\top\right) \underset{\sim}{x} && \text{(Theorem 0.19)} \\
&= \left(\underset{\sim}{u}^\top \mathbf{A}\right) \underset{\sim}{x} && \text{(transposing twice returns the original matrix)} \\
&= \underset{\sim}{u}^\top \left(\mathbf{A}\underset{\sim}{x}\right) && \text{(Theorem 0.7)} \\
&= \underset{\sim}{u}^\top \underset{\sim}{0}_m && \text{(substitute } \mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m) \\
&= 0. && \text{(every term of the inner product is 0)}
\end{aligned}$$

But $\underset{\sim}{x}^\top \underset{\sim}{x} = x_1^2 + \cdots + x_n^2$, which is 0 only when every $x_i = 0$. So $\underset{\sim}{x} = \underset{\sim}{0}_n$. □

Example 0.6 (The row space and null space in Example 0.3)**.** For $\mathbf{A}$ in Example 0.1, the row space is $\text{span}\{(1, -2, -2)\}$ and the null space is $\text{span}\{(2, 1, 0), (2, 0, 1)\}$ (Example 0.3). A vector $c\,(1, -2, -2)$ of the row space is in the null space only if

$$\begin{aligned}\mathbf{A}\,(c\,(1,-2,-2)) &= c\,(1-2\cdot(-2)-2\cdot(-2),\ 3\,(1-2\cdot(-2)-2\cdot(-2))) && \text{(Example 0.3's formula for } \mathbf{A}\underset{\sim}{x}\text{)}\\ &= c\,(9,27) && \text{(arithmetic)}\end{aligned}$$

is $\underset{\sim}{0}_2$, that is, only if $c = 0$. So the two subspaces share only $\underset{\sim}{0}_3$.

Exercise 0.8 (Testing a row-space vector against the null space)**.** Theorem 0.4 says that a nonzero vector in the row space is never in the null space. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \qquad \underset{\sim}{v} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix},$$

so that $\underset{\sim}{v} = (1,1,0) + (0,1,1)$ adds the two rows of $\mathbf{A}$.
Check that $\underset{\sim}{v}$ is not in $\mathcal{N}(\mathbf{A})$.

Hint

Multiply each row of $\mathbf{A}$ by $\underset{\sim}{v}$.

Solution

Solution 0.8. Here

$$\mathbf{A}\underset{\sim}{v} = \begin{bmatrix} 1+2+0 \\ 0+2+1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} \neq \underset{\sim}{0}_2,$$

so $\underset{\sim}{v}$ is not in $\mathcal{N}(\mathbf{A})$, as Theorem 0.4 requires of a nonzero vector in the row space.

Definition 0.5 (Cross-product matrix)**.** Let $\mathbf{A}$ be an $m \times n$ matrix with columns $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_n$ and $\mathbf{B}$ an $m \times q$ matrix with columns $\underset{\sim}{b}_1, \ldots, \underset{\sim}{b}_q$. The **cross-product matrix** of $\mathbf{A}$ and $\mathbf{B}$ is the $n \times q$ matrix $\mathbf{A}^\top\mathbf{B}$, whose entry (i, j) is the dot product $\underset{\sim}{a}_i \cdot \underset{\sim}{b}_j$.

This is what R's `crossprod(A, B)` computes, and `crossprod(A)` is $\mathbf{A}^\top\mathbf{A}$. In NumPy it has no name of its own: write `A.T @ B`.

For two column vectors it is the 1×1 matrix holding their dot product. It is unrelated to the cross product $\underset{\sim}{u} \times \underset{\sim}{v}$ of two vectors in $\mathbb{R}^3$, and differs from the Frobenius inner product $\operatorname{tr}(\mathbf{A}^\top\mathbf{B})$, which sums the diagonal of this matrix into a single number.

Example 0.7 (A cross-product matrix)**.** Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Then $\mathbf{A}^\top \mathbf{B}$ is 2×1, and

$$\begin{aligned} \mathbf{A}^\top \mathbf{B} &= \begin{bmatrix} \underset{\sim}{a}_1 \cdot \underset{\sim}{b}_1 \\ \underset{\sim}{a}_2 \cdot \underset{\sim}{b}_1 \end{bmatrix} && \text{(Definition 0.5)} \\ &= \begin{bmatrix} 1+2+0 \\ 0+2+3 \end{bmatrix} && \text{(multiply and add)} \\ &= \begin{bmatrix} 3 \\ 5 \end{bmatrix}. && \text{(add)} \end{aligned}$$

Exercise 0.9 (Computing a cross-product matrix)**.** In R, `crossprod(A, B)` computes the cross-product matrix $\mathbf{A}^\top \mathbf{B}$. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Compute $\mathbf{A}^\top \mathbf{B}$.

Hint

Entry i is the dot product of column i of $\mathbf{A}$ with $(2, 3)$.

Solution

Solution 0.9. The columns of $\mathbf{A}$ are $(1, 1)$ and $(0, 1)$, so

$$\mathbf{A}^\top \mathbf{B} = \begin{bmatrix} 1 \cdot 2 + 1 \cdot 3 \\ 0 \cdot 2 + 1 \cdot 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}.$$

Remark (Column-column products)

Remark 0.1 (Column-column products). Ordinary matrix multiplication pairs a row of the left factor with a column of the right factor, which makes each entry a row-by-column product [38, Sec. 2]. Each entry of a cross-product matrix instead pairs a column of $\mathbf{A}$ with a column of $\mathbf{B}$, so in these notes we call it a column-column product.

Definition 0.6 (Gram matrix)**.** The **Gram matrix** [39] of an $m \times n$ matrix $\mathbf{A}$ is the $n \times n$ matrix $\mathbf{A}^\top \mathbf{A}$, the cross-product matrix of $\mathbf{A}$ with itself.

Example 0.8 (A Gram matrix and its entries)**.** Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$, with columns $\underset{\sim}{a}_1 = (1, 1, 0)$ and $\underset{\sim}{a}_2 = (0, 1, 1)$. Then

$$\begin{aligned} \mathbf{A}^\top \mathbf{A} &= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} && \text{(Definition 0.6)} \\ &= \begin{bmatrix} 1\cdot 1 + 1\cdot 1 + 0\cdot 0 & 1\cdot 0 + 1\cdot 1 + 0\cdot 1 \\ 0\cdot 1 + 1\cdot 1 + 1\cdot 0 & 0\cdot 0 + 1\cdot 1 + 1\cdot 1 \end{bmatrix} && \text{(Definition 0.10)} \\ &= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}. && \text{(multiply and add)} \end{aligned}$$

Row i of $\mathbf{A}^\top$ is column i of $\mathbf{A}$, so entry (i, j) of the Gram matrix is the dot product $\underset{\sim}{a}_i \cdot \underset{\sim}{a}_j$: for instance,

$$\begin{aligned} \underset{\sim}{a}_1 \cdot \underset{\sim}{a}_2 &= 0 + 1 + 0 \\ &= 1 \end{aligned}$$

is entry $(1, 2)$.

Exercise 0.10 (Computing a Gram matrix)**.** The Gram matrix holds the dot products of every pair of columns of $\mathbf{A}$. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Compute $\mathbf{A}^\top \mathbf{A}$.

Hint

Entry (i, j) is $\underset{\sim}{a}_i \cdot \underset{\sim}{a}_j$, so entries $(1, 2)$ and $(2, 1)$ agree.

Solution

Solution 0.10. The columns of $\mathbf{A}$ are $\underset{\sim}{a}_1 = (1, 0, 1)$ and $\underset{\sim}{a}_2 = (2, 1, 0)$, so

$$\begin{aligned} \underset{\sim}{a}_1 \cdot \underset{\sim}{a}_1 &= 1 + 0 + 1 = 2, \\ \underset{\sim}{a}_1 \cdot \underset{\sim}{a}_2 &= 2 + 0 + 0 = 2, \\ \underset{\sim}{a}_2 \cdot \underset{\sim}{a}_2 &= 4 + 1 + 0 = 5, \end{aligned}$$

and

$$\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 2 & 2 \\ 2 & 5 \end{bmatrix}.$$

Theorem 0.5 (The cross-product matrix is the sum of the outer products of the rows)**.** *Let* $\mathbf{A}$ *be an* $m \times n$ *matrix with rows* $\underset{\sim}{r}_1{}^\top, \dots, \underset{\sim}{r}_m{}^\top$*. Let* $\mathbf{B}$ *be an* $m \times q$ *matrix with rows* $\underset{\sim}{s}_1{}^\top, \dots, \underset{\sim}{s}_m{}^\top$*. Then the cross-product matrix of* $\mathbf{A}$ *with* $\mathbf{B}$ *is*

$$\mathbf{A}^\top \mathbf{B} = \sum_{i=1}^{m} \underset{\sim}{r}_i \, \underset{\sim}{s}_i{}^\top.$$

Proof

Proof. Compare entry (j, k) of each side.

$$\begin{aligned}
(\mathbf{A}^\top \mathbf{B})_{jk} &= \sum_{i=1}^{m} (\mathbf{A})_{ij} (\mathbf{B})_{ik} && \text{(Definition 0.10)} \\
&= \sum_{i=1}^{m} (\underset{\sim}{r}_i)_j (\underset{\sim}{s}_i)_k && \text{(entries of the rows)} \\
&= \sum_{i=1}^{m} (\underset{\sim}{r}_i \, \underset{\sim}{s}_i{}^\top)_{jk} && \text{(Definition 0.1)} \\
&= \left(\sum_{i=1}^{m} \underset{\sim}{r}_i \, \underset{\sim}{s}_i{}^\top \right)_{jk}. && \text{(entries of a sum)}
\end{aligned}$$

Every entry agrees, so the two matrices agree. □

Example 0.9 (The Gram matrix and a cross-product as sums of outer products)**.** Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ be the matrices of Example 0.7. The rows of $\mathbf{A}$ are $\underset{\sim}{r}_1 = (1, 0)$, $\underset{\sim}{r}_2 = (1, 1)$, $\underset{\sim}{r}_3 = (0, 1)$. Taking $\mathbf{B} = \mathbf{A}$ in Theorem 0.5, the Gram matrix is the sum of the outer products of the rows with themselves:

$$\begin{aligned}
\mathbf{A}^\top \mathbf{A} &= \underset{\sim}{r}_1 \, \underset{\sim}{r}_1{}^\top + \underset{\sim}{r}_2 \, \underset{\sim}{r}_2{}^\top + \underset{\sim}{r}_3 \, \underset{\sim}{r}_3{}^\top && \text{(Theorem 0.5)} \\
&= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} && \text{(Definition 0.1)} \\
&= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}. && \text{(add)}
\end{aligned}$$

This is the Gram matrix of Example 0.8. Now keep the column $\mathbf{B}$ above. Its rows are

the 1-vectors $\underset{\sim}{s}_1 = (1)$, $\underset{\sim}{s}_2 = (2)$, $\underset{\sim}{s}_3 = (3)$, so

$$\begin{aligned} \mathbf{A}^\top \mathbf{B} &= 1\underset{\sim}{r}_1 + 2\underset{\sim}{r}_2 + 3\underset{\sim}{r}_3 && \text{(Theorem 0.5)} \\ &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 3 \end{bmatrix} && \text{(scalar multiples)} \\ &= \begin{bmatrix} 3 \\ 5 \end{bmatrix}, && \text{(add)} \end{aligned}$$

the same as in Example 0.7.

Exercise 0.11 (Computing a cross-product matrix one row at a time)**.** Theorem 0.5 builds a cross-product matrix by adding one outer product per row, which is how it can be updated as observations arrive. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Compute $\mathbf{A}^\top \mathbf{B}$ by adding two outer products.

Hint

Rows of $\mathbf{A}$ are $(1, 2)$ and $(0, 1)$, and rows of $\mathbf{B}$ are 3 and 1.

Solution

Solution 0.11. By Theorem 0.5,

$$\mathbf{A}^\top \mathbf{B} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} [3] + \begin{bmatrix} 0 \\ 1 \end{bmatrix} [1] = \begin{bmatrix} 3 \\ 6 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}.$$

Multiplying directly gives the same: $\mathbf{A}^\top \mathbf{B} = (1 \cdot 3 + 0 \cdot 1,\ 2 \cdot 3 + 1 \cdot 1) = (3, 7)$.

Theorem 0.6 ($\mathbf{A}^\top \mathbf{A}$ has the same null space as $\mathbf{A}$)**.** *For any $m \times n$ matrix $\mathbf{A}$,*

$$\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A}).$$

Proof

Proof. $\mathcal{N}(\mathbf{A}) \subseteq \mathcal{N}(\mathbf{A}^\top \mathbf{A})$**.** If $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$, then

$$\begin{aligned} \mathbf{A}^\top \mathbf{A}\underset{\sim}{x} &= \mathbf{A}^\top \underset{\sim}{0}_m \\ &= \underset{\sim}{0}_n. \end{aligned}$$

$\mathcal{N}(\mathbf{A}^\top\mathbf{A}) \subseteq \mathcal{N}(\mathbf{A})$. If $\mathbf{A}^\top\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_n$, then

$$\begin{aligned}
0 &= \underset{\sim}{x}^\top \underset{\sim}{0}_n && \text{(every term of the inner product is 0)} \\
&= \underset{\sim}{x}^\top (\mathbf{A}^\top\mathbf{A}\underset{\sim}{x}) && \text{(substitute } \underset{\sim}{0}_n = \mathbf{A}^\top\mathbf{A}\underset{\sim}{x}\text{)} \\
&= (\underset{\sim}{x}^\top\mathbf{A}^\top)(\mathbf{A}\underset{\sim}{x}) && \text{(Theorem 0.7)} \\
&= (\mathbf{A}\underset{\sim}{x})^\top(\mathbf{A}\underset{\sim}{x}) && \text{(Theorem 0.19)} \\
&= (\mathbf{A}\underset{\sim}{x})\cdot(\mathbf{A}\underset{\sim}{x}) && \text{(Example 0.8)} \\
&= \|\mathbf{A}\underset{\sim}{x}\|^2. && \text{(square both sides of Equation 2)}
\end{aligned}$$

A vector whose Euclidean norm is 0 has every entry 0, so $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$. □

Example 0.10 (The null space of a Gram matrix)**.** For $\mathbf{A}$ in Example 0.1,

$$\begin{aligned}
\mathbf{A}^\top\mathbf{A} &= \begin{bmatrix} 1 & 3 \\ -2 & -6 \\ -2 & -6 \end{bmatrix}\begin{bmatrix} 1 & -2 & -2 \\ 3 & -6 & -6 \end{bmatrix} \\
&= \begin{bmatrix} 10 & -20 & -20 \\ -20 & 40 & 40 \\ -20 & 40 & 40 \end{bmatrix}.
\end{aligned}$$

Every row is a multiple of $(1, -2, -2)$, so $\mathbf{A}^\top\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_3$ exactly when $x_1 - 2x_2 - 2x_3 = 0$, the same condition as for $\mathcal{N}(\mathbf{A})$ in Example 0.3.

Exercise 0.12 (Finding the null space of a Gram matrix)**.** Theorem 0.6 lets the null space of a Gram matrix be read off from the original matrix. Let $\mathbf{A}$ be the 3×2 matrix whose every entry is 1.
Find $\mathcal{N}(\mathbf{A}^\top\mathbf{A})$.

Hint

Find $\mathcal{N}(\mathbf{A})$ first. Each entry of $\mathbf{A}\underset{\sim}{x}$ is $x_1 + x_2$.

Solution

Solution 0.12. Each entry of $\mathbf{A}\underset{\sim}{x}$ is $x_1 + x_2$, so $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_3$ exactly when $x_2 = -x_1$, and $\mathcal{N}(\mathbf{A}) = \text{span}\{(1, -1)\}$. By Theorem 0.6,

$$\mathcal{N}(\mathbf{A}^\top\mathbf{A}) = \text{span}\{(1, -1)\}.$$

As a check, $\mathbf{A}^\top\mathbf{A} = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}$, which sends $(1, -1)$ to $\underset{\sim}{0}_2$.

Rank-nullity and rank factorization

Source

This section is adapted from Zhou [40], used under the MIT License (see the license text in Section). Theorem 0.5 and Corollary 0.2 restore the source's statements that a product has rank at most the rank of either factor, and that multiplying by a matrix of full column rank or by a matrix whose rank is its number of rows does not change the rank. These parts of the source are not part of this section:

- the names "full row rank", "full rank" and "singular"
- the statement that $\mathcal{C}(\mathbf{C}) \subseteq \mathcal{C}(\mathbf{A})$ exactly when $\mathbf{C} = \mathbf{AB}$ for some $\mathbf{B}$, and its row space version, and the containment $\mathcal{N}(\mathbf{B}) \subseteq \mathcal{N}(\mathbf{AB})$
- the dimension of a subspace as the largest number of linearly independent vectors in it, or the smallest number of vectors that span it
- the properties of a rank factorization $\mathbf{A} = \mathbf{CR}$ ($\operatorname{rank}(\mathbf{C}) = \operatorname{rank}(\mathbf{R}) = r$, $\mathcal{C}(\mathbf{A}) = \mathcal{C}(\mathbf{C})$, $\mathcal{N}(\mathbf{A}) = \mathcal{N}(\mathbf{R})$), the fact that it is not unique, and the second proof of $\operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{A}^\top)$ that it gives
- calling $\mathbf{AA}^\top$ a Gram matrix, and the equation $\mathcal{N}(\mathbf{AA}^\top) = \mathcal{N}(\mathbf{A}^\top)$ (Theorem 0.6 applied to $\mathbf{A}^\top$ gives it)
- the statement that $\mathcal{C}(\mathbf{A})$ and $\mathcal{N}(\mathbf{A}^\top)$ share only the zero vector (it follows from Theorem 0.1)

Theorem 0.1 (The rank is the dimension of the column space)**.** *For any $m \times n$ matrix $\mathbf{A}$,*

$$\operatorname{rank}(\mathbf{A}) = \dim(\mathcal{C}(\mathbf{A})).$$

Proof

Proof. Let $r = \operatorname{rank}(\mathbf{A})$ (Definition 0.2), and choose r linearly independent columns of $\mathbf{A}$; by Definition 0.2, no larger set of columns is linearly independent. We show the chosen columns are a basis of $\mathcal{C}(\mathbf{A})$ (Definition 0.3).

Every column is in the span of the chosen columns. A chosen column is in that span trivially. If some unchosen column $\underset{\sim}{a}_j$ were not in the span, then adding it to the chosen columns would give $r+1$ linearly independent columns (Theorem 0.8), more than r. So every column of $\mathbf{A}$ is in the span of the chosen columns.

The chosen columns span $\mathcal{C}(\mathbf{A})$. The span of the chosen columns is a subspace (Theorem 0.2), so it is closed under addition and scalar multiplication, and it contains every linear combination of the columns of $\mathbf{A}$. By Theorem 0.1, those combinations make

up $\mathcal{C}(\mathbf{A})$, so $\mathcal{C}(\mathbf{A})$ is contained in the span of the chosen columns. The chosen columns are themselves columns of $\mathbf{A}$, so their span is contained in $\mathcal{C}(\mathbf{A})$, and the two sets are equal.
The chosen columns are linearly independent and span $\mathcal{C}(\mathbf{A})$, so they are a basis of it, and $\dim(\mathcal{C}(\mathbf{A})) = r$ (Definition 0.6). If $r = 0$, every column is $\underset{\sim}{0}_m$, $\mathcal{C}(\mathbf{A}) = \{\underset{\sim}{0}_m\}$, and both sides are 0. □

Example 0.1 (The rank of the matrix in Example 0.1)**.** For $\mathbf{A}$ in Example 0.1, $\mathcal{C}(\mathbf{A}) = \text{span}\{(1, 3)\}$ (1), and $(1, 3)$ is linearly independent because it is nonzero, so it is a basis and $\mathcal{C}(\mathbf{A})$ has dimension 1. So $\text{rank}(\mathbf{A}) = 1$, by Theorem 0.1. Directly: the first column $(1, 3)$ is linearly independent on its own, and any two columns are dependent, because each column is a multiple of $(1, 3)$.

Exercise 0.1 (Finding a rank from the column space)**.** Theorem 0.1 reads the rank of a matrix off the dimension of its column space. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}.$$

Find $\text{rank}(\mathbf{A})$ by finding $\dim(\mathcal{C}(\mathbf{A}))$.

Hint

In $\mathbf{A}$, the second column is a multiple of the first.

Solution

Solution 0.1. In $\mathbf{A}$, the second column is $(2, 4, 6) = 2\,(1, 2, 3)$, so by Theorem 0.1

$$\mathcal{C}(\mathbf{A}) = \text{span}\{(1, 2, 3), (2, 4, 6)\} = \text{span}\{(1, 2, 3)\}.$$

The single nonzero vector $(1, 2, 3)$ is a basis of that span, so $\dim(\mathcal{C}(\mathbf{A})) = 1$, and by Theorem 0.1, $\text{rank}(\mathbf{A}) = 1$.

Definition 0.1 (Nullity)**.** The **nullity** [41] of a matrix $\mathbf{A}$ is the dimension of its null space (Definition 0.3):

$$\text{nullity}(\mathbf{A}) \stackrel{\text{def}}{=} \dim(\mathcal{N}(\mathbf{A})).$$

Example 0.2 (The nullity of the matrix in Example 0.1)**.** For $\mathbf{A}$ in Example 0.1, $\mathcal{N}(\mathbf{A}) = \text{span}\{(2,1,0),(2,0,1)\}$ (Example 0.3). Those two vectors are linearly independent: $c_1(2,1,0) + c_2(2,0,1) = (2c_1 + 2c_2, c_1, c_2)$, which is $\underset{\sim}{0}_3$ only if

$$\begin{aligned} c_1 &= c_2 \\ &= 0. \end{aligned}$$

So they are a basis of $\mathcal{N}(\mathbf{A})$, and $\text{nullity}(\mathbf{A}) = 2$. The 2×2 identity matrix has nullity 0, because $\mathbf{I}\underset{\sim}{x} = \underset{\sim}{x}$ is $\underset{\sim}{0}_2$ only when $\underset{\sim}{x} = \underset{\sim}{0}_2$.

Theorem 0.2 (Rank-nullity theorem)**.** *For any $m \times n$ matrix $\mathbf{A}$,*

$$\text{rank}(\mathbf{A}) + \text{nullity}(\mathbf{A}) = n.$$

Proof

Proof. Let $\nu = \text{nullity}(\mathbf{A})$. Choose a basis $\underset{\sim}{x}_1, \dots, \underset{\sim}{x}_\nu$ of $\mathcal{N}(\mathbf{A})$, and extend it to a basis $\underset{\sim}{x}_1, \dots, \underset{\sim}{x}_\nu, \underset{\sim}{y}_1, \dots, \underset{\sim}{y}_{n-\nu}$ of $\mathbb{R}^n$ (Theorem 0.9); it has n vectors, because every basis of $\mathbb{R}^n$ has $\dim(\mathbb{R}^n) = n$ vectors (Theorem 0.7, Example 0.14). We show that $\mathbf{A}\underset{\sim}{y}_1, \dots, \mathbf{A}\underset{\sim}{y}_{n-\nu}$ are a basis of $\mathcal{C}(\mathbf{A})$, using Theorem 0.4 to apply $\mathbf{A}$ to linear combinations.

They are linearly independent. Suppose $\sum_{i=1}^{n-\nu} v_i \mathbf{A}\underset{\sim}{y}_i = \underset{\sim}{0}_m$. Then $\mathbf{A}\left(\sum_i v_i \underset{\sim}{y}_i\right) = \underset{\sim}{0}_m$ (Theorem 0.4), so $\sum_i v_i \underset{\sim}{y}_i \in \mathcal{N}(\mathbf{A})$, and it equals $\sum_{j=1}^{\nu} u_j \underset{\sim}{x}_j$ for some numbers u_j, since the $\underset{\sim}{x}_j$ span $\mathcal{N}(\mathbf{A})$. Moving everything to one side,

$$\sum_{j=1}^{\nu} (-u_j)\, \underset{\sim}{x}_j + \sum_{i=1}^{n-\nu} v_i \underset{\sim}{y}_i = \underset{\sim}{0}_n,$$

and because the $\underset{\sim}{x}$'s and $\underset{\sim}{y}$'s together form a basis of $\mathbb{R}^n$, which is linearly independent, every coefficient is 0; in particular every $v_i = 0$.

They span $\mathcal{C}(\mathbf{A})$. Take any $\underset{\sim}{w} \in \mathcal{C}(\mathbf{A})$, so $\underset{\sim}{w} = \mathbf{A}\underset{\sim}{z}$ for some $\underset{\sim}{z} \in \mathbb{R}^n$ (Definition 0.1). Write $\underset{\sim}{z} = \sum_j a_j \underset{\sim}{x}_j + \sum_i b_i \underset{\sim}{y}_i$ in the basis of $\mathbb{R}^n$. Then

$$\begin{aligned} \underset{\sim}{w} &= \mathbf{A}\left(\sum_{j=1}^{\nu} a_j \underset{\sim}{x}_j + \sum_{i=1}^{n-\nu} b_i \underset{\sim}{y}_i\right) && \text{(substitute } \underset{\sim}{z}\text{)} \\ &= \sum_{j=1}^{\nu} a_j\, \mathbf{A}\underset{\sim}{x}_j + \sum_{i=1}^{n-\nu} b_i\, \mathbf{A}\underset{\sim}{y}_i && \text{(Theorem 0.4)} \\ &= \sum_{j=1}^{\nu} a_j\, \underset{\sim}{0}_m + \sum_{i=1}^{n-\nu} b_i\, \mathbf{A}\underset{\sim}{y}_i && \text{(each } \underset{\sim}{x}_j \in \mathcal{N}(\mathbf{A})\text{)} \\ &= \sum_{i=1}^{n-\nu} b_i\, \mathbf{A}\underset{\sim}{y}_i, && \text{(drop the zero terms)} \end{aligned}$$

a linear combination of $\mathbf{A}\underset{\sim}{y}_1, \ldots, \mathbf{A}\underset{\sim}{y}_{n-\nu}$. These vectors are in $\mathcal{C}(\mathbf{A})$, so their span is exactly $\mathcal{C}(\mathbf{A})$.
So $\dim(\mathcal{C}(\mathbf{A})) = n - \nu$, and $\operatorname{rank}(\mathbf{A}) = n - \nu$ by Theorem 0.1. □

Example 0.3 (Checking rank-nullity on the matrix in Example 0.1)**.** The matrix $\mathbf{A}$ in Example 0.1 has $n = 3$ columns, rank 1 (Example 0.1) and nullity 2 (Example 0.2), and $1+2 = 3$. The 2×2 identity matrix has rank 2 and nullity 0 (Example 0.2), and $2+0 = 2$.

Exercise 0.2 (Finding the nullity of a 3×3 matrix)**.** The nullity of a matrix is the dimension of its null space. Let

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

Find $\operatorname{nullity}(\mathbf{A})$.

Hint

Write out $\mathbf{A}\underset{\sim}{x}$, find the null space, and then find a basis of it.

Solution

Solution 0.2. For $\underset{\sim}{x} = (x_1, x_2, x_3)$, $\mathbf{A}\underset{\sim}{x} = (2x_1, 0, 5x_3)$, which is $\underset{\sim}{0}_3$ exactly when $x_1 = 0$ and $x_3 = 0$. So $\mathcal{N}(\mathbf{A}) = \{(0, x_2, 0) : x_2 \in \mathbb{R}\} = \operatorname{span}\{\underset{\sim}{e}_2\}$. The single nonzero vector $\underset{\sim}{e}_2$ is a basis of it, so $\operatorname{nullity}(\mathbf{A}) = 1$.

Exercise 0.3 (Finding a nullity from a rank)**.** Suppose that a 5×8 matrix $\mathbf{A}$ has rank 3.
Find $\operatorname{nullity}(\mathbf{A})$.

Hint

Use Theorem 0.2 with $n = 8$.

Solution

Solution 0.3. By Theorem 0.2, $\operatorname{rank}(\mathbf{A}) + \operatorname{nullity}(\mathbf{A}) = 8$, so

$$\operatorname{nullity}(\mathbf{A}) = 8 - 3 = 5.$$

Theorem 0.3 (Multiplying on the right cannot increase the rank)**.** *If* $\mathbf{A}$ *is* $m \times n$ *and* $\mathbf{B}$ *is* $n \times k$*, then*

$$\operatorname{rank}(\mathbf{AB}) \leq \operatorname{rank}(\mathbf{A}).$$

Proof

Proof. $\mathcal{C}(\mathbf{AB}) \subseteq \mathcal{C}(\mathbf{A})$**.** Any vector in $\mathcal{C}(\mathbf{AB})$ is $(\mathbf{AB})\,\underset{\sim}{x}$ for some $\underset{\sim}{x} \in \mathbb{R}^k$ (Definition 0.1), and

$$(\mathbf{AB})\,\underset{\sim}{x} = \mathbf{A}\,(\mathbf{B}\underset{\sim}{x}) \quad \text{(Theorem 0.7)}$$

is $\mathbf{A}$ times the vector $\mathbf{B}\underset{\sim}{x} \in \mathbb{R}^n$, so it is in $\mathcal{C}(\mathbf{A})$.
Compare dimensions. Both column spaces are subspaces (Theorem 0.1), so

$$\begin{aligned} \operatorname{rank}(\mathbf{AB}) &= \dim(\mathcal{C}(\mathbf{AB})) && \text{(Theorem 0.1)} \\ &\leq \dim(\mathcal{C}(\mathbf{A})) && \text{(Theorem 0.10, part 2)} \\ &= \operatorname{rank}(\mathbf{A}). && \text{(Theorem 0.1)} \end{aligned}$$

□

Example 0.4 (A product can lose rank)**.** Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \qquad \mathbf{AB} = \begin{bmatrix} 1\cdot 0 + 0\cdot 1 & 1\cdot 0 + 0\cdot 0 \\ 0\cdot 0 + 0\cdot 1 & 0\cdot 0 + 0\cdot 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

$\mathbf{A}$ has rank 1, but $\mathbf{AB}$ has rank 0, so the inequality in Theorem 0.3 can be strict. With the 2×2 identity in place of $\mathbf{B}$, $\mathbf{AI} = \mathbf{A}$ keeps rank 1, and the inequality is an equality.

Exercise 0.4 (Finding how large the rank of a matrix product can be)**.** Theorem 0.3 shows how large the rank of a matrix product can be. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}.$$

Find the number that, by Theorem 0.3, $\operatorname{rank}(\mathbf{AB})$ cannot exceed.

Hint

Find $\operatorname{rank}(\mathbf{A})$ first.

Solution

Solution 0.4. Columns of $\mathbf{A}$ are $(1,0)$ and $\underset{\sim}{0}_2$, so its column space is $\operatorname{span}\{(1,0)\}$, and Theorem 0.1 gives $\operatorname{rank}(\mathbf{A}) = 1$. By Theorem 0.3, $\operatorname{rank}(\mathbf{AB}) \leq 1$. Indeed, $\mathbf{AB} = \begin{bmatrix} 2 & 3 \\ 0 & 0 \end{bmatrix}$, which has rank 1.

Theorem 0.4 (Fundamental theorem of ranks)**.** *For any $m \times n$ matrix $\mathbf{A}$,*

$$\begin{aligned} \operatorname{rank}(\mathbf{A}) &= \operatorname{rank}(\mathbf{A}^\top) \\ &= \operatorname{rank}(\mathbf{A}^\top \mathbf{A}) \\ &= \operatorname{rank}(\mathbf{A}\mathbf{A}^\top). \end{aligned}$$

In particular, the largest number of linearly independent rows of $\mathbf{A}$ equals the largest number of linearly independent columns.

Proof

Proof. $\operatorname{rank}(\mathbf{A}^\top \mathbf{A}) = \operatorname{rank}(\mathbf{A})$**.** Both matrices have n columns, and $\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A})$ (Theorem 0.6), so they have the same nullity (Definition 0.1). Then

$$\begin{aligned} \operatorname{rank}(\mathbf{A}^\top \mathbf{A}) &= n - \operatorname{nullity}(\mathbf{A}^\top \mathbf{A}) && \text{(Theorem 0.2)} \\ &= n - \operatorname{nullity}(\mathbf{A}) && \text{(equal null spaces)} \\ &= \operatorname{rank}(\mathbf{A}). && \text{(Theorem 0.2)} \end{aligned}$$

$\operatorname{rank}(\mathbf{A}\mathbf{A}^\top) = \operatorname{rank}(\mathbf{A}^\top)$**.** Apply the previous step to the $n \times m$ matrix $\mathbf{A}^\top$, using $(\mathbf{A}^\top)^\top = \mathbf{A}$.

Chain the inequalities.

$$\begin{aligned} \operatorname{rank}(\mathbf{A}) &= \operatorname{rank}(\mathbf{A}^\top \mathbf{A}) && \text{(first step)} \\ &\leq \operatorname{rank}(\mathbf{A}^\top) && \text{(Theorem 0.3, with } \mathbf{A}^\top \text{ on the left)} \\ &= \operatorname{rank}(\mathbf{A}\mathbf{A}^\top) && \text{(second step)} \\ &\leq \operatorname{rank}(\mathbf{A}). && \text{(Theorem 0.3, with } \mathbf{A} \text{ on the left)} \end{aligned}$$

The chain starts and ends at $\operatorname{rank}(\mathbf{A})$, so every quantity in it equals $\operatorname{rank}(\mathbf{A})$. The columns of $\mathbf{A}^\top$ are the rows of $\mathbf{A}$, which gives the statement about rows. □

Example 0.5 (Four equal ranks)**.** For $\mathbf{A}$ in Example 0.1, $\operatorname{rank}(\mathbf{A}) = 1$ (Example 0.1). Its transpose $\mathbf{A}^\top = \begin{bmatrix} 1 & 3 \\ -2 & -6 \\ -2 & -6 \end{bmatrix}$ has second column 3 times its first, so its rank is 1. $\mathbf{A}^\top \mathbf{A}$

in Example 0.10 has every row a multiple of $(1, -2, -2)$, so its rank is 1 too, and

$$\mathbf{A}\mathbf{A}^\top = \begin{bmatrix} 1\cdot 1 + (-2)(-2) + (-2)(-2) & 1\cdot 3 + (-2)(-6) + (-2)(-6) \\ 3\cdot 1 + (-6)(-2) + (-6)(-2) & 3\cdot 3 + (-6)(-6) + (-6)(-6) \end{bmatrix} = \begin{bmatrix} 9 & 27 \\ 27 & 81 \end{bmatrix},$$

whose second column is 3 times its first, so its rank is 1.

Corollary 0.1 (The rank is at most the smaller dimension)**.** *For any $m \times n$ matrix $\mathbf{A}$,* $\operatorname{rank}(\mathbf{A}) \leq \min\{m, n\}$.

Proof

Proof. $\mathbf{A}$ has n columns, so at most n of them are linearly independent, and $\operatorname{rank}(\mathbf{A}) \leq n$ (Definition 0.2). Likewise $\mathbf{A}^\top$ has m columns, so $\operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{A}^\top) \leq m$ (Theorem 0.4). □

Example 0.6 (A 2×3 matrix has rank at most 2)**.** By Corollary 0.1, every 2×3 matrix has rank at most $\min\{2, 3\} = 2$, even though it has 3 columns. The matrix $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ of Remark 0.1 reaches that bound: its first two columns $(1, 0)$ and $(0, 1)$ are linearly independent, so its rank is 2.

Theorem 0.5 (Multiplying on the left cannot increase the rank)**.** *If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times k$, then*

$$\operatorname{rank}(\mathbf{AB}) \leq \operatorname{rank}(\mathbf{B}).$$

Proof

Proof.

$$\begin{aligned} \operatorname{rank}(\mathbf{AB}) &= \operatorname{rank}((\mathbf{AB})^\top) && \text{(Theorem 0.4)} \\ &= \operatorname{rank}(\mathbf{B}^\top \mathbf{A}^\top) && \text{(Theorem 0.19)} \\ &\leq \operatorname{rank}(\mathbf{B}^\top) && \text{(Theorem 0.3, with } \mathbf{B}^\top \text{ on the left)} \\ &= \operatorname{rank}(\mathbf{B}). && \text{(Theorem 0.4)} \end{aligned}$$

□

Example 0.7 (A product of rank 1 with a factor of rank 2)**.** Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \qquad \mathbf{AB} = \begin{bmatrix} 1\cdot 1 + 1\cdot 1 & 1\cdot 0 + 1\cdot 0 \\ 0\cdot 1 + 1\cdot 1 & 0\cdot 0 + 1\cdot 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & 0 \end{bmatrix}.$$

$\mathbf{A}$ has rank 2, because its columns $(1, 0)$ and $(1, 1)$ are linearly independent. $\mathbf{B}$ has rank 1, because its second column is $\underset{\sim}{0}_2$ and its first is not, and $\mathbf{AB}$ has rank 1 for the same reason. So $\text{rank}(\mathbf{AB}) = 1 \leq \text{rank}(\mathbf{B}) = 1$ (Theorem 0.5), and the bound of Theorem 0.3, 2, is not the tighter one.

Corollary 0.2 (Multiplying by a matrix of full rank does not change the rank)**.** *Let* $\mathbf{A}$ *be* $m \times n$ *and* $\mathbf{B}$ *be* $n \times k$*.*

1. *If* $\mathbf{A}$ *has full column rank (Definition 0.3), then* $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{B})$*.*
2. *If* $\text{rank}(\mathbf{B}) = n$*, then* $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{A})$*.*

Proof

Proof. **Part 1.** Because $\mathbf{A}$ has full column rank, $\mathbf{A}\underset{\sim}{y} = \underset{\sim}{0}_m$ only for $\underset{\sim}{y} = \underset{\sim}{0}_n$ (Theorem 0.3). So for $\underset{\sim}{x} \in \mathbb{R}^k$,

$$\mathbf{AB}\underset{\sim}{x} = \underset{\sim}{0}_m \iff \mathbf{B}\underset{\sim}{x} = \underset{\sim}{0}_n \quad \text{(Theorem 0.3, with } \underset{\sim}{y} = \mathbf{B}\underset{\sim}{x})$$

so $\mathcal{N}(\mathbf{AB}) = \mathcal{N}(\mathbf{B})$ (Definition 0.3), and the two matrices have the same nullity (Definition 0.1). Both have k columns, so by Theorem 0.2 they have the same rank.
Part 2. $\mathcal{C}(\mathbf{B})$ is a subspace of $\mathbb{R}^n$ (Theorem 0.1) of dimension n (Theorem 0.1), so $\mathcal{C}(\mathbf{B}) = \mathbb{R}^n$ (Theorem 0.11). Every vector of $\mathcal{C}(\mathbf{AB})$ is $\mathbf{A}\,(\mathbf{B}\underset{\sim}{x})$ for some $\underset{\sim}{x}$ (Definition 0.1), and $\mathbf{B}\underset{\sim}{x}$ takes every vector of $\mathbb{R}^n$, so $\mathcal{C}(\mathbf{AB}) = \mathcal{C}(\mathbf{A})$ and the two matrices have the same rank (Theorem 0.1). □

Example 0.8 (Ranks of products with a full-rank factor)**.** Let $\mathbf{B} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$, which has rank 1.

- Part 1: $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$ has two linearly independent columns, so it has full column rank, and $\mathbf{AB} = \begin{bmatrix} 2 & 0 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}$ has rank $1 = \text{rank}(\mathbf{B})$.

- Part 2: let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, which has rank 1, and let $\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, which has rank 2. Then $\mathbf{AB} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ has rank $1 = \operatorname{rank}(\mathbf{A})$.

Theorem 0.6 (A centered data matrix has rank at most $n-1$)**.** *Let $\mathbf{X}$ be an $n \times p$ matrix with rows $\underset{\sim}{x}_1^\top, \ldots, \underset{\sim}{x}_n^\top$, let $\bar{\underset{\sim}{x}} = \frac{1}{n}\sum_{i=1}^n \underset{\sim}{x}_i$ be the mean of the rows, and let $\mathbf{X}_c = \mathbf{X} - \underset{\sim}{1}\bar{\underset{\sim}{x}}^\top$ be the matrix whose row i is $(\underset{\sim}{x}_i - \bar{\underset{\sim}{x}})^\top$, where $\underset{\sim}{1}$ is the ones vector of length n (Definition 0.16). Then* $\operatorname{rank}(\mathbf{X}_c) \le n-1$.

Proof

Proof. The entries of each column of $\mathbf{X}_c$ sum to zero:

$$\begin{aligned} \underset{\sim}{1}^\top \mathbf{X}_c &= \underset{\sim}{1}^\top \mathbf{X} - (\underset{\sim}{1}^\top \underset{\sim}{1})\bar{\underset{\sim}{x}}^\top && \text{(Theorem 0.8)} \\ &= \sum_{i=1}^n \underset{\sim}{x}_i^\top - n\,\bar{\underset{\sim}{x}}^\top && (\underset{\sim}{1}^\top \underset{\sim}{1} = n) \\ &= n\,\bar{\underset{\sim}{x}}^\top - n\,\bar{\underset{\sim}{x}}^\top && \text{(definition of } \bar{\underset{\sim}{x}}) \\ &= \underset{\sim}{0}_p^\top. \end{aligned}$$

Transposing, $\mathbf{X}_c^\top \underset{\sim}{1} = \underset{\sim}{0}_p$. The ones vector is nonzero, so it is in the null space of the $p \times n$ matrix $\mathbf{X}_c^\top$, and $\operatorname{nullity}(\mathbf{X}_c^\top) \ge 1$. By Theorem 0.2, applied to the n columns of $\mathbf{X}_c^\top$,

$$\begin{aligned} \operatorname{rank}(\mathbf{X}_c) &= \operatorname{rank}(\mathbf{X}_c^\top) && \text{(Theorem 0.4)} \\ &= n - \operatorname{nullity}(\mathbf{X}_c^\top) && \text{(Theorem 0.2)} \\ &\le n - 1. && (\operatorname{nullity}(\mathbf{X}_c^\top) \ge 1) \end{aligned}$$

□

Example 0.9 (Two points in three dimensions)**.** $\mathbf{X} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix}$ has $\bar{\underset{\sim}{x}} = (2, 2, 2)$, so

$$\mathbf{X}_c = \begin{bmatrix} -1 & 0 & 1 \\ 1 & 0 & -1 \end{bmatrix}.$$

The second row is -1 times the first, so $\operatorname{rank}(\mathbf{X}_c) = 1 = n - 1$, and the column sums are zero: $\underset{\sim}{1}^\top \mathbf{X}_c = \underset{\sim}{0}_3^\top$.

Corollary 0.3 (A sample covariance matrix from $n \leq p$ points is singular)**.** *With* $\mathbf{X}_c$ *as in Theorem 0.6, the sample covariance matrix*

$$\mathbf{S} = \frac{1}{n-1} \mathbf{X}_c{}^\top \mathbf{X}_c$$

is $p \times p$ *with* $\operatorname{rank}(\mathbf{S}) \leq n-1$*. So if* $n \leq p$*, then* $\mathbf{S}$ *is not invertible.*

Proof

Proof.

$$\begin{aligned} \operatorname{rank}(\mathbf{S}) &= \operatorname{rank}(\mathbf{X}_c{}^\top \mathbf{X}_c) && \text{(multiplying by } \tfrac{1}{n-1} \neq 0 \text{ does not change the rank)} \\ &= \operatorname{rank}(\mathbf{X}_c) && \text{(Theorem 0.4)} \\ &\leq n-1. && \text{(Theorem 0.6)} \end{aligned}$$

If $n \leq p$, then $\operatorname{rank}(\mathbf{S}) \leq n-1 < p$, so $\mathbf{S}$ does not have full column rank, and it is not invertible by Theorem 0.1. □

Example 0.10 (As many points as dimensions)**.** The $n = 3$ points $(1,0,0)$, $(0,1,0)$ and $(0,0,1)$ in $p = 3$ dimensions have $\underset{\sim}{\bar{x}} = \frac{1}{3}(1,1,1)$, so

$$\mathbf{X}_c = \mathbf{I}_3 - \tfrac{1}{3} \underset{\sim}{1}\underset{\sim}{1}^\top, \qquad \mathbf{S} = \tfrac{1}{2} \mathbf{X}_c = \tfrac{1}{6} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix},$$

using $\mathbf{X}_c{}^\top \mathbf{X}_c = \mathbf{X}_c$ (the centering matrix is a symmetric projection). Then $\mathbf{S}\underset{\sim}{1} = \underset{\sim}{0}_3$ and $\operatorname{rank}(\mathbf{S}) = 2 = n - 1 < 3$. With one more point than dimensions ($n = p + 1$), the rank bound allows $\mathbf{S}$ to have full rank p.

Definition 0.2 (Affine subspace)**.** An **affine subspace** [42] of $\mathbb{R}^p$ is a set of the form

$$\{\underset{\sim}{x}_0 + \underset{\sim}{s} : \underset{\sim}{s} \in \mathcal{S}\}$$

for some vector $\underset{\sim}{x}_0 \in \mathbb{R}^p$ and some subspace $\mathcal{S}$ of $\mathbb{R}^p$ (Definition 0.1): the image of $\mathcal{S}$ under the translation by $\underset{\sim}{x}_0$ (Definition 0.23).

Example 0.11 (A line that misses the origin is an affine subspace)**.** The line $\mathcal{T} = \{(x,1) : x \in \mathbb{R}\}$ of Example 0.2 is not a subspace, but it is an affine subspace, with $\underset{\sim}{x}_0 = (0,1)$ and $\mathcal{S} = \operatorname{span}\{(1,0)\}$ (Definition 0.2):

$$
\begin{aligned}
(0,1) + c\,(1,0) &= (0 + c,\ 1 + 0) && \text{(Definition 0.9, Definition 0.7)} \\
&= (c, 1), && \text{(add)}
\end{aligned}
$$

and as c ranges over $\mathbb{R}$, $(c, 1)$ ranges over all of $\mathcal{T}$. Every subspace $\mathcal{S}$ is also an affine subspace, with $\underset{\sim}{x}_0 = \underset{\sim}{0}$.

Theorem 0.7 (A hyperplane is a translated subspace of dimension $p-1$)**.** *Let $\underset{\sim}{w} \in \mathbb{R}^p$ be nonzero, let $b \in \mathbb{R}$, let $\mathcal{H} = \{\underset{\sim}{x} \in \mathbb{R}^p : \underset{\sim}{w}^\top \underset{\sim}{x} + b = 0\}$ be the hyperplane with normal vector $\underset{\sim}{w}$ and offset b (Definition 0.27), and let $\mathcal{H}_0 = \{\underset{\sim}{x} \in \mathbb{R}^p : \underset{\sim}{w}^\top \underset{\sim}{x} = 0\}$ be the hyperplane with the same normal vector and offset 0.*

1. *$\mathcal{H}_0$ is a subspace of $\mathbb{R}^p$ (Definition 0.1) of dimension $p - 1$ (Definition 0.6).*
2. *$\mathcal{H}$ is not empty, and for any $\underset{\sim}{x}_0 \in \mathcal{H}$, $\mathcal{H} = \{\underset{\sim}{x}_0 + \underset{\sim}{s} : \underset{\sim}{s} \in \mathcal{H}_0\}$, so $\mathcal{H}$ is an affine subspace (Definition 0.2).*
3. *For any $\underset{\sim}{x}, \underset{\sim}{y} \in \mathcal{H}$, the normal vector is orthogonal to their difference: $\underset{\sim}{w} \perp (\underset{\sim}{x} - \underset{\sim}{y})$ (Definition 0.19).*

Proof

Proof. Here $\underset{\sim}{w}^\top$ is a $1 \times p$ matrix, and $\underset{\sim}{w}^\top \underset{\sim}{x} = \underset{\sim}{w} \cdot \underset{\sim}{x}$ (Example 0.8).
Part 1. $\mathcal{H}_0$ is the null space $\mathcal{N}(\underset{\sim}{w}^\top)$ (Definition 0.3), so it is a subspace (Theorem 0.2). The p columns of $\underset{\sim}{w}^\top$ are the numbers $w_1, \ldots, w_p$. Some $w_j \neq 0$, because $\underset{\sim}{w} \neq \underset{\sim}{0}$, and that column on its own is linearly independent ($c\,w_j = 0$ forces $c = 0$), so $\operatorname{rank}(\underset{\sim}{w}^\top) \geq 1$ (Definition 0.2); and $\operatorname{rank}(\underset{\sim}{w}^\top) \leq \min\{1, p\} = 1$ (Corollary 0.1). So

$$
\begin{aligned}
\dim(\mathcal{H}_0) &= \operatorname{nullity}(\underset{\sim}{w}^\top) && \text{(Definition 0.1)} \\
&= p - \operatorname{rank}(\underset{\sim}{w}^\top) && \text{(Theorem 0.2, for the } p \text{ columns of } \underset{\sim}{w}^\top) \\
&= p - 1. && \text{(the rank is 1)}
\end{aligned}
$$

Part 2. $\underset{\sim}{w} \cdot \underset{\sim}{w} = w_1^2 + \cdots + w_p^2 > 0$, because some $w_j \neq 0$. So $\underset{\sim}{x}^* \stackrel{\text{def}}{=} -\frac{b}{\underset{\sim}{w} \cdot \underset{\sim}{w}}\, \underset{\sim}{w}$ is defined, and

$$
\begin{aligned}
\underset{\sim}{w}^\top \underset{\sim}{x}^* + b &= -\frac{b}{\underset{\sim}{w} \cdot \underset{\sim}{w}}\, (\underset{\sim}{w}^\top \underset{\sim}{w}) + b && \text{(homogeneity of } \underset{\sim}{x} \mapsto \underset{\sim}{w}^\top \underset{\sim}{x}\text{, Theorem 0.12)} \\
&= -b + b && (\underset{\sim}{w}^\top \underset{\sim}{w} = \underset{\sim}{w} \cdot \underset{\sim}{w}) \\
&= 0, && \text{(arithmetic)}
\end{aligned}
$$

so $\underset{\sim}{x}^* \in \mathcal{H}$, and $\mathcal{H}$ is not empty. Now take any $\underset{\sim}{x}_0 \in \mathcal{H}$, so $\underset{\sim}{w}^\top \underset{\sim}{x}_0 = -b$. For any $\underset{\sim}{x} \in \mathbb{R}^p$,

$$
\begin{aligned}
\underset{\sim}{w}^\top (\underset{\sim}{x} - \underset{\sim}{x}_0) &= \underset{\sim}{w}^\top \underset{\sim}{x} - \underset{\sim}{w}^\top \underset{\sim}{x}_0 && \text{(Theorem 0.4, with coefficients 1 and } -1) \\
&= \underset{\sim}{w}^\top \underset{\sim}{x} - (-b) && (\underset{\sim}{x}_0 \in \mathcal{H}) \\
&= \underset{\sim}{w}^\top \underset{\sim}{x} + b, && \text{(arithmetic)}
\end{aligned}
$$

so $\underset{\sim}{x} \in \mathcal{H}$ exactly when $\underset{\sim}{x} - \underset{\sim}{x}_0 \in \mathcal{H}_0$, that is, exactly when $\underset{\sim}{x} = \underset{\sim}{x}_0 + \underset{\sim}{s}$ with $\underset{\sim}{s} = \underset{\sim}{x} - \underset{\sim}{x}_0 \in \mathcal{H}_0$. $\mathcal{H}_0$ is a subspace (part 1), so $\mathcal{H}$ has the form in Definition 0.2.

Part 3. For $\underset{\sim}{x}, \underset{\sim}{y} \in \mathcal{H}$, both $\underset{\sim}{w}^\top \underset{\sim}{x}$ and $\underset{\sim}{w}^\top \underset{\sim}{y}$ equal $-b$, so

$$\begin{aligned} \underset{\sim}{w} \cdot (\underset{\sim}{x} - \underset{\sim}{y}) &= \underset{\sim}{w}^\top \underset{\sim}{x} - \underset{\sim}{w}^\top \underset{\sim}{y} && \text{(Theorem 0.4, with coefficients 1 and } -1) \\ &= (-b) - (-b) && (\underset{\sim}{x}, \underset{\sim}{y} \in \mathcal{H}) \\ &= 0. && \text{(arithmetic)} \end{aligned}$$

□

Example 0.12 (The plane $x_1 + x_2 + x_3 = 1$ in $\mathbb{R}^3$)**.** Take $\underset{\sim}{w} = (1,1,1)$ and $b = -1$, so $\mathcal{H} = \{\underset{\sim}{x} : x_1 + x_2 + x_3 - 1 = 0\}$, the plane that Example 0.3 showed is not a subspace. By part 1, $\mathcal{H}_0 = \{\underset{\sim}{x} : x_1 + x_2 + x_3 = 0\}$ is a subspace of dimension $3 - 1 = 2$.

- $\underset{\sim}{x}_0 = (1,0,0)$ is in $\mathcal{H}$, since $1 + 0 + 0 - 1 = 0$, and so is $\underset{\sim}{y} = (0,1,0)$.
- Their difference $\underset{\sim}{y} - \underset{\sim}{x}_0 = (-1,1,0)$ has entries summing to $-1 + 1 + 0 = 0$, so it is in $\mathcal{H}_0$, and $\underset{\sim}{y} = \underset{\sim}{x}_0 + (-1,1,0)$, as part 2 says.
- $$\begin{aligned} \underset{\sim}{w} \cdot (-1,1,0) &= -1 + 1 + 0 \\ &= 0, \end{aligned}$$

  as part 3 says.
- The point built in the proof is

  $$\begin{aligned} \underset{\sim}{x}^* &= -\frac{-1}{3}\,(1,1,1) \\ &= (\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}), \end{aligned}$$

  and $\frac{1}{3} + \frac{1}{3} + \frac{1}{3} - 1 = 0$.

Theorem 0.8 (Where a hyperplane passes, and its shape in $\mathbb{R}^2$ and $\mathbb{R}^3$)**.** *Let $\mathcal{H} = \{\underset{\sim}{x} \in \mathbb{R}^p : \underset{\sim}{w} \cdot \underset{\sim}{x} + b = 0\}$ be a hyperplane (Definition 0.27).*

1. *$\mathcal{H}$ contains the origin $\underset{\sim}{0}$ exactly when $b = 0$.*
2. *If $p = 2$, then $\mathcal{H}$ is a line: $\mathcal{H} = \{\underset{\sim}{x}_0 + t\,\underset{\sim}{v} : t \in \mathbb{R}\}$ for some $\underset{\sim}{x}_0 \in \mathbb{R}^2$ and some nonzero $\underset{\sim}{v} \in \mathbb{R}^2$.*
3. *If $p = 3$, then $\mathcal{H}$ is a plane: $\mathcal{H} = \{\underset{\sim}{x}_0 + s\,\underset{\sim}{v}_1 + t\,\underset{\sim}{v}_2 : s,t \in \mathbb{R}\}$ for some $\underset{\sim}{x}_0 \in \mathbb{R}^3$ and some linearly independent $\underset{\sim}{v}_1, \underset{\sim}{v}_2 \in \mathbb{R}^3$.*

Proof

Proof. **Part 1.** Since $\underset{\sim}{w} \cdot \underset{\sim}{0} = 0$,

$$\begin{aligned} \underset{\sim}{w} \cdot \underset{\sim}{0} + b &= 0 + b && \text{(the dot product with } \underset{\sim}{0} \text{ is 0)} \\ &= b, && \text{(Theorem 4 in Convexity, Infimum and Sums)} \end{aligned}$$

so $\underset{\sim}{0} \in \mathcal{H}$ exactly when $b = 0$.

Parts 2 and 3. By Theorem 0.7, $\mathcal{H} = \{\underset{\sim}{x}_0 + \underset{\sim}{s} : \underset{\sim}{s} \in \mathcal{H}_0\}$ for any $\underset{\sim}{x}_0 \in \mathcal{H}$, where $\mathcal{H}_0$ is a subspace of $\mathbb{R}^p$ of dimension $p-1$.

For $p = 2$, $\mathcal{H}_0$ has dimension 1, so it has a basis (Definition 0.3) with one vector $\underset{\sim}{v}$. That vector is nonzero, and $\mathcal{H}_0 = \{t\,\underset{\sim}{v} : t \in \mathbb{R}\}$.

For $p = 3$, $\mathcal{H}_0$ has dimension 2, so it has a basis with two vectors $\underset{\sim}{v}_1$ and $\underset{\sim}{v}_2$. They are linearly independent, and $\mathcal{H}_0 = \{s\,\underset{\sim}{v}_1 + t\,\underset{\sim}{v}_2 : s, t \in \mathbb{R}\}$. □

Example 0.13 (A line in $\mathbb{R}^2$ and a plane in $\mathbb{R}^3$, checked on a grid of points)**.** In $\mathbb{R}^2$, take $\underset{\sim}{w} = (1, 1)$ and $b = -1$, the line of Example 0.24. The point $\underset{\sim}{x}_0 = -\frac{b}{\underset{\sim}{w} \cdot \underset{\sim}{w}}\,\underset{\sim}{w}$ is on it, and the vector $\underset{\sim}{v} = (-w_2, w_1)$ is nonzero with $\underset{\sim}{w} \cdot \underset{\sim}{v} = 0$, so it spans the subspace $\mathcal{H}_0$ of Theorem 0.7. In $\mathbb{R}^3$, take $\underset{\sim}{w} = (1, 1, 1)$ and $b = -1$, the plane of Example 0.12, with the independent vectors $\underset{\sim}{v}_1 = (1, -1, 0)$ and $\underset{\sim}{v}_2 = (0, 1, -1)$ in $\mathcal{H}_0$. The code evaluates $\underset{\sim}{w} \cdot \underset{\sim}{x} + b$ at the origin and at points $\underset{\sim}{x}_0 + t\,\underset{\sim}{v}$ and $\underset{\sim}{x}_0 + s\,\underset{\sim}{v}_1 + t\,\underset{\sim}{v}_2$.

```r
level <- function(w, b, x) sum(w * x) + b

w2 <- c(1, 1)
b2 <- -1
point_line <- -b2 / sum(w2 * w2) * w2
v <- c(-w2[2], w2[1])
line_levels <- sapply(-2:2, function(t) level(w2, b2, point_line + t * v))

w3 <- c(1, 1, 1)
b3 <- -1
point_plane <- -b3 / sum(w3 * w3) * w3
v1 <- c(1, -1, 0)
v2 <- c(0, 1, -1)
grid <- expand.grid(s = -2:2, t = -2:2)
plane_levels <- mapply(
  function(s, t) level(w3, b3, point_plane + s * v1 + t * v2),
  grid$s, grid$t
)

c(
  origin_in_R2 = level(w2, b2, c(0, 0)),
  origin_in_R3 = level(w3, b3, c(0, 0, 0)),
  origin_with_b0 = level(w3, 0, c(0, 0, 0))
)
#>   origin_in_R2   origin_in_R3 origin_with_b0
#>             -1             -1              0
```

At the origin the left side equals the offset, so the origin is on the line only if b is 0: it gives -1 for the line and -1 for the plane, and 0 once the offset is changed to 0. All 5 points $\underset{\sim}{x}_0 + t\,\underset{\sim}{v}$ on the line and all 25 points $\underset{\sim}{x}_0 + s\,\underset{\sim}{v}_1 + t\,\underset{\sim}{v}_2$ on the plane satisfy the equation up to rounding error: 30 of the 30 points have $|\underset{\sim}{w} \cdot \underset{\sim}{x} + b| < 10^{-10}$.

Theorem 0.9 (The graph of an affine function is a hyperplane)**.** *Let* $\underset{\sim}{w} \in \mathbb{R}^p$, $b \in \mathbb{R}$. *The graph of the function* $\underset{\sim}{x} \mapsto \underset{\sim}{w} \cdot \underset{\sim}{x} + b$ *is the set*

$$\left\{ \begin{bmatrix} \underset{\sim}{x} \\ y \end{bmatrix} \in \mathbb{R}^{p+1} : y = \underset{\sim}{w} \cdot \underset{\sim}{x} + b \right\}.$$

This set is a hyperplane in $\mathbb{R}^{p+1}$*, with normal vector* $\begin{bmatrix} \underset{\sim}{w} \\ -1 \end{bmatrix}$.

Proof

Proof. For $\begin{bmatrix} \underset{\sim}{x} \\ y \end{bmatrix} \in \mathbb{R}^{p+1}$,

$$\begin{aligned} y = \underset{\sim}{w} \cdot \underset{\sim}{x} + b &\iff \underset{\sim}{w} \cdot \underset{\sim}{x} - y + b = 0 && \text{(subtract } y \text{ from both sides)} \\ &\iff \begin{bmatrix} \underset{\sim}{w} \\ -1 \end{bmatrix} \cdot \begin{bmatrix} \underset{\sim}{x} \\ y \end{bmatrix} + b = 0. && \text{(dot product of stacked vectors)} \end{aligned}$$

So the set is the solution set of the equation of a hyperplane, with normal vector $\begin{bmatrix} \underset{\sim}{w} \\ -1 \end{bmatrix}$. That vector is nonzero, because its last entry is -1. □

Example 0.14 (The graph of a line is a line in the plane)**.** Take $p = 1$, $\underset{\sim}{w} = 2$, $b = 1$. The graph of $y = 2x + 1$ is the set of points (x, y) in $\mathbb{R}^2$ with $2x - y + 1 = 0$. By Theorem 0.9, this is a hyperplane with normal vector $(2, -1)$. The point $(0, 1)$ is on it, since $2 \cdot 0 - 1 + 1 = 0$. The point $(0, 0)$ is off it, since $2 \cdot 0 - 0 + 1 = 1 \neq 0$. Figure 1 shows the line, the two points, and the normal vector drawn from $(0, 1)$.

```
arrow_length <- grid::unit(0.15, "inches")
normal_arrow <- grid::arrow(length = arrow_length)
point_x <- c(0, 0)
point_y <- c(1, 0)
point_label <- c("on the line", "off the line")
points <- tibble::tibble(x = point_x, y = point_y, label = point_label)
normal <- tibble::tibble(x = 0, y = 1, xend = 2, yend = 0)
normal_aes <- ggplot2::aes(x = x, y = y, xend = xend, yend = yend)
x_limits <- c(-2, 3)
y_limits <- c(-2, 4)
ggplot2::ggplot(points) +
  ggplot2::aes(x = x, y = y, colour = label) +
  ggplot2::geom_abline(intercept = 1, slope = 2) +
  ggplot2::geom_segment(
    mapping = normal_aes,
    data = normal,
    arrow = normal_arrow,
    inherit.aes = FALSE
  ) +
  ggplot2::geom_point(size = 3) +
  ggplot2::coord_fixed(xlim = x_limits, ylim = y_limits) +
  ggplot2::labs(x = "x", y = "y", colour = NULL)
```

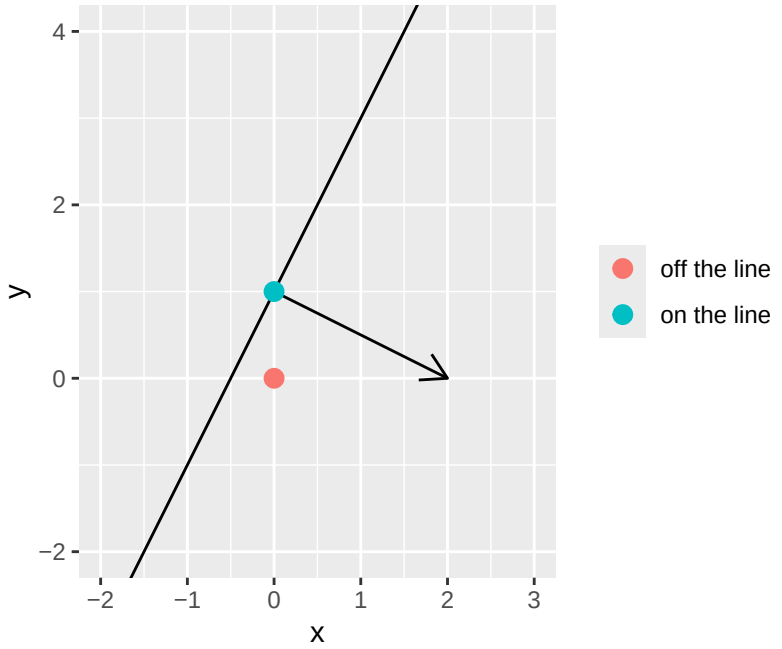


Figure 1: The graph of $y = 2x + 1$, a line in the plane, with its normal vector $(2, -1)$ drawn from the point $(0, 1)$ on it and the point $(0, 0)$ off it

Definition 0.3 (Open half-space)**.** Let $\mathcal{H} = \{\underset{\sim}{x} \in \mathbb{R}^p : \underset{\sim}{w} \cdot \underset{\sim}{x} + b = 0\}$ be a hyperplane (Definition 0.27). The two **open half-spaces** [43] of $\mathcal{H}$ are

$$\mathcal{H}^+ \stackrel{\text{def}}{=} \{\underset{\sim}{x} \in \mathbb{R}^p : \underset{\sim}{w} \cdot \underset{\sim}{x} + b > 0\} \quad \text{and} \quad \mathcal{H}^- \stackrel{\text{def}}{=} \{\underset{\sim}{x} \in \mathbb{R}^p : \underset{\sim}{w} \cdot \underset{\sim}{x} + b < 0\}. \tag{5}$$

Definition 0.4 (Closed half-space)**.** The **closed half-spaces** of the hyperplane $\mathcal{H}$ of Definition 0.3 also include $\mathcal{H}$ itself: $\mathcal{H}^+ \cup \mathcal{H}$ and $\mathcal{H}^- \cup \mathcal{H}$.

Remark

Remark. Replacing $(\underset{\sim}{w}, b)$ by $(-\underset{\sim}{w}, -b)$ describes the same hyperplane but swaps the names $\mathcal{H}^+$ and $\mathcal{H}^-$. The sign of $\underset{\sim}{w} \cdot \underset{\sim}{x} + b$ tells you which side of the hyperplane $\underset{\sim}{x}$ is on. A linear classifier uses this sign to assign one of two labels.

Example 0.15 (Half-spaces of a line in the plane)**.** Let $\mathcal{H}$ be the line in $\mathbb{R}^2$ with $\underset{\sim}{w} = (1, 2)$ and $b = -2$ (Definition 0.27), so $\mathcal{H} = \{\underset{\sim}{x} : x_1 + 2x_2 - 2 = 0\}$. Evaluate $\underset{\sim}{w} \cdot \underset{\sim}{x} + b$ at three points (Definition 0.3):

- At $(0,0)$ it equals $0+0-2=-2<0$, so $(0,0)\in\mathcal{H}^-$.
- At $(2,0)$ it equals $2+0-2=0$, so $(2,0)\in\mathcal{H}$.
- At $(2,1)$ it equals $2+2-2=2>0$, so $(2,1)\in\mathcal{H}^+$.

Moving from $(2,0)$ in the direction $\underset{\sim}{w}=(1,2)$ gives $(3,2)$, where the value is $3+4-2=5>0$. So the direction $\underset{\sim}{w}$ leads into $\mathcal{H}^+$.

Remark (The two sides of a hyperplane)

Remark 0.1 (The two sides of a hyperplane). Every point of $\mathbb{R}^p$ is in exactly one of $\mathcal{H}^+$, $\mathcal{H}$ and $\mathcal{H}^-$, because the number $\underset{\sim}{w}\cdot\underset{\sim}{x}+b$ is positive, zero or negative. The vector $\underset{\sim}{w}$ points into $\mathcal{H}^+$: the number $\underset{\sim}{w}\cdot\underset{\sim}{x}+b$ goes up when $\underset{\sim}{x}$ moves in the direction $\underset{\sim}{w}$.

Theorem 0.10 (Differences of points in a hyperplane and in a half-space)**.** *Let $\underset{\sim}{w}\in\mathbb{R}^p$ be nonzero, let $b\in\mathbb{R}$, and let $\mathcal{H}$, $\mathcal{H}^+$ be the hyperplane and open half-space of Definition 0.3. For a set $S\subseteq\mathbb{R}^p$, let $D(S)\overset{def}{=}\{\underset{\sim}{y}-\underset{\sim}{x}:\underset{\sim}{x},\underset{\sim}{y}\in S\}$ be the set of differences of its points.*

1. *$D(\mathcal{H})=\{\underset{\sim}{s}:\underset{\sim}{w}\cdot\underset{\sim}{s}=0\}$, a subspace of dimension $p-1$.*
2. *$D(\mathcal{H}^+)=\mathbb{R}^p$, a subspace of dimension p.*

The same holds for $\mathcal{H}^-$.

Proof

Proof. **Part 1.** Let $\underset{\sim}{x},\underset{\sim}{y}\in\mathcal{H}$. By Theorem 0.7 part 3, $\underset{\sim}{w}\cdot(\underset{\sim}{y}-\underset{\sim}{x})=0$, so $\underset{\sim}{y}-\underset{\sim}{x}\in\mathcal{H}_0=\{\underset{\sim}{s}:\underset{\sim}{w}\cdot\underset{\sim}{s}=0\}$. Conversely, take any $\underset{\sim}{x}_0\in\mathcal{H}$ (Theorem 0.7 part 2 gives such a point). For any $\underset{\sim}{s}\in\mathcal{H}_0$, the point $\underset{\sim}{x}_0+\underset{\sim}{s}$ is in $\mathcal{H}$ by the same part, and $(\underset{\sim}{x}_0+\underset{\sim}{s})-\underset{\sim}{x}_0=\underset{\sim}{s}$. So $D(\mathcal{H})=\mathcal{H}_0$, which has dimension $p-1$ by Theorem 0.7 part 1.
Part 2. Every difference is in $\mathbb{R}^p$, so $D(\mathcal{H}^+)\subseteq\mathbb{R}^p$. For the reverse inclusion, take any $\underset{\sim}{x}_0\in\mathcal{H}$. Take any $\underset{\sim}{v}\in\mathbb{R}^p$. For a number $t>0$ to be chosen, let $\underset{\sim}{x}_t\overset{\text{def}}{=}\underset{\sim}{x}_0+t\,\underset{\sim}{w}$. Then

$$\begin{aligned}\underset{\sim}{w}\cdot\underset{\sim}{x}_t+b&=(\underset{\sim}{w}\cdot\underset{\sim}{x}_0+b)+t\,\underset{\sim}{w}\cdot\underset{\sim}{w} && \text{(linearity of } \underset{\sim}{x}\mapsto\underset{\sim}{w}\cdot\underset{\sim}{x})\\ &=t\,\underset{\sim}{w}\cdot\underset{\sim}{w}, && (\underset{\sim}{x}_0\in\mathcal{H})\end{aligned}$$

which is positive, so $\underset{\sim}{x}_t\in\mathcal{H}^+$. Next,

$$\begin{aligned}\underset{\sim}{w}\cdot(\underset{\sim}{x}_t+\underset{\sim}{v})+b&=(\underset{\sim}{w}\cdot\underset{\sim}{x}_t+b)+\underset{\sim}{w}\cdot\underset{\sim}{v} && \text{(linearity of } \underset{\sim}{x}\mapsto\underset{\sim}{w}\cdot\underset{\sim}{x})\\ &=t\,\underset{\sim}{w}\cdot\underset{\sim}{w}+\underset{\sim}{w}\cdot\underset{\sim}{v}. && \text{(the display above)}\end{aligned}$$

Because $\underset{\sim}{w}\cdot\underset{\sim}{w}>0$, this is positive once $t>-\underset{\sim}{w}\cdot\underset{\sim}{v}/\underset{\sim}{w}\cdot\underset{\sim}{w}$. Choose such a $t>0$. Then $\underset{\sim}{x}_t$ and $\underset{\sim}{x}_t+\underset{\sim}{v}$ are both in $\mathcal{H}^+$, and their difference is $\underset{\sim}{v}$. So $\underset{\sim}{v}\in D(\mathcal{H}^+)$. The whole space $\mathbb{R}^p$ is a subspace of dimension p (Definition 0.6).

The proof for $\mathcal{H}^-$ is the same with $-\underset{\sim}{w}$ and $-b$ in place of $\underset{\sim}{w}$ and b. □

Example 0.16 (Two iris species on either side of a line)**.** The `iris` data set in R has the petal length and petal width, in centimeters, of 150 irises. Take the 100 flowers of the species versicolor and virginica, and let $\underset{\sim}{x} = (x_1, x_2)$ be the petal length and petal width of one flower. A logistic regression of the species on $\underset{\sim}{x}$ gives a coefficient vector $\underset{\sim}{w}$ and an intercept b. The line $\underset{\sim}{w} \cdot \underset{\sim}{x} + b = 0$ is a hyperplane in $\mathbb{R}^2$ (Definition 0.27), and the two species mostly fall in its two open half-spaces (Definition 0.3). Figure 2 shows the line and the flowers.

```
two_species <- dplyr::filter(iris, Species != "setosa") |> droplevels()
fit <- glm(
  Species ~ Petal.Length + Petal.Width,
  family = binomial,
  data = two_species
)
w <- coef(fit)[c("Petal.Length", "Petal.Width")]
b <- coef(fit)[["(Intercept)"]]
two_species$score <- as.vector(
  as.matrix(two_species[c("Petal.Length", "Petal.Width")]) %*% w + b
)
two_species$side <- ifelse(two_species$score > 0, "positive", "negative")
ggplot2::ggplot(two_species) +
  ggplot2::aes(x = Petal.Length, y = Petal.Width, colour = Species) +
  ggplot2::geom_point() +
  ggplot2::geom_abline(intercept = -b / w[[2]], slope = -w[[1]] / w[[2]]) +
  ggplot2::labs(x = "Petal length (cm)", y = "Petal width (cm)")
```

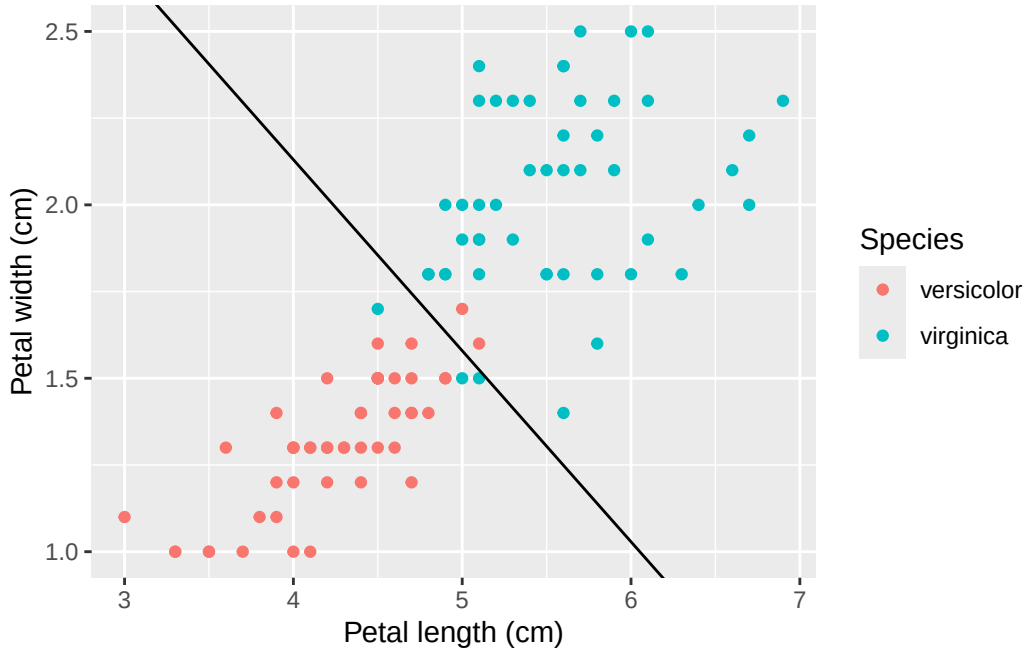


Figure 2: Petal length and width of versicolor and virginica irises, with the line where a logistic regression is indifferent between the two species

```
table(two_species$Species, two_species$side)
#>
#>              negative positive
#>   versicolor       47        3
#>   virginica         3       47
```

The vector $\underset{\sim}{w}$ points toward the virginica side, so the positive half-space $\mathcal{H}^+$ holds 50 of the 100 flowers, and 47 of them are virginica. The line is a set of dimension $p-1=1$, and each side is a region of the plane of dimension $p=2$ (Theorem 0.10).

Exercise 0.5 (Which side of the plane?)**.** Let $\mathcal{H}$ be the plane in $\mathbb{R}^3$ with $\underset{\sim}{w}=(1,1,1)$ and $b=-1$ (Example 0.12).

1. For each point, say whether it is in $\mathcal{H}^+$, $\mathcal{H}$ or $\mathcal{H}^-$ (Definition 0.3): $(0,0,0)$, $(1,0,0)$ and $(1,1,1)$.
2. Write the plane and its half-spaces using $-\underset{\sim}{w}$ and $-b$. Which points from part 1 change sides?
3. State the dimension of $D(\mathcal{H})$ and of $D(\mathcal{H}^+)$ (Theorem 0.10).

Solution

Solution 0.5.

1. Compute $\underset{\sim}{w} \cdot \underset{\sim}{x} + b = x_1 + x_2 + x_3 - 1$ at each point.
   - At $(0,0,0)$ it equals $0+0+0-1 = -1 < 0$, so the point is in $\mathcal{H}^-$.
   - At $(1,0,0)$ it equals $1+0+0-1 = 0$, so the point is in $\mathcal{H}$.
   - At $(1,1,1)$ it equals $1+1+1-1 = 2 > 0$, so the point is in $\mathcal{H}^+$.
2. With $-\underset{\sim}{w} = (-1,-1,-1)$ and $-b = 1$, the equation is $-x_1 - x_2 - x_3 + 1 = 0$. Multiplying both sides by -1 gives the original equation, so it is the same plane. The new expression is the negative of the old one at every point, so the new positive half-space is the old $\mathcal{H}^-$ and the new negative half-space is the old $\mathcal{H}^+$. The origin $(0,0,0)$ is now in the positive half-space, and $(1,1,1)$ is now in the negative half-space. The point $(1,0,0)$ stays on the plane. So the origin and $(1,1,1)$ both change names, and the point on the plane does not.
3. Here $p = 3$. $D(\mathcal{H})$ is the subspace $\{\underset{\sim}{s} : s_1 + s_2 + s_3 = 0\}$, which has dimension $p - 1 = 2$. $D(\mathcal{H}^+) = \mathbb{R}^3$ has dimension $p = 3$.

Definition 0.5 (Rank factorization)**.** Let $\mathbf{A}$ be an $m \times n$ matrix with $\operatorname{rank}(\mathbf{A}) = r \geq 1$. A **rank factorization** [44] of $\mathbf{A}$ is a product

$$\underset{m\times n}{\mathbf{A}} = \underset{m\times r}{\mathbf{C}}\ \underset{r\times n}{\mathbf{R}}\ .$$

Example 0.17 (A rank factorization of a rank-one matrix)**.** The matrix $\mathbf{A}$ in Example 0.1 has rank 1 (Example 0.1), and

$$\begin{bmatrix}1\\3\end{bmatrix}\begin{bmatrix}1 & -2 & -2\end{bmatrix} = \begin{bmatrix}1\cdot 1 & 1\cdot(-2) & 1\cdot(-2)\\ 3\cdot 1 & 3\cdot(-2) & 3\cdot(-2)\end{bmatrix} = \begin{bmatrix}1 & -2 & -2\\ 3 & -6 & -6\end{bmatrix},$$

so $\mathbf{C} = \begin{bmatrix}1\\3\end{bmatrix}$ (2×1) and $\mathbf{R} = \begin{bmatrix}1 & -2 & -2\end{bmatrix}$ (1×3) are a rank factorization. It is not unique: $\mathbf{C} = \begin{bmatrix}2\\6\end{bmatrix}$ and $\mathbf{R} = \begin{bmatrix}\frac{1}{2} & -1 & -1\end{bmatrix}$ give the same product. A product $\mathbf{CR}$ with $\mathbf{C}$ of size 2×2 is not a rank factorization of this $\mathbf{A}$, because the inner dimension must equal $\operatorname{rank}(\mathbf{A}) = 1$.

Theorem 0.11 (Every nonzero matrix has a rank factorization)**.** *Every $m \times n$ matrix $\mathbf{A}$ with $\operatorname{rank}(\mathbf{A}) = r \geq 1$ has a rank factorization (Definition 0.5). One is given by taking the columns of $\mathbf{C}$ to be any basis of $\mathcal{C}(\mathbf{A})$.*

Proof

Proof. $\mathcal{C}(\mathbf{A})$ has dimension r (Theorem 0.1), so it has a basis $\underset{\sim}{c}_1, \dots, \underset{\sim}{c}_r$ (Theorem 0.9); let $\mathbf{C}$ be the $m \times r$ matrix with these columns. Each column $\underset{\sim}{a}_j$ of $\mathbf{A}$ is in $\mathcal{C}(\mathbf{A})$, so $\underset{\sim}{a}_j = r_{1j}\underset{\sim}{c}_1 + \cdots + r_{rj}\underset{\sim}{c}_r$ for some numbers r_{ij}, because the basis spans $\mathcal{C}(\mathbf{A})$. Let $\mathbf{R}$ be the $r \times n$ matrix with entries r_{ij}, whose column j is $\underset{\sim}{r}_j = (r_{1j}, \dots, r_{rj})$. Then column j of $\mathbf{CR}$ is

$$\begin{aligned} \mathbf{C}\underset{\sim}{r}_j &= r_{1j}\underset{\sim}{c}_1 + \cdots + r_{rj}\underset{\sim}{c}_r && \text{(Theorem 0.3)} \\ &= \underset{\sim}{a}_j, && \text{(choice of the } r_{ij}\text{)} \end{aligned}$$

so $\mathbf{CR} = \mathbf{A}$. □

Example 0.18 (Building a rank factorization from a basis)**.** For $\mathbf{A}$ in Example 0.1, $(1, 3)$ spans $\mathcal{C}(\mathbf{A})$ (1), and it is linearly independent because it is nonzero ($c\,(1, 3) = \underset{\sim}{0}_2$ forces $c = 0$), so it is a basis of $\mathcal{C}(\mathbf{A})$. The columns of $\mathbf{A}$ are $(1, 3) = 1 \cdot (1, 3)$, $(-2, -6) = -2 \cdot (1, 3)$ and $(-2, -6) = -2 \cdot (1, 3)$, so the construction in Theorem 0.11 gives $\mathbf{C} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\mathbf{R} = \begin{bmatrix} 1 & -2 & -2 \end{bmatrix}$, the factorization in Example 0.17.

Sums and direct sums

Source

This section, Section and Section are adapted from Zhou [45], used under the MIT License (see the license text in Section). The source proves that a subspace and its orthogonal complement together make up $\mathbb{R}^p$ by extending an orthonormal basis; the proof here uses the rank-nullity theorem (Theorem 0.2) instead. The source defines the orthogonal complement inside an ambient vector space $V \subseteq \mathbb{R}^m$; here the ambient space is always $\mathbb{R}^p$, which is all that the fundamental theorem and the projections need.

Definition 0.1 (Sum of subspaces)**.** The **sum** of two subspaces $\mathcal{S}_1$ and $\mathcal{S}_2$ of $\mathbb{R}^p$ (Definition 0.1) is the set of all sums of a vector from $\mathcal{S}_1$ and a vector from $\mathcal{S}_2$:

$$\mathcal{S}_1 + \mathcal{S}_2 \stackrel{\text{def}}{=} \{\underset{\sim}{x}_1 + \underset{\sim}{x}_2 : \underset{\sim}{x}_1 \in \mathcal{S}_1,\ \underset{\sim}{x}_2 \in \mathcal{S}_2\}.$$

Example 0.1 (Two lines sum to a plane)**.** Let $\mathcal{S}_1 = \text{span}\{(1,0,0)\}$ and $\mathcal{S}_2 = \text{span}\{(0,1,0)\}$, two lines in $\mathbb{R}^3$. A vector of $\mathcal{S}_1 + \mathcal{S}_2$ has the form $a\,(1,0,0) + b\,(0,1,0) = (a,b,0)$, and a and b can be any numbers, so $\mathcal{S}_1 + \mathcal{S}_2 = \{(a,b,0) : a,b \in \mathbb{R}\}$, the plane $z = 0$.

Example 0.2 (The sum is not the union)**.** For the two lines in Example 0.1, the union $\mathcal{S}_1 \cup \mathcal{S}_2$ contains $(1,0,0)$ and $(0,1,0)$ but not their sum $(1,1,0)$: every multiple of $(1,0,0)$ has second entry 0, and every multiple of $(0,1,0)$ has first entry 0. So the union is not closed under addition and is not a subspace, while the sum, the whole plane $z = 0$, is.

Theorem 0.1 (Sums and intersections of subspaces are subspaces)**.** *If $\mathcal{S}_1$ and $\mathcal{S}_2$ are subspaces of $\mathbb{R}^p$ (Definition 0.1), then $\mathcal{S}_1 + \mathcal{S}_2$ (Definition 0.1) and $\mathcal{S}_1 \cap \mathcal{S}_2$ are subspaces of $\mathbb{R}^p$.*

Proof

Proof. Both $\mathcal{S}_1$ and $\mathcal{S}_2$ contain $\underset{\sim}{0}$ (Theorem 0.1).
The sum. It is not empty: it contains $\underset{\sim}{0} + \underset{\sim}{0} = \underset{\sim}{0}$. Take two of its vectors, $\underset{\sim}{u} = \underset{\sim}{u}_1 + \underset{\sim}{u}_2$ and $\underset{\sim}{w} = \underset{\sim}{w}_1 + \underset{\sim}{w}_2$, with $\underset{\sim}{u}_1, \underset{\sim}{w}_1 \in \mathcal{S}_1$ and $\underset{\sim}{u}_2, \underset{\sim}{w}_2 \in \mathcal{S}_2$, and a number c. Then

$$\begin{aligned} \underset{\sim}{u} + \underset{\sim}{w} &= (\underset{\sim}{u}_1 + \underset{\sim}{u}_2) + (\underset{\sim}{w}_1 + \underset{\sim}{w}_2) && \text{(substitute)} \\ &= (\underset{\sim}{u}_1 + \underset{\sim}{w}_1) + (\underset{\sim}{u}_2 + \underset{\sim}{w}_2), && \text{(regroup; vector addition is entrywise)} \end{aligned}$$

where $\underset{\sim}{u}_1 + \underset{\sim}{w}_1 \in \mathcal{S}_1$ and $\underset{\sim}{u}_2 + \underset{\sim}{w}_2 \in \mathcal{S}_2$, because each subspace is closed under addition. Likewise

$$\begin{aligned} c\,\underset{\sim}{u} &= c\,(\underset{\sim}{u}_1 + \underset{\sim}{u}_2) && \text{(substitute)} \\ &= c\,\underset{\sim}{u}_1 + c\,\underset{\sim}{u}_2, && \text{(distribute } c \text{ entrywise)} \end{aligned}$$

where $c\,\underset{\sim}{u}_1 \in \mathcal{S}_1$ and $c\,\underset{\sim}{u}_2 \in \mathcal{S}_2$, because each subspace is closed under scalar multiplication. So $\underset{\sim}{u} + \underset{\sim}{w}$ and $c\,\underset{\sim}{u}$ are in $\mathcal{S}_1 + \mathcal{S}_2$.
The intersection. It is not empty: it contains $\underset{\sim}{0}$. If $\underset{\sim}{u}$ and $\underset{\sim}{w}$ are in both $\mathcal{S}_1$ and $\mathcal{S}_2$, then $\underset{\sim}{u} + \underset{\sim}{w}$ and $c\,\underset{\sim}{u}$ are in $\mathcal{S}_1$, because $\mathcal{S}_1$ is a subspace, and in $\mathcal{S}_2$ for the same reason, so they are in $\mathcal{S}_1 \cap \mathcal{S}_2$. □

Example 0.3 (Two planes in $\mathbb{R}^3$)**.** Let $\mathcal{S}_1 = \{(a,b,0) : a,b \in \mathbb{R}\}$, the plane $z = 0$, and $\mathcal{S}_2 = \{(0,b,c) : b,c \in \mathbb{R}\}$, the plane $x = 0$.

- **Intersection:** a vector in both has third entry 0 and first entry 0, so $\mathcal{S}_1 \cap \mathcal{S}_2 = \{(0, b, 0) : b \in \mathbb{R}\}$, the y-axis, which is a line through the origin.
- **Sum:** $(a, b, 0)+(0, b', c) = (a, b+b', c)$, and any (x, y, z) arises this way, with $a = x$, $b = y$, $b' = 0$ and $c = z$. So $\mathcal{S}_1 + \mathcal{S}_2 = \mathbb{R}^3$.

Theorem 0.2 (The dimension of a sum of subspaces)**.** *If $\mathcal{S}_1$ and $\mathcal{S}_2$ are subspaces of $\mathbb{R}^p$, then*

$$\dim(\mathcal{S}_1 + \mathcal{S}_2) = \dim(\mathcal{S}_1) + \dim(\mathcal{S}_2) - \dim(\mathcal{S}_1 \cap \mathcal{S}_2).$$

Proof

Proof. All three sets in the formula are subspaces (Theorem 0.1), so each has a dimension (Definition 0.6). Let $k = \dim(\mathcal{S}_1 \cap \mathcal{S}_2)$, $d_1 = \dim(\mathcal{S}_1)$ and $d_2 = \dim(\mathcal{S}_2)$.

Build a list. Choose a basis $\underset{\sim}{z}_1, \ldots, \underset{\sim}{z}_k$ of $\mathcal{S}_1 \cap \mathcal{S}_2$ (Theorem 0.9). These vectors are linearly independent and lie in $\mathcal{S}_1$, so Theorem 0.9 extends them to a basis $\underset{\sim}{z}_1, \ldots, \underset{\sim}{z}_k, \underset{\sim}{x}_1, \ldots, \underset{\sim}{x}_{d_1-k}$ of $\mathcal{S}_1$, which has d_1 vectors (Theorem 0.7). In the same way, extend them to a basis $\underset{\sim}{z}_1, \ldots, \underset{\sim}{z}_k, \underset{\sim}{y}_1, \ldots, \underset{\sim}{y}_{d_2-k}$ of $\mathcal{S}_2$. We show that the $\underset{\sim}{z}$'s, $\underset{\sim}{x}$'s and $\underset{\sim}{y}$'s together, $k + (d_1 - k) + (d_2 - k) = d_1 + d_2 - k$ vectors, are a basis of $\mathcal{S}_1 + \mathcal{S}_2$ (Definition 0.3).

They span $\mathcal{S}_1 + \mathcal{S}_2$. Any $\underset{\sim}{v} \in \mathcal{S}_1 + \mathcal{S}_2$ is $\underset{\sim}{v}_1 + \underset{\sim}{v}_2$ with $\underset{\sim}{v}_1 \in \mathcal{S}_1$ and $\underset{\sim}{v}_2 \in \mathcal{S}_2$. $\underset{\sim}{v}_1$ is a linear combination of the $\underset{\sim}{z}$'s and $\underset{\sim}{x}$'s, and $\underset{\sim}{v}_2$ is a linear combination of the $\underset{\sim}{z}$'s and $\underset{\sim}{y}$'s, so their sum is a linear combination of all three groups. Conversely, each vector in the list is in $\mathcal{S}_1 + \mathcal{S}_2$ (for example, $\underset{\sim}{x}_1 = \underset{\sim}{x}_1 + \underset{\sim}{0}$), and $\mathcal{S}_1 + \mathcal{S}_2$ is a subspace, so it contains every linear combination of them.

They are linearly independent. Suppose

$$\sum_{i=1}^{k} a_i \underset{\sim}{z}_i + \sum_{j=1}^{d_1-k} b_j \underset{\sim}{x}_j + \sum_{l=1}^{d_2-k} c_l \underset{\sim}{y}_l = \underset{\sim}{0}.$$

Let $\underset{\sim}{w} \overset{\text{def}}{=} \sum_l c_l \underset{\sim}{y}_l$. Then $\underset{\sim}{w} \in \mathcal{S}_2$, because it is a linear combination of vectors of $\mathcal{S}_2$. Subtracting $\sum_i a_i \underset{\sim}{z}_i + \sum_j b_j \underset{\sim}{x}_j$ from both sides of the supposed equation gives

$$\underset{\sim}{w} = \sum_{i=1}^{k} (-a_i)\, \underset{\sim}{z}_i + \sum_{j=1}^{d_1-k} (-b_j)\, \underset{\sim}{x}_j,$$

a linear combination of vectors of $\mathcal{S}_1$, so $\underset{\sim}{w} \in \mathcal{S}_1$ too. So $\underset{\sim}{w} \in \mathcal{S}_1 \cap \mathcal{S}_2$, and $\underset{\sim}{w} = \sum_i e_i \underset{\sim}{z}_i$ for some numbers e_i, since the $\underset{\sim}{z}$'s span the intersection. Then

$$\begin{aligned}
\underset{\sim}{0} &= \underset{\sim}{w} - \underset{\sim}{w} && \text{(a vector minus itself)} \\
&= \sum_{i=1}^{k} e_i \underset{\sim}{z}_i - \sum_{l=1}^{d_2-k} c_l \underset{\sim}{y}_l && \text{(substitute the two expressions for } \underset{\sim}{w}\text{)} \\
&= \sum_{i=1}^{k} e_i \underset{\sim}{z}_i + \sum_{l=1}^{d_2-k} (-c_l)\, \underset{\sim}{y}_l, && \text{(write each subtraction as adding } -1 \text{ times)}
\end{aligned}$$

and because the z's and y's are a basis of $\mathcal{S}_2$, every $c_l = 0$. The supposed equation then reads $\sum_i a_i \underset{\sim}{z}_i + \sum_j b_j \underset{\sim}{x}_j \mathrel{\tilde{=}} \underset{\sim}{0}$, and because the $\underset{\sim}{z}$'s and $\underset{\sim}{x}$'s are a basis of $\mathcal{S}_1$, every a_i and b_j is 0.
So $\dim(\mathcal{S}_1 + \mathcal{S}_2) = d_1 + d_2 - k$. When some of the groups are empty (for example $k = 0$, or $\mathcal{S}_1 = \mathcal{S}_1 \cap \mathcal{S}_2$), the sums over them are $\underset{\sim}{0}$ and the argument goes through unchanged. □

Example 0.4 (Two planes in $\mathbb{R}^3$, by dimension)**.** For the planes $z = 0$ and $x = 0$ in Example 0.3, each has dimension 2: the plane $z = 0$ has basis $(1,0,0), (0,1,0)$ (Example 0.14), and the plane $x = 0$ has basis $(0,1,0), (0,0,1)$ by the same argument. Their intersection, the y-axis $\operatorname{span}\{(0,1,0)\}$, has dimension 1, because the single nonzero vector $(0,1,0)$ is linearly independent ($c\,(0,1,0) = (0,c,0)$ is $\underset{\sim}{0}$ only if $c = 0$) and so is a basis of it. Theorem 0.2 gives

$$\begin{aligned}
\dim(\mathcal{S}_1 + \mathcal{S}_2) &= 2 + 2 - 1 \\
&= 3,
\end{aligned}$$

which agrees with $\mathcal{S}_1 + \mathcal{S}_2 = \mathbb{R}^3$ (Example 0.3) and $\dim(\mathbb{R}^3) = 3$ (Example 0.14).

Corollary 0.1 (The dimension of a sum is at most the sum of the dimensions)**.** *If $\mathcal{S}_1$ and $\mathcal{S}_2$ are subspaces of $\mathbb{R}^p$, then*

$$\dim(\mathcal{S}_1 + \mathcal{S}_2) \le \dim(\mathcal{S}_1) + \dim(\mathcal{S}_2),$$

with equality exactly when $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\underset{\sim}{0}\}$.

Proof

Proof. By Theorem 0.2, the two sides differ by $\dim(\mathcal{S}_1 \cap \mathcal{S}_2) \ge 0$. That dimension is 0 exactly when a basis of the intersection is the empty list, that is, when the intersection is the span of the empty list, $\{\underset{\sim}{0}\}$ (Definition 0.2, Example 0.14). □

Example 0.5 (Equality and strict inequality)**.**

- For the lines $\mathcal{S}_1 = \operatorname{span}\{(1,0,0)\}$ and $\mathcal{S}_2 = \operatorname{span}\{(0,1,0)\}$ in Example 0.1, a com-

mon vector satisfies $a\,(1,0,0) = b\,(0,1,0)$, so $(a,-b,0) = \underset{\sim}{0}$ and

$$\begin{aligned} a &= b \\ &= 0. \end{aligned}$$

The intersection is $\{\underset{\sim}{0}\}$, and indeed

$$\begin{aligned} \dim(\mathcal{S}_1 + \mathcal{S}_2) &= 2 \\ &= 1 + 1 : \end{aligned}$$

the sum is the plane $z = 0$ (Example 0.1), which has dimension 2 (Example 0.14), and each line has dimension 1, with its spanning vector as a basis.

- For

$$\begin{aligned} \mathcal{S}_1 &= \mathcal{S}_2 \\ &= \text{span}\{(1,0,0)\}, \end{aligned}$$

the sum is the same line, because $a\,(1,0,0)+b\,(1,0,0) = (a+b)\,(1,0,0)$, so $\dim(\mathcal{S}_1+\mathcal{S}_2) = 1 < 1+1$, a strict inequality.

Theorem 0.3 (Rank is subadditive)**.** *For $m \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$,*

$$\text{rank}(\mathbf{A}+\mathbf{B}) \leq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B}).$$

Proof

Proof. $\mathcal{C}(\mathbf{A}+\mathbf{B}) \subseteq \mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{B})$**.** Any vector of $\mathcal{C}(\mathbf{A}+\mathbf{B})$ is $(\mathbf{A}+\mathbf{B})\,\underset{\sim}{x}$ for some $\underset{\sim}{x} \in \mathbb{R}^n$ (Definition 0.1), and $(\mathbf{A}+\mathbf{B})\,\underset{\sim}{x} = \mathbf{A}\underset{\sim}{x} + \mathbf{B}\underset{\sim}{x}$ (Theorem 0.8, with $\underset{\sim}{x}$ as an $n \times 1$ matrix), a vector of $\mathcal{C}(\mathbf{A})$ plus a vector of $\mathcal{C}(\mathbf{B})$.
Compare dimensions. All the sets involved are subspaces of $\mathbb{R}^m$ (Theorem 0.1, Theorem 0.1), so

$$\begin{aligned} \text{rank}(\mathbf{A}+\mathbf{B}) &= \dim(\mathcal{C}(\mathbf{A}+\mathbf{B})) && \text{(Theorem 0.1)} \\ &\leq \dim(\mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{B})) && \text{(the containment, and part 2 of Theorem 0.10)} \\ &\leq \dim(\mathcal{C}(\mathbf{A})) + \dim(\mathcal{C}(\mathbf{B})) && \text{(Corollary 0.1)} \\ &= \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B}). && \text{(Theorem 0.1)} \end{aligned}$$

□

Example 0.6 (Subadditivity with equality and without)**.** Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$. Each has one nonzero column and one zero column. The nonzero column on its own is linearly independent ($c\,\underset{\sim}{v} = \underset{\sim}{0}_2$ with $\underset{\sim}{v} \neq \underset{\sim}{0}_2$ forces $c = 0$), and the two columns together are not, since 1 times the zero column is $\underset{\sim}{0}_2$; so each matrix has rank 1 (Definition 0.2). By the same argument, $-\mathbf{A}$ has rank 1.

- $\mathbf{A} + \mathbf{B}$ is the 2×2 identity matrix, which has rank 2 (Example 0.3), so the bound $2 \leq 1 + 1$ holds with equality.
- $\mathbf{A} + (-\mathbf{A})$ is the 2×2 zero matrix, which has rank 0: any nonempty list of its columns contains $\underset{\sim}{0}_2$, and $1 \cdot \underset{\sim}{0}_2 = \underset{\sim}{0}_2$. So the bound $0 \leq \operatorname{rank}(\mathbf{A}) + \operatorname{rank}(-\mathbf{A}) = 1 + 1$ is strict.

Definition 0.2 (Complementary subspaces)**.** Let $\mathcal{S}_1$, $\mathcal{S}_2$ and $\mathcal{V}$ be subspaces of $\mathbb{R}^p$. $\mathcal{S}_1$ and $\mathcal{S}_2$ are **complementary** in $\mathcal{V}$ if

$$\mathcal{V} = \mathcal{S}_1 + \mathcal{S}_2 \quad \text{and} \quad \mathcal{S}_1 \cap \mathcal{S}_2 = \{\underset{\sim}{0}\}$$

(Definition 0.1).

Definition 0.3 (Direct sum)**.** If $\mathcal{S}_1$ and $\mathcal{S}_2$ are complementary in $\mathcal{V}$ (Definition 0.2), then $\mathcal{V}$ is the **direct sum** [46] of $\mathcal{S}_1$ and $\mathcal{S}_2$, written $\mathcal{V} = \mathcal{S}_1 \oplus \mathcal{S}_2$.

Example 0.7 ($\mathbb{R}^3$ is a plane plus a line)**.** Let $\mathcal{S}_1 = \{(a, b, 0) : a, b \in \mathbb{R}\}$, the plane $z = 0$, and $\mathcal{S}_2 = \{(0, 0, c) : c \in \mathbb{R}\}$, the z-axis.

- **Sum:** any $(x, y, z) \in \mathbb{R}^3$ is $(x, y, 0) + (0, 0, z)$, so $\mathcal{S}_1 + \mathcal{S}_2 = \mathbb{R}^3$.
- **Intersection:** a vector in both has third entry 0 and first and second entries 0, so $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\underset{\sim}{0}\}$.

So $\mathbb{R}^3 = \mathcal{S}_1 \oplus \mathcal{S}_2$.

Example 0.8 (Two planes that sum to $\mathbb{R}^3$ but not directly)**.** The planes $z = 0$ and $x = 0$ sum to $\mathbb{R}^3$ (Example 0.3), but they share the y-axis, so they are not complementary, and $\mathbb{R}^3$ is not their direct sum. The shared line shows up as more than one way to split a vector:

$$\begin{aligned}(0,1,0) &= (0,1,0) + \underset{\sim}{0} \\ &= \underset{\sim}{0} + (0,1,0),\end{aligned}$$

with the first term in the plane $z = 0$ and the second in the plane $x = 0$ both times.

Theorem 0.4 (Three ways to recognize a direct sum)**.** *Let $\mathcal{S}_1$ and $\mathcal{S}_2$ be subspaces of $\mathbb{R}^p$, and let $\mathcal{V} = \mathcal{S}_1 + \mathcal{S}_2$. The following statements are equivalent:*

1. *$\mathcal{V} = \mathcal{S}_1 \oplus \mathcal{S}_2$ (Definition 0.3).*
2. *$\dim(\mathcal{V}) = \dim(\mathcal{S}_1) + \dim(\mathcal{S}_2)$.*
3. *Every $\underset{\sim}{x} \in \mathcal{V}$ can be written as $\underset{\sim}{x} = \underset{\sim}{x}_1 + \underset{\sim}{x}_2$ with $\underset{\sim}{x}_1 \in \mathcal{S}_1$ and $\underset{\sim}{x}_2 \in \mathcal{S}_2$ in only one way.*

Proof

Proof. Since $\mathcal{V} = \mathcal{S}_1 + \mathcal{S}_2$ is given, statement 1 says exactly that $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\underset{\sim}{0}\}$.
1 and 2 are equivalent. By Corollary 0.1, $\dim(\mathcal{S}_1 + \mathcal{S}_2) = \dim(\mathcal{S}_1) + \dim(\mathcal{S}_2)$ exactly when $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\underset{\sim}{0}\}$.
1 implies 3. Every $\underset{\sim}{x} \in \mathcal{V}$ has at least one such expression, because $\mathcal{V} = \mathcal{S}_1 + \mathcal{S}_2$. Suppose

$$\begin{aligned}\underset{\sim}{x} &= \underset{\sim}{u}_1 + \underset{\sim}{u}_2 \\ &= \underset{\sim}{v}_1 + \underset{\sim}{v}_2,\end{aligned}$$

with $\underset{\sim}{u}_1, \underset{\sim}{v}_1 \in \mathcal{S}_1$ and $\underset{\sim}{u}_2, \underset{\sim}{v}_2 \in \mathcal{S}_2$. Then

$$\begin{aligned}\underset{\sim}{u}_1 - \underset{\sim}{v}_1 &= (\underset{\sim}{u}_1 + \underset{\sim}{u}_2) - \underset{\sim}{u}_2 - \underset{\sim}{v}_1 && \text{(add and subtract } \underset{\sim}{u}_2\text{)} \\ &= (\underset{\sim}{v}_1 + \underset{\sim}{v}_2) - \underset{\sim}{u}_2 - \underset{\sim}{v}_1 && \text{(substitute } \underset{\sim}{u}_1 + \underset{\sim}{u}_2 = \underset{\sim}{v}_1 + \underset{\sim}{v}_2\text{)} \\ &= \underset{\sim}{v}_2 - \underset{\sim}{u}_2. && \text{(cancel } \underset{\sim}{v}_1\text{)}\end{aligned}$$

The left side is in $\mathcal{S}_1$ and the right side is in $\mathcal{S}_2$, because each subspace is closed under addition and scalar multiplication (the difference is $\underset{\sim}{u}_1 + (-1)\,\underset{\sim}{v}_1$). So both sides are in $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\underset{\sim}{0}\}$, which gives $\underset{\sim}{u}_1 = \underset{\sim}{v}_1$ and $\underset{\sim}{u}_2 = \underset{\sim}{v}_2$.
3 implies 1. Take any $\underset{\sim}{x} \in \mathcal{S}_1 \cap \mathcal{S}_2$. Then $\underset{\sim}{x} = \underset{\sim}{x} + \underset{\sim}{0}$, with $\underset{\sim}{x} \in \mathcal{S}_1$ and $\underset{\sim}{0} \in \mathcal{S}_2$, and $\underset{\sim}{x} = \underset{\sim}{0} + \underset{\sim}{x}$, with $\underset{\sim}{0} \in \mathcal{S}_1$ and $\underset{\sim}{x} \in \mathcal{S}_2$ (Theorem 0.1). By statement 3 these two expressions are the same, so $\underset{\sim}{x} = \underset{\sim}{0}$, and $\mathcal{S}_1 \cap \mathcal{S}_2 = \{\underset{\sim}{0}\}$. □

Example 0.9 (Checking the three statements)**.** For the plane $z = 0$ and the z-axis in Example 0.7:

- the dimensions add up:

$$\begin{aligned} 2+1 &= 3 \\ &= \dim(\mathbb{R}^3), \end{aligned}$$

since the plane has dimension 2 (Example 0.14), the z-axis has the single nonzero vector $(0,0,1)$ as a basis, and $\dim(\mathbb{R}^3) = 3$ (Example 0.14);

- the split is unique: if

$$\begin{aligned} (x,y,z) &= (a,b,0) + (0,0,c) \\ &= (a,b,c), \end{aligned}$$

then $a = x$, $b = y$ and $c = z$.

For the planes $z = 0$ and $x = 0$ in Example 0.8, both statements fail:

$$\begin{aligned} 2+2 &= 4 \\ &\neq 3 \\ &= \dim(\mathcal{S}_1 + \mathcal{S}_2) \end{aligned}$$

(Example 0.4), and $(0,1,0)$ splits in two ways.

Definition 0.4 (Projection onto a subspace along a complement)**.** Let $\mathcal{S}$ and $\mathcal{T}$ be subspaces of $\mathbb{R}^p$ with $\mathbb{R}^p = \mathcal{S} \oplus \mathcal{T}$ (Definition 0.3), and let $\underset{\sim}{y} \in \mathbb{R}^p$. By Theorem 0.4, $\underset{\sim}{y} = \underset{\sim}{u} + \underset{\sim}{v}$ for exactly one $\underset{\sim}{u} \in \mathcal{S}$ and $\underset{\sim}{v} \in \mathcal{T}$. The vector $\underset{\sim}{u}$ is the **projection** [47] of $\underset{\sim}{y}$ onto $\mathcal{S}$ along $\mathcal{T}$ [48], Definition 6.3, p. 163.

Example 0.10 (The projection depends on the complement)**.** Let $\mathcal{S} = \text{span}\{(1,0)\}$, the horizontal axis in $\mathbb{R}^2$, and let $\underset{\sim}{y} = (3,2)$.

- **Along $\mathcal{T}_1 = \text{span}\{(1,-1)\}$:** $(3,2) = (5,0) + (-2,2)$, with $(5,0) \in \mathcal{S}$ and $(-2,2) = -2 \cdot (1,-1) \in \mathcal{T}_1$, so the projection of $(3,2)$ onto $\mathcal{S}$ along $\mathcal{T}_1$ is $(5,0)$.
- **Along $\mathcal{T}_2 = \text{span}\{(0,1)\}$:** $(3,2) = (3,0) + (0,2)$, with $(3,0) \in \mathcal{S}$ and $(0,2) \in \mathcal{T}_2$, so the projection of $(3,2)$ onto $\mathcal{S}$ along $\mathcal{T}_2$ is $(3,0)$.

Each of $\mathcal{T}_1$ and $\mathcal{T}_2$ is a line that meets $\mathcal{S}$ only at $\underset{\sim}{0}$, so

$$\begin{aligned} \mathbb{R}^2 &= \mathcal{S} \oplus \mathcal{T}_1 \\ &= \mathcal{S} \oplus \mathcal{T}_2 \end{aligned}$$

(Definition 0.3, Theorem 0.4). The same vector has a different projection onto the same subspace when the complement changes.

Orthogonal complements

Theorem 0.1 (The dot product is linear in each slot)**.** *For vectors* $\underset{\sim}{x}, \underset{\sim}{u}, \underset{\sim}{w} \in \mathbb{R}^p$ *and numbers* a, b*,*

$$\underset{\sim}{x} \cdot (a\,\underset{\sim}{u} + b\,\underset{\sim}{w}) = a\,(\underset{\sim}{x} \cdot \underset{\sim}{u}) + b\,(\underset{\sim}{x} \cdot \underset{\sim}{w})$$

and

$$(a\,\underset{\sim}{u} + b\,\underset{\sim}{w}) \cdot \underset{\sim}{x} = a\,(\underset{\sim}{u} \cdot \underset{\sim}{x}) + b\,(\underset{\sim}{w} \cdot \underset{\sim}{x}).$$

Proof

Proof. **Second slot.**

$$\begin{aligned}
\underset{\sim}{x} \cdot (a\,\underset{\sim}{u} + b\,\underset{\sim}{w}) &= \sum_{i=1}^{p} x_i\,(a\,\underset{\sim}{u} + b\,\underset{\sim}{w})_i && \text{(Definition 0.8)} \\
&= \sum_{i=1}^{p} x_i\,(au_i + bw_i) && \text{(Definition 0.9, Definition 0.7)} \\
&= \sum_{i=1}^{p} (x_i a u_i + x_i b w_i) && \text{(distribute each } x_i\text{)} \\
&= \sum_{i=1}^{p} (a\,x_i u_i + b\,x_i w_i) && \text{(commute the factors in each product)} \\
&= \sum_{i=1}^{p} a\,x_i u_i + \sum_{i=1}^{p} b\,x_i w_i && \text{(split the finite sum)} \\
&= a \sum_{i=1}^{p} x_i u_i + b \sum_{i=1}^{p} x_i w_i && \text{(factor } a \text{ and } b \text{ out of the sums)} \\
&= a\,(\underset{\sim}{x} \cdot \underset{\sim}{u}) + b\,(\underset{\sim}{x} \cdot \underset{\sim}{w}). && \text{(Definition 0.8)}
\end{aligned}$$

First slot.

$$\begin{aligned}
(a\,\underset{\sim}{u} + b\,\underset{\sim}{w}) \cdot \underset{\sim}{x} &= \underset{\sim}{x} \cdot (a\,\underset{\sim}{u} + b\,\underset{\sim}{w}) && \text{(Theorem 0.1)} \\
&= a\,(\underset{\sim}{x} \cdot \underset{\sim}{u}) + b\,(\underset{\sim}{x} \cdot \underset{\sim}{w}) && \text{(second slot)} \\
&= a\,(\underset{\sim}{u} \cdot \underset{\sim}{x}) + b\,(\underset{\sim}{w} \cdot \underset{\sim}{x}). && \text{(Theorem 0.1, twice)}
\end{aligned}$$

Two special cases are used often. With

$$\begin{aligned} a &= b \\ &= 1, \end{aligned}$$

and $1\,\underset{\sim}{u} = \underset{\sim}{u}$, the theorem gives $\underset{\sim}{x} \cdot (\underset{\sim}{u} + \underset{\sim}{w}) = \underset{\sim}{x} \cdot \underset{\sim}{u} + \underset{\sim}{x} \cdot \underset{\sim}{w}$. With $\underset{\sim}{w} = \underset{\sim}{u}$ and $b = 0$, and $0\,\underset{\sim}{u} = \underset{\sim}{0}$, it gives $\underset{\sim}{x} \cdot (a\,\underset{\sim}{u}) = a\,(\underset{\sim}{x} \cdot \underset{\sim}{u})$, and likewise in the first slot. □

Example 0.1 (Splitting a dot product)**.** Let $\underset{\sim}{x} = (1, 2)$, $\underset{\sim}{u} = (3, 0)$, $\underset{\sim}{w} = (0, 1)$, $a = 2$ and $b = -1$. Directly, $a\,\underset{\sim}{u} + b\,\underset{\sim}{w} = (6, -1)$ and

$$\begin{aligned} \underset{\sim}{x} \cdot (6, -1) &= 6 - 2 \\ &= 4. \end{aligned}$$

By Theorem 0.1,

$$\begin{aligned} a\,(\underset{\sim}{x} \cdot \underset{\sim}{u}) + b\,(\underset{\sim}{x} \cdot \underset{\sim}{w}) &= 2 \cdot 3 + (-1) \cdot 2 \\ &= 4, \end{aligned}$$

the same number.

Definition 0.1 (Orthogonal complement)**.** The **orthogonal complement** [49] of a set $\mathcal{X}$ of vectors in $\mathbb{R}^p$ is the set of vectors orthogonal (Definition 0.19) to every vector of $\mathcal{X}$:

$$\mathcal{X}^{\perp} \stackrel{\text{def}}{=} \{\underset{\sim}{u} \in \mathbb{R}^p : \underset{\sim}{x} \cdot \underset{\sim}{u} = 0 \text{ for all } \underset{\sim}{x} \in \mathcal{X}\}.$$

$\mathcal{X}^{\perp}$ is read "$\mathcal{X}$ perp".

Example 0.2 (Orthogonal complements in $\mathbb{R}^3$)**.**

- **A plane.** Let $\mathcal{X} = \{(a, b, 0) : a, b \in \mathbb{R}\}$, the plane $z = 0$. A vector $\underset{\sim}{u}$ is in $\mathcal{X}^{\perp}$ when

  $$\begin{aligned} (a, b, 0) \cdot \underset{\sim}{u} &= au_1 + bu_2 \\ &= 0 \end{aligned}$$

  for all a and b. Taking $(a, b) = (1, 0)$ forces $u_1 = 0$, and taking $(a, b) = (0, 1)$ forces $u_2 = 0$; conversely, if

  $$\begin{aligned} u_1 &= u_2 \\ &= 0, \end{aligned}$$

  then $au_1 + bu_2 = 0$ for all a and b. So $\mathcal{X}^{\perp} = \{(0, 0, c) : c \in \mathbb{R}\}$, the z-axis.

- **Two vectors.** The two-element set $\{(1, 0, 0), (0, 1, 0)\}$ is not a subspace, and its orthogonal complement is the z-axis too: the two conditions $u_1 = 0$ and $u_2 = 0$ are the same as for the plane.

- **The extremes.** $\{\underset{\sim}{0}\}^\perp = \mathbb{R}^3$, because $\underset{\sim}{0} \cdot \underset{\sim}{u} = 0$ for every $\underset{\sim}{u}$. $(\mathbb{R}^3)^\perp = \{\underset{\sim}{0}\}$, because a $\underset{\sim}{u}$ in it is orthogonal to itself, and

$$\begin{aligned} \underset{\sim}{u} \cdot \underset{\sim}{u} &= u_1^2 + u_2^2 + u_3^2 \\ &= 0 \end{aligned}$$

forces $\underset{\sim}{u} = \underset{\sim}{0}$.

i Remark (The orthogonal complement of a set that is not a subspace)

Remark 0.1 (The orthogonal complement of a set that is not a subspace). $\mathcal{X}$ in Definition 0.1 need not be a subspace.

Theorem 0.2 (An orthogonal complement is a subspace)**.** *For any set $\mathcal{X}$ of vectors in $\mathbb{R}^p$, $\mathcal{X}^\perp$ (Definition 0.1) is a subspace of $\mathbb{R}^p$ (Definition 0.1).*

i Proof

Proof. $\mathcal{X}^\perp$ is not empty:

$$\begin{aligned} \underset{\sim}{x} \cdot \underset{\sim}{0} &= \sum_i x_i \cdot 0 \\ &= 0 \end{aligned}$$

for every $\underset{\sim}{x}$, so $\underset{\sim}{0} \in \mathcal{X}^\perp$. Let $\underset{\sim}{u}, \underset{\sim}{w} \in \mathcal{X}^\perp$, let $c \in \mathbb{R}$, and let $\underset{\sim}{x} \in \mathcal{X}$.
Addition.

$$\begin{aligned} \underset{\sim}{x} \cdot (\underset{\sim}{u} + \underset{\sim}{w}) &= \underset{\sim}{x} \cdot \underset{\sim}{u} + \underset{\sim}{x} \cdot \underset{\sim}{w} && \text{(Theorem 0.1, special case } a \text{ and } b \text{ both equal to 1)} \\ &= 0 + 0 && (\underset{\sim}{u}, \underset{\sim}{w} \in \mathcal{X}^\perp) \\ &= 0. && \text{(arithmetic)} \end{aligned}$$

Scalar multiplication.

$$\begin{aligned} \underset{\sim}{x} \cdot (c\,\underset{\sim}{u}) &= c\,(\underset{\sim}{x} \cdot \underset{\sim}{u}) && \text{(Theorem 0.1, special case } b = 0) \\ &= c \cdot 0 && (\underset{\sim}{u} \in \mathcal{X}^\perp) \\ &= 0. && \text{(arithmetic)} \end{aligned}$$

These two equations hold for every $\underset{\sim}{x} \in \mathcal{X}$, so $\underset{\sim}{u} + \underset{\sim}{w}$ and $c\,\underset{\sim}{u}$ are in $\mathcal{X}^\perp$. □

Example 0.3 (The complement of a single vector is a line)**.** The one-element set $\mathcal{X} = \{(1,2)\}$ in $\mathbb{R}^2$ is not a subspace: $2\,(1,2) = (2,4)$ is not in it. Its orthogonal complement is

$$\begin{aligned}\mathcal{X}^\perp &= \{\underset{\sim}{u} \in \mathbb{R}^2 : u_1 + 2u_2 = 0\} \\ &= \{c\,(-2,1) : c \in \mathbb{R}\},\end{aligned}$$

since $u_1 + 2u_2 = 0$ means

$$\begin{aligned}\underset{\sim}{u} &= (-2u_2, u_2) \\ &= u_2\,(-2,1).\end{aligned}$$

That set is $\text{span}\{(-2,1)\}$, a subspace by Theorem 0.2, as Theorem 0.2 says it must be. It contains $(-2,1)$, the vector Remark 0.8 found perpendicular to $(1,2)$.

Theorem 0.3 (The orthogonal complement of a column space is a null space)**.** *For any $m \times n$ matrix* $\mathbf{A}$*,*

$$\mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top).$$

Proof

Proof. Let $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_n$ be the columns of $\mathbf{A}$, so $\underset{\sim}{a}_j = (a_{1j}, \ldots, a_{mj})$. For $\underset{\sim}{x} \in \mathbb{R}^m$, entry j of $\mathbf{A}^\top \underset{\sim}{x}$ is

$$\begin{aligned}(\mathbf{A}^\top \underset{\sim}{x})_j &= \sum_{i=1}^{m} (\mathbf{A}^\top)_{ji}\, x_i && \text{(Definition 0.16)} \\ &= \sum_{i=1}^{m} a_{ij}\, x_i && \text{(Definition 0.6)} \\ &= \underset{\sim}{a}_j \cdot \underset{\sim}{x}. && \text{(Definition 0.8)}\end{aligned}$$

So $\underset{\sim}{x} \in \mathcal{N}(\mathbf{A}^\top)$ (Definition 0.3) exactly when $\underset{\sim}{a}_j \cdot \underset{\sim}{x} = 0$ for every column $\underset{\sim}{a}_j$.
$\mathcal{C}(\mathbf{A})^\perp \subseteq \mathcal{N}(\mathbf{A}^\top)$**.** Each column $\underset{\sim}{a}_j$ is in $\mathcal{C}(\mathbf{A})$ (Theorem 0.1), so a vector orthogonal to all of $\mathcal{C}(\mathbf{A})$ is orthogonal to every column.
$\mathcal{N}(\mathbf{A}^\top) \subseteq \mathcal{C}(\mathbf{A})^\perp$**.** Suppose $\underset{\sim}{a}_j \cdot \underset{\sim}{x} = 0$ for every j, and take any $\underset{\sim}{y} \in \mathcal{C}(\mathbf{A})$, so $\underset{\sim}{y} = \mathbf{A}\underset{\sim}{z}$ for some $\underset{\sim}{z} \in \mathbb{R}^n$ (Definition 0.1). Then

$$
\begin{aligned}
\underset{\sim}{y} \cdot \underset{\sim}{x} &= \sum_{i=1}^{m} (\mathbf{A}\underset{\sim}{z})_i \, x_i && \text{(Definition 0.8)} \\
&= \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} z_j \right) x_i && \text{(Definition 0.16)} \\
&= \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} \, z_j \, x_i && \text{(distribute each } x_i \text{ over the inner sum)} \\
&= \sum_{j=1}^{n} \sum_{i=1}^{m} a_{ij} \, z_j \, x_i && \text{(swap the order of the finite sums)} \\
&= \sum_{j=1}^{n} z_j \sum_{i=1}^{m} a_{ij} \, x_i && \text{(commute, then factor } z_j \text{ out of the inner sum)} \\
&= \sum_{j=1}^{n} z_j \, (\underset{\sim}{a}_j \cdot \underset{\sim}{x}) && \text{(Definition 0.8)} \\
&= \sum_{j=1}^{n} z_j \cdot 0 && \text{(each } \underset{\sim}{a}_j \cdot \underset{\sim}{x} = 0\text{)} \\
&= 0. && \text{(arithmetic)}
\end{aligned}
$$

So $\underset{\sim}{x}$ is orthogonal to every vector of $\mathcal{C}(\mathbf{A})$. □

Example 0.4 (The complement of the column space in Example 0.1)**.** For $\mathbf{A}$ in Example 0.1, $\mathcal{C}(\mathbf{A}) = \text{span}\{(1,3)\}$ (1). Example 0.3 found $\mathcal{N}(\mathbf{A}^\top) = \text{span}\{(-3,1)\}$, so by Theorem 0.3 this line is $\mathcal{C}(\mathbf{A})^\perp$. Directly, for any c,

$$
\begin{aligned}
(c\,(1,3)) \cdot (-3,1) &= c\,((1,3) \cdot (-3,1)) && \text{(Theorem 0.1, special case } b = 0\text{, first slot)} \\
&= c\,(-3+3) && \text{(Definition 0.8)} \\
&= 0. && \text{(arithmetic)}
\end{aligned}
$$

Theorem 0.4 (A subspace and its orthogonal complement make up the whole space)**.** *Let $\mathcal{S}$ be a subspace of $\mathbb{R}^p$. Then*

1. $\mathcal{S} \cap \mathcal{S}^\perp = \{\underset{\sim}{0}\}$;
2. $\dim(\mathcal{S}) + \dim(\mathcal{S}^\perp) = p$;
3. $\mathbb{R}^p = \mathcal{S} \oplus \mathcal{S}^\perp$ *(Definition 0.3).*

So every $\underset{\sim}{y} \in \mathbb{R}^p$ can be written as $\underset{\sim}{y} = \underset{\sim}{u} + \underset{\sim}{v}$ with $\underset{\sim}{u} \in \mathcal{S}$ and $\underset{\sim}{v} \in \mathcal{S}^\perp$ in only one way (Theorem 0.4).

Proof

Proof. $\mathcal{S}^\perp$ is a subspace (Theorem 0.2), so it has a dimension (Definition 0.6).
Part 1. Both $\mathcal{S}$ and $\mathcal{S}^\perp$ contain $\underset{\sim}{0}$ (Theorem 0.1). If $\underset{\sim}{x}$ is in both, then $\underset{\sim}{x}$ is orthogonal to every vector of $\mathcal{S}$, $\underset{\sim}{x}$ itself included, so

$$\begin{aligned}\underset{\sim}{x} \cdot \underset{\sim}{x} &= x_1^2 + \cdots + x_p^2 \\ &= 0,\end{aligned}$$

which forces every $x_i = 0$.
Part 2. Let $d = \dim(\mathcal{S})$. If $d = 0$, then $\mathcal{S}$ has a basis (Theorem 0.9) with 0 vectors (Definition 0.6), the empty list, whose span is $\{\underset{\sim}{0}\}$ (Definition 0.2); so $\mathcal{S} = \{\underset{\sim}{0}\}$, $\mathcal{S}^\perp = \mathbb{R}^p$ (every $\underset{\sim}{u}$ satisfies $\underset{\sim}{0} \cdot \underset{\sim}{u} = 0$), and $0 + p = p$ (Example 0.14). If $d \geq 1$, choose a basis $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_d$ of $\mathcal{S}$ (Theorem 0.9), and let $\mathbf{A}$ be the $p \times d$ matrix with these columns. The basis spans $\mathcal{S}$, so $\mathcal{S} = \mathcal{C}(\mathbf{A})$ (Theorem 0.1), and $\mathcal{S}^\perp = \mathcal{N}(\mathbf{A}^\top)$ (Theorem 0.3). $\mathbf{A}^\top$ is $d \times p$, so

$$\begin{aligned}\dim(\mathcal{S}^\perp) &= \dim(\mathcal{N}(\mathbf{A}^\top)) && \text{(Theorem 0.3)} \\ &= \text{nullity}(\mathbf{A}^\top) && \text{(Definition 0.1)} \\ &= p - \text{rank}(\mathbf{A}^\top) && \text{(Theorem 0.2, for the } p \text{ columns of } \mathbf{A}^\top) \\ &= p - \text{rank}(\mathbf{A}) && \text{(Theorem 0.4)} \\ &= p - \dim(\mathcal{C}(\mathbf{A})) && \text{(Theorem 0.1)} \\ &= p - d. && (\mathcal{C}(\mathbf{A}) = \mathcal{S})\end{aligned}$$

Part 3. $\mathcal{S} + \mathcal{S}^\perp$ is a subspace of $\mathbb{R}^p$ (Theorem 0.1), and

$$\begin{aligned}\dim(\mathcal{S} + \mathcal{S}^\perp) &= \dim(\mathcal{S}) + \dim(\mathcal{S}^\perp) - \dim(\mathcal{S} \cap \mathcal{S}^\perp) && \text{(Theorem 0.2)} \\ &= p - \dim(\mathcal{S} \cap \mathcal{S}^\perp) && \text{(part 2)} \\ &= p - 0 && \text{(part 1, and } \dim(\{\underset{\sim}{0}\}) = 0 \text{ by Example 0.14)} \\ &= p. && \text{(arithmetic)}\end{aligned}$$

Since $\mathcal{S} + \mathcal{S}^\perp \subseteq \mathbb{R}^p$, $\mathbb{R}^p$ is a subspace (Example 0.1) and $\dim(\mathbb{R}^p) = p$ (Example 0.14), Theorem 0.11 gives $\mathcal{S} + \mathcal{S}^\perp = \mathbb{R}^p$. Together with part 1, $\mathbb{R}^p = \mathcal{S} \oplus \mathcal{S}^\perp$. The uniqueness of the split then follows from Theorem 0.4 (statement 1 implies statement 3). □

Example 0.5 (Splitting a vector of $\mathbb{R}^3$ along a line and its complement)**.** Let $\mathcal{S} = \text{span}\{(1, 1, 0)\}$ in $\mathbb{R}^3$. A single nonzero vector is linearly independent ($c\,\underset{\sim}{v} = \underset{\sim}{0}$ with $\underset{\sim}{v} \neq \underset{\sim}{0}$ forces $c = 0$), so $(1, 1, 0)$ is a basis of $\mathcal{S}$ and $\dim(\mathcal{S}) = 1$. A vector $\underset{\sim}{u}$ is in $\mathcal{S}^\perp$ exactly when $u_1 + u_2 = 0$: that condition says $(1, 1, 0) \cdot \underset{\sim}{u} = 0$, and then

$$\begin{aligned} c\,(1,1,0)\cdot \underset{\sim}{u} &= c\,(u_1+u_2) \\ &= 0 \end{aligned}$$

for every c. So

$$\begin{aligned} \mathcal{S}^{\perp} &= \{(a,-a,c) : a,c \in \mathbb{R}\} \\ &= \operatorname{span}\{(1,-1,0),(0,0,1)\}. \end{aligned}$$

Those two vectors are linearly independent, because $a\,(1,-1,0)+c\,(0,0,1)=(a,-a,c)$ is $\underset{\sim}{0}$ only if

$$\begin{aligned} a &= c \\ &= 0, \end{aligned}$$

so $\dim(\mathcal{S}^{\perp})=2$, and $1+2=3$, as part 2 says.
To split $\underset{\sim}{y}=(3,1,2)$, look for $\underset{\sim}{u}=t\,(1,1,0)$ with

$$\begin{aligned} \underset{\sim}{v} &= \underset{\sim}{y}-\underset{\sim}{u} \\ &= (3-t,1-t,2) \end{aligned}$$

in $\mathcal{S}^{\perp}$: that membership needs $(3-t)+(1-t)=0$, so $t=2$. Then $\underset{\sim}{u}=(2,2,0)$ and $\underset{\sim}{v}=(1,-1,2)$, with $\underset{\sim}{u}+\underset{\sim}{v}=(3,1,2)$ and

$$\begin{aligned} (1,1,0)\cdot(1,-1,2) &= 1-1+0 \\ &= 0. \end{aligned}$$

Theorem 0.5 (The orthogonal complement of the orthogonal complement)**.** *For any subspace $\mathcal{S}$ of $\mathbb{R}^p$,*

$$(\mathcal{S}^{\perp})^{\perp} = \mathcal{S}.$$

Proof

Proof. **$\mathcal{S} \subseteq (\mathcal{S}^{\perp})^{\perp}$.** Take $\underset{\sim}{s} \in \mathcal{S}$ and any $\underset{\sim}{v} \in \mathcal{S}^{\perp}$. By Definition 0.1, $\underset{\sim}{s}\cdot\underset{\sim}{v}=0$, so $\underset{\sim}{v}\cdot\underset{\sim}{s}=0$ too (Theorem 0.1). So $\underset{\sim}{s}$ is orthogonal to every vector of $\mathcal{S}^{\perp}$, that is, $\underset{\sim}{s}\in(\mathcal{S}^{\perp})^{\perp}$.
Equal dimensions. $\mathcal{S}^{\perp}$ is a subspace (Theorem 0.2), so part 2 of Theorem 0.4 applies to it as well as to $\mathcal{S}$:

$$\begin{aligned} \dim((\mathcal{S}^{\perp})^{\perp}) &= p-\dim(\mathcal{S}^{\perp}) && \text{(Theorem 0.4, applied to } \mathcal{S}^{\perp}) \\ &= p-(p-\dim(\mathcal{S})) && \text{(Theorem 0.4, applied to } \mathcal{S}) \\ &= \dim(\mathcal{S}). && \text{(arithmetic)} \end{aligned}$$

$(\mathcal{S}^\perp)^\perp$ is a subspace (Theorem 0.2) that contains $\mathcal{S}$ and has the same dimension, so it equals $\mathcal{S}$ (Theorem 0.11). □

Example 0.6 (Back to the line)**.** In Example 0.5, $\mathcal{S} = \text{span}\{(1,1,0)\}$ and $\mathcal{S}^\perp = \text{span}\{(1,-1,0),(0,0,1)\}$. Put those two spanning vectors in the columns of the 3×2 matrix $\mathbf{B} = \begin{bmatrix} 1 & 0 \\ -1 & 0 \\ 0 & 1 \end{bmatrix}$, so $\mathcal{S}^\perp = \mathcal{C}(\mathbf{B})$ (Theorem 0.1) and $(\mathcal{S}^\perp)^\perp = \mathcal{N}(\mathbf{B}^\top)$ (Theorem 0.3). Since $\mathbf{B}^\top \underset{\sim}{w} = (w_1 - w_2,\ w_3)$ (Definition 0.16), $\underset{\sim}{w} \in (\mathcal{S}^\perp)^\perp$ exactly when $w_1 - w_2 = 0$ and $w_3 = 0$. So

$$\begin{aligned}(\mathcal{S}^\perp)^\perp &= \{(a,a,0) : a \in \mathbb{R}\} \\ &= \mathcal{S}.\end{aligned}$$

Example 0.7 (A set that is not a subspace does not come back)**.** The one-element set $\mathcal{X} = \{(1,1,0)\}$ is not a subspace. Its orthogonal complement is the same as that of $\mathcal{S}$ in Example 0.6, since both are defined by the single condition $u_1 + u_2 = 0$, so $(\mathcal{X}^\perp)^\perp = \text{span}\{(1,1,0)\}$, which contains $(2,2,0) \notin \mathcal{X}$. Theorem 0.5 needs $\mathcal{S}$ to be a subspace.

The fundamental theorem of linear algebra

Theorem 0.1 (Fundamental theorem of linear algebra)**.** *Let* $\mathbf{A}$ *be an* $m \times n$ *matrix with* $\text{rank}(\mathbf{A}) = r$*. Then*

1. $\mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top)$*, and* $\mathbb{R}^m = \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A}^\top)$*;*
2. $\mathcal{C}(\mathbf{A}) = \mathcal{N}(\mathbf{A}^\top)^\perp$*;*
3. $\mathcal{N}(\mathbf{A})^\perp = \mathcal{C}(\mathbf{A}^\top)$*, and* $\mathbb{R}^n = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$*;*
4. $$\begin{aligned}\dim(\mathcal{C}(\mathbf{A})) &= \dim(\mathcal{C}(\mathbf{A}^\top)) \\ &= r,\end{aligned}$$

   $\dim(\mathcal{N}(\mathbf{A})) = n - r$*, and* $\dim(\mathcal{N}(\mathbf{A}^\top)) = m - r$*.*

Proof

Proof. **Part 1.** The first equation is Theorem 0.3. $\mathcal{C}(\mathbf{A})$ is a subspace of $\mathbb{R}^m$ (Theorem 0.1), so

$$\begin{aligned}\mathbb{R}^m &= \mathcal{C}(\mathbf{A}) \oplus \mathcal{C}(\mathbf{A})^\perp && \text{(Theorem 0.4)}\\ &= \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A}^\top). && \text{(first equation)}\end{aligned}$$

Part 2.

$$\begin{aligned}\mathcal{C}(\mathbf{A}) &= (\mathcal{C}(\mathbf{A})^\perp)^\perp && \text{(Theorem 0.5)}\\ &= \mathcal{N}(\mathbf{A}^\top)^\perp. && \text{(part 1)}\end{aligned}$$

Part 3. Apply parts 2 and 1 to the $n \times m$ matrix $\mathbf{A}^\top$, using $(\mathbf{A}^\top)^\top = \mathbf{A}$: part 2 gives $\mathcal{C}(\mathbf{A}^\top) = \mathcal{N}(\mathbf{A})^\perp$, and part 1 gives $\mathbb{R}^n = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$.
Part 4. $\dim(\mathcal{C}(\mathbf{A})) = r$ by Theorem 0.1. $\dim(\mathcal{C}(\mathbf{A}^\top)) = \operatorname{rank}(\mathbf{A}^\top)$ by Theorem 0.1, and that is r by Theorem 0.4. $\mathbf{A}$ has n columns, so $\dim(\mathcal{N}(\mathbf{A})) = n - r$ (Definition 0.1, Theorem 0.2). $\mathbf{A}^\top$ has m columns and rank r, so $\dim(\mathcal{N}(\mathbf{A}^\top)) = m - r$ in the same way. □

Example 0.1 (The four subspaces of the matrix in Example 0.1)**.** For $\mathbf{A}$ in Example 0.1, $m = 2$, $n = 3$ and $r = 1$ (Example 0.1).

- **In $\mathbb{R}^3$:** the row space $\mathcal{C}(\mathbf{A}^\top) = \operatorname{span}\{(1, -2, -2)\}$

  (1) has dimension $1 = r$, and $\mathcal{N}(\mathbf{A}) = \operatorname{span}\{(2, 1, 0), (2, 0, 1)\}$ has dimension $2 = n - r$ (Example 0.2). The spanning vectors are orthogonal:

$$\begin{aligned}(1, -2, -2) \cdot (2, 1, 0) &= 2 - 2 + 0\\ &= 0\end{aligned}$$

  and

$$\begin{aligned}(1, -2, -2) \cdot (2, 0, 1) &= 2 + 0 - 2\\ &= 0.\end{aligned}$$

- **In $\mathbb{R}^2$:** $\mathcal{C}(\mathbf{A}) = \operatorname{span}\{(1, 3)\}$ has dimension $1 = r$, and $\mathcal{N}(\mathbf{A}^\top) = \operatorname{span}\{(-3, 1)\}$ has dimension $1 = m - r$ (Example 0.4).

By part 3, $y = (1, 0, 0)$ splits uniquely into a row-space part $t\,(1, -2, -2)$ and a null-space part. The null-space part $y - t\,(1, -2, -2)$ lies in $\mathcal{N}(\mathbf{A})$ (part 3), and $\mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)^\perp$ (Theorem 0.3 applied to $\mathbf{A}^\top$, with $(\mathbf{A}^\top)^\top = \mathbf{A}$), so it must be orthogonal to $(1, -2, -2)$:

$$
\begin{aligned}
0 &= (1,-2,-2)\cdot((1,0,0)-t\,(1,-2,-2)) && \text{(orthogonality)}\\
&= (1,-2,-2)\cdot(1\cdot(1,0,0)+(-t)\,(1,-2,-2)) && \text{(write the difference as a linear combination)}\\
&= 1\cdot((1,-2,-2)\cdot(1,0,0))+(-t)\,((1,-2,-2)\cdot(1,-2,-2)) && \text{(Theorem 0.1)}\\
&= 1\cdot 1+(-t)\cdot 9 && \text{(Definition 0.8)}\\
&= 1-9t, && \text{(arithmetic)}
\end{aligned}
$$

so $t=\frac{1}{9}$. The row-space part is $\left(\frac{1}{9},-\frac{2}{9},-\frac{2}{9}\right)$ and the null-space part is $\left(\frac{8}{9},\frac{2}{9},\frac{2}{9}\right)$; as a check, row 1 of $\mathbf{A}$ gives $\frac{8}{9}-2\cdot\frac{2}{9}-2\cdot\frac{2}{9}=0$, and row 2 is 3 times row 1, so $\mathbf{A}$ sends the null-space part to $\underset{\sim}{0}_2$.
These four statements are the fundamental theorem of linear algebra [50].

Inner products

Definition 0.1 (Positivity)**.** A function $\langle\cdot,\cdot\rangle$ that assigns a real number to each ordered pair of vectors in $\mathbb{R}^p$ has **positivity** if

$$\langle \underset{\sim}{x}, \underset{\sim}{x}\rangle \geq 0$$

for all $\underset{\sim}{x}\in\mathbb{R}^p$.

Definition 0.2 (Definiteness)**.** A function $\langle\cdot,\cdot\rangle$ that assigns a real number to each ordered pair of vectors in $\mathbb{R}^p$ has **definiteness** if

$$\langle \underset{\sim}{x}, \underset{\sim}{x}\rangle = 0 \quad \text{if and only if} \quad \underset{\sim}{x} = \underset{\sim}{0}_{p\times 1}$$

for all $\underset{\sim}{x}\in\mathbb{R}^p$.

Definition 0.3 (Linearity in the first slot)**.** A function $\langle\cdot,\cdot\rangle$ that assigns a real number to each ordered pair of vectors in $\mathbb{R}^p$ has **linearity in the first slot** if

$$\langle a\,\underset{\sim}{x}+b\,\underset{\sim}{y}, \underset{\sim}{z}\rangle = a\,\langle \underset{\sim}{x}, \underset{\sim}{z}\rangle + b\,\langle \underset{\sim}{y}, \underset{\sim}{z}\rangle$$

for all $\underset{\sim}{x}, \underset{\sim}{y}, \underset{\sim}{z}\in\mathbb{R}^p$ and all real numbers a, b.

Definition 0.4 (Symmetry)**.** A function $\langle\cdot,\cdot\rangle$ that assigns a real number to each ordered pair of vectors in $\mathbb{R}^p$ has **symmetry** if

$$\langle \underset{\sim}{x}, \underset{\sim}{y} \rangle = \langle \underset{\sim}{y}, \underset{\sim}{x} \rangle$$

for all $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^p$.

Definition 0.5 (Inner product)**.** An **inner product** [51] on $\mathbb{R}^p$ is a function $\langle \cdot, \cdot \rangle$ that assigns a real number to each ordered pair of vectors in $\mathbb{R}^p$ and that has the following four properties:

- positivity;
- definiteness;
- linearity in the first slot;
- symmetry.

Remark (The definition in other sources)

Remark 0.1 (The definition in other sources). Axler [52], Definition 6.2, p. 183 states the same definition for any vector space over the real or complex numbers. Axler splits linearity in the first slot into additivity and homogeneity, and his last property is *conjugate* symmetry, $\langle \underset{\sim}{u}, \underset{\sim}{v} \rangle = \overline{\langle \underset{\sim}{v}, \underset{\sim}{u} \rangle}$, which for real numbers is the same as symmetry. Symmetry and linearity in the first slot together give linearity in the second slot.

Example 0.1 (The dot product is an inner product)**.** The dot product (2) is an inner product on $\mathbb{R}^p$, called the **Euclidean** or **standard** inner product. Each of the four properties in Definition 0.5 holds:

- **positivity**: $\underset{\sim}{x} \cdot \underset{\sim}{x} = \sum_{i=1}^{p} x_i^2$ is a sum of squares, so it is at least 0;
- **definiteness**: a sum of squares is 0 exactly when every term is 0, so $\underset{\sim}{x} \cdot \underset{\sim}{x} = 0$ exactly when every $x_i = 0$, that is, when $\underset{\sim}{x} = \underset{\sim}{0}_{p \times 1}$;
- **linearity in the first slot**: this property is the first-slot half of Theorem 0.1;
- **symmetry**: this property is Theorem 0.1.

For instance, with $\underset{\sim}{x} = (1, -2, 2)$,

$$\begin{aligned} \underset{\sim}{x} \cdot \underset{\sim}{x} &= 1 \cdot 1 + (-2) \cdot (-2) + 2 \cdot 2 && \text{(definition of the dot product)} \\ &= 1 + 4 + 4 && \text{(multiply)} \\ &= 9 && \text{(add)} \end{aligned}$$

which is positive, as positivity and definiteness require of a nonzero vector.

Definition 0.6 (Inner product space)**.** A **real inner product space** [51] is a pair $(V, \langle \cdot, \cdot \rangle)$ of a real vector space V and an inner product $\langle \cdot, \cdot \rangle$ on it, a function with the four properties of Definition 0.5. We say that V is **equipped with** that inner product: the inner product is part of the data of the space, so the same set of vectors with a different inner product is a different inner product space. Definition 0.5 states the properties for $V = \mathbb{R}^p$ (Definition 0.2), and the same four properties define an inner product on other real vector spaces, such as spaces of matrices or functions.

Example 0.2 (Real coordinate space as an inner product space)**.** The Euclidean space $\mathbb{R}^p$ equipped with the standard dot product (2), that is, the pair $(\mathbb{R}^p, \langle \cdot, \cdot \rangle)$ with $\langle \underset{\sim}{x}, \underset{\sim}{y} \rangle = \underset{\sim}{x} \cdot \underset{\sim}{y}$, is an inner product space, because the dot product satisfies the four inner product axioms (Example 0.1). For $p = 2$, any vector $\underset{\sim}{x} = (x_1, x_2) \in \mathbb{R}^2$ has squared norm $\langle \underset{\sim}{x}, \underset{\sim}{x} \rangle = x_1^2 + x_2^2 \geq 0$, which is zero if and only if

$$\begin{aligned} x_1 &= x_2 \\ &= 0. \end{aligned}$$

Equipping the same vector space $\mathbb{R}^p$ with different inner products yields different inner product spaces with different geometric notions of angle and distance on the same set of vectors.

Example 0.3 (A weighted inner product)**.** Let $\underset{\sim}{c} = (c_1, \ldots, c_p)$ be a vector of positive numbers, and define

$$\langle \underset{\sim}{x}, \underset{\sim}{y} \rangle_{\underset{\sim}{c}} \overset{\text{def}}{=} \sum_{i=1}^{p} c_i \, x_i y_i.$$

This function is also an inner product on $\mathbb{R}^p$ [52], Example 6.3(b), p. 184:

- **positivity**: $\langle \underset{\sim}{x}, \underset{\sim}{x} \rangle_{\underset{\sim}{c}} = \sum_{i=1}^{p} c_i \, x_i^2$, and each term $c_i \, x_i^2$ is a positive number times a square, so the sum is at least 0;
- **definiteness**: a sum of nonnegative terms is 0 exactly when every term is 0, and $c_i \, x_i^2 = 0$ exactly when $x_i = 0$, because $c_i > 0$; so $\langle \underset{\sim}{x}, \underset{\sim}{x} \rangle_{\underset{\sim}{c}} = 0$ exactly when $\underset{\sim}{x} = \underset{\sim}{0}_{p \times 1}$;
- **linearity in the first slot**: see the derivation after this list;
- **symmetry**: $c_i \, x_i y_i = c_i \, y_i x_i$ for each i, so $\langle \underset{\sim}{x}, \underset{\sim}{y} \rangle_{\underset{\sim}{c}} = \langle \underset{\sim}{y}, \underset{\sim}{x} \rangle_{\underset{\sim}{c}}$.

For linearity in the first slot, take $\underset{\sim}{x}, \underset{\sim}{y}, \underset{\sim}{z} \in \mathbb{R}^p$ and real numbers a, b:

$$
\begin{aligned}
\langle a\,\underset{\sim}{x} + b\,\underset{\sim}{y}, \underset{\sim}{z}\rangle_{\underset{\sim}{c}} &= \sum_{i=1}^{p} c_i\,(a\,\underset{\sim}{x} + b\,\underset{\sim}{y})_i\,z_i && \text{(definition of } \langle\cdot,\cdot\rangle_{\underset{\sim}{c}}\text{)} \\
&= \sum_{i=1}^{p} c_i\,(ax_i + by_i)\,z_i && \text{(Definition 0.9, Definition 0.7)} \\
&= \sum_{i=1}^{p} (a\,c_i x_i z_i + b\,c_i y_i z_i) && \text{(distribute, and commute the factors in each product)} \\
&= \sum_{i=1}^{p} a\,c_i x_i z_i + \sum_{i=1}^{p} b\,c_i y_i z_i && \text{(split the finite sum)} \\
&= a\sum_{i=1}^{p} c_i x_i z_i + b\sum_{i=1}^{p} c_i y_i z_i && \text{(factor } a \text{ and } b \text{ out of the sums)} \\
&= a\,\langle \underset{\sim}{x}, \underset{\sim}{z}\rangle_{\underset{\sim}{c}} + b\,\langle \underset{\sim}{y}, \underset{\sim}{z}\rangle_{\underset{\sim}{c}} && \text{(definition of } \langle\cdot,\cdot\rangle_{\underset{\sim}{c}}\text{)}
\end{aligned}
$$

Different inner products can give different numbers for the same pair of vectors. With $p = 2$, $\underset{\sim}{c} = (2, 1)$, $\underset{\sim}{x} = (1, 3)$ and $\underset{\sim}{y} = (4, -1)$:

$$
\begin{aligned}
\langle \underset{\sim}{x}, \underset{\sim}{y}\rangle_{\underset{\sim}{c}} &= 2 \cdot 1 \cdot 4 + 1 \cdot 3 \cdot (-1) && \text{(definition of } \langle\cdot,\cdot\rangle_{\underset{\sim}{c}}\text{)} \\
&= 8 - 3 && \text{(multiply)} \\
&= 5 && \text{(subtract)}
\end{aligned}
$$

while the dot product of the same vectors is

$$
\begin{aligned}
\underset{\sim}{x} \cdot \underset{\sim}{y} &= 1 \cdot 4 + 3 \cdot (-1) && \text{(definition of the dot product)} \\
&= 4 - 3 && \text{(multiply)} \\
&= 1 && \text{(subtract)}
\end{aligned}
$$

Example 0.4 (Functions that are not inner products)**.** Each of these functions on $\mathbb{R}^2$ has some of the properties in Definition 0.5, but not all four:

- $f(\underset{\sim}{x}, \underset{\sim}{y}) \overset{\text{def}}{=} x_1 y_1 - x_2 y_2$ is symmetric, because $x_i y_i = y_i x_i$, and linear in the first slot, because it is a linear combination of x_1 and x_2 with coefficients y_1 and $-y_2$. It fails positivity: for $\underset{\sim}{x} = (0, 1)$,

$$
\begin{aligned}
f(\underset{\sim}{x}, \underset{\sim}{x}) &= 0 \cdot 0 - 1 \cdot 1 \\
&= -1 \\
&< 0.
\end{aligned}
$$

  It also fails definiteness: for $\underset{\sim}{x} = (1, 1)$,

$$f(\underset{\sim}{x}, \underset{\sim}{x}) = 1 \cdot 1 - 1 \cdot 1$$
$$= 0$$

although $\underset{\sim}{x} \neq \underset{\sim}{0}_{2\times 1}$.

- $g(\underset{\sim}{x}, \underset{\sim}{y}) \overset{\text{def}}{=} x_1 y_1$ is symmetric and linear in the first slot, for the same reasons as f, and has positivity, because $g(\underset{\sim}{x}, \underset{\sim}{x}) = x_1^2 \geq 0$. It fails definiteness: for $\underset{\sim}{x} = (0, 1)$,

$$g(\underset{\sim}{x}, \underset{\sim}{x}) = 0 \cdot 0$$
$$= 0$$

although $\underset{\sim}{x} \neq \underset{\sim}{0}_{2\times 1}$.

- $h(\underset{\sim}{x}, \underset{\sim}{y}) \overset{\text{def}}{=} |x_1 y_1| + |x_2 y_2|$, where $|\cdot|$ is the absolute value, is symmetric, because $\widetilde{|x_i y_i|} = |y_i x_i|$, and has positivity and definiteness, because $h(\underset{\sim}{x}, \underset{\sim}{x}) = x_1^2 + x_2^2$ is a sum of squares. It is not linear in the first slot: take

$$\underset{\sim}{x} = \underset{\sim}{y}$$
$$= (1, 0),$$

$a = -1$ and $b = 0$; then

$$h(a\,\underset{\sim}{x}, \underset{\sim}{y}) = |(-1) \cdot 1| + |0 \cdot 0|$$
$$= 1,$$

while

$$a\,h(\underset{\sim}{x}, \underset{\sim}{y}) = (-1) \cdot (|1 \cdot 1| + |0 \cdot 0|)$$
$$= -1.$$

Definition 0.7 (Orthogonality in an inner product space)**.** In an inner product space (Definition 0.6) with inner product $\langle \cdot, \cdot \rangle$, two vectors $\underset{\sim}{x}$ and $\underset{\sim}{y}$ are **orthogonal with respect to** $\langle \cdot, \cdot \rangle$, written $\underset{\sim}{x} \perp \underset{\sim}{y}$, if

$$\langle \underset{\sim}{x}, \underset{\sim}{y} \rangle = 0$$

[48], Definition 7.3, p. 181.

Example 0.5 (Orthogonality depends on the inner product)**.** Take the weighted inner product of Example 0.3 with $\underset{\sim}{c} = (2, 1)$, and let $\underset{\sim}{x} = (1, 2)$ and $\underset{\sim}{y} = (1, -1)$. Then

$$\begin{aligned}\langle \underset{\sim}{x}, \underset{\sim}{y} \rangle_{\underset{\sim}{c}} &= 2 \cdot 1 \cdot 1 + 1 \cdot 2 \cdot (-1) && \text{(definition of } \langle \cdot, \cdot \rangle_{\underset{\sim}{c}}\text{)} \\ &= 2 - 2 && \text{(multiply)} \\ &= 0, && \text{(subtract)}\end{aligned}$$

so $\underset{\sim}{x}$ and $\underset{\sim}{y}$ are orthogonal with respect to $\langle \cdot, \cdot \rangle_{\underset{\sim}{c}}$. They are not orthogonal with respect to the dot product, since

$$\begin{aligned}\underset{\sim}{x} \cdot \underset{\sim}{y} &= 1 \cdot 1 + 2 \cdot (-1) \\ &= -1 \\ &\neq 0.\end{aligned}$$

i Remark (Orthogonality for the dot product)

Remark 0.2 (Orthogonality for the dot product). With the dot product as the inner product (Example 0.1), the condition of Definition 0.7 is orthogonality as in Definition 0.19.

Cauchy-Schwarz and the triangle inequality

i Source

This section is adapted from Zhou [53], used under the MIT License (see the license text in Section). The source leaves the proof of the triangle inequality to class; it is written out here. These parts of the source are not part of this section:

- its table of names for the angle between two vectors (aligned, acute, orthogonal, obtuse, opposed)
- the triangle inequality for distances
- the identity relating the average, the standard deviation and the root mean square of a vector

Theorem 0.1 (Basic properties of the Euclidean norm)**.** *For any $\underset{\sim}{x} \in \mathbb{R}^p$ and any number c:*

1. $\|\underset{\sim}{x}\| \geq 0$, *and* $\|\underset{\sim}{x}\| = 0$ *if and only if* $\underset{\sim}{x} = \underset{\sim}{0}$;
2. $\|c\,\underset{\sim}{x}\| = |c|\,\|\underset{\sim}{x}\|$.

Proof

Proof. **Part 1.** $\|\underset{\sim}{x}\| = \sqrt{x_1^2 + \cdots + x_p^2}$ (Equation 2) is the square root of a sum of squares, so it is at least 0. It is 0 exactly when $x_1^2 + \cdots + x_p^2 = 0$, and a sum of squares is 0 exactly when every term is 0, that is, when every $x_i = 0$: if some $x_i \neq 0$, its square is positive and the sum is positive, and if every $x_i = 0$, the sum is 0.
Part 2.

$$\begin{aligned}
\|c\,\underset{\sim}{x}\| &= \sqrt{\sum_{i=1}^{p} (c\,x_i)^2} && \text{(Equation 2, and Definition 0.9)} \\
&= \sqrt{\sum_{i=1}^{p} c^2 x_i^2} && ((c\,x_i)^2 = c^2 x_i^2) \\
&= \sqrt{c^2 \sum_{i=1}^{p} x_i^2} && \text{(factor } c^2 \text{ out of the sum)} \\
&= \sqrt{c^2}\sqrt{\sum_{i=1}^{p} x_i^2} && \text{(the square root of a product of nonnegative numbers)} \\
&= |c|\,\|\underset{\sim}{x}\|. && (\sqrt{c^2} = |c|, \text{ and Equation 2})
\end{aligned}$$

□

Example 0.1 (Scaling a vector scales its length)**.** For $\underset{\sim}{x} = (3, 4)$, $\|\underset{\sim}{x}\| = 5$ (Example 0.15). Then $-2\,\underset{\sim}{x} = (-6, -8)$ has norm

$$\begin{aligned}
\sqrt{36 + 64} &= 10 \\
&= |-2| \cdot 5,
\end{aligned}$$

as part 2 says. Dividing by the norm gives a vector of length 1:

$$\begin{aligned}
\left\|\tfrac{1}{5}\,\underset{\sim}{x}\right\| &= \tfrac{1}{5} \cdot 5 \\
&= 1,
\end{aligned}$$

and $\frac{1}{5}\,\underset{\sim}{x} = (0.6, 0.8)$ is the unit vector in Example 0.15.

Theorem 0.2 (The squared norm of a sum)**.** *For any* $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^p$,

$$\|\underset{\sim}{x} + \underset{\sim}{y}\|^2 = \|\underset{\sim}{x}\|^2 + 2\,(\underset{\sim}{x} \cdot \underset{\sim}{y}) + \|\underset{\sim}{y}\|^2.$$

In particular, if $\underset{\sim}{x} \perp \underset{\sim}{y}$ *(Definition 0.19), then* $\|\underset{\sim}{x} + \underset{\sim}{y}\|^2 = \|\underset{\sim}{x}\|^2 + \|\underset{\sim}{y}\|^2$ *(the* ***Pythagorean theorem****) [54].*

Proof

Proof.

$$\begin{aligned}
\|\underset{\sim}{x}+\underset{\sim}{y}\|^2 &= (\underset{\sim}{x}+\underset{\sim}{y})\cdot(\underset{\sim}{x}+\underset{\sim}{y}) && \text{(square both sides of Equation 2)}\\
&= \underset{\sim}{x}\cdot(\underset{\sim}{x}+\underset{\sim}{y})+\underset{\sim}{y}\cdot(\underset{\sim}{x}+\underset{\sim}{y}) && \text{(Theorem 0.1, first slot, } a \text{ and } b \text{ both equal to 1)}\\
&= \underset{\sim}{x}\cdot\underset{\sim}{x}+\underset{\sim}{x}\cdot\underset{\sim}{y}+\underset{\sim}{y}\cdot\underset{\sim}{x}+\underset{\sim}{y}\cdot\underset{\sim}{y} && \text{(Theorem 0.1, second slot, } a \text{ and } b \text{ both equal to 1, twice)}\\
&= \underset{\sim}{x}\cdot\underset{\sim}{x}+\underset{\sim}{x}\cdot\underset{\sim}{y}+\underset{\sim}{x}\cdot\underset{\sim}{y}+\underset{\sim}{y}\cdot\underset{\sim}{y} && \text{(Theorem 0.1)}\\
&= \underset{\sim}{x}\cdot\underset{\sim}{x}+2\,(\underset{\sim}{x}\cdot\underset{\sim}{y})+\underset{\sim}{y}\cdot\underset{\sim}{y} && \text{(combine like terms)}\\
&= \|\underset{\sim}{x}\|^2+2\,(\underset{\sim}{x}\cdot\underset{\sim}{y})+\|\underset{\sim}{y}\|^2. && \text{(Equation 2, squared)}
\end{aligned}$$

If $\underset{\sim}{x}\perp\underset{\sim}{y}$, then $\underset{\sim}{x}\cdot\underset{\sim}{y}=0$, and the middle term drops out. □

Example 0.2 (Checking the expansion)**.** For $\underset{\sim}{x}=(3,0)$ and $\underset{\sim}{y}=(1,4)$: $\underset{\sim}{x}+\underset{\sim}{y}=(4,4)$, so

$$\begin{aligned}
\|\underset{\sim}{x}+\underset{\sim}{y}\|^2 &= 16+16\\
&= 32,
\end{aligned}$$

and

$$\begin{aligned}
\|\underset{\sim}{x}\|^2+2\,(\underset{\sim}{x}\cdot\underset{\sim}{y})+\|\underset{\sim}{y}\|^2 &= 9+2\cdot 3+17\\
&= 32.
\end{aligned}$$

For the orthogonal vectors $(3,0)$ and $(0,4)$,

$$\begin{aligned}
\|(3,4)\|^2 &= 25\\
&= 9+16,
\end{aligned}$$

the 3-4-5 right triangle.

Theorem 0.3 (Cauchy-Schwarz inequality)**.** *For any* $\underset{\sim}{x},\underset{\sim}{y}\in\mathbb{R}^p$,

$$|\underset{\sim}{x}\cdot\underset{\sim}{y}|\le\|\underset{\sim}{x}\|\,\|\underset{\sim}{y}\|,$$

with equality exactly when $\underset{\sim}{x}$ *and* $\underset{\sim}{y}$ *are linearly dependent (Definition 0.1) [55].*

Proof

Proof. **If** $\underset{\sim}{y}=\underset{\sim}{0}$**.** Then

$$\begin{aligned} \underset{\sim}{x} \cdot \underset{\sim}{0} &= \sum_i x_i \cdot 0 \\ &= 0 \end{aligned}$$

(2) and $\|\underset{\sim}{0}\| = 0$ (Theorem 0.1, part 1), so both sides are 0 and equality holds; and $\underset{\sim}{x}, \underset{\sim}{y}$ are linearly dependent, because $0\,\underset{\sim}{x} + 1\,\underset{\sim}{y} = \underset{\sim}{0}$.

If $\underset{\sim}{y} \neq \underset{\sim}{0}$. Then $\|\underset{\sim}{y}\| > 0$ (Theorem 0.1, part 1). Let $t \stackrel{\text{def}}{=} (\underset{\sim}{x} \cdot \underset{\sim}{y})/\|\underset{\sim}{y}\|^2$. Then

$$\begin{aligned}
\|\underset{\sim}{x} - t\,\underset{\sim}{y}\|^2 &= \|\underset{\sim}{x} + (-1)\,(t\,\underset{\sim}{y})\|^2 && (\underset{\sim}{x} - \underset{\sim}{v} = \underset{\sim}{x} + (-1)\,\underset{\sim}{v}) \\
&= \|\underset{\sim}{x} + (-t)\,\underset{\sim}{y}\|^2 && ((-1)\,(t\,\underset{\sim}{y}) = (-t)\,\underset{\sim}{y}, \text{ entrywise}) \\
&= \|\underset{\sim}{x}\|^2 + 2\,(\underset{\sim}{x} \cdot ((-t)\,\underset{\sim}{y})) + \|(-t)\,\underset{\sim}{y}\|^2 && (\text{Theorem 0.2}) \\
&= \|\underset{\sim}{x}\|^2 + 2\,(-t)\,(\underset{\sim}{x} \cdot \underset{\sim}{y}) + \|(-t)\,\underset{\sim}{y}\|^2 && (\text{Theorem 0.1, special case } b = 0) \\
&= \|\underset{\sim}{x}\|^2 + 2\,(-t)\,(\underset{\sim}{x} \cdot \underset{\sim}{y}) + (|-t|\,\|\underset{\sim}{y}\|)^2 && (\text{Theorem 0.1, part 2}) \\
&= \|\underset{\sim}{x}\|^2 + 2\,(-t)\,(\underset{\sim}{x} \cdot \underset{\sim}{y}) + |-t|^2\,\|\underset{\sim}{y}\|^2 && ((ab)^2 = a^2b^2) \\
&= \|\underset{\sim}{x}\|^2 - 2t\,(\underset{\sim}{x} \cdot \underset{\sim}{y}) + t^2\,\|\underset{\sim}{y}\|^2 && (2\,(-t) = -2t \text{ and } |-t|^2 = t^2) \\
&= \|\underset{\sim}{x}\|^2 - 2\,\frac{(\underset{\sim}{x} \cdot \underset{\sim}{y})^2}{\|\underset{\sim}{y}\|^2} + \frac{(\underset{\sim}{x} \cdot \underset{\sim}{y})^2}{\|\underset{\sim}{y}\|^4}\,\|\underset{\sim}{y}\|^2 && (\text{substitute } t) \\
&= \|\underset{\sim}{x}\|^2 - 2\,\frac{(\underset{\sim}{x} \cdot \underset{\sim}{y})^2}{\|\underset{\sim}{y}\|^2} + \frac{(\underset{\sim}{x} \cdot \underset{\sim}{y})^2}{\|\underset{\sim}{y}\|^2} && (\text{cancel one factor } \|\underset{\sim}{y}\|^2) \\
&= \|\underset{\sim}{x}\|^2 - \frac{(\underset{\sim}{x} \cdot \underset{\sim}{y})^2}{\|\underset{\sim}{y}\|^2}. && (\text{combine the last two terms})
\end{aligned} \tag{6}$$

The left side is at least 0 (Theorem 0.1, part 1), so

$$\begin{aligned}
0 &\leq \|\underset{\sim}{x}\|^2 - \frac{(\underset{\sim}{x} \cdot \underset{\sim}{y})^2}{\|\underset{\sim}{y}\|^2} && (\text{Equation 6}) \\
\frac{(\underset{\sim}{x} \cdot \underset{\sim}{y})^2}{\|\underset{\sim}{y}\|^2} &\leq \|\underset{\sim}{x}\|^2 && (\text{add } (\underset{\sim}{x} \cdot \underset{\sim}{y})^2/\|\underset{\sim}{y}\|^2 \text{ to both sides}) \\
(\underset{\sim}{x} \cdot \underset{\sim}{y})^2 &\leq \|\underset{\sim}{x}\|^2\,\|\underset{\sim}{y}\|^2, && (\text{multiply both sides by } \|\underset{\sim}{y}\|^2 > 0)
\end{aligned}$$

and taking nonnegative square roots of both sides, which preserves the inequality, gives $|\underset{\sim}{x} \cdot \underset{\sim}{y}| \leq \|\underset{\sim}{x}\|\,\|\underset{\sim}{y}\|$.

When equality holds ($\underset{\sim}{y} \neq \underset{\sim}{0}$). Both sides of the inequality are nonnegative, so equality holds exactly when $(\underset{\sim}{x} \cdot \underset{\sim}{y})^2 = \|\underset{\sim}{x}\|^2\,\|\underset{\sim}{y}\|^2$, that is (dividing by $\|\underset{\sim}{y}\|^2 > 0$ and rearranging), when $\|\underset{\sim}{x}\|^2 - (\underset{\sim}{x} \cdot \underset{\sim}{y})^2/\|\underset{\sim}{y}\|^2 = 0$. By Equation 6, that quantity is $\|\underset{\sim}{x} - t\,\underset{\sim}{y}\|^2$, so equality holds exactly when $\underset{\sim}{x} - t\,\underset{\sim}{y} = \underset{\sim}{0}$ (Theorem 0.1, part 1), that is, when $\underset{\sim}{x} = t\,\underset{\sim}{y}$.

- If $\underset{\sim}{x} = t\,\underset{\sim}{y}$, then $1\,\underset{\sim}{x} + (-t)\,\underset{\sim}{y} = \underset{\sim}{0}$, so the two vectors are linearly dependent.

- Conversely, suppose $a\,\underset{\sim}{x} + b\,\underset{\sim}{y} = \underset{\sim}{0}$ with a, b not both 0. Then $a \neq 0$: otherwise $b \neq 0$ and $b\,\underset{\sim}{y} = \underset{\sim}{0}$, so

$$\begin{aligned} \underset{\sim}{y} &= \tfrac{1}{b}\,(b\,\underset{\sim}{y}) \\ &= \underset{\sim}{0}, \end{aligned}$$

contrary to assumption. Subtracting $b\,\underset{\sim}{y}$ from both sides gives $a\,\underset{\sim}{x} = -b\,\underset{\sim}{y}$, and multiplying by $\frac{1}{a}$ gives $\underset{\sim}{x} = s\,\underset{\sim}{y}$ with $s = -b/a$. Then

$$\begin{aligned} |\underset{\sim}{x} \cdot \underset{\sim}{y}| &= |(s\,\underset{\sim}{y}) \cdot \underset{\sim}{y}| && \text{(substitute } \underset{\sim}{x} = s\,\underset{\sim}{y}) \\ &= |s\,(\underset{\sim}{y} \cdot \underset{\sim}{y})| && \text{(Theorem 0.1, special case } b = 0, \text{ first slot)} \\ &= |s|\,|\underset{\sim}{y} \cdot \underset{\sim}{y}| && (|ab| = |a|\,|b|) \\ &= |s|\,\|\underset{\sim}{y}\|^2 && (\underset{\sim}{y} \cdot \underset{\sim}{y} = \|\underset{\sim}{y}\|^2 \geq 0, \text{ Equation 2}) \\ &= (|s|\,\|\underset{\sim}{y}\|)\,\|\underset{\sim}{y}\| && \text{(regroup)} \\ &= \|s\,\underset{\sim}{y}\|\,\|\underset{\sim}{y}\| && \text{(Theorem 0.1, part 2)} \\ &= \|\underset{\sim}{x}\|\,\|\underset{\sim}{y}\|, && (s\,\underset{\sim}{y} = \underset{\sim}{x}) \end{aligned}$$

so equality holds.

□

Example 0.3 (A strict case and an equality case)**.**

- For $\underset{\sim}{x} = (1, 2)$ and $\underset{\sim}{y} = (3, 4)$:

$$\begin{aligned} \underset{\sim}{x} \cdot \underset{\sim}{y} &= 3 + 8 \\ &= 11, \end{aligned}$$

while $\|\underset{\sim}{x}\|\,\|\underset{\sim}{y}\| = \sqrt{5} \cdot 5 \approx 11.18$. The inequality is strict, and the vectors are linearly independent: $(3, 4)$ is not a multiple of $(1, 2)$, since $3 \cdot 2 \neq 4$.

- For $\underset{\sim}{x} = (1, 2)$ and

$$\begin{aligned} \underset{\sim}{y} &= (2, 4) \\ &= 2\,\underset{\sim}{x}: \end{aligned}$$

$$\begin{aligned} \underset{\sim}{x} \cdot \underset{\sim}{y} &= 2 + 8 \\ &= 10 \end{aligned}$$

and

$$\begin{aligned}\|\underset{\sim}{x}\|\,\|\underset{\sim}{y}\| &= \sqrt{5}\,\sqrt{20}\\ &= \sqrt{100}\\ &= 10,\end{aligned}$$

so equality holds.

Theorem 0.4 (Triangle inequality)**.** *For any* $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^p$*,*

$$\|\underset{\sim}{x} + \underset{\sim}{y}\| \le \|\underset{\sim}{x}\| + \|\underset{\sim}{y}\|.$$

Proof

Proof.

$$\begin{aligned}\|\underset{\sim}{x} + \underset{\sim}{y}\|^2 &= \|\underset{\sim}{x}\|^2 + 2\,(\underset{\sim}{x} \cdot \underset{\sim}{y}) + \|\underset{\sim}{y}\|^2 && \text{(Theorem 0.2)}\\ &\le \|\underset{\sim}{x}\|^2 + 2\,|\underset{\sim}{x} \cdot \underset{\sim}{y}| + \|\underset{\sim}{y}\|^2 && \text{(a number is at most its absolute value)}\\ &\le \|\underset{\sim}{x}\|^2 + 2\,\|\underset{\sim}{x}\|\,\|\underset{\sim}{y}\| + \|\underset{\sim}{y}\|^2 && \text{(Theorem 0.3)}\\ &= (\|\underset{\sim}{x}\| + \|\underset{\sim}{y}\|)^2. && \text{(expand the square)}\end{aligned}$$

Both $\|\underset{\sim}{x} + \underset{\sim}{y}\|$ and $\|\underset{\sim}{x}\| + \|\underset{\sim}{y}\|$ are nonnegative (Theorem 0.1), and for nonnegative numbers $a^2 \le b^2$ implies $a \le b$. □

Example 0.4 (A strict case and an equality case)**.**

- For $\underset{\sim}{x} = (3, 0)$ and $\underset{\sim}{y} = (0, 4)$:

$$\begin{aligned}\|\underset{\sim}{x} + \underset{\sim}{y}\| &= \|(3, 4)\|\\ &= 5,\end{aligned}$$

while

$$\begin{aligned}\|\underset{\sim}{x}\| + \|\underset{\sim}{y}\| &= 3 + 4\\ &= 7.\end{aligned}$$

- For $\underset{\sim}{x} = (1, 0)$ and $\underset{\sim}{y} = (2, 0)$, which point the same way:

$$\begin{aligned}\|(3, 0)\| &= 3\\ &= 1 + 2,\end{aligned}$$

so equality holds.

Definition 0.1 (Norm (normed vector space, normed space))**.** A **norm** [56] on a real vector space V is a function $\|\cdot\| : V \to [0, \infty)$ that assigns to each vector $\underset{\sim}{x} \in V$ a nonnegative real number $\|\underset{\sim}{x}\|$, and satisfies the following three axioms for all $\underset{\sim}{x}, \underset{\sim}{y} \in V$ and all scalars $c \in \mathbb{R}$:

1. **Positivity and definiteness**: $\|\underset{\sim}{x}\| \geq 0$, and $\|\underset{\sim}{x}\| = 0$ if and only if $\underset{\sim}{x} = \underset{\sim}{0}$;
2. **Absolute homogeneity**: $\|c\,\underset{\sim}{x}\| = |c|\,\|\underset{\sim}{x}\|$;
3. **Subadditivity (triangle inequality)**: $\|\underset{\sim}{x} + \underset{\sim}{y}\| \leq \|\underset{\sim}{x}\| + \|\underset{\sim}{y}\|$ [57].

A vector space equipped with a norm is called a **normed vector space** (or **normed space**).

Example 0.5 (Euclidean space as a normed vector space)**.** The Euclidean space $\mathbb{R}^p$ equipped with the Euclidean norm (Definition 0.20) is a normed vector space:

- Positivity and absolute homogeneity hold by Theorem 0.1.
- The triangle inequality holds by Theorem 0.4.

Every inner product space $(\mathbb{R}^p, \langle \cdot, \cdot \rangle)$ (Definition 0.6) naturally defines an induced norm $\|\underset{\sim}{x}\| \stackrel{\text{def}}{=} \sqrt{\langle \underset{\sim}{x}, \underset{\sim}{x} \rangle}$, which satisfies all three norm axioms.

Definition 0.2 (Vector p-norm (ℓ_p norm, ℓ_1 norm, taxicab norm, Manhattan norm, ℓ_2 norm, infinity norm))**.** For a vector $\underset{\sim}{x} = (x_1, \ldots, x_p) \in \mathbb{R}^p$ and any real number $k \geq 1$, the **vector p-norm** [58] (or ℓ_p **norm**, traditionally named with parameter p but denoted here with exponent k to avoid clashing with the ambient vector dimension p) is:

$$\|\underset{\sim}{x}\|_k \stackrel{\text{def}}{=} \left(\sum_{i=1}^{p} |x_i|^k \right)^{1/k}.$$

The cases $k = 1$ and $k = 2$ are named the ℓ_1 **norm** (also the **taxicab norm** or **Manhattan norm**) and the ℓ_2 **norm**.

Example 0.6 (Calculating ℓ_1, ℓ_2, and ℓ_∞ norms)**.** For $\underset{\sim}{x} = (3, -4) \in \mathbb{R}^2$:

- ℓ_1 norm:

$$\begin{aligned} \|\underset{\sim}{x}\|_1 &= |3| + |-4| \\ &= 3 + 4 \\ &= 7. \end{aligned}$$

- ℓ_2 norm:

$$\begin{aligned}\|\underset{\sim}{x}\|_2 &= \sqrt{3^2 + (-4)^2} \\ &= \sqrt{9 + 16} \\ &= \sqrt{25} \\ &= 5.\end{aligned}$$

- ℓ_∞ norm:

$$\begin{aligned}\|\underset{\sim}{x}\|_\infty &= \max(|3|, |-4|) \\ &= 4.\end{aligned}$$

Notice that $4 \leq 5 \leq 7$. For any vector $\underset{\sim}{x} \in \mathbb{R}^p$, the norms satisfy the chain of inequalities:

$$\|\underset{\sim}{x}\|_\infty \leq \|\underset{\sim}{x}\|_2 \leq \|\underset{\sim}{x}\|_1 \leq \sqrt{p}\,\|\underset{\sim}{x}\|_2 \leq p\,\|\underset{\sim}{x}\|_\infty.$$

ℹ Remark (Unit balls and regularization in machine learning)

Remark 0.1 (Unit balls and regularization in machine learning). The unit ball in $\mathbb{R}^2$ under the ℓ_p norm is the set $\{\underset{\sim}{x} \in \mathbb{R}^2 : \|\underset{\sim}{x}\|_k \leq 1\}$:

- For $k = 1$, the unit ball is a diamond (cross-polytope) with sharp vertices along the coordinate axes $(\pm 1, 0)$ and $(0, \pm 1)$.
- For $k = 2$, the unit ball is the smooth round Euclidean unit disk.
- For $k = \infty$, the unit ball is an axis-aligned square with vertices $(\pm 1, \pm 1)$.

In regression and machine learning, adding an ℓ_1 penalty (Lasso regularization) to an objective function encourages sparse solutions: the level curves of the loss function tend to make contact with the sharp corners of the ℓ_1 unit ball on the coordinate axes, setting irrelevant feature weights exactly to zero. In contrast, the smooth spherical ℓ_2 penalty (Ridge regularization) shrinks all weights toward zero without setting them exactly to zero.

Definition 0.3 (Infinity norm (ℓ_∞ norm, maximum norm, Chebyshev norm))**.** For a vector $\underset{\sim}{x} = (x_1, \ldots, x_p) \in \mathbb{R}^p$, the **infinity norm** [59] (also the ℓ_∞ **norm**, **maximum norm**, or **Chebyshev norm**) is defined separately from the p-norms (Definition 0.2) by

$$\|\underset{\sim}{x}\|_\infty \stackrel{\text{def}}{=} \max_{1 \leq i \leq p} |x_i|.$$

Theorem 0.5 (The ℓ_1, ℓ_2 and infinity norms, and the norm axioms)**.** *Let* $\underset{\sim}{x} = (x_1, \ldots, x_p) \in \mathbb{R}^p$.

1. $\|\underset{\sim}{x}\|_1 = \sum_{i=1}^p |x_i|$, *and* $\|\underset{\sim}{x}\|_2 = \sqrt{\sum_{i=1}^p x_i^2}$, *which is the Euclidean norm (Definition 0.20).*
2. $\|\underset{\sim}{x}\|_\infty = \lim_{k\to\infty} \|\underset{\sim}{x}\|_k$.
3. *For every real number* $k \geq 1$, *and for* $k = \infty$, $\|\cdot\|_k$ *is a norm on* $\mathbb{R}^p$ *(Definition 0.1).*

Proof

Proof. **Part 1.** Setting $k = 1$ in Definition 0.2 gives $\left(\sum_{i=1}^p |x_i|\right)^{1/1} = \sum_{i=1}^p |x_i|$. Setting $k = 2$ gives $\left(\sum_{i=1}^p |x_i|^2\right)^{1/2}$, and $|x_i|^2 = x_i^2$ for every real number x_i.

Part 2. If $\underset{\sim}{x} = \underset{\sim}{0}$, every $\|\underset{\sim}{x}\|_k$ is 0, and so is $\|\underset{\sim}{x}\|_\infty$. Otherwise, let $M \overset{\text{def}}{=} \max_i |x_i| > 0$, and pick j with $|x_j| = M$. Every term of $\sum_{i=1}^p |x_i|^k$ is at most M^k, and the jth term equals M^k, so

$$M^k \leq \sum_{i=1}^p |x_i|^k \leq p\, M^k.$$

Taking the $1/k$ power, which preserves order on $[0, \infty)$, gives

$$M \leq \|\underset{\sim}{x}\|_k \leq p^{1/k} M.$$

As $k \to \infty$, $p^{1/k} = \exp\left\{\frac{\ln p}{k}\right\} \to 1$, so $\|\underset{\sim}{x}\|_k$ is squeezed between M and a quantity that tends to M. Thus $\|\underset{\sim}{x}\|_k \to M = \|\underset{\sim}{x}\|_\infty$.

Part 3, finite k**.**

- *Positivity and definiteness.* Each term $|x_i|^k$ is at least 0, so $\|\underset{\sim}{x}\|_k \geq 0$. The sum $\sum_{i=1}^p |x_i|^k$ is 0 exactly when every term is 0, that is, when $\underset{\sim}{x} = \underset{\sim}{0}$.
- *Absolute homogeneity.* For a number c,

$$\begin{aligned} \|c\,\underset{\sim}{x}\|_k &= \left(\sum_{i=1}^p |c\,x_i|^k\right)^{1/k} && \text{(definition)} \\ &= \left(|c|^k \sum_{i=1}^p |x_i|^k\right)^{1/k} && (|c\,x_i| = |c|\,|x_i|\text{, and factor out } |c|^k) \\ &= |c|\,\|\underset{\sim}{x}\|_k. && \text{(take the } 1/k \text{ power of each factor)} \end{aligned}$$

- *Triangle inequality.* Let $\underset{\sim}{y} \in \mathbb{R}^p$. If $\underset{\sim}{x} = \underset{\sim}{0}$ or $\underset{\sim}{y} = \underset{\sim}{0}$, both sides of $\|\underset{\sim}{x} + \underset{\sim}{y}\|_k \leq \|\underset{\sim}{x}\|_k + \|\underset{\sim}{y}\|_k$ are equal. Otherwise, let $a \overset{\text{def}}{=} \|\underset{\sim}{x}\|_k > 0$, $b \overset{\text{def}}{=} \|\underset{\sim}{y}\|_k > 0$, $\lambda \overset{\text{def}}{=} \frac{a}{a+b}$, $\underset{\sim}{u} \overset{\text{def}}{=} \underset{\sim}{x}/a$ and $\underset{\sim}{v} \overset{\text{def}}{=} \underset{\sim}{y}/b$. By absolute homogeneity, $\|\underset{\sim}{u}\|_k = \|\underset{\sim}{v}\|_k = 1$, and

$$\frac{\underset{\sim}{x}+\underset{\sim}{y}}{a+b} = \lambda\,\underset{\sim}{u} + (1-\lambda)\,\underset{\sim}{v}.$$

The function $s \mapsto s^k$ is non-decreasing and convex on $[0,\infty)$ (its second derivative $k(k-1)s^{k-2}$ is nonnegative). So for each coordinate i,

$$\begin{aligned} |\lambda u_i + (1-\lambda)v_i|^k &\le (\lambda|u_i| + (1-\lambda)|v_i|)^k && \text{(triangle inequality for numbers; } s^k \text{ is non-decreasing)} \\ &\le \lambda|u_i|^k + (1-\lambda)|v_i|^k. && (s^k \text{ is convex}) \end{aligned}$$

Summing over i, and using $\sum_{i=1}^{p}|u_i|^k = \|\underset{\sim}{u}\|_k^k = 1$ and likewise for $\underset{\sim}{v}$,

$$\|\lambda\,\underset{\sim}{u} + (1-\lambda)\,\underset{\sim}{v}\|_k^k \le \lambda + (1-\lambda) = 1.$$

So $\|\underset{\sim}{x}+\underset{\sim}{y}\|_k/(a+b) \le 1$ by absolute homogeneity, that is, $\|\underset{\sim}{x}+\underset{\sim}{y}\|_k \le a+b$.

Part 3, $k=\infty$. Every $|x_i|$ is at least 0, and the largest of them is 0 exactly when $\underset{\sim}{x} = \underset{\sim}{0}$. For a number c, $\max_i|c\,x_i| = |c|\max_i|x_i|$. For each i,

$$\begin{aligned} |x_i + y_i| &\le |x_i| + |y_i| && \text{(triangle inequality for numbers)} \\ &\le \|\underset{\sim}{x}\|_\infty + \|\underset{\sim}{y}\|_\infty, && \text{(each is at most the maximum)} \end{aligned}$$

and taking the maximum over i on the left gives $\|\underset{\sim}{x}+\underset{\sim}{y}\|_\infty \le \|\underset{\sim}{x}\|_\infty + \|\underset{\sim}{y}\|_\infty$. □

Example 0.7 (The ℓ_k norm approaches the maximum norm as k grows)**.** Take $\underset{\sim}{x} = (3,-4)$ and $\underset{\sim}{y} = (-1,5)$ in $\mathbb{R}^2$. The code computes $\|\underset{\sim}{x}\|_k$ for several values of k, including $k=\infty$ as the maximum, and checks the triangle inequality for $\underset{\sim}{x}$ and $\underset{\sim}{y}$ at each of them.

```r
lk_norm <- function(x, k) {
  if (is.infinite(k)) max(abs(x)) else sum(abs(x)^k)^(1 / k)
}

x <- c(3, -4)
y <- c(-1, 5)
ks <- c(1, 2, 4, 8, 16, 32, 64, Inf)
norms <- sapply(ks, function(k) lk_norm(x, k))
triangle <- sapply(
  ks,
  function(k) lk_norm(x + y, k) <= lk_norm(x, k) + lk_norm(y, k)
)
data.frame(k = ks, norm = norms, triangle_holds = triangle)
#>     k     norm triangle_holds
#> 1   1 7.000000           TRUE
#> 2   2 5.000000           TRUE
#> 3   4 4.284572           TRUE
#> 4   8 4.047992           TRUE
#> 5  16 4.002494           TRUE
#> 6  32 4.000013           TRUE
#> 7  64 4.000000           TRUE
#> 8 Inf 4.000000           TRUE
```

By part 1 of Theorem 0.5, the ℓ_1 norm 7 equals $|3| + |-4| = 7$, and the ℓ_2 norm 5 equals $\sqrt{3^2 + (-4)^2} = 5$. By part 2, the norms approach the maximum norm 4: at $k = 64$ the ℓ_k norm is within 0.00000000063 of it. By part 3, the triangle inequality holds at all 8 of the 8 values of k.

Definition 0.4 (Cosine similarity)**.** The **cosine similarity** [60] between two nonzero vectors $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^p$ is their dot product divided by the product of their lengths:

$$\operatorname{cossim}(\underset{\sim}{x}, \underset{\sim}{y}) \stackrel{\text{def}}{=} \frac{\underset{\sim}{x} \cdot \underset{\sim}{y}}{\|\underset{\sim}{x}\| \, \|\underset{\sim}{y}\|}.$$

Example 0.8 (Cosine similarity of parallel, orthogonal, and arbitrary vectors)**.** For vectors in $\mathbb{R}^2$:

- Parallel vectors $\underset{\sim}{x} = (1, 2)$ and $\underset{\sim}{y} = (2, 4)$:

$$\begin{aligned}\operatorname{cossim}(\underset{\sim}{x}, \underset{\sim}{y}) &= \frac{2+8}{\sqrt{5}\sqrt{20}} \\ &= \frac{10}{10} \\ &= 1.\end{aligned}$$

- Orthogonal vectors $\underset{\sim}{x} = (1, 2)$ and $\underset{\sim}{y} = (-2, 1)$:

$$\begin{aligned}\operatorname{cossim}(\underset{\sim}{x}, \underset{\sim}{y}) &= \frac{-2+2}{\sqrt{5}\sqrt{5}} \\ &= \frac{0}{5} \\ &= 0.\end{aligned}$$

- Arbitrary vectors $\underset{\sim}{x} = (1, 0)$ and $\underset{\sim}{y} = (1, 1)$:

$$\begin{aligned}\operatorname{cossim}(\underset{\sim}{x}, \underset{\sim}{y}) &= \frac{1 \cdot 1 + 0 \cdot 1}{1 \cdot \sqrt{2}} \\ &= \frac{1}{\sqrt{2}}.\end{aligned}$$

When both vectors have unit Euclidean norm ($\|\underset{\sim}{x}\| = 1$ and $\|\underset{\sim}{y}\| = 1$), cosine similarity equals their inner product: $\operatorname{cossim}(\underset{\sim}{x}, \underset{\sim}{y}) = \langle \underset{\sim}{x}, \underset{\sim}{y} \rangle$.

Definition 0.5 (Angle between two vectors)**.** The **angle** between two nonzero vectors, $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^p$, is

$$\theta \stackrel{\text{def}}{=} \arccos(\operatorname{cossim}(\underset{\sim}{x}, \underset{\sim}{y})).$$

In words, the angle is the arccosine of the cosine similarity of the two vectors.

ℹ Remark (Why the angle is well defined)

Remark 0.2 (Why the angle is well defined). The cosine similarity is defined, because both vectors are nonzero (Theorem 0.1). By Theorem 0.3, it lies in the closed interval $[-1, 1]$, which is where the arccosine, arccos, is defined. So the angle θ lies in $[0, \pi]$, with cosine $\cos\theta = \operatorname{cossim}(\underset{\sim}{x}, \underset{\sim}{y})$.

Corollary 0.1 (A right angle means orthogonal)**.** *For nonzero vectors $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^p$ with angle θ (Definition 0.5),*

$$\theta = \frac{\pi}{2} \quad \iff \quad \underset{\sim}{x} \cdot \underset{\sim}{y} = 0,$$

so for nonzero vectors, orthogonal (Definition 0.19) means at a right angle.

Proof

Proof. Since $\|\underset{\sim}{x}\|\,\|\underset{\sim}{y}\| > 0$ (Theorem 0.1), $\operatorname{cossim}(\underset{\sim}{x}, \underset{\sim}{y}) = 0$ exactly when $\underset{\sim}{x} \cdot \underset{\sim}{y} = 0$ (Definition 0.4), so $\cos\theta = 0$ exactly when $\underset{\sim}{x} \cdot \underset{\sim}{y} = 0$ (Remark 0.2). On $[0, \pi]$, $\cos$ vanishes only at $\pi/2$. □

Example 0.9 (Some angles in the plane)**.**

- $(1, 0)$ and $(1, 1)$: $\cos\theta = \frac{1}{1 \cdot \sqrt{2}}$, so $\theta = \pi/4$.
- $(1, 2)$ and $(-2, 1)$:

$$\begin{aligned} \cos\theta &= \frac{-2 + 2}{\sqrt{5}\,\sqrt{5}} \\ &= 0, \end{aligned}$$

  so $\theta = \pi/2$.

- $(1, 0)$ and $(-3, 0)$:

$$\begin{aligned} \cos\theta &= \frac{-3}{1 \cdot 3} \\ &= -1, \end{aligned}$$

  so $\theta = \pi$.

- $(1, 0)$ and $\underset{\sim}{0}$: there is no angle, because the ratio would divide by $\|\underset{\sim}{0}\| = 0$.

Theorem 0.6 (The sign of the inner product)**.** *For nonzero $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^p$ with angle $\theta \in [0, \pi]$ between them,*

$$\underset{\sim}{x} \cdot \underset{\sim}{y} = \|\underset{\sim}{x}\|_2 \|\underset{\sim}{y}\|_2 \cos\theta,$$

so:

- $\underset{\sim}{x} \cdot \underset{\sim}{y} > 0$ *when the vectors point broadly in the same direction ($\theta < \pi/2$).*
- $\underset{\sim}{x} \cdot \underset{\sim}{y} = 0$ *exactly when the vectors are orthogonal ($\theta = \pi/2$).*
- $\underset{\sim}{x} \cdot \underset{\sim}{y} < 0$ *when the vectors point broadly in opposing directions ($\theta > \pi/2$).*

Proof

Proof. By the definition of the angle, $\cos\theta$ is the cosine similarity $\underset{\sim}{x}\cdot\underset{\sim}{y}/(\|\underset{\sim}{x}\|_2\|\underset{\sim}{y}\|_2)$, which gives the displayed equation. Both vectors are nonzero, so their norms are positive, and $\underset{\sim}{x}\cdot\underset{\sim}{y}$ has the sign of $\cos\theta$. On $[0,\pi]$, the cosine is positive for $\theta<\pi/2$, zero at $\pi/2$ and negative for $\theta>\pi/2$. □

Example 0.10 (The sign of the inner product in the plane)**.**

- $(1,0)$ and $(1,1)$ have inner product $1>0$ and angle $\pi/4<\pi/2$.
- $(1,2)$ and $(-2,1)$ have inner product $-2+2=0$ and angle $\pi/2$.
- $(1,0)$ and $(-3,0)$ have inner product $-3<0$ and angle $\pi>\pi/2$.

These are the pairs of Example 0.9.

Remark (How the two angles relate)

Remark 0.3 (How the two angles relate). For nonzero vectors in the plane, $\underset{\sim}{x},\underset{\sim}{y}$, the angle between the vectors is the angle on the unit circle between the points $\underset{\sim}{x}/\|\underset{\sim}{x}\|,\underset{\sim}{y}/\|\underset{\sim}{y}\|$. The cosine similarity of the vectors is the dot product of these points. Rotating the plane puts the first point at $(1,0)$, with no change to the distance between the points along the circle or to their dot product. For the angle on the unit circle, θ, the second point is then $(\cos\theta,\pm\sin\theta)$, so the dot product of the points is $\cos\theta$. Applying arccos to it returns θ, which lies in $[0,\pi]$.

Definition 0.6 (Cosine similarity in an inner product space)**.** In a general real inner product space (Definition 0.6) with inner product $\langle\cdot,\cdot\rangle$ and induced norm $\|\cdot\|$, the **cosine similarity** is

$$\operatorname{cossim}(\underset{\sim}{x},\underset{\sim}{y})\overset{\text{def}}{=}\frac{\langle\underset{\sim}{x},\underset{\sim}{y}\rangle}{\|\underset{\sim}{x}\|\,\|\underset{\sim}{y}\|}.$$

Corollary 0.2 (Cosine similarity of unit vectors is their inner product)**.** *In a real inner product space (Definition 0.6), for all vectors* $\underset{\sim}{x}$ *and* $\underset{\sim}{y}$ *with* $\|\underset{\sim}{x}\|=\|\underset{\sim}{y}\|=1$*,*

$$\operatorname{cossim}(\underset{\sim}{x},\underset{\sim}{y})=\langle\underset{\sim}{x},\underset{\sim}{y}\rangle.$$

Proof

Proof. Both norms in the denominator of Definition 0.6 equal 1, so

$$\operatorname{cossim}(\underset{\sim}{x}, \underset{\sim}{y}) = \frac{\langle \underset{\sim}{x}, \underset{\sim}{y} \rangle}{1 \cdot 1} = \langle \underset{\sim}{x}, \underset{\sim}{y} \rangle.$$

□

Theorem 0.7 (Pearson correlation is the cosine similarity of the centered vectors)**.** *Let $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^n$ be two vectors that are not constant, with means $\bar{x} = \frac{1}{n}\sum_{i=1}^n x_i$ and $\bar{y} = \frac{1}{n}\sum_{i=1}^n y_i$. Write $\underset{\sim}{x}^{\mathrm{c}} = \underset{\sim}{x} - \bar{x}\,\underset{\sim}{1}$ and $\underset{\sim}{y}^{\mathrm{c}} = \underset{\sim}{y} - \bar{y}\,\underset{\sim}{1}$ for the **centered** vectors, whose means are zero. Then the Pearson correlation coefficient r of the n pairs (x_i, y_i) is the cosine similarity (Definition 0.4) of the centered vectors:*

$$r = \operatorname{cossim}(\underset{\sim}{x}^{\mathrm{c}}, \underset{\sim}{y}^{\mathrm{c}}).$$

Proof

Proof. Entry i of $\underset{\sim}{x}^{\mathrm{c}}$ is $x_i - \bar{x}$, and entry i of $\underset{\sim}{y}^{\mathrm{c}}$ is $y_i - \bar{y}$. Neither centered vector is zero, because neither $\underset{\sim}{x}$ nor $\underset{\sim}{y}$ is constant, so the cosine similarity is defined. The three pieces of the cosine similarity are

$$\underset{\sim}{x}^{\mathrm{c}} \cdot \underset{\sim}{y}^{\mathrm{c}} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}),$$
$$\|\underset{\sim}{x}^{\mathrm{c}}\| = \sqrt{\sum_{i=1}^n (x_i - \bar{x})^2},$$
$$\|\underset{\sim}{y}^{\mathrm{c}}\| = \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2},$$

by the definitions of the dot product and the Euclidean norm. Dividing the first by the product of the other two gives the formula for r. □

Example 0.11 (Centering changes the cosine similarity)**.** Take $\underset{\sim}{x} = (1, 2, 3)$ and $\underset{\sim}{y} = (1, 3, 2)$. Their means are $\bar{x} = \bar{y} = 2$, so $\underset{\sim}{x}^{\mathrm{c}} = (-1, 0, 1)$ and $\underset{\sim}{y}^{\mathrm{c}} = (-1, 1, 0)$. Then

$$\underset{\sim}{x}^{\mathrm{c}} \cdot \underset{\sim}{y}^{\mathrm{c}} = 1 + 0 + 0 = 1,$$
$$\|\underset{\sim}{x}^{\mathrm{c}}\| = \|\underset{\sim}{y}^{\mathrm{c}}\| = \sqrt{2},$$

so $r = \operatorname{cossim}(\underset{\sim}{x}^{\mathrm{c}}, \underset{\sim}{y}^{\mathrm{c}}) = 1/2$.
Without centering, $\underset{\sim}{x} \cdot \underset{\sim}{y} = 1 + 6 + 6 = 13$ and $\|\underset{\sim}{x}\| = \|\underset{\sim}{y}\| = \sqrt{14}$, so $\operatorname{cossim}(\underset{\sim}{x}, \underset{\sim}{y}) = 13/14$.

The uncentered value is close to 1 mostly because both vectors have positive entries and the same mean, not because they move together.

Corollary 0.3 (Pearson correlation is the cosine of the angle between the centered vectors)**.** *For the two data vectors $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^n$ of Theorem 0.7, the centered vectors $\underset{\sim}{x}^{\mathrm{c}}$, $\underset{\sim}{y}^{\mathrm{c}}$ are nonzero, so they have an angle θ. The Pearson correlation coefficient r satisfies*

$$r = \cos\theta.$$

Proof

Proof. By Theorem 0.7, $r = \operatorname{cossim}(\underset{\sim}{x}^{\mathrm{c}}, \underset{\sim}{y}^{\mathrm{c}})$. By Definition 0.5, $\theta = \arccos(\operatorname{cossim}(\underset{\sim}{x}^{\mathrm{c}}, \underset{\sim}{y}^{\mathrm{c}}))$, so $\cos\theta = \operatorname{cossim}(\underset{\sim}{x}^{\mathrm{c}}, \underset{\sim}{y}^{\mathrm{c}})$ (Remark 0.2). □

Example 0.12 (A correlation of one half is an angle of sixty degrees)**.** In Example 0.11, the correlation is $r = 1/2$, for the data vectors $\underset{\sim}{x} = (1, 2, 3)$, $y = (1, 3, 2)$. By Corollary 0.3, the angle θ between the centered vectors $\underset{\sim}{x}^{\mathrm{c}} = (-1, 0, 1)$, $\underset{\sim}{y}^{\mathrm{c}} = (-1, 1, 0)$ satisfies

$$\cos\theta = \frac{1}{2},$$

so the angle is $\theta = \pi/3$.

Orthonormal bases and Gram-Schmidt

Source

Like Section , this section is adapted from Zhou [53], used under the MIT License (see the license text in Section). The source leaves the proof that orthonormal vectors are linearly independent to class; it is written out here.

Theorem 0.1 (The dot product of a vector with a linear combination)**.** *For vectors $\underset{\sim}{x}, \underset{\sim}{u}_1, \ldots, \underset{\sim}{u}_k \in \mathbb{R}^p$ and numbers $c_1, \ldots, c_k$, with $k \geq 1$,*

$$\underset{\sim}{x} \cdot \left(\sum_{j=1}^{k} c_j \underset{\sim}{u}_j \right) = \sum_{j=1}^{k} c_j \, (\underset{\sim}{x} \cdot \underset{\sim}{u}_j).$$

Proof

Proof. Write u_{ji} for entry i of $\underset{\sim}{u}_j$. Then

$$\begin{aligned}
\underset{\sim}{x} \cdot \left(\sum_{j=1}^{k} c_j \underset{\sim}{u}_j \right) &= \sum_{i=1}^{p} x_i \sum_{j=1}^{k} c_j u_{ji} && \text{(Definition 0.8, Definition 0.9)} \\
&= \sum_{i=1}^{p} \sum_{j=1}^{k} x_i c_j u_{ji} && \text{(distribute each } x_i \text{ over the inner sum)} \\
&= \sum_{j=1}^{k} \sum_{i=1}^{p} x_i c_j u_{ji} && \text{(swap the order of the finite sums)} \\
&= \sum_{j=1}^{k} \sum_{i=1}^{p} c_j x_i u_{ji} && \text{(commute the factors in each product)} \\
&= \sum_{j=1}^{k} c_j \sum_{i=1}^{p} x_i u_{ji} && \text{(factor } c_j \text{ out of the inner sum)} \\
&= \sum_{j=1}^{k} c_j \, (\underset{\sim}{x} \cdot \underset{\sim}{u}_j). && \text{(Definition 0.8)}
\end{aligned}$$

□

Example 0.1 (Dotting with a combination of three vectors)**.** Let $\underset{\sim}{x} = (1, 2, 3)$ and take the combination $2\,(1, 0, 0) - (0, 1, 0) + 4\,(0, 0, 1) = (2, -1, 4)$. Directly,

$$\begin{aligned}
\underset{\sim}{x} \cdot (2, -1, 4) &= 2 - 2 + 12 \\
&= 12.
\end{aligned}$$

By Theorem 0.1, $2 \cdot 1 - 1 \cdot 2 + 4 \cdot 3 = 12$, the same number.

Theorem 0.2 (Orthonormal vectors are linearly independent)**.** *If $\underset{\sim}{q}_1, \ldots, \underset{\sim}{q}_k$ are orthonormal (Definition 0.23), then they are linearly independent (Definition 0.1).*

Proof

Proof. Suppose $\sum_{j=1}^{k} c_j \underset{\sim}{q}_j = \underset{\sim}{0}$. Take any $i \in \{1, \ldots, k\}$. Then

$$
\begin{aligned}
0 &= \underset{\sim}{q}_i \cdot \underset{\sim}{0} && \text{(every term of the dot product is 0)} \\
&= \underset{\sim}{q}_i \cdot \left(\sum_{j=1}^{k} c_j \underset{\sim}{q}_j \right) && \text{(substitute the supposed equation)} \\
&= \sum_{j=1}^{k} c_j \left(\underset{\sim}{q}_i \cdot \underset{\sim}{q}_j \right) && \text{(Theorem 0.1)} \\
&= c_i. && \left(\underset{\sim}{q}_i \cdot \underset{\sim}{q}_j \text{ is 1 if } j = i \text{ and 0 otherwise} \right)
\end{aligned}
$$

So every $c_i = 0$. □

Example 0.2 (Three orthonormal vectors in $\mathbb{R}^3$)**.** Let

$$
\underset{\sim}{q}_1 = (0, 0, 1), \quad \underset{\sim}{q}_2 = \tfrac{1}{\sqrt{2}} (1, 1, 0), \quad \underset{\sim}{q}_3 = \tfrac{1}{\sqrt{2}} (1, -1, 0).
$$

Each has norm 1:

$$
\begin{aligned}
\left\| \underset{\sim}{q}_2 \right\|^2 &= \tfrac{1}{2} (1 + 1 + 0) \\
&= 1,
\end{aligned}
$$

and likewise for the others. They are mutually orthogonal: $\underset{\sim}{q}_1 \cdot \underset{\sim}{q}_2 = 0$ and $\underset{\sim}{q}_1 \cdot \underset{\sim}{q}_3 = 0$, because $\underset{\sim}{q}_1$ has its only nonzero entry where the others have $\underset{\sim}{0}$, and

$$
\begin{aligned}
\underset{\sim}{q}_2 \cdot \underset{\sim}{q}_3 &= \tfrac{1}{2} (1 - 1 + 0) \\
&= 0.
\end{aligned}
$$

So by Theorem 0.2 they are linearly independent, with no need to solve $c_1 \underset{\sim}{q}_1 + c_2 \underset{\sim}{q}_2 + c_3 \underset{\sim}{q}_3 = \underset{\sim}{0}$.
The converse fails: $(1, 0)$ and $(1, 1)$ are linearly independent, since $c_1(1, 0) + c_2(1, 1) = (c_1 + c_2, c_2)$ is $\underset{\sim}{0}$ only if $c_2 = 0$ and then $c_1 = 0$, but they are not orthonormal, since $(1, 0) \cdot (1, 1) = 1 \neq 0$.

Definition 0.1 (Orthonormal basis)**.** An **orthonormal basis** [61] of a subspace $\mathcal{S}$ of $\mathbb{R}^p$ is a basis of $\mathcal{S}$ (Definition 0.3) whose vectors are orthonormal (Definition 0.23).

Example 0.3 (Orthonormal bases of $\mathbb{R}^3$)**.**

- The indicator vectors $\underset{\sim}{e}_1, \underset{\sim}{e}_2, \underset{\sim}{e}_3$ are orthonormal

  (3) and a basis of $\mathbb{R}^3$ (Example 0.14).

- The vectors $\underset{\sim}{q}_1, \underset{\sim}{q}_2, \underset{\sim}{q}_3$ of Example 0.2 are orthonormal and linearly independent, so they are a basis of their span (Definition 0.3), which therefore has dimension

3 (Definition 0.6) and is all of $\mathbb{R}^3$ (Theorem 0.11, Example 0.14). So they are an orthonormal basis of $\mathbb{R}^3$ too.

Example 0.4 (Bases that are not orthonormal, and orthonormal lists that are not bases)**.**

- $(1,0),(1,1)$ is a basis of $\mathbb{R}^2$ (Example 0.9). It is not an orthonormal basis, since $(1,0)\cdot(1,1)=1$.
- $(1,0,0),(0,1,0)$ is orthonormal but not a basis of $\mathbb{R}^3$: its span is the plane $z=0$ (Example 0.4), which does not contain $(0,0,1)$.

Theorem 0.3 (Coordinates in an orthonormal basis are dot products)**.** *If $\underset{\sim}{q}_1,\ldots,\underset{\sim}{q}_k$ is an orthonormal basis of a subspace $\mathcal{S}$ (Definition 0.1), then every $\underset{\sim}{x}\in\mathcal{S}$ satisfies*

$$\underset{\sim}{x}=\sum_{i=1}^{k}(\underset{\sim}{q}_i\cdot\underset{\sim}{x})\,\underset{\sim}{q}_i.$$

Proof

Proof. The basis spans $\mathcal{S}$, so $\underset{\sim}{x}=\sum_{j=1}^{k}c_j\underset{\sim}{q}_j$ for some numbers c_j. For each i,

$$\begin{aligned}\underset{\sim}{q}_i\cdot\underset{\sim}{x}&=\underset{\sim}{q}_i\cdot\left(\sum_{j=1}^{k}c_j\underset{\sim}{q}_j\right) && \text{(substitute)}\\ &=\sum_{j=1}^{k}c_j\,(\underset{\sim}{q}_i\cdot\underset{\sim}{q}_j) && \text{(Theorem 0.1)}\\ &=c_i, && \text{(Definition 0.23)}\end{aligned}$$

so each coefficient c_i is $\underset{\sim}{q}_i\cdot\underset{\sim}{x}$. □

Example 0.5 (Expanding a vector in an orthonormal basis)**.** With the orthonormal basis $\underset{\sim}{q}_1,\underset{\sim}{q}_2,\underset{\sim}{q}_3$ of $\mathbb{R}^3$ from Example 0.3 and $\underset{\sim}{x}=(1,2,3)$: $\underset{\sim}{q}_1\cdot\underset{\sim}{x}=3$,

$$\begin{aligned}\underset{\sim}{q}_2\cdot\underset{\sim}{x}&=\tfrac{1}{\sqrt{2}}\,(1+2)\\ &=\tfrac{3}{\sqrt{2}}\end{aligned}$$

and

$$\begin{aligned} \underset{\sim}{q}_3 \cdot \underset{\sim}{x} &= \tfrac{1}{\sqrt{2}}(1-2) \\ &= -\tfrac{1}{\sqrt{2}}. \end{aligned}$$

Then

$$\begin{aligned} \sum_{i=1}^{3} (\underset{\sim}{q}_i \cdot \underset{\sim}{x})\, \underset{\sim}{q}_i &= 3\,(0,0,1) + \tfrac{3}{\sqrt{2}} \cdot \tfrac{1}{\sqrt{2}}\,(1,1,0) - \tfrac{1}{\sqrt{2}} \cdot \tfrac{1}{\sqrt{2}}\,(1,-1,0) && \text{(substitute)} \\ &= (0,0,3) + \left(\tfrac{3}{2}, \tfrac{3}{2}, 0\right) + \left(-\tfrac{1}{2}, \tfrac{1}{2}, 0\right) && \text{(multiply out each term)} \\ &= (1,2,3), && \text{(add entrywise)} \end{aligned}$$

which is $\underset{\sim}{x}$, as Theorem 0.3 says.

Definition 0.2 (Gram-Schmidt process)**.** The **Gram-Schmidt process** [62] takes vectors $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_k \in \mathbb{R}^p$ and, for $i = 1, 2, \ldots, k$ in turn:

1. **Orthogonalize:** $\underset{\sim}{q}'_i \stackrel{\text{def}}{=} \underset{\sim}{a}_i - \sum_{j=1}^{i-1} (\underset{\sim}{q}_j \cdot \underset{\sim}{a}_i)\, \underset{\sim}{q}_j$ (for $i = 1$ the sum is empty, so $\underset{\sim}{q}'_1 = \underset{\sim}{a}_1$);
2. **Test:** if $\underset{\sim}{q}'_i = \underset{\sim}{0}$, stop;
3. **Normalize:** $\underset{\sim}{q}_i \stackrel{\text{def}}{=} \underset{\sim}{q}'_i / \|\underset{\sim}{q}'_i\|$.

The output is the list $\underset{\sim}{q}_1, \underset{\sim}{q}_2, \ldots$ produced before the process stops, or all k of them if it never stops.

Example 0.6 (Two steps of Gram-Schmidt in $\mathbb{R}^3$)**.** Let $\underset{\sim}{a}_1 = (1,1,0)$ and $\underset{\sim}{a}_2 = (1,0,1)$.

- **Step 1.** $\underset{\sim}{q}'_1 = (1,1,0)$, with norm $\sqrt{2}$, so $\underset{\sim}{q}_1 = \frac{1}{\sqrt{2}}(1,1,0)$.
- **Step 2.**

  $$\begin{aligned} \underset{\sim}{q}_1 \cdot \underset{\sim}{a}_2 &= \tfrac{1}{\sqrt{2}}(1+0+0) \\ &= \tfrac{1}{\sqrt{2}}, \end{aligned}$$

  so

  $$\begin{aligned} \underset{\sim}{q}'_2 &= (1,0,1) - \tfrac{1}{\sqrt{2}} \cdot \tfrac{1}{\sqrt{2}}\,(1,1,0) && \text{(orthogonalize)} \\ &= (1,0,1) - \left(\tfrac{1}{2}, \tfrac{1}{2}, 0\right) && \text{(multiply out)} \\ &= \left(\tfrac{1}{2}, -\tfrac{1}{2}, 1\right), && \text{(subtract entrywise)} \end{aligned}$$

  with norm $\sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{3}{2}}$, so

$$\begin{aligned} \underset{\sim}{q}_2 &= \sqrt{\tfrac{2}{3}}\left(\tfrac{1}{2}, -\tfrac{1}{2}, 1\right) \\ &= \tfrac{1}{\sqrt{6}}\,(1, -1, 2). \end{aligned}$$

As a check,

$$\begin{aligned} \underset{\sim}{q}_1 \cdot \underset{\sim}{q}_2 &= \tfrac{1}{\sqrt{12}}\,(1 - 1 + 0) \\ &= 0. \end{aligned}$$

Example 0.7 (Gram-Schmidt stops on dependent vectors)**.** Let $\underset{\sim}{a}_1 = (1, 2)$ and $\underset{\sim}{a}_2 = (2, 4)$. Step 1 gives $\underset{\sim}{q}_1 = \frac{1}{\sqrt{5}}\,(1, 2)$. In step 2,

$$\begin{aligned} \underset{\sim}{q}_1 \cdot \underset{\sim}{a}_2 &= \tfrac{1}{\sqrt{5}}\,(2 + 8) \\ &= \tfrac{10}{\sqrt{5}}, \end{aligned}$$

and

$$\begin{aligned} \underset{\sim}{q}_2' &= (2, 4) - \tfrac{10}{\sqrt{5}} \cdot \tfrac{1}{\sqrt{5}}\,(1, 2) \\ &= (2, 4) - 2\,(1, 2) \\ &= \underset{\sim}{0}, \end{aligned}$$

so the process stops: $\underset{\sim}{a}_2$ is a multiple of $\underset{\sim}{a}_1$.

Theorem 0.4 (What Gram-Schmidt produces)**.** *Run the Gram-Schmidt process (Definition 0.2) on* $\underset{\sim}{a}_1, \dots, \underset{\sim}{a}_k \in \mathbb{R}^p$.

1. *If it has produced* $\underset{\sim}{q}_1, \dots, \underset{\sim}{q}_i$ *without stopping, then* $\underset{\sim}{q}_1, \dots, \underset{\sim}{q}_i$ *are orthonormal and* $\operatorname{span}\{\underset{\sim}{q}_1, \dots, \underset{\sim}{q}_i\} = \operatorname{span}\{\underset{\sim}{a}_1, \dots, \underset{\sim}{a}_i\}$.
2. *If it reaches step* i *(that is, it did not stop at steps* $1, \dots, i-1$*), it stops there exactly when* $\underset{\sim}{a}_i \in \operatorname{span}\{\underset{\sim}{a}_1, \dots, \underset{\sim}{a}_{i-1}\}$.
3. *It completes all* k *steps exactly when* $\underset{\sim}{a}_1, \dots, \underset{\sim}{a}_k$ *are linearly independent; then* $\underset{\sim}{q}_1, \dots, \underset{\sim}{q}_k$ *is an orthonormal basis of* $\operatorname{span}\{\underset{\sim}{a}_1, \dots, \underset{\sim}{a}_k\}$ *(Definition 0.1).*

Proof

Proof. **Parts 1 and 2, by induction on** i**.** Before step 1, the list of q's is empty, which is orthonormal, and both spans are $\{\underset{\sim}{0}\}$ (Definition 0.2). Suppose part 1 holds after step $i - 1$, and write

$$\begin{aligned}\mathcal{S}_{i-1} &= \operatorname{span}\{\underset{\sim}{a}_1, \dots, \underset{\sim}{a}_{i-1}\} \\ &= \operatorname{span}\{\underset{\sim}{q}_1, \dots, \underset{\sim}{q}_{i-1}\}.\end{aligned}$$

The orthonormal vectors $\underset{\sim}{q}_1, \dots, \underset{\sim}{q}_{i-1}$ are linearly independent (Theorem 0.2) and span $\mathcal{S}_{i-1}$, so they are an orthonormal basis of it.

$\underset{\sim}{q}'_i$ is orthogonal to the earlier $\underset{\sim}{q}$'s. For each $l < i$,

$$\begin{aligned}\underset{\sim}{q}_l \cdot \underset{\sim}{q}'_i &= \underset{\sim}{q}_l \cdot \left(1\,\underset{\sim}{a}_i + \sum_{j=1}^{i-1}\left(-\underset{\sim}{q}_j \cdot \underset{\sim}{a}_i\right)\underset{\sim}{q}_j\right) && \text{(Definition 0.2, as a linear combination)} \\ &= 1\,(\underset{\sim}{q}_l \cdot \underset{\sim}{a}_i) + \sum_{j=1}^{i-1}\left(-\underset{\sim}{q}_j \cdot \underset{\sim}{a}_i\right)(\underset{\sim}{q}_l \cdot \underset{\sim}{q}_j) && \text{(Theorem 0.1)} \\ &= \underset{\sim}{q}_l \cdot \underset{\sim}{a}_i - \sum_{j=1}^{i-1}(\underset{\sim}{q}_j \cdot \underset{\sim}{a}_i)(\underset{\sim}{q}_l \cdot \underset{\sim}{q}_j) && \text{($1\,z = z$, and pull the minus signs out of the sum)} \\ &= \underset{\sim}{q}_l \cdot \underset{\sim}{a}_i - \underset{\sim}{q}_l \cdot \underset{\sim}{a}_i && \text{(only the $j = l$ term survives, Definition 0.23)} \\ &= 0. && \text{(arithmetic)}\end{aligned}$$

Part 2 at step i. If $\underset{\sim}{q}'_i = \underset{\sim}{0}$, then $\underset{\sim}{a}_i = \sum_{j<i}(\underset{\sim}{q}_j \cdot \underset{\sim}{a}_i)\,\underset{\sim}{q}_j \in \mathcal{S}_{i-1}$. Conversely, if $\underset{\sim}{a}_i \in \mathcal{S}_{i-1}$, then $\underset{\sim}{a}_i = \sum_{j<i}(\underset{\sim}{q}_j \cdot \underset{\sim}{a}_i)\,\underset{\sim}{q}_j$ (Theorem 0.3), so $\underset{\sim}{q}'_i = \underset{\sim}{0}$.

Part 1 at step i, if the process does not stop. Then $\underset{\sim}{q}'_i \neq \underset{\sim}{0}$, so $\|\underset{\sim}{q}'_i\| > 0$, and $\underset{\sim}{q}_i = \underset{\sim}{q}'_i / \|\underset{\sim}{q}'_i\|$ has norm $\|\underset{\sim}{q}'_i\|/\|\underset{\sim}{q}'_i\| = 1$ (Theorem 0.1) and for $l < i$

$$\begin{aligned}\underset{\sim}{q}_l \cdot \underset{\sim}{q}_i &= \frac{1}{\|\underset{\sim}{q}'_i\|}\,(\underset{\sim}{q}_l \cdot \underset{\sim}{q}'_i) && \text{(Theorem 0.1, special case $b = 0$)} \\ &= 0, && (\underset{\sim}{q}'_i \text{ is orthogonal to } \underset{\sim}{q}_l)\end{aligned}$$

and $\underset{\sim}{q}_i \cdot \underset{\sim}{q}_l = 0$ too (Theorem 0.1). So $\underset{\sim}{q}_1, \dots, \underset{\sim}{q}_i$ are orthonormal. For the spans:

- $\underset{\sim}{a}_i = \|\underset{\sim}{q}'_i\|\,\underset{\sim}{q}_i + \sum_{j<i}(\underset{\sim}{q}_j \cdot \underset{\sim}{a}_i)\,\underset{\sim}{q}_j$ is in $\operatorname{span}\{\underset{\sim}{q}_1, \dots, \underset{\sim}{q}_i\}$, and so is each of $\underset{\sim}{a}_1, \dots, \underset{\sim}{a}_{i-1}$, which lie in $\mathcal{S}_{i-1} = \operatorname{span}\{\underset{\sim}{q}_1, \dots, \underset{\sim}{q}_{i-1}\} \subseteq \operatorname{span}\{\underset{\sim}{q}_1, \dots, \underset{\sim}{q}_i\}$.
- $\underset{\sim}{q}_i = \left(\underset{\sim}{a}_i - \sum_{j<i}(\underset{\sim}{q}_j \cdot \underset{\sim}{a}_i)\,\underset{\sim}{q}_j\right)/\|\underset{\sim}{q}'_i\|$ is in $\operatorname{span}\{\underset{\sim}{a}_1, \dots, \underset{\sim}{a}_i\}$, because each $\underset{\sim}{q}_j$ with $j < i$ lies in $\mathcal{S}_{i-1}$; and so is each of $\underset{\sim}{q}_1, \dots, \underset{\sim}{q}_{i-1}$.

A span is a subspace (Theorem 0.2), so it contains every linear combination of vectors in it; each span therefore contains the other, and they are equal.

Part 3. If the process completes, then $\underset{\sim}{a}_1 \notin$ span of the empty list, and for each $i \geq 2$, $\underset{\sim}{a}_i \notin \operatorname{span}\{\underset{\sim}{a}_1, \dots, \underset{\sim}{a}_{i-1}\}$ (part 2). Applying Theorem 0.8 k times, starting from the empty list, shows that $\underset{\sim}{a}_1, \dots, \underset{\sim}{a}_k$ are linearly independent. Conversely, if they are linearly independent, no $\underset{\sim}{a}_i$ is a linear combination $\sum_{j<i} c_j \underset{\sim}{a}_j$ of the earlier ones, because then

$\sum_{j<i} c_j \underset{\sim}{a}_j - \underset{\sim}{a}_i = \underset{\sim}{0}$ would be a combination with a nonzero coefficient; so by part 2 the process never stops. When it completes, $\underset{\sim}{q}_1, \dots, \underset{\sim}{q}_k$ are orthonormal, hence linearly independent (Theorem 0.2), and span $\operatorname{span}\{\underset{\sim}{a}_1, \dots, \underset{\sim}{a}_k\}$ (part 1), so they are an orthonormal basis of it. □

Example 0.8 (Finishing an orthonormal basis of $\mathbb{R}^3$)**.** Continue Example 0.6 with $\underset{\sim}{a}_3 = (0,1,1)$. $\underset{\sim}{q}_1 \cdot \underset{\sim}{a}_3 = \frac{1}{\sqrt{2}}$ and

$$\begin{aligned} \underset{\sim}{q}_2 \cdot \underset{\sim}{a}_3 &= \tfrac{1}{\sqrt{6}}(0 - 1 + 2) \\ &= \tfrac{1}{\sqrt{6}}, \end{aligned}$$

so

$$\begin{aligned} \underset{\sim}{q}_3' &= (0,1,1) - \tfrac{1}{\sqrt{2}} \cdot \tfrac{1}{\sqrt{2}}(1,1,0) - \tfrac{1}{\sqrt{6}} \cdot \tfrac{1}{\sqrt{6}}(1,-1,2) && \text{(orthogonalize)} \\ &= (0,1,1) - \tfrac{1}{2}(1,1,0) - \tfrac{1}{6}(1,-1,2) && \text{(multiply the scalars)} \\ &= \left(0 - \tfrac{1}{2} - \tfrac{1}{6},\ 1 - \tfrac{1}{2} + \tfrac{1}{6},\ 1 - 0 - \tfrac{2}{6}\right) && \text{(subtract entrywise)} \\ &= \left(-\tfrac{2}{3}, \tfrac{2}{3}, \tfrac{2}{3}\right), && \text{(arithmetic)} \end{aligned}$$

with norm $\sqrt{3 \cdot \frac{4}{9}} = \frac{2}{\sqrt{3}}$, so $\underset{\sim}{q}_3 = \frac{1}{\sqrt{3}}(-1,1,1)$. The process did not stop, so by Theorem 0.4 $\underset{\sim}{a}_1, \underset{\sim}{a}_2, \underset{\sim}{a}_3$ are linearly independent and $\underset{\sim}{q}_1, \underset{\sim}{q}_2, \underset{\sim}{q}_3$ is an orthonormal basis of their span. That span therefore has dimension 3 (Definition 0.6), so it is $\mathbb{R}^3$ (Theorem 0.11, Example 0.14). As a check,

$$\begin{aligned} \underset{\sim}{q}_1 \cdot \underset{\sim}{q}_3 &= \tfrac{1}{\sqrt{6}}(-1 + 1 + 0) \\ &= 0 \end{aligned}$$

and

$$\begin{aligned} \underset{\sim}{q}_2 \cdot \underset{\sim}{q}_3 &= \tfrac{1}{\sqrt{18}}(-1 - 1 + 2) \\ &= 0. \end{aligned}$$

Corollary 0.1 (Every subspace has an orthonormal basis)**.** *Let $\mathcal{S}$ be a subspace of $\mathbb{R}^p$. Every orthonormal list of vectors in $\mathcal{S}$ extends to an orthonormal basis of $\mathcal{S}$ (Definition 0.1). Starting from the empty list shows that $\mathcal{S}$ has an orthonormal basis.*

ℹ Proof

Proof. Let $\underset{\sim}{u}_1, \dots, \underset{\sim}{u}_r \in \mathcal{S}$ be orthonormal. They are linearly independent (Theorem 0.2), so they extend to a basis $\underset{\sim}{u}_1, \dots, \underset{\sim}{u}_r, \underset{\sim}{a}_{r+1}, \dots, \underset{\sim}{a}_d$ of $\mathcal{S}$ (Theorem 0.9). Run the Gram-Schmidt process on this basis. It is linearly independent, so the process completes, and

the output is an orthonormal basis of its span, which is $\mathcal{S}$ (Theorem 0.4, part 3).
The first r outputs are $\underset{\sim}{u}_1, \ldots, \underset{\sim}{u}_r$ themselves, by induction on $i \leq r$. Suppose the outputs before step i are $\underset{\sim}{u}_1, \ldots, \underset{\sim}{u}_{i-1}$ (for $i = 1$ there are none, and the sum below is empty). Then

$$\begin{aligned} \underset{\sim}{q}'_i &= \underset{\sim}{u}_i - \sum_{j<i} (\underset{\sim}{u}_j \cdot \underset{\sim}{u}_i)\, \underset{\sim}{u}_j \\ &= \underset{\sim}{u}_i, \end{aligned}$$

because each $\underset{\sim}{u}_j \cdot \underset{\sim}{u}_i = 0$; $\underset{\sim}{q}'_i = \underset{\sim}{u}_i \neq \underset{\sim}{0}$, because $\|\underset{\sim}{u}_i\| = 1$, so the process does not stop; and

$$\begin{aligned} \underset{\sim}{q}_i &= \underset{\sim}{u}_i / \|\underset{\sim}{u}_i\| \\ &= \underset{\sim}{u}_i. \end{aligned}$$

So the orthonormal basis contains the starting list. □

Example 0.9 (Extending one unit vector to an orthonormal basis of $\mathbb{R}^3$)**.** Start from the unit vector $\underset{\sim}{u}_1 = \frac{1}{\sqrt{2}}(1,1,0)$. Extend it to a basis of $\mathbb{R}^3$ as in Example 0.16, starting from the empty list: $\underset{\sim}{u}_1$ is not in the span of the empty list, $\{\underset{\sim}{0}\}$, because $\|\underset{\sim}{u}_1\| = 1$; $(1,0,0)$ is not in $\text{span}\{\underset{\sim}{u}_1\}$, because every multiple of $\underset{\sim}{u}_1$ has equal first and second entries, and $(0,0,1)$ is not in $\text{span}\{\underset{\sim}{u}_1, (1,0,0)\}$, because every combination of those two vectors has third entry 0. So $\underset{\sim}{u}_1, (1,0,0), (0,0,1)$ are linearly independent (Theorem 0.8), and three linearly independent vectors in $\mathbb{R}^3$ are a basis of it (they are a basis of their span, which has dimension 3 (Definition 0.6) and so is $\mathbb{R}^3$ (Theorem 0.11, Example 0.14)). Gram-Schmidt on $\underset{\sim}{u}_1, (1,0,0), (0,0,1)$:

1. $\underset{\sim}{q}_1 = \underset{\sim}{u}_1$.
2. $\underset{\sim}{q}_1 \cdot (1,0,0) = \frac{1}{\sqrt{2}}$, so

$$\begin{aligned} \underset{\sim}{q}'_2 &= (1,0,0) - \tfrac{1}{2}(1,1,0) \\ &= (\tfrac{1}{2}, -\tfrac{1}{2}, 0), \end{aligned}$$

   with norm $\frac{1}{\sqrt{2}}$, and $\underset{\sim}{q}_2 = \frac{1}{\sqrt{2}}(1,-1,0)$.
3. $(0,0,1)$ is orthogonal to $\underset{\sim}{q}_1$ and $\underset{\sim}{q}_2$, so $\underset{\sim}{q}'_3 = (0,0,1)$, which already has norm 1, and $\underset{\sim}{q}_3 = (0,0,1)$.

The result is the orthonormal basis of Example 0.3, in a different order.

Outer product

Definition 0.1 (Outer product)**.** The **outer product** [63] of a vector $\underset{\sim}{u}$ of length m and a vector $\underset{\sim}{v}$ of length n is the $m \times n$ matrix product (4) $\underset{\sim}{u}\,\underset{\sim}{v}^{\top}$ of the column vector $\underset{\sim}{u}$ with the row vector $\underset{\sim}{v}^{\top}$. Its entries are

$$\left(\underset{m\times 1}{\underset{\sim}{u}}\ \underset{1\times n}{\underset{\sim}{v}^{\top}}\right)_{ij} = u_i v_j \qquad \text{(definition of matrix multiplication; the sum has one term)}$$

Example 0.1 (An outer product)**.** For $\underset{\sim}{u} = (1, 2, 3)$ and $\underset{\sim}{v} = (4, 5)$:

$$\begin{aligned}
\underset{\sim}{u}\,\underset{\sim}{v}^{\top} &= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 4 & 5 \end{bmatrix} && \text{(definition of the outer product)} \\
&= \begin{bmatrix} 1\cdot 4 & 1\cdot 5 \\ 2\cdot 4 & 2\cdot 5 \\ 3\cdot 4 & 3\cdot 5 \end{bmatrix} && \text{(entry } (i,j) \text{ is } u_i v_j) \\
&= \begin{bmatrix} 4 & 5 \\ 8 & 10 \\ 12 & 15 \end{bmatrix} && \text{(multiply)}
\end{aligned}$$

Remark (Outer product and dot product)

Remark 0.1 (Outer product and dot product). The dot product $\underset{\sim}{u}^{\top}\underset{\sim}{v}$ (Example 0.8) needs two vectors of the same length and gives a number. The outer product $\underset{\sim}{u}\,\underset{\sim}{v}^{\top}$ takes vectors of any two lengths and gives a matrix [48, Ch. 1], p. 11.
For example, $\underset{\sim}{u} = (1, 2, 3)$ and $\underset{\sim}{v} = (4, 5)$ in Example 0.1 have different lengths, so they have no dot product, but their outer product is a 3×2 matrix. For $\underset{\sim}{a} = (1, 2)$ and $\underset{\sim}{b} = (3, 4)$, which have the same length, both products exist:

$$\begin{aligned}
\underset{\sim}{a}^{\top}\underset{\sim}{b} &= 1\cdot 3 + 2\cdot 4 \\
&= 11 \\
\underset{\sim}{a}\,\underset{\sim}{b}^{\top} &= \begin{bmatrix} 1\cdot 3 & 1\cdot 4 \\ 2\cdot 3 & 2\cdot 4 \end{bmatrix} \\
&= \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}
\end{aligned}$$

Theorem 0.1 (An outer product has rank one)**.** *If $\underset{\sim}{u}$ is a nonzero vector of length m and $\underset{\sim}{v}$ is a nonzero vector of length n, then* $\operatorname{rank}(\underset{\sim}{u}\,\underset{\sim}{v}^{\top}) = 1$ *(Definition 0.2).*

Proof

Proof. Column j of $\underset{\sim}{u}\,\underset{\sim}{v}^{\top}$ has entries $u_1 v_j, \ldots, u_m v_j$, so it is the vector $v_j \underset{\sim}{u}$.
The rank is at least one. Because $\underset{\sim}{v} \neq \underset{\sim}{0}$, some entry v_j is nonzero. Then column j, $v_j \underset{\sim}{u}$, is a nonzero vector, and a single nonzero vector is linearly independent (Definition 0.1): $c\,(v_j \underset{\sim}{u}) = \underset{\sim}{0}$ forces $c = 0$.
The rank is at most one. Take any two columns $j \neq k$. If

$$\begin{aligned} v_j &= v_k \\ &= 0, \end{aligned}$$

both columns are $\underset{\sim}{0}$, and $1 \cdot (v_j \underset{\sim}{u}) + 1 \cdot (v_k \underset{\sim}{u}) = \underset{\sim}{0}$. Otherwise, use the coefficients $c_j = v_k$ and $c_k = -v_j$, which are not both zero:

$$\begin{aligned} v_k\,(v_j \underset{\sim}{u}) - v_j\,(v_k \underset{\sim}{u}) &= (v_k v_j - v_j v_k)\,\underset{\sim}{u} && \text{(collect the scalar multiples of } \underset{\sim}{u}) \\ &= 0 \cdot \underset{\sim}{u} && \text{(multiplication of numbers is commutative)} \\ &= \underset{\sim}{0} && \text{(a zero multiple of a vector is } \underset{\sim}{0}) \end{aligned}$$

Either way, every pair of columns has a combination with coefficients that are not all zero and that equals $\underset{\sim}{0}$. Now take any set of two or more columns. It contains a pair $j \neq k$. Use that pair's coefficients for columns j and k, and the coefficient 0 for every other column in the set. This combination equals $\underset{\sim}{0}$, and its coefficients are not all zero, so the set is not linearly independent (Definition 0.1). The largest linearly independent set of columns therefore has one column. □

Example 0.2 (The rank of an outer product)**.** In Example 0.1, the second column $(5, 10, 15)$ of $\underset{\sim}{u}\,\underset{\sim}{v}^{\top}$ is $\frac{5}{4}$ times the first column $(4, 8, 12)$, because the columns are $v_1 \underset{\sim}{u} = 4\underset{\sim}{u}$ and $v_2 \underset{\sim}{u} = 5\underset{\sim}{u}$. So the two columns are not linearly independent, and the 3×2 matrix has rank 1.

Kronecker product

Definition 0.1 (Kronecker product)**.** The **Kronecker product** of an $m \times n$ matrix $\mathbf{A}$ and a $p \times q$ matrix $\mathbf{B}$ is the $mp \times nq$ block matrix whose block in row i and column j is $a_{ij}\mathbf{B}$, where a_{ij} is the (i, j) entry of $\mathbf{A}$:

$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11}\mathbf{B} & \cdots & a_{1n}\mathbf{B} \\ \vdots & \ddots & \vdots \\ a_{m1}\mathbf{B} & \cdots & a_{mn}\mathbf{B} \end{bmatrix}.$$

Example 0.1 (A Kronecker product)**.** For $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 0 & 1 \end{bmatrix}$:

$$\begin{aligned} \mathbf{A} \otimes \mathbf{B} &= \begin{bmatrix} 1 \cdot \mathbf{B} & 2 \cdot \mathbf{B} \\ 3 \cdot \mathbf{B} & 4 \cdot \mathbf{B} \end{bmatrix} && \text{(definition of the Kronecker product)} \\ &= \begin{bmatrix} 0 & 1 & 0 & 2 \\ 0 & 3 & 0 & 4 \end{bmatrix} && \text{(multiply each entry of } \mathbf{A} \text{ by } \mathbf{B}\text{)} \end{aligned}$$

The result is 2×4: $\mathbf{A}$ is 2×2 and $\mathbf{B}$ is 1×2, so $\mathbf{A} \otimes \mathbf{B}$ is $(2 \cdot 1) \times (2 \cdot 2)$.

Hilbert spaces

Definition 0.1 (Cauchy sequence)**.** A sequence $(\underset{\sim}{x}_n)$ of vectors in a real inner product space V is a **Cauchy sequence** [64] if for every $\varepsilon > 0$ there is a natural number N such that

$$\|\underset{\sim}{x}_m - \underset{\sim}{x}_n\| < \varepsilon \quad \text{for all } m, n \geq N,$$

where $\|\underset{\sim}{x}\| = \sqrt{\langle \underset{\sim}{x}, \underset{\sim}{x} \rangle}$ for every $\underset{\sim}{x} \in V$.

Example 0.1 (A Cauchy sequence of real numbers)**.** The real numbers $\mathbb{R}$ with the inner product $\langle x, y \rangle = x\,y$ form an inner product space, and $\|x\| = |x|$ in it. The sequence $x_n = \frac{1}{n}$ is a Cauchy sequence. Given $\varepsilon > 0$, take N to be any natural number larger than $\frac{1}{\varepsilon}$. For all $m, n \geq N$, both $\frac{1}{m}$ and $\frac{1}{n}$ lie in $(0, \frac{1}{N}]$, so $\left|\frac{1}{m} - \frac{1}{n}\right| < \frac{1}{N} < \varepsilon$. For example, with $\varepsilon = 0.01$, take $N = 101$: all $m, n \geq 101$ have $\left|\frac{1}{m} - \frac{1}{n}\right| < \frac{1}{101} < 0.01$.

Definition 0.2 (Complete inner product space)**.** A real inner product space V is **complete** [65] if for every Cauchy sequence $(\underset{\sim}{x}_n)$ in V there is a vector $\underset{\sim}{y} \in V$ such that for

every $\varepsilon > 0$ there is a natural number N with

$$\|\underset{\sim}{x}_n - \underset{\sim}{y}\| < \varepsilon \quad \text{for every } n \geq N.$$

Example 0.2 (Polynomials are not complete)**.** Let V be the set of polynomials on $[0, 1]$, with the inner product $\langle f, g \rangle = \int_0^1 f(t)\, g(t)\, dt$. The partial sums $p_n(t) = \sum_{k=0}^{n} \frac{t^k}{k!}$ are a Cauchy sequence in V, because they converge uniformly on $[0, 1]$ to e^t. The only candidate for a limit is e^t, which is not a polynomial, so no polynomial $\underset{\sim}{y} \in V$ satisfies the condition of Definition 0.2, and V is not complete.

Definition 0.3 (Hilbert space)**.** A **Hilbert space** [66] is a complete real inner product space [48, Sec. 16.8], Definition 16.8, p. 547.

Example 0.3 (Real coordinate space is a Hilbert space)**.** The Euclidean space $\mathbb{R}^2$ with the dot product (Example 0.2) is a Hilbert space. Let $(\underset{\sim}{x}_n)$ with $\underset{\sim}{x}_n = (x_{n,1}, x_{n,2})$ be a Cauchy sequence in it. Each coordinate sequence is a Cauchy sequence of real numbers, because $|x_{m,j} - x_{n,j}| \leq \|\underset{\sim}{x}_m - \underset{\sim}{x}_n\|$ for $j = 1, 2$. Each coordinate sequence then converges to a real number y_j, because every Cauchy sequence of real numbers converges [65], and $y = (y_1, y_2)$ satisfies the condition of Definition 0.2. For example, $\underset{\sim}{x}_n = (\frac{1}{n}, 1 + \frac{1}{n})$ is a Cauchy sequence with limit $\underset{\sim}{y} = (0, 1)$.

Special Matrices

Some special matrices appeared earlier:

- the zero matrix (Definition 0.7), in the section on matrix addition
- square matrices (Definition 0.11), in the section on square and identity matrices
- the identity matrix (Definition 0.15), in that same section

Definition 0.1 (Order of a square matrix)**.** The **order** of a square matrix (Definition 0.11) is its number of rows, which equals its number of columns.

Example 0.1 (Orders of square matrices)**.** $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ has order 2, and $[5]$ has order 1. A 2×3 matrix has no order, since it is not square (Example 0.10).

Definition 0.2 (Matrix power)**.** For a square matrix $\mathbf{A}$ of order p and a positive integer k, the k-th **power** of $\mathbf{A}$ is:

$$\mathbf{A}^k = \underbrace{\mathbf{A}\,\mathbf{A}\cdots\mathbf{A}}_{k \text{ copies}}$$

In particular, $\mathbf{A}^2 = \mathbf{A}\mathbf{A}$.

Example 0.2 (Powers of a 2×2 matrix)**.** Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Then

$$\begin{aligned}
\mathbf{A}^2 &= \begin{bmatrix} 1\cdot 1 + 1\cdot 0 & 1\cdot 1 + 1\cdot 1 \\ 0\cdot 1 + 1\cdot 0 & 0\cdot 1 + 1\cdot 1 \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, && \text{(multiply and add)} \\
\mathbf{A}^3 = \mathbf{A}^2\mathbf{A} &= \begin{bmatrix} 1\cdot 1 + 2\cdot 0 & 1\cdot 1 + 2\cdot 1 \\ 0\cdot 1 + 1\cdot 0 & 0\cdot 1 + 1\cdot 1 \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}. && \text{(multiply and add)}
\end{aligned}$$

A 2×3 matrix $\mathbf{B}$ is not square (Definition 0.11), so $\mathbf{B}^2$ is not defined; indeed $\mathbf{B}\mathbf{B}$ would multiply a matrix with 3 columns by one with 2 rows, which 4 does not allow.

Definition 0.3 (Symmetric matrix)**.** A square matrix $\mathbf{A}$ is **symmetric** [67] if $\mathbf{A}^\top = \mathbf{A}$, i.e., $a_{ij} = a_{ji}$ for all i and j.

Remark (Symmetric matrices in statistics)

Remark 0.1 (Symmetric matrices in statistics)*.* Covariance matrices are symmetric: entry (i, j) of the covariance matrix of a random vector Y is $\mathrm{Cov}(Y_i, Y_j)$, and $\mathrm{Cov}(Y_i, Y_j) = \mathrm{Cov}(Y_j, Y_i)$. For example, if $\mathrm{Var}(Y_1) = 4$, $\mathrm{Var}(Y_2) = 9$, and $\mathrm{Cov}(Y_1, Y_2) = 1$, the covariance matrix of (Y_1, Y_2) is

$$\begin{bmatrix} 4 & 1 \\ 1 & 9 \end{bmatrix}.$$

Example 0.3 (A matrix that is not symmetric)**.** $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is not symmetric:

$$\begin{aligned} a_{12} &= 2 \\ &\neq 3 \\ &= a_{21}. \end{aligned}$$

Definition 0.4 (Diagonal matrix)**.** A square matrix $\mathbf{D}$ is a **diagonal matrix** [68] if all off-diagonal entries are zero: $d_{ij} = 0$ whenever $i \neq j$:

$$\mathbf{D} = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_p \end{bmatrix}$$

Remark (diag notation)

Remark 0.2 (diag notation). For numbers $d_1, \ldots, d_p$, $\text{diag}(d_1, \ldots, d_p)$ is the $p \times p$ diagonal matrix with diagonal entries $d_1, \ldots, d_p$. For example,

$$\text{diag}(2, 5) = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix},$$

and the $p \times p$ identity matrix is $\mathbf{I}_p = \text{diag}(1, \ldots, 1)$.

Example 0.4 (A matrix that is not diagonal)**.** $\mathbf{D} = \begin{bmatrix} 2 & 1 \\ 0 & 5 \end{bmatrix}$ is not diagonal: its off-diagonal entry $d_{12} = 1 \neq 0$.

Definition 0.5 (Matrix inverse)**.** For a square $p \times p$ matrix $\mathbf{A}$, the **inverse** $\mathbf{A}^{-1}$ (if it exists) is the unique matrix satisfying:

$$\begin{aligned} \mathbf{A}\,\mathbf{A}^{-1} &= \mathbf{A}^{-1}\,\mathbf{A} \\ &= \mathbf{I}_p \end{aligned}$$

Definition 0.6 (Invertible matrix, non-singular matrix)**.** A $p \times p$ matrix $\mathbf{A}$ is **invertible** [69] (or **non-singular**) if some $p \times p$ matrix $\mathbf{B}$ satisfies

$$\begin{aligned} \mathbf{AB} &= \mathbf{BA} \\ &= \mathbf{I}_p, \end{aligned}$$

where $\mathbf{I}_p$ is the identity matrix (Definition 0.15). A square matrix that is not invertible

is **singular**.

Example 0.5 (An invertible matrix and a singular one)**.** The matrix $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ is invertible, with $\mathbf{A}^{-1} = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 1 \end{bmatrix}$: multiplying out gives

$$\begin{aligned} \mathbf{A}\mathbf{A}^{-1} &= \mathbf{A}^{-1}\mathbf{A} \\ &= \mathbf{I}_2. \end{aligned}$$

The matrix $\mathbf{M} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ is singular: for any 2×2 matrix $\mathbf{C}$, the two rows of $\mathbf{MC}$ are equal, so $\mathbf{MC}$ can never be $\mathbf{I}_2$, whose two rows differ.

Remark (Invertible matrices and inverses)

Remark 0.3 (Invertible matrices and inverses). Let $\mathbf{A}$ be a $p \times p$ matrix, and suppose that two $p \times p$ matrices $\mathbf{B}$ and $\mathbf{C}$ satisfy

$$\begin{aligned} \mathbf{A}\mathbf{B} &= \mathbf{B}\mathbf{A} \\ &= \mathbf{I}_p \end{aligned}$$

and

$$\begin{aligned} \mathbf{A}\mathbf{C} &= \mathbf{C}\mathbf{A} \\ &= \mathbf{I}_p. \end{aligned}$$

Then

$$\begin{aligned} \mathbf{B} &= \mathbf{B}\,\mathbf{I}_p && \text{(identity matrix)} \\ &= \mathbf{B}(\mathbf{A}\mathbf{C}) && (\mathbf{A}\mathbf{C} = \mathbf{I}_p) \\ &= (\mathbf{B}\mathbf{A})\mathbf{C} && \text{(matrix multiplication is associative)} \\ &= \mathbf{I}_p\,\mathbf{C} && (\mathbf{B}\mathbf{A} = \mathbf{I}_p) \\ &= \mathbf{C} && \text{(identity matrix)} \end{aligned}$$

The steps use Theorem 0.7 and Theorem 0.10. Since $\mathbf{B} = \mathbf{C}$, at most one matrix satisfies these equations, which is the uniqueness that 5 asserts. When $\mathbf{A}$ is invertible, the matrix $\mathbf{B}$ in Definition 0.6 is therefore the inverse $\mathbf{A}^{-1}$ of $\mathbf{A}$, and $\mathbf{A}$ is invertible exactly when it has an inverse. For example, for $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ in Example 0.5, $\begin{bmatrix} 0.5 & -0.5 \\ 0 & 1 \end{bmatrix}$ is the only 2×2 matrix $\mathbf{B}$ with

$$\begin{aligned}\mathbf{AB} &= \mathbf{BA} \\ &= \mathbf{I}_2,\end{aligned}$$

so it is $\mathbf{A}^{-1}$.

Theorem 0.1 (Inverse of a product)**.** *If* $\mathbf{A}$ *and* $\mathbf{B}$ *are invertible* $p \times p$ *matrices, then* $\mathbf{AB}$ *is invertible, and*

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$$

Proof

Proof. Multiply $\mathbf{AB}$ by the candidate inverse on the right:

$$\begin{aligned}(\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}) &= \mathbf{A}(\mathbf{B}\mathbf{B}^{-1})\mathbf{A}^{-1} && \text{(matrix multiplication is associative)} \\ &= \mathbf{A}\,\mathbf{I}_p\,\mathbf{A}^{-1} && \text{(definition of } \mathbf{B}^{-1}) \\ &= \mathbf{A}\mathbf{A}^{-1} && \text{(identity matrix)} \\ &= \mathbf{I}_p && \text{(definition of } \mathbf{A}^{-1})\end{aligned}$$

and on the left:

$$\begin{aligned}(\mathbf{B}^{-1}\mathbf{A}^{-1})(\mathbf{AB}) &= \mathbf{B}^{-1}(\mathbf{A}^{-1}\mathbf{A})\mathbf{B} && \text{(matrix multiplication is associative)} \\ &= \mathbf{B}^{-1}\,\mathbf{I}_p\,\mathbf{B} && \text{(definition of } \mathbf{A}^{-1}) \\ &= \mathbf{B}^{-1}\mathbf{B} && \text{(identity matrix)} \\ &= \mathbf{I}_p && \text{(definition of } \mathbf{B}^{-1})\end{aligned}$$

So $\mathbf{B}^{-1}\mathbf{A}^{-1}$ satisfies 5 for $\mathbf{AB}$. The steps use Theorem 0.7 and Theorem 0.10. □

Theorem 0.2 (Inverse of a transpose)**.** *If* $\mathbf{A}$ *is an invertible* $p \times p$ *matrix, then* $\mathbf{A}^\top$ *is invertible, and*

$$(\mathbf{A}^\top)^{-1} = (\mathbf{A}^{-1})^\top$$

Proof

Proof. Multiply $\mathbf{A}^\top$ by the candidate inverse on each side:

$$\begin{aligned}
\mathbf{A}^\top \left(\mathbf{A}^{-1}\right)^\top &= \left(\mathbf{A}^{-1}\mathbf{A}\right)^\top && \text{(transpose of a product)} \\
&= \mathbf{I}_p{}^\top && \text{(definition of } \mathbf{A}^{-1}\text{)} \\
&= \mathbf{I}_p && (\mathbf{I}_p \text{ is symmetric)} \\
\left(\mathbf{A}^{-1}\right)^\top \mathbf{A}^\top &= \left(\mathbf{A}\mathbf{A}^{-1}\right)^\top && \text{(transpose of a product)} \\
&= \mathbf{I}_p{}^\top && \text{(definition of } \mathbf{A}^{-1}\text{)} \\
&= \mathbf{I}_p && (\mathbf{I}_p \text{ is symmetric)}
\end{aligned}$$

So $\left(\mathbf{A}^{-1}\right)^\top$ satisfies 5 for $\mathbf{A}^\top$. The first step of each display is Theorem 0.19. □

Example 0.6 (Inverting a transpose)**.** For $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$, Example 0.5 gives $\mathbf{A}^{-1} = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 1 \end{bmatrix}$, so Theorem 0.2 says

$$\begin{aligned}
\left(\mathbf{A}^\top\right)^{-1} &= \left(\begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}\right)^{-1} \\
&= \left(\mathbf{A}^{-1}\right)^\top \\
&= \begin{bmatrix} 0.5 & 0 \\ -0.5 & 1 \end{bmatrix},
\end{aligned}$$

and multiplying $\begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0.5 & 0 \\ -0.5 & 1 \end{bmatrix}$ out does give $\mathbf{I}_2$.

Corollary 0.1 (The inverse of a symmetric matrix is symmetric)**.** *If* $\mathbf{A}$ *is symmetric and invertible, then* $\mathbf{A}^{-1}$ *is symmetric.*

Proof

Proof.

$$\begin{aligned}
\left(\mathbf{A}^{-1}\right)^\top &= \left(\mathbf{A}^\top\right)^{-1} && \text{(inverse of a transpose)} \\
&= \mathbf{A}^{-1} && (\mathbf{A} \text{ is symmetric)}
\end{aligned}$$

The first step is Theorem 0.2. □

Example 0.7 (Inverting a symmetric matrix)**.** $\mathbf{S} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ is symmetric, and $\mathbf{S}^{-1} =$

$\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$ is symmetric too, as Corollary 0.1 says.

Theorem 0.3 (An invertible factor can be cancelled)**.** *Let* $\mathbf{A}$ *be an invertible* $p \times p$ *matrix (Definition 0.6), and let* $\mathbf{B}$ *and* $\mathbf{C}$ *be* $p \times n$ *matrices. If* $\mathbf{AB} = \mathbf{AC}$*, then* $\mathbf{B} = \mathbf{C}$*.*

Proof

Proof.

$$\begin{aligned} \mathbf{B} &= \mathbf{I}_p \mathbf{B} && \text{(Theorem 0.10)} \\ &= (\mathbf{A}^{-1}\mathbf{A})\,\mathbf{B} && \text{(Definition 0.5)} \\ &= \mathbf{A}^{-1}(\mathbf{AB}) && \text{(Theorem 0.7)} \\ &= \mathbf{A}^{-1}(\mathbf{AC}) && (\mathbf{AB} = \mathbf{AC}) \\ &= (\mathbf{A}^{-1}\mathbf{A})\,\mathbf{C} && \text{(Theorem 0.7)} \\ &= \mathbf{I}_p \mathbf{C} && \text{(Definition 0.5)} \\ &= \mathbf{C}. && \text{(Theorem 0.10)} \end{aligned}$$

□ □

Corollary 0.2 (An invertible factor can be cancelled on the right)**.** *Let* $\mathbf{A}$ *be an invertible* $p \times p$ *matrix, and let* $\mathbf{B}$ *and* $\mathbf{C}$ *be* $n \times p$ *matrices. If* $\mathbf{BA} = \mathbf{CA}$*, then* $\mathbf{B} = \mathbf{C}$*.*
The proof of Theorem 0.3 works with every product reversed: multiply both sides on the right by $\mathbf{A}^{-1}$*.*

Example 0.8 (Solving for a matrix by cancelling an invertible factor)**.** Let $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$, which is invertible with $\mathbf{A}^{-1} = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 1 \end{bmatrix}$ (Example 0.5). If $\mathbf{AB} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ for a 2×1 matrix $\mathbf{B}$, then multiplying both sides on the left by $\mathbf{A}^{-1}$, as in the proof of Theorem 0.3,

$$\begin{aligned} \mathbf{B} &= \mathbf{A}^{-1} \begin{bmatrix} 3 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \end{aligned}$$

Check: $\mathbf{A} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2+1 \\ 0+1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$.

Example 0.9 (A product of two nonzero matrices can be the zero matrix)**.** With

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix},$$

neither matrix is the zero matrix (Definition 0.7), but

$$\begin{aligned} \mathbf{AB} &= \begin{bmatrix} 1 \cdot 0 + 0 \cdot 0 & 1 \cdot 0 + 0 \cdot 1 \\ 0 \cdot 0 + 0 \cdot 0 & 0 \cdot 0 + 0 \cdot 1 \end{bmatrix} \\ &= \mathbf{0}_{2\times 2}. \end{aligned}$$

For numbers, $ab = 0$ forces $a = 0$ or $b = 0$. For matrices it does not.
Because $\mathbf{A0}_{2\times 2} = \mathbf{0}_{2\times 2} = \mathbf{AB}$ while $\mathbf{B} \neq \mathbf{0}_{2\times 2}$, the factor $\mathbf{A}$ cannot be cancelled from $\mathbf{AB} = \mathbf{A0}_{2\times 2}$. This does not contradict Theorem 0.3, because $\mathbf{A}$ is singular: its second row is zero, so $\mathbf{AC}$ has a zero second row for every 2×2 matrix $\mathbf{C}$, and $\mathbf{AC}$ can never be $\mathbf{I}_2$.

Remark (Matrix arithmetic compared with scalar arithmetic)

Remark 0.4 (Matrix arithmetic compared with scalar arithmetic). **Sums and differences.** Matrix addition has every property that addition of numbers has: it is commutative (Theorem 0.3), associative (Theorem 0.4), has the zero matrix as its identity (Theorem 0.5), and gives each matrix a negative (Theorem 0.6). The only new condition is that $\mathbf{A} + \mathbf{B}$ is defined only when $\mathbf{A}$ and $\mathbf{B}$ have the same dimensions.
Products. Matrix multiplication keeps three properties of multiplication of numbers: it is associative (Theorem 0.7), it is distributive over addition (Theorem 0.8), and it has an identity, $\mathbf{I}_p$ (Theorem 0.10). It differs from multiplication of numbers in four ways:

- $\mathbf{AB}$ is defined only when the number of columns of $\mathbf{A}$ equals the number of rows of $\mathbf{B}$ (4).
- It is not commutative: $\mathbf{AB} \neq \mathbf{BA}$ in general (Remark 0.2).
- A nonzero square matrix need not have an inverse (Example 0.5).
- $\mathbf{AB} = \mathbf{0}$ can hold with $\mathbf{A} \neq \mathbf{0}$ and $\mathbf{B} \neq \mathbf{0}$ (Example 0.9), so a factor cannot be cancelled from $\mathbf{AB} = \mathbf{AC}$ unless it is invertible (Theorem 0.3).

Definition 0.7 (Idempotent matrix)**.** A square matrix $\mathbf{A}$ is **idempotent** [70] if

$$\mathbf{A}^2 = \mathbf{A}$$

Example 0.10 (An idempotent matrix)**.** For $\mathbf{P} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$,

$$\begin{aligned} \mathbf{P}^2 &= \begin{bmatrix} 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 0 + 0 \cdot 0 \\ 0 \cdot 1 + 0 \cdot 0 & 0 \cdot 0 + 0 \cdot 0 \end{bmatrix} && \text{(definition of matrix multiplication)} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \\ &= \mathbf{P}, && \text{(multiply and add)} \end{aligned}$$

so $\mathbf{P}$ is idempotent.

Example 0.11 (A matrix that is not idempotent)**.** For $\mathbf{A} = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$,

$$\begin{aligned} \mathbf{A}^2 &= \begin{bmatrix} 2 \cdot 2 + 0 \cdot 0 & 2 \cdot 0 + 0 \cdot 0 \\ 0 \cdot 2 + 0 \cdot 0 & 0 \cdot 0 + 0 \cdot 0 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 0 \\ 0 & 0 \end{bmatrix} \\ &\neq \mathbf{A}, \end{aligned}$$

so $\mathbf{A}$ is not idempotent.

Definition 0.8 (Orthogonal projection matrix)**.** A square matrix $\mathbf{P}$ is an **orthogonal projection matrix** (also called an **orthogonal projector**) if it is both symmetric (Definition 0.3) and idempotent (Definition 0.7):

$$\mathbf{P}^\top = \mathbf{P} \qquad \text{and} \qquad \mathbf{P}^2 = \mathbf{P}$$

Definition 0.9 (Oblique projection)**.** A square matrix is an **oblique projection** if it is idempotent (Definition 0.7) but not symmetric (Definition 0.3).

Example 0.12 (An orthogonal projection and an oblique one)**.** $\mathbf{P} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ is symmetric, and $\mathbf{P}^2 = \mathbf{P}$, so $\mathbf{P}$ is an orthogonal projection matrix; it maps (v_1, v_2) to $(v_1, 0)$.
$\mathbf{Q} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ is idempotent:

$$\begin{aligned}\mathbf{Q}^2 &= \begin{bmatrix} 1\cdot 1 + 1\cdot 0 & 1\cdot 1 + 1\cdot 0 \\ 0\cdot 1 + 0\cdot 0 & 0\cdot 1 + 0\cdot 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \\ &= \mathbf{Q},\end{aligned}$$

but $\mathbf{Q}^\top \neq \mathbf{Q}$, so $\mathbf{Q}$ is an oblique projection, not an orthogonal projection matrix.

Remark (What "projection matrix" means in these notes)

Remark 0.5 (What "projection matrix" means in these notes). Some texts call any idempotent matrix a projection matrix, so that both $\mathbf{P}$ and $\mathbf{Q}$ in Example 0.12 would count as one. Regression texts often say "projection matrix" when they mean an orthogonal one. In these notes, "projection matrix" always means an orthogonal projection matrix in the sense of Definition 0.8, such as $\mathbf{P}$ in Example 0.12, and never an oblique projection (Definition 0.9) such as $\mathbf{Q}$.

Theorem 0.4 (Complement of a projection matrix)**.** *If $\mathbf{P}$ is a $p \times p$ orthogonal projection matrix (Definition 0.8), then $\mathbf{I}_p - \mathbf{P}$ is also an orthogonal projection matrix.*

Proof

Proof. We verify symmetry and idempotency.

Symmetry:

$$\begin{aligned}(\mathbf{I} - \mathbf{P})^\top &= \mathbf{I}^\top - \mathbf{P}^\top \\ &= \mathbf{I} - \mathbf{P}\end{aligned}$$

Idempotency:

$$\begin{aligned}(\mathbf{I} - \mathbf{P})^2 &= (\mathbf{I} - \mathbf{P})(\mathbf{I} - \mathbf{P}) \\ &= \mathbf{I} - \mathbf{P} - \mathbf{P} + \mathbf{P}^2 \\ &= \mathbf{I} - \mathbf{P} - \mathbf{P} + \mathbf{P} \\ &= \mathbf{I} - \mathbf{P}\end{aligned}$$

□

Theorem 0.5 (Projection matrices produce orthogonal decompositions)**.** *If $\mathbf{P}$ is a $p \times p$ orthogonal projection matrix (Definition 0.8) and $\underset{\sim}{v}$ is any vector of length p, then the two components of the decomposition*

$$\underset{\sim}{v} = \underbrace{\mathbf{P}\underset{\sim}{v}}_{\textit{projected}} + \underbrace{(\mathbf{I}_p - \mathbf{P})\underset{\sim}{v}}_{\textit{residual}}$$

are orthogonal:

$$\mathbf{P}\underset{\sim}{v} \perp (\mathbf{I}_p - \mathbf{P})\underset{\sim}{v}$$

Proof

Proof.

$$\begin{aligned}
(\mathbf{P}\underset{\sim}{v})^\top (\mathbf{I} - \mathbf{P})\underset{\sim}{v} &= \underset{\sim}{v}^\top \mathbf{P}^\top (\mathbf{I} - \mathbf{P})\underset{\sim}{v} \\
&= \underset{\sim}{v}^\top \mathbf{P} (\mathbf{I} - \mathbf{P})\underset{\sim}{v} \\
&= \underset{\sim}{v}^\top (\mathbf{P} - \mathbf{P}^2)\underset{\sim}{v} \\
&= \underset{\sim}{v}^\top (\mathbf{P} - \mathbf{P})\underset{\sim}{v} \\
&= \underset{\sim}{v}^\top \mathbf{0}\, \underset{\sim}{v} \\
&= 0
\end{aligned}$$

where the second line uses symmetry ($\mathbf{P}^\top = \mathbf{P}$) and the fourth line uses idempotency ($\mathbf{P}^2 = \mathbf{P}$). □

Theorem 0.6 (The diagonal of an orthogonal projection matrix lies between 0 and 1)**.** *Let* $\mathbf{P}$ *be an* $n \times n$ *orthogonal projection matrix with entries* p_{ij}*. Then*

$$0 \le p_{ii} \le 1 \qquad \textit{for every } i = 1, \dots, n.$$

Proof

Proof. Since $\mathbf{P}$ is idempotent and symmetric, $p_{ij} = p_{ji}$ and $\mathbf{P}^2 = \mathbf{P}$. Compare the (i, i) entries of $\mathbf{P}^2$ and $\mathbf{P}$:

$$\begin{aligned}
p_{ii} &= (\mathbf{P}^2)_{ii} && \text{(idempotent)} \\
&= \sum_{j=1}^{n} p_{ij}\, p_{ji} && \text{(definition of matrix multiplication)} \\
&= \sum_{j=1}^{n} p_{ij}^2 && \text{(symmetric)} \\
&= p_{ii}^2 + \sum_{j \neq i} p_{ij}^2. && \text{(separate the term } j = i\text{)}
\end{aligned}$$

Each p_{ij}^2 is non-negative, so $p_{ii} \ge 0$. The same equation also gives $p_{ii} \ge p_{ii}^2$, so $p_{ii}(1 -$

$p_{ii}) \geq 0$. Since $p_{ii} \geq 0$, this requires $p_{ii} \leq 1$. □

Example 0.13 (Diagonal of a projection onto a line)**.** Let $\underset{\sim}{u} = (1, 1, 0)$. The matrix $\mathbf{P} = \frac{\underset{\sim}{u}\underset{\sim}{u}^\top}{\underset{\sim}{u}^\top \underset{\sim}{u}}$ is the orthogonal projection onto the line through $\underset{\sim}{u}$. To see this, take the unit vector $\underset{\sim}{q} = \underset{\sim}{u}/\sqrt{2}$, for which $\underset{\sim}{q}\underset{\sim}{q}^\top = \mathbf{P}$, and apply projecting with an orthonormal basis:

$$\mathbf{P} = \frac{1}{2}\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Its diagonal is $(\frac{1}{2}, \frac{1}{2}, 0)$. Each entry is between 0 and 1, as Theorem 0.6 says. The entry 0 belongs to a coordinate that $\underset{\sim}{u}$ does not involve. For $\mathbf{P} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ from Example 0.12 the diagonal is $(1, 0)$, so p_{ii} can be 0 and can be 1.

Definition 0.10 (Orthogonal matrix)**.** A $p \times p$ matrix $\mathbf{Q}$ is **orthogonal** [71] if

$$\underset{p\times p}{\underset{\sim}{\mathbf{Q}}^\top}\ \underset{p\times p}{\underset{\sim}{\mathbf{Q}}} = \mathbf{I}_p$$

Example 0.14 (A rotation matrix)**.** The matrix

$$\mathbf{Q} = \begin{bmatrix} 0.6 & -0.8 \\ 0.8 & 0.6 \end{bmatrix}$$

rotates each vector in the plane counterclockwise by the angle θ with $\cos\theta = 0.6$ and $\sin\theta = 0.8$ (about 53 degrees) [48, Ch. 8], Example 8.1, p. 207. It is orthogonal:

$$\begin{aligned} \mathbf{Q}^\top\mathbf{Q} &= \begin{bmatrix} 0.6 & 0.8 \\ -0.8 & 0.6 \end{bmatrix}\begin{bmatrix} 0.6 & -0.8 \\ 0.8 & 0.6 \end{bmatrix} && \text{(definition of the transpose)} \\ &= \begin{bmatrix} 0.36 + 0.64 & -0.48 + 0.48 \\ -0.48 + 0.48 & 0.64 + 0.36 \end{bmatrix} && \text{(definition of matrix multiplication)} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} && \text{(add)} \end{aligned}$$

Example 0.15 (Orthogonal columns are not enough)**.** $\mathbf{Q} = 2\mathbf{I}_2$ has orthogonal columns $(2, 0)$ and $(0, 2)$, but

$$\begin{aligned}\mathbf{Q}^\top\mathbf{Q} &= \begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix}\begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix}\\ &= \begin{bmatrix}4 & 0\\ 0 & 4\end{bmatrix}\\ &\neq \mathbf{I}_2,\end{aligned}$$

so $\mathbf{Q}$ is not an orthogonal matrix: its columns have norm 2, not 1.

Remark (The columns of an orthogonal matrix are orthonormal)

Remark 0.6 (The columns of an orthogonal matrix are orthonormal). Entry (i, j) of $\mathbf{Q}^\top\mathbf{Q}$ is the dot product of column i and column j of $\mathbf{Q}$, so $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_p$ says that the columns of $\mathbf{Q}$ are orthonormal (Definition 0.23). For a square matrix, $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_p$ also implies $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}_p$, so $\mathbf{Q}^{-1} = \mathbf{Q}^\top$ (5) [48, Ch. 8], Theorem 8.1 and Definition 8.1, p. 209.
For example, the columns of $\mathbf{Q}$ in Example 0.14 are $(0.6, 0.8)$ and $(-0.8, 0.6)$. The diagonal entries $0.36 + 0.64 = 1$ of $\mathbf{Q}^\top\mathbf{Q}$ are their squared norms, and the off-diagonal entries $-0.48 + 0.48 = 0$ are their dot product. Multiplying in the other order also gives $\mathbf{I}_2$:

$$\begin{aligned}\mathbf{Q}\mathbf{Q}^\top &= \begin{bmatrix}0.6 & -0.8\\ 0.8 & 0.6\end{bmatrix}\begin{bmatrix}0.6 & 0.8\\ -0.8 & 0.6\end{bmatrix} && \text{(definition of the transpose)}\\ &= \begin{bmatrix}0.36+0.64 & 0.48-0.48\\ 0.48-0.48 & 0.64+0.36\end{bmatrix} && \text{(definition of matrix multiplication)}\\ &= \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} && \text{(add)}\end{aligned}$$

So $\mathbf{Q}^{-1} = \mathbf{Q}^\top$.

Theorem 0.7 (Orthogonal matrices preserve length)**.** *If* $\mathbf{Q}$ *is a* $p \times p$ *orthogonal matrix (Definition 0.10) and* $\underset{\sim}{x}$ *is a vector of length* p*, then*

$$\|\mathbf{Q}\underset{\sim}{x}\| = \|\underset{\sim}{x}\|$$

Proof

Proof.

$$\begin{aligned}
\|\mathbf{Q}\underset{\sim}{x}\|^2 &= (\mathbf{Q}\underset{\sim}{x}) \cdot (\mathbf{Q}\underset{\sim}{x}) && \text{(definition of the Euclidean norm)} \\
&= (\mathbf{Q}\underset{\sim}{x})^\top (\mathbf{Q}\underset{\sim}{x}) && \text{(dot product as a matrix product)} \\
&= \underset{\sim}{x}^\top \mathbf{Q}^\top (\mathbf{Q}\underset{\sim}{x}) && \text{(transpose of a product)} \\
&= \underset{\sim}{x}^\top (\mathbf{Q}^\top \mathbf{Q})\, \underset{\sim}{x} && \text{(regroup; matrix multiplication is associative)} \\
&= \underset{\sim}{x}^\top \mathbf{I}_p\, \underset{\sim}{x} && \text{(definition of an orthogonal matrix)} \\
&= \underset{\sim}{x}^\top \underset{\sim}{x} && \text{(identity matrix)} \\
&= \|\underset{\sim}{x}\|^2 && \text{(definition of the Euclidean norm)}
\end{aligned}$$

Both norms are nonnegative square roots, so equal squares give $\|\mathbf{Q}\underset{\sim}{x}\| = \|\underset{\sim}{x}\|$. The second step is Example 0.8, the third is Theorem 0.19, and the fourth is Theorem 0.7. □

Example 0.16 (Rotating a vector keeps its length)**.** With $\mathbf{Q}$ from Example 0.14 and $\underset{\sim}{x} = (3, 4)$ from Example 0.15:

$$\begin{aligned}
\mathbf{Q}\underset{\sim}{x} &= \begin{bmatrix} 0.6 \cdot 3 - 0.8 \cdot 4 \\ 0.8 \cdot 3 + 0.6 \cdot 4 \end{bmatrix} && \text{(definition of matrix-vector multiplication)} \\
&= \begin{bmatrix} 1.8 - 3.2 \\ 2.4 + 2.4 \end{bmatrix} && \text{(multiply)} \\
&= \begin{bmatrix} -1.4 \\ 4.8 \end{bmatrix} && \text{(add)}
\end{aligned}$$

$$\begin{aligned}
\|\mathbf{Q}\underset{\sim}{x}\| &= \sqrt{(-1.4)^2 + 4.8^2} && \text{(definition of the norm)} \\
&= \sqrt{1.96 + 23.04} && \text{(square)} \\
&= \sqrt{25} && \text{(add)} \\
&= 5 && \text{(take the square root)}
\end{aligned}$$

which is $\|\underset{\sim}{x}\| = 5$.

Quadratic Forms

Definition 0.1 (Quadratic form)**.** A **quadratic form** [72] is a mathematical expression of the structure

$$\underset{\sim}{x}^{\top}\,\mathbf{S}\,\underset{\sim}{x}$$

where $\underset{\sim}{x}$ is a $p \times 1$ vector and $\mathbf{S}$ is a $p \times p$ matrix.

Remark (Quadratic forms in statistics)

Remark 0.1 (Quadratic forms in statistics). A quadratic form extends the scalar expression cx^2 to vectors: with $p = 1$ and $\mathbf{S} = [c]$, $\underset{\sim}{x}^{\top}\,\mathbf{S}\,\underset{\sim}{x} = cx^2$. With $p = 2$,

$$\underset{\sim}{x}^{\top}\begin{bmatrix}1 & 2\\ 0 & 3\end{bmatrix}\underset{\sim}{x} = x_1^2 + 2x_1x_2 + 3x_2^2,$$

which is $1 + 4 + 12 = 17$ at $\underset{\sim}{x} = (1, 2)$.
Quadratic forms occur often in statistics:

- The residual sum of squares in linear regression (see Vector Calculus) is a quadratic form.
- The variance of a linear combination of estimates (see Inference about Gaussian Linear Regression Models) is a quadratic form: $\operatorname{Var}\left(\underset{\sim}{x}\cdot\underset{\sim}{\hat{\beta}}\right) = \underset{\sim}{x}^{\top}\operatorname{Var}\left(\underset{\sim}{\hat{\beta}}\right)\underset{\sim}{x}$.

Theorem 0.1 (A quadratic form is additive in its matrix)**.** *If* $\mathbf{A}$ *and* $\mathbf{B}$ *are* $p \times p$ *matrices,* c *is a number and* $\underset{\sim}{x}$ *is a vector of length* p*, then*

$$\underset{\sim}{x}^{\top}(\mathbf{A}+\mathbf{B})\underset{\sim}{x} = \underset{\sim}{x}^{\top}\mathbf{A}\underset{\sim}{x} + \underset{\sim}{x}^{\top}\mathbf{B}\underset{\sim}{x},$$
$$\underset{\sim}{x}^{\top}(c\,\mathbf{A})\underset{\sim}{x} = c\,\underset{\sim}{x}^{\top}\mathbf{A}\underset{\sim}{x}.$$

So for a fixed $\underset{\sim}{x}$*, the map* $\mathbf{S} \mapsto \underset{\sim}{x}^{\top}\mathbf{S}\underset{\sim}{x}$ *is linear in* $\mathbf{S}$*.*

Proof

Proof. By the distributive law (Theorem 0.8), applied on the left and then on the right,

$$\begin{aligned}\underset{\sim}{x}^{\top}(\mathbf{A}+\mathbf{B})\underset{\sim}{x} &= (\underset{\sim}{x}^{\top}\mathbf{A} + \underset{\sim}{x}^{\top}\mathbf{B})\underset{\sim}{x} && \text{(distribute } \underset{\sim}{x}^{\top} \text{ over the sum)}\\ &= \underset{\sim}{x}^{\top}\mathbf{A}\underset{\sim}{x} + \underset{\sim}{x}^{\top}\mathbf{B}\underset{\sim}{x} && \text{(distribute } \underset{\sim}{x} \text{ over the sum)}\end{aligned}$$

For the scalar, a number can be moved out of any product, so $\underset{\sim}{x}^{\top}(c\,\mathbf{A})\underset{\sim}{x} = c\,\underset{\sim}{x}^{\top}\mathbf{A}\underset{\sim}{x}$. □

Example 0.1 (Combining two quadratic forms into one)**.** Take $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$, $\mathbf{B} = \mathbf{I}$ and $\underset{\sim}{x} = (1, 2)$. Then

$$\begin{aligned} \underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{x} &= 1 + 4 + 12 = 17, \\ \underset{\sim}{x}^\top \mathbf{I} \underset{\sim}{x} &= 1 + 4 = 5, \\ \underset{\sim}{x}^\top (\mathbf{A} + \mathbf{I}) \underset{\sim}{x} &= \underset{\sim}{x}^\top \begin{bmatrix} 2 & 2 \\ 0 & 4 \end{bmatrix} \underset{\sim}{x} \\ &= 2 + 4 + 16 \\ &= 22, \end{aligned}$$

which equals $17 + 5$. In particular, $\underset{\sim}{x}^\top (\mathbf{A} + c\,\mathbf{I}) \underset{\sim}{x} = \underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{x} + c\,\|\underset{\sim}{x}\|^2$ for every number c, because $\underset{\sim}{x}^\top \mathbf{I} \underset{\sim}{x} = \underset{\sim}{x}^\top \underset{\sim}{x} = \|\underset{\sim}{x}\|^2$. This is how two quadratic terms in the same vector combine into a single quadratic form.

Definition 0.2 (Symmetric part of a square matrix)**.** The **symmetric part** of a $p \times p$ matrix $\mathbf{S}$ is

$$\frac{1}{2}(\mathbf{S} + \mathbf{S}^\top)$$

Example 0.2 (The symmetric part of a 2×2 matrix)**.** For $\mathbf{S} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$, the symmetric part is

$$\begin{aligned} \frac{1}{2}\left(\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} \right) &= \frac{1}{2} \begin{bmatrix} 2 & 2 \\ 2 & 6 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}, \end{aligned}$$

which is symmetric.

Theorem 0.2 (A quadratic form depends only on the symmetric part)**.** *If* $\mathbf{S}$ *is a* $p \times p$ *matrix and* $\underset{\sim}{x}$ *is a vector of length* p*, then*

$$\underset{\sim}{x}^\top \mathbf{S} \underset{\sim}{x} = \underset{\sim}{x}^\top \left(\frac{1}{2}(\mathbf{S} + \mathbf{S}^\top) \right) \underset{\sim}{x}.$$

So the value of a quadratic form depends only on the symmetric part (Definition 0.2) of $\mathbf{S}$.

Proof

Proof. The quadratic form $\underset{\sim}{x}^\top \mathbf{S} \underset{\sim}{x}$ is a 1×1 matrix, so it equals its own transpose:

$$\begin{aligned} \underset{\sim}{x}^\top \mathbf{S} \underset{\sim}{x} &= \left(\underset{\sim}{x}^\top \mathbf{S} \underset{\sim}{x}\right)^\top && \text{(a } 1 \times 1 \text{ matrix is symmetric)} \\ &= \underset{\sim}{x}^\top \mathbf{S}^\top \left(\underset{\sim}{x}^\top\right)^\top && \text{(transpose of a product, twice)} \\ &= \underset{\sim}{x}^\top \mathbf{S}^\top \underset{\sim}{x} && \text{(transposing twice changes nothing)} \end{aligned}$$

Averaging the two expressions for $\underset{\sim}{x}^\top \mathbf{S} \underset{\sim}{x}$:

$$\begin{aligned} \underset{\sim}{x}^\top \mathbf{S} \underset{\sim}{x} &= \frac{1}{2}\left(\underset{\sim}{x}^\top \mathbf{S} \underset{\sim}{x} + \underset{\sim}{x}^\top \mathbf{S}^\top \underset{\sim}{x}\right) && \text{(average of two equal numbers)} \\ &= \frac{1}{2}\, \underset{\sim}{x}^\top (\mathbf{S} + \mathbf{S}^\top) \underset{\sim}{x} && \text{(distributive law)} \\ &= \underset{\sim}{x}^\top \left(\frac{1}{2}(\mathbf{S} + \mathbf{S}^\top)\right) \underset{\sim}{x} && \text{(move the scalar } \tfrac{1}{2} \text{ inside)} \end{aligned}$$

The distributive step is Theorem 0.8, and the transpose step is Theorem 0.19. □

Example 0.3 (Replacing a matrix by its symmetric part)**.** With $\mathbf{S} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ from Example 0.2 and $\underset{\sim}{x} = (1, 1)$:

$$\begin{aligned} \underset{\sim}{x}^\top \mathbf{S} \underset{\sim}{x} &= 1 + 2 + 0 + 3 \\ &= 6 \\ \underset{\sim}{x}^\top \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix} \underset{\sim}{x} &= 1 + 1 + 1 + 3 \\ &= 6 \end{aligned}$$

Both quadratic forms equal the sum of their matrix's entries, because every entry of $\underset{\sim}{x}$ is 1.

Trace and Matrix Inner Product

Definition 0.1 (Trace)**.** The **trace** [73] of a $p \times p$ matrix $\mathbf{M}$ is the sum of its diagonal entries:

$$\operatorname{tr}(\mathbf{M}) \overset{\text{def}}{=} \sum_{i=1}^{p} M_{ii}$$

Example 0.1 (The trace of a 2×2 matrix)**.**

$$\begin{aligned}\operatorname{tr}\left(\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}\right) &= 1 + 4\\ &= 5\end{aligned}$$

Theorem 0.1 (The trace of a product does not depend on the order)**.** *For an $m \times n$ matrix $\mathbf{A}$ and an $n \times m$ matrix $\mathbf{B}$:*

$$\operatorname{tr}\left(\underbrace{\mathbf{AB}}_{m\times m}\right) = \operatorname{tr}\left(\underbrace{\mathbf{BA}}_{n\times n}\right)$$

Proof

Proof.

$$\begin{aligned}
\operatorname{tr}(\mathbf{AB}) &= \sum_{i=1}^{m} (\mathbf{AB})_{ii} && \text{(definition of the trace)}\\
&= \sum_{i=1}^{m}\sum_{s=1}^{n} a_{is}\, b_{si} && \text{(definition of matrix multiplication)}\\
&= \sum_{s=1}^{n}\sum_{i=1}^{m} a_{is}\, b_{si} && \text{(swap the order of two finite sums)}\\
&= \sum_{s=1}^{n}\sum_{i=1}^{m} b_{si}\, a_{is} && \text{(multiplication of numbers is commutative)}\\
&= \sum_{s=1}^{n} (\mathbf{BA})_{ss} && \text{(definition of matrix multiplication)}\\
&= \operatorname{tr}(\mathbf{BA}) && \text{(definition of the trace)}
\end{aligned}$$

□

Example 0.2 (Traces of the two products of a 2×3 and a 3×2 matrix)**.** Let

$$\mathbf{A} = \begin{bmatrix}1 & 2 & 0\\ 0 & 1 & 3\end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix}1 & 0\\ 2 & 1\\ 0 & 1\end{bmatrix}$$

The products are

$$\mathbf{AB} = \begin{bmatrix} 5 & 2 \\ 2 & 4 \end{bmatrix} \qquad \mathbf{BA} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & 3 \\ 0 & 1 & 3 \end{bmatrix}$$

The trace needs only the diagonal entries. For $\mathbf{AB}$, which is 2×2:

$$\begin{aligned} \operatorname{tr}(\mathbf{AB}) &= (\mathbf{AB})_{11} + (\mathbf{AB})_{22} && \text{(definition of the trace)} \\ &= (1 \cdot 1 + 2 \cdot 2 + 0 \cdot 0) + (0 \cdot 0 + 1 \cdot 1 + 3 \cdot 1) && \text{(definition of matrix multiplication)} \\ &= (1 + 4 + 0) + (0 + 1 + 3) && \text{(multiply)} \\ &= 5 + 4 && \text{(add within each entry)} \\ &= 9 && \text{(add)} \end{aligned}$$

For $\mathbf{BA}$, which is 3×3:

$$\begin{aligned} \operatorname{tr}(\mathbf{BA}) &= (\mathbf{BA})_{11} + (\mathbf{BA})_{22} + (\mathbf{BA})_{33} && \text{(definition of the trace)} \\ &= (1 \cdot 1 + 0 \cdot 0) + (2 \cdot 2 + 1 \cdot 1) + (0 \cdot 0 + 1 \cdot 3) && \text{(definition of matrix multiplication)} \\ &= (1 + 0) + (4 + 1) + (0 + 3) && \text{(multiply)} \\ &= 1 + 5 + 3 && \text{(add within each entry)} \\ &= 9 && \text{(add)} \end{aligned}$$

The products have different sizes, but their traces agree.

i Remark (Moving a factor of a triple product)

Remark 0.1 (Moving a factor of a triple product). Let $\mathbf{A}$ be $m \times n$, $\mathbf{B}$ be $n \times k$, and $\mathbf{C}$ be $k \times m$, so that $\mathbf{AB}$ is $m \times k$ and $\mathbf{ABC}$ is square. Applying Theorem 0.1 to the $m \times k$ matrix $\mathbf{AB}$ and the $k \times m$ matrix $\mathbf{C}$ moves the last factor of the triple product to the front [48, Ch. 1], Theorem 1.5 and eq. 1.17, p. 19:

$$\begin{aligned} \operatorname{tr}(\mathbf{ABC}) &= \operatorname{tr}((\mathbf{AB})\,\mathbf{C}) && \text{(matrix multiplication is associative)} \\ &= \operatorname{tr}(\mathbf{C}\,(\mathbf{AB})) && \text{(the theorem, applied to the two factors } \mathbf{AB} \text{ and } \mathbf{C}) \\ &= \operatorname{tr}(\mathbf{CAB}) && \text{(matrix multiplication is associative)} \end{aligned}$$

For example, let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \qquad \mathbf{C} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Then

$$\mathbf{AB} = \begin{bmatrix} 1\cdot 1 + 2\cdot 1 & 1\cdot 0 + 2\cdot 1 \\ 0\cdot 1 + 1\cdot 1 & 0\cdot 0 + 1\cdot 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} \qquad \mathbf{CA} = \begin{bmatrix} 0\cdot 1 + 1\cdot 0 & 0\cdot 2 + 1\cdot 1 \\ 1\cdot 1 + 0\cdot 0 & 1\cdot 2 + 0\cdot 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}$$

so

$$(\mathbf{AB})\,\mathbf{C} = \begin{bmatrix} 3\cdot 0 + 2\cdot 1 & 3\cdot 1 + 2\cdot 0 \\ 1\cdot 0 + 1\cdot 1 & 1\cdot 1 + 1\cdot 0 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix} \qquad (\mathbf{CA})\,\mathbf{B} = \begin{bmatrix} 0\cdot 1 + 1\cdot 1 & 0\cdot 0 + 1\cdot 1 \\ 1\cdot 1 + 2\cdot 1 & 1\cdot 0 + 2\cdot 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}$$

and both traces are the same:

$$\begin{aligned} \operatorname{tr}(\mathbf{ABC}) &= 2 + 1 \\ &= 3 \end{aligned}$$

and

$$\begin{aligned} \operatorname{tr}(\mathbf{CAB}) &= 1 + 2 \\ &= 3. \end{aligned}$$

Definition 0.2 (Matrix inner product)**.** The **inner product** of two $n \times p$ matrices $\mathbf{A}$ and $\mathbf{B}$ is

$$\langle \mathbf{A}, \mathbf{B} \rangle \stackrel{\text{def}}{=} \operatorname{tr}\left(\underbrace{\mathbf{A}^\top \mathbf{B}}_{p \times p} \right)$$

Banerjee and Roy [48, Ch. 15], eq. 15.7, p. 491 defines the same inner product on $n \times p$ matrices with the trace.

Theorem 0.2 (The matrix inner product multiplies matching entries)**.** *For two $n \times p$ matrices $\mathbf{A}$ and $\mathbf{B}$:*

$$\langle \mathbf{A}, \mathbf{B} \rangle = \sum_{i=1}^{n} \sum_{j=1}^{p} a_{ij}\, b_{ij}$$

Proof

Proof.

$$
\begin{aligned}
\langle \mathbf{A}, \mathbf{B} \rangle &= \operatorname{tr}(\mathbf{A}^\top \mathbf{B}) && \text{(definition of the inner product)} \\
&= \sum_{j=1}^{p} (\mathbf{A}^\top \mathbf{B})_{jj} && \text{(definition of the trace)} \\
&= \sum_{j=1}^{p} \sum_{i=1}^{n} (\mathbf{A}^\top)_{ji} \, b_{ij} && \text{(definition of matrix multiplication)} \\
&= \sum_{j=1}^{p} \sum_{i=1}^{n} a_{ij} \, b_{ij} && \text{(definition of the transpose)} \\
&= \sum_{i=1}^{n} \sum_{j=1}^{p} a_{ij} \, b_{ij} && \text{(swap the order of two finite sums)}
\end{aligned}
$$

□

Example 0.3 (The inner product of two 2×2 matrices)**.** Let

$$
\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \qquad \mathbf{B} = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}
$$

From Definition 0.2:

$$
\begin{aligned}
\langle \mathbf{A}, \mathbf{B} \rangle &= \operatorname{tr}(\mathbf{A}^\top \mathbf{B}) && \text{(definition of the inner product)} \\
&= \operatorname{tr}\left(\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^\top \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix} \right) && \text{(substitute)} \\
&= \operatorname{tr}\left(\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix} \right) && \text{(definition of the transpose)} \\
&= \operatorname{tr}\left(\begin{bmatrix} -3 & 7 \\ -4 & 10 \end{bmatrix} \right) && \text{(multiply)} \\
&= -3 + 10 && \text{(definition of the trace)} \\
&= 7 && \text{(add)}
\end{aligned}
$$

From Theorem 0.2:

$$
\begin{aligned}
\langle \mathbf{A}, \mathbf{B} \rangle &= 1 \cdot 0 + 2 \cdot 1 + 3 \cdot (-1) + 4 \cdot 2 && \text{(multiply matching entries and add)} \\
&= 0 + 2 - 3 + 8 && \text{(multiply)} \\
&= 7 && \text{(add)}
\end{aligned}
$$

Remark (The matrix inner product as a dot product)

Remark 0.2 (The matrix inner product as a dot product). Theorem 0.2 says that the matrix inner product is the dot product (2) of the two matrices' entries, each listed as one vector of length np, with both matrices' entries listed in the same order.
For example, listing the entries of $\mathbf{A}$ and $\mathbf{B}$ in Example 0.3 row by row gives $(1, 2, 3, 4)$ and $(0, 1, -1, 2)$, and

$$\begin{aligned}(1, 2, 3, 4) \cdot (0, 1, -1, 2) &= 7 \\ &= \langle \mathbf{A}, \mathbf{B} \rangle.\end{aligned}$$

Definition 0.3 (Frobenius norm)**.** The **Frobenius norm** [74] of an $n \times p$ matrix $\mathbf{A}$ is

$$\|\mathbf{A}\|_F \stackrel{\text{def}}{=} \sqrt{\langle \mathbf{A}, \mathbf{A} \rangle}$$

where $\langle \cdot, \cdot \rangle$ is the matrix inner product (Definition 0.2).

Example 0.4 (The Frobenius norm of a 2×2 matrix)**.** For $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$:

$$\begin{aligned}\|\mathbf{A}\|_F &= \sqrt{\langle \mathbf{A}, \mathbf{A} \rangle} && \text{(definition of the Frobenius norm)} \\ &= \sqrt{1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3 + 4 \cdot 4} && \text{(sum of products of matching entries)} \\ &= \sqrt{1 + 4 + 9 + 16} && \text{(multiply)} \\ &= \sqrt{30} && \text{(add)}\end{aligned}$$

The second step is Theorem 0.2.

Remark (The Frobenius norm is the Euclidean norm of the entries)

Remark 0.3 (The Frobenius norm is the Euclidean norm of the entries). By Theorem 0.2, $\|\mathbf{A}\|_F^2 = \sum_{i=1}^{n} \sum_{j=1}^{p} a_{ij}^2$, a sum of squares, so the square root is always defined. $\|\mathbf{A}\|_F$ is the Euclidean norm (Definition 0.20) of the entries of $\mathbf{A}$, listed as one vector of length np [48, Ch. 15], Definition 15.4, p. 492.
For example, listing the entries of $\mathbf{A}$ in Example 0.4 row by row gives the vector $(1, 2, 3, 4)$ of length $2 \cdot 2 = 4$, and

$$\begin{aligned}\|(1, 2, 3, 4)\| &= \sqrt{1 + 4 + 9 + 16} \\ &= \sqrt{30} \\ &= \|\mathbf{A}\|_F.\end{aligned}$$

Theorem 0.3 (The squared Frobenius norm adds up the squared column norms)**.** *If* $\mathbf{A}$ *is an* $n \times p$ *matrix with columns* $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_p$*, then*

$$\|\mathbf{A}\|_F^2 = \sum_{j=1}^{p} \|\underset{\sim}{a}_j\|^2.$$

Proof

Proof. Entry i of $\underset{\sim}{a}_j$ is a_{ij}, so

$$\begin{aligned}
\|\mathbf{A}\|_F^2 &= \sum_{i=1}^{n}\sum_{j=1}^{p} a_{ij}^2 && \text{(Remark 0.3)} \\
&= \sum_{j=1}^{p}\sum_{i=1}^{n} a_{ij}^2 && \text{(swap the order of two finite sums)} \\
&= \sum_{j=1}^{p} \|\underset{\sim}{a}_j\|^2 && \text{(Definition 0.20, applied to each column)}
\end{aligned}$$

□

Example 0.5 (The Frobenius norm of a 2×2 matrix, column by column)**.** The matrix $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ in Example 0.4 has columns $\underset{\sim}{a}_1 = (1, 3)$ and $\underset{\sim}{a}_2 = (2, 4)$, so

$$\begin{aligned}
\|\mathbf{A}\|_F^2 &= \|\underset{\sim}{a}_1\|^2 + \|\underset{\sim}{a}_2\|^2 && \text{(Theorem 0.3)} \\
&= (1^2 + 3^2) + (2^2 + 4^2) && \text{(definition of the norm)} \\
&= 10 + 20 && \text{(square and add)} \\
&= 30 && \text{(add)}
\end{aligned}$$

which matches $\|\mathbf{A}\|_F = \sqrt{30}$ from Example 0.4.

Matrix Decompositions

Definition 0.1 (Eigenvalue)**.** Let $\mathbf{A}$ be a $p \times p$ matrix. A real number λ is an **eigenvalue** of $\mathbf{A}$ if some real vector $\underset{\sim}{v} \neq \underset{\sim}{0}$ of length p satisfies

$$\underset{p\times p}{\mathbf{A}} \; \underset{p\times 1}{\underset{\sim}{v}} = \lambda \; \underset{p\times 1}{\underset{\sim}{v}}$$

Definition 0.2 (Eigenvector)**.** In Definition 0.1, any such $\underset{\sim}{v}$ is an **eigenvector** of $\mathbf{A}$ for λ.

Example 0.1 (Eigenvectors of a 2×2 matrix)**.** Let $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

$(1, 1)$ is an eigenvector for the eigenvalue 3:

$$\begin{aligned} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} &= \begin{bmatrix} 2+1 \\ 1+2 \end{bmatrix} && \text{(definition of matrix-vector multiplication)} \\ &= 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} && \text{(factor out 3)} \end{aligned}$$

$(1, -1)$ is an eigenvector for the eigenvalue 1:

$$\begin{aligned} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} &= \begin{bmatrix} 2-1 \\ 1-2 \end{bmatrix} && \text{(definition of matrix-vector multiplication)} \\ &= 1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} && \text{(factor out 1)} \end{aligned}$$

Remark (Multiples of an eigenvector)

Remark 0.1 (Multiples of an eigenvector). Multiplying by $\mathbf{A}$ rescales an eigenvector by λ and does not change the line it lies on. Any nonzero multiple $c\,\underset{\sim}{v}$ of an eigenvector is also an eigenvector for the same λ:

$$\begin{aligned} \mathbf{A}(c\,\underset{\sim}{v}) &= c\,\mathbf{A}\underset{\sim}{v} && \text{(move the scalar } c \text{ to the front)} \\ &= c\,\lambda\underset{\sim}{v} && (\underset{\sim}{v} \text{ is an eigenvector for } \lambda) \\ &= \lambda\,(c\,\underset{\sim}{v}) && \text{(multiplication of numbers is commutative)} \end{aligned}$$

For example, in Example 0.1, $2\,(1, 1) = (2, 2)$ is also an eigenvector for the eigenvalue 3:

$$\begin{aligned} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} &= \begin{bmatrix} 2 \cdot 2 + 1 \cdot 2 \\ 1 \cdot 2 + 2 \cdot 2 \end{bmatrix} && \text{(definition of matrix-vector multiplication)} \\ &= \begin{bmatrix} 4+2 \\ 2+4 \end{bmatrix} && \text{(multiply)} \\ &= \begin{bmatrix} 6 \\ 6 \end{bmatrix} && \text{(add)} \\ &= 3 \begin{bmatrix} 2 \\ 2 \end{bmatrix} && \text{(factor out 3)} \end{aligned}$$

Example 0.2 (A vector that is not an eigenvector)**.** For the same $\mathbf{A}$, $(1, 0)$ is not an eigenvector:

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix},$$

and no number λ gives $(2, 1) = \lambda (1, 0)$, because the second entries would need $1 = \lambda \cdot 0$.

Remark (Real and complex eigenvalues)

Remark 0.2 (Real and complex eigenvalues). These notes take λ and $\underset{\sim}{v}$ to be real. Banerjee and Roy [48, Ch. 11], Definition 11.1, pp. 312-313 also allows complex eigenvalues and eigenvectors, because some real matrices, such as a rotation by 90 degrees, have no real eigenvalues [48, Ch. 11], Example 11.2, p. 312.

For example, $\mathbf{R} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ rotates each vector in the plane counterclockwise by 90 degrees: $\mathbf{R}(v_1, v_2) = (-v_2, v_1)$. Suppose $\mathbf{R}\underset{\sim}{v} = \lambda \underset{\sim}{v}$ for a real number λ. Matching entries gives $-v_2 = \lambda v_1$ and $v_1 = \lambda v_2$. Substituting the second equation into the first gives $-v_2 = \lambda^2 v_2$, so $(1 + \lambda^2) v_2 = 0$. Since $1 + \lambda^2 > 0$, $v_2 = 0$, and then

$$\begin{aligned} v_1 &= \lambda v_2 \\ &= 0. \end{aligned}$$

So the only solution is $\underset{\sim}{v} = \underset{\sim}{0}$, and $\mathbf{R}$ has no real eigenvalue.

Video

Chapter 14 of the *Essence of linear algebra* series by 3Blue1Brown shows the geometric meaning of eigenvectors and eigenvalues:

- Eigenvectors and eigenvalues shows how certain vectors maintain their direction during a linear transformation and are merely stretched or squished by an eigenvalue factor (Definition 0.1).

Theorem 0.1 (Spectral theorem for symmetric matrices)**.** *If $\mathbf{A}$ is a $p \times p$ symmetric matrix (Definition 0.3) with real entries, then there are a $p \times p$ orthogonal matrix $\mathbf{Q}$ (Definition 0.10) and a $p \times p$ diagonal matrix $\mathbf{\Lambda}$ (Definition 0.4) with real diagonal entries $\lambda_1, \ldots, \lambda_p$ such that*

$$\underset{p \times p}{\mathbf{A}} = \underset{p \times p}{\mathbf{Q}} \; \underset{p \times p}{\mathbf{\Lambda}} \; \underset{p \times p}{\mathbf{Q}^\top}$$

Each λ_i is an eigenvalue of $\mathbf{A}$ (Definition 0.1), and column i of $\mathbf{Q}$ is an eigenvector of

$\mathbf{A}$ *for* λ_i.

Remark (An equivalent form of the spectral theorem)

Remark 0.3 (An equivalent form of the spectral theorem). The proof, by induction on p, is outside the scope of these notes; see Banerjee and Roy [48, Ch. 11], Theorem 11.27, p. 349, which states the result in the equivalent form $\mathbf{Q}^\top \mathbf{A}\mathbf{Q} = \mathbf{\Lambda}$. The two forms are equivalent because

$$\begin{aligned}\mathbf{Q}^\top \mathbf{Q} &= \mathbf{Q}\mathbf{Q}^\top \\ &= \mathbf{I}_p\end{aligned}$$

(Remark 0.6). Multiplying $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$ by $\mathbf{Q}^\top$ on the left and by $\mathbf{Q}$ on the right gives

$$\begin{aligned}\mathbf{Q}^\top \mathbf{A}\mathbf{Q} &= \mathbf{Q}^\top(\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top)\mathbf{Q} && \text{(substitute)} \\ &= (\mathbf{Q}^\top\mathbf{Q})\mathbf{\Lambda}(\mathbf{Q}^\top\mathbf{Q}) && \text{(regroup; matrix multiplication is associative)} \\ &= \mathbf{I}_p\mathbf{\Lambda}\mathbf{I}_p && (\mathbf{Q} \text{ is orthogonal}) \\ &= \mathbf{\Lambda} && \text{(multiplying by } \mathbf{I}_p \text{ changes nothing)}\end{aligned}$$

Multiplying $\mathbf{Q}^\top \mathbf{A}\mathbf{Q} = \mathbf{\Lambda}$ by $\mathbf{Q}$ on the left and by $\mathbf{Q}^\top$ on the right gives back the first form:

$$\begin{aligned}\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top &= \mathbf{Q}(\mathbf{Q}^\top \mathbf{A}\mathbf{Q})\mathbf{Q}^\top && \text{(substitute)} \\ &= (\mathbf{Q}\mathbf{Q}^\top)\mathbf{A}(\mathbf{Q}\mathbf{Q}^\top) && \text{(regroup; matrix multiplication is associative)} \\ &= \mathbf{I}_p\mathbf{A}\mathbf{I}_p && (\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}_p) \\ &= \mathbf{A} && \text{(multiplying by } \mathbf{I}_p \text{ changes nothing)}\end{aligned}$$

For example, let $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and $\mathbf{Q} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, which is symmetric, so $\mathbf{Q}^\top = \mathbf{Q}$. Then

$$
\begin{aligned}
\mathbf{Q}^\top \mathbf{A} \mathbf{Q} &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(substitute)} \\
&= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(move the scalars to the front)} \\
&= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(multiply the scalars)} \\
&= \frac{1}{2} \begin{bmatrix} 1 \cdot 2 + 1 \cdot 1 & 1 \cdot 1 + 1 \cdot 2 \\ 1 \cdot 2 + (-1) \cdot 1 & 1 \cdot 1 + (-1) \cdot 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(definition of matrix multiplication, first two matric} \\
&= \frac{1}{2} \begin{bmatrix} 2+1 & 1+2 \\ 2-1 & 1-2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{2} \begin{bmatrix} 3 & 3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(add)} \\
&= \frac{1}{2} \begin{bmatrix} 3 \cdot 1 + 3 \cdot 1 & 3 \cdot 1 + 3 \cdot (-1) \\ 1 \cdot 1 + (-1) \cdot 1 & 1 \cdot 1 + (-1) \cdot (-1) \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \frac{1}{2} \begin{bmatrix} 3+3 & 3-3 \\ 1-1 & 1+1 \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{2} \begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix} && \text{(add)} \\
&= \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} && \text{(multiply by } \tfrac{1}{2}\text{)}
\end{aligned}
$$

a diagonal matrix with the eigenvalues 3 and 1 of $\mathbf{A}$ from Example 0.1 on its diagonal.

Theorem 0.2 (Eigenvectors of a symmetric matrix for distinct eigenvalues are orthogonal)**.** *Let* $\mathbf{A}$ *be a symmetric* $p \times p$ *matrix (Definition 0.3), and let* $\underset{\sim}{u}$ *and* $\underset{\sim}{v}$ *be eigenvectors of* $\mathbf{A}$ *(Definition 0.2) for eigenvalues* $\lambda_1 \neq \lambda_2$*. Then* $\underset{\sim}{u} \perp \underset{\sim}{v}$ *(Definition 0.19).*

Proof

Proof.

$$\begin{aligned}
\lambda_1 (\underset{\sim}{u} \cdot \underset{\sim}{v}) &= (\lambda_1 \underset{\sim}{u}) \cdot \underset{\sim}{v} && \text{(Definition 0.8)} \\
&= (\mathbf{A}\underset{\sim}{u}) \cdot \underset{\sim}{v} && (\underset{\sim}{u} \text{ is an eigenvector for } \lambda_1) \\
&= (\mathbf{A}\underset{\sim}{u})^\top \underset{\sim}{v} && \text{(Definition 0.8)} \\
&= \underset{\sim}{u}^\top \mathbf{A}^\top \underset{\sim}{v} && \text{(Theorem 0.19)} \\
&= \underset{\sim}{u}^\top \mathbf{A}\, \underset{\sim}{v} && (\mathbf{A} \text{ is symmetric}) \\
&= \underset{\sim}{u}^\top (\lambda_2 \underset{\sim}{v}) && (\underset{\sim}{v} \text{ is an eigenvector for } \lambda_2) \\
&= \lambda_2 (\underset{\sim}{u} \cdot \underset{\sim}{v}). && \text{(Definition 0.8)}
\end{aligned}$$

So $(\lambda_1 - \lambda_2)(\underset{\sim}{u} \cdot \underset{\sim}{v}) = 0$, and $\lambda_1 - \lambda_2 \neq 0$, so $\underset{\sim}{u} \cdot \underset{\sim}{v} = 0$. □

Example 0.3 (Eigenvectors of a symmetric 2×2 matrix)**.** $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ is symmetric. Its eigenvectors $(1, 1)$ for $\lambda_1 = 3$ and $(1, -1)$ for $\lambda_2 = 1$ (Example 0.1) satisfy

$$\begin{aligned}
(1, 1) \cdot (1, -1) &= 1 \cdot 1 + 1 \cdot (-1) \\
&= 0.
\end{aligned}$$

Definition 0.3 (Eigendecomposition, spectral decomposition)**.** Let $\mathbf{A}$ be a $p \times p$ symmetric matrix with real entries. An **eigendecomposition** [75], or **spectral decomposition**, of $\mathbf{A}$ is a factorization

$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$$

with $\mathbf{Q}$ a $p \times p$ orthogonal matrix (Definition 0.10) and $\mathbf{\Lambda}$ a $p \times p$ diagonal matrix (Definition 0.4).

Example 0.4 (An eigendecomposition of a 2×2 symmetric matrix)**.** For $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, Example 0.1 found the eigenvectors $(1, 1)$ for 3 and $(1, -1)$ for 1. They are orthogonal (Definition 0.19):

$$\begin{aligned}
(1, 1) \cdot (1, -1) &= 1 \cdot 1 + 1 \cdot (-1) && \text{(definition of the dot product)} \\
&= 1 - 1 && \text{(multiply)} \\
&= 0 && \text{(add)}
\end{aligned}$$

Each has norm $\sqrt{1^2 + 1^2} = \sqrt{2}$ (Definition 0.20), so dividing each by $\sqrt{2}$ gives two orthonormal eigenvectors (Definition 0.23). Put them in the columns of $\mathbf{Q}$, and the

matching eigenvalues on the diagonal of $\mathbf{\Lambda}$:

$$\mathbf{Q} = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} \qquad \mathbf{\Lambda} = \begin{bmatrix}3 & 0\\ 0 & 1\end{bmatrix}$$

This $\mathbf{Q}$ is symmetric, so $\mathbf{Q}^\top = \mathbf{Q}$. Then

$$\begin{aligned}
\mathbf{Q\Lambda Q}^\top &= \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}3 & 0\\ 0 & 1\end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} && \text{(substitute)}\\
&= \frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}3 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} && \text{(move the scalars to the front)}\\
&= \frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}3 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} && \text{(multiply the scalars)}\\
&= \frac{1}{2}\begin{bmatrix}1\cdot 3+1\cdot 0 & 1\cdot 0+1\cdot 1\\ 1\cdot 3+(-1)\cdot 0 & 1\cdot 0+(-1)\cdot 1\end{bmatrix}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} && \text{(definition of matrix multiplication, first two matrice}\\
&= \frac{1}{2}\begin{bmatrix}3+0 & 0+1\\ 3+0 & 0-1\end{bmatrix}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} && \text{(multiply)}\\
&= \frac{1}{2}\begin{bmatrix}3 & 1\\ 3 & -1\end{bmatrix}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} && \text{(add)}\\
&= \frac{1}{2}\begin{bmatrix}3\cdot 1+1\cdot 1 & 3\cdot 1+1\cdot(-1)\\ 3\cdot 1+(-1)\cdot 1 & 3\cdot 1+(-1)\cdot(-1)\end{bmatrix} && \text{(definition of matrix multiplication)}\\
&= \frac{1}{2}\begin{bmatrix}3+1 & 3-1\\ 3-1 & 3+1\end{bmatrix} && \text{(multiply)}\\
&= \frac{1}{2}\begin{bmatrix}4 & 2\\ 2 & 4\end{bmatrix} && \text{(add)}\\
&= \begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix} && \text{(multiply by } \tfrac{1}{2}\text{)}
\end{aligned}$$

which is $\mathbf{A}$.

Remark (An eigendecomposition is not unique)

Remark 0.4 (An eigendecomposition is not unique). A symmetric matrix can have more than one eigendecomposition. Reordering the eigenvalues on the diagonal of $\mathbf{\Lambda}$, and the columns of $\mathbf{Q}$ with them, gives another one, and so does multiplying a column of $\mathbf{Q}$ by -1.
For example, take $\mathbf{A}$ from Example 0.4. Swapping the two eigenvalues and the two columns gives

$$\mathbf{Q}_1 = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix} \qquad \mathbf{\Lambda}_1 = \begin{bmatrix}1 & 0\\ 0 & 3\end{bmatrix}$$

and multiplying the second column of $\mathbf{Q}$ by -1 gives

$$\begin{aligned}\mathbf{Q}_2 &= \frac{1}{\sqrt{2}}\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix}\\ \mathbf{\Lambda}_2 &= \mathbf{\Lambda}\\ &= \begin{bmatrix}3 & 0\\ 0 & 1\end{bmatrix}\end{aligned}$$

The columns of $\mathbf{Q}_1$ and $\mathbf{Q}_2$ are the columns of $\mathbf{Q}$, reordered or multiplied by -1, so they are still orthonormal, and $\mathbf{Q}_1$ and $\mathbf{Q}_2$ are orthogonal. Both products give back $\mathbf{A}$:

$$\begin{aligned}
\mathbf{Q}_1\mathbf{\Lambda}_1{\mathbf{Q}_1}^\top &= \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 3\end{bmatrix}\left(\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix}\right)^\top && \text{(substitute)}\\
&= \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 3\end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix} && \text{(definition of the transpose)}\\
&= \frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 3\end{bmatrix}\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix} && \text{(move the scalars to the front)}\\
&= \frac{1}{2}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 3\end{bmatrix}\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix} && \text{(multiply the scalars)}\\
&= \frac{1}{2}\begin{bmatrix}1\cdot 1+1\cdot 0 & 1\cdot 0+1\cdot 3\\ -1\cdot 1+1\cdot 0 & -1\cdot 0+1\cdot 3\end{bmatrix}\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix} && \text{(definition of matrix multiplication, first two matrice}\\
&= \frac{1}{2}\begin{bmatrix}1+0 & 0+3\\ -1+0 & 0+3\end{bmatrix}\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix} && \text{(multiply)}\\
&= \frac{1}{2}\begin{bmatrix}1 & 3\\ -1 & 3\end{bmatrix}\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix} && \text{(add)}\\
&= \frac{1}{2}\begin{bmatrix}1\cdot 1+3\cdot 1 & 1\cdot(-1)+3\cdot 1\\ -1\cdot 1+3\cdot 1 & -1\cdot(-1)+3\cdot 1\end{bmatrix} && \text{(definition of matrix multiplication)}\\
&= \frac{1}{2}\begin{bmatrix}1+3 & -1+3\\ -1+3 & 1+3\end{bmatrix} && \text{(multiply)}\\
&= \frac{1}{2}\begin{bmatrix}4 & 2\\ 2 & 4\end{bmatrix} && \text{(add)}\\
&= \begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix} && \text{(multiply by } \tfrac{1}{2}\text{)}
\end{aligned}$$

$$
\begin{aligned}
\mathbf{Q}_2\mathbf{\Lambda}_2{\mathbf{Q}_2}^\top &= \frac{1}{\sqrt{2}}\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix}\begin{bmatrix}3 & 0\\ 0 & 1\end{bmatrix}\left(\frac{1}{\sqrt{2}}\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix}\right)^\top && \text{(substitute)}\\
&= \frac{1}{\sqrt{2}}\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix}\begin{bmatrix}3 & 0\\ 0 & 1\end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix} && \text{(definition of the transpose)}\\
&= \frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}}\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix}\begin{bmatrix}3 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix} && \text{(move the scalars to the front)}\\
&= \frac{1}{2}\begin{bmatrix}1 & -1\\ 1 & 1\end{bmatrix}\begin{bmatrix}3 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix} && \text{(multiply the scalars)}\\
&= \frac{1}{2}\begin{bmatrix}1\cdot 3+(-1)\cdot 0 & 1\cdot 0+(-1)\cdot 1\\ 1\cdot 3+1\cdot 0 & 1\cdot 0+1\cdot 1\end{bmatrix}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix} && \text{(definition of matrix multiplication, first two mat}\\
&= \frac{1}{2}\begin{bmatrix}3+0 & 0-1\\ 3+0 & 0+1\end{bmatrix}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix} && \text{(multiply)}\\
&= \frac{1}{2}\begin{bmatrix}3 & -1\\ 3 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\ -1 & 1\end{bmatrix} && \text{(add)}\\
&= \frac{1}{2}\begin{bmatrix}3\cdot 1+(-1)\cdot(-1) & 3\cdot 1+(-1)\cdot 1\\ 3\cdot 1+1\cdot(-1) & 3\cdot 1+1\cdot 1\end{bmatrix} && \text{(definition of matrix multiplication)}\\
&= \frac{1}{2}\begin{bmatrix}3+1 & 3-1\\ 3-1 & 3+1\end{bmatrix} && \text{(multiply)}\\
&= \frac{1}{2}\begin{bmatrix}4 & 2\\ 2 & 4\end{bmatrix} && \text{(add)}\\
&= \begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix} && \text{(multiply by } \tfrac{1}{2}\text{)}
\end{aligned}
$$

Theorem 0.3 (Singular value decomposition)**.** *Let* $\mathbf{A}$ *be an* $n\times p$ *matrix with real entries and* $\operatorname{rank}(\mathbf{A}) = r$ *(Definition 0.2). Then there are an* $n\times n$ *orthogonal matrix* $\mathbf{U}$*, a* $p\times p$ *orthogonal matrix* $\mathbf{V}$ *(Definition 0.10), and numbers* $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$ *such that*

$$\underset{n\times p}{\mathbf{A}} = \underset{n\times n}{\mathbf{U}}\ \underset{n\times p}{\mathbf{D}}\ \underset{p\times p}{\mathbf{V}^\top}$$

where $\mathbf{D}$ *has entries* $d_{ii} = \sigma_i$ *for* $i = 1, \ldots, r$ *and every other entry* 0*.*

i Remark (The SVD applies to every real matrix)

Remark 0.5 (The SVD applies to every real matrix). The proof is outside the scope of these notes; see Banerjee and Roy [48, Ch. 12], Theorem 12.1, p. 373. Unlike the spectral

theorem (Theorem 0.1), Theorem 0.3 applies to every real matrix, including one that is not square or not symmetric.

For example, the 1×2 matrix $\mathbf{A} = \begin{bmatrix} 1 & 1 \end{bmatrix}$ is not square, so Theorem 0.1 does not apply to it. It has rank 1, and Theorem 0.3 holds with $\mathbf{U} = \begin{bmatrix} 1 \end{bmatrix}$, $\mathbf{D} = \begin{bmatrix} \sqrt{2} & 0 \end{bmatrix}$, and $\mathbf{V}$ the orthogonal matrix $\mathbf{Q}$ from Example 0.4, which is symmetric, so $\mathbf{V}^\top = \mathbf{V}$:

$$
\begin{aligned}
\mathbf{U}\mathbf{D}\mathbf{V}^\top &= \begin{bmatrix} 1 \end{bmatrix} \begin{bmatrix} \sqrt{2} & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(substitute)} \\
&= \begin{bmatrix} \sqrt{2} & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(multiplying by the } 1 \times 1 \text{ identity changes nothing)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(move the scalar to the front)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} \cdot 1 + 0 \cdot 1 & \sqrt{2} \cdot 1 + 0 \cdot (-1) \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} + 0 & \sqrt{2} + 0 \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} & \sqrt{2} \end{bmatrix} && \text{(add)} \\
&= \begin{bmatrix} 1 & 1 \end{bmatrix} && \text{(divide by } \sqrt{2}\text{)}
\end{aligned}
$$

$\mathbf{U}$ is orthogonal because

$$
\begin{aligned}
\mathbf{U}^\top \mathbf{U} &= \begin{bmatrix} 1 \end{bmatrix} \\
&= \mathbf{I}_1,
\end{aligned}
$$

and the single singular value is $\sigma_1 = \sqrt{2}$.

Definition 0.4 (Singular value decomposition)**.** Let $\mathbf{A}$ be an $n \times p$ matrix with real entries and $\operatorname{rank}(\mathbf{A}) = r$. A **singular value decomposition** [76] (SVD) of $\mathbf{A}$ is a factorization $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$ with $\mathbf{U}$, $\mathbf{D}$ and $\mathbf{V}$ as in Theorem 0.3.

Definition 0.5 (Singular values)**.** The numbers $\sigma_1 \geq \cdots \geq \sigma_r > 0$ on the diagonal of $\mathbf{D}$ in a singular value decomposition of $\mathbf{A}$ (Definition 0.4) are the **singular values** of $\mathbf{A}$.

Example 0.5 (An SVD of a 3×2 matrix)**.** Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{bmatrix} \qquad \mathbf{U} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} \qquad \mathbf{D} = \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \qquad \mathbf{V} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$\mathbf{V}$ is the orthogonal matrix $\mathbf{Q}$ from Example 0.4. $\mathbf{U}$ is orthogonal too. Entry (i, j) of $\mathbf{U}^\top\mathbf{U}$ is the dot product of columns i and j of $\mathbf{U}$, and those columns are $\underset{\sim}{u}_1 = (\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})$, $\underset{\sim}{u}_2 = (0, 1, 0)$ and $\underset{\sim}{u}_3 = (\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}})$:

$$\begin{aligned} \underset{\sim}{u}_1 \cdot \underset{\sim}{u}_1 &= \tfrac{1}{\sqrt{2}} \cdot \tfrac{1}{\sqrt{2}} + 0 \cdot 0 + \tfrac{1}{\sqrt{2}} \cdot \tfrac{1}{\sqrt{2}} && \text{(definition of the dot product)} \\ &= \tfrac{1}{2} + 0 + \tfrac{1}{2} && \text{(multiply)} \\ &= 1 && \text{(add)} \end{aligned}$$

$$\begin{aligned} \underset{\sim}{u}_2 \cdot \underset{\sim}{u}_2 &= 0 \cdot 0 + 1 \cdot 1 + 0 \cdot 0 && \text{(definition of the dot product)} \\ &= 0 + 1 + 0 && \text{(multiply)} \\ &= 1 && \text{(add)} \end{aligned}$$

$$\begin{aligned} \underset{\sim}{u}_3 \cdot \underset{\sim}{u}_3 &= \tfrac{1}{\sqrt{2}} \cdot \tfrac{1}{\sqrt{2}} + 0 \cdot 0 + \left(-\tfrac{1}{\sqrt{2}}\right) \cdot \left(-\tfrac{1}{\sqrt{2}}\right) && \text{(definition of the dot product)} \\ &= \tfrac{1}{2} + 0 + \tfrac{1}{2} && \text{(multiply)} \\ &= 1 && \text{(add)} \end{aligned}$$

$$\begin{aligned} \underset{\sim}{u}_1 \cdot \underset{\sim}{u}_2 &= \tfrac{1}{\sqrt{2}} \cdot 0 + 0 \cdot 1 + \tfrac{1}{\sqrt{2}} \cdot 0 && \text{(definition of the dot product)} \\ &= 0 + 0 + 0 && \text{(multiply)} \\ &= 0 && \text{(add)} \end{aligned}$$

$$\begin{aligned} \underset{\sim}{u}_1 \cdot \underset{\sim}{u}_3 &= \tfrac{1}{\sqrt{2}} \cdot \tfrac{1}{\sqrt{2}} + 0 \cdot 0 + \tfrac{1}{\sqrt{2}} \cdot \left(-\tfrac{1}{\sqrt{2}}\right) && \text{(definition of the dot product)} \\ &= \tfrac{1}{2} + 0 - \tfrac{1}{2} && \text{(multiply)} \\ &= 0 && \text{(add)} \end{aligned}$$

$$\begin{aligned} \underset{\sim}{u}_2 \cdot \underset{\sim}{u}_3 &= 0 \cdot \tfrac{1}{\sqrt{2}} + 1 \cdot 0 + 0 \cdot \left(-\tfrac{1}{\sqrt{2}}\right) && \text{(definition of the dot product)} \\ &= 0 + 0 + 0 && \text{(multiply)} \\ &= 0 && \text{(add)} \end{aligned}$$

The dot product is symmetric, so these six values fill in all nine entries, and $\mathbf{U}^\top\mathbf{U} = \mathbf{I}_3$. $\mathbf{V}$ is symmetric, so $\mathbf{V}^\top = \mathbf{V}$. Then

$$
\begin{aligned}
\mathbf{DV}^\top &= \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(substitute)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(move the scalar to the front)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} 2 \cdot 1 + 0 \cdot 1 & 2 \cdot 1 + 0 \cdot (-1) \\ 0 \cdot 1 + \sqrt{2} \cdot 1 & 0 \cdot 1 + \sqrt{2} \cdot (-1) \\ 0 \cdot 1 + 0 \cdot 1 & 0 \cdot 1 + 0 \cdot (-1) \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} 2 + 0 & 2 + 0 \\ 0 + \sqrt{2} & 0 - \sqrt{2} \\ 0 + 0 & 0 + 0 \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} 2 & 2 \\ \sqrt{2} & -\sqrt{2} \\ 0 & 0 \end{bmatrix} && \text{(add)} \\
&= \begin{bmatrix} \sqrt{2} & \sqrt{2} \\ 1 & -1 \\ 0 & 0 \end{bmatrix} && \text{(divide by } \sqrt{2}\text{)}
\end{aligned}
$$

and

$$
\begin{aligned}
\mathbf{UDV}^\top &= \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \sqrt{2} & \sqrt{2} \\ 1 & -1 \\ 0 & 0 \end{bmatrix} && \text{(substitute)} \\
&= \begin{bmatrix} \frac{1}{\sqrt{2}} \cdot \sqrt{2} + 0 \cdot 1 + \frac{1}{\sqrt{2}} \cdot 0 & \frac{1}{\sqrt{2}} \cdot \sqrt{2} + 0 \cdot (-1) + \frac{1}{\sqrt{2}} \cdot 0 \\ 0 \cdot \sqrt{2} + 1 \cdot 1 + 0 \cdot 0 & 0 \cdot \sqrt{2} + 1 \cdot (-1) + 0 \cdot 0 \\ \frac{1}{\sqrt{2}} \cdot \sqrt{2} + 0 \cdot 1 - \frac{1}{\sqrt{2}} \cdot 0 & \frac{1}{\sqrt{2}} \cdot \sqrt{2} + 0 \cdot (-1) - \frac{1}{\sqrt{2}} \cdot 0 \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \begin{bmatrix} 1 + 0 + 0 & 1 + 0 + 0 \\ 0 + 1 + 0 & 0 - 1 + 0 \\ 1 + 0 - 0 & 1 + 0 - 0 \end{bmatrix} && \text{(multiply)} \\
&= \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(add)}
\end{aligned}
$$

which is $\mathbf{A}$. $\mathbf{A}$ has rank 2, because its two columns are linearly independent, and its singular values are $\sigma_1 = 2$ and $\sigma_2 = \sqrt{2}$.

Remark (An SVD is not unique)

Remark 0.6 (An SVD is not unique). An SVD is not unique [48, Ch. 12], Examples 12.2 and 12.3, p. 378. For any $i \leq r$, multiplying column i of both $\mathbf{U}$ and $\mathbf{V}$ by -1 gives another one. The singular values do not depend on which SVD is chosen [48, Ch. 12], pp. 371 and 378.
For example, in Example 0.5, multiplying the first columns of $\mathbf{U}$ and $\mathbf{V}$ by -1 gives

$$\mathbf{U}_1 = \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} \qquad \mathbf{V}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix}$$

Their columns are the columns of $\mathbf{U}$ and $\mathbf{V}$, some multiplied by -1, so they are still orthonormal. With the same $\mathbf{D}$:

$$\begin{aligned}
\mathbf{D}{\mathbf{V}_1}^\top &= \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \left(\frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix} \right)^\top && \text{(substitute)} \\
&= \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} && \text{(definition of the transpose)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} 2 & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} && \text{(move the scalar to the front)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} 2 \cdot (-1) + 0 \cdot 1 & 2 \cdot (-1) + 0 \cdot (-1) \\ 0 \cdot (-1) + \sqrt{2} \cdot 1 & 0 \cdot (-1) + \sqrt{2} \cdot (-1) \\ 0 \cdot (-1) + 0 \cdot 1 & 0 \cdot (-1) + 0 \cdot (-1) \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} -2 + 0 & -2 + 0 \\ 0 + \sqrt{2} & 0 - \sqrt{2} \\ 0 + 0 & 0 + 0 \end{bmatrix} && \text{(multiply)} \\
&= \frac{1}{\sqrt{2}} \begin{bmatrix} -2 & -2 \\ \sqrt{2} & -\sqrt{2} \\ 0 & 0 \end{bmatrix} && \text{(add)} \\
&= \begin{bmatrix} -\sqrt{2} & -\sqrt{2} \\ 1 & -1 \\ 0 & 0 \end{bmatrix} && \text{(divide by } \sqrt{2}\text{)}
\end{aligned}$$

and

$$
\begin{aligned}
\mathbf{U}_1\mathbf{D}\mathbf{V}_1{}^\top &= \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} -\sqrt{2} & -\sqrt{2} \\ 1 & -1 \\ 0 & 0 \end{bmatrix} && \text{(substitute)} \\
&= \begin{bmatrix} -\frac{1}{\sqrt{2}} \cdot \left(-\sqrt{2}\right) + 0 \cdot 1 + \frac{1}{\sqrt{2}} \cdot 0 & -\frac{1}{\sqrt{2}} \cdot \left(-\sqrt{2}\right) + 0 \cdot (-1) + \frac{1}{\sqrt{2}} \cdot 0 \\ 0 \cdot \left(-\sqrt{2}\right) + 1 \cdot 1 + 0 \cdot 0 & 0 \cdot \left(-\sqrt{2}\right) + 1 \cdot (-1) + 0 \cdot 0 \\ -\frac{1}{\sqrt{2}} \cdot \left(-\sqrt{2}\right) + 0 \cdot 1 - \frac{1}{\sqrt{2}} \cdot 0 & -\frac{1}{\sqrt{2}} \cdot \left(-\sqrt{2}\right) + 0 \cdot (-1) - \frac{1}{\sqrt{2}} \cdot 0 \end{bmatrix} && \text{(definition of matrix multipl} \\
&= \begin{bmatrix} 1+0+0 & 1+0+0 \\ 0+1+0 & 0-1+0 \\ 1+0-0 & 1+0-0 \end{bmatrix} && \text{(multiply)} \\
&= \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(add)}
\end{aligned}
$$

which is $\mathbf{A}$ again. Both SVDs have the singular values $\sigma_1 = 2$ and $\sigma_2 = \sqrt{2}$.

i Remark (Counting zero singular values)

Remark 0.7 (Counting zero singular values). Some texts, and R's `svd()`, also count $\min(n,p) - r$ singular values equal to 0, so that every $n \times p$ matrix has $\min(n,p)$ singular values.

For example, $\mathbf{B} = \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}$ has rank $r = 1$. Taking

$$
\begin{aligned}
\mathbf{U} &= \mathbf{V} \\
&= \mathbf{I}_2
\end{aligned}
$$

and $\mathbf{D} = \mathbf{B}$ gives an SVD $\mathbf{B} = \mathbf{I}_2\mathbf{B}\mathbf{I}_2{}^\top$, so by Definition 0.5, $\mathbf{B}$ has one singular value, $\sigma_1 = 3$. R's `svd()` reports $\min(2,2) = 2$ singular values for $\mathbf{B}$: 3 and 0.

Theorem 0.4 (An SVD gives an eigendecomposition of $\mathbf{A}^\top\mathbf{A}$). *Let $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$ be a singular value decomposition (Definition 0.4) of an $n \times p$ matrix $\mathbf{A}$ with singular values $\sigma_1, \ldots, \sigma_r$. Then $\mathbf{A}^\top\mathbf{A}$ is symmetric (Definition 0.3), and*

$$
\underbrace{\mathbf{A}^\top\mathbf{A}}_{p\times p} = \underbrace{\mathbf{V}}_{p\times p}\ \underbrace{\mathbf{\Lambda}}_{p\times p}\ \underbrace{\mathbf{V}^\top}_{p\times p}
$$

where $\mathbf{\Lambda} \overset{def}{=} \mathbf{D}^\top\mathbf{D}$ is the $p \times p$ diagonal matrix whose i-th diagonal entry λ_i is σ_i^2 for $i \le r$ and 0 for $i > r$. So $\mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$ is an eigendecomposition (Definition 0.3) of $\mathbf{A}^\top\mathbf{A}$: column i of $\mathbf{V}$ is an eigenvector of $\mathbf{A}^\top\mathbf{A}$ for the eigenvalue λ_i.

Proof

Proof. **Symmetry.**

$$\begin{aligned}(\mathbf{A}^\top\mathbf{A})^\top &= \mathbf{A}^\top(\mathbf{A}^\top)^\top && \text{(transpose of a product)}\\ &= \mathbf{A}^\top\mathbf{A} && \text{(transposing twice changes nothing)}\end{aligned}$$

The first step is Theorem 0.19, and the second follows from Definition 0.6, which swaps rows and columns, so swapping them again restores $\mathbf{A}$.
The factorization.

$$\begin{aligned}\mathbf{A}^\top\mathbf{A} &= (\mathbf{U}\mathbf{D}\mathbf{V}^\top)^\top\mathbf{U}\mathbf{D}\mathbf{V}^\top && \text{(substitute the SVD)}\\ &= (\mathbf{D}\mathbf{V}^\top)^\top\mathbf{U}^\top\mathbf{U}\mathbf{D}\mathbf{V}^\top && \text{(transpose of the product of } \mathbf{U} \text{ and } \mathbf{D}\mathbf{V}^\top)\\ &= (\mathbf{V}^\top)^\top\mathbf{D}^\top\mathbf{U}^\top\mathbf{U}\mathbf{D}\mathbf{V}^\top && \text{(transpose of the product of } \mathbf{D} \text{ and } \mathbf{V}^\top)\\ &= \mathbf{V}\mathbf{D}^\top\mathbf{U}^\top\mathbf{U}\mathbf{D}\mathbf{V}^\top && \text{(transposing twice changes nothing)}\\ &= \mathbf{V}\mathbf{D}^\top(\mathbf{U}^\top\mathbf{U})\mathbf{D}\mathbf{V}^\top && \text{(regroup; matrix multiplication is associative)}\\ &= \mathbf{V}\mathbf{D}^\top\mathbf{I}_n\mathbf{D}\mathbf{V}^\top && (\mathbf{U} \text{ is orthogonal})\\ &= \mathbf{V}(\mathbf{D}^\top\mathbf{D})\mathbf{V}^\top && \text{(identity matrix)}\\ &= \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top && \text{(definition of } \mathbf{\Lambda})\end{aligned}$$

The entries of $\mathbf{\Lambda}$. Write λ_{ij} for entry (i,j) of $\mathbf{\Lambda}$, so $\lambda_i = \lambda_{ii}$. Only the entries d_{kk} with $k \le r$ of $\mathbf{D}$ can be nonzero.

$$\begin{aligned}\lambda_{ij} &= \sum_{k=1}^{n}(\mathbf{D}^\top)_{ik}\,d_{kj} && \text{(definition of matrix multiplication)}\\ &= \sum_{k=1}^{n} d_{ki}\,d_{kj} && \text{(definition of the transpose)}\end{aligned}$$

The term for k is nonzero only if $k = i \le r$ and $k = j$. So $\lambda_{ij} = 0$ when $i \neq j$. On the diagonal, only the term $k = i$ can be nonzero, and for $i \le r$

$$\begin{aligned}\lambda_i &= \lambda_{ii} && \text{(notation)}\\ &= d_{ii}\,d_{ii} && \text{(the only term that can be nonzero)}\\ &= \sigma_i \cdot \sigma_i && \text{(definition of } \mathbf{D})\\ &= \sigma_i^2 && \text{(notation for a square)}\end{aligned}$$

For $i > r$, every term is 0, so $\lambda_i = 0$.
The eigenvectors. Write $\underset{\sim}{v}_i$ for column i of $\mathbf{V}$, and $\underset{\sim}{e}_i$ for the indicator vector (Definition 0.18) of length p. Column i of $\mathbf{V}^\top\mathbf{V} = \mathbf{I}_p$ says $\mathbf{V}^\top\underset{\sim}{v}_i = \underset{\sim}{e}_i$. Then

$$
\begin{aligned}
\mathbf{A}^\top \mathbf{A}\, \underset{\sim}{v}_i &= \mathbf{V}\mathbf{\Lambda}\,\mathbf{V}^\top \underset{\sim}{v}_i && \text{(the factorization)} \\
&= \mathbf{V}\mathbf{\Lambda}\, \underset{\sim}{e}_i && (\mathbf{V}^\top \underset{\sim}{v}_i = \underset{\sim}{e}_i) \\
&= \mathbf{V}\,(\lambda_i\, \underset{\sim}{e}_i) && \text{(column } i \text{ of the diagonal matrix } \mathbf{\Lambda}) \\
&= \lambda_i\, \mathbf{V}\underset{\sim}{e}_i && \text{(move the scalar } \lambda_i \text{ to the front)} \\
&= \lambda_i\, \underset{\sim}{v}_i && (\mathbf{V}\underset{\sim}{e}_i \text{ is column } i \text{ of } \mathbf{V})
\end{aligned}
$$

and $\underset{\sim}{v}_i \neq \underset{\sim}{0}$ because it has norm 1. □

Example 0.6 ($\mathbf{A}^\top\mathbf{A}$ for the matrix of the SVD example)**.** For $\mathbf{A}$ in Example 0.5:

$$
\begin{aligned}
\mathbf{A}^\top\mathbf{A} &= \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{bmatrix} && \text{(definition of the transpose)} \\
&= \begin{bmatrix} 1+1+1 & 1-1+1 \\ 1-1+1 & 1+1+1 \end{bmatrix} && \text{(definition of matrix multiplication)} \\
&= \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} && \text{(add)}
\end{aligned}
$$

Theorem 0.4 says its eigenvalues are $\sigma_1^2 = 4$ and $\sigma_2^2 = 2$, with the columns of $\mathbf{V}$ as eigenvectors. Checking the first column, up to its factor $\frac{1}{\sqrt{2}}$:

$$
\begin{aligned}
\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} &= \begin{bmatrix} 3\cdot 1 + 1\cdot 1 \\ 1\cdot 1 + 3\cdot 1 \end{bmatrix} && \text{(definition of matrix-vector multiplication)} \\
&= \begin{bmatrix} 3+1 \\ 1+3 \end{bmatrix} && \text{(multiply)} \\
&= \begin{bmatrix} 4 \\ 4 \end{bmatrix} && \text{(add)} \\
&= 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix} && \text{(factor out 4)}
\end{aligned}
$$

and the second:

$$
\begin{aligned}
\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} &= \begin{bmatrix} 3\cdot 1 + 1\cdot (-1) \\ 1\cdot 1 + 3\cdot (-1) \end{bmatrix} && \text{(definition of matrix-vector multiplication)} \\
&= \begin{bmatrix} 3-1 \\ 1-3 \end{bmatrix} && \text{(multiply)} \\
&= \begin{bmatrix} 2 \\ -2 \end{bmatrix} && \text{(add)} \\
&= 2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} && \text{(factor out 2)}
\end{aligned}
$$

Remark (Building an SVD from an eigendecomposition)

Remark 0.8 (Building an SVD from an eigendecomposition). Theorem 0.4 matches Banerjee and Roy [48, Ch. 12], pp. 372-373, which works in the other direction: it constructs an SVD from a spectral decomposition of $\mathbf{A}^\top\mathbf{A}$ and sets $\sigma_i = \sqrt{\lambda_i}$.
For example, in Example 0.6 the eigenvalues of $\mathbf{A}^\top\mathbf{A}$ are 4 and 2, and $\sqrt{4} = 2$ and $\sqrt{2}$ are the singular values σ_1 and σ_2 of $\mathbf{A}$ from Example 0.5.

Definite Matrices

Definition 0.1 (Positive semidefinite matrix)**.** A $p \times p$ matrix $\mathbf{A}$ is **positive semidefinite** [77] if it satisfies both conditions:

- $\mathbf{A}$ is symmetric (Definition 0.3).
- Every quadratic form (Definition 0.1) in $\mathbf{A}$ is non-negative: $\underset{\sim}{x}^\top\mathbf{A}\underset{\sim}{x} \geq 0$ for every vector $\underset{\sim}{x}$ of length p.

Example 0.1 (A positive semidefinite matrix)**.** Let $\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, which is symmetric. For any $\underset{\sim}{x} = (x_1, x_2)$:

$$\begin{aligned} \underset{\sim}{x}^\top\mathbf{B}\underset{\sim}{x} &= x_1^2 + x_1x_2 + x_2x_1 + x_2^2 && \text{(multiply out the quadratic form)} \\ &= (x_1 + x_2)^2 && \text{(complete the square)} \\ &\geq 0 && \text{(a square is non-negative)} \end{aligned}$$

So $\mathbf{B}$ is positive semidefinite.

Theorem 0.1 (Entries and symmetry of a Gram matrix)**.** *Let $\mathbf{A}$ be an $m \times n$ matrix with columns $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_n$, and let $\mathbf{G} = \mathbf{A}^\top\mathbf{A}$ (Definition 0.6).*

1. *Entry (i, j) of $\mathbf{G}$ is $\underset{\sim}{a}_i \cdot \underset{\sim}{a}_j$.*
2. *$\mathbf{G}$ is symmetric (Definition 0.3).*
3. *$\underset{\sim}{x}^\top\mathbf{G}\underset{\sim}{x} = \|\mathbf{A}\underset{\sim}{x}\|^2 \geq 0$ for every $\underset{\sim}{x} \in \mathbb{R}^n$, so $\mathbf{G}$ is positive semidefinite (Definition 0.1).*

Proof

Proof. **Entries.** Row i of $\mathbf{A}^\top$ is $\underset{\sim}{a}_i^\top$, so entry (i, j) of $\mathbf{A}^\top\mathbf{A}$ is the dot product of that row with column j of $\mathbf{A}$, namely $\underset{\sim}{a}_i^\top\underset{\sim}{a}_j = \underset{\sim}{a}_i \cdot \underset{\sim}{a}_j$.

Symmetry. The dot product is commutative, so

$$\begin{aligned} g_{ij} &= \underset{\sim}{a}_i \cdot \underset{\sim}{a}_j \\ &= \underset{\sim}{a}_j \cdot \underset{\sim}{a}_i \\ &= g_{ji}. \end{aligned}$$

Positive semidefiniteness. For any $\underset{\sim}{x}$,

$$\begin{aligned} \underset{\sim}{x}^\top \mathbf{G} \underset{\sim}{x} &= \underset{\sim}{x}^\top \mathbf{A}^\top \mathbf{A} \underset{\sim}{x} && \text{(substitute } \mathbf{G}\text{)} \\ &= (\mathbf{A}\underset{\sim}{x})^\top (\mathbf{A}\underset{\sim}{x}) && \text{(Theorem 0.19)} \\ &= \|\mathbf{A}\underset{\sim}{x}\|^2 && \text{(Equation 2, squared)} \\ &\geq 0. && \text{(a squared norm is nonnegative)} \end{aligned}$$

□

Example 0.2 (Checking the properties on a small Gram matrix)**.** For $\mathbf{A}$ in Example 0.8, $\mathbf{G} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. It is symmetric: $g_{12} = g_{21} = 1$. Take $\underset{\sim}{x} = (1, -1)$. Then

$$\begin{aligned} \underset{\sim}{x}^\top \mathbf{G} \underset{\sim}{x} &= \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} && (\mathbf{G}\underset{\sim}{x} = (1, -1)) \\ &= 2, && \text{(multiply and add)} \end{aligned}$$

and $\mathbf{A}\underset{\sim}{x} = (1, 0, -1)$ has $\|\mathbf{A}\underset{\sim}{x}\|^2 = 1 + 0 + 1 = 2$, as the theorem says. The converse (every positive semidefinite matrix is a Gram matrix) is this theorem.

Definition 0.2 (Positive definite matrix)**.** A $p \times p$ matrix $\mathbf{A}$ is **positive definite** [77] if it satisfies both conditions:

- $\mathbf{A}$ is symmetric (Definition 0.3).
- Every quadratic form (Definition 0.1) in $\mathbf{A}$ at a nonzero vector is positive: $\underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{x} > 0$ for every vector $\underset{\sim}{x} \neq \underset{\sim}{0}$ of length p.

Example 0.3 (Positive definite, semidefinite, and neither)**.**

- The identity matrix $\mathbf{I}_p$ (Definition 0.15) is positive definite: $\underset{\sim}{x}^\top \mathbf{I}_p \underset{\sim}{x} = \sum_{i=1}^{p} x_i^2$, which is positive unless every x_i is 0.
- $\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ from Example 0.1 is positive semidefinite but not positive definite: at $\underset{\sim}{x} = (1, -1) \neq \underset{\sim}{0}$,

$$\begin{aligned} \underset{\sim}{x}^\top \mathbf{B} \underset{\sim}{x} &= (1-1)^2 \\ &= 0. \end{aligned}$$

- $\mathbf{D} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ is symmetric but not positive semidefinite: at $\underset{\sim}{x} = (1, -1)$,

$$\begin{aligned} \underset{\sim}{x}^\top \mathbf{D} \underset{\sim}{x} &= 1 - 2 - 2 + 1 \\ &= -2 \\ &< 0. \end{aligned}$$

i Remark (Why the definition requires symmetry)

Remark 0.1 (Why the definition requires symmetry). A positive definite matrix is positive semidefinite (Definition 0.1): $\underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{x} > 0$ for every $\underset{\sim}{x} \neq \underset{\sim}{0}$, and $\underset{\sim}{0}^\top \mathbf{A} \underset{\sim}{0} = 0$. For example, $\mathbf{I}_p$ in Example 0.3 is both.
Some sources drop the symmetry condition from both definitions. These notes keep it, because without it a matrix can pass the quadratic-form condition and still have no real eigenvalues (Definition 0.1). For example, $\mathbf{C} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$ is not symmetric, and for any $\underset{\sim}{x} = (x_1, x_2)$:

$$\begin{aligned} \underset{\sim}{x}^\top \mathbf{C} \underset{\sim}{x} &= x_1(x_1 + x_2) + x_2(-x_1 + x_2) && \text{(multiply out the quadratic form)} \\ &= x_1^2 + x_1 x_2 - x_2 x_1 + x_2^2 && \text{(distribute)} \\ &= x_1^2 + x_2^2 && \text{(the middle terms cancel)} \end{aligned}$$

which is positive for every $\underset{\sim}{x} \neq \underset{\sim}{0}$. But suppose $\mathbf{C} \underset{\sim}{v} = \lambda \underset{\sim}{v}$ for a real number λ. Matching entries gives $v_1 + v_2 = \lambda v_1$ and $-v_1 + v_2 = \lambda v_2$, so $v_2 = (\lambda - 1)\, v_1$ and $-v_1 = (\lambda - 1)\, v_2$. Substituting the first into the second gives $-v_1 = (\lambda - 1)^2 v_1$, so $(1 + (\lambda - 1)^2) v_1 = 0$. Since $1 + (\lambda - 1)^2 > 0$, $v_1 = 0$, and then

$$\begin{aligned} v_2 &= (\lambda - 1)\, v_1 \\ &= 0. \end{aligned}$$

So no real λ has an eigenvector $\underset{\sim}{v} \neq \underset{\sim}{0}$.

See also [77].

Theorem 0.2 (Definiteness and eigenvalues)**.** *Let* $\mathbf{A}$ *be a* $p \times p$ *symmetric matrix with real entries, with eigendecomposition* $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$ *(Definition 0.3) and eigenvalues* $\lambda_1, \ldots, \lambda_p$ *on the diagonal of* $\mathbf{\Lambda}$*. Then:*

- $\mathbf{A}$ *is positive semidefinite (Definition 0.1) if and only if every* $\lambda_i \geq 0$*.*

- $\mathbf{A}$ *is positive definite (Definition 0.2) if and only if every* $\lambda_i > 0$.

Proof

Proof. For any vector $\underset{\sim}{x}$ of length p, let $\underset{\sim}{y} = \mathbf{Q}^\top \underset{\sim}{x}$. Then:

$$\begin{aligned}
\underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{x} &= \underset{\sim}{x}^\top \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top \underset{\sim}{x} && \text{(substitute the eigendecomposition)} \\
&= (\mathbf{Q}^\top \underset{\sim}{x})^\top \mathbf{\Lambda} (\mathbf{Q}^\top \underset{\sim}{x}) && \text{(transpose of a product)} \\
&= \underset{\sim}{y}^\top \mathbf{\Lambda} \underset{\sim}{y} && \text{(definition of } \underset{\sim}{y}) \\
&= \sum_{i=1}^{p} \lambda_i y_i^2 && (\mathbf{\Lambda} \text{ is diagonal})
\end{aligned}$$

The second step is Theorem 0.19. Also, $\underset{\sim}{x} = \underset{\sim}{0}$ exactly when $\underset{\sim}{y} = \underset{\sim}{0}$: $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}_p$ for an orthogonal matrix (Definition 0.10), so $\underset{\sim}{x} = \mathbf{Q}\underset{\sim}{y}$.
If every $\lambda_i \geq 0$, then every term $\lambda_i y_i^2 \geq 0$, so $\underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{x} \geq 0$. *If every* $\lambda_i > 0$ and $\underset{\sim}{x} \neq \underset{\sim}{0}$, then some $y_i \neq 0$, so at least one term is positive and the rest are non-negative, and $\underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{x} > 0$.
Conversely, take $\underset{\sim}{x} = \underset{\sim}{q}_i$, column i of $\mathbf{Q}$, which is not $\underset{\sim}{0}$ because it has norm 1. Then $\underset{\sim}{y} = \mathbf{Q}^\top \underset{\sim}{q}_i$ is column i of $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$, so $y_i = 1$ and every other entry of y is 0, and the display gives $\underset{\sim}{q}_i^\top \mathbf{A} \underset{\sim}{q}_i = \lambda_i$. So if $\mathbf{A}$ is positive semidefinite, $\lambda_i \geq 0$, and if $\mathbf{A}$ is positive definite, $\lambda_i > 0$. □

Example 0.4 (Reading definiteness off the eigenvalues)**.**

- $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ has eigenvalues 3 and 1 (Example 0.1), both positive, so it is positive definite.
- $\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ has eigenvalues 2, for the eigenvector $(1, 1)$, and 0, for the eigenvector $(1, -1)$, so it is positive semidefinite but not positive definite, as Example 0.3 found directly.

Definition 0.3 (Condition number of a positive definite matrix)**.** Let $\mathbf{S}$ be a $p \times p$ positive definite matrix with eigenvalues $\lambda_{\min} \leq \cdots \leq \lambda_{\max}$, all positive by definiteness and eigenvalues. The **condition number** of $\mathbf{S}$ is

$$\kappa(\mathbf{S}) \stackrel{\text{def}}{=} \frac{\lambda_{\max}}{\lambda_{\min}}.$$

The condition number is at least 1, that is, $\kappa(\mathbf{S}) \geq 1$, because $\lambda_{\max} \geq \lambda_{\min} > 0$, and $\kappa(\mathbf{S}) = 1$ exactly when all eigenvalues coincide.

Example 0.5 (A condition number of 3)**.** $\mathbf{S} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ has eigenvalues 3 and 1 (eigenvalue example), so $\kappa(\mathbf{S}) = 3/1 = 3$. The identity matrix $\mathbf{I}_2$ has eigenvalues 1 and 1, so $\kappa(\mathbf{I}_2) = 1$.

Definition 0.4 (Condition number of a matrix)**.** The **condition number** of an $n \times p$ matrix $\mathbf{A}$ is the ratio of its largest to its smallest singular value, counting all $\min(n, p)$ singular values of $\mathbf{A}$, zeros included (Remark 0.7) [78]:

$$\kappa(\mathbf{A}) \stackrel{\text{def}}{=} \frac{\sigma_{\max}}{\sigma_{\min}} = \frac{\sigma_1}{\sigma_{\min(n,p)}}.$$

This generalizes Definition 0.3, which covers positive definite matrices. For a nonzero matrix, $\kappa(\mathbf{A}) \geq 1$. When the columns of a tall matrix $\mathbf{A}$ ($n \geq p$) are linearly dependent, its smallest singular value is 0, and $\kappa(\mathbf{A}) = \infty$.

Example 0.6 (The condition number of a matrix with nearly parallel columns)**.** Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1.01 \end{bmatrix}$. Its columns, $(1, 1)$ and $(1, 1.01)$, are nearly parallel. We find its singular values and their ratio, then repeat after multiplying every entry by 1000:

R

```
entries_mcn <- c(1, 1, 1, 1.01)
a_mcn <- matrix(entries_mcn, nrow = 2)
svd_mcn <- svd(a_mcn)
sv_mcn <- svd_mcn$d
a_mcn_scaled <- 1000 * a_mcn
svd_mcn_scaled <- svd(a_mcn_scaled)
sv_mcn_scaled <- svd_mcn_scaled$d
sv_mcn
#> [1] 2.0050125 0.0049875
sv_mcn[1] / sv_mcn[2]
#> [1] 402.0075
sv_mcn_scaled[1] / sv_mcn_scaled[2]
#> [1] 402.0075
```

Python

```python
import numpy as np

a_mcn = np.array([[1.0, 1.0], [1.0, 1.01]])
sv_mcn = np.linalg.svd(a_mcn, compute_uv=False)
sv_mcn_scaled = np.linalg.svd(1000 * a_mcn, compute_uv=False)
print(sv_mcn)
#> [2.0050125 0.0049875]
print(sv_mcn[0] / sv_mcn[1])
#> 402.0075124842948
print(sv_mcn_scaled[0] / sv_mcn_scaled[1])
#> 402.00751248429236
```

The singular values are 2.005 and 0.005, so $\kappa(\mathbf{A}) = 402$. Multiplying every entry by 1000 multiplies both singular values by 1000, so the condition number is still 402.

i Remark (Other condition numbers)

Remark 0.2 (Other condition numbers). Other norms give other condition numbers. This definition uses the 2-norm, which is the one built from the singular values. For a positive definite matrix, the singular values are the eigenvalues, and this definition agrees with the ratio of its largest to its smallest eigenvalue in Definition 0.3.

Theorem 0.3 (Forming $\mathbf{A}^\top\mathbf{A}$ squares the condition number)**.** *Let* $\mathbf{A}$ *be a nonzero* $n \times p$ *matrix with* $n \geq p$*. Then*

$$\kappa(\mathbf{A}^\top\mathbf{A}) = \kappa(\mathbf{A})^2.$$

i Proof

Proof. By Theorem 0.4, the eigenvalues of the symmetric matrix $\mathbf{A}^\top\mathbf{A}$ are $\sigma_1^2 \geq \cdots \geq \sigma_p^2$, where $\sigma_1 \geq \cdots \geq \sigma_p$ are the singular values of $\mathbf{A}$, with zeros counted (Remark 0.7). These eigenvalues are non-negative, so for the symmetric matrix $\mathbf{A}^\top\mathbf{A}$ they are also its singular values. Therefore

$$\kappa(\mathbf{A}^\top\mathbf{A}) = \frac{\sigma_1^2}{\sigma_p^2} = \left(\frac{\sigma_1}{\sigma_p}\right)^2 = \kappa(\mathbf{A})^2.$$

If $\sigma_p = 0$, both sides are ∞. □

Example 0.7 (Squaring the condition number)**.** For the matrix $\mathbf{A}$ from Example 0.6, compare $\kappa(\mathbf{A})$ with $\kappa(\mathbf{A}^\top\mathbf{A})$:

R

```
# Uses a_mcn and sv_mcn from the earlier example
gram_cng <- crossprod(a_mcn)
kappa_a_cng <- sv_mcn[1] / sv_mcn[2]
svd_gram_cng <- svd(gram_cng)
sv_gram_cng <- svd_gram_cng$d
sv_gram_cng[1] / sv_gram_cng[2]
#> [1] 161610
kappa_a_cng^2
#> [1] 161610
```

Python

```
# Uses a_mcn and sv_mcn from the earlier example
gram_cng = a_mcn.T @ a_mcn
kappa_a_cng = sv_mcn[0] / sv_mcn[1]
sv_gram_cng = np.linalg.svd(gram_cng, compute_uv=False)
print(sv_gram_cng[0] / sv_gram_cng[1])
#> 161610.0400912735
print(kappa_a_cng ** 2)
#> 161610.04009381044
```

The condition number of $\mathbf{A}^\top\mathbf{A}$ is 161610, and $\kappa(\mathbf{A})^2 = 161610$, as Theorem 0.3 says.

Definition 0.5 (Ill-conditioned matrix)**.** A matrix $\mathbf{A}$ is **ill-conditioned** when its condition number is so large that small relative changes in its entries, such as those from rounding, can change the solution of a linear system by a much larger relative amount. The relative change in the solution can be up to $\kappa(\mathbf{A})$ times the relative change in the entries. A solution computed with $\kappa(\mathbf{A}) \approx 10^k$ therefore loses about k of the digits that the floating-point arithmetic carries [79], Lecture 12.
There is no sharp threshold: how large is too large depends on how many accurate digits the answer needs.

Definition 0.6 (Well-conditioned matrix)**.** A matrix $\mathbf{A}$ is **well-conditioned** when its condition number is close to 1, so it is not ill-conditioned.

Example 0.8 (A small change in the right-hand side moves the solution a long way)**.** Let $\mathbf{A}$ be the matrix from Example 0.6, and let $\underset{\sim}{b} = (2, 2.01)$. The system $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ has solution $\underset{\sim}{x} = (1, 1)$. Change the second entry of $\underset{\sim}{b}$ from 2.01 to 2.011, and solve again:

R

```
# Uses a_mcn and sv_mcn from the previous example
b_ic <- c(2, 2.01)
b_new_ic <- c(2, 2.011)
x_ic <- solve(a_mcn, b_ic)
x_new_ic <- solve(a_mcn, b_new_ic)
x_ic
#> [1] 1 1
x_new_ic
#> [1] 0.9 1.1

# Relative changes, measured with the Euclidean length
db_ic <- (b_new_ic - b_ic)^2 |> sum() |> sqrt()
nb_ic <- b_ic^2 |> sum() |> sqrt()
dx_ic <- (x_new_ic - x_ic)^2 |> sum() |> sqrt()
nx_ic <- x_ic^2 |> sum() |> sqrt()
rel_b_ic <- db_ic / nb_ic
rel_x_ic <- dx_ic / nx_ic
c(rel_b_ic, rel_x_ic, rel_x_ic / rel_b_ic)
#> [1] 3.526706e-04 1.000000e-01 2.835507e+02
```

Python

```
# Uses a_mcn and sv_mcn from the previous example
b_ic = np.array([2.0, 2.01])
b_new_ic = np.array([2.0, 2.011])
x_ic = np.linalg.solve(a_mcn, b_ic)
x_new_ic = np.linalg.solve(a_mcn, b_new_ic)
print(x_ic)
#> [1. 1.]
print(x_new_ic)
#> [0.9 1.1]

# Relative changes, measured with the Euclidean length
rel_b_ic = np.linalg.norm(b_new_ic - b_ic) / np.linalg.norm(b_ic)
rel_x_ic = np.linalg.norm(x_new_ic - x_ic) / np.linalg.norm(x_ic)
print(rel_b_ic, rel_x_ic, rel_x_ic / rel_b_ic)
#> 0.0003526706147110207 0.1000000000000334 283.55070093371313
```

The right-hand side changes by a relative amount of 0.00035, and the solution changes by a relative amount of 0.100, about 284 times as much. That amplification is what Definition 0.5 describes, and it cannot exceed $\kappa(\mathbf{A}) = 402.0$.

Theorem 0.4 (A positive definite matrix has a positive definite inverse)**.** *Let* $\mathbf{A}$ *be a* $p \times p$ *positive definite matrix (Definition 0.2) with real entries, with eigendecomposition* $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$ *(Definition 0.3) and eigenvalues* $\lambda_1, \ldots, \lambda_p$*. Then* $\mathbf{A}$ *is invertible (Definition 0.6),*

$$\mathbf{A}^{-1} = \mathbf{Q}\,\mathbf{\Lambda}^{-1}\,\mathbf{Q}^\top, \qquad \mathbf{\Lambda}^{-1} = \begin{bmatrix} 1/\lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1/\lambda_p \end{bmatrix},$$

and $\mathbf{A}^{-1}$ *is positive definite.*

Proof

Proof. By Theorem 0.2, every $\lambda_i > 0$, so $\mathbf{\Lambda}^{-1}$ is well defined, and multiplying the two diagonal matrices entry by entry gives

$$\begin{aligned} \mathbf{\Lambda}\mathbf{\Lambda}^{-1} &= \mathbf{\Lambda}^{-1}\mathbf{\Lambda} \\ &= \mathbf{I}_p. \end{aligned}$$

Write $\mathbf{B} = \mathbf{Q}\mathbf{\Lambda}^{-1}\mathbf{Q}^\top$. Then:

$$\begin{aligned}
\mathbf{AB} &= \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top\mathbf{Q}\mathbf{\Lambda}^{-1}\mathbf{Q}^\top && \text{(substitute)}\\
&= \mathbf{Q}\mathbf{\Lambda}\mathbf{I}_p\mathbf{\Lambda}^{-1}\mathbf{Q}^\top && (\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_p)\\
&= \mathbf{Q}\mathbf{\Lambda}\mathbf{\Lambda}^{-1}\mathbf{Q}^\top && \text{(identity matrix)}\\
&= \mathbf{Q}\mathbf{Q}^\top && (\mathbf{\Lambda}\mathbf{\Lambda}^{-1} = \mathbf{I}_p)\\
&= \mathbf{I}_p && (\mathbf{Q} \text{ is orthogonal})
\end{aligned}$$

The same steps, with $\mathbf{\Lambda}^{-1}$ and $\mathbf{\Lambda}$ swapped, give $\mathbf{BA} = \mathbf{I}_p$, so $\mathbf{B} = \mathbf{A}^{-1}$ (5). The steps with $\mathbf{Q}$ use Definition 0.10, whose remark records that $\mathbf{QQ}^\top = \mathbf{I}_p$ too, and the identity step uses Theorem 0.10.

$\mathbf{A}^{-1}$ is symmetric by Corollary 0.1. For $\underset{\sim}{x} \neq \underset{\sim}{0}$, let $\underset{\sim}{y} = \mathbf{Q}^\top\underset{\sim}{x}$, which is not $\underset{\sim}{0}$ because $\underset{\sim}{x} = \mathbf{Q}\underset{\sim}{y}$. As in the proof of Theorem 0.2:

$$\begin{aligned}
\underset{\sim}{x}^\top\mathbf{A}^{-1}\underset{\sim}{x} &= \underset{\sim}{y}^\top\mathbf{\Lambda}^{-1}\underset{\sim}{y} && (\text{substitute } \mathbf{A}^{-1} = \mathbf{Q}\mathbf{\Lambda}^{-1}\mathbf{Q}^\top)\\
&= \sum_{i=1}^{p}\frac{y_i^2}{\lambda_i} && (\mathbf{\Lambda}^{-1} \text{ is diagonal})\\
&> 0 && (\text{each } \lambda_i > 0, \text{ and some } y_i \neq 0)
\end{aligned}$$

So $\mathbf{A}^{-1}$ is positive definite. □

Example 0.9 (Inverting a positive definite matrix)**.** For $\mathbf{A} = \begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$, Example 0.4 gives $\mathbf{Q} = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$ and eigenvalues 3 and 1, so:

$$\begin{aligned}
\mathbf{A}^{-1} &= \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}1/3 & 0\\ 0 & 1\end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} && (\text{substitute; } \mathbf{Q} \text{ is symmetric})\\
&= \frac{1}{2}\begin{bmatrix}1/3 & 1\\ 1/3 & -1\end{bmatrix}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} && (\text{multiply the first two matrices})\\
&= \frac{1}{2}\begin{bmatrix}4/3 & -2/3\\ -2/3 & 4/3\end{bmatrix} && (\text{multiply})\\
&= \frac{1}{3}\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix} && (\text{simplify})
\end{aligned}$$

Multiplying out,

$$\begin{aligned}
\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}\cdot\frac{1}{3}\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix} &= \frac{1}{3}\begin{bmatrix}3 & 0\\ 0 & 3\end{bmatrix}\\
&= \mathbf{I}_2.
\end{aligned}$$

Definition 0.7 (Paraboloid)**.** A **paraboloid** [80] in $\mathbb{R}^{p+1}$ is the graph $\{(\underset{\sim}{x}, f(\underset{\sim}{x})) \mid \underset{\sim}{x} \in \mathbb{R}^p\}$ of a function

$$f(\underset{\sim}{x}) = \underset{\sim}{x}^\top \mathbf{S} \underset{\sim}{x} + \underset{\sim}{b} \cdot \underset{\sim}{x} + c,$$

where $\mathbf{S}$ is a nonzero symmetric $p \times p$ matrix (Definition 0.3), $\underset{\sim}{b} \in \mathbb{R}^p$, and $c \in \mathbb{R}$. The first term is a quadratic form (Definition 0.1).

Definition 0.8 (Elliptic paraboloid)**.** The paraboloid of Definition 0.7 is **elliptic** if $\mathbf{S}$ is positive definite (Definition 0.2); it is then a bowl that opens upward.

Example 0.10 (A bowl with its lowest point at the origin)**.** Take $p = 2$, $\mathbf{S} = \mathbf{I}_2$, $\underset{\sim}{b} = \underset{\sim}{0}_{2\times 1}$ and $c = 0$. Then $\mathbf{S}$ is positive definite (Example 0.3), so the graph of

$$\begin{aligned} f(\underset{\sim}{x}) &= \underset{\sim}{x}^\top \mathbf{I}_2 \underset{\sim}{x} \\ &= x_1^2 + x_2^2 \end{aligned}$$

is an elliptic paraboloid. At $\underset{\sim}{x} = \underset{\sim}{0}$, $f(\underset{\sim}{x}) = 0$; at $\underset{\sim}{x} = (1, 1)$, $f(\underset{\sim}{x}) = 2$; and $f(\underset{\sim}{x}) > 0$ at every $\underset{\sim}{x} \neq \underset{\sim}{0}$, so the origin is the only lowest point of the bowl. Figure 3 shows the surface.

(This 3D surface is interactive; see the HTML or slide version of this page to view it.)

Figure 3: The surface $z = x_1^2 + x_2^2$ over $[-2, 2]^2$: a bowl that opens upward, with its lowest point above the origin. Horizontal slices of the bowl are circles around the origin.

Definition 0.9 (Matrix-induced inner product ($\mathbf{A}$-inner product))**.** Let $\mathbf{A}$ be a $p \times p$ symmetric positive definite matrix (Definition 0.2) ($\mathbf{A} = \mathbf{A}^\top$ and $\underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{x} > 0$ for all nonzero $\underset{\sim}{x} \in \mathbb{R}^p$). The **matrix-induced inner product** (or **$\mathbf{A}$-inner product**) on $\mathbb{R}^p$ is defined for all $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^p$ by

$$\langle \underset{\sim}{x}, \underset{\sim}{y} \rangle_{\mathbf{A}} \stackrel{\text{def}}{=} \underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{y}.$$

Example 0.11 (A matrix-induced inner product calculation)**.** Consider the symmetric matrix

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

For any nonzero vector $\underset{\sim}{x} = (x_1, x_2) \in \mathbb{R}^2$:

$$\begin{aligned} \underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{x} &= 2x_1^2 + 2x_1x_2 + 2x_2^2 && \text{(expand the quadratic form)} \\ &= (x_1 + x_2)^2 + x_1^2 + x_2^2 && \text{(complete the square)} \end{aligned}$$

which is strictly positive because $x_1^2 + x_2^2 > 0$ when $\underset{\sim}{x} \neq \underset{\sim}{0}$. Thus $\mathbf{A}$ is positive definite (Definition 0.2).

For vectors $\underset{\sim}{x} = (1, 2)$ and $\underset{\sim}{y} = (3, -1)$, their inner product induced by $\mathbf{A}$ is:

$$\begin{aligned} \langle \underset{\sim}{x}, \underset{\sim}{y} \rangle_{\mathbf{A}} &= \underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{y} && \text{(definition of } \langle \cdot, \cdot \rangle_{\mathbf{A}}) \\ &= \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} && \text{(substitute vectors and matrix)} \\ &= \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \cdot 3 + 1 \cdot (-1) \\ 1 \cdot 3 + 2 \cdot (-1) \end{bmatrix} && \text{(Definition 0.16)} \\ &= \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \end{bmatrix} && \text{(multiply and add)} \\ &= 1 \cdot 5 + 2 \cdot 1 && \text{(Definition 0.16)} \\ &= 7. && \text{(multiply and add)} \end{aligned}$$

In contrast, the standard dot product yields:

$$\begin{aligned} \underset{\sim}{x} \cdot \underset{\sim}{y} &= 1 \cdot 3 + 2 \cdot (-1) \\ &= 1. \end{aligned}$$

The off-diagonal entries

$$\begin{aligned} A_{12} &= A_{21} \\ &= 1 \end{aligned}$$

couple the coordinate directions, altering lengths and angles from their Euclidean values. When $\mathbf{A} = \mathbf{I}_p$, the matrix-induced inner product recovers the standard dot product $\langle \underset{\sim}{x}, \underset{\sim}{y} \rangle_{\mathbf{I}_p} = \underset{\sim}{x} \cdot \underset{\sim}{y}$.

Theorem 0.5 (Properties of the matrix-induced inner product)**.** *For every $p \times p$ symmetric positive definite matrix $\mathbf{A}$, the function $\langle \underset{\sim}{x}, \underset{\sim}{y} \rangle_{\mathbf{A}} = \underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{y}$ satisfies the four inner product axioms (Definition 0.5): positivity, definiteness, linearity in the first slot, and symmetry. Thus $(\mathbb{R}^p, \langle \cdot, \cdot \rangle_{\mathbf{A}})$ is an inner product space (Definition 0.6).*

Taking $\mathbf{A} = \mathbf{I}_p$ recovers the standard dot product (Example 0.1), while taking $\mathbf{A} = \operatorname{diag}(c_1, \ldots, c_p)$ recovers the weighted inner product (Example 0.3).

Conversely, every inner product on $\mathbb{R}^p$ has this form for a unique symmetric positive definite matrix $\mathbf{A}$, whose (i, j) entry is $A_{ij} = \langle \underset{\sim}{e}_i, \underset{\sim}{e}_j \rangle$, where $\underset{\sim}{e}_1, \ldots, \underset{\sim}{e}_p$ are the standard basis vectors (Definition 0.18).

Proof

Proof.

- **positivity**: $\langle \underset{\sim}{x}, \underset{\sim}{x} \rangle_{\mathbf{A}} = \underset{\sim}{x}^{\top} \mathbf{A} \underset{\sim}{x} \geq 0$ for all $\underset{\sim}{x} \in \mathbb{R}^p$, because $\mathbf{A}$ is positive definite (Definition 0.2);
- **definiteness**: $\langle \underset{\sim}{x}, \underset{\sim}{x} \rangle_{\mathbf{A}} = 0$ if and only if $\underset{\sim}{x} = \underset{\sim}{0}_{p \times 1}$, by definition of a positive definite matrix (Definition 0.2);
- **linearity in the first slot**: for all $\underset{\sim}{x}, \underset{\sim}{y}, \underset{\sim}{z} \in \mathbb{R}^p$ and real numbers a, b,

$$\begin{aligned}
\langle a \underset{\sim}{x} + b \underset{\sim}{y}, \underset{\sim}{z} \rangle_{\mathbf{A}} &= (a \underset{\sim}{x} + b \underset{\sim}{y})^{\top} \mathbf{A} \underset{\sim}{z} && \text{(definition of } \langle \cdot, \cdot \rangle_{\mathbf{A}}\text{)} \\
&= (a \underset{\sim}{x}^{\top} + b \underset{\sim}{y}^{\top}) \mathbf{A} \underset{\sim}{z} && \text{(Theorem 0.18, Definition 0.9, Definition 0.6)} \\
&= a \underset{\sim}{x}^{\top} \mathbf{A} \underset{\sim}{z} + b \underset{\sim}{y}^{\top} \mathbf{A} \underset{\sim}{z} && \text{(Theorem 0.8)} \\
&= a \langle \underset{\sim}{x}, \underset{\sim}{z} \rangle_{\mathbf{A}} + b \langle \underset{\sim}{y}, \underset{\sim}{z} \rangle_{\mathbf{A}} && \text{(definition of } \langle \cdot, \cdot \rangle_{\mathbf{A}}\text{)}
\end{aligned}$$

- **symmetry**: because $\underset{\sim}{y}^{\top} \mathbf{A} \underset{\sim}{x}$ is a 1×1 scalar and $\mathbf{A}$ is symmetric ($\mathbf{A}^{\top} = \mathbf{A}$),

$$\begin{aligned}
\langle \underset{\sim}{y}, \underset{\sim}{x} \rangle_{\mathbf{A}} &= \underset{\sim}{y}^{\top} \mathbf{A} \underset{\sim}{x} && \text{(definition of } \langle \cdot, \cdot \rangle_{\mathbf{A}}\text{)} \\
&= (\underset{\sim}{y}^{\top} \mathbf{A} \underset{\sim}{x})^{\top} && \text{(a scalar equals its transpose)} \\
&= \underset{\sim}{x}^{\top} \mathbf{A}^{\top} \underset{\sim}{y} && \text{(Theorem 0.19, twice)} \\
&= \underset{\sim}{x}^{\top} \mathbf{A} \underset{\sim}{y} && (\mathbf{A} \text{ is symmetric}) \\
&= \langle \underset{\sim}{x}, \underset{\sim}{y} \rangle_{\mathbf{A}} && \text{(definition of } \langle \cdot, \cdot \rangle_{\mathbf{A}}\text{)}
\end{aligned}$$

For the converse, let $\langle \cdot, \cdot \rangle$ be any inner product on $\mathbb{R}^p$. Any vectors $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^p$ can be written in the standard basis as $\underset{\sim}{x} = \sum_{i=1}^{p} x_i \, \underset{\sim}{e}_i$ and $\underset{\sim}{y} = \sum_{j=1}^{p} y_j \, \underset{\sim}{e}_j$ (Definition 0.3, Definition 0.18). By linearity in both slots (Remark 0.1):

$$\begin{aligned}
\langle \underset{\sim}{x}, \underset{\sim}{y} \rangle &= \left\langle \sum_{i=1}^{p} x_i \, \underset{\sim}{e}_i, \sum_{j=1}^{p} y_j \, \underset{\sim}{e}_j \right\rangle && \text{(expand in the standard basis)} \\
&= \sum_{i=1}^{p} \sum_{j=1}^{p} x_i y_j \, \langle \underset{\sim}{e}_i, \underset{\sim}{e}_j \rangle && \text{(linearity in both slots)} \\
&= \sum_{i=1}^{p} x_i \sum_{j=1}^{p} A_{ij} y_j && (\text{define } A_{ij} \overset{\text{def}}{=} \langle \underset{\sim}{e}_i, \underset{\sim}{e}_j \rangle) \\
&= \underset{\sim}{x}^{\top} \mathbf{A} \underset{\sim}{y}. && \text{(Definition 0.16)}
\end{aligned}$$

Symmetry of $\langle \cdot, \cdot \rangle$ ensures

$$\begin{aligned}
A_{ji} &= \langle \underset{\sim}{e}_j, \underset{\sim}{e}_i \rangle \\
&= \langle \underset{\sim}{e}_i, \underset{\sim}{e}_j \rangle \\
&= A_{ij},
\end{aligned}$$

so $\mathbf{A}$ is symmetric (Definition 0.3). For any $\underset{\sim}{x} \neq \underset{\sim}{0}$, $\underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{x} = \langle \underset{\sim}{x}, \underset{\sim}{x} \rangle > 0$ by positivity and definiteness of the inner product, so $\mathbf{A}$ is positive definite (Definition 0.2).
For uniqueness, if another matrix $\mathbf{B}$ satisfies $\langle \underset{\sim}{x}, \underset{\sim}{y} \rangle = \underset{\sim}{x}^\top \mathbf{B} \underset{\sim}{y}$ for all $\underset{\sim}{x}, \underset{\sim}{y}$, evaluating at $\underset{\sim}{x} = \underset{\sim}{e}_i$ and $\underset{\sim}{y} = \underset{\sim}{e}_j$ gives

$$\begin{aligned} B_{ij} &= \underset{\sim}{e}_i{}^\top \mathbf{B} \underset{\sim}{e}_j \\ &= \langle \underset{\sim}{e}_i, \underset{\sim}{e}_j \rangle \\ &= A_{ij}, \end{aligned}$$

so $\mathbf{B} = \mathbf{A}$. □

Example 0.12 (Checking the inner product axioms, and recovering the matrix from the inner product)**.** Take $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ from Example 0.11, the vectors $\underset{\sim}{x} = (1, 2)$, $\underset{\sim}{y} = (3, -1)$ and $\underset{\sim}{z} = (0, 4)$, and the numbers $a = 2$ and $b = -3$. The code computes both sides of each axiom in Theorem 0.5, and then the numbers $\langle \underset{\sim}{e}_i, \underset{\sim}{e}_j \rangle_{\mathbf{A}}$ for the standard basis vectors.

```r
A <- matrix(c(2, 1, 1, 2), nrow = 2)
ip <- function(x, y) drop(t(x) %*% A %*% y)

x <- c(1, 2)
y <- c(3, -1)
z <- c(0, 4)
a <- 2
b <- -3

c(
  positivity = ip(x, x),
  symmetry_xy = ip(x, y),
  symmetry_yx = ip(y, x),
  linearity_left = ip(a * x + b * y, z),
  linearity_right = a * ip(x, z) + b * ip(y, z)
)
#>      positivity     symmetry_xy     symmetry_yx  linearity_left linearity_right
#>              14               7               7              28              28

e <- diag(2)
recovered <- outer(1:2, 1:2, Vectorize(function(i, j) ip(e[, i], e[, j])))
recovered
#>      [,1] [,2]
#> [1,]    2    1
#> [2,]    1    2
```

The inner product of $\underset{\sim}{x}$ with itself is 14, which is positive. The two orders give the same number, 7 and 7. Both sides of the linearity axiom equal 28. The matrix of numbers $\left\langle \underset{\sim}{e}_i, \underset{\sim}{e}_j \right\rangle_{\mathbf{A}}$ is equal to $\mathbf{A}$, as the converse in Theorem 0.5 says.

Corollary 0.1 (A positive definite matrix is a transpose product)**.** *Let* $\mathbf{A}$ *be a* $p \times p$ *symmetric positive definite matrix (Definition 0.2). Then* $\mathbf{A} = \mathbf{W}^\top \mathbf{W}$ *for some invertible* $p \times p$ *matrix* $\mathbf{W}$*.*

Proof

Proof. By the spectral theorem (Theorem 0.1), $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$ with $\mathbf{Q}$ orthogonal and $\mathbf{\Lambda}$ diagonal with the eigenvalues λ_i of $\mathbf{A}$ on its diagonal. These are positive (Theorem 0.2), so $\mathbf{\Lambda}^{1/2}$, the diagonal matrix with entries $\sqrt{\lambda_i}$, is defined. Take $\mathbf{W} = \mathbf{\Lambda}^{1/2}\mathbf{Q}^\top$. Since $\mathbf{\Lambda}^{1/2}$ is diagonal, $\left(\mathbf{\Lambda}^{1/2}\right)^\top = \mathbf{\Lambda}^{1/2}$, and

$$\begin{aligned}\mathbf{W}^\top\mathbf{W} &= \mathbf{Q}\mathbf{\Lambda}^{1/2}\mathbf{\Lambda}^{1/2}\mathbf{Q}^\top \\ &= \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top \\ &= \mathbf{A}.\end{aligned}$$

$\mathbf{W}$ is invertible, because $\mathbf{Q}^\top$ has inverse $\mathbf{Q}$ and $\mathbf{\Lambda}^{1/2}$ is diagonal with positive entries, so it has the diagonal inverse with entries $1/\sqrt{\lambda_i}$, and a product of invertible matrices is invertible (Theorem 0.1). □

Theorem 0.6 (A matrix-induced inner product is a dot product of transformed vectors)**.** *Let* $\mathbf{A}$ *be a* $p \times p$ *symmetric positive definite matrix, and let* $\mathbf{W}$ *be any* $p \times p$ *matrix with* $\mathbf{A} = \mathbf{W}^\top\mathbf{W}$*. Then for all* $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^p$*, the* $\mathbf{A}$*-inner product (Definition 0.9) satisfies*

$$\langle \underset{\sim}{x}, \underset{\sim}{y} \rangle_{\mathbf{A}} = (\mathbf{W}\underset{\sim}{x}) \cdot (\mathbf{W}\underset{\sim}{y}).$$

Proof

Proof.

$$\begin{aligned}\langle \underset{\sim}{x}, \underset{\sim}{y} \rangle_{\mathbf{A}} &= \underset{\sim}{x}^\top\mathbf{W}^\top\mathbf{W}\underset{\sim}{y} \\ &= (\mathbf{W}\underset{\sim}{x})^\top(\mathbf{W}\underset{\sim}{y}) \\ &= (\mathbf{W}\underset{\sim}{x}) \cdot (\mathbf{W}\underset{\sim}{y}).\end{aligned}$$

□

Remark (Inner product spaces in machine learning)

Remark 0.3 (Inner product spaces in machine learning)*.* In machine learning and statistics, choosing or learning an inner product space tailors geometric notions of distance, angle, and projection to the structure of the data:

1. **Mahalanobis metric and whitening**: Standard Euclidean distance treats all coordinate axes identically, ignoring feature correlations and variance differences. If data features have a symmetric positive definite covariance matrix $\mathbf{\Sigma}$ (Definition 0.2), the precision matrix $\mathbf{A} = \mathbf{\Sigma}^{-1}$ (Theorem 0.4) defines the Mahalanobis inner product $\langle \underset{\sim}{x}, \underset{\sim}{y} \rangle_{\mathbf{\Sigma}^{-1}} = \underset{\sim}{x}^\top\mathbf{\Sigma}^{-1}\underset{\sim}{y}$. The induced distance $d_{\mathbf{\Sigma}^{-1}}(\underset{\sim}{x}, \underset{\sim}{y}) = \sqrt{(\underset{\sim}{x} - \underset{\sim}{y})^\top\mathbf{\Sigma}^{-1}(\underset{\sim}{x} - \underset{\sim}{y})}$ measures statistical distance normalized by variance along each principal direction. Factoring $\mathbf{\Sigma}^{-1} = \mathbf{W}^\top\mathbf{W}$ (via Cholesky or spectral decomposition, Corollary 0.1), this inner product is the Euclidean dot product of the whitened vectors $\mathbf{W}\underset{\sim}{x}$ and $\mathbf{W}\underset{\sim}{y}$ (Theorem 0.6).
2. **The kernel trick and reproducing kernel Hilbert spaces (RKHS)**: Al-

gorithms such as support vector machines (SVMs), kernel ridge regression, and Gaussian processes depend on training vectors solely through pairwise inner products $\langle \underset{\sim}{x}_i, \underset{\sim}{x}_j \rangle$. By mapping data into a higher- or infinite-dimensional inner product space via a feature map $D : \mathbb{R}^p \to \mathcal{H}$, the kernel trick replaces the computation of $D(\underset{\sim}{x}_i)$ with an evaluation of a positive definite kernel function:

$$k(\underset{\sim}{x}_i, \underset{\sim}{x}_j) \stackrel{\text{def}}{=} \left\langle D(\underset{\sim}{x}_i), D(\underset{\sim}{x}_j) \right\rangle_{\mathcal{H}}.$$

Mercer's theorem guarantees that every symmetric positive definite kernel function implicitly defines a valid inner product space $\mathcal{H}$ (a reproducing kernel Hilbert space) without requiring explicit coordinate representations.

3. **Embeddings and cosine similarity**: In natural language processing and recommendation systems, data items are embedded as vectors in high-dimensional representations. The angle between embeddings captures semantic similarity through cosine similarity (Definition 0.6):

$$\operatorname{cossim}(\underset{\sim}{x}, \underset{\sim}{y}) = \frac{\langle \underset{\sim}{x}, \underset{\sim}{y} \rangle}{\|\underset{\sim}{x}\| \, \|\underset{\sim}{y}\|}.$$

When embedding vectors are normalized to unit Euclidean norm ($\|\underset{\sim}{x}\| = 1$ and $\|\underset{\sim}{y}\| = 1$), cosine similarity equals the inner product $\langle \underset{\sim}{x}, \underset{\sim}{y} \rangle$ (Corollary 0.2). Similarly, the self-attention mechanism in Transformers computes query-key alignment scores through scaled inner products $\underset{\sim}{q} \cdot \underset{\sim}{k} / \sqrt{d_k}$.

4. **Weighted and generalized least squares**: In linear regression $y \approx \mathbf{X}\beta$, when observation errors have unequal variances or known correlations with positive definite covariance matrix $\mathbf{\Omega}$, ordinary least squares is inefficient. Generalized least squares minimizes the residual norm in the inner product space induced by $\mathbf{W} = \mathbf{\Omega}^{-1}$:

$$\min_{\underset{\sim}{\beta}} \|\underset{\sim}{y} - \mathbf{X}\underset{\sim}{\beta}\|_{\mathbf{W}}^2 = \min_{\underset{\sim}{\beta}} \langle \underset{\sim}{y} - \mathbf{X}\underset{\sim}{\beta}, \underset{\sim}{y} - \mathbf{X}\underset{\sim}{\beta} \rangle_{\mathbf{W}}.$$

Determinants

Definition 0.1 (Determinant)**.** The **determinant** [81] of a $p \times p$ matrix $\mathbf{A}$ with entries a_{ij}, written $\det(\mathbf{A})$ or $|\mathbf{A}|$, is the number defined recursively in p:

- For $p = 1$, $\det(\mathbf{A}) = a_{11}$.

- For $p \geq 2$,
$$\det(\mathbf{A}) = \sum_{j=1}^{p} (-1)^{1+j}\, a_{1j} \det\big(\mathbf{A}_{(1j)}\big),$$
where $\mathbf{A}_{(1j)}$ is the $(p-1) \times (p-1)$ matrix left after deleting row 1 and column j of $\mathbf{A}$.

Example 0.1 (Determinant of a 2×2 matrix)**.** For $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, deleting row 1 and column 1 leaves $\big[d\big]$, and deleting row 1 and column 2 leaves $\big[c\big]$. So:

$$\begin{aligned} \det(\mathbf{A}) &= (-1)^{1+1}\, a \det\big(\big[d\big]\big) + (-1)^{1+2}\, b \det\big(\big[c\big]\big) && \text{(definition, } p = 2\text{)} \\ &= ad - bc && \text{(definition, } p = 1\text{)} \end{aligned}$$

For example,

$$\begin{aligned} \det\left(\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\right) &= 2 \cdot 2 - 1 \cdot 1 \\ &= 3. \end{aligned}$$

Remark (Cofactor expansion)

Remark 0.1 (Cofactor expansion). The recursive formula in Definition 0.1 is the *cofactor expansion* along the first row. Other sources define the determinant as a sum over all orderings of the columns and derive this expansion from it; the two definitions agree [48]. Each ordering picks one entry from each row, from the column that the ordering assigns to that row, and multiplies them. Its sign is $+1$ if the ordering takes an even number of swaps of two columns to reach from $(1, 2, \dots, p)$, and -1 if it takes an odd number.
For $p = 2$, the columns have two orderings. The ordering $(1, 2)$ keeps each column in place, picks the entries a_{11} and a_{22}, and has sign $+1$. The ordering $(2, 1)$ swaps the two columns, picks the entries a_{12} and a_{21}, and has sign -1. The sum $a_{11}a_{22} - a_{12}a_{21}$ is the value $ad - bc$ from Example 0.1.

See also [81].

Video

Chapter 6 of the *Essence of linear algebra* series by 3Blue1Brown shows the geometric meaning of the determinant:

- The determinant shows how the determinant measures the factor by which a linear transformation scales areas and volumes, and why a zero determinant corresponds

to compressing space into a lower dimension.

Theorem 0.1 (Determinant of a diagonal matrix)**.** *The determinant (Definition 0.1) of a $p \times p$ diagonal matrix (Definition 0.4) is the product of its diagonal entries:*

$$\det\left(\begin{bmatrix} d_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & d_p \end{bmatrix}\right) = \prod_{i=1}^{p} d_i$$

Proof

Proof. By induction on p. For $p = 1$, the matrix is $\begin{bmatrix} d_1 \end{bmatrix}$, and its determinant is d_1 by definition.
For $p \geq 2$, suppose the result holds for $(p-1) \times (p-1)$ diagonal matrices, and let $\mathbf{D}$ be $p \times p$ and diagonal. Row 1 of $\mathbf{D}$ is $(d_1, 0, \ldots, 0)$, so only the $j = 1$ term of the definition can be nonzero, and deleting row 1 and column 1 of $\mathbf{D}$ leaves the diagonal matrix with diagonal entries $d_2, \ldots, d_p$:

$$\begin{aligned}
\det(\mathbf{D}) &= (-1)^{1+1} d_1 \det\left(\mathbf{D}_{(11)}\right) + \sum_{j=2}^{p} (-1)^{1+j} \cdot 0 \cdot \det\left(\mathbf{D}_{(1j)}\right) && \text{(definition)} \\
&= d_1 \det\left(\mathbf{D}_{(11)}\right) && \text{(drop the zero terms)} \\
&= d_1 \prod_{i=2}^{p} d_i && \text{(induction hypothesis)} \\
&= \prod_{i=1}^{p} d_i && \text{(combine)}
\end{aligned}$$

□

Example 0.2 (Determinant of a scaled identity matrix)**.** For a number c, $c\,\mathbf{I}_p$ is diagonal with every diagonal entry equal to c, so $\det(c\,\mathbf{I}_p) = c^p$. In particular, $\det(\mathbf{I}_p) = 1$.

Theorem 0.2 (Determinant of a product)**.** *For $p \times p$ matrices $\mathbf{A}$ and $\mathbf{B}$,*

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B}).$$

Example 0.3 (Checking the product rule)**.** For $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$, Example 0.1 gives

$$\begin{aligned} \det(\mathbf{A}) &= 2 \cdot 1 - 1 \cdot 0 \\ &= 2 \end{aligned}$$

and

$$\begin{aligned} \det(\mathbf{B}) &= 1 \cdot 1 - 0 \cdot 3 \\ &= 1. \end{aligned}$$

The product is

$$\begin{aligned} \mathbf{AB} &= \begin{bmatrix} 2 \cdot 1 + 1 \cdot 3 & 2 \cdot 0 + 1 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 3 & 0 \cdot 0 + 1 \cdot 1 \end{bmatrix} \\ &= \begin{bmatrix} 5 & 1 \\ 3 & 1 \end{bmatrix}, \end{aligned}$$

with

$$\begin{aligned} \det(\mathbf{AB}) &= 5 \cdot 1 - 1 \cdot 3 \\ &= 2 \\ &= \det(\mathbf{A}) \det(\mathbf{B}). \end{aligned}$$

Remark (An example is not a proof)

Remark 0.2 (An example is not a proof). Example 0.3 checks Theorem 0.2 for one pair of 2×2 matrices, which shows the theorem holds there but does not prove it for every pair. The proof needs properties of the determinant that these notes don't develop; see Banerjee and Roy [48].

See also [81].

Theorem 0.3 (The determinant of a symmetric matrix is the product of its eigenvalues)**.** *Let* $\mathbf{A}$ *be a* $p \times p$ *symmetric matrix with real entries, with eigendecomposition* $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$ *(Definition 0.3) and eigenvalues* $\lambda_1, \ldots, \lambda_p$ *on the diagonal of* $\mathbf{\Lambda}$*. Then*

$$\det(\mathbf{A}) = \prod_{i=1}^{p} \lambda_i.$$

Proof

Proof. First, $\det(\mathbf{Q}^\top)\det(\mathbf{Q}) = 1$:

$$\begin{aligned}
\det(\mathbf{Q}^\top)\det(\mathbf{Q}) &= \det(\mathbf{Q}^\top\mathbf{Q}) && \text{(determinant of a product)} \\
&= \det(\mathbf{I}_p) && (\mathbf{Q} \text{ is orthogonal}) \\
&= 1 && \text{(determinant of the identity)}
\end{aligned}$$

Then:

$$\begin{aligned}
\det(\mathbf{A}) &= \det(\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top) && \text{(substitute the eigendecomposition)} \\
&= \det(\mathbf{Q})\det(\mathbf{\Lambda})\det(\mathbf{Q}^\top) && \text{(determinant of a product, twice)} \\
&= \det(\mathbf{\Lambda}) \cdot \det(\mathbf{Q}^\top)\det(\mathbf{Q}) && \text{(reorder the three numbers)} \\
&= \det(\mathbf{\Lambda}) && \text{(the first display)} \\
&= \prod_{i=1}^{p} \lambda_i && \text{(determinant of a diagonal matrix)}
\end{aligned}$$

The product steps are Theorem 0.2, the orthogonality step is Definition 0.10, and the identity and diagonal steps are Theorem 0.1 and Example 0.2. □

Corollary 0.1 (A positive definite matrix has a positive determinant)**.** *If* $\mathbf{A}$ *is positive definite (Definition 0.2), then* $\det(\mathbf{A}) > 0$.

Proof

Proof. By Theorem 0.2, every eigenvalue of $\mathbf{A}$ is positive, so their product, which is $\det(\mathbf{A})$ by Theorem 0.3, is positive. □

Example 0.4 (Two ways to the same determinant)**.** $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ has eigenvalues 3 and 1 (Example 0.1), so Theorem 0.3 gives

$$\begin{aligned}
\det(\mathbf{A}) &= 3 \cdot 1 \\
&= 3,
\end{aligned}$$

matching $2 \cdot 2 - 1 \cdot 1 = 3$ from Example 0.1.

Design Matrix

Definition 0.1.

Design matrix, model matrix
In a regression model with n observations and p predictors, the **design matrix** [82] (or **model matrix**) $\mathbf{X}$ is the $n \times p$ matrix whose i-th row $\underset{\sim}{x}_i^\top$ holds, for observation i, the p entries the model uses. These can be covariates, a 1 for the intercept, 0/1 columns for categories, or functions of covariates, such as x^2. For a straight-line fit, for example, the design matrix has a first column of 1s, which allows the intercept to be estimated [3, Sec. 2.4], [82]. In symbols,

$$\mathbf{X} = \begin{bmatrix} \underset{\sim}{x}_1^\top \\ \underset{\sim}{x}_2^\top \\ \vdots \\ \underset{\sim}{x}_n^\top \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

Remark (Products with the design matrix)

Remark 0.1 (Products with the design matrix). The product $\mathbf{X}\underset{\sim}{\beta}$ collects the values $\underset{\sim}{x}_i^\top \underset{\sim}{\beta}$ for all n observations into a single $n \times 1$ vector:

$$\mathbf{X}\underset{\sim}{\beta} = \begin{bmatrix} \underset{\sim}{x}_1^\top \underset{\sim}{\beta} \\ \vdots \\ \underset{\sim}{x}_n^\top \underset{\sim}{\beta} \end{bmatrix}$$

For example, take the rank-2 matrix from Example 0.2 as $\mathbf{X}$, with $n = 3$ observations and $p = 2$ columns: an intercept column and one covariate. With $\underset{\sim}{\beta} = (1, 2)$,

$$\mathbf{X}\underset{\sim}{\beta} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}.$$

The matrix $\mathbf{X}^\top\mathbf{X}$ is $p \times p$ and symmetric, since

$$
\begin{aligned}
\left(\mathbf{X}^\top \mathbf{X}\right)^\top &= \mathbf{X}^\top \left(\mathbf{X}^\top\right)^\top \\
&= \mathbf{X}^\top \mathbf{X}
\end{aligned}
$$

(Theorem 0.19). When $\mathbf{X}^\top \mathbf{X}$ is invertible, it appears in the least squares solution (Definition 0.10, defined later in this chapter) $\underset{\sim}{\hat{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \underset{\sim}{y}$.

Theorem 0.1 ($\mathbf{X}^\top \mathbf{X}$ is invertible when $\mathbf{X}$ has full column rank)**.** *If $\mathbf{X}$ is an $n \times p$ matrix with $\operatorname{rank}(\mathbf{X}) = p$ (Definition 0.2), then the $p \times p$ matrix $\mathbf{X}^\top \mathbf{X}$ is invertible.*

Proof

Proof. Let $\underset{\sim}{c}$ be a vector of length p with $\mathbf{X}^\top \mathbf{X} \underset{\sim}{c} = \underset{\sim}{0}$. Then

$$
\begin{aligned}
0 = \underset{\sim}{c}^\top \mathbf{X}^\top \mathbf{X} \underset{\sim}{c} \quad & \text{(multiply } \mathbf{X}^\top \mathbf{X} \underset{\sim}{c} = \underset{\sim}{0} \text{ on the left by } \underset{\sim}{c}^\top\text{)} \\
= (\mathbf{X} \underset{\sim}{c})^\top (\mathbf{X} \underset{\sim}{c}) \quad & \text{(transpose of a product)} \\
= \|\mathbf{X} \underset{\sim}{c}\|^2 \quad & \text{(definition of the Euclidean norm)}
\end{aligned}
$$

so $\mathbf{X} \underset{\sim}{c} = \underset{\sim}{0}$. Because $\mathbf{X} \underset{\sim}{c} = c_1(\text{column } 1) + \cdots + c_p(\text{column } p)$ and the columns of $\mathbf{X}$ are linearly independent, $\underset{\sim}{c} = \underset{\sim}{0}$. So the only solution of $\mathbf{X}^\top \mathbf{X} \underset{\sim}{c} = \underset{\sim}{0}$ is $\underset{\sim}{c} = \underset{\sim}{0}$, which says that the columns of the square matrix $\mathbf{X}^\top \mathbf{X}$ are linearly independent, and a square matrix with linearly independent columns is invertible [48], Corollary 5.7, p. 143. □

Example 0.1 (Inverting $\mathbf{X}^\top \mathbf{X}$)**.** For the rank-2 matrix $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ from Example 0.2:

$$
\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}, \qquad \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix},
$$

and multiplying the two out gives $\mathbf{I}_2$.

Definition 0.2 (Hat matrix)**.** For an $n \times p$ design matrix $\mathbf{X}$ (Definition 0.1) with $\operatorname{rank}(\mathbf{X}) = p$, the **hat matrix** [83] is the $n \times n$ matrix

$$
\underset{n \times n}{\mathbf{H}} \overset{\text{def}}{=} \underset{n \times p}{\mathbf{X}} \underbrace{(\mathbf{X}^\top \mathbf{X})^{-1}}_{p \times p} \underset{p \times n}{\mathbf{X}^\top}
$$

The inverse exists by Theorem 0.1.

Example 0.2 (The hat matrix of an intercept-only model)**.** With $n = 2$ observations and only an intercept, $\mathbf{X} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ (2×1, rank 1), so $\mathbf{X}^\top \mathbf{X} = 2$ and

$$\begin{aligned} \mathbf{H} &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \frac{1}{2} \cdot \begin{bmatrix} 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}. \end{aligned}$$

Then $\mathbf{H}y = (\bar{y}, \bar{y})$, where $\bar{y} = (y_1 + y_2)/2$: the fitted values of an intercept-only model are the sample mean.

Theorem 0.2 (Hat matrix is a projection matrix)**.** *If* $\mathbf{X}$ *is an* $n \times p$ *design matrix with* $\operatorname{rank}(\mathbf{X}) = p$*, then the hat matrix* $\mathbf{H}$ *(Definition 0.2) is an orthogonal projection matrix (Definition 0.8).*

Proof

Proof. We verify symmetry and idempotency. Both use that $\mathbf{X}^\top \mathbf{X}$ is symmetric,

$$\begin{aligned} \left(\mathbf{X}^\top \mathbf{X}\right)^\top &= \mathbf{X}^\top \left(\mathbf{X}^\top\right)^\top \\ &= \mathbf{X}^\top \mathbf{X} \end{aligned}$$

(Theorem 0.19), so its inverse is symmetric too (Corollary 0.1).
Symmetry:

$$\begin{aligned} \mathbf{H}^\top &= \left(\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top\right)^\top && \text{(definition of } \mathbf{H}) \\ &= \left(\mathbf{X}^\top\right)^\top \cdot \left((\mathbf{X}^\top \mathbf{X})^{-1}\right)^\top \cdot \mathbf{X}^\top && \text{(transpose of a product, twice)} \\ &= \mathbf{X} \cdot \left((\mathbf{X}^\top \mathbf{X})^{-1}\right)^\top \cdot \mathbf{X}^\top && \text{(transposing twice changes nothing)} \\ &= \mathbf{X} \cdot (\mathbf{X}^\top \mathbf{X})^{-1} \cdot \mathbf{X}^\top && \text{(the inverse of a symmetric matrix is symmetric)} \\ &= \mathbf{H} && \text{(definition of } \mathbf{H}) \end{aligned}$$

Idempotency:

$$\begin{aligned} \mathbf{H}^2 &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \cdot \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(definition of } \mathbf{H}) \\ &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{X}^\top \mathbf{X}) (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(regroup; matrix multiplication is associative)} \\ &= \mathbf{X}\, \mathbf{I}_p\, (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(definition of the inverse)} \\ &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(identity matrix)} \\ &= \mathbf{H} && \text{(definition of } \mathbf{H}) \end{aligned}$$

□

Example 0.3 (The intercept-only hat matrix is a projection)**.** For $\mathbf{H} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$ from Example 0.2, $\mathbf{H}^\top = \mathbf{H}$, and

$$\begin{aligned} \mathbf{H}^2 &= \begin{bmatrix} 0.5 \cdot 0.5 + 0.5 \cdot 0.5 & 0.5 \cdot 0.5 + 0.5 \cdot 0.5 \\ 0.5 \cdot 0.5 + 0.5 \cdot 0.5 & 0.5 \cdot 0.5 + 0.5 \cdot 0.5 \end{bmatrix} \\ &= \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} \\ &= \mathbf{H}, \end{aligned}$$

as Theorem 0.2 says.

Remark (Why it is called the hat matrix)

Remark 0.2 (Why it is called the hat matrix). The hat matrix gives the fitted values of the least squares solution (Definition 0.10, defined later in this chapter):

$$\begin{aligned} \widehat{\underset{\sim}{y}} &= \mathbf{X}\widehat{\beta} \\ &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \underset{\sim}{y} \\ &= \mathbf{H}\underset{\sim}{y}. \end{aligned}$$

Multiplying by $\mathbf{H}$ "puts a hat" on y, which is where the name comes from. In Example 0.2, $\mathbf{H}$ puts a hat on $\underset{\sim}{y}$ by replacing each y_i with the sample mean $\bar{y}$.

Orthogonal projection onto a subspace

Source

This section is adapted from the second half of Zhou [45], used under the MIT License (see the license text in Section). It connects the orthogonal projection matrices of Definition 0.8, defined by an algebraic property, to the geometric decomposition of Theorem 0.4.

Definition 0.3 (Orthogonal projection onto a subspace)**.** Let $\mathcal{S}$ be a subspace of $\mathbb{R}^p$ and $\underset{\sim}{y} \in \mathbb{R}^p$. By Theorem 0.4, $\underset{\sim}{y} = \underset{\sim}{u} + \underset{\sim}{v}$ for exactly one $\underset{\sim}{u} \in \mathcal{S}$ and $\underset{\sim}{v} \in \mathcal{S}^\perp$. The vector $\underset{\sim}{u}$ is the **orthogonal projection** of $\underset{\sim}{y}$ onto $\mathcal{S}$. It is the projection of $\underset{\sim}{y}$ onto $\mathcal{S}$ along $\mathcal{S}^\perp$ (Definition 0.4).

Example 0.4 (Projecting onto a line in $\mathbb{R}^3$)**.** In Example 0.5, $y = (3,1,2)$ splits as $(2,2,0) + (1,-1,2)$, with $(2,2,0) \in \mathcal{S} = \text{span}\{(1,1,0)\}$ and $(1,-1,2) \in \mathcal{S}^\perp$. So the orthogonal projection of $(3,1,2)$ onto $\mathcal{S}$ is $(2,2,0)$. A vector already in $\mathcal{S}$, such as $(5,5,0)$, is its own projection, since $(5,5,0) = (5,5,0) + \underset{\sim}{0}$ and $\underset{\sim}{0} \in \mathcal{S}^\perp$ (Theorem 0.2, Theorem 0.1); a vector of $\mathcal{S}^\perp$, such as $(1,-1,2)$, projects to $\underset{\sim}{0}$, since $(1,-1,2) = \underset{\sim}{0} + (1,-1,2)$ and $\underset{\sim}{0} \in \mathcal{S}$ (Theorem 0.1).

Example 0.5 (A split that is not the orthogonal one)**.** $(3,1,2) = (3,3,0) + (0,-2,2)$ also writes $(3,1,2)$ as a vector of $\mathcal{S} = \text{span}\{(1,1,0)\}$ plus a remainder, but $(1,1,0)\cdot(0,-2,2) = -2 \neq 0$, so the remainder is not in $\mathcal{S}^\perp$, and $(3,3,0)$ is not the orthogonal projection of $(3,1,2)$ onto $\mathcal{S}$.

Theorem 0.3 (The orthogonal projection is the closest point)**.** *Let $\mathcal{S}$ be a subspace of $\mathbb{R}^p$, let $\underset{\sim}{y} \in \mathbb{R}^p$, and let $\underset{\sim}{u}$ be the orthogonal projection of $\underset{\sim}{y}$ onto $\mathcal{S}$ (Definition 0.3). Then for every $\underset{\sim}{w} \in \mathcal{S}$,*

$$\|\underset{\sim}{y} - \underset{\sim}{u}\| \leq \|\underset{\sim}{y} - \underset{\sim}{w}\|,$$

with equality only when $\underset{\sim}{w} = \underset{\sim}{u}$.

Proof

Proof. By Definition 0.3, $\underset{\sim}{y} - \underset{\sim}{u} \in \mathcal{S}^\perp$. Both $\underset{\sim}{u}$ and $\underset{\sim}{w}$ are in $\mathcal{S}$, so $\underset{\sim}{u} - \underset{\sim}{w} = \underset{\sim}{u} + (-1)\,\underset{\sim}{w} \in \mathcal{S}$ (Definition 0.1), and therefore $(\underset{\sim}{y} - \underset{\sim}{u}) \perp (\underset{\sim}{u} - \underset{\sim}{w})$ (Definition 0.1). Then

$$\begin{aligned}
\|\underset{\sim}{y} - \underset{\sim}{w}\|^2 &= \|(\underset{\sim}{y} - \underset{\sim}{u}) + (\underset{\sim}{u} - \underset{\sim}{w})\|^2 && \text{(add and subtract } \underset{\sim}{u}\text{)} \\
&= \|\underset{\sim}{y} - \underset{\sim}{u}\|^2 + \|\underset{\sim}{u} - \underset{\sim}{w}\|^2 && \text{(Pythagorean theorem, Theorem 0.2)} \\
&\geq \|\underset{\sim}{y} - \underset{\sim}{u}\|^2, && (\|\underset{\sim}{u} - \underset{\sim}{w}\|^2 \geq 0, \text{ Theorem 0.1})
\end{aligned}$$

with equality exactly when $\|\underset{\sim}{u} - \underset{\sim}{w}\| = 0$, that is, when $\underset{\sim}{w} = \underset{\sim}{u}$ (Theorem 0.1, part 1). Taking nonnegative square roots preserves the inequality. □

Example 0.6 (No point of the line is closer)**.** In Example 0.4, $y = (3,1,2)$ has projection $\underset{\sim}{u} = (2,2,0)$ onto $\mathcal{S} = \text{span}\{(1,1,0)\}$, at distance $\|(1,-1,2)\| = \sqrt{6}$. Another point of $\mathcal{S}$, such as $\underset{\sim}{w} = (3,3,0)$, is farther:

$$\begin{aligned}
\|\underset{\sim}{y} - \underset{\sim}{w}\| &= \|(0,-2,2)\| \\
&= \sqrt{8} \\
&> \sqrt{6},
\end{aligned}$$

and the difference of squares is

$$\begin{aligned}\|\underset{\sim}{u} - \underset{\sim}{w}\|^2 &= \|(-1,-1,0)\|^2 \\ &= 2 \\ &= 8 - 6,\end{aligned}$$

by the Pythagorean theorem (Theorem 0.2), since $\underset{\sim}{y} - \underset{\sim}{u} \in \mathcal{S}^\perp$ and $\underset{\sim}{u} - \underset{\sim}{w} \in \mathcal{S}$.

Theorem 0.4 (Projecting with an orthonormal basis)**.** *Let $\mathcal{S}$ be a subspace of $\mathbb{R}^p$ with $\dim(\mathcal{S}) = r \geq 1$, and let $\mathbf{Q}$ be the $p \times r$ matrix whose columns $\underset{\sim}{q}_1, \ldots, \underset{\sim}{q}_r$ are an orthonormal basis of $\mathcal{S}$ (Definition 0.1, Corollary 0.1). Then for every $\underset{\sim}{y} \in \mathbb{R}^p$, $\mathbf{Q}\mathbf{Q}^\top \underset{\sim}{y}$ is the orthogonal projection of $\underset{\sim}{y}$ onto $\mathcal{S}$, and $\mathbf{Q}\mathbf{Q}^\top$ is an orthogonal projection matrix (Definition 0.8).*

Proof

Proof. **$\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_r$.** Write q_{ki} for entry k of $\underset{\sim}{q}_i$, which is entry (k,i) of $\mathbf{Q}$. Then

$$\begin{aligned}(\mathbf{Q}^\top\mathbf{Q})_{ij} &= \sum_{k=1}^{p} (\mathbf{Q}^\top)_{ik}\, q_{kj} && \text{(Definition 0.10)} \\ &= \sum_{k=1}^{p} q_{ki}\, q_{kj} && \text{(Definition 0.6)} \\ &= \underset{\sim}{q}_i \cdot \underset{\sim}{q}_j, && \text{(Definition 0.8)}\end{aligned}$$

which is 1 if $i = j$ and 0 otherwise (Definition 0.23), so $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_r$ (Definition 0.15).
The split. Let $\underset{\sim}{u} = \mathbf{Q}\mathbf{Q}^\top \underset{\sim}{y}$ and $\underset{\sim}{v} = \underset{\sim}{y} - \underset{\sim}{u}$, so $\underset{\sim}{y} = \underset{\sim}{u} + \underset{\sim}{v}$. $\underset{\sim}{u} = \mathbf{Q}\,(\mathbf{Q}^\top \underset{\sim}{y})$ (Theorem 0.7) is in $\mathcal{C}(\mathbf{Q})$ (Definition 0.1), which is $\operatorname{span}\{\underset{\sim}{q}_1, \ldots, \underset{\sim}{q}_r\} = \mathcal{S}$ (Theorem 0.1). For $\underset{\sim}{v}$,

$$\begin{aligned}\mathbf{Q}^\top \underset{\sim}{v} &= \mathbf{Q}^\top (\underset{\sim}{y} - \mathbf{Q}\mathbf{Q}^\top \underset{\sim}{y}) && \text{(substitute)} \\ &= \mathbf{Q}^\top \underset{\sim}{y} - \mathbf{Q}^\top (\mathbf{Q}\mathbf{Q}^\top \underset{\sim}{y}) && \text{(Theorem 0.4, with coefficients 1 and } -1) \\ &= \mathbf{Q}^\top \underset{\sim}{y} - (\mathbf{Q}^\top\mathbf{Q})\, \mathbf{Q}^\top \underset{\sim}{y} && \text{(Theorem 0.7)} \\ &= \mathbf{Q}^\top \underset{\sim}{y} - \mathbf{I}_r\, \mathbf{Q}^\top \underset{\sim}{y} && (\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_r, \text{ from the first part of the proof}) \\ &= \mathbf{Q}^\top \underset{\sim}{y} - \mathbf{Q}^\top \underset{\sim}{y} && \text{(Theorem 0.10)} \\ &= \underset{\sim}{0}_r, && \text{(arithmetic)}\end{aligned}$$

so $\underset{\sim}{v} \in \mathcal{N}(\mathbf{Q}^\top)$, and

$$\begin{aligned}\mathcal{N}(\mathbf{Q}^\top) &= \mathcal{C}(\mathbf{Q})^\perp \\ &= \mathcal{S}^\perp\end{aligned}$$

(Theorem 0.3). By the uniqueness in Theorem 0.4, $\underset{\sim}{u}$ is the orthogonal projection of $\underset{\sim}{y}$ onto $\mathcal{S}$.

$\mathbf{QQ}^\top$ is symmetric and idempotent.

$$\begin{aligned}(\mathbf{QQ}^\top)^\top &= (\mathbf{Q}^\top)^\top \mathbf{Q}^\top && \text{(Theorem 0.19)}\\ &= \mathbf{QQ}^\top, && \text{(transposing twice, Definition 0.6)}\end{aligned}$$

and

$$\begin{aligned}(\mathbf{QQ}^\top)(\mathbf{QQ}^\top) &= \mathbf{Q}\,(\mathbf{Q}^\top\mathbf{Q})\,\mathbf{Q}^\top && \text{(Theorem 0.7)}\\ &= \mathbf{Q}\,\mathbf{I}_r\,\mathbf{Q}^\top && (\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_r\text{, from the first part of the proof})\\ &= \mathbf{QQ}^\top. && \text{(Theorem 0.10)}\end{aligned}$$

□

Example 0.7 (The projection matrix onto a line)**.** For $\mathcal{S} = \text{span}\{(1,1,0)\}$, the single vector $\underset{\sim}{q}_1 = \frac{1}{\sqrt{2}}(1,1,0)$ is an orthonormal basis: it spans $\mathcal{S}$, it is nonzero and so linearly independent, and

$$\begin{aligned}\|\underset{\sim}{q}_1\| &= \sqrt{\tfrac{1}{2} + \tfrac{1}{2} + 0}\\ &= 1.\end{aligned}$$

With $\mathbf{Q} = [\underset{\sim}{q}_1]$,

$$\begin{aligned}\mathbf{QQ}^\top &= \tfrac{1}{2}\begin{bmatrix}1\\1\\0\end{bmatrix}\begin{bmatrix}1 & 1 & 0\end{bmatrix}\\ &= \begin{bmatrix}\frac{1}{2} & \frac{1}{2} & 0\\ \frac{1}{2} & \frac{1}{2} & 0\\ 0 & 0 & 0\end{bmatrix}.\end{aligned}$$

Applied to $\underset{\sim}{y} = (3,1,2)$, it gives $\left(\frac{3+1}{2}, \frac{3+1}{2}, 0\right) = (2,2,0)$, the projection found in Example 0.4.

Theorem 0.5 (A projection matrix projects onto its column space, and is the only one that does)**.**

1. *If $\mathbf{P}$ is a $p \times p$ orthogonal projection matrix (Definition 0.8), then for every $\underset{\sim}{y} \in \mathbb{R}^p$, $\mathbf{P}\underset{\sim}{y}$ is the orthogonal projection of $\underset{\sim}{y}$ onto $\mathcal{C}(\mathbf{P})$ (Definition 0.3).*
2. *If $\mathbf{P}_1$ and $\mathbf{P}_2$ are $p \times p$ matrices such that, for every $\underset{\sim}{y}$, both $\mathbf{P}_1\underset{\sim}{y}$ and $\mathbf{P}_2\underset{\sim}{y}$ are the orthogonal projection of $\underset{\sim}{y}$ onto the same subspace $\mathcal{S}$, then $\mathbf{P}_1 = \mathbf{P}_2$.*

Proof

Proof. **Part 1.** Write $\underset{\sim}{y} = \mathbf{P}\underset{\sim}{y} + (\underset{\sim}{y} - \mathbf{P}\underset{\sim}{y})$. The first term is in $\mathcal{C}(\mathbf{P})$ (Definition 0.1). For the second,

$$\begin{aligned}
\mathbf{P}(\underset{\sim}{y} - \mathbf{P}\underset{\sim}{y}) &= \mathbf{P}\underset{\sim}{y} - \mathbf{P}(\mathbf{P}\underset{\sim}{y}) && \text{(Theorem 0.4, with coefficients 1 and } -1) \\
&= \mathbf{P}\underset{\sim}{y} - (\mathbf{P}\mathbf{P})\underset{\sim}{y} && \text{(Theorem 0.7)} \\
&= \mathbf{P}\underset{\sim}{y} - \mathbf{P}^2\underset{\sim}{y} && \text{(Definition 0.2)} \\
&= \mathbf{P}\underset{\sim}{y} - \mathbf{P}\underset{\sim}{y} && (\mathbf{P} \text{ is idempotent}) \\
&= \underset{\sim}{0}_p, && \text{(arithmetic)}
\end{aligned}$$

so $\underset{\sim}{y} - \mathbf{P}\underset{\sim}{y} \in \mathcal{N}(\mathbf{P})$. Since $\mathbf{P} = \mathbf{P}^\top$,

$$\begin{aligned}
\mathcal{N}(\mathbf{P}) &= \mathcal{N}(\mathbf{P}^\top) \\
&= \mathcal{C}(\mathbf{P})^\perp
\end{aligned}$$

(Theorem 0.3). By the uniqueness in Theorem 0.4, $\mathbf{P}\underset{\sim}{y}$ is the orthogonal projection of $\underset{\sim}{y}$ onto $\mathcal{C}(\mathbf{P})$.

Part 2. The orthogonal projection of a vector onto $\mathcal{S}$ is unique (Definition 0.3), so $\mathbf{P}_1\underset{\sim}{e}_j = \mathbf{P}_2\underset{\sim}{e}_j$ for each indicator vector $\underset{\sim}{e}_j$ (Definition 0.18). $\mathbf{P}_1\underset{\sim}{e}_j$ is column j of $\mathbf{P}_1$: by Theorem 0.3 it is the combination of the columns of $\mathbf{P}_1$ with coefficients the entries of $\underset{\sim}{e}_j$, which are 1 in place j and 0 elsewhere (Definition 0.18). Likewise for $\mathbf{P}_2$, so the two matrices have the same columns. □

Example 0.8 (Two routes to the same projection matrix)**.** The matrix $\mathbf{P} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ of Example 0.12 is an orthogonal projection matrix, and its column space is $\text{span}\{(1,0)\}$, the horizontal axis. By part 1, $\mathbf{P}(v_1, v_2) = (v_1, 0)$ is the orthogonal projection onto that axis: the remainder $(0, v_2)$ is orthogonal to $(1,0)$. The vector $\underset{\sim}{q}_1 = (1,0)$ has length 1 and spans the axis, so it is an orthonormal basis of it, and with $\mathbf{B} = [\underset{\sim}{q}_1]$,

$$\begin{aligned}
\mathbf{B}\mathbf{B}^\top &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} \\
&= \mathbf{P},
\end{aligned}$$

as part 2 and Theorem 0.4 require.

The oblique projection $\mathbf{Q} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ of Example 0.12 also maps every vector into the horizontal axis. It sends $(0,1)$ to $(1,0)$, but the remainder $(0,1) - (1,0) = (-1,1)$ is not orthogonal to the axis, so $(1,0)$ is not the orthogonal projection of $(0,1)$ onto the axis, which is $\mathbf{P}(0,1) = (0,0)$.

Theorem 0.6 (The hat matrix projects onto the column space of the design matrix)**.** *If* $\mathbf{X}$ *is an* $n \times p$ *design matrix with* $\operatorname{rank}(\mathbf{X}) = p$*, then for every* $y \in \mathbb{R}^n$*,* $\mathbf{H}y$ *(Definition 0.2) is the orthogonal projection of* y *onto* $\mathcal{C}(\mathbf{X})$*. So* $\mathbf{H} = \mathbf{Q}\mathbf{Q}^\top$ *for any* $n \times \widetilde{p}$ *matrix* $\mathbf{Q}$ *whose columns are an orthonormal basis of* $\mathcal{C}(\mathbf{X})$*.*

Proof

Proof. $\mathbf{H}$ is an orthogonal projection matrix (Theorem 0.2), so $\mathbf{H}y$ is the orthogonal projection of y onto $\mathcal{C}(\mathbf{H})$ (Theorem 0.5, part 1). It remains to show $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$.
$\mathcal{C}(\mathbf{H}) \subseteq \mathcal{C}(\mathbf{X})$**.** $\mathbf{H}\underset{\sim}{z} = \mathbf{X}\left((\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\underset{\sim}{z}\right)$ (Theorem 0.7), which is in $\mathcal{C}(\mathbf{X})$ (Definition 0.1).
$\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{H})$**.** First,

$$\begin{aligned}
\mathbf{H}\mathbf{X} &= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X} && \text{(Definition 0.2)}\\
&= \mathbf{X}\left((\mathbf{X}^\top\mathbf{X})^{-1}(\mathbf{X}^\top\mathbf{X})\right) && \text{(Theorem 0.7)}\\
&= \mathbf{X}\,\mathbf{I}_p && \text{(Definition 0.5)}\\
&= \mathbf{X}. && \text{(Theorem 0.10)}
\end{aligned}$$

So for every $\underset{\sim}{b} \in \mathbb{R}^p$, $\mathbf{X}\underset{\sim}{b} = (\mathbf{H}\mathbf{X})\,\underset{\sim}{b}$, since $\mathbf{H}\mathbf{X} = \mathbf{X}$, and $(\mathbf{H}\mathbf{X})\,\underset{\sim}{b} = \mathbf{H}\,(\mathbf{X}\underset{\sim}{b})$ (Theorem 0.7), which is in $\mathcal{C}(\mathbf{H})$.
$\mathcal{C}(\mathbf{X})$ has dimension $p \geq 1$ (Theorem 0.1), so an orthonormal basis of it has p vectors (Definition 0.6), and $\mathbf{Q}\mathbf{Q}^\top$ also gives the orthogonal projection onto $\mathcal{C}(\mathbf{X})$ (Theorem 0.4, with n in place of p and $r = p$). By Theorem 0.5, part 2, $\mathbf{H} = \mathbf{Q}\mathbf{Q}^\top$. □

Example 0.9 (The intercept-only hat matrix, from an orthonormal basis)**.** For the intercept-only design $\mathbf{X} = \begin{bmatrix}1\\1\end{bmatrix}$ of Example 0.2, $\mathcal{C}(\mathbf{X}) = \operatorname{span}\{(1,1)\}$, with orthonormal basis $\underset{\sim}{q}_1 = \frac{1}{\sqrt{2}}(1,1)$, since

$$\begin{aligned}
\|\underset{\sim}{q}_1\| &= \sqrt{\tfrac{1}{2} + \tfrac{1}{2}}\\
&= 1.
\end{aligned}$$

With $\mathbf{Q} = [\underset{\sim}{q}_1]$,

$$\begin{aligned}
\mathbf{Q}\mathbf{Q}^\top &= \tfrac{1}{2}\begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}1 & 1\end{bmatrix}\\
&= \begin{bmatrix}0.5 & 0.5\\ 0.5 & 0.5\end{bmatrix},
\end{aligned}$$

the hat matrix found in Example 0.2. The fitted values $(\bar{y}, \bar{y})$ are the closest point to y on the line of constant vectors (Theorem 0.3), and the residuals $(y_1 - \bar{y}, y_2 - \bar{y})$ are orthogonal to $(1,1)$:

$$
\begin{aligned}
(1,1)\cdot(y_1-\bar{y},y_2-\bar{y}) &= y_1+y_2-2\bar{y}\\
&= 0,
\end{aligned}
$$

since $2\bar{y}=y_1+y_2$.

Generalized inverses and linear systems

i Source

This section is adapted from Zhou [84], used under the MIT License (see the license text in Section). The source defers the existence of the Moore-Penrose inverse to the singular value decomposition; the proof here builds it from a rank factorization (Theorem 0.11) instead, and the section adds the projection $\mathbf{X}\mathbf{G}\mathbf{X}^{\top}$ for any generalized inverse $\mathbf{G}$ of $\mathbf{X}^{\top}\mathbf{X}$, which the source does not cover. Theorem 0.13 restores the source's formulas for the Moore-Penrose inverse of a matrix of full column rank or full row rank, and the fourth equivalent statement in Theorem 0.16 restores the source's rank test for consistency. These parts of the source are not part of this section:

- the QR forms of the full-rank formulas
- $\operatorname{rank}(\mathbf{A}^{-})\geq\operatorname{rank}(\mathbf{A})$, $\mathcal{C}(\mathbf{A})=\mathcal{C}(\mathbf{A}\mathbf{A}^{-})$, $\operatorname{rank}(\mathbf{A})=\operatorname{rank}(\mathbf{A}\mathbf{A}^{-})=\operatorname{rank}(\mathbf{A}^{-}\mathbf{A})$, and $\operatorname{rank}(\mathbf{A}^{+})=\operatorname{rank}(\mathbf{A})$
- the remarks that $\mathbf{A}\mathbf{A}^{-}$ and $\mathbf{A}^{-}\mathbf{A}$ are projections, oblique in general and orthogonal for $\mathbf{A}^{+}$
- the reflexive generalized inverse (conditions 1 and 2 alone), and the generalized inverses of $\mathbf{A}^{\top}$ and $k\mathbf{A}$
- the operation counts for the triangular solves, LU and QR, and the comparison of the Cholesky factorization with LU

Theorem 0.7 (A scalar factor passes through a matrix product)**.** *For an $m\times k$ matrix $\mathbf{A}$, a $k\times n$ matrix $\mathbf{B}$ and a number c,*

$$
\begin{aligned}
\mathbf{A}\,(c\,\mathbf{B}) &= c\,(\mathbf{A}\mathbf{B})\\
&= (c\,\mathbf{A})\,\mathbf{B}.
\end{aligned}
$$

In particular, $\mathbf{A}\,(\mathbf{B}-\mathbf{C})=\mathbf{A}\mathbf{B}-\mathbf{A}\mathbf{C}$ and $(\mathbf{B}-\mathbf{C})\,\mathbf{D}=\mathbf{B}\mathbf{D}-\mathbf{C}\mathbf{D}$ for matrices of compatible sizes, where $\mathbf{B}-\mathbf{C}=\mathbf{B}+(-1)\,\mathbf{C}$ (Theorem 0.6, Definition 0.9).

Proof

Proof. $\mathbf{A}(c\,\mathbf{B}) = c(\mathbf{AB})$. For each entry (i, j),

$$\begin{aligned}
(\mathbf{A}(c\,\mathbf{B}))_{ij} &= \sum_{l=1}^{k} a_{il}\,(c\,\mathbf{B})_{lj} && \text{(Definition 0.10)}\\
&= \sum_{l=1}^{k} a_{il}\,(c\,b_{lj}) && \text{(Definition 0.9)}\\
&= \sum_{l=1}^{k} c\,(a_{il}\,b_{lj}) && \text{(reorder the product of numbers)}\\
&= c\sum_{l=1}^{k} a_{il}\,b_{lj} && \text{(factor } c \text{ out of the sum)}\\
&= c\,(\mathbf{AB})_{ij} && \text{(Definition 0.10)}\\
&= (c\,(\mathbf{AB}))_{ij}. && \text{(Definition 0.9)}
\end{aligned}$$

$(c\,\mathbf{A})\,\mathbf{B} = c(\mathbf{AB})$. For each entry (i, j),

$$\begin{aligned}
((c\,\mathbf{A})\,\mathbf{B})_{ij} &= \sum_{l=1}^{k}(c\,a_{il})\,b_{lj} && \text{(Definition 0.10, Definition 0.9)}\\
&= c\sum_{l=1}^{k} a_{il}\,b_{lj} && \text{(factor } c \text{ out of the sum)}\\
&= (c\,(\mathbf{AB}))_{ij}. && \text{(Definition 0.10, Definition 0.9)}
\end{aligned}$$

Differences.

$$\begin{aligned}
\mathbf{A}(\mathbf{B}-\mathbf{C}) &= \mathbf{A}(\mathbf{B}+(-1)\,\mathbf{C}) && \text{(the difference as a sum)}\\
&= \mathbf{AB}+\mathbf{A}((-1)\,\mathbf{C}) && \text{(Theorem 0.8)}\\
&= \mathbf{AB}+(-1)\,(\mathbf{AC}) && \text{(first part)}\\
&= \mathbf{AB}-\mathbf{AC}, && \text{(the difference as a sum)}
\end{aligned}$$

and $(\mathbf{B}-\mathbf{C})\,\mathbf{D} = \mathbf{BD}-\mathbf{CD}$ the same way, using the second law in Theorem 0.8 and the second part. A column vector is a matrix with one column, so all of this applies with $\mathbf{B}$, $\mathbf{C}$ or $\mathbf{D}$ a vector. □

Example 0.10 (Pulling a scalar out of a product)**.** With $\mathbf{A} = \begin{bmatrix}1 & 2\end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix}3\\4\end{bmatrix}$:

$$\begin{aligned}
\mathbf{A}(2\,\mathbf{B}) &= \begin{bmatrix}1 & 2\end{bmatrix}\begin{bmatrix}6\\8\end{bmatrix}\\
&= 22,
\end{aligned}$$

and

$$\begin{aligned} 2(\mathbf{AB}) &= 2(3+8) \\ &= 22. \end{aligned}$$

Definition 0.4 (Generalized inverse)**.** A **generalized inverse** [85] of an $m \times n$ matrix $\mathbf{A}$ is an $n \times m$ matrix $\mathbf{G}$ such that

$$\underset{m\times n}{\mathbf{A}} \; \underset{n\times m}{\mathbf{G}} \; \underset{m\times n}{\mathbf{A}} = \mathbf{A}.$$

Example 0.11 (A matrix with many generalized inverses)**.** Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $\mathbf{G} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$. Then

$$\begin{aligned} \mathbf{AGA} &= \begin{bmatrix} g_{11} & g_{12} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} && \text{(Definition 0.10, for } \mathbf{AG}) \\ &= \begin{bmatrix} g_{11} & 0 \\ 0 & 0 \end{bmatrix}, && \text{(Definition 0.10)} \end{aligned}$$

which equals $\mathbf{A}$ exactly when $g_{11} = 1$. So every matrix $\begin{bmatrix} 1 & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$ is a generalized inverse of $\mathbf{A}$: a generalized inverse need not be unique. The zero matrix is not one, since its $(1,1)$ entry is 0.

Theorem 0.8 (An invertible matrix has only one generalized inverse)**.** *If $\mathbf{A}$ is an invertible $p \times p$ matrix (Definition 0.6), its only generalized inverse is $\mathbf{A}^{-1}$.*

Proof

Proof.

$$\begin{aligned} \mathbf{A}\mathbf{A}^{-1}\mathbf{A} &= \mathbf{I}_p\mathbf{A} \\ &= \mathbf{A} \end{aligned}$$

(5, Theorem 0.10), so $\mathbf{A}^{-1}$ is a generalized inverse. If $\mathbf{G}$ is any generalized inverse, then

$$
\begin{aligned}
\mathbf{G} &= \mathbf{I}_p\,\mathbf{G}\,\mathbf{I}_p && \text{(Theorem 0.10)}\\
&= (\mathbf{A}^{-1}\mathbf{A})\,\mathbf{G}\,(\mathbf{A}\mathbf{A}^{-1}) && \text{(Definition 0.5)}\\
&= \mathbf{A}^{-1}\,(\mathbf{A}\mathbf{G}\mathbf{A})\,\mathbf{A}^{-1} && \text{(Theorem 0.7)}\\
&= \mathbf{A}^{-1}\mathbf{A}\mathbf{A}^{-1} && \text{(Definition 0.4)}\\
&= \mathbf{I}_p\,\mathbf{A}^{-1} && \text{(Definition 0.5)}\\
&= \mathbf{A}^{-1}. && \text{(Theorem 0.10)}
\end{aligned}
$$

□

Example 0.12 (The generalized inverse of an invertible matrix)**.** For $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ of Example 0.5, the only generalized inverse is $\mathbf{A}^{-1} = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 1 \end{bmatrix}$. The matrix of Example 0.11, by contrast, has infinitely many, so by Theorem 0.8 it is singular.

Theorem 0.9 (A generalized inverse solves every solvable system)**.** *Let* $\mathbf{A}$ *be* $m \times n$*, let* $\mathbf{G}$ *be a generalized inverse of* $\mathbf{A}$ *(Definition 0.4), and let* $\underset{\sim}{b} \in \mathbb{R}^m$*. If* $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ *has a solution, then* $\mathbf{G}\underset{\sim}{b}$ *is a solution.*

Proof

Proof. Let $\underset{\sim}{x}_0$ be a solution, so $\mathbf{A}\underset{\sim}{x}_0 = \underset{\sim}{b}$. Then

$$
\begin{aligned}
\mathbf{A}\,(\mathbf{G}\underset{\sim}{b}) &= \mathbf{A}\mathbf{G}\,(\mathbf{A}\underset{\sim}{x}_0) && \text{(substitute } \underset{\sim}{b} = \mathbf{A}\underset{\sim}{x}_0)\\
&= (\mathbf{A}\mathbf{G}\mathbf{A})\,\underset{\sim}{x}_0 && \text{(Theorem 0.7)}\\
&= \mathbf{A}\underset{\sim}{x}_0 && \text{(Definition 0.4)}\\
&= \underset{\sim}{b}. && (\underset{\sim}{x}_0 \text{ is a solution})
\end{aligned}
$$

□

Example 0.13 (Solving with a generalized inverse)**.** With $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and its generalized inverse $\mathbf{G} = \begin{bmatrix} 1 & 5 \\ 7 & 0 \end{bmatrix}$ (Example 0.11), the system $\mathbf{A}\underset{\sim}{x} = (2, 0)$ has the solution $(2, 0)$, and $\mathbf{G}\,(2, 0) = (2, 14)$ is a solution too: $\mathbf{A}\,(2, 14) = (2, 0)$. The system $\mathbf{A}\underset{\sim}{x} = (2, 1)$ has no solution, since the second entry of $\mathbf{A}\underset{\sim}{x}$ is always 0; there $\mathbf{G}\,(2, 1) = (7, 14)$ gives $\mathbf{A}\,(7, 14) = (7, 0) \neq (2, 1)$.

Definition 0.5 (Moore-Penrose inverse)**.** A **Moore-Penrose inverse** [86] of an $m \times n$ matrix $\mathbf{A}$ is an $n \times m$ matrix $\mathbf{G}$ satisfying all four conditions

1. $\mathbf{AGA} = \mathbf{A}$;
2. $\mathbf{GAG} = \mathbf{G}$;
3. $(\mathbf{AG})^\top = \mathbf{AG}$;
4. $(\mathbf{GA})^\top = \mathbf{GA}$.

Example 0.14 (Checking the four conditions)**.** For $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, the matrix $\mathbf{G} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ satisfies all four:

$$\begin{aligned} \mathbf{AG} &= \mathbf{GA} \\ &= \mathbf{A} \end{aligned}$$

and $\mathbf{A}^2 = \mathbf{A}$, so conditions 1 and 2 read $\mathbf{A} = \mathbf{A}$, and conditions 3 and 4 hold because $\mathbf{A}$ is symmetric.

The generalized inverse $\mathbf{G}' = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ of $\mathbf{A}$ (Example 0.11) is not a Moore-Penrose inverse: $\mathbf{AG}' = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, which is not symmetric, so condition 3 fails.

Theorem 0.10 (A Moore-Penrose inverse is a generalized inverse)**.** *If* $\mathbf{G}$ *is a Moore-Penrose inverse of an* $m \times n$ *matrix* $\mathbf{A}$ *(Definition 0.5), then* $\mathbf{G}$ *is a generalized inverse of* $\mathbf{A}$ *(Definition 0.4).*

Proof

Proof. A generalized inverse of $\mathbf{A}$ is an $n \times m$ matrix $\mathbf{G}$ with $\mathbf{AGA} = \mathbf{A}$. A Moore-Penrose inverse is an $n \times m$ matrix, and condition 1 of Definition 0.5 is $\mathbf{AGA} = \mathbf{A}$. □

Example 0.15 (The Moore-Penrose inverse of a rank-one matrix)**.** The 3×2 matrix $\mathbf{A}$ below has rank 1, because its second column is twice its first, so it has no ordinary inverse. The function `MASS::ginv()` computes its Moore-Penrose inverse $\mathbf{G}$. The code checks the four conditions of Definition 0.5, of which the first is the equation that makes $\mathbf{G}$ a generalized inverse.

```r
A <- cbind(c(1, 2, 3), c(2, 4, 6))
G <- MASS::ginv(A)
conditions <- c(
  "1: AGA = A" = isTRUE(all.equal(A %*% G %*% A, A)),
  "2: GAG = G" = isTRUE(all.equal(G %*% A %*% G, G)),
  "3: AG symmetric" = isTRUE(all.equal(A %*% G, t(A %*% G))),
  "4: GA symmetric" = isTRUE(all.equal(G %*% A, t(G %*% A)))
)
conditions
#>      1: AGA = A      2: GAG = G 3: AG symmetric 4: GA symmetric
#>            TRUE            TRUE            TRUE            TRUE
round(G, 4)
#>        [,1]   [,2]   [,3]
#> [1,] 0.0143 0.0286 0.0429
#> [2,] 0.0286 0.0571 0.0857
```

All 4 of 4 conditions hold, so $\mathbf{G}$ is a Moore-Penrose inverse of $\mathbf{A}$. By Theorem 0.10 it is also a generalized inverse, which is condition 1: $\mathbf{AGA} = \mathbf{A}$ up to rounding error.

Theorem 0.11 (A matrix has at most one Moore-Penrose inverse)**.** *If $\mathbf{G}_1$ and $\mathbf{G}_2$ are both Moore-Penrose inverses of an $m \times n$ matrix $\mathbf{A}$ (Definition 0.5), then $\mathbf{G}_1 = \mathbf{G}_2$.*

Proof

Proof. $\mathbf{AG}_1 = \mathbf{AG}_2$**.**

$$
\begin{aligned}
\mathbf{AG}_1 &= (\mathbf{AG}_1)^\top && \text{(condition 3 for } \mathbf{G}_1\text{)} \\
&= \mathbf{G}_1{}^\top \mathbf{A}^\top && \text{(Theorem 0.19)} \\
&= \mathbf{G}_1{}^\top (\mathbf{AG}_2\mathbf{A})^\top && \text{(condition 1 for } \mathbf{G}_2\text{)} \\
&= \mathbf{G}_1{}^\top \mathbf{A}^\top \mathbf{G}_2{}^\top \mathbf{A}^\top && \text{(Theorem 0.19, twice)} \\
&= (\mathbf{G}_1{}^\top \mathbf{A}^\top)(\mathbf{G}_2{}^\top \mathbf{A}^\top) && \text{(Theorem 0.7)} \\
&= (\mathbf{AG}_1)^\top (\mathbf{AG}_2)^\top && \text{(Theorem 0.19, twice)} \\
&= (\mathbf{AG}_1)(\mathbf{AG}_2) && \text{(condition 3 for } \mathbf{G}_1 \text{ and } \mathbf{G}_2\text{)} \\
&= (\mathbf{AG}_1\mathbf{A})\,\mathbf{G}_2 && \text{(Theorem 0.7)} \\
&= \mathbf{AG}_2. && \text{(condition 1 for } \mathbf{G}_1\text{)}
\end{aligned}
$$

$\mathbf{G}_1\mathbf{A} = \mathbf{G}_2\mathbf{A}$**.**

$$
\begin{aligned}
\mathbf{G}_1\mathbf{A} &= (\mathbf{G}_1\mathbf{A})^\top && \text{(condition 4 for } \mathbf{G}_1\text{)}\\
&= \mathbf{A}^\top \mathbf{G}_1{}^\top && \text{(Theorem 0.19)}\\
&= (\mathbf{A}\mathbf{G}_2\mathbf{A})^\top \mathbf{G}_1{}^\top && \text{(condition 1 for } \mathbf{G}_2\text{)}\\
&= \mathbf{A}^\top \mathbf{G}_2{}^\top \mathbf{A}^\top \mathbf{G}_1{}^\top && \text{(Theorem 0.19, twice)}\\
&= (\mathbf{A}^\top \mathbf{G}_2{}^\top)(\mathbf{A}^\top \mathbf{G}_1{}^\top) && \text{(Theorem 0.7)}\\
&= (\mathbf{G}_2\mathbf{A})^\top (\mathbf{G}_1\mathbf{A})^\top && \text{(Theorem 0.19, twice)}\\
&= (\mathbf{G}_2\mathbf{A})(\mathbf{G}_1\mathbf{A}) && \text{(condition 4 for } \mathbf{G}_2 \text{ and } \mathbf{G}_1\text{)}\\
&= \mathbf{G}_2(\mathbf{A}\mathbf{G}_1\mathbf{A}) && \text{(Theorem 0.7)}\\
&= \mathbf{G}_2\mathbf{A}. && \text{(condition 1 for } \mathbf{G}_1\text{)}
\end{aligned}
$$

Conclusion.

$$
\begin{aligned}
\mathbf{G}_1 &= \mathbf{G}_1\mathbf{A}\mathbf{G}_1 && \text{(condition 2 for } \mathbf{G}_1\text{)}\\
&= \mathbf{G}_1(\mathbf{A}\mathbf{G}_1) && \text{(Theorem 0.7)}\\
&= \mathbf{G}_1(\mathbf{A}\mathbf{G}_2) && \text{(the first step)}\\
&= (\mathbf{G}_1\mathbf{A})\mathbf{G}_2 && \text{(Theorem 0.7)}\\
&= (\mathbf{G}_2\mathbf{A})\mathbf{G}_2 && \text{(the second step)}\\
&= \mathbf{G}_2\mathbf{A}\mathbf{G}_2 && \text{(Theorem 0.7)}\\
&= \mathbf{G}_2. && \text{(condition 2 for } \mathbf{G}_2\text{)}
\end{aligned}
$$

□

Example 0.16 (Only one of the many generalized inverses qualifies)**.** The matrix $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ has infinitely many generalized inverses $\begin{bmatrix} 1 & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$ (Example 0.11), and $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ is a Moore-Penrose inverse (Example 0.14). By Theorem 0.11 it is the only one, so every other choice of g_{12}, g_{21}, g_{22} breaks condition 2, 3 or 4. For instance, $g_{21} = 1$ (the rest 0) gives $\mathbf{G}\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$, which is not symmetric.

Theorem 0.12 (Every matrix has a Moore-Penrose inverse)**.** *Every $m \times n$ matrix $\mathbf{A}$ has a Moore-Penrose inverse (Definition 0.5), written $\mathbf{A}^+$; it is unique by Theorem 0.11.*

- *If $\mathbf{A} = \mathbf{0}_{m\times n}$, then $\mathbf{A}^+ = \mathbf{0}_{n\times m}$.*
- *If* $\operatorname{rank}(\mathbf{A}) = r \geq 1$ *and $\mathbf{A} = \mathbf{C}\mathbf{R}$ is a rank factorization (Definition 0.5), then*

$$
\underbrace{\mathbf{A}^+}_{n\times m} = \underbrace{\mathbf{R}^\top}_{n\times r} \underbrace{(\mathbf{R}\mathbf{R}^\top)^{-1}}_{r\times r} \underbrace{(\mathbf{C}^\top\mathbf{C})^{-1}}_{r\times r} \underbrace{\mathbf{C}^\top}_{r\times m}.
$$

Proof

Proof. **The two cases cover every matrix.** If $\mathbf{A} \neq \mathbf{0}_{m\times n}$, it has a nonzero column, which is linearly independent on its own ($c\,\underset{\sim}{v} = \underset{\sim}{0}$ with $\underset{\sim}{v} \neq \underset{\sim}{0}$ forces $c = 0$), so $\operatorname{rank}(\mathbf{A}) \geq 1$ (Definition 0.2).
The zero matrix. With $\mathbf{A} = \mathbf{0}_{m\times n}$ and $\mathbf{G} = \mathbf{0}_{n\times m}$, every product in conditions 1 to 4 is a zero matrix, and a zero matrix equals its own transpose, so all four conditions hold.
The two inverses exist. Now suppose $\operatorname{rank}(\mathbf{A}) = r \geq 1$. A rank factorization exists (Theorem 0.11). $\mathbf{C}$ is $m \times r$ and $\mathbf{R}$ is $r \times n$. Both have rank r:

$$\begin{aligned} r &= \operatorname{rank}(\mathbf{CR}) && (\mathbf{A} = \mathbf{CR}) \\ &\leq \operatorname{rank}(\mathbf{C}) && (\text{Theorem } 0.3) \\ &\leq r, && (\text{Corollary } 0.1,\ \mathbf{C} \text{ has } r \text{ columns}) \end{aligned}$$

and, using $(\mathbf{CR})^\top = \mathbf{R}^\top\mathbf{C}^\top$ (Theorem 0.19),

$$\begin{aligned} r &= \operatorname{rank}(\mathbf{R}^\top\mathbf{C}^\top) && (\text{Theorem } 0.4) \\ &\leq \operatorname{rank}(\mathbf{R}^\top) && (\text{Theorem } 0.3) \\ &\leq r. && (\text{Corollary } 0.1,\ \mathbf{R}^\top \text{ has } r \text{ columns}) \end{aligned}$$

So $\mathbf{C}$ and $\mathbf{R}^\top$ have full column rank r, and $\mathbf{C}^\top\mathbf{C}$ and $\mathbf{RR}^\top = (\mathbf{R}^\top)^\top\mathbf{R}^\top$ are invertible (Theorem 0.1). They are symmetric:

$$\begin{aligned} (\mathbf{C}^\top\mathbf{C})^\top &= \mathbf{C}^\top(\mathbf{C}^\top)^\top \\ &= \mathbf{C}^\top\mathbf{C} \end{aligned}$$

(Theorem 0.19, Definition 0.6), and likewise for $\mathbf{RR}^\top$. Write $\mathbf{S} \overset{\text{def}}{=} (\mathbf{RR}^\top)^{-1}$ and $\mathbf{T} \overset{\text{def}}{=} (\mathbf{C}^\top\mathbf{C})^{-1}$, both symmetric (Corollary 0.1), and $\mathbf{G} \overset{\text{def}}{=} \mathbf{R}^\top\mathbf{STC}^\top$.
Two products.

$$\begin{aligned} \mathbf{AG} &= \mathbf{CR}\,\mathbf{R}^\top\mathbf{STC}^\top && (\text{substitute } \mathbf{A} \text{ and } \mathbf{G}) \\ &= \mathbf{C}\,(\mathbf{RR}^\top)\,\mathbf{S}\,\mathbf{TC}^\top && (\text{Theorem } 0.7) \\ &= \mathbf{C}\,\mathbf{I}_r\,\mathbf{TC}^\top && (\text{Definition } 0.5) \\ &= \mathbf{CTC}^\top, && (\text{Theorem } 0.10) \end{aligned}$$

$$\begin{aligned} \mathbf{GA} &= \mathbf{R}^\top\mathbf{STC}^\top\,\mathbf{CR} && (\text{substitute } \mathbf{G} \text{ and } \mathbf{A}) \\ &= \mathbf{R}^\top\mathbf{S}\,\mathbf{T}\,(\mathbf{C}^\top\mathbf{C})\,\mathbf{R} && (\text{Theorem } 0.7) \\ &= \mathbf{R}^\top\mathbf{S}\,\mathbf{I}_r\,\mathbf{R} && (\text{Definition } 0.5) \\ &= \mathbf{R}^\top\mathbf{SR}. && (\text{Theorem } 0.10) \end{aligned}$$

Conditions 3 and 4.

$$
\begin{aligned}
(\mathbf{AG})^\top &= (\mathbf{CTC}^\top)^\top && \text{(first product)} \\
&= (\mathbf{C}(\mathbf{TC}^\top))^\top && \text{(Theorem 0.7)} \\
&= (\mathbf{TC}^\top)^\top \mathbf{C}^\top && \text{(Theorem 0.19)} \\
&= (\mathbf{C}^\top)^\top \mathbf{T}^\top \mathbf{C}^\top && \text{(Theorem 0.19)} \\
&= \mathbf{C}\mathbf{T}^\top \mathbf{C}^\top && \text{(Definition 0.6)} \\
&= \mathbf{CTC}^\top, && (\mathbf{T} \text{ is symmetric})
\end{aligned}
$$

$$
\begin{aligned}
(\mathbf{GA})^\top &= (\mathbf{R}^\top \mathbf{SR})^\top && \text{(second product)} \\
&= (\mathbf{R}^\top(\mathbf{SR}))^\top && \text{(Theorem 0.7)} \\
&= (\mathbf{SR})^\top (\mathbf{R}^\top)^\top && \text{(Theorem 0.19)} \\
&= \mathbf{R}^\top \mathbf{S}^\top (\mathbf{R}^\top)^\top && \text{(Theorem 0.19)} \\
&= \mathbf{R}^\top \mathbf{S}^\top \mathbf{R} && \text{(Definition 0.6)} \\
&= \mathbf{R}^\top \mathbf{SR}. && (\mathbf{S} \text{ is symmetric})
\end{aligned}
$$

Condition 1.

$$
\begin{aligned}
\mathbf{AGA} &= \mathbf{CTC}^\top \mathbf{CR} && \text{(first product)} \\
&= \mathbf{C}\mathbf{T}(\mathbf{C}^\top \mathbf{C})\mathbf{R} && \text{(Theorem 0.7)} \\
&= \mathbf{C}\mathbf{I}_r \mathbf{R} && \text{(Definition 0.5)} \\
&= \mathbf{A}. && \text{(Theorem 0.10)}
\end{aligned}
$$

Condition 2.

$$
\begin{aligned}
\mathbf{GAG} &= \mathbf{R}^\top \mathbf{SR}\,\mathbf{R}^\top \mathbf{STC}^\top && \text{(second product)} \\
&= \mathbf{R}^\top \mathbf{S}(\mathbf{RR}^\top)\mathbf{S}\,\mathbf{TC}^\top && \text{(Theorem 0.7)} \\
&= \mathbf{R}^\top \mathbf{I}_r \mathbf{S}\,\mathbf{TC}^\top && \text{(Definition 0.5)} \\
&= \mathbf{G}. && \text{(Theorem 0.10)}
\end{aligned}
$$

□

Example 0.17 (The Moore-Penrose inverse of a rank-one matrix)**.** For $\mathbf{A}$ in Example 0.1, the rank factorization $\mathbf{C} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$, $\mathbf{R} = \begin{bmatrix} 1 & -2 & -2 \end{bmatrix}$ (Example 0.17) gives

$$
\begin{aligned}
\mathbf{C}^\top \mathbf{C} &= 1 + 9 \\
&= 10
\end{aligned}
$$

and

$$\begin{aligned}\mathbf{R}\mathbf{R}^\top &= 1+4+4\\ &= 9,\end{aligned}$$

so

$$\begin{aligned}\mathbf{A}^+ &= \frac{1}{9\cdot 10}\begin{bmatrix}1\\-2\\-2\end{bmatrix}\begin{bmatrix}1 & 3\end{bmatrix}\\ &= \frac{1}{90}\begin{bmatrix}1 & 3\\-2 & -6\\-2 & -6\end{bmatrix}.\end{aligned}$$

As a check on condition 1,

$$\begin{aligned}\mathbf{A}\mathbf{A}^+ &= \mathbf{C}\mathbf{T}\mathbf{C}^\top\\ &= \frac{1}{10}\begin{bmatrix}1 & 3\\3 & 9\end{bmatrix},\end{aligned}$$

and its first row times $\mathbf{A}$ is $\frac{1}{10}(1\cdot(1,-2,-2)+3\cdot(3,-6,-6)) = (1,-2,-2)$, the first row of $\mathbf{A}$; the second row is 3 times the first, like the second row of $\mathbf{A}$.

Theorem 0.13 (The Moore-Penrose inverse of a matrix of full column rank or full row rank)**.** *Let* $\mathbf{A}$ *be an* $m\times n$ *matrix.*

1. *If* $\mathbf{A}$ *has full column rank (Definition 0.3), then* $\mathbf{A}^\top\mathbf{A}$ *is invertible and* $\mathbf{A}^+ = \left((\mathbf{A}^\top\mathbf{A})\right)^{-1}\mathbf{A}^\top$.
2. *If* $\operatorname{rank}(\mathbf{A}) = m$*, then* $\mathbf{A}\mathbf{A}^\top$ *is invertible and* $\mathbf{A}^+ = \mathbf{A}^\top\left((\mathbf{A}\mathbf{A}^\top)\right)^{-1}$.

Proof

Proof. By Theorem 0.11, it is enough to show that the matrix $\mathbf{G}$ named in each part satisfies the four conditions of Definition 0.5.
Part 1. $\mathbf{A}^\top\mathbf{A}$ is invertible by Theorem 0.1. Let $\mathbf{G} = \left((\mathbf{A}^\top\mathbf{A})\right)^{-1}\mathbf{A}^\top$. Then

$$\begin{aligned}\mathbf{G}\mathbf{A} &= \left((\mathbf{A}^\top\mathbf{A})\right)^{-1}\mathbf{A}^\top\mathbf{A} && \text{(definition of } \mathbf{G})\\ &= \mathbf{I}_n. && \text{(Definition 0.5)}\end{aligned}$$

- Condition 4: $(\mathbf{G}\mathbf{A})^\top = {\mathbf{I}_n}^\top = \mathbf{I}_n = \mathbf{G}\mathbf{A}$.
- Condition 1: $\mathbf{A}\mathbf{G}\mathbf{A} = \mathbf{A}\mathbf{I}_n = \mathbf{A}$.
- Condition 2: $\mathbf{G}\mathbf{A}\mathbf{G} = \mathbf{I}_n\mathbf{G} = \mathbf{G}$.
- Condition 3: $(\mathbf{A}\mathbf{G})^\top = \mathbf{A}\left(\left((\mathbf{A}^\top\mathbf{A})\right)^{-1}\right)^\top\mathbf{A}^\top = \mathbf{A}\left((\mathbf{A}^\top\mathbf{A})^\top\right)^{-1}\mathbf{A}^\top =$

$\mathbf{A}\left(\left(\mathbf{A}^\top\mathbf{A}\right)\right)^{-1}\mathbf{A}^\top = \mathbf{AG}$, by Theorem 0.19, Theorem 0.2, and because $\mathbf{A}^\top\mathbf{A}$ is symmetric.

Part 2. $\text{rank}(\mathbf{AA}^\top) = \text{rank}(\mathbf{A}) = m$ (Theorem 0.4), and $\mathbf{AA}^\top$ is $m \times m$, so it is invertible (Theorem 0.1). Let $\mathbf{G} = \mathbf{A}^\top\left(\left(\mathbf{AA}^\top\right)\right)^{-1}$. Then $\mathbf{AG} = \mathbf{AA}^\top\left(\left(\mathbf{AA}^\top\right)\right)^{-1} = \mathbf{I}_m$ (5).

- Condition 3: $(\mathbf{AG})^\top = {\mathbf{I}_m}^\top = \mathbf{I}_m = \mathbf{AG}$.
- Condition 1: $\mathbf{AGA} = \mathbf{I}_m\mathbf{A} = \mathbf{A}$.
- Condition 2: $\mathbf{GAG} = \mathbf{GI}_m = \mathbf{G}$.
- Condition 4: $(\mathbf{GA})^\top = \mathbf{A}^\top\left(\left(\left(\mathbf{AA}^\top\right)\right)^{-1}\right)^\top\mathbf{A} = \mathbf{A}^\top\left(\left(\mathbf{AA}^\top\right)\right)^{-1}\mathbf{A} = \mathbf{GA}$, by Theorem 0.19, Theorem 0.2, and because $\mathbf{AA}^\top$ is symmetric.

□

Example 0.18 (Moore-Penrose inverses of a column and a row)**.** The 2×1 matrix $\mathbf{A} = \begin{bmatrix}1\\1\end{bmatrix}$ has full column rank, and $\mathbf{A}^\top\mathbf{A} = \begin{bmatrix}2\end{bmatrix}$, so

$$\begin{aligned}\mathbf{A}^+ &= \tfrac{1}{2}\begin{bmatrix}1 & 1\end{bmatrix}\\ &= \begin{bmatrix}\frac{1}{2} & \frac{1}{2}\end{bmatrix}.\end{aligned}$$

Then $\mathbf{A}^+\mathbf{A} = \begin{bmatrix}1\end{bmatrix}$, and $\mathbf{AA}^+ = \begin{bmatrix}\frac{1}{2} & \frac{1}{2}\\ \frac{1}{2} & \frac{1}{2}\end{bmatrix}$ is symmetric.

The 1×2 matrix $\mathbf{B} = \begin{bmatrix}1 & 1\end{bmatrix}$ has rank 1, which is its count of rows, and $\mathbf{BB}^\top = \begin{bmatrix}2\end{bmatrix}$, so

$$\begin{aligned}\mathbf{B}^+ &= \begin{bmatrix}1\\1\end{bmatrix}\tfrac{1}{2}\\ &= \begin{bmatrix}\frac{1}{2}\\ \frac{1}{2}\end{bmatrix}.\end{aligned}$$

Theorem 0.14 (The Moore-Penrose inverse of a symmetric matrix from its eigendecomposition)**.** *Let $\mathbf{S} = \mathbf{Q\Lambda Q}^\top$ be an eigendecomposition (Definition 0.3) of a symmetric matrix $\mathbf{S}$, with diagonal entries $\lambda_1, \ldots, \lambda_p$ in the diagonal matrix $\mathbf{\Lambda}$. Let $\mathbf{\Lambda}^+$ be the diagonal matrix with jth diagonal entry $1/\lambda_j$ if $\lambda_j \neq 0$ and 0 if $\lambda_j = 0$. Then*

$$\mathbf{S}^+ = \mathbf{Q\Lambda}^+\mathbf{Q}^\top.$$

Proof

Proof. Let $\mathbf{G} = \mathbf{Q}\mathbf{\Lambda}^+\mathbf{Q}^\top$. By Theorem 0.11, it is enough to show that $\mathbf{G}$ satisfies the four conditions of Definition 0.5 for $\mathbf{S}$.

Two facts about the diagonal matrices. The jth diagonal entry of $\mathbf{\Lambda}\mathbf{\Lambda}^+\mathbf{\Lambda}$ is $\lambda_j \cdot (1/\lambda_j) \cdot \lambda_j = \lambda_j$ if $\lambda_j \neq 0$, and $0 = \lambda_j$ if $\lambda_j = 0$. So $\mathbf{\Lambda}\mathbf{\Lambda}^+\mathbf{\Lambda} = \mathbf{\Lambda}$. In the same way, $\mathbf{\Lambda}^+\mathbf{\Lambda}\mathbf{\Lambda}^+ = \mathbf{\Lambda}^+$. Both $\mathbf{\Lambda}\mathbf{\Lambda}^+$ and $\mathbf{\Lambda}^+\mathbf{\Lambda}$ are diagonal, so each is symmetric.

Conditions 1 and 2.

$$\begin{aligned}
\mathbf{SGS} &= \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top\,\mathbf{Q}\mathbf{\Lambda}^+\mathbf{Q}^\top\,\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top && \text{(definitions of } \mathbf{S} \text{ and } \mathbf{G}\text{)} \\
&= \mathbf{Q}\mathbf{\Lambda}\mathbf{\Lambda}^+\mathbf{\Lambda}\mathbf{Q}^\top && (\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}, \text{ twice}) \\
&= \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top && (\mathbf{\Lambda}\mathbf{\Lambda}^+\mathbf{\Lambda} = \mathbf{\Lambda}) \\
&= \mathbf{S}, && \text{(definition of } \mathbf{S}\text{)}
\end{aligned}$$

and likewise

$$\begin{aligned}
\mathbf{GSG} &= \mathbf{Q}\mathbf{\Lambda}^+\mathbf{\Lambda}\mathbf{\Lambda}^+\mathbf{Q}^\top && (\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}, \text{ twice}) \\
&= \mathbf{Q}\mathbf{\Lambda}^+\mathbf{Q}^\top && (\mathbf{\Lambda}^+\mathbf{\Lambda}\mathbf{\Lambda}^+ = \mathbf{\Lambda}^+) \\
&= \mathbf{G}. && \text{(definition of } \mathbf{G}\text{)}
\end{aligned}$$

Conditions 3 and 4.

$$\begin{aligned}
\mathbf{SG} &= \mathbf{Q}\mathbf{\Lambda}\mathbf{\Lambda}^+\mathbf{Q}^\top && (\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}) \\
&= (\mathbf{Q}\mathbf{\Lambda}\mathbf{\Lambda}^+\mathbf{Q}^\top)^\top && (\mathbf{\Lambda}\mathbf{\Lambda}^+ \text{ is diagonal, so equals its transpose}) \\
&= (\mathbf{SG})^\top, && \text{(just shown, in the other order)}
\end{aligned}$$

and the same steps with $\mathbf{\Lambda}^+\mathbf{\Lambda}$ in place of $\mathbf{\Lambda}\mathbf{\Lambda}^+$ give $\mathbf{GS} = (\mathbf{GS})^\top$. □

Example 0.19 (The Moore-Penrose inverse of a rank-one symmetric matrix)**.** For the symmetric matrix $\mathbf{S} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, an eigendecomposition is

$$\mathbf{S} = \underbrace{\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}}_{\mathbf{Q}} \underbrace{\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}}_{\mathbf{\Lambda}} \mathbf{Q}^\top.$$

Inverting only the nonzero eigenvalue gives $\mathbf{\Lambda}^+ = \begin{bmatrix} 1/2 & 0 \\ 0 & 0 \end{bmatrix}$, so

$$\begin{aligned}
\mathbf{S}^+ &= \mathbf{Q}\mathbf{\Lambda}^+\mathbf{Q}^\top && \text{(Theorem 0.14)} \\
&= \frac{1}{2}\cdot\frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix} && \text{(only the first column of } \mathbf{Q} \text{ contributes)} \\
&= \frac{1}{4}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}.
\end{aligned}$$

Condition 1 of Definition 0.5 holds: $\mathbf{S}\mathbf{S}^+ = \frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$, and multiplying by $\mathbf{S}$ again gives $\mathbf{S}$. An inverse of $\mathbf{S}$ itself does not exist, because $\mathbf{S}$ has a zero eigenvalue.

Theorem 0.15 (The Moore-Penrose inverse of a Gram matrix)**.** *For every $m \times n$ matrix $\mathbf{A}$,*

$$(\mathbf{A}^\top\mathbf{A})^+ = \mathbf{A}^+(\mathbf{A}^+)^\top.$$

Proof

Proof. Let $\mathbf{M} = \mathbf{A}^\top\mathbf{A}$ and $\mathbf{G} = \mathbf{A}^+(\mathbf{A}^+)^\top$. By Theorem 0.11, it is enough to show that $\mathbf{G}$ satisfies the four conditions of Definition 0.5 for $\mathbf{M}$.

Two identities. Conditions 3 and 4 for $\mathbf{A}^+$ say that $\mathbf{A}\mathbf{A}^+$ and $\mathbf{A}^+\mathbf{A}$ are symmetric. Using condition 1 for $\mathbf{A}^+$,

$$\begin{aligned}
\mathbf{A}^\top\mathbf{A}\mathbf{A}^+ &= \mathbf{A}^\top(\mathbf{A}\mathbf{A}^+)^\top && \text{(condition 3 for } \mathbf{A}^+) \\
&= (\mathbf{A}\mathbf{A}^+\mathbf{A})^\top && \text{(Theorem 0.19)} \\
&= \mathbf{A}^\top. && \text{(condition 1 for } \mathbf{A}^+)
\end{aligned}$$

So

$$\begin{aligned}
\mathbf{M}\mathbf{G} &= \mathbf{A}^\top\mathbf{A}\mathbf{A}^+(\mathbf{A}^+)^\top && \text{(definitions of } \mathbf{M} \text{ and } \mathbf{G}) \\
&= \mathbf{A}^\top(\mathbf{A}^+)^\top && (\mathbf{A}^\top\mathbf{A}\mathbf{A}^+ = \mathbf{A}^\top) \\
&= (\mathbf{A}^+\mathbf{A})^\top, && \text{(Theorem 0.19)}
\end{aligned}$$

which is $\mathbf{A}^+\mathbf{A}$ by condition 4 for $\mathbf{A}^+$. Hence $\mathbf{M}\mathbf{G} = \mathbf{A}^+\mathbf{A}$, and it is symmetric, which is condition 3 for $\mathbf{G}$.

For the other order,

$$\begin{aligned}
\mathbf{G}\mathbf{M} &= \mathbf{A}^+(\mathbf{A}^+)^\top\mathbf{A}^\top\mathbf{A} && \text{(definitions of } \mathbf{G} \text{ and } \mathbf{M}) \\
&= \mathbf{A}^+(\mathbf{A}\mathbf{A}^+)^\top\mathbf{A} && \text{(Theorem 0.19)} \\
&= \mathbf{A}^+\mathbf{A}\mathbf{A}^+\mathbf{A} && \text{(condition 3 for } \mathbf{A}^+) \\
&= \mathbf{A}^+\mathbf{A}, && \text{(condition 1 for } \mathbf{A}^+)
\end{aligned}$$

so $\mathbf{GM} = \mathbf{A}^+\mathbf{A}$ is symmetric, which is condition 4 for $\mathbf{G}$.
Conditions 1 and 2.

$$\begin{aligned}
\mathbf{MGM} &= \mathbf{A}^+\mathbf{A}\,\mathbf{A}^\top\mathbf{A} && (\mathbf{MG} = \mathbf{A}^+\mathbf{A}) \\
&= \left(\mathbf{A}^+\mathbf{A}\right)^\top \mathbf{A}^\top\mathbf{A} && (\text{condition 4 for } \mathbf{A}^+) \\
&= \left(\mathbf{A}\mathbf{A}^+\mathbf{A}\right)^\top \mathbf{A} && (\text{Theorem 0.19}) \\
&= \mathbf{A}^\top\mathbf{A} && (\text{condition 1 for } \mathbf{A}^+) \\
&= \mathbf{M}, && (\text{definition of } \mathbf{M}) \\
\mathbf{GMG} &= \mathbf{A}^+\mathbf{A}\,\mathbf{A}^+\left(\mathbf{A}^+\right)^\top && (\mathbf{GM} = \mathbf{A}^+\mathbf{A}) \\
&= \mathbf{A}^+\left(\mathbf{A}^+\right)^\top && (\text{condition 2 for } \mathbf{A}^+) \\
&= \mathbf{G}. && (\text{definition of } \mathbf{G})
\end{aligned}$$

□

Example 0.20 (The Moore-Penrose inverse of the Gram matrix of a column of ones)**.** For $\mathbf{A} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, the Gram matrix is $\mathbf{A}^\top\mathbf{A} = \begin{bmatrix} 2 \end{bmatrix}$, so $\left(\mathbf{A}^\top\mathbf{A}\right)^+ = \begin{bmatrix} 1/2 \end{bmatrix}$. By Theorem 0.13, $\mathbf{A}^+ = \left(\left(\mathbf{A}^\top\mathbf{A}\right)\right)^{-1}\mathbf{A}^\top = \begin{bmatrix} 1/2 & 1/2 \end{bmatrix}$. Then

$$\begin{aligned}
\mathbf{A}^+\left(\mathbf{A}^+\right)^\top &= \begin{bmatrix} 1/2 & 1/2 \end{bmatrix}\begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix} \\
&= \frac{1}{4} + \frac{1}{4} \\
&= \frac{1}{2},
\end{aligned}$$

which is $\left(\mathbf{A}^\top\mathbf{A}\right)^+$, as Theorem 0.15 states.

Corollary 0.1 (The Moore-Penrose inverse from the Gram matrix)**.** *For every $m \times n$ matrix* $\mathbf{A}$*,*

$$\mathbf{A}^+ = \left(\mathbf{A}^\top\mathbf{A}\right)^+\mathbf{A}^\top.$$

Proof

Proof.

$$\begin{aligned}
\left(\mathbf{A}^\top \mathbf{A}\right)^+ \mathbf{A}^\top &= \mathbf{A}^+ \left(\mathbf{A}^+\right)^\top \mathbf{A}^\top && \text{(Theorem 0.15)} \\
&= \mathbf{A}^+ \left(\mathbf{A}\mathbf{A}^+\right)^\top && \text{(Theorem 0.19)} \\
&= \mathbf{A}^+ \mathbf{A}\mathbf{A}^+ && \text{(condition 3 of Definition 0.5)} \\
&= \mathbf{A}^+. && \text{(condition 2 of Definition 0.5)}
\end{aligned}$$

□

Example 0.21 (Recovering the Moore-Penrose inverse of a column of ones)**.** For $\mathbf{A} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, Example 0.20 found $\left(\mathbf{A}^\top \mathbf{A}\right)^+ = \begin{bmatrix} 1/2 \end{bmatrix}$. Then

$$\begin{aligned}
\left(\mathbf{A}^\top \mathbf{A}\right)^+ \mathbf{A}^\top &= \begin{bmatrix} 1/2 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1/2 & 1/2 \end{bmatrix},
\end{aligned}$$

which is $\mathbf{A}^+$, as Corollary 0.1 states.

Definition 0.6 (Consistent linear system)**.** A linear system (Definition 0.17) $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$, with $\mathbf{A}$ an $m \times n$ matrix and $\underset{\sim}{b} \in \mathbb{R}^m$, is **consistent** [87] if it has at least one solution $\underset{\sim}{x} \in \mathbb{R}^n$, and **inconsistent** otherwise.

Example 0.22 (A consistent system and an inconsistent one)**.** With $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\mathbf{A}\left(x_1, x_2\right) = \left(x_1, 0\right)$ (Definition 0.16).

- $\mathbf{A}\underset{\sim}{x} = (2, 0)$ is consistent: $\underset{\sim}{x} = (2, 0)$ is a solution.
- $\mathbf{A}\underset{\sim}{x} = (2, 1)$ is inconsistent: the second entry of $\mathbf{A}\underset{\sim}{x}$ is always 0, never 1.

Theorem 0.16 (When a linear system has a solution)**.** *Let $\mathbf{A}$ be $m \times n$, let $\underset{\sim}{b} \in \mathbb{R}^m$, and let $\mathbf{G}$ be any generalized inverse of $\mathbf{A}$ (Definition 0.4). The following statements are equivalent:*

1. *$\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ is consistent (Definition 0.6);*
2. *$\underset{\sim}{b} \in \mathcal{C}(\mathbf{A})$ (Definition 0.1);*
3. *$\mathbf{A}\mathbf{G}\underset{\sim}{b} = \underset{\sim}{b}$;*
4. *$\operatorname{rank}\left(\begin{bmatrix} \mathbf{A} & \underset{\sim}{b} \end{bmatrix}\right) = \operatorname{rank}(\mathbf{A})$, where $\begin{bmatrix} \mathbf{A} & \underset{\sim}{b} \end{bmatrix}$ is the $m \times (n+1)$ matrix whose last column is $\underset{\sim}{b}$.*

Proof

Proof. **1 and 2 are equivalent.** Statement 1 says $\underset{\sim}{b} = \mathbf{A}\underset{\sim}{x}$ for some $\underset{\sim}{x}$ (Definition 0.6), and $\mathcal{C}(\mathbf{A})$ is the set of vectors $\mathbf{A}\underset{\sim}{x}$ (Definition 0.1), so $\underset{\sim}{b}$ is in it exactly when $\underset{\sim}{b} = \mathbf{A}\underset{\sim}{x}$ for some $\underset{\sim}{x}$.
1 implies 3. By Theorem 0.9, $\mathbf{G}\underset{\sim}{b}$ is a solution, so $\mathbf{A}(\mathbf{G}\underset{\sim}{b}) = \underset{\sim}{b}$, which is statement 3 (Theorem 0.7).
3 implies 1. If $\mathbf{AG}\underset{\sim}{b} = \underset{\sim}{b}$, then $\underset{\sim}{x} = \mathbf{G}\underset{\sim}{b}$ is a solution, since $\mathbf{A}(\mathbf{G}\underset{\sim}{b}) = (\mathbf{AG})\underset{\sim}{b}$ (Theorem 0.7).
2 and 4 are equivalent. Every column of $\mathbf{A}$ is a column of $\begin{bmatrix}\mathbf{A} & \underset{\sim}{b}\end{bmatrix}$, so $\mathcal{C}(\mathbf{A}) \subseteq \mathcal{C}\left(\begin{bmatrix}\mathbf{A} & \underset{\sim}{b}\end{bmatrix}\right)$.

- If $\underset{\sim}{b} \in \mathcal{C}(\mathbf{A})$, every combination of the columns of $\begin{bmatrix}\mathbf{A} & \underset{\sim}{b}\end{bmatrix}$ is a combination of the columns of $\mathbf{A}$ (Definition 0.1), so the two column spaces (Definition 0.1) coincide and their ranks agree (Theorem 0.1).
- If the ranks agree, the two column spaces have the same dimension (Theorem 0.1), so they coincide (Theorem 0.11), and $\underset{\sim}{b} \in \mathcal{C}\left(\begin{bmatrix}\mathbf{A} & \underset{\sim}{b}\end{bmatrix}\right) = \mathcal{C}(\mathbf{A})$.

□

Example 0.23 (Testing consistency with a generalized inverse)**.** For $\mathbf{A} = \begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}$ and the generalized inverse $\mathbf{G} = \begin{bmatrix}1 & 5\\ 7 & 0\end{bmatrix}$, $\mathbf{AG} = \begin{bmatrix}1 & 5\\ 0 & 0\end{bmatrix}$.

- $\underset{\sim}{b} = (2, 0)$:

$$\begin{aligned}\mathbf{AG}\underset{\sim}{b} &= (2, 0)\\ &= \underset{\sim}{b},\end{aligned}$$

  so the system is consistent.

- $\underset{\sim}{b} = (2, 1)$: $\mathbf{AG}\underset{\sim}{b} = (7, 0) \neq \underset{\sim}{b}$, so it is not, as Example 0.13 found directly.

Item 4 agrees. $\mathbf{A}$ has rank 1. For $\underset{\sim}{b} = (2, 0)$, $\begin{bmatrix}\mathbf{A} & \underset{\sim}{b}\end{bmatrix} = \begin{bmatrix}1 & 0 & 2\\ 0 & 0 & 0\end{bmatrix}$ also has rank 1. For $\underset{\sim}{b} = (2, 1)$, $\begin{bmatrix}\mathbf{A} & \underset{\sim}{b}\end{bmatrix} = \begin{bmatrix}1 & 0 & 2\\ 0 & 0 & 1\end{bmatrix}$ has rank 2, because $(1, 0)$ and $(2, 1)$ are linearly independent.

Theorem 0.17 (All solutions of a linear system)**.** *Let* $\mathbf{A}$ *be* $m \times n$ *with a generalized inverse* $\mathbf{G}$ *(Definition 0.4).*

1. $\mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{I}_n - \mathbf{GA})$*:* $\underset{\sim}{x}$ *solves* $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$ *exactly when* $\underset{\sim}{x} = (\mathbf{I}_n - \mathbf{GA})\,\underset{\sim}{q}$ *for some* $q \in \mathbb{R}^n$*.*
2. *If* $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ *is consistent (Definition 0.6), then* $\underset{\sim}{x}$ *is a solution exactly when*

$$\underset{\sim}{x} = \mathbf{G}\underset{\sim}{b} + (\mathbf{I}_n - \mathbf{GA})\,\underset{\sim}{q} \quad \text{for some } \underset{\sim}{q} \in \mathbb{R}^n :$$

one particular solution plus a vector of $\mathcal{N}(\mathbf{A})$*.*

Proof

Proof. **Part 1.** First,

$$\begin{aligned}
\mathbf{A}\,(\mathbf{I}_n - \mathbf{GA}) &= \mathbf{AI}_n - \mathbf{A}\,(\mathbf{GA}) && \text{(Theorem 0.7)}\\
&= \mathbf{AI}_n - \mathbf{AGA} && \text{(Theorem 0.7)}\\
&= \mathbf{A} - \mathbf{AGA} && \text{(Theorem 0.10)}\\
&= \mathbf{A} - \mathbf{A} && \text{(Definition 0.4)}\\
&= \mathbf{0}_{m\times n}, && \text{(arithmetic)}
\end{aligned}$$

so $\mathbf{A}\,(\mathbf{I}_n - \mathbf{GA})\,\underset{\sim}{q} = \underset{\sim}{0}_m$ for every $\underset{\sim}{q}$ (Theorem 0.7), and $\mathcal{C}(\mathbf{I}_n - \mathbf{GA}) \subseteq \mathcal{N}(\mathbf{A})$. Conversely, if $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m$, then

$$\begin{aligned}
(\mathbf{I}_n - \mathbf{GA})\,\underset{\sim}{x} &= \mathbf{I}_n\underset{\sim}{x} - (\mathbf{GA})\,\underset{\sim}{x} && \text{(Theorem 0.7, with } \underset{\sim}{x} \text{ as the last factor)}\\
&= \underset{\sim}{x} - (\mathbf{GA})\,\underset{\sim}{x} && \text{(Theorem 0.10)}\\
&= \underset{\sim}{x} - \mathbf{G}\,(\mathbf{A}\underset{\sim}{x}) && \text{(Theorem 0.7)}\\
&= \underset{\sim}{x} - \mathbf{G}\,\underset{\sim}{0}_m && (\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_m)\\
&= \underset{\sim}{x}, && (\mathbf{G}\,\underset{\sim}{0}_m = \underset{\sim}{0}_n)
\end{aligned}$$

so $\underset{\sim}{x}$ is in $\mathcal{C}(\mathbf{I}_n - \mathbf{GA})$, with $\underset{\sim}{q} = \underset{\sim}{x}$.
Part 2. $\mathbf{G}\underset{\sim}{b}$ is a solution (Theorem 0.9). For any $\underset{\sim}{x}$,

$$\begin{aligned}
\mathbf{A}\,(\underset{\sim}{x} - \mathbf{G}\underset{\sim}{b}) &= \mathbf{A}\underset{\sim}{x} - \mathbf{A}\,(\mathbf{G}\underset{\sim}{b}) && \text{(Theorem 0.4, with coefficients 1 and } -1)\\
&= \mathbf{A}\underset{\sim}{x} - \mathbf{AG}\underset{\sim}{b} && \text{(Theorem 0.7)}\\
&= \mathbf{A}\underset{\sim}{x} - \underset{\sim}{b}, && \text{(Theorem 0.16, statement 3)}
\end{aligned}$$

so $\underset{\sim}{x}$ solves $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ exactly when $\underset{\sim}{x} - \mathbf{G}\underset{\sim}{b} \in \mathcal{N}(\mathbf{A})$, that is, by part 1, exactly when $\underset{\sim}{x} - \mathbf{G}\underset{\sim}{b} = (\mathbf{I}_n - \mathbf{GA})\,\underset{\sim}{q}$ for some $\underset{\sim}{q}$. □

Example 0.24 (All solutions of a consistent system)**.** For $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\mathbf{G} = \begin{bmatrix} 1 & 5 \\ 7 & 0 \end{bmatrix}$ and $\underset{\sim}{b} = (2, 0)$: $\mathbf{G}\underset{\sim}{b} = (2, 14)$, $\mathbf{GA} = \begin{bmatrix} 1 & 0 \\ 7 & 0 \end{bmatrix}$, and $\mathbf{I}_2 - \mathbf{GA} = \begin{bmatrix} 0 & 0 \\ -7 & 1 \end{bmatrix}$, so

$$\underset{\sim}{x} = (2, 14) + (0, -7q_1 + q_2), \qquad q_1, q_2 \in \mathbb{R}.$$

Since $-7q_1 + q_2$ can be any number, the solutions are exactly the vectors $(2, t)$, $t \in \mathbb{R}$, which is what

$$\begin{aligned} \mathbf{A}\,(x_1, x_2) &= (x_1, 0) \\ &= (2, 0) \end{aligned}$$

says directly.

Corollary 0.2 (Solvable for every right-hand side, and solvable uniquely)**.** *Let* $\mathbf{A}$ *be* $m \times n$*.*

1. $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ *is consistent for every* $\underset{\sim}{b} \in \mathbb{R}^m$ *exactly when* $\text{rank}(\mathbf{A}) = m$*.*
2. *A consistent system* $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ *has exactly one solution exactly when* $\text{rank}(\mathbf{A}) = n$*.*

Proof

Proof. **Part 1.** By Theorem 0.16, the system is consistent for every $\underset{\sim}{b}$ exactly when $\mathcal{C}(\mathbf{A}) = \mathbb{R}^m$. $\mathcal{C}(\mathbf{A})$ is a subspace of $\mathbb{R}^m$ (Theorem 0.1) with dimension $\text{rank}(\mathbf{A})$ (Theorem 0.1), and $\dim(\mathbb{R}^m) = m$ (Example 0.14). If the dimension is m, then $\mathcal{C}(\mathbf{A}) = \mathbb{R}^m$ (Theorem 0.11); if $\mathcal{C}(\mathbf{A}) = \mathbb{R}^m$, the dimension is m.
Part 2. By Theorem 0.17, the solutions are a particular solution plus the vectors of $\mathcal{N}(\mathbf{A})$, so there is exactly one solution exactly when $\mathcal{N}(\mathbf{A}) = \{\underset{\sim}{0}_n\}$, that is, when $\text{nullity}(\mathbf{A}) = 0$ (a subspace of dimension 0 has the empty list as a basis, whose span is $\{\underset{\sim}{0}\}$, by Definition 0.6 and Definition 0.2; and $\{\underset{\sim}{0}\}$ has dimension 0, by Example 0.14). By Theorem 0.2, that is when $\text{rank}(\mathbf{A}) = n$. □

Example 0.25 (Three systems)**.**

- The 2×3 matrix $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ of Example 0.6 has rank $2 = m$, so every system with it is consistent, but $2 < 3 = n$, so the solutions are never unique.
- The 3×2 matrix $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ of Example 0.1 has rank $2 = n$, so a consistent system has one solution, but $2 < 3 = m$, so some systems are inconsistent: $(1, 0, 0) \notin \mathcal{C}(\mathbf{X})$,

because $a\,(1,1,1)+b\,(1,2,3)$ has equal differences between consecutive entries, and $(1,0,0)$ does not.

- An invertible $p \times p$ matrix $\mathbf{A}$ has rank p: if $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}$ then

$$\begin{aligned} \underset{\sim}{x} &= \mathbf{A}^{-1}\mathbf{A}\underset{\sim}{x} \\ &= \underset{\sim}{0}, \end{aligned}$$

so its nullity is 0 and its rank is p (Theorem 0.2). So every system with it is consistent (part 1, with $p = m$) and has exactly one solution (part 2, with $p = n$), namely $\mathbf{A}^{-1}\underset{\sim}{b}$, since $\mathbf{A}\,(\mathbf{A}^{-1}\underset{\sim}{b}) = \underset{\sim}{b}$.

Theorem 0.18 (The projection onto a column space, with any generalized inverse)**.** *Let* $\mathbf{X}$ *be an* $n \times p$ *matrix of any rank, and let* $\mathbf{G}$ *be any generalized inverse of* $\mathbf{X}^\top\mathbf{X}$ *(Definition 0.4). Then for every* $y \in \mathbb{R}^n$, $\mathbf{X}\mathbf{G}\mathbf{X}^\top y$ *is the orthogonal projection of* y *onto* $\mathcal{C}(\mathbf{X})$ *(Definition 0.3). So the matrix* $\mathbf{X}\mathbf{G}\mathbf{X}^\top$ *is the same for every choice of* $\mathbf{G}$*, and when* $\operatorname{rank}(\mathbf{X}) = p$ *it is the hat matrix* $\mathbf{H}$ *(Definition 0.2).*

Proof

Proof. **$\mathbf{X}^\top y$ is in $\mathcal{C}(\mathbf{X}^\top\mathbf{X})$.** $\mathcal{C}(\mathbf{X}^\top\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X}^\top)$, since $(\mathbf{X}^\top\mathbf{X})\,\underset{\sim}{z} = \mathbf{X}^\top(\mathbf{X}\underset{\sim}{z})$ (Theorem 0.7). Both are subspaces of $\mathbb{R}^p$ (Theorem 0.1) with the same dimension: $\operatorname{rank}(\mathbf{X}^\top\mathbf{X}) = \operatorname{rank}(\mathbf{X}^\top)$ (Theorem 0.4, Theorem 0.1). So $\mathcal{C}(\mathbf{X}^\top\mathbf{X}) = \mathcal{C}(\mathbf{X}^\top)$ (Theorem 0.11), and since $\mathbf{X}^\top\underset{\sim}{y} \in \mathcal{C}(\mathbf{X}^\top)$ (Definition 0.1), $\mathbf{X}^\top\underset{\sim}{y} = \mathbf{X}^\top\mathbf{X}\underset{\sim}{z}$ for some $\underset{\sim}{z} \in \mathbb{R}^p$.
The split. Let

$$\begin{aligned} \underset{\sim}{u} &= \mathbf{X}\mathbf{G}\mathbf{X}^\top y \\ &= \mathbf{X}\,(\mathbf{G}\mathbf{X}^\top\underset{\sim}{y}) \end{aligned}$$

(Theorem 0.7), which is in $\mathcal{C}(\mathbf{X})$ (Definition 0.1). For $\underset{\sim}{y} - \underset{\sim}{u}$,

$$\begin{aligned} \mathbf{X}^\top(\underset{\sim}{y} - \underset{\sim}{u}) &= \mathbf{X}^\top\underset{\sim}{y} - \mathbf{X}^\top\underset{\sim}{u} && \text{(Theorem 0.4, with coefficients 1 and } -1) \\ &= \mathbf{X}^\top\underset{\sim}{y} - \mathbf{X}^\top(\mathbf{X}\mathbf{G}\mathbf{X}^\top\underset{\sim}{y}) && \text{(substitute } \underset{\sim}{u}) \\ &= \mathbf{X}^\top\underset{\sim}{y} - (\mathbf{X}^\top\mathbf{X})\,\mathbf{G}\,(\mathbf{X}^\top\underset{\sim}{y}) && \text{(Theorem 0.7)} \\ &= \mathbf{X}^\top\underset{\sim}{y} - (\mathbf{X}^\top\mathbf{X})\,\mathbf{G}\,(\mathbf{X}^\top\mathbf{X})\,\underset{\sim}{z} && \text{(first step)} \\ &= \mathbf{X}^\top\underset{\sim}{y} - ((\mathbf{X}^\top\mathbf{X})\,\mathbf{G}\,(\mathbf{X}^\top\mathbf{X}))\,\underset{\sim}{z} && \text{(Theorem 0.7)} \\ &= \mathbf{X}^\top\underset{\sim}{y} - (\mathbf{X}^\top\mathbf{X})\,\underset{\sim}{z} && \text{(Definition 0.4)} \\ &= \mathbf{X}^\top\underset{\sim}{y} - \mathbf{X}^\top\underset{\sim}{y} && \text{(first step)} \\ &= \underset{\sim}{0}_p, && \text{(arithmetic)} \end{aligned}$$

so $y - \underset{\sim}{u} \in \mathcal{N}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X})^\perp$ (Theorem 0.3). By the uniqueness in Theorem 0.4, $\underset{\sim}{u}$ is the orthogonal projection of y onto $\mathcal{C}(\mathbf{X})$.

Same matrix. Every choice of $\mathbf{G}$ gives a matrix that projects every y onto $\mathcal{C}(\mathbf{X})$, so all choices give the same matrix (Theorem 0.5, part 2). When $\operatorname{rank}(\mathbf{X}) = p$, $(\mathbf{X}^\top\mathbf{X})^{-1}$ is one choice (Theorem 0.1, Theorem 0.8), and it gives $\mathbf{H}$. □

Example 0.26 (A design matrix with a repeated column)**.** Let $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, an intercept column entered twice, so $\operatorname{rank}(\mathbf{X}) = 1 < 2$ and $\mathbf{X}^\top\mathbf{X} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$ is not invertible: its rank is $\operatorname{rank}(\mathbf{X}) = 1 < 2$ (Theorem 0.4), while an invertible 2×2 matrix has rank 2 (Example 0.25). The matrix $\mathbf{G} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 0 \end{bmatrix}$ is a generalized inverse of it:

$$
\begin{aligned}
(\mathbf{X}^\top\mathbf{X})\,\mathbf{G}\,(\mathbf{X}^\top\mathbf{X}) &= \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} && \text{(Definition 0.10, for the first product)} \\
&= \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}. && \text{(Definition 0.10)}
\end{aligned}
$$

Then

$$
\begin{aligned}
\mathbf{X}\mathbf{G}\mathbf{X}^\top &= \begin{bmatrix} \frac{1}{2} & 0 \\ \frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} && \text{(Definition 0.10, for } \mathbf{X}\mathbf{G}\text{)} \\
&= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}, && \text{(Definition 0.10)}
\end{aligned}
$$

the hat matrix of the intercept-only model (Example 0.2). Another generalized inverse, $\mathbf{G}' = \begin{bmatrix} 0 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$, which is one because $(\mathbf{X}^\top\mathbf{X})\,\mathbf{G}' = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$ and that times $\mathbf{X}^\top\mathbf{X}$ is again $\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$, gives $\mathbf{X}\mathbf{G}' = \begin{bmatrix} 0 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix}$ and the same $\mathbf{X}\mathbf{G}'\mathbf{X}^\top = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$, as Theorem 0.18 says it must. Both project onto $\mathcal{C}(\mathbf{X}) = \operatorname{span}\{(1,1)\}$, the column space of the intercept-only design.

Solving linear systems and least squares

Source

This section is adapted from Zhou [84] and Zhou [88], used under the MIT License (see the license text in Section). The first source states that every invertible matrix has an LU factorization; that is false without reordering rows (the 2×2 row swap $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ has none), so this section states only how an LU factorization, when one exists, solves a system. The least squares results are proved here from the closest-point theorem (Theorem 0.3) rather than by differentiating, and the QR factorization is derived from Gram-Schmidt (Theorem 0.4). These parts of the source are not part of this section:

- the operation counts for the triangular solves, LU and QR
- solving a system by inverting a matrix through its QR factorization, and the comparison of that with solving it directly
- the pivoted factorization $\mathbf{PA} = \mathbf{LU}$, which Example 0.36 only touches

Definition 0.7 (Upper triangular matrix)**.** A square matrix $\mathbf{U}$ is **upper triangular** [89] if every entry below the diagonal is 0: $u_{ij} = 0$ whenever $i > j$.

Definition 0.8 (Lower triangular matrix)**.** A square matrix $\mathbf{L}$ is **lower triangular** if every entry above the diagonal is 0: $\ell_{ij} = 0$ whenever $i < j$.

Definition 0.9 (Unit lower triangular matrix)**.** A lower triangular matrix (Definition 0.8) whose diagonal entries are all 1 is **unit lower triangular**.

Example 0.27 (Triangular and not)**.**

- $\begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix}$ is upper triangular: its entries below the diagonal, in positions $(2,1)$, $(3,1)$ and $(3,2)$, are 0.
- $\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$ is unit lower triangular.
- A diagonal matrix (Definition 0.4) is both upper and lower triangular.
- $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is neither: its $(2,1)$ entry is not 0, and neither is its $(1,2)$ entry.

Theorem 0.19 (Solving a triangular system by substitution)**.** *Let* $\mathbf{U}$ *be an* $n \times n$ *upper triangular matrix (Definition 0.7) whose diagonal entries* $u_{11}, \dots, u_{nn}$ *are all nonzero. For every* $\underset{\sim}{b} \in \mathbb{R}^n$*, the system* $\mathbf{U}\underset{\sim}{x} = \underset{\sim}{b}$ *has exactly one solution, given for* $i = n, n-1, \dots, 1$ *in turn by*

$$x_i = \frac{1}{u_{ii}}\left(b_i - \sum_{j=i+1}^{n} u_{ij}\, x_j\right)$$

*(**back substitution**). Likewise, if* $\mathbf{L}$ *is lower triangular with nonzero diagonal,* $\mathbf{L}\underset{\sim}{y} = \underset{\sim}{b}$ *has exactly one solution, given for* $i = 1, \dots, n$ *by* $y_i = \frac{1}{\ell_{ii}}\left(b_i - \sum_{j=1}^{i-1} \ell_{ij}\, y_j\right)$ *(**forward substitution**).*

Proof

Proof. Row i of $\mathbf{U}\underset{\sim}{x} = \underset{\sim}{b}$ reads

$$\begin{aligned} b_i &= \sum_{j=1}^{n} u_{ij}\, x_j && \text{(Definition 0.16)} \\ &= \sum_{j=i}^{n} u_{ij}\, x_j && (u_{ij} = 0 \text{ for } j < i) \\ &= u_{ii}\, x_i + \sum_{j=i+1}^{n} u_{ij}\, x_j. && \text{(split off the } j = i \text{ term)} \end{aligned}$$

This equation is equivalent to each of

$$\begin{aligned} u_{ii}\, x_i &= b_i - \sum_{j=i+1}^{n} u_{ij}\, x_j && \text{(subtract the sum from both sides)} \\ x_i &= \frac{1}{u_{ii}}\left(b_i - \sum_{j=i+1}^{n} u_{ij}\, x_j\right). && \text{(divide by } u_{ii} \neq 0) \end{aligned}$$

Row n involves only x_n, so it fixes x_n; once $x_{i+1}, \dots, x_n$ are fixed, row i fixes x_i. So the rows, taken from the last to the first, hold exactly when $\underset{\sim}{x}$ is the vector the formula builds: there is one solution, and only one. For $\mathbf{L}$, row i reads $\sum_{j=1}^{i-1} \ell_{ij}\, y_j + \ell_{ii}\, y_i = b_i$, and the same argument runs from the first row to the last. □

Example 0.28 (Back substitution on a 3×3 system)**.** Solve $\begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix} \underset{\sim}{x} = \begin{bmatrix} 8 \\ 1 \\ 1 \end{bmatrix}$:

1. $$\begin{aligned} x_3 &= \frac{1}{-1} \cdot 1 \\ &= -1; \end{aligned}$$

2. $$\begin{aligned} x_2 &= \frac{1}{1/2}\left(1 - \tfrac{1}{2} \cdot (-1)\right) \\ &= 2 \cdot \tfrac{3}{2} \\ &= 3; \end{aligned}$$

3. $$\begin{aligned} x_1 &= \frac{1}{2}(8 - 1 \cdot 3 - (-1)(-1)) \\ &= \frac{1}{2} \cdot 4 \\ &= 2. \end{aligned}$$

So $\underset{\sim}{x} = (2, 3, -1)$. Without a nonzero diagonal the method fails: $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \underset{\sim}{x} = (1, 1)$ has no solution, since its second row reads $0 = 1$.

Theorem 0.20 (QR factorization)**.** *Let* $\mathbf{A}$ *be an* $m \times n$ *matrix with* $\operatorname{rank}(\mathbf{A}) = n$*. Then*

$$\underset{m\times n}{\mathbf{A}} = \underset{m\times n}{\mathbf{Q}} \; \underset{n\times n}{\mathbf{R}},$$

where the columns of $\mathbf{Q}$ *are orthonormal (Definition 0.23) and* $\mathbf{R}$ *is upper triangular (Definition 0.7) with positive diagonal entries.*

Proof

Proof. The n columns $\underset{\sim}{a}_1, \ldots, \underset{\sim}{a}_n$ of $\mathbf{A}$ are linearly independent (Definition 0.3), so the Gram-Schmidt process (Definition 0.2) runs all n steps and gives orthonormal $\underset{\sim}{q}_1, \ldots, \underset{\sim}{q}_n$ (Theorem 0.4, parts 1 and 3); let $\mathbf{Q}$ have these columns. At step i, $\underset{\sim}{q}'_i \neq \underset{\sim}{0}$, and rearranging the orthogonalize step,

$$\begin{aligned} \underset{\sim}{a}_i &= \underset{\sim}{q}'_i + \sum_{j=1}^{i-1} (\underset{\sim}{q}_j \cdot \underset{\sim}{a}_i)\, \underset{\sim}{q}_j && \text{(add the sum to both sides of the orthogonalize step)} \\ &= \left\|\underset{\sim}{q}'_i\right\| \underset{\sim}{q}_i + \sum_{j=1}^{i-1} (\underset{\sim}{q}_j \cdot \underset{\sim}{a}_i)\, \underset{\sim}{q}_j. && \text{(normalize step: } \underset{\sim}{q}'_i = \left\|\underset{\sim}{q}'_i\right\| \underset{\sim}{q}_i) \end{aligned}$$

Let $\mathbf{R}$ be the $n \times n$ matrix with $r_{ji} = \underset{\sim}{q}_j \cdot \underset{\sim}{a}_i$ for $j < i$, $r_{ii} = \|\underset{\sim}{q}'_i\| > 0$, and $r_{ji} = 0$ for $j > i$; it is upper triangular with positive diagonal. Column i of $\mathbf{QR}$ is $\mathbf{Q}$ times column i of $\mathbf{R}$ (4), and

$$\begin{aligned}
\mathbf{Q}\,(r_{1i}, \ldots, r_{ni}) &= \sum_{j=1}^{n} r_{ji}\, \underset{\sim}{q}_j && \text{(Theorem 0.3)} \\
&= \sum_{j=1}^{i} r_{ji}\, \underset{\sim}{q}_j && (r_{ji} = 0 \text{ for } j > i) \\
&= r_{ii}\, \underset{\sim}{q}_i + \sum_{j=1}^{i-1} r_{ji}\, \underset{\sim}{q}_j && \text{(split off the } j = i \text{ term)} \\
&= \|\underset{\sim}{q}'_i\|\, \underset{\sim}{q}_i + \sum_{j=1}^{i-1} (\underset{\sim}{q}_j \cdot \underset{\sim}{a}_i)\, \underset{\sim}{q}_j && \text{(the entries of } \mathbf{R}) \\
&= \underset{\sim}{a}_i. && \text{(the display)}
\end{aligned}$$

So $\mathbf{QR} = \mathbf{A}$. □

Example 0.29 (A QR factorization from Gram-Schmidt)**.** For $\mathbf{A}$ with columns $\underset{\sim}{a}_1 = (1,1,0)$, $\underset{\sim}{a}_2 = (1,0,1)$, $\underset{\sim}{a}_3 = (0,1,1)$, Example 0.6 and Example 0.8 found $\underset{\sim}{q}_1 = \frac{1}{\sqrt{2}}(1,1,0)$, $\underset{\sim}{q}_2 = \frac{1}{\sqrt{6}}(1,-1,2)$, $\underset{\sim}{q}_3 = \frac{1}{\sqrt{3}}(-1,1,1)$, with $\|\underset{\sim}{q}'_1\| = \sqrt{2}$, $\|\underset{\sim}{q}'_2\| = \sqrt{3/2}$, $\|\underset{\sim}{q}'_3\| = 2/\sqrt{3}$,

$$\begin{aligned}
\underset{\sim}{q}_1 \cdot \underset{\sim}{a}_2 &= \underset{\sim}{q}_1 \cdot \underset{\sim}{a}_3 \\
&= \tfrac{1}{\sqrt{2}}
\end{aligned}$$

and $\underset{\sim}{q}_2 \cdot \underset{\sim}{a}_3 = \frac{1}{\sqrt{6}}$. So

$$\mathbf{R} = \begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \sqrt{3/2} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{2}{\sqrt{3}} \end{bmatrix}.$$

As a check on the second column,

$$\begin{aligned}
\tfrac{1}{\sqrt{2}}\, \underset{\sim}{q}_1 + \sqrt{3/2}\, \underset{\sim}{q}_2 &= \tfrac{1}{2}(1,1,0) + \tfrac{1}{2}(1,-1,2) \\
&= (1,0,1) \\
&= \underset{\sim}{a}_2.
\end{aligned}$$

Definition 0.10 (Least squares solution)**.** Let $\mathbf{A}$ be $m \times n$ and $\underset{\sim}{b} \in \mathbb{R}^m$. A **least squares solution** [90] of $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ is a vector $\underset{\sim}{\hat{x}} \in \mathbb{R}^n$ such that

$$\|\underset{\sim}{b} - \mathbf{A}\underset{\sim}{\hat{x}}\| \le \|\underset{\sim}{b} - \mathbf{A}\underset{\sim}{x}\| \quad \text{for every } \underset{\sim}{x} \in \mathbb{R}^n.$$

Example 0.30 (Fitting a constant)**.** Let $\mathbf{A} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\underset{\sim}{b} = (1, 3)$. The system $x\,(1, 1) = (1, 3)$ is inconsistent (Definition 0.6): its first entry needs $x = 1$ and its second needs $x = 3$. For any number x,

$$\begin{aligned} \|\underset{\sim}{b} - \mathbf{A}x\|^2 &= (1 - x)^2 + (3 - x)^2 \\ &= 2\,(x - 2)^2 + 2, \end{aligned}$$

which is smallest at $x = 2$, so $\hat{x} = 2$ is the least squares solution. The value $x = 1$ is not one: it gives $0 + 4 = 4 > 2$. The normal equations read $2x = 4$, and their solution is $x = 2$ too.

Definition 0.11 (Normal equations)**.** For $\mathbf{A}$ and $\underset{\sim}{b}$ as in Definition 0.10, the equations $\mathbf{A}^\top\mathbf{A}\underset{\sim}{x} = \mathbf{A}^\top\underset{\sim}{b}$ are the **normal equations** [91].

Theorem 0.21 (Least squares solutions solve the normal equations)**.** *Let* $\mathbf{A}$ *be* $m \times n$ *and* $\underset{\sim}{b} \in \mathbb{R}^m$*.*

1. $\underset{\sim}{\hat{x}}$ *is a least squares solution (Definition 0.10) exactly when it solves the normal equations* $\mathbf{A}^\top\mathbf{A}\underset{\sim}{\hat{x}} = \mathbf{A}^\top\underset{\sim}{b}$*.*
2. *The normal equations are consistent (Definition 0.6).*
3. *Every least squares solution gives the same fitted vector* $\mathbf{A}\underset{\sim}{\hat{x}}$*: the orthogonal projection of* $\underset{\sim}{b}$ *onto* $\mathcal{C}(\mathbf{A})$ *(Definition 0.3).*
4. *The least squares solution is unique exactly when* $\operatorname{rank}(\mathbf{A}) = n$*, and then* $\underset{\sim}{\hat{x}} = (\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top\underset{\sim}{b}$*.*

Proof

Proof. Let $\underset{\sim}{u}$ be the orthogonal projection of $\underset{\sim}{b}$ onto $\mathcal{C}(\mathbf{A})$. The vectors $\mathbf{A}\underset{\sim}{x}$, as $\underset{\sim}{x}$ ranges over $\mathbb{R}^n$, are exactly the points of $\mathcal{C}(\mathbf{A})$ (Definition 0.1). **Least squares solutions are the solutions of** $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{u}$**.** If $\mathbf{A}\underset{\sim}{\hat{x}} = \underset{\sim}{u}$, then for every $\underset{\sim}{x}$, $\mathbf{A}\underset{\sim}{x} \in \mathcal{C}(\mathbf{A})$, so $\|\underset{\sim}{b} - \mathbf{A}\underset{\sim}{\hat{x}}\| = \|\underset{\sim}{b} - \underset{\sim}{u}\| \le \|\underset{\sim}{b} - \mathbf{A}\underset{\sim}{x}\|$ (Theorem 0.3), and $\underset{\sim}{\hat{x}}$ is a least squares solution. Conversely, let $\underset{\sim}{\hat{x}}$ be a least squares solution. $\underset{\sim}{u} \in \mathcal{C}(\mathbf{A})$, so $\underset{\sim}{u} = \mathbf{A}\underset{\sim}{x_0}$ for some $\underset{\sim}{x_0}$ (Definition 0.1), and

$$\begin{aligned} \|\underset{\sim}{b} - \mathbf{A}\underset{\sim}{\hat{x}}\| &\le \|\underset{\sim}{b} - \mathbf{A}\underset{\sim}{x_0}\| && \text{(Definition 0.10)} \\ &= \|\underset{\sim}{b} - \underset{\sim}{u}\| && (\underset{\sim}{u} = \mathbf{A}\underset{\sim}{x_0}) \\ &\le \|\underset{\sim}{b} - \mathbf{A}\underset{\sim}{\hat{x}}\|, && \text{(Theorem 0.3, with } \underset{\sim}{w} = \mathbf{A}\underset{\sim}{\hat{x}} \in \mathcal{C}(\mathbf{A})) \end{aligned}$$

so the last inequality is an equality, and the equality case of Theorem 0.3 gives $\mathbf{A}\hat{\underset{\sim}{x}} = \underset{\sim}{u}$.
Parts 1 and 3. $\mathbf{A}\hat{\underset{\sim}{x}} = \underset{\sim}{u}$ holds exactly when $\underset{\sim}{b} - \mathbf{A}\hat{\underset{\sim}{x}} \in \mathcal{C}(\mathbf{A})^{\perp}$: if $\mathbf{A}\hat{\underset{\sim}{x}} = \underset{\sim}{u}$, then $\underset{\sim}{b} - \underset{\sim}{u} \in \mathcal{C}(\mathbf{A})^{\perp}$ (Definition 0.3); conversely, if $\underset{\sim}{b} - \mathbf{A}\hat{\underset{\sim}{x}} \in \mathcal{C}(\mathbf{A})^{\perp}$, then $\underset{\sim}{b} = \mathbf{A}\hat{\underset{\sim}{x}} + (\underset{\sim}{b} - \mathbf{A}\hat{\underset{\sim}{x}})$ is a split into $\mathcal{C}(\mathbf{A})$ and $\mathcal{C}(\mathbf{A})^{\perp}$, so $\mathbf{A}\hat{\underset{\sim}{x}} = \underset{\sim}{u}$ by the uniqueness in Theorem 0.4. And $\mathcal{C}(\mathbf{A})^{\perp} = \mathcal{N}(\mathbf{A}^{\top})$ (Theorem 0.3), where

$$\begin{aligned} \mathbf{A}^{\top}(\underset{\sim}{b} - \mathbf{A}\hat{\underset{\sim}{x}}) &= \mathbf{A}^{\top}\underset{\sim}{b} - \mathbf{A}^{\top}(\mathbf{A}\hat{\underset{\sim}{x}}) && \text{(Theorem 0.7)} \\ &= \mathbf{A}^{\top}\underset{\sim}{b} - \mathbf{A}^{\top}\mathbf{A}\hat{\underset{\sim}{x}}, && \text{(Theorem 0.7)} \end{aligned}$$

which is $\underset{\sim}{0}_n$ exactly when the normal equations hold.
Part 2. The vector $\underset{\sim}{x}_0$ above satisfies $\mathbf{A}\underset{\sim}{x}_0 = \underset{\sim}{u}$, so it is a least squares solution, and it solves the normal equations by part 1.
Part 4. The least squares solutions are exactly the solutions of $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{u}$, a consistent system, so there is exactly one when $\operatorname{rank}(\mathbf{A}) = n$ and more than one otherwise (Corollary 0.2). When $\operatorname{rank}(\mathbf{A}) = n$, $\mathbf{A}^{\top}\mathbf{A}$ is invertible (Theorem 0.1), and with $\mathbf{M} \overset{\text{def}}{=} \mathbf{A}^{\top}\mathbf{A}$,

$$\begin{aligned} \hat{\underset{\sim}{x}} &= \mathbf{I}_n\,\hat{\underset{\sim}{x}} && \text{(Theorem 0.10)} \\ &= (\mathbf{M}^{-1}\mathbf{M})\,\hat{\underset{\sim}{x}} && \text{(Definition 0.5)} \\ &= \mathbf{M}^{-1}\,(\mathbf{M}\hat{\underset{\sim}{x}}) && \text{(Theorem 0.7)} \\ &= \mathbf{M}^{-1}\,\mathbf{A}^{\top}\underset{\sim}{b}. && \text{(the normal equations)} \end{aligned}$$

□

Example 0.31 (A least squares line)**.** Fit $y = x_1 + x_2 t$ to the points $(t, y) = (1,1), (2,2), (3,2)$: $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ (the $\mathbf{X}$ of Example 0.1) and $\underset{\sim}{b} = (1,2,2)$. $\mathbf{A}$ has rank 2 (Example 0.2), so the solution is unique (part 4). $\mathbf{A}^{\top}\mathbf{A} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$ has inverse $\frac{1}{6}\begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}$ (Example 0.1), and

$$\begin{aligned} \mathbf{A}^{\top}\underset{\sim}{b} &= (1+2+2,\ 1+4+6) \\ &= (5, 11), \end{aligned}$$

so

$$\begin{aligned} \hat{\underset{\sim}{x}} &= \frac{1}{6}(14 \cdot 5 - 6 \cdot 11,\ -6 \cdot 5 + 3 \cdot 11) \\ &= \frac{1}{6}\,(4, 3) \\ &= \left(\tfrac{2}{3}, \tfrac{1}{2}\right). \end{aligned}$$

The fitted values are $\mathbf{A}\hat{\underset{\sim}{x}} = \left(\frac{7}{6}, \frac{5}{3}, \frac{13}{6}\right)$, and the residuals $\underset{\sim}{b} - \mathbf{A}\hat{\underset{\sim}{x}} = \left(-\frac{1}{6}, \frac{1}{3}, -\frac{1}{6}\right)$ are

orthogonal to both columns of $\mathbf{A}$: $-\frac{1}{6}+\frac{1}{3}-\frac{1}{6}=0$ and $-\frac{1}{6}+\frac{2}{3}-\frac{1}{2}=0$.

Example 0.32 (Many least squares solutions, one fitted vector)**.** With the rank-1 matrix $\mathbf{A}=\begin{bmatrix}1 & 2\\ 1 & 2\\ 1 & 2\end{bmatrix}$ (Example 0.2) and $\underset{\sim}{b}=(1,2,2)$, $\mathbf{A}^\top\mathbf{A}=\begin{bmatrix}3 & 6\\ 6 & 12\end{bmatrix}$ and $\mathbf{A}^\top\underset{\sim}{b}=(5,10)$, so both normal equations say $3x_1+6x_2=5$, that is, $x_1+2x_2=\frac{5}{3}$. There are infinitely many least squares solutions, such as $(\frac{5}{3},0)$ and $(0,\frac{5}{6})$, as part 4 predicts for $\operatorname{rank}(\mathbf{A})=1<2$; but every one gives the same fitted vector

$$\begin{aligned}\mathbf{A}\underset{\sim}{\hat{x}} &= (x_1+2x_2)\,(1,1,1)\\ &= \tfrac{5}{3}\,(1,1,1),\end{aligned}$$

as part 3 says.

Theorem 0.22 (Solving least squares by QR)**.** *Let* $\mathbf{A}$ *be* $m\times n$ *with* $\operatorname{rank}(\mathbf{A})=n$*, with QR factorization* $\mathbf{A}=\mathbf{QR}$ *(Theorem 0.20), and let* $\underset{\sim}{b}\in\mathbb{R}^m$*. The least squares solution (Definition 0.10) is the unique solution of*

$$\underset{n\times n}{\mathbf{R}}\ \underset{\sim}{\hat{x}}=\underset{n\times m}{\mathbf{Q}^\top}\ \underset{\sim}{b},$$

which back substitution computes (Theorem 0.19). When $m=n$*, it is the unique solution of* $\mathbf{A}\underset{\sim}{x}=\underset{\sim}{b}$*.*

Proof

Proof. $\mathbf{R}$ is upper triangular with positive diagonal, so $\mathbf{R}\underset{\sim}{x}=\mathbf{Q}^\top\underset{\sim}{b}$ has exactly one solution $\underset{\sim}{\hat{x}}$ (Theorem 0.19). $\mathbf{Q}^\top\mathbf{Q}=\mathbf{I}_n$, as in the proof of Theorem 0.4. Then

$$\begin{aligned}
\mathbf{A}^\top\mathbf{A}\underset{\sim}{\hat{x}} &= (\mathbf{QR})^\top(\mathbf{QR})\,\underset{\sim}{\hat{x}} && \text{(substitute } \mathbf{A}=\mathbf{QR})\\
&= \mathbf{R}^\top\mathbf{Q}^\top(\mathbf{QR})\,\underset{\sim}{\hat{x}} && \text{(Theorem 0.19)}\\
&= \mathbf{R}^\top(\mathbf{Q}^\top\mathbf{Q})\,(\mathbf{R}\underset{\sim}{\hat{x}}) && \text{(Theorem 0.7)}\\
&= \mathbf{R}^\top\mathbf{I}_n\,(\mathbf{R}\underset{\sim}{\hat{x}}) && (\mathbf{Q}^\top\mathbf{Q}=\mathbf{I}_n)\\
&= \mathbf{R}^\top(\mathbf{R}\underset{\sim}{\hat{x}}) && \text{(Theorem 0.10)}\\
&= \mathbf{R}^\top(\mathbf{Q}^\top\underset{\sim}{b}) && (\mathbf{R}\underset{\sim}{\hat{x}}=\mathbf{Q}^\top\underset{\sim}{b})\\
&= (\mathbf{R}^\top\mathbf{Q}^\top)\,\underset{\sim}{b} && \text{(Theorem 0.7)}\\
&= (\mathbf{QR})^\top\underset{\sim}{b} && \text{(Theorem 0.19)}\\
&= \mathbf{A}^\top\underset{\sim}{b}, && (\mathbf{A}=\mathbf{QR})
\end{aligned}$$

so $\hat{\underset{\sim}{x}}$ solves the normal equations and is the least squares solution (Theorem 0.21), which is unique because $\operatorname{rank}(\mathbf{A}) = n$. When $m = n$,

$$\begin{aligned}\operatorname{rank}(\mathbf{A}) &= n \\ &= m,\end{aligned}$$

so $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ is consistent for every $\underset{\sim}{b}$ (Corollary 0.2, part 1), so its solution leaves residual $\underset{\sim}{0}$, is a least squares solution, and is therefore $\hat{\underset{\sim}{x}}$. □

Example 0.33 (The least squares line by QR)**.** For $\mathbf{A}$ and $\underset{\sim}{b}$ of Example 0.31, Gram-Schmidt gives $\underset{\sim}{q}_1 = \frac{1}{\sqrt{3}}(1,1,1)$; then

$$\begin{aligned}\underset{\sim}{q}_1 \cdot (1,2,3) &= \tfrac{6}{\sqrt{3}} \\ &= 2\sqrt{3},\end{aligned}$$

$$\begin{aligned}\underset{\sim}{q}_2' &= (1,2,3) - 2\,(1,1,1) \\ &= (-1,0,1)\end{aligned}$$

with norm $\sqrt{2}$, and $\underset{\sim}{q}_2 = \frac{1}{\sqrt{2}}(-1,0,1)$. So

$$\begin{aligned}\mathbf{R} &= \begin{bmatrix}\sqrt{3} & 2\sqrt{3} \\ 0 & \sqrt{2}\end{bmatrix}, \\ \mathbf{Q}^\top \underset{\sim}{b} &= \left(\tfrac{1+2+2}{\sqrt{3}},\ \tfrac{-1+0+2}{\sqrt{2}}\right) \\ &= \left(\tfrac{5}{\sqrt{3}}, \tfrac{1}{\sqrt{2}}\right).\end{aligned}$$

Back substitution gives

$$\begin{aligned}x_2 &= \tfrac{1}{\sqrt{2}} \cdot \tfrac{1}{\sqrt{2}} \\ &= \tfrac{1}{2}\end{aligned}$$

and

$$\begin{aligned}x_1 &= \tfrac{1}{\sqrt{3}}\left(\tfrac{5}{\sqrt{3}} - 2\sqrt{3} \cdot \tfrac{1}{2}\right) \\ &= \tfrac{5}{3} - 1 \\ &= \tfrac{2}{3},\end{aligned}$$

the solution found in Example 0.31.

Theorem 0.23 (The Moore-Penrose inverse gives a least squares solution)**.** *Let* $\mathbf{A}$ *be* $m \times n$ *and* $\underset{\sim}{b} \in \mathbb{R}^m$*. Then* $\mathbf{A}^+\underset{\sim}{b}$ *is a least squares solution of* $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$*, and a vector* $\underset{\sim}{x} \in \mathbb{R}^n$ *is a least squares solution if and only if*

$$\mathbf{A}\underset{\sim}{x} = \mathbf{A}\mathbf{A}^{+}\underset{\sim}{b}.$$

Proof

Proof. Let $\mathbf{P} = \mathbf{A}\mathbf{A}^{+}$. By conditions 3 and 1 of Definition 0.5, $\mathbf{P}^{\top} = \mathbf{P}$ and $\mathbf{P}\mathbf{A} = \mathbf{A}$. Hence

$$\begin{aligned}
\mathbf{A}^{\top}(\mathbf{I}-\mathbf{P}) &= \mathbf{A}^{\top} - \mathbf{A}^{\top}\mathbf{P} && \text{(distributing)}\\
&= \mathbf{A}^{\top} - \mathbf{A}^{\top}\mathbf{P}^{\top} && (\mathbf{P}^{\top} = \mathbf{P})\\
&= \mathbf{A}^{\top} - (\mathbf{P}\mathbf{A})^{\top} && \text{(Theorem 0.19)}\\
&= \mathbf{A}^{\top} - \mathbf{A}^{\top} && (\mathbf{P}\mathbf{A} = \mathbf{A})\\
&= \mathbf{0}.
\end{aligned}$$

For every $\underset{\sim}{x} \in \mathbb{R}^n$, write $\underset{\sim}{b} - \mathbf{A}\underset{\sim}{x} = \underset{\sim}{u} + \underset{\sim}{v}$ with

$$\underset{\sim}{u} = \underset{\sim}{b} - \mathbf{P}\underset{\sim}{b} \qquad \text{and} \qquad \underset{\sim}{v} = \mathbf{P}\underset{\sim}{b} - \mathbf{A}\underset{\sim}{x} = \mathbf{A}(\mathbf{A}^{+}\underset{\sim}{b} - \underset{\sim}{x}).$$

Their inner product is

$$\begin{aligned}
\underset{\sim}{v}\cdot\underset{\sim}{u} &= (\mathbf{A}^{+}\underset{\sim}{b} - \underset{\sim}{x})^{\top}\,\mathbf{A}^{\top}(\mathbf{I}-\mathbf{P})\underset{\sim}{b} && \text{(Theorem 0.19)}\\
&= 0, && (\mathbf{A}^{\top}(\mathbf{I}-\mathbf{P}) = \mathbf{0})
\end{aligned}$$

so $\|\underset{\sim}{b} - \mathbf{A}\underset{\sim}{x}\|^2 = \|\underset{\sim}{u}\|^2 + \|\underset{\sim}{v}\|^2$. That is,

$$\|\underset{\sim}{b} - \mathbf{A}\underset{\sim}{x}\|^2 = \|\underset{\sim}{b} - \mathbf{P}\underset{\sim}{b}\|^2 + \|\mathbf{A}(\mathbf{A}^{+}\underset{\sim}{b} - \underset{\sim}{x})\|^2.$$

The first summand does not involve $\underset{\sim}{x}$, and the second is nonnegative, with a zero at $\underset{\sim}{x} = \mathbf{A}^{+}\underset{\sim}{b}$. So $\mathbf{A}^{+}\underset{\sim}{b}$ minimizes $\|\underset{\sim}{b} - \mathbf{A}\underset{\sim}{x}\|$, and the smallest norm is $\|\underset{\sim}{b} - \mathbf{P}\underset{\sim}{b}\|$. A vector $\underset{\sim}{x}$ attains that norm if and only if $\mathbf{A}(\mathbf{A}^{+}\underset{\sim}{b} - \underset{\sim}{x}) = \underset{\sim}{0}$, which is $\mathbf{A}\underset{\sim}{x} = \mathbf{A}\mathbf{A}^{+}\underset{\sim}{b}$. □

Theorem 0.24 (The Moore-Penrose inverse gives the least squares solution of smallest norm)**.** *Let* $\mathbf{A}$ *be* $m \times n$ *and* $\underset{\sim}{b} \in \mathbb{R}^m$*. If* $\underset{\sim}{x}$ *is a least squares solution (Definition 0.10) of* $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$*, then*

$$\|\underset{\sim}{x}\|^2 = \|\mathbf{A}^{+}\underset{\sim}{b}\|^2 + \|\underset{\sim}{x} - \mathbf{A}^{+}\underset{\sim}{b}\|^2.$$

So $\|\underset{\sim}{x}\| \geq \|\mathbf{A}^{+}\underset{\sim}{b}\|$*, with equality only if* $\underset{\sim}{x} = \mathbf{A}^{+}\underset{\sim}{b}$*.*

Proof

Proof. Let $\underset{\sim}{d} = \underset{\sim}{x} - \mathbf{A}^{+}\underset{\sim}{b}$. By Theorem 0.23, $\mathbf{A}\underset{\sim}{x} = \mathbf{A}\mathbf{A}^{+}\underset{\sim}{b}$, so $\mathbf{A}\underset{\sim}{d} = \underset{\sim}{0}$.
Let $\underset{\sim}{w} = (\mathbf{A}^{+})^{\top}\mathbf{A}^{+}\underset{\sim}{b}$. Then

$$\begin{aligned}
\mathbf{A}^{+}\underset{\sim}{b} &= \mathbf{A}^{+}\mathbf{A}\mathbf{A}^{+}\underset{\sim}{b} && \text{(condition 2 of Definition 0.5)}\\
&= (\mathbf{A}^{+}\mathbf{A})^{\top}\mathbf{A}^{+}\underset{\sim}{b} && \text{(condition 4 of Definition 0.5)}\\
&= \mathbf{A}^{\top}(\mathbf{A}^{+})^{\top}\mathbf{A}^{+}\underset{\sim}{b} && \text{(Theorem 0.19)}\\
&= \mathbf{A}^{\top}\underset{\sim}{w}. && \text{(definition of } \underset{\sim}{w})
\end{aligned}$$

Therefore

$$\begin{aligned}
\mathbf{A}^{+}\underset{\sim}{b}\cdot\underset{\sim}{d} &= \mathbf{A}^{\top}\underset{\sim}{w}\cdot\underset{\sim}{d} && (\mathbf{A}^{+}\underset{\sim}{b} = \mathbf{A}^{\top}\underset{\sim}{w})\\
&= \underset{\sim}{w}\cdot\mathbf{A}\underset{\sim}{d} && \text{(Theorem 0.19)}\\
&= 0. && (\mathbf{A}\underset{\sim}{d} = \underset{\sim}{0})
\end{aligned}$$

Expanding $\|\underset{\sim}{x}\|^2 = \|\mathbf{A}^{+}\underset{\sim}{b} + \underset{\sim}{d}\|^2$ then gives

$$\begin{aligned}
\|\underset{\sim}{x}\|^2 &= \|\mathbf{A}^{+}\underset{\sim}{b}\|^2 + 2\,\mathbf{A}^{+}\underset{\sim}{b}\cdot\underset{\sim}{d} + \|\underset{\sim}{d}\|^2 && \text{(expanding)}\\
&= \|\mathbf{A}^{+}\underset{\sim}{b}\|^2 + \|\underset{\sim}{d}\|^2. && (\mathbf{A}^{+}\underset{\sim}{b}\cdot\underset{\sim}{d} = 0)
\end{aligned}$$

Equality in $\|\underset{\sim}{x}\| \geq \|\mathbf{A}^{+}\underset{\sim}{b}\|$ holds only if $\|\underset{\sim}{d}\| = 0$, that is, $\underset{\sim}{x} = \mathbf{A}^{+}\underset{\sim}{b}$. □

Example 0.34 (A rank-deficient least squares problem)**.** Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $\underset{\sim}{b} = (2, 3)$.
By Example 0.14, $\mathbf{A}^{+} = \mathbf{A}$, so $\mathbf{A}^{+}\underset{\sim}{b} = (2, 0)$ and $\mathbf{A}\mathbf{A}^{+}\underset{\sim}{b} = (2, 0)$.
By Theorem 0.23, $\underset{\sim}{x} = (x_1, x_2)$ is a least squares solution if and only if $\mathbf{A}\underset{\sim}{x} = (x_1, 0) = (2, 0)$, that is, $x_1 = 2$ with x_2 arbitrary. Each of them leaves the same residual $\underset{\sim}{b} - \mathbf{A}\underset{\sim}{x} = (0, 3)$, of norm 3.
For each such $\underset{\sim}{x}$,

$$\|\underset{\sim}{x}\|^2 = 4 + x_2^2 = \|\mathbf{A}^{+}\underset{\sim}{b}\|^2 + \|\underset{\sim}{x} - \mathbf{A}^{+}\underset{\sim}{b}\|^2,$$

as Theorem 0.24 states. The smallest is 4, at $x_2 = 0$, which is $\mathbf{A}^{+}\underset{\sim}{b} = (2, 0)$.

Definition 0.12 (LU factorization)**.** An **LU factorization** [92] of an $n \times n$ matrix $\mathbf{A}$ is a product $\mathbf{A} = \mathbf{L}\mathbf{U}$ with $\mathbf{L}$ unit lower triangular (Definition 0.9) and $\mathbf{U}$ upper triangular (Definition 0.7).

Example 0.35 (An LU factorization of a 3×3 matrix)**.** Let $\mathbf{A} = \begin{bmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{bmatrix}$, $\mathbf{L} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 1 & 0 \\ -1 & 4 & 1 \end{bmatrix}$ (unit lower triangular) and $\mathbf{U} = \begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix}$ (upper triangular). By 4, entry (i, j) of $\mathbf{LU}$ is $\sum_k \ell_{ik}\, u_{kj}$, so row i of $\mathbf{LU}$ is $\sum_k \ell_{ik}$ times row k of $\mathbf{U}$. Row 1 is $1 \cdot (2, 1, -1) = (2, 1, -1)$; row 2 is $-\frac{3}{2}(2, 1, -1) + (0, \frac{1}{2}, \frac{1}{2}) = (-3, -1, 2)$, and row 3 is $-(2, 1, -1) + 4(0, \frac{1}{2}, \frac{1}{2}) + (0, 0, -1) = (-2, 1, 2)$. These are the rows of $\mathbf{A}$, so $\mathbf{LU} = \mathbf{A}$ is an LU factorization. The below-diagonal entries of $\mathbf{L}$ are the negatives of the multiples of earlier rows that Gaussian elimination adds to reduce $\mathbf{A}$ to $\mathbf{U}$.

Example 0.36 (An invertible matrix with no LU factorization)**.** $\mathbf{P} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is invertible: $\mathbf{P}^2 = \mathbf{I}_2$, so $\mathbf{P}^{-1} = \mathbf{P}$ (5); but it has no LU factorization. If $\mathbf{P} = \mathbf{LU}$, the $(1, 1)$ entry gives $0 = 1 \cdot u_{11}$, so $u_{11} = 0$, and then the $(2, 1)$ entry gives

$$\begin{aligned} 1 &= \ell_{21}\, u_{11} \\ &= 0, \end{aligned}$$

which is impossible. Swapping the two rows first removes the obstacle: the swapped matrix is $\mathbf{I}_2 = \mathbf{I}_2\, \mathbf{I}_2$.

Theorem 0.25 (Solving a system with an LU factorization)**.** *If $\mathbf{A} = \mathbf{LU}$ is an LU factorization (Definition 0.12) and the diagonal entries of $\mathbf{U}$ are all nonzero, then for every $\underset{\sim}{b} \in \mathbb{R}^n$ the system $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$ has exactly one solution: solve $\mathbf{L}\underset{\sim}{y} = \underset{\sim}{b}$ by forward substitution, then $\mathbf{U}\underset{\sim}{x} = \underset{\sim}{y}$ by back substitution (Theorem 0.19).*

Proof

Proof. $\mathbf{L}$ has diagonal entries 1 and $\mathbf{U}$ has nonzero ones, so each of the two triangular systems has exactly one solution (Theorem 0.19). If $\underset{\sim}{y}$ and $\underset{\sim}{x}$ are those solutions, then

$$\begin{aligned} \mathbf{A}\underset{\sim}{x} &= \mathbf{L}\,(\mathbf{U}\underset{\sim}{x}) \\ &= \mathbf{L}\underset{\sim}{y} \\ &= \underset{\sim}{b} \end{aligned}$$

(Theorem 0.7), so $\underset{\sim}{x}$ is a solution. Conversely, if $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{b}$, then $\underset{\sim}{y} \overset{\text{def}}{=} \mathbf{U}\underset{\sim}{x}$ satisfies

$$\begin{aligned}\mathbf{L}\underset{\sim}{y} &= \mathbf{L}\left(\mathbf{U}\underset{\sim}{x}\right)\\ &= (\mathbf{LU})\,\underset{\sim}{x}\\ &= \mathbf{A}\underset{\sim}{x}\\ &= \underset{\sim}{b}\end{aligned}$$

(Theorem 0.7), so $\underset{\sim}{y}$ is the unique solution of that system, and $\underset{\sim}{x}$ is then the unique solution of $\mathbf{U}\underset{\sim}{x} = \underset{\sim}{y}$. □

Example 0.37 (Two triangular solves)**.** With $\mathbf{L}$ and $\mathbf{U}$ of Example 0.35, solve $\mathbf{A}\underset{\sim}{x} = (8, -11, -3)$. Forward substitution on $\mathbf{L}\underset{\sim}{y} = (8, -11, -3)$ gives $y_1 = 8$,

$$\begin{aligned}y_2 &= -11 + \tfrac{3}{2}\cdot 8\\ &= 1,\end{aligned}$$

$$\begin{aligned}y_3 &= -3 + 8 - 4\cdot 1\\ &= 1.\end{aligned}$$

Back substitution on $\mathbf{U}\underset{\sim}{x} = (8, 1, 1)$ is Example 0.28, which gives $\underset{\sim}{x} = (2, 3, -1)$. As a check,

$$\begin{aligned}\mathbf{A}\,(2, 3, -1) &= (4 + 3 + 1,\ -6 - 3 - 2,\ -4 + 3 - 2)\\ &= (8, -11, -3).\end{aligned}$$

Similarity and diagonalization

i Source

This section is adapted from Zhou [93], used under the MIT License (see the license text in Section). These notes take eigenvalues to be real numbers (Definition 0.1), so the source's complex eigenvalues are left out, and eigenvalues are characterized through null spaces rather than through the characteristic polynomial. Only the source's material on similarity, diagonalization and the basic eigenvalue properties is adapted; its characteristic polynomial and algebraic multiplicity, its trace and determinant identities, and the part of its section on symmetric matrices that builds bases of the column space and null space from eigenvectors are left out. The rest of that section is covered by Theorem 0.1 and Theorem 0.2. Its result that eigenvalues of orthogonal matrices have modulus 1 appears here in the real form ± 1. Three other parts of the source are not part of this section:

- its statement that the eigenvalues of a lower triangular matrix are also its diagonal entries (Theorem 0.4 states the upper triangular case)
- its three further examples of matrices with geometric multiplicity smaller than algebraic multiplicity (Example 0.2 gives one nilpotent example)
- its code cells The source's two-vector argument that eigenvectors for distinct eigenvalues are independent is extended here to any number of eigenvectors.

Theorem 0.1 (A square matrix is invertible exactly when it has full column rank)**.** *An $n \times n$ matrix $\mathbf{A}$ is invertible (Definition 0.6) exactly when* $\operatorname{rank}(\mathbf{A}) = n$ *(full column rank, Definition 0.3), that is, exactly when* $\mathcal{N}(\mathbf{A}) = \{\underset{\sim}{0}_n\}$.

Proof

Proof. **Invertible implies rank n.** If $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_n$, then

$$\begin{aligned} \underset{\sim}{x} &= \mathbf{I}_n \underset{\sim}{x} && \text{(Theorem 0.10)} \\ &= (\mathbf{A}^{-1}\mathbf{A})\,\underset{\sim}{x} && \text{(Definition 0.5)} \\ &= \mathbf{A}^{-1}\,(\mathbf{A}\underset{\sim}{x}) && \text{(Theorem 0.7)} \\ &= \mathbf{A}^{-1}\,\underset{\sim}{0}_n && (\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_n) \\ &= \underset{\sim}{0}_n, && \text{(Definition 0.16)} \end{aligned}$$

so $\mathcal{N}(\mathbf{A}) = \{\underset{\sim}{0}_n\}$, the nullity is 0, and $\operatorname{rank}(\mathbf{A}) = n$ (Theorem 0.2).
Rank n implies invertible. Every system $\mathbf{A}\underset{\sim}{y} = \underset{\sim}{b}$ has exactly one solution (Corollary 0.2, both parts, with $m = n$). Let $\underset{\sim}{y}_j$ solve $\mathbf{A}\underset{\sim}{y}_j = \underset{\sim}{e}_j$ (Definition 0.18), and let $\mathbf{B}$ have columns $\underset{\sim}{y}_1, \ldots, \underset{\sim}{y}_n$. Column j of $\mathbf{AB}$ is $\mathbf{A}\underset{\sim}{y}_j = \underset{\sim}{e}_j$ (4), so $\mathbf{AB} = \mathbf{I}_n$. For the other order,

$$\begin{aligned} \mathbf{A}\,(\mathbf{BA} - \mathbf{I}_n) &= \mathbf{ABA} - \mathbf{AI}_n && \text{(Theorem 0.7, Theorem 0.7)} \\ &= \mathbf{I}_n\mathbf{A} - \mathbf{AI}_n && (\mathbf{AB} = \mathbf{I}_n) \\ &= \mathbf{A} - \mathbf{A} && \text{(Theorem 0.10)} \\ &= \mathbf{0}_{n\times n}, && \text{(arithmetic)} \end{aligned}$$

so every column of $\mathbf{BA} - \mathbf{I}_n$ is in $\mathcal{N}(\mathbf{A}) = \{\underset{\sim}{0}_n\}$ (4), and $\mathbf{BA} = \mathbf{I}_n$. So $\mathbf{B}$ satisfies Definition 0.6. The two conditions in the statement are equivalent by Theorem 0.2: the nullity is 0 exactly when $\mathcal{N}(\mathbf{A}) = \{\underset{\sim}{0}_n\}$, because a subspace of dimension 0 has the empty list as a basis, whose span is $\{\underset{\sim}{0}\}$ (Definition 0.6, Definition 0.2), and $\{\underset{\sim}{0}\}$ has dimension 0 (Example 0.14). □

Example 0.1 (Rank decides invertibility)**.**

- $\begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ of Example 0.5 has rank 2: $c_1(2,0) + c_2(1,1) = (2c_1 + c_2, c_2)$ is $\underset{\sim}{0}$ only if $c_2 = 0$ and then $c_1 = 0$. So it is invertible, as that example found by exhibiting the inverse.
- $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ has rank less than 2: $(1, -1)$ is a nonzero vector in its null space. So it is singular, as Example 0.5 found.

Definition 0.1 (Eigenspace)**.** Let λ be an eigenvalue of a $p \times p$ matrix $\mathbf{A}$ (Definition 0.1). The **eigenspace** [94] of λ is $\mathcal{E}_\lambda \overset{\text{def}}{=} \mathcal{N}(\mathbf{A} - \lambda \mathbf{I}_p)$. It is a subspace of $\mathbb{R}^p$ (Theorem 0.2).

Definition 0.2 (Geometric multiplicity)**.** The dimension (Definition 0.6) of the eigenspace $\mathcal{E}_\lambda$ (Definition 0.1) of an eigenvalue λ is the **geometric multiplicity** of λ.

Example 0.2 (Eigenspaces of two matrices)**.**

- For $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ of Example 0.1, $\mathbf{A} - 3\,\mathbf{I}_2 = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$ sends $\underset{\sim}{v}$ to $(v_2 - v_1)\,(1, -1)$, so $\mathcal{E}_3 = \text{span}\{(1,1)\}$, with geometric multiplicity 1.
- For $\mathbf{J} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $\mathbf{J}\underset{\sim}{v} = (v_2, 0)$.

$$\begin{aligned} \mathbf{J}\,(1,0) &= (0,0) \\ &= 0\,(1,0), \end{aligned}$$

so 0 is an eigenvalue. If $\mathbf{J}\underset{\sim}{v} = \lambda \underset{\sim}{v}$ with $\lambda \neq 0$, the second entry gives $\lambda v_2 = 0$, so $v_2 = 0$, and then the first gives $\lambda v_1 = 0$, so $\underset{\sim}{v} = \underset{\sim}{0}$. So 0 is the only eigenvalue, and

$$\begin{aligned} \mathcal{E}_0 &= \mathcal{N}(\mathbf{J}) \\ &= \{(t,0) : t \in \mathbb{R}\}, \end{aligned}$$

with geometric multiplicity 1.

Example 0.3 (A number that is not an eigenvalue has no eigenspace)**.** For $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

and $\lambda = 2$, $\mathbf{A} - 2\,\mathbf{I}_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ sends $\underset{\sim}{v}$ to (v_2, v_1), which is $\underset{\sim}{0}$ only for $\underset{\sim}{v} = \underset{\sim}{0}$, so $\mathcal{N}(\mathbf{A} - 2\,\mathbf{I}_2) = \{\underset{\sim}{0}\}$. And 2 is not an eigenvalue (Definition 0.1): $\mathbf{A}\underset{\sim}{v} = 2\underset{\sim}{v}$ reads $(2v_1 + v_2, v_1 + 2v_2) = (2v_1, 2v_2)$, so $v_2 = 0$ and $v_1 = 0$. So Definition 0.1 does not apply to 2.

Theorem 0.2 (Eigenvalues are where $\mathbf{A} - \lambda\mathbf{I}$ is singular)**.** *Let* $\mathbf{A}$ *be a* $p \times p$ *matrix and* λ *a real number. Then* λ *is an eigenvalue of* $\mathbf{A}$ *(Definition 0.1) exactly when* $\mathbf{A} - \lambda\,\mathbf{I}_p$ *is singular (Definition 0.6). In particular,* $\mathbf{A}$ *is singular exactly when* 0 *is one of its eigenvalues; and each eigenspace* $\mathcal{E}_\lambda$ *(Definition 0.1) is a subspace of* $\mathbb{R}^p$ *consisting of* $\underset{\sim}{0}$ *and the eigenvectors for* λ*.*

Proof

Proof. For any $\underset{\sim}{v} \in \mathbb{R}^p$,

$$\begin{aligned} (\mathbf{A} - \lambda\,\mathbf{I}_p)\,\underset{\sim}{v} &= \mathbf{A}\underset{\sim}{v} - (\lambda\,\mathbf{I}_p)\,\underset{\sim}{v} && \text{(Theorem 0.7, the difference law)} \\ &= \mathbf{A}\underset{\sim}{v} - \lambda\,(\mathbf{I}_p\underset{\sim}{v}) && \text{(Theorem 0.7, the scalar factor)} \\ &= \mathbf{A}\underset{\sim}{v} - \lambda\underset{\sim}{v}, && \text{(Theorem 0.10)} \end{aligned}$$

so $\mathbf{A}\underset{\sim}{v} = \lambda\underset{\sim}{v}$ exactly when $\underset{\sim}{v} \in \mathcal{N}(\mathbf{A} - \lambda\,\mathbf{I}_p)$; the nonzero such $\underset{\sim}{v}$ are the eigenvectors for λ. So λ is an eigenvalue exactly when that null space contains a nonzero vector, which by Theorem 0.1 is exactly when $\mathbf{A} - \lambda\,\mathbf{I}_p$ is singular. With $\lambda = 0$, $\mathbf{A} - 0\,\mathbf{I}_p = \mathbf{A}$. An eigenspace is a null space, so it is a subspace (Theorem 0.2). □

Example 0.4 (Eigenvalues and singular matrices)**.** For $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $\mathbf{A} - 3\,\mathbf{I}_2 = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$ is singular (its columns are negatives of each other, so its rank is 1), matching the eigenvalue 3 of Example 0.1. $\mathbf{A}$ itself has rank 2, since $c_1(2,1) + c_2(1,2) = (2c_1 + c_2, c_1 + 2c_2)$ is $\underset{\sim}{0}$ only if

$$\begin{aligned} c_1 &= c_2 \\ &= 0 \end{aligned}$$

(subtract twice the second entry from the first: $-3c_2 = 0$, and then the second entry gives $c_1 = 0$); so it is invertible (Theorem 0.1), and 0 is not an eigenvalue. The singular matrix $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ has eigenvalue 0, with eigenvector $(1, -1)$.

Theorem 0.3 (Eigenvalues of shifts and powers)**.** *Let $\underset{\sim}{v}$ be an eigenvector of a $p \times p$ matrix $\mathbf{A}$ for the eigenvalue λ (Definition 0.1). Then*

1. *$\underset{\sim}{v}$ is an eigenvector of $\mathbf{A} + s\,\mathbf{I}_p$ for $\lambda + s$, for every number s;*
2. *$\underset{\sim}{v}$ is an eigenvector of $\mathbf{A}^k$ (Definition 0.2) for λ^k, for every positive integer k.*

Proof

Proof. **Part 1.**

$$\begin{aligned}
(\mathbf{A} + s\,\mathbf{I}_p)\,\underset{\sim}{v} &= \mathbf{A}\underset{\sim}{v} + (s\,\mathbf{I}_p)\,\underset{\sim}{v} && \text{(Theorem 0.8)} \\
&= \mathbf{A}\underset{\sim}{v} + s\,(\mathbf{I}_p\underset{\sim}{v}) && \text{(Theorem 0.7)} \\
&= \lambda\underset{\sim}{v} + s\,(\mathbf{I}_p\underset{\sim}{v}) && \text{(eigenvector)} \\
&= \lambda\underset{\sim}{v} + s\underset{\sim}{v} && \text{(Theorem 0.10)} \\
&= (\lambda + s)\,\underset{\sim}{v}. && \text{(add entrywise)}
\end{aligned}$$

Part 2, by induction on k. For $k = 1$ it is the assumption. If $\mathbf{A}^{k-1}\underset{\sim}{v} = \lambda^{k-1}\underset{\sim}{v}$, then

$$\begin{aligned}
\mathbf{A}^k\underset{\sim}{v} &= \mathbf{A}\,(\mathbf{A}^{k-1}\underset{\sim}{v}) && \text{(Definition 0.2, Theorem 0.7)} \\
&= \mathbf{A}\,(\lambda^{k-1}\underset{\sim}{v}) && \text{(induction hypothesis)} \\
&= \lambda^{k-1}\,(\mathbf{A}\underset{\sim}{v}) && \text{(Theorem 0.7)} \\
&= \lambda^{k-1}\,(\lambda\underset{\sim}{v}) && \text{(eigenvector)} \\
&= \lambda^k\underset{\sim}{v}. && \text{(multiply the numbers)}
\end{aligned}$$

In both parts $\underset{\sim}{v} \neq \underset{\sim}{0}$, so $\underset{\sim}{v}$ is an eigenvector. □

Example 0.5 (Shifting and squaring)**.** For $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ with eigenvector $(1, 1)$ for 3 (Example 0.1):

- $\mathbf{A} - 2\,\mathbf{I}_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ sends $(1, 1)$ to $(1, 1) = (3 - 2)\,(1, 1)$;
- $$\begin{aligned}
\mathbf{A}^2 &= \begin{bmatrix} 2 \cdot 2 + 1 \cdot 1 & 2 \cdot 1 + 1 \cdot 2 \\ 1 \cdot 2 + 2 \cdot 1 & 1 \cdot 1 + 2 \cdot 2 \end{bmatrix} \\
&= \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}
\end{aligned}$$

  (4) sends $(1, 1)$ to $(9, 9) = 3^2\,(1, 1)$.

Theorem 0.4 (The eigenvalues of an upper triangular matrix are its diagonal entries)**.** *The eigenvalues (Definition 0.1) of an upper triangular $p \times p$ matrix $\mathbf{U}$ (Definition 0.7) are exactly its diagonal entries $u_{11}, \dots, u_{pp}$.*

Proof

Proof. $\mathbf{U} - \lambda \mathbf{I}_p$ is upper triangular, with diagonal entries $u_{ii} - \lambda$ (Definition 0.15, Definition 0.8).
If λ is not a diagonal entry, every $u_{ii} - \lambda \neq 0$, so $(\mathbf{U} - \lambda \mathbf{I}_p)\, \underset{\sim}{v} = \underset{\sim}{0}$ has exactly one solution, $\underset{\sim}{v} = \underset{\sim}{0}$ (Theorem 0.19), and λ is not an eigenvalue (Theorem 0.2).
If λ is a diagonal entry, let k be the smallest index with $u_{kk} = \lambda$, and write $\mathbf{M} = \mathbf{U} - \lambda \mathbf{I}_p$, so $m_{kk} = 0$ and $m_{ii} \neq 0$ for $i < k$. Look for $\underset{\sim}{v}$ with $v_k = 1$ and $v_j = 0$ for $j > k$. Entry i of $\mathbf{M}\underset{\sim}{v}$ is

$$\begin{aligned} \sum_{j=1}^{p} m_{ij}\, v_j &= \sum_{j=i}^{p} m_{ij}\, v_j \quad (m_{ij} = 0 \text{ for } j < i) \\ &= \sum_{j=i}^{k} m_{ij}\, v_j, \quad (v_j = 0 \text{ for } j > k) \end{aligned}$$

where the last sum is empty, and so 0, when $i > k$. So:

- for $i > k$ entry i is 0;
- for $i = k$ it is $m_{kk}\, v_k = 0$;
- for $i < k$ it is 0 exactly when $v_i = -\frac{1}{m_{ii}} \sum_{j=i+1}^{k} m_{ij}\, v_j$, and these equations fix $v_{k-1}, \dots, v_1$ in turn, as in Theorem 0.19.

So $\mathbf{M}\underset{\sim}{v} = \underset{\sim}{0}$ with $v_k = 1$, hence $\underset{\sim}{v} \neq \underset{\sim}{0}$, and λ is an eigenvalue. □

Example 0.6 (Eigenvalues read off the diagonal)**.** $\mathbf{U} = \begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix}$ of Example 0.27 has eigenvalues 2, $\frac{1}{2}$ and -1. For $\lambda = \frac{1}{2}$ (so $k = 2$), take $v_2 = 1$, $v_3 = 0$; row 1 of $(\mathbf{U} - \frac{1}{2}\mathbf{I}_3)\, \underset{\sim}{v}$ is $\frac{3}{2} v_1 + 1 = 0$, so $v_1 = -\frac{2}{3}$. Check:

$$\begin{aligned} \mathbf{U}\,(-\tfrac{2}{3}, 1, 0) &= (-\tfrac{4}{3} + 1, \tfrac{1}{2}, 0) \\ &= \tfrac{1}{2}\,(-\tfrac{2}{3}, 1, 0). \end{aligned}$$

The matrix $\mathbf{J}$ of Example 0.2 is upper triangular with both diagonal entries 0, and 0 is its only eigenvalue.

Theorem 0.5 (Eigenvalues of idempotent and orthogonal matrices)**.**

1. *Every eigenvalue of an idempotent matrix (Definition 0.7) is* 0 *or* 1*.*
2. *Every eigenvalue of an orthogonal matrix (Definition 0.10) is* 1 *or* -1*.*

Proof

Proof. Let $\underset{\sim}{v} \neq \underset{\sim}{0}$ with $\mathbf{A}\underset{\sim}{v} = \lambda \underset{\sim}{v}$.
Part 1. If $\mathbf{A}^2 = \mathbf{A}$, then

$$\begin{aligned} \lambda \underset{\sim}{v} &= \mathbf{A}\underset{\sim}{v} && \text{(eigenvector)} \\ &= \mathbf{A}^2 \underset{\sim}{v} && \text{(idempotent)} \\ &= \lambda^2 \underset{\sim}{v}, && \text{(Theorem 0.3, part 2)} \end{aligned}$$

and subtracting $\lambda \underset{\sim}{v}$ from both sides gives $(\lambda^2 - \lambda)\,\underset{\sim}{v} = \underset{\sim}{0}$. Some entry of $\underset{\sim}{v}$ is nonzero, so

$$\begin{aligned} \lambda^2 - \lambda &= \lambda\,(\lambda - 1) \\ &= 0, \end{aligned}$$

and a product of two numbers is 0 only when one of them is: $\lambda = 0$ or $\lambda = 1$.
Part 2. If $\mathbf{A}$ is orthogonal, then

$$\begin{aligned} \|\underset{\sim}{v}\| &= \|\mathbf{A}\underset{\sim}{v}\| && \text{(Theorem 0.7)} \\ &= \|\lambda \underset{\sim}{v}\| && \text{(eigenvector)} \\ &= |\lambda|\,\|\underset{\sim}{v}\|, && \text{(Theorem 0.1, part 2)} \end{aligned}$$

and $\|\underset{\sim}{v}\| > 0$ (Theorem 0.1, part 1), so dividing by $\|\underset{\sim}{v}\|$ gives $|\lambda| = 1$; since λ is real (Definition 0.1), $\lambda = 1$ or $\lambda = -1$. □

Example 0.7 (Projections and reflections)**.**

- The idempotent matrix $\begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$ of Example 0.3 sends $(1, 1)$ to $(1, 1)$ and $(1, -1)$ to $(0, 0)$: eigenvalues 1 and 0.
- The matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, which swaps the two entries, is orthogonal: its columns $(0, 1)$ and $(1, 0)$ are orthonormal (Remark 0.6). It sends $(1, 1)$ to $(1, 1)$ and $(1, -1)$ to $(-1, 1)$: eigenvalues 1 and -1.
- The rotation $\mathbf{Q}$ of Example 0.14 has no real eigenvalue. By part 2 the only candidates are ± 1. $\mathbf{Q}\underset{\sim}{v} = \underset{\sim}{v}$ reads $-0.4\,v_1 - 0.8\,v_2 = 0$ and $0.8\,v_1 - 0.4\,v_2 = 0$, so $v_1 = -2v_2$ and

$$\begin{aligned} v_2 &= 2v_1 \\ &= -4v_2, \end{aligned}$$

forcing $\underset{\sim}{v} = \underset{\sim}{0}$; $\mathbf{Q}\underset{\sim}{v} = -\underset{\sim}{v}$ reads $1.6\,v_1 - 0.8\,v_2 = 0$ and $0.8\,v_1 + 1.6\,v_2 = 0$, so $v_2 = 2v_1$ and $0.8\,v_1 + 3.2\,v_1 = 0$, forcing $\underset{\sim}{v} = \underset{\sim}{0}$ again.

Definition 0.3 (Similar matrices)**.** Two $p \times p$ matrices $\mathbf{A}$ and $\mathbf{B}$ are **similar** [95] if $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ for some invertible $p \times p$ matrix $\mathbf{P}$ (Definition 0.6).

Remark (Similarity read both ways)

Remark 0.1 (Similarity read both ways). If $\mathbf{P}$ is invertible, then $\mathbf{P}^{-1}$ is invertible with inverse $\mathbf{P}$:

$$\begin{aligned} \mathbf{P}^{-1}\mathbf{P} &= \mathbf{P}\mathbf{P}^{-1} \\ &= \mathbf{I}_p \end{aligned}$$

(5) is the condition of Definition 0.6 for $\mathbf{P}^{-1}$ with $\mathbf{P}$ as the other factor, and that factor is unique (Remark 0.3). If $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$, then

$$\begin{aligned} \mathbf{P}\mathbf{B}\mathbf{P}^{-1} &= \mathbf{P}\left(\mathbf{P}^{-1}\mathbf{A}\mathbf{P}\right)\mathbf{P}^{-1} && \text{(substitute } \mathbf{B}\text{)} \\ &= \left(\mathbf{P}\mathbf{P}^{-1}\right)\mathbf{A}\left(\mathbf{P}\mathbf{P}^{-1}\right) && \text{(Theorem 0.7)} \\ &= \mathbf{I}_p\,\mathbf{A}\,\mathbf{I}_p && \text{(Definition 0.5)} \\ &= \mathbf{A}, && \text{(Theorem 0.10)} \end{aligned}$$

and the same steps with $\mathbf{P}$ and $\mathbf{P}^{-1}$ exchanged turn $\mathbf{A} = \mathbf{P}\mathbf{B}\mathbf{P}^{-1}$ back into $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$. So $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ exactly when $\mathbf{A} = \mathbf{P}\mathbf{B}\mathbf{P}^{-1}$, and similarity goes both ways: $\mathbf{A}$ is similar to $\mathbf{B}$ through $\mathbf{P}^{-1}$, whose inverse is $\mathbf{P}$.

Example 0.8 (A matrix similar to a diagonal one)**.** Let $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and $\mathbf{P} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. Then

$$\begin{aligned} \mathbf{P}^2 &= \begin{bmatrix} 1\cdot 1 + 1\cdot 1 & 1\cdot 1 + 1\cdot(-1) \\ 1\cdot 1 + (-1)\cdot 1 & 1\cdot 1 + (-1)(-1) \end{bmatrix} && \text{(Definition 0.10)} \\ &= \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} && \text{(arithmetic)} \\ &= 2\,\mathbf{I}_2, && \text{(Definition 0.9, Definition 0.15)} \end{aligned}$$

so

$$\begin{aligned}\mathbf{P}\left(\tfrac{1}{2}\mathbf{P}\right) &= \tfrac{1}{2}\mathbf{P}^2 \\ &= \mathbf{I}_2\end{aligned}$$

and likewise $\left(\frac{1}{2}\mathbf{P}\right)\mathbf{P} = \mathbf{I}_2$ (Theorem 0.7): $\mathbf{P}^{-1} = \frac{1}{2}\mathbf{P}$ (5). Then

$$\begin{aligned}
\mathbf{P}^{-1}\mathbf{A}\mathbf{P} &= \left(\tfrac{1}{2}\mathbf{P}\right)\mathbf{A}\mathbf{P} && \text{(substitute } \mathbf{P}^{-1}\text{)} \\
&= \tfrac{1}{2}\left(\mathbf{P}\mathbf{A}\mathbf{P}\right) && \text{(Theorem 0.7)} \\
&= \tfrac{1}{2}\mathbf{P}\left(\mathbf{A}\mathbf{P}\right) && \text{(Theorem 0.7)} \\
&= \tfrac{1}{2}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}3 & 1\\ 3 & -1\end{bmatrix} && \text{(Definition 0.10, for } \mathbf{AP}\text{)} \\
&= \tfrac{1}{2}\begin{bmatrix}6 & 0\\ 0 & 2\end{bmatrix} && \text{(Definition 0.10)} \\
&= \begin{bmatrix}3 & 0\\ 0 & 1\end{bmatrix}, && \text{(Definition 0.9)}
\end{aligned}$$

so $\mathbf{A}$ is similar to $\mathrm{diag}(3, 1)$. At the other extreme, the identity matrix is similar only to itself:

$$\begin{aligned}\mathbf{P}^{-1}\mathbf{I}_p\mathbf{P} &= \mathbf{P}^{-1}\mathbf{P} \\ &= \mathbf{I}_p\end{aligned}$$

for every invertible $\mathbf{P}$ (Theorem 0.10, 5).

Theorem 0.6 (Similar matrices have the same eigenvalues)**.** *If* $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ *(Definition 0.3), then* $\underset{\sim}{v}$ *is an eigenvector of* $\mathbf{A}$ *for* λ *exactly when* $\mathbf{P}^{-1}\underset{\sim}{v}$ *is an eigenvector of* $\mathbf{B}$ *for* λ*. So* $\mathbf{A}$ *and* $\mathbf{B}$ *have the same eigenvalues.*

Proof

Proof. **From A to B.** Let $\mathbf{A}\underset{\sim}{v} = \lambda\underset{\sim}{v}$ with $\underset{\sim}{v} \neq \underset{\sim}{0}$, and $\underset{\sim}{w} = \mathbf{P}^{-1}\underset{\sim}{v}$. Then

$$\begin{aligned}
\mathbf{B}\underset{\sim}{w} &= \left(\mathbf{P}^{-1}\mathbf{A}\mathbf{P}\right)\left(\mathbf{P}^{-1}\underset{\sim}{v}\right) && \text{(substitute } \mathbf{B} \text{ and } \underset{\sim}{w}\text{)} \\
&= \mathbf{P}^{-1}\mathbf{A}\left(\mathbf{P}\mathbf{P}^{-1}\right)\underset{\sim}{v} && \text{(Theorem 0.7)} \\
&= \mathbf{P}^{-1}\mathbf{A}\,\mathbf{I}_p\,\underset{\sim}{v} && \text{(Definition 0.5)} \\
&= \mathbf{P}^{-1}\mathbf{A}\underset{\sim}{v} && \text{(Theorem 0.10)} \\
&= \mathbf{P}^{-1}\left(\mathbf{A}\underset{\sim}{v}\right) && \text{(Theorem 0.7)} \\
&= \mathbf{P}^{-1}\left(\lambda\underset{\sim}{v}\right) && \text{(eigenvector)} \\
&= \lambda\,\underset{\sim}{w}. && \text{(Theorem 0.7)}
\end{aligned}$$

And $\underset{\sim}{w} \neq \underset{\sim}{0}$: if $\underset{\sim}{w} = \underset{\sim}{0}$, then

$$\begin{aligned}\underset{\sim}{v} &= \mathbf{I}_p \underset{\sim}{v} \\ &= \mathbf{P}\left(\mathbf{P}^{-1}\underset{\sim}{v}\right) \\ &= \mathbf{P}\,\underset{\sim}{0} \\ &= \underset{\sim}{0}\end{aligned}$$

(Theorem 0.10, 5, Theorem 0.7, Definition 0.16), which is false.
From B to A. By Remark 0.1,

$$\begin{aligned}\mathbf{A} &= \mathbf{P}\mathbf{B}\mathbf{P}^{-1} \\ &= (\mathbf{P}^{-1})^{-1}\mathbf{B}\,\mathbf{P}^{-1},\end{aligned}$$

so the first part, with $\mathbf{P}^{-1}$ in place of $\mathbf{P}$ and the roles of $\mathbf{A}$ and $\mathbf{B}$ exchanged, shows that if $\underset{\sim}{w} = \mathbf{P}^{-1}\underset{\sim}{v}$ is an eigenvector of $\mathbf{B}$ for λ, then $\mathbf{P}\underset{\sim}{w}$ is an eigenvector of $\mathbf{A}$ for λ; and

$$\begin{aligned}\mathbf{P}\underset{\sim}{w} &= \mathbf{P}\mathbf{P}^{-1}\underset{\sim}{v} \\ &= \underset{\sim}{v}\end{aligned}$$

(Theorem 0.7, 5, Theorem 0.10). □

Example 0.9 (Eigenvectors carried across a similarity)**.** In Example 0.8, $\mathbf{B} = \mathrm{diag}(3, 1)$ has eigenvectors $(1, 0)$ for 3 and $(0, 1)$ for 1: $\mathrm{diag}(3, 1)\,(1, 0) = (3, 0)$ and $\mathrm{diag}(3, 1)\,(0, 1) = (0, 1)$. $\mathbf{P}\,(1, 0) = (1, 1)$ and $\mathbf{P}\,(0, 1) = (1, -1)$ are the eigenvectors of $\mathbf{A}$ for 3 and 1 found in Example 0.1.

Theorem 0.7 (Eigenvectors for distinct eigenvalues are linearly independent)**.** *If $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k$ are eigenvectors of a $p \times p$ matrix $\mathbf{A}$ for eigenvalues $\lambda_1, \ldots, \lambda_k$ that are all different, then $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_k$ are linearly independent (Definition 0.1).*

Proof

Proof. By induction on k. For $k = 1$, $\underset{\sim}{v}_1 \neq \underset{\sim}{0}$ is linearly independent on its own. Suppose the result holds for $k - 1$ eigenvectors, and $\sum_{i=1}^{k} c_i \underset{\sim}{v}_i = \underset{\sim}{0}$. Then

$$
\begin{aligned}
\underset{\sim}{0} &= (\mathbf{A} - \lambda_k \mathbf{I}_p) \sum_{i=1}^{k} c_i \underset{\sim}{v}_i && \text{(multiply the supposed equation by } \mathbf{A} - \lambda_k \mathbf{I}_p) \\
&= \sum_{i=1}^{k} c_i \left(\mathbf{A} - \lambda_k \mathbf{I}_p\right) \underset{\sim}{v}_i && \text{(Theorem 0.4)} \\
&= \sum_{i=1}^{k} c_i \left(\lambda_i - \lambda_k\right) \underset{\sim}{v}_i && \text{(Theorem 0.3, part 1, with } s = -\lambda_k) \\
&= \sum_{i=1}^{k-1} c_i \left(\lambda_i - \lambda_k\right) \underset{\sim}{v}_i. && \text{(the } i = k \text{ term is } \underset{\sim}{0})
\end{aligned}
$$

By the induction hypothesis $\underset{\sim}{v}_1, \ldots, \underset{\sim}{v}_{k-1}$ are linearly independent, so every $c_i \left(\lambda_i - \lambda_k\right) = 0$ for $i < k$, and since $\lambda_i \neq \lambda_k$, every $c_i = 0$ for $i < k$. The supposed equation then reads $c_k \underset{\sim}{v}_k = \underset{\sim}{0}$ with $\underset{\sim}{v}_k \neq \underset{\sim}{0}$, so $c_k = 0$ too. □

Example 0.10 (Independent eigenvectors)**.** The eigenvectors $(1, 1)$ for 3 and $(1, -1)$ for 1 of Example 0.1 are linearly independent, as Theorem 0.7 says they must be: $c_1(1, 1) + c_2(1, -1) = (c_1 + c_2, c_1 - c_2)$ is $\underset{\sim}{0}$ only if

$$
\begin{aligned}
c_1 &= c_2 \\
&= 0.
\end{aligned}
$$

Two eigenvectors for the same eigenvalue need not be independent: $(1, 1)$ and $(2, 2)$ are both eigenvectors for 3.

Definition 0.4 (Diagonalizable matrix)**.** A $p \times p$ matrix $\mathbf{A}$ is **diagonalizable** [96] if $\mathbf{A} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1}$ for some invertible $p \times p$ matrix $\mathbf{X}$ and diagonal $p \times p$ matrix $\mathbf{\Lambda}$ (Definition 0.4); by Remark 0.1, that is the same as $\mathbf{\Lambda} = \mathbf{X}^{-1}\mathbf{A}\mathbf{X}$, so a diagonalizable matrix is one similar to a diagonal matrix (Definition 0.3).

Example 0.11 (Diagonalizable and not)**.**

- $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ is diagonalizable: with $\mathbf{P}$ as in Example 0.8, $\operatorname{diag}(3, 1) = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$.
- $\mathbf{J} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ is not. Suppose $\mathbf{J} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1}$, so $\mathbf{\Lambda} = \mathbf{X}^{-1}\mathbf{J}\mathbf{X}$ (Remark 0.1). The eigenvalues of $\mathbf{\Lambda}$ are its diagonal entries (Theorem 0.4, since a diagonal matrix is upper triangular) and are the eigenvalues of $\mathbf{J}$ (Theorem 0.6), of which 0 is the only one (Example 0.2). So $\mathbf{\Lambda} = \mathbf{0}_{2\times 2}$, and then

$$\begin{aligned}\mathbf{J} &= \mathbf{X}\,\mathbf{0}_{2\times 2}\,\mathbf{X}^{-1}\\ &= \mathbf{0}_{2\times 2}\end{aligned}$$

(4), which it is not.

Theorem 0.8 (Diagonalizable means a basis of eigenvectors)**.** *A $p \times p$ matrix $\mathbf{A}$ is diagonalizable (Definition 0.4) exactly when it has p linearly independent eigenvectors. Then $\mathbf{A} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1}$ with the eigenvectors as the columns of $\mathbf{X}$ and their eigenvalues, in the same order, on the diagonal of $\mathbf{\Lambda}$.*

Proof

Proof. Let $\mathbf{X}$ have columns $\underset{\sim}{x}_1, \ldots, \underset{\sim}{x}_p$ and $\mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_p)$. Column j of $\mathbf{AX}$ is $\mathbf{A}\underset{\sim}{x}_j$, and column j of $\mathbf{X\Lambda}$ is $\mathbf{X}$ times column j of $\mathbf{\Lambda}$ (4), which is

$$\begin{aligned}\mathbf{X}\,(\lambda_j \underset{\sim}{e}_j) &= \lambda_j\,(\mathbf{X}\underset{\sim}{e}_j)\\ &= \lambda_j \underset{\sim}{x}_j\end{aligned}$$

(Theorem 0.7, Theorem 0.3), so

$$\mathbf{AX} = \mathbf{X\Lambda} \iff \mathbf{A}\underset{\sim}{x}_j = \lambda_j \underset{\sim}{x}_j \text{ for every } j. \tag{7}$$

Also, $\mathbf{X}$ has rank p exactly when its p columns are linearly independent (Definition 0.2), which by Theorem 0.1 is exactly when $\mathbf{X}$ is invertible.

Eigenvectors give a diagonalization. If $\underset{\sim}{x}_1, \ldots, \underset{\sim}{x}_p$ are linearly independent eigenvectors, then $\mathbf{X}$ is invertible and Equation 7 gives $\mathbf{AX} = \mathbf{X\Lambda}$, so

$$\begin{aligned}\mathbf{A} &= \mathbf{A}\,\mathbf{I}_p && \text{(Theorem 0.10)}\\ &= \mathbf{A}\,(\mathbf{X}\mathbf{X}^{-1}) && \text{(Definition 0.5)}\\ &= (\mathbf{AX})\,\mathbf{X}^{-1} && \text{(Theorem 0.7)}\\ &= \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1}. && (\mathbf{AX} = \mathbf{X\Lambda})\end{aligned}$$

A diagonalization gives eigenvectors. If $\mathbf{A} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1}$, then

$$\begin{aligned}\mathbf{AX} &= (\mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1})\,\mathbf{X} && \text{(substitute } \mathbf{A})\\ &= \mathbf{X\Lambda}\,(\mathbf{X}^{-1}\mathbf{X}) && \text{(Theorem 0.7)}\\ &= \mathbf{X\Lambda}\,\mathbf{I}_p && \text{(Definition 0.5)}\\ &= \mathbf{X\Lambda}, && \text{(Theorem 0.10)}\end{aligned}$$

so each column satisfies $\mathbf{A}\underset{\sim}{x}_j = \lambda_j \underset{\sim}{x}_j$ (Equation 7). $\mathbf{X}$ is invertible, so its columns are linearly independent, and in particular nonzero; they are p linearly independent eigenvectors. □

Example 0.12 (Building the diagonalization from eigenvectors)**.** The eigenvectors $(1,1)$ and $(1,-1)$ of $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ are linearly independent (Example 0.10), so with $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ and $\mathbf{\Lambda} = \operatorname{diag}(3,1)$, $\mathbf{A} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1}$; by Remark 0.1 this is the similarity of Example 0.8, with $\mathbf{X} = \mathbf{P}$. Because $\mathbf{A}$ is symmetric, Theorem 0.1 even gives such a factorization with orthonormal eigenvectors (Definition 0.3): dividing each column of $\mathbf{X}$ by its length $\sqrt{2}$ gives one. By contrast, every eigenvector of $\mathbf{J}$ in Example 0.2 has the form $(t,0)$, so any two of them are multiples of each other, and $\mathbf{J}$ has no two linearly independent eigenvectors: another way to see that it is not diagonalizable.

Corollary 0.1 (Distinct eigenvalues make a matrix diagonalizable)**.** *A $p \times p$ matrix with p different eigenvalues is diagonalizable.*

Proof

Proof. Choose one eigenvector for each eigenvalue. These p eigenvectors are linearly independent (Theorem 0.7), so the matrix is diagonalizable (Theorem 0.8). □

Example 0.13 (A triangular matrix with distinct diagonal entries)**.** $\mathbf{U}$ of Example 0.6 has the three different eigenvalues 2, $\frac{1}{2}$ and -1 (Theorem 0.4), so it is diagonalizable, even though it is not symmetric, so Theorem 0.1 does not apply to it. The converse fails: $\mathbf{I}_2$ is diagonal, hence diagonalizable ($\mathbf{I}_2 = \mathbf{I}_2\,\mathbf{I}_2\,\mathbf{I}_2^{-1}$), but has only the eigenvalue 1.

Theorem 0.9 (Powers of a diagonalizable matrix)**.** *If $\mathbf{A} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1}$ (Definition 0.4) with $\mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \dots, \lambda_p)$, then for every positive integer k, $\mathbf{A}^k = \mathbf{X}\mathbf{\Lambda}^k\mathbf{X}^{-1}$ and $\mathbf{\Lambda}^k = \operatorname{diag}(\lambda_1^k, \dots, \lambda_p^k)$.*

Proof

Proof. **Powers of $\mathbf{\Lambda}$.** If $\mathbf{D} = \operatorname{diag}(d_1, \dots, d_p)$ and $\mathbf{E} = \operatorname{diag}(e_1, \dots, e_p)$, with entries d_{il} and e_{lj} (so $d_{ii} = d_i$ and $e_{jj} = e_j$), entry (i,j) of $\mathbf{DE}$ is $\sum_l d_{il}\, e_{lj}$ (4), where $d_{il} = 0$ unless $l = i$ and $e_{lj} = 0$ unless $l = j$. For $i \neq j$ no term survives, so the entry is 0; for $i = j$ only $l = i$ survives, giving $d_i\, e_i$. So $\mathbf{DE} = \operatorname{diag}(d_1 e_1, \dots, d_p e_p)$, and induction on k gives $\mathbf{\Lambda}^k = \operatorname{diag}(\lambda_1^k, \dots, \lambda_p^k)$.
Powers of $\mathbf{A}$, by induction on k. $k = 1$ is the assumption. If it holds for $k-1$,

$$
\begin{aligned}
\mathbf{A}^k &= \mathbf{A}^{k-1}\mathbf{A} && \text{(Definition 0.2, Theorem 0.7)}\\
&= \mathbf{X}\mathbf{\Lambda}^{k-1}\mathbf{X}^{-1}\,\mathbf{A} && \text{(induction hypothesis)}\\
&= \mathbf{X}\mathbf{\Lambda}^{k-1}\mathbf{X}^{-1}\,\mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1} && \text{(the assumption)}\\
&= \mathbf{X}\mathbf{\Lambda}^{k-1}\,(\mathbf{X}^{-1}\mathbf{X})\,\mathbf{\Lambda}\mathbf{X}^{-1} && \text{(Theorem 0.7)}\\
&= \mathbf{X}\mathbf{\Lambda}^{k-1}\,\mathbf{I}_p\,\mathbf{\Lambda}\mathbf{X}^{-1} && \text{(Definition 0.5)}\\
&= \mathbf{X}\mathbf{\Lambda}^{k-1}\mathbf{\Lambda}\mathbf{X}^{-1} && \text{(Theorem 0.10)}\\
&= \mathbf{X}\mathbf{\Lambda}^{k}\mathbf{X}^{-1}. && \text{(Definition 0.2)}
\end{aligned}
$$

□

Example 0.14 (A closed form for the powers)**.** For $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, with $\mathbf{X}$ as in Example 0.12 and $\mathbf{X}^{-1} = \frac{1}{2}\,\mathbf{X}$ (Example 0.8),

$$
\begin{aligned}
\mathbf{A}^k &= \mathbf{X}\ \mathrm{diag}(3^k, 1)\,\mathbf{X}^{-1} && \text{(Theorem 0.9)}\\
&= \mathbf{X}\ \mathrm{diag}(3^k, 1)\,(\tfrac{1}{2}\,\mathbf{X}) && \text{(substitute } \mathbf{X}^{-1}\text{)}\\
&= \tfrac{1}{2}\,\mathbf{X}\ \mathrm{diag}(3^k, 1)\,\mathbf{X} && \text{(Theorem 0.7)}\\
&= \tfrac{1}{2}\begin{bmatrix} 3^k & 1 \\ 3^k & -1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} && \text{(Definition 0.10, for } \mathbf{X}\ \mathrm{diag}(3^k, 1)\text{)}\\
&= \tfrac{1}{2}\begin{bmatrix} 3^k + 1 & 3^k - 1 \\ 3^k - 1 & 3^k + 1 \end{bmatrix}. && \text{(Definition 0.10)}
\end{aligned}
$$

For $k = 2$ this gives $\frac{1}{2}\begin{bmatrix} 10 & 8 \\ 8 & 10 \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$, which matches $\mathbf{A}^2$ in Example 0.5.

Gram matrices, Schur complements and Cholesky

i Source

This section is adapted from Zhou [97], used under the MIT License (see the license text in Section). The source's eigenvalue and quadratic-form characterizations of definiteness are covered in these notes already (Theorem 0.2). These parts of the source are left out:

- its covariance test, which needs probability
- its material on minimization
- its positive semidefinite version of the Schur complement test
- its pivot test
- its Hadamard-product "Schur lemma"

- its notes on the LU factorization and computational cost
- its statement that a nonnegative combination of positive semidefinite matrices is positive semidefinite (Theorem 0.2 states the positive definite version)
- its section on the ellipse $ax^2+2bxy+cy^2=1$, with its worked example of a positive definite and an indefinite matrix
- its numerical example of the Cholesky factorization

The source leaves the Schur complement test unproved; it is proved here from a block expansion of the quadratic form, and the Cholesky factorization is derived from it.

Theorem 0.1 (Definite matrices are Gram matrices)**.** *Let* $\mathbf{A}$ *be a* $p \times p$ *matrix.*

1. $\mathbf{A}$ *is positive semidefinite (Definition 0.1) exactly when* $\mathbf{A} = \mathbf{B}^\top\mathbf{B}$ *for some matrix* $\mathbf{B}$ *with* p *columns.*
2. $\mathbf{A}$ *is positive definite (Definition 0.2) exactly when* $\mathbf{A} = \mathbf{B}^\top\mathbf{B}$ *for some matrix* $\mathbf{B}$ *with* p *linearly independent columns.*

Proof

Proof. **If $\mathbf{A} = \mathbf{B}^\top\mathbf{B}$.** $\mathbf{A}$ is symmetric, since

$$\begin{aligned}(\mathbf{B}^\top\mathbf{B})^\top &= \mathbf{B}^\top(\mathbf{B}^\top)^\top \\ &= \mathbf{B}^\top\mathbf{B}\end{aligned}$$

(Theorem 0.19, Definition 0.6). For any $\underset{\sim}{x}$,

$$\begin{aligned}\underset{\sim}{x}^\top\mathbf{A}\underset{\sim}{x} &= \underset{\sim}{x}^\top\mathbf{B}^\top\mathbf{B}\underset{\sim}{x} && \text{(substitute } \mathbf{A}\text{)} \\ &= (\mathbf{B}\underset{\sim}{x})^\top(\mathbf{B}\underset{\sim}{x}) && \text{(Theorem 0.19, Theorem 0.7)} \\ &= \|\mathbf{B}\underset{\sim}{x}\|^2 && \text{(Equation 2, squared)} \\ &\geq 0. && \text{(Theorem 0.1)}\end{aligned}$$

If the columns of $\mathbf{B}$ are linearly independent, then $\mathcal{N}(\mathbf{B}) = \{\underset{\sim}{0}\}$ ($\mathbf{B}\underset{\sim}{x}$ is the combination of the columns with coefficients x_j, Theorem 0.3, and only the zero combination is $\underset{\sim}{0}$, Definition 0.1), so for $\underset{\sim}{x} \neq \underset{\sim}{0}$, $\mathbf{B}\underset{\sim}{x} \neq \underset{\sim}{0}$ and $\|\mathbf{B}\underset{\sim}{x}\|^2 > 0$ (Theorem 0.1, part 1).
Only if. Let $\mathbf{A}$ be positive semidefinite. It is symmetric, so $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$ (Theorem 0.1), and every $\lambda_i \geq 0$ (Theorem 0.2). Let $\mathbf{\Lambda}^{1/2} \stackrel{\text{def}}{=} \operatorname{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_p})$, which is symmetric, with $\mathbf{\Lambda}^{1/2}\mathbf{\Lambda}^{1/2} = \mathbf{\Lambda}$ (the product of diagonal matrices multiplies their diagonals, as in the proof of Theorem 0.9), and let $\mathbf{B} \stackrel{\text{def}}{=} \mathbf{\Lambda}^{1/2}\mathbf{Q}^\top$. Then

$$\begin{aligned}
\mathbf{B}^\top\mathbf{B} &= \left(\mathbf{Q}^\top\right)^\top \left(\mathbf{\Lambda}^{1/2}\right)^\top \mathbf{\Lambda}^{1/2}\mathbf{Q}^\top && \text{(Theorem 0.19)}\\
&= \mathbf{Q}\,\mathbf{\Lambda}^{1/2}\mathbf{\Lambda}^{1/2}\,\mathbf{Q}^\top && \text{(Definition 0.6; } \mathbf{\Lambda}^{1/2} \text{ is symmetric)}\\
&= \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top && (\mathbf{\Lambda}^{1/2}\mathbf{\Lambda}^{1/2} = \mathbf{\Lambda})\\
&= \mathbf{A}. && \text{(Theorem 0.1)}
\end{aligned}$$

If $\mathbf{A}$ is positive definite, it is positive semidefinite (Remark 0.1), so the same $\mathbf{B}$ gives $\mathbf{A} = \mathbf{B}^\top\mathbf{B}$; and every $\lambda_i > 0$ (Theorem 0.2), so $\mathbf{\Lambda}^{1/2}$ is diagonal with nonzero diagonal, and invertible (its inverse is $\text{diag}(1/\sqrt{\lambda_1}, \ldots, 1/\sqrt{\lambda_p})$); $\mathbf{Q}^\top$ is invertible with inverse $\mathbf{Q}$ (Remark 0.6). So $\mathbf{B}$ is invertible (Theorem 0.1), and its p columns are linearly independent (Theorem 0.1, Definition 0.2). □

Example 0.1 (Gram factorizations)**.**

- With $\mathbf{R} = \begin{bmatrix}1 & 1\end{bmatrix}$ (1×2), $\mathbf{R}^\top\mathbf{R} = \begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$, the positive semidefinite matrix of Example 0.1; the two columns of $\mathbf{R}$, 1 and 1, are dependent, so part 2 does not apply to $\mathbf{R}$; $\mathbf{R}^\top\mathbf{R}$ is in fact not positive definite (Example 0.3).
- With $\mathbf{X} = \begin{bmatrix}1 & 1\\ 1 & 2\\ 1 & 3\end{bmatrix}$, whose columns are independent (Example 0.2), $\mathbf{X}^\top\mathbf{X} = \begin{bmatrix}3 & 6\\ 6 & 14\end{bmatrix}$ (Example 0.1) is positive definite.
- $\mathbf{D} = \begin{bmatrix}1 & 2\\ 2 & 1\end{bmatrix}$ of Example 0.3 is symmetric but not positive semidefinite, so by part 1 it is not $\mathbf{B}^\top\mathbf{B}$ for any $\mathbf{B}$.

Theorem 0.2 (Operations that preserve definiteness)**.**

1. *If $\mathbf{C}$ is a $p \times p$ positive definite matrix (Definition 0.2) and $\mathbf{A}$ is a $p \times k$ matrix with k linearly independent columns, then the $k \times k$ matrix $\mathbf{A}^\top\mathbf{C}\mathbf{A}$ is positive definite.*
2. *If $\mathbf{A}_1$ and $\mathbf{A}_2$ are positive definite $p \times p$ matrices and $\alpha_1, \alpha_2 > 0$, then $\alpha_1\mathbf{A}_1 + \alpha_2\mathbf{A}_2$ is positive definite.*

Proof

Proof. **Part 1.** $\mathbf{A}^\top\mathbf{C}\mathbf{A}$ is symmetric:

$$\begin{aligned}
\left(\mathbf{A}^\top\mathbf{C}\mathbf{A}\right)^\top &= \mathbf{A}^\top\,\mathbf{C}^\top\left(\mathbf{A}^\top\right)^\top\\
&= \mathbf{A}^\top\mathbf{C}\mathbf{A}
\end{aligned}$$

(Theorem 0.19, twice; $\mathbf{C}$ symmetric; Definition 0.6). For $\underset{\sim}{x} \neq \underset{\sim}{0}_k$, $\mathbf{A}\underset{\sim}{x} \neq \underset{\sim}{0}_p$, since the

columns of $\mathbf{A}$ are independent (Theorem 0.3, Definition 0.1), and

$$\begin{aligned} \underset{\sim}{x}^\top (\mathbf{A}^\top \mathbf{C} \mathbf{A}) \underset{\sim}{x} &= (\mathbf{A}\underset{\sim}{x})^\top \mathbf{C} (\mathbf{A}\underset{\sim}{x}) && \text{(Theorem 0.19, Theorem 0.7)} \\ &> 0. && (\mathbf{C} \text{ positive definite, } \mathbf{A}\underset{\sim}{x} \neq \underset{\sim}{0}) \end{aligned}$$

Part 2. $\alpha_1 \mathbf{A}_1 + \alpha_2 \mathbf{A}_2$ is symmetric: its transpose is $\alpha_1 \mathbf{A}_1{}^\top + \alpha_2 \mathbf{A}_2{}^\top = \alpha_1 \mathbf{A}_1 + \alpha_2 \mathbf{A}_2$ (Theorem 0.18, Definition 0.9, Definition 0.6). For $\underset{\sim}{x} \neq \underset{\sim}{0}$,

$$\begin{aligned} \underset{\sim}{x}^\top (\alpha_1 \mathbf{A}_1 + \alpha_2 \mathbf{A}_2) \underset{\sim}{x} &= \underset{\sim}{x}^\top (\alpha_1 \mathbf{A}_1) \underset{\sim}{x} + \underset{\sim}{x}^\top (\alpha_2 \mathbf{A}_2) \underset{\sim}{x} && \text{(Theorem 0.8, twice)} \\ &= \alpha_1 \, \underset{\sim}{x}^\top \mathbf{A}_1 \underset{\sim}{x} + \alpha_2 \, \underset{\sim}{x}^\top \mathbf{A}_2 \underset{\sim}{x} && \text{(Theorem 0.7)} \\ &> 0. && \text{(each term is positive)} \end{aligned}$$

□

Example 0.2 (Building positive definite matrices)**.**

- With $\mathbf{C} = \mathbf{I}_3$ and $\mathbf{X}$ of Example 0.1, part 1 gives again that $\mathbf{X}^\top \mathbf{X}$ is positive definite.
- With $\alpha_1 = 2$, $\mathbf{A}_1 = \mathbf{I}_2$ (Example 0.3), $\alpha_2 = 1$ and

$$\begin{aligned} \mathbf{A}_2 &= \mathbf{X}^\top \mathbf{X} \\ &= \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \end{aligned}$$

  (Example 0.1), part 2 gives that $2\mathbf{I}_2 + \mathbf{X}^\top \mathbf{X} = \begin{bmatrix} 5 & 6 \\ 6 & 16 \end{bmatrix}$ is positive definite.
- An aside beyond part 2: $\mathbf{I}_2 + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ is positive definite plus positive semidefinite; its quadratic form is $x_1^2 + x_2^2 + (x_1 + x_2)^2 > 0$ for $\underset{\sim}{x} \neq \underset{\sim}{0}$, so it is positive definite, in line with its eigenvalues 3 and 1 (Example 0.1).
- Independence is needed in part 1: with $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, whose columns are equal, $\mathbf{A}^\top \mathbf{I}_2 \mathbf{A} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$, and $(1, -1)$ gives quadratic form 0.

Theorem 0.3 (A positive definite matrix is invertible)**.** *Every positive definite matrix (Definition 0.2) is invertible (Definition 0.6).*

Proof

Proof. Let $\mathbf{A}$ be a $p \times p$ positive definite matrix, and let $\underset{\sim}{x}$ be a vector of length p with $\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_p$. Then

$$\begin{aligned} \underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{x} &= \underset{\sim}{x}^\top (\mathbf{A}\underset{\sim}{x}) && \text{(Definition 0.1)} \\ &= \underset{\sim}{x}^\top \underset{\sim}{0}_p && (\mathbf{A}\underset{\sim}{x} = \underset{\sim}{0}_p) \\ &= 0. && \text{(every term of the inner product is 0)} \end{aligned}$$

If $\underset{\sim}{x} \neq \underset{\sim}{0}_p$, then $\underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{x} > 0$ because $\mathbf{A}$ is positive definite, which contradicts $\underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{x} = 0$. So $\underset{\sim}{x} = \underset{\sim}{0}_p$, the null space $\mathcal{N}(\mathbf{A})$ is $\left\{\underset{\sim}{0}_p\right\}$, and $\mathbf{A}$ is invertible by Theorem 0.1. □

Example 0.3 (Positive definite matrices are invertible; the converse fails)**.** $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ is positive definite, since

$$\begin{aligned} \underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{x} &= 2x_1^2 + 2x_1x_2 + 2x_2^2 && \text{(multiply out)} \\ &= x_1^2 + x_2^2 + (x_1 + x_2)^2, && \text{(expand the square)} \end{aligned}$$

which is 0 only when $x_1 = x_2 = 0$. It is invertible, with $(\mathbf{A})^{-1} = \frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$.

The converse fails. $\mathbf{D} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ of Example 0.3 is not positive definite, yet it is invertible: $(\mathbf{D})^{-1} = \frac{1}{3}\begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}$.

Theorem 0.4 (Adding a positive multiple of the identity makes a matrix positive definite)**.** *Let* $\mathbf{A}$ *be a* $p \times p$ *positive semidefinite matrix (Definition 0.1) and let* $\alpha > 0$*. Then* $\mathbf{A} + \alpha\mathbf{I}_p$ *is positive definite (Definition 0.2), so it is invertible (Theorem 0.3).*

Proof

Proof. $\mathbf{A} + \alpha\mathbf{I}_p$ is symmetric: its transpose is $\mathbf{A}^\top + \alpha\mathbf{I}_p = \mathbf{A} + \alpha\mathbf{I}_p$ (Theorem 0.18, Definition 0.9, Definition 0.6). For $\underset{\sim}{x} \neq \underset{\sim}{0}_p$,

$$\begin{aligned} \underset{\sim}{x}^\top (\mathbf{A} + \alpha\mathbf{I}_p)\, \underset{\sim}{x} &= \underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{x} + \alpha\, \underset{\sim}{x}^\top \mathbf{I}_p\, \underset{\sim}{x} && \text{(Theorem 0.8, Theorem 0.7)} \\ &= \underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{x} + \alpha \left\|\underset{\sim}{x}\right\|^2 && \text{(Theorem 0.10, Equation 2, squared)} \\ &> 0. && (\underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{x} \geq 0,\ \alpha > 0,\ \underset{\sim}{x} \neq \underset{\sim}{0}_p) \end{aligned}$$

□

Example 0.4 (Repairing a singular matrix)**.** $\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ of Example 0.1 is positive semidefinite and singular: $\mathbf{B}(1, -1) = \underset{\sim}{0}_2$. With $\alpha = 1$,

$$\mathbf{B} + \mathbf{I}_2 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$

which is positive definite and invertible by Example 0.3.

Corollary 0.1 ($\mathbf{X}^\top\mathbf{X}$ is invertible exactly when $\mathbf{X}$ has full column rank)**.** *For an $n \times p$ matrix $\mathbf{X}$, the $p \times p$ matrix $\mathbf{X}^\top\mathbf{X}$ is invertible (Definition 0.6) exactly when* $\operatorname{rank}(\mathbf{X}) = p$ *(Definition 0.3).*

Proof

Proof. The following statements are equivalent:

$$\begin{aligned} \mathbf{X}^\top\mathbf{X} \text{ is invertible} &\iff \mathcal{N}(\mathbf{X}^\top\mathbf{X}) = \{\underset{\sim}{0}_p\} && \text{(Theorem 0.1)} \\ &\iff \mathcal{N}(\mathbf{X}) = \{\underset{\sim}{0}_p\} && \text{(Theorem 0.6)} \\ &\iff \mathbf{X} \text{ has full column rank.} && \text{(Theorem 0.3)} \end{aligned}$$

□

Example 0.5 (A design matrix with a repeated column)**.** $\mathbf{X} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix}$ of Example 0.2 has rank $1 < 2$. Its cross-product matrix is

$$\mathbf{X}^\top\mathbf{X} = \begin{bmatrix} 3 & 6 \\ 6 & 12 \end{bmatrix},$$

which is singular: $(2, -1)$ is in its null space, because $\mathbf{X}(2, -1) = \underset{\sim}{0}_3$ (Example 0.5). For the full-column-rank matrix of Example 0.1, $\mathbf{X}^\top\mathbf{X}$ is invertible.

Definition 0.1 (Block matrix)**.** Split the rows of an $m \times n$ matrix $\mathbf{X}$ into the first k and the last $m - k$, and its columns into the first l and the last $n - l$, with $1 \le k < m$ and $1 \le l < n$. This cuts $\mathbf{X}$ into four smaller matrices, its **blocks**:

$$\mathbf{X} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix},$$

where $\mathbf{A}$ $(k \times l)$ holds the entries x_{ij} with $i \le k$ and $j \le l$, $\mathbf{B}$ $(k \times (n - l))$ those with

$i \le k$ and $j > l$, $\mathbf{C}$ ($(m-k) \times l$) those with $i > k$ and $j \le l$, and $\mathbf{D}$ ($(m-k) \times (n-l)$) those with $i > k$ and $j > l$, each in its original order. A matrix written this way is a **block matrix** [98].

Example 0.6 (Cutting a 3×3 matrix into blocks)**.** With

$$\begin{aligned} k &= l \\ &= 1, \end{aligned}$$

$$\mathbf{X} = \begin{bmatrix} 4 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 3 & 6 \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix}, \qquad \mathbf{A} = \begin{bmatrix} 4 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 2 & 1 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 5 & 3 \\ 3 & 6 \end{bmatrix}.$$

For instance, entry $(1, 2)$ of $\mathbf{D}$ is

$$\begin{aligned} x_{1+1,2+1} &= x_{23} \\ &= 3. \end{aligned}$$

With

$$\begin{aligned} k &= l \\ &= 2 \end{aligned}$$

instead, the leading 2×2 block is $\begin{bmatrix} 4 & 2 \\ 2 & 5 \end{bmatrix}$.

Definition 0.2 (Leading block of a square matrix, top-left block)**.** When $\mathbf{X}$ is $n \times n$ and $1 \le k \le n$, the $k \times k$ matrix of the entries x_{ij} with $i, j \le k$ is the **leading** (or **top-left**) $k \times k$ **block** of $\mathbf{X}$; for $k < n$ it is the block $\mathbf{A}$ of Definition 0.1 with $l = k$, and for $k = n$ it is $\mathbf{X}$ itself.

Theorem 0.5 (Leading blocks of a positive definite matrix are positive definite)**.** *Let $\mathbf{X}$ be a $p \times p$ positive definite matrix (Definition 0.2), and for $1 \le k \le p$ let $\mathbf{X}_k$ be its leading $k \times k$ block (Definition 0.1), with entries x_{ij} for $i, j \le k$. Then $\mathbf{X}_k$ is positive definite. Also, every diagonal entry of a positive definite matrix is positive.*

Proof

Proof. $\mathbf{X}_k$ is symmetric, because $\mathbf{X}$ is. For $\underset{\sim}{u} \in \mathbb{R}^k$ with $\underset{\sim}{u} \neq \underset{\sim}{0}$, let $\underset{\sim}{y} = (u_1, \ldots, u_k, 0, \ldots, 0) \in \mathbb{R}^p$, which is nonzero, with entries $y_i = u_i$ for $i \le k$ and $y_i = 0$

for $i > k$. Then

$$\begin{aligned}
\underset{\sim}{u}^{\top}\mathbf{X}_k\underset{\sim}{u} &= \sum_{i=1}^{k}\sum_{j=1}^{k} u_i\, x_{ij}\, u_j && \text{(Definition 0.16, Definition 0.8)}\\
&= \sum_{i=1}^{k}\sum_{j=1}^{k} y_i\, x_{ij}\, y_j && (y_i = u_i \text{ for } i \le k)\\
&= \sum_{i=1}^{p}\sum_{j=1}^{p} y_i\, x_{ij}\, y_j && \text{(each added term has a factor } y_i = 0 \text{ or } y_j = 0)\\
&= \underset{\sim}{y}^{\top}\mathbf{X}\underset{\sim}{y} && \text{(Definition 0.16, Definition 0.8)}\\
&> \underset{\sim}{0}. && (\mathbf{X} \text{ positive definite, } \underset{\sim}{y} \neq \underset{\sim}{0})
\end{aligned}$$

For the diagonal entry x_{ii}, take $\underset{\sim}{e}_i$ (Definition 0.18), which is nonzero:

$$\begin{aligned}
\underset{\sim}{e}_i{}^{\top}\mathbf{X}\underset{\sim}{e}_i &= \sum_{r=1}^{p}\sum_{s=1}^{p}(\underset{\sim}{e}_i)_r\, x_{rs}\, (\underset{\sim}{e}_i)_s && \text{(Definition 0.16, Definition 0.8)}\\
&= x_{ii} && \text{(only the term with } r \text{ and } s \text{ both equal to } i \text{ is nonzero)}\\
&> 0. && (\mathbf{X} \text{ positive definite})
\end{aligned}$$

□

Example 0.7 (A quick necessary test)**.** $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ is positive definite (Example 0.2), and its diagonal entries $2, 2$ and its 1×1 leading block $[2]$ are positive. The test can only rule matrices out: $-\mathbf{I}_2$ has negative diagonal entries, so it is not positive definite, even though

$$\begin{aligned}
\det(-\mathbf{I}_2) &= (-1)(-1) - 0\\
&= 1\\
&> 0
\end{aligned}$$

(Definition 0.1); a positive determinant alone does not make a matrix positive definite. And positive diagonal entries are not enough: $\mathbf{D} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ of Example 0.3 has them but is not positive definite.

Theorem 0.6 (The quadratic form of a block matrix)**.** *Let $k, m \ge 1$, let $\mathbf{A}$ be a symmetric $k \times k$ matrix, $\mathbf{B}$ a $k \times m$ matrix and $\mathbf{C}$ a symmetric $m \times m$ matrix, and let $\mathbf{X}$ be the $(k+m) \times (k+m)$ block matrix (Definition 0.1)*

$$\mathbf{X} = \begin{bmatrix} \underset{\sim}{\mathbf{A}} & \underset{\sim}{\mathbf{B}} \\ {\scriptstyle k\times k} & {\scriptstyle k\times m} \\ \underset{\sim}{\mathbf{B}}^{\top} & \underset{\sim}{\mathbf{C}} \\ {\scriptstyle m\times k} & {\scriptstyle m\times m} \end{bmatrix},$$

that is, $x_{ij} = a_{ij}$, $x_{i,k+j} = b_{ij}$, $x_{k+i,j} = b_{ji}$ *and* $x_{k+i,k+j} = c_{ij}$ *for indices in range.* $\mathbf{X}$ *is symmetric (Definition 0.3):*

$$\begin{aligned} x_{k+i,j} &= b_{ji} \\ &= x_{j,k+i}, \end{aligned}$$

and $\mathbf{A}$ *and* $\mathbf{C}$ *are symmetric. Split* $\underset{\sim}{v} \in \mathbb{R}^{k+m}$ *as* $\underset{\sim}{v} = (\underset{\sim}{u}, \underset{\sim}{w})$*, with* $\underset{\sim}{u} \in \mathbb{R}^{k}$ *its first* k *entries and* $\underset{\sim}{w} \in \mathbb{R}^{m}$ *the rest. Then*

$$\underset{\sim}{v}^{\top}\mathbf{X}\underset{\sim}{v} = \underset{\sim}{u}^{\top}\mathbf{A}\underset{\sim}{u} + 2\,\underset{\sim}{u}^{\top}\mathbf{B}\underset{\sim}{w} + \underset{\sim}{w}^{\top}\mathbf{C}\underset{\sim}{w}, \tag{8}$$

and if $\mathbf{A}$ *is invertible, with* $\mathbf{S} \overset{def}{=} \mathbf{C} - \mathbf{B}^{\top}\mathbf{A}^{-1}\mathbf{B}$ *and* $\underset{\sim}{z} \overset{def}{=} \underset{\sim}{u} + \mathbf{A}^{-1}\mathbf{B}\underset{\sim}{w}$*,*

$$\underset{\sim}{v}^{\top}\mathbf{X}\underset{\sim}{v} = \underset{\sim}{z}^{\top}\mathbf{A}\underset{\sim}{z} + \underset{\sim}{w}^{\top}\mathbf{S}\underset{\sim}{w}.$$

Proof

Proof. **$\mathbf{X}\underset{\sim}{v}$ in blocks.** For $i \le k$,

$$\begin{aligned} (\mathbf{X}\underset{\sim}{v})_i &= \sum_{j=1}^{k+m} x_{ij}\, v_j && \text{(Definition 0.16)} \\ &= \sum_{j=1}^{k} x_{ij}\, v_j + \sum_{j=1}^{m} x_{i,k+j}\, v_{k+j} && \text{(split the sum at } j = k\text{)} \\ &= \sum_{j=1}^{k} a_{ij}\, u_j + \sum_{j=1}^{m} b_{ij}\, w_j && \text{(the blocks of } \mathbf{X} \text{ and of } \underset{\sim}{v}\text{)} \\ &= (\mathbf{A}\underset{\sim}{u})_i + (\mathbf{B}\underset{\sim}{w})_i, && \text{(Definition 0.16)} \end{aligned}$$

and in the same way $(\mathbf{X}\underset{\sim}{v})_{k+i} = (\mathbf{B}^{\top}\underset{\sim}{u})_i + (\mathbf{C}\underset{\sim}{w})_i$ for $i \le m$.
The expansion.

$$\begin{aligned} \underset{\sim}{v}^{\top}\mathbf{X}\underset{\sim}{v} &= \sum_{i=1}^{k} u_i\, (\mathbf{X}\underset{\sim}{v})_i + \sum_{i=1}^{m} w_i\, (\mathbf{X}\underset{\sim}{v})_{k+i} && \text{(Definition 0.8, split the sum at } k\text{)} \\ &= \underset{\sim}{u} \cdot (\mathbf{A}\underset{\sim}{u} + \mathbf{B}\underset{\sim}{w}) + \underset{\sim}{w} \cdot (\mathbf{B}^{\top}\underset{\sim}{u} + \mathbf{C}\underset{\sim}{w}) && \text{(the blocks of } \mathbf{X}\underset{\sim}{v}\text{)} \\ &= \underset{\sim}{u}^{\top}\mathbf{A}\underset{\sim}{u} + \underset{\sim}{u}^{\top}\mathbf{B}\underset{\sim}{w} + \underset{\sim}{w}^{\top}\mathbf{B}^{\top}\underset{\sim}{u} + \underset{\sim}{w}^{\top}\mathbf{C}\underset{\sim}{w} && \text{(Theorem 0.1)} \\ &= \underset{\sim}{u}^{\top}\mathbf{A}\underset{\sim}{u} + \underset{\sim}{u}^{\top}\mathbf{B}\underset{\sim}{w} + \underset{\sim}{u}^{\top}\mathbf{B}\underset{\sim}{w} + \underset{\sim}{w}^{\top}\mathbf{C}\underset{\sim}{w} && (\underset{\sim}{w}^{\top}\mathbf{B}^{\top}\underset{\sim}{u} = (\underset{\sim}{w}^{\top}\mathbf{B}^{\top}\underset{\sim}{u})^{\top}\text{, which equals } \underset{\sim}{u}^{\top}\mathbf{B}\underset{\sim}{w}\text{; } 1\times 1\text{, Definition 0} \\ &= \underset{\sim}{u}^{\top}\mathbf{A}\underset{\sim}{u} + 2\,\underset{\sim}{u}^{\top}\mathbf{B}\underset{\sim}{w} + \underset{\sim}{w}^{\top}\mathbf{C}\underset{\sim}{w}. && \text{(combine like terms)} \end{aligned}$$

Completing the square. Write $\underset{\sim}{d} \stackrel{\text{def}}{=} \mathbf{A}^{-1}\mathbf{B}\underset{\sim}{w}$, so $\underset{\sim}{z} = \underset{\sim}{u}+\underset{\sim}{d}$ and $\mathbf{A}\underset{\sim}{d} = \mathbf{B}\underset{\sim}{w}$ (Theorem 0.7, 5, Theorem 0.10); $\mathbf{A}^{-1}$ is symmetric (Corollary 0.1). Then

$$\begin{aligned}
\underset{\sim}{z}^\top \mathbf{A} \underset{\sim}{z} &= (\underset{\sim}{u}+\underset{\sim}{d}) \cdot \mathbf{A}\,(\underset{\sim}{u}+\underset{\sim}{d}) && \text{(substitute } \underset{\sim}{z} = \underset{\sim}{u}+\underset{\sim}{d}\text{)} \\
&= (\underset{\sim}{u}+\underset{\sim}{d}) \cdot (\mathbf{A}\underset{\sim}{u}+\mathbf{A}\underset{\sim}{d}) && \text{(Theorem 0.8)} \\
&= \underset{\sim}{u}^\top \mathbf{A}\underset{\sim}{u} + \underset{\sim}{u}^\top \mathbf{A}\underset{\sim}{d} + \underset{\sim}{d}^\top \mathbf{A}\underset{\sim}{u} + \underset{\sim}{d}^\top \mathbf{A}\underset{\sim}{d} && \text{(Theorem 0.1, both slots)} \\
&= \underset{\sim}{u}^\top \mathbf{A}\underset{\sim}{u} + \underset{\sim}{u}^\top \mathbf{A}\underset{\sim}{d} + \underset{\sim}{u}^\top \mathbf{A}\underset{\sim}{d} + \underset{\sim}{d}^\top \mathbf{A}\underset{\sim}{d} && \text{(as in the expansion, with } \mathbf{A}^\top = \mathbf{A}\text{)} \\
&= \underset{\sim}{u}^\top \mathbf{A}\underset{\sim}{u} + 2\,\underset{\sim}{u}^\top \mathbf{A}\underset{\sim}{d} + \underset{\sim}{d}^\top \mathbf{A}\underset{\sim}{d} && \text{(combine like terms)} \\
&= \underset{\sim}{u}^\top \mathbf{A}\underset{\sim}{u} + 2\,\underset{\sim}{u}^\top \mathbf{B}\underset{\sim}{w} + \underset{\sim}{d}^\top \mathbf{B}\underset{\sim}{w} && \text{(}\mathbf{A}\underset{\sim}{d} = \mathbf{B}\underset{\sim}{w}\text{, twice)} \\
&= \underset{\sim}{u}^\top \mathbf{A}\underset{\sim}{u} + 2\,\underset{\sim}{u}^\top \mathbf{B}\underset{\sim}{w} + \underset{\sim}{w}^\top \mathbf{B}^\top \mathbf{A}^{-1}\mathbf{B}\underset{\sim}{w}, && \text{(}\underset{\sim}{d}^\top = \underset{\sim}{w}^\top \mathbf{B}^\top \mathbf{A}^{-1}\text{, Theorem 0.19, Corollary 0.1)}
\end{aligned}$$

and

$$\begin{aligned}
\underset{\sim}{w}^\top \mathbf{S} \underset{\sim}{w} &= \underset{\sim}{w}^\top (\mathbf{C} - \mathbf{B}^\top \mathbf{A}^{-1}\mathbf{B})\,\underset{\sim}{w} && \text{(substitute } \mathbf{S}\text{)} \\
&= \underset{\sim}{w}^\top (\mathbf{C}\underset{\sim}{w} - \mathbf{B}^\top \mathbf{A}^{-1}\mathbf{B}\underset{\sim}{w}) && \text{(Theorem 0.7, difference rule on the right)} \\
&= \underset{\sim}{w}^\top \mathbf{C}\underset{\sim}{w} - \underset{\sim}{w}^\top \mathbf{B}^\top \mathbf{A}^{-1}\mathbf{B}\underset{\sim}{w}. && \text{(Theorem 0.7, difference rule on the left)}
\end{aligned}$$

Adding the two displays, the $\underset{\sim}{w}^\top \mathbf{B}^\top \mathbf{A}^{-1}\mathbf{B}\underset{\sim}{w}$ terms cancel, leaving $\underset{\sim}{u}^\top \mathbf{A}\underset{\sim}{u} + 2\,\underset{\sim}{u}^\top \mathbf{B}\underset{\sim}{w} + \underset{\sim}{w}^\top \mathbf{C}\underset{\sim}{w}$, which is $\underset{\sim}{v}^\top \mathbf{X}\underset{\sim}{v}$ by Equation 8. □

Example 0.8 (The 2×2 case)**.** For $\mathbf{X} = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ ($k = 1$ and $m = 1$) and $a \neq 0$, the theorem reads

$$au^2 + 2buw + cw^2 = a\left(u + \tfrac{b}{a}\,w\right)^2 + \left(c - \tfrac{b^2}{a}\right) w^2,$$

with $\underset{\sim}{v} = (u, w)$: the familiar completing of the square. With

$$\begin{aligned} a &= c \\ &= 5 \end{aligned}$$

and $b = 4$:

$$\begin{aligned}
5\,(u + 0.8\,w)^2 + 1.8\,w^2 &= 5\,(u^2 + 1.6\,uw + 0.64\,w^2) + 1.8\,w^2 \\
&= 5u^2 + 8uw + 3.2\,w^2 + 1.8\,w^2 \\
&= 5u^2 + 8uw + 5w^2.
\end{aligned}$$

Definition 0.3 (Schur complement)**.** For a block matrix $\mathbf{X} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{bmatrix}$ as in Theorem 0.6 with $\mathbf{A}$ invertible, the **Schur complement** [99] of $\mathbf{A}$ in $\mathbf{X}$ is the $m \times m$ matrix $\mathbf{S} = \mathbf{C} - \mathbf{B}^\top \mathbf{A}^{-1} \mathbf{B}$ defined in Theorem 0.6.

Example 0.9 (Schur complements of 2×2 matrices)**.**

- For $\begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$ with $\mathbf{A} = [5]$:

$$\begin{aligned} \mathbf{S} &= 5 - 4 \cdot \tfrac{1}{5} \cdot 4 \\ &= \tfrac{25}{5} - \tfrac{16}{5} \\ &= \tfrac{9}{5}, \end{aligned}$$

  that is, $\mathbf{S} = [\frac{9}{5}]$.

- For $\mathbf{D} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ with $\mathbf{A} = [1]$:

$$\begin{aligned} \mathbf{S} &= [1 - 2 \cdot 1 \cdot 2] \\ &= [-3]. \end{aligned}$$

- For $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ the top-left block $[0]$ is not invertible, so there is no Schur complement of it.

Theorem 0.7 (Schur complement test)**.** *Let* $\mathbf{X} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{bmatrix}$ *be as in Theorem 0.6. Then* $\mathbf{X}$ *is positive definite (Definition 0.2) exactly when* $\mathbf{A}$ *is positive definite and its Schur complement* $\mathbf{S}$ *(Definition 0.3) is positive definite.*

Proof

Proof. **If.** $\mathbf{A}$ is positive definite, so invertible (Theorem 0.4), and $\mathbf{S}$ is defined. $\mathbf{X}$ is symmetric. Let $\underset{\sim}{v} = (\underset{\sim}{u}, \underset{\sim}{w}) \neq \underset{\sim}{0}$ and $\underset{\sim}{z} = \underset{\sim}{u} + \mathbf{A}^{-1}\mathbf{B}\underset{\sim}{w}$. By Theorem 0.6, $\underset{\sim}{v}^\top \mathbf{X} \underset{\sim}{v} = \underset{\sim}{z}^\top \mathbf{A} \underset{\sim}{z} + \underset{\sim}{w}^\top \mathbf{S} \underset{\sim}{w}$, and both terms are at least 0. If $\underset{\sim}{w} \neq \underset{\sim}{0}$, the second term is positive. If $\underset{\sim}{w} = \underset{\sim}{0}$, then $\underset{\sim}{u} \neq \underset{\sim}{0}$ and $\underset{\sim}{z} = \underset{\sim}{u}$, so the first term is positive. Either way $\underset{\sim}{v}^\top \mathbf{X} \underset{\sim}{v} > 0$.
Only if. $\mathbf{A}$ is the top-left $k \times k$ block of $\mathbf{X}$, so it is positive definite (Theorem 0.5), and invertible (Theorem 0.4). $\mathbf{S}$ is symmetric:

$$\begin{aligned}\left(\mathbf{B}^\top\mathbf{A}^{-1}\mathbf{B}\right)^\top &= \mathbf{B}^\top\left(\mathbf{A}^{-1}\right)^\top\mathbf{B}\\ &= \mathbf{B}^\top\mathbf{A}^{-1}\mathbf{B}\end{aligned}$$

(Theorem 0.19, Definition 0.6 for $\left(\mathbf{B}^\top\right)^\top = \mathbf{B}$, Corollary 0.1), and $\mathbf{C}$ is symmetric, so $\mathbf{S} = \mathbf{C} - \mathbf{B}^\top\mathbf{A}^{-1}\mathbf{B}$ is symmetric (the transpose of a difference is the difference of the transposes, entry by entry, Definition 0.6). For $\underset{\sim}{w} \neq \underset{\sim}{0}$, take $\underset{\sim}{u} = -\mathbf{A}^{-1}\mathbf{B}\underset{\sim}{w}$, so $\underset{\sim}{z} = \underset{\sim}{0}$ and $\underset{\sim}{v} = (\underset{\sim}{u}, \underset{\sim}{w}) \neq \underset{\sim}{0}$; then $\underset{\sim}{w}^\top\mathbf{S}\underset{\sim}{w} = \underset{\sim}{v}^\top\mathbf{X}\underset{\sim}{v} > 0$ (Theorem 0.6). □

Example 0.10 (Testing 2×2 matrices)**.** A symmetric $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$ is positive definite exactly when $a > 0$ and $c - \frac{b^2}{a} > 0$, that is, $a > 0$ and $ac > b^2$ (a 1×1 matrix $[s]$ is positive definite exactly when $s > 0$, since $x^\top sx = sx^2$).

- $\begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$: $5 > 0$ and $\frac{9}{5} > 0$ (Example 0.9), so it is positive definite.
- $\mathbf{D} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$: $1 > 0$ but $\mathbf{S} = -3$, so it is not, as Example 0.3 found directly.

Theorem 0.8 (The inverse of a positive definite block matrix)**.** *Let* $\mathbf{X} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{bmatrix}$ *be as in Theorem 0.6, with* $\mathbf{X}$ *positive definite. Then* $\mathbf{X}$ *is invertible, and with the Schur complement* $\mathbf{S}$ *of* $\mathbf{A}$*, its inverse is the following, where* $\mathbf{P}$ *is the block of the first* k *rows and columns of* $\mathbf{X}^{-1}$*:*

$$\mathbf{X}^{-1} = \begin{bmatrix} \mathbf{P} & -\mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1} \\ -\mathbf{S}^{-1}\mathbf{B}^\top\mathbf{A}^{-1} & \mathbf{S}^{-1} \end{bmatrix},$$

Proof

Proof. $\mathbf{X}$ is invertible, because it is positive definite. By the Schur complement test, $\mathbf{A}$ and $\mathbf{S}$ are positive definite as well, and so both are invertible. Let $\mathbf{W} = \begin{bmatrix} -\mathbf{A}^{-1}\mathbf{B} \\ \mathbf{I}_m \end{bmatrix}$. Each entry of $\mathbf{XW}$ adds one group of k summands to one group of m summands, so $\mathbf{XW}$ can be computed one block row at a time:

$$\begin{aligned}
\mathbf{XW} &= \begin{bmatrix} -\mathbf{A}\mathbf{A}^{-1}\mathbf{B} + \mathbf{B} \\ -\mathbf{B}^\top\mathbf{A}^{-1}\mathbf{B} + \mathbf{C} \end{bmatrix} && \text{(one block row at a time)} \\
&= \begin{bmatrix} -\mathbf{I}_k\mathbf{B} + \mathbf{B} \\ \mathbf{S} \end{bmatrix} && (\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}_k, \text{ and definition of } \mathbf{S}) \\
&= \begin{bmatrix} \mathbf{0} \\ \mathbf{S} \end{bmatrix} && (\mathbf{I}_k\mathbf{B} = \mathbf{B})
\end{aligned}$$

Multiplying both sides on the left by $\mathbf{X}^{-1}$ gives $\mathbf{W}$ on the left, since $\mathbf{X}^{-1}\mathbf{X} = \mathbf{I}_{k+m}$. Cut $\mathbf{X}^{-1}$ into the four pieces $\mathbf{P}$, $\mathbf{Q}$, $\mathbf{R}$ and $\mathbf{T}$, with $\mathbf{Q}$ and $\mathbf{T}$ in the last m columns. Then the right side is $\mathbf{X}^{-1}\begin{bmatrix} \mathbf{0} \\ \mathbf{S} \end{bmatrix}$, and the last m columns of $\mathbf{X}^{-1}$ multiply $\mathbf{S}$:

$$\begin{bmatrix} -\mathbf{A}^{-1}\mathbf{B} \\ \mathbf{I}_m \end{bmatrix} = \begin{bmatrix} \mathbf{QS} \\ \mathbf{TS} \end{bmatrix}.$$

So $\mathbf{TS} = \mathbf{I}_m$, and $\mathbf{QS} = -\mathbf{A}^{-1}\mathbf{B}$. Multiplying both equations on the right by $\mathbf{S}^{-1}$ gives $\mathbf{T} = \mathbf{S}^{-1}$, and $\mathbf{Q} = -\mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1}$. $\mathbf{X}^{-1}$ is symmetric, by the inverse of a symmetric matrix is symmetric, so $\mathbf{R} = \mathbf{Q}^\top = -\mathbf{S}^{-1}\mathbf{B}^\top\mathbf{A}^{-1}$, using $(\mathbf{S}^{-1})^\top = \mathbf{S}^{-1}$ and $(\mathbf{A}^{-1})^\top = \mathbf{A}^{-1}$. □

Example 0.11 (The bottom-right entry of the inverse of a 2×2 matrix)**.**
Let $\mathbf{X} = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$ with $\mathbf{A} = [5]$, whose Schur complement is $\mathbf{S} = [\frac{9}{5}]$ (Example 0.9). It is positive definite (Example 0.10), so by Theorem 0.8:

$$\begin{aligned}
\mathbf{T} &= \mathbf{S}^{-1} \\
&= [\tfrac{5}{9}].
\end{aligned}$$

Directly, $\mathbf{X}^{-1} = \frac{1}{9}\begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}$, because

$$\begin{aligned}
\begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} \cdot \tfrac{1}{9}\begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix} &= \tfrac{1}{9}\begin{bmatrix} 25-16 & -20+20 \\ 20-20 & -16+25 \end{bmatrix} \\
&= \mathbf{I}_2,
\end{aligned}$$

and its bottom-right entry is $\frac{5}{9}$. The top-right entry is $-\mathbf{A}^{-1}\mathbf{B}\mathbf{S}^{-1}$:

$$-\tfrac{1}{5} \cdot 4 \cdot \tfrac{5}{9} = -\tfrac{4}{9}.$$

Theorem 0.9 (Cholesky factorization)**.** *A $p \times p$ matrix $\mathbf{A}$ is positive definite (Definition 0.2) exactly when $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$ for some lower triangular $\mathbf{L}$ (Definition 0.8) with positive diagonal entries. For a positive definite $\mathbf{A}$ that $\mathbf{L}$ is unique; it is the **Cholesky factor** [100] of $\mathbf{A}$.*

Proof

Proof. **If.** $\mathbf{L}^\top$ is upper triangular with positive diagonal (Definition 0.6), so $\mathbf{L}^\top \underset{\sim}{x} = \underset{\sim}{0}$ only for $\underset{\sim}{x} = \underset{\sim}{0}$ (Theorem 0.19), and the columns of $\mathbf{L}^\top$ are linearly independent (Theorem 0.3, Definition 0.1). So $\mathbf{A} = (\mathbf{L}^\top)^\top \mathbf{L}^\top$ (Definition 0.6) is positive definite (Theorem 0.1, part 2).

Only if, with uniqueness, by induction on p. For $p = 1$, $\mathbf{A} = [a]$ with $a > 0$ (Theorem 0.5), and $[c]\,[c] = [a]$ with $c > 0$ holds exactly for $c = \sqrt{a}$. For $p > 1$, assume every $(p-1) \times (p-1)$ positive definite matrix has exactly one such factor, and split off the first row and column. $\mathbf{A}$ is symmetric, so its first row is the transpose of its first column:

$$\mathbf{A} = \begin{bmatrix} a_{11} & \underset{\sim}{b}^\top \\ \underset{\sim}{b} & \mathbf{A}_{22} \end{bmatrix}, \qquad \mathbf{L} = \begin{bmatrix} \ell_{11} & \underset{\sim}{0}_{1\times(p-1)} \\ \underset{\sim}{g} & \mathbf{L}_{22} \end{bmatrix}, \qquad \mathbf{L}^\top = \begin{bmatrix} \ell_{11} & \underset{\sim}{g}^\top \\ \underset{\sim}{0}_{(p-1)\times 1} & \mathbf{L}_{22}{}^\top \end{bmatrix},$$

with $\underset{\sim}{b}, \underset{\sim}{g} \in \mathbb{R}^{p-1}$, where b_i and g_i are entries $(i+1, 1)$ of $\mathbf{A}$ and $\mathbf{L}$, and $(\mathbf{A}_{22})_{ij}$ and $(\mathbf{L}_{22})_{ij}$ are their entries $(i+1, j+1)$; $\mathbf{L}^\top$ has this form by Definition 0.6. A $p \times p$ matrix $\mathbf{L}$ is lower triangular with positive diagonal exactly when $\ell_{11} > 0$, its first row is otherwise zero, and $\mathbf{L}_{22}$ is lower triangular with positive diagonal; $\underset{\sim}{g}$ can be anything. So choosing $\mathbf{L}$ means choosing $\ell_{11} > 0$, $\underset{\sim}{g}$ and such an $\mathbf{L}_{22}$.

Entry (i, j) of $\mathbf{L}\mathbf{L}^\top$ is $\sum_{r=1}^{p} \ell_{ir}\, \ell_{jr}$, with ℓ_{ir} entry (i, r) of $\mathbf{L}$ (4, Definition 0.6). Row 1 of $\mathbf{L}$ is $(\ell_{11}, 0, \ldots, 0)$, and row $i+1$ is $(g_i, (\mathbf{L}_{22})_{i1}, \ldots, (\mathbf{L}_{22})_{i,p-1})$. So, for $i, j = 1, \ldots, p-1$:

$$\begin{aligned}
(\mathbf{L}\mathbf{L}^\top)_{11} &= \ell_{11}\,\ell_{11} + \sum_{r=2}^{p} 0 \cdot 0 \\
&= \ell_{11}^2, && \text{(rows 1 and 1)} \\
(\mathbf{L}\mathbf{L}^\top)_{i+1,1} &= g_i\,\ell_{11} + \sum_{r=2}^{p} (\mathbf{L}_{22})_{i,r-1} \cdot 0 \\
&= \ell_{11}\, g_i, && \text{(rows } i+1 \text{ and 1)} \\
(\mathbf{L}\mathbf{L}^\top)_{i+1,j+1} &= g_i\, g_j + \sum_{r=2}^{p} (\mathbf{L}_{22})_{i,r-1}\,(\mathbf{L}_{22})_{j,r-1} && \text{(rows } i+1 \text{ and } j+1) \\
&= (\underset{\sim}{g}\underset{\sim}{g}^\top)_{ij} + (\mathbf{L}_{22}\mathbf{L}_{22}{}^\top)_{ij}. && \text{(Definition 0.10, with } s = r-1)
\end{aligned}$$

Entry $(1, j+1)$ is $\ell_{11}\, g_j$ by the same computation with the roles swapped, the transpose of the $(2, 1)$ block; it matches $\underset{\sim}{b}^\top$ exactly when the $(2, 1)$ block matches $\underset{\sim}{b}$, which is where the symmetry of $\mathbf{A}$ is used. So $\mathbf{L}\mathbf{L}^\top = \mathbf{A}$ says exactly

$$\ell_{11}^2 = a_{11}, \qquad \ell_{11}\,\underset{\sim}{g} = \underset{\sim}{b}, \qquad \underset{\sim}{g}\underset{\sim}{g}^\top + \mathbf{L}_{22}\mathbf{L}_{22}{}^\top = \mathbf{A}_{22}.$$

$a_{11} > 0$ (Theorem 0.5), so the first equation with $\ell_{11} > 0$ forces $\ell_{11} = \sqrt{a_{11}}$; the second then forces $\underset{\sim}{g} = \frac{1}{\ell_{11}}\,\underset{\sim}{b}$; and the third becomes

$$\begin{aligned}
\mathbf{L}_{22}\mathbf{L}_{22}{}^{\top} &= \mathbf{A}_{22} - \underset{\sim}{g}\underset{\sim}{g}^{\top} && \text{(subtract } \underset{\sim}{g}\underset{\sim}{g}^{\top}\text{)} \\
&= \mathbf{A}_{22} - \left(\tfrac{1}{\ell_{11}}\,\underset{\sim}{b}\right)\left(\tfrac{1}{\ell_{11}}\,\underset{\sim}{b}\right)^{\top} && \text{(substitute } \underset{\sim}{g}\text{)} \\
&= \mathbf{A}_{22} - \tfrac{1}{\ell_{11}^2}\,\underset{\sim}{b}\underset{\sim}{b}^{\top} && \text{(Definition 0.6, Theorem 0.7)} \\
&= \mathbf{A}_{22} - \tfrac{1}{a_{11}}\,\underset{\sim}{b}\underset{\sim}{b}^{\top}. && (\ell_{11}^2 = a_{11})
\end{aligned}$$

Here $[a_{11}]$ is invertible with inverse $[1/a_{11}]$, so with $\mathbf{B} = \underset{\sim}{b}^{\top}$ in Definition 0.3,

$$\begin{aligned}
\mathbf{B}^{\top}\,[a_{11}]^{-1}\mathbf{B} &= \underset{\sim}{b}\,\tfrac{1}{a_{11}}\,\underset{\sim}{b}^{\top} \\
&= \tfrac{1}{a_{11}}\,\underset{\sim}{b}\underset{\sim}{b}^{\top}
\end{aligned}$$

(Theorem 0.7), and the right side is the Schur complement of $[a_{11}]$ in $\mathbf{A}$. It is positive definite by Theorem 0.7 (with $k = 1$; $\mathbf{A}_{22}$ is symmetric because $\mathbf{A}$ is). By the induction hypothesis it has exactly one factor $\mathbf{L}_{22}$ of the required kind. So $\mathbf{L}$ exists and is unique. □

Example 0.12 (Two Cholesky factors)**.**

- $\mathbf{A} = \begin{bmatrix} 4 & 2 \\ 2 & 5 \end{bmatrix}$:

$$\begin{aligned}
\ell_{11} &= \sqrt{4} \\
&= 2,
\end{aligned}$$

$$\begin{aligned}
\ell_{21} &= 2/2 \\
&= 1,
\end{aligned}$$

and

$$\begin{aligned}
\ell_{22}^2 &= 5 - 1^2 \\
&= 4,
\end{aligned}$$

so $\ell_{22} = 2$: $\mathbf{L} = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$, and

$$\begin{aligned}
\mathbf{L}\mathbf{L}^{\top} &= \begin{bmatrix} 2 \cdot 2 & 2 \cdot 1 \\ 1 \cdot 2 & 1 \cdot 1 + 2 \cdot 2 \end{bmatrix} \\
&= \mathbf{A}.
\end{aligned}$$

- $\begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$: $\ell_{11} = \sqrt{5}$, $\ell_{21} = 4/\sqrt{5}$, and

$$\begin{aligned}\ell_{22}^2 &= 5 - \tfrac{16}{5} \\ &= \tfrac{9}{5},\end{aligned}$$

the Schur complement of Example 0.9, so $\ell_{22} = \frac{3}{\sqrt{5}}$.

- For $\mathbf{D} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ the method breaks down:

$$\begin{aligned}\ell_{22}^2 &= 1 - 4 \\ &= -3\end{aligned}$$

has no real solution, matching the failure of the Schur complement test (Example 0.10).

Complex vectors and matrices

Definition 0.1 (Complex vector)**.** A **complex vector** of length p is a column vector whose entries are complex numbers; the set of all of them is denoted $\mathbb{C}^p$.

Definition 0.2 (Complex matrix)**.** A **complex matrix** of size $m \times n$ is a matrix with complex entries; the set of all of them is denoted $\mathbb{C}^{m \times n}$.

Example 0.1 (A complex vector and its conjugate)**.**

$$\underset{3\times 1}{z} = \begin{bmatrix} 1+i \\ 2 \\ -3\,i \end{bmatrix} \in \mathbb{C}^3, \qquad \underset{3\times 1}{\overline{z}} = \begin{bmatrix} 1-i \\ 2 \\ 3\,i \end{bmatrix}.$$

The real entry 2 is its own conjugate. Every real vector is also a complex vector, since each real number a is the complex number $a + 0\,i$, so $\mathbb{R}^p$ is a subset of $\mathbb{C}^p$.

Definition 0.3 (Complex conjugate of a complex vector or matrix)**.** The **complex conjugate** of a complex vector or matrix takes the complex conjugate of each entry: $(\overline{\mathbf{A}})_{jk} \stackrel{\text{def}}{=} \overline{a_{jk}}$.

Definition 0.4 (Conjugate transpose, Hermitian adjoint, Hermitian conjugate, transjugate)**.** The **conjugate transpose** [101] (also called the **Hermitian adjoint**, **Hermitian**

conjugate, or **transjugate**) of an $m \times n$ complex matrix $\mathbf{A}$ is the $n \times m$ matrix $\mathbf{A}^{\mathsf{H}}$ obtained by transposing $\mathbf{A}$ and then taking the complex conjugate of each entry:

$$(\mathbf{A}^{\mathsf{H}})_{jk} \overset{\text{def}}{=} \overline{a_{kj}} \quad \text{for } j \in \{1, \dots, n\},\ k \in \{1, \dots, m\}.$$

Remark (Notation for the conjugate transpose)

Remark 0.1 (Notation for the conjugate transpose). The operation is denoted variously across mathematics and physics: $\mathbf{A}^*$ in pure mathematics (Axler [52], Definition 7.7, p. 231), $\mathbf{A}^\dagger$ ("A-dagger") in physics and quantum mechanics, and $\mathbf{A}^{\mathsf{H}}$ in engineering, numerical linear algebra, and statistics. These notes write $\mathbf{A}^{\mathsf{H}}$ because * is reserved for optimal points (as in $\underset{\sim}{x}^*$) and † frequently denotes the Moore-Penrose pseudoinverse in the broader literature.
The Hermitian adjoint matrix should not be confused with the classical *adjugate matrix* (the transpose of the matrix of cofactors), which older texts occasionally referred to as the "adjoint."

Example 0.2 (The conjugate transpose of a 2×3 matrix)**.** For

$$\underset{2\times 3}{\underset{\sim}{\mathbf{A}}} = \begin{bmatrix} 1+i & 2 & 0 \\ -i & 3-2\,i & 4 \end{bmatrix},$$

transposing gives a 3×2 matrix, and conjugating each of its entries gives

$$\underset{3\times 2}{\underset{\sim}{\mathbf{A}}^{\mathsf{H}}} = \begin{bmatrix} 1-i & i \\ 2 & 3+2\,i \\ 0 & 4 \end{bmatrix}.$$

For instance,

$$\begin{aligned} (\mathbf{A}^{\mathsf{H}})_{12} &= \overline{a_{21}} \\ &= \overline{-i} \\ &= i. \end{aligned}$$

The transpose alone would have $-i$ in that position, so for a matrix with an entry that is not real, $\mathbf{A}^{\mathsf{H}}$ and $\mathbf{A}^\top$ differ. For a real matrix, conjugating changes nothing, so $\mathbf{A}^{\mathsf{H}} = \mathbf{A}^\top$.

Remark (Conjugate transpose of a complex vector)

Remark 0.2 (Conjugate transpose of a complex vector). The conjugate transpose of a column vector $\underset{\sim}{z} \in \mathbb{C}^p$ is the $1 \times p$ row vector $\underset{\sim}{z}^{\mathsf{H}} = [\overline{z_1},\ \dots,\ \overline{z_p}]$.

Theorem 0.1 (Properties of the Hermitian adjoint)**.** *Let* $\mathbf{A}$ *and* $\mathbf{B}$ *be complex matrices of compatible dimensions, and let* $c \in \mathbb{C}$ *be a scalar. The Hermitian adjoint (Definition 0.4) satisfies:*

1. ***Involution****:*
$$(\mathbf{A}^{\mathsf{H}})^{\mathsf{H}} = \mathbf{A}$$
2. ***Additivity****:*
$$(\mathbf{A} + \mathbf{B})^{\mathsf{H}} = \mathbf{A}^{\mathsf{H}} + \mathbf{B}^{\mathsf{H}}$$
3. ***Conjugate-homogeneity****:*
$$(c\,\mathbf{A})^{\mathsf{H}} = \overline{c}\,\mathbf{A}^{\mathsf{H}}$$
4. ***Reversal of products****:*
$$(\mathbf{AB})^{\mathsf{H}} = \mathbf{B}^{\mathsf{H}}\mathbf{A}^{\mathsf{H}}$$

If $\mathbf{A}$ *is invertible, then* $\mathbf{A}^{\mathsf{H}}$ *is also invertible, and*

$$(\mathbf{A}^{\mathsf{H}})^{-1} = (\mathbf{A}^{-1})^{\mathsf{H}}.$$

Proof

Proof. Properties 1-3 follow directly from entrywise conjugation and transposition:

- For involution:
$$\begin{aligned}[(\mathbf{A}^{\mathsf{H}})^{\mathsf{H}}]_{jk} &= \overline{(\mathbf{A}^{\mathsf{H}})_{kj}} \\ &= \overline{\overline{a_{jk}}} \\ &= a_{jk}.\end{aligned}$$
- For additivity:
$$\begin{aligned}[(\mathbf{A} + \mathbf{B})^{\mathsf{H}}]_{jk} &= \overline{(\mathbf{A} + \mathbf{B})_{kj}} \\ &= \overline{a_{kj} + b_{kj}} \\ &= \overline{a_{kj}} + \overline{b_{kj}} \\ &= (\mathbf{A}^{\mathsf{H}})_{jk} + (\mathbf{B}^{\mathsf{H}})_{jk}.\end{aligned}$$
- For conjugate-homogeneity:

$$\begin{aligned}
[(c\,\mathbf{A})^{\mathsf{H}}]_{jk} &= \overline{(c\,\mathbf{A})_{kj}} \\
&= \overline{c\,a_{kj}} \\
&= \overline{c}\,\overline{a_{kj}} \\
&= \overline{c}\,(\mathbf{A}^{\mathsf{H}})_{jk}.
\end{aligned}$$

Unlike the real transpose where $(c\,\mathbf{A})^{\top} = c\,\mathbf{A}^{\top}$, scaling conjugates the scalar.

For property 4 (reversal of products), let $\mathbf{A}$ be an $m \times k$ matrix and $\mathbf{B}$ a $k \times n$ matrix. Entry (i, j) of the product's Hermitian adjoint (for $i \in \{1, \ldots, n\}$ and $j \in \{1, \ldots, m\}$) is:

$$\begin{aligned}
[(\mathbf{AB})^{\mathsf{H}}]_{ij} &= \overline{(\mathbf{AB})_{ji}} && \text{(Definition 0.4)} \\
&= \overline{\sum_{s=1}^{k} a_{js}\,b_{si}} && \text{(definition of matrix multiplication)} \\
&= \sum_{s=1}^{k} \overline{a_{js}}\,\overline{b_{si}} && \text{(conjugate of sums and products)} \\
&= \sum_{s=1}^{k} \overline{b_{si}}\,\overline{a_{js}} && \text{(complex multiplication is commutative)} \\
&= \sum_{s=1}^{k} (\mathbf{B}^{\mathsf{H}})_{is}\,(\mathbf{A}^{\mathsf{H}})_{sj} && \text{(Definition 0.4)} \\
&= (\mathbf{B}^{\mathsf{H}}\mathbf{A}^{\mathsf{H}})_{ij} && \text{(definition of matrix multiplication).}
\end{aligned}$$

For the inverse, applying property 4 to $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$ gives:

$$\begin{aligned}
(\mathbf{A}^{-1})^{\mathsf{H}}\mathbf{A}^{\mathsf{H}} &= (\mathbf{A}\mathbf{A}^{-1})^{\mathsf{H}} \\
&= \mathbf{I}^{\mathsf{H}} \\
&= \mathbf{I},
\end{aligned}$$

so $(\mathbf{A}^{\mathsf{H}})^{-1} = (\mathbf{A}^{-1})^{\mathsf{H}}$. □

Theorem 0.2 (The adjoint property for inner products)**.** *Let* $\mathbf{A}$ *be an* $m \times n$ *complex matrix. Under the standard inner product on complex Euclidean space,* $\langle \underset{\sim}{u}, \underset{\sim}{v} \rangle \overset{def}{=} \underset{\sim}{u}^{\mathsf{H}} \underset{\sim}{v}$*,*

$$\langle \mathbf{A}\underset{\sim}{u}, \underset{\sim}{v} \rangle = \langle \underset{\sim}{u}, \mathbf{A}^{\mathsf{H}}\underset{\sim}{v} \rangle \quad \textit{for all } \underset{\sim}{u} \in \mathbb{C}^n \textit{ and } \underset{\sim}{v} \in \mathbb{C}^m.$$

Proof

Proof. By the definition of the standard complex inner product and the product rule for the Hermitian adjoint (Theorem 0.1):

$$\begin{aligned}
\langle \mathbf{A}\underset{\sim}{u}, \underset{\sim}{v}\rangle &= (\mathbf{A}\underset{\sim}{u})^{\mathsf{H}}\underset{\sim}{v} && \text{(definition of standard inner product)}\\
&= (\underset{\sim}{u}^{\mathsf{H}}\mathbf{A}^{\mathsf{H}})\underset{\sim}{v} && \text{(Theorem 0.1, reversal of products)}\\
&= \underset{\sim}{u}^{\mathsf{H}}(\mathbf{A}^{\mathsf{H}}\underset{\sim}{v}) && \text{(associativity of matrix multiplication)}\\
&= \langle \underset{\sim}{u}, \mathbf{A}^{\mathsf{H}}\underset{\sim}{v}\rangle && \text{(definition of standard inner product).}
\end{aligned}$$

□

Remark (Where the name "adjoint" comes from)

Remark 0.3 (Where the name "adjoint" comes from). This identity is the origin of the name **adjoint**: in functional analysis and operator theory, the *adjoint* of a linear map $T : V \to W$ between inner product spaces is defined as the unique operator $T^* : W \to V$ satisfying $\langle T\underset{\sim}{u}, \underset{\sim}{v}\rangle = \langle \underset{\sim}{u}, T^*\underset{\sim}{v}\rangle$ (Axler [52], Definition 7.1, p. 226). When V and W are finite-dimensional spaces equipped with their standard orthonormal bases, the matrix representing the adjoint operator T^* is precisely the Hermitian adjoint matrix $\mathbf{A}^{\mathsf{H}}$ (Axler [52], Proposition 7.9, p. 231).

Theorem 0.3 ($\underset{\sim}{z}^{\mathsf{H}}\underset{\sim}{z}$ is a sum of squared absolute values)**.** *For every* $\underset{\sim}{z} \in \mathbb{C}^p$*,*

$$\underset{1\times p}{\underset{\sim}{z}^{\mathsf{H}}}\ \underset{p\times 1}{\underset{\sim}{z}} = \sum_{j=1}^{p} |z_j|^2,$$

which is a nonnegative real number, and is 0 *only when* $\underset{\sim}{z} = \underset{\sim}{0}_{p\times 1}$*.*

Proof

Proof. The last step uses the identity for a complex number times its conjugate.

$$\begin{aligned}
\underset{\sim}{z}^{\mathsf{H}}\underset{\sim}{z} &= \sum_{j=1}^{p} \overline{z_j}\, z_j && \text{(Definition 0.4, and the row-times-column product)}\\
&= \sum_{j=1}^{p} z_j\, \overline{z_j} && \text{(complex multiplication is commutative)}\\
&= \sum_{j=1}^{p} |z_j|^2 && \text{(a number times its conjugate)}
\end{aligned}$$

By the definition of the absolute value, each $|z_j|^2 = (\operatorname{Re} z_j)^2 + (\operatorname{Im} z_j)^2$ is a nonnegative real number, so their sum is too. A sum of nonnegative numbers is 0 only when every term is 0, and $|z_j|^2 = 0$ only when

$$\begin{aligned} \operatorname{Re} z_j &= \operatorname{Im} z_j \\ &= 0, \end{aligned}$$

that is, when $z_j = 0$. □

Example 0.3 (Computing $\underset{\sim}{z}^{\mathsf{H}}\underset{\sim}{z}$)**.** Let $\underset{\sim}{z} = (1+i, 2)^\top$. By Theorem 0.3,

$$\begin{aligned} \underbrace{\underset{\sim}{z}^{\mathsf{H}}}_{1\times 2}\,\underset{2\times 1}{\underset{\sim}{z}} &= |1+i|^2 + |2|^2 && \text{(Theorem 0.3)} \\ &= (1^2 + 1^2) + (2^2 + 0^2) && \text{(definition of the absolute value, squared)} \\ &= 2 + 4 && \text{(add inside each group)} \\ &= 6 && \text{(add)} \end{aligned}$$

Multiplying out directly gives the same number: $\overline{2} \cdot 2 = 4$, and

$$\begin{aligned} \overline{(1+i)}\,(1+i) &= (1-i)(1+i) && \text{(conjugate)} \\ &= 1 + i - i - i^2 && \text{(distribute)} \\ &= 1 - i^2 && \text{(cancel } i\text{)} \\ &= 2 && (i^2 = -1) \end{aligned}$$

Example 0.4 (Why complex vectors use the conjugate transpose)**.** For a real vector $\underset{\sim}{x}$, $\underset{\sim}{x}^\top\underset{\sim}{x} = \sum_{j=1}^{p} x_j^2$ is 0 only when $\underset{\sim}{x}$ is the zero vector. The same formula fails for complex vectors. Let $\underset{\sim}{z} = (1, i)^\top$. Then

$$\begin{aligned} \underbrace{\underset{\sim}{z}^{\top}}_{1\times 2}\,\underset{2\times 1}{\underset{\sim}{z}} &= 1 \cdot 1 + i \cdot i && \text{(multiply entry by entry and add)} \\ &= 1 + i^2 && \text{(multiply)} \\ &= 0 && (i^2 = -1) \end{aligned}$$

although $\underset{\sim}{z} \neq \underset{\sim}{0}_{2\times 1}$. Using the conjugate transpose instead,

$$\begin{aligned} \underbrace{\underset{\sim}{z}^{\mathsf{H}}}_{1\times 2}\,\underset{2\times 1}{\underset{\sim}{z}} &= \overline{1} \cdot 1 + \overline{i} \cdot i && \text{(Definition 0.4)} \\ &= 1 + (-i) \cdot i && \text{(conjugate each entry)} \\ &= 1 - i^2 && \text{(multiply)} \\ &= 2 && (i^2 = -1) \end{aligned}$$

which is positive, as Theorem 0.3 guarantees.

Definition 0.5 (Hermitian matrix, self-adjoint matrix)**.** A square complex matrix $\mathbf{A} \in \mathbb{C}^{n\times n}$ is **Hermitian** [102] (or **self-adjoint**) if it equals its conjugate transpose (Definition 0.4):

$$\mathbf{A}^{\mathsf{H}} = \mathbf{A}.$$

Remark (Why "self-adjoint")

Remark 0.4 (Why "self-adjoint"). The term *self-adjoint* comes directly from Theorem 0.2: when $\mathbf{A}$ is Hermitian, $\langle \mathbf{A}\underset{\sim}{u}, \underset{\sim}{v}\rangle = \langle \underset{\sim}{u}, \mathbf{A}\underset{\sim}{v}\rangle$ for all $\underset{\sim}{u}, \underset{\sim}{v} \in \mathbb{C}^n$. The linear operator moves across the inner product without change.

Example 0.5 (A Hermitian matrix)**.**

$$\underset{\sim}{\underset{2\times 2}{\mathbf{A}}} = \begin{bmatrix} 2 & 1-i \\ 1+i & 3 \end{bmatrix}$$

is Hermitian: transposing swaps the off-diagonal entries, and conjugating turns $1+i$ back into $1-i$ and $1-i$ back into $1+i$, so $\mathbf{A}^{\mathsf{H}} = \mathbf{A}$. The diagonal entries of a Hermitian matrix must be real, because $a_{jj} = \overline{a_{jj}}$, and only a real number is its own conjugate: if $a + b\,i = a - b\,i$, then the imaginary parts give $b = -b$, so $b = 0$.
Every real symmetric matrix is Hermitian, because for a real matrix $\mathbf{A}^{\mathsf{H}} = \mathbf{A}^{\top}$.

Example 0.6 (A symmetric complex matrix that is not Hermitian)**.**

$$\underset{\sim}{\underset{2\times 2}{\mathbf{B}}} = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$$

is symmetric, $\mathbf{B}^{\top} = \mathbf{B}$, but it is not Hermitian:

$$\underset{\sim}{\underset{2\times 2}{\mathbf{B}^{\mathsf{H}}}} = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} \neq \mathbf{B}.$$

Definition 0.6 (Skew-Hermitian matrix, anti-Hermitian matrix)**.** A square complex matrix $\mathbf{A} \in \mathbb{C}^{n\times n}$ is **skew-Hermitian** [103] (or **anti-Hermitian**) if it equals the negative of its conjugate transpose (Definition 0.4):

$$\mathbf{A}^{\mathsf{H}} = -\mathbf{A}.$$

Remark (The diagonal of a skew-Hermitian matrix)

Remark 0.5 (The diagonal of a skew-Hermitian matrix). Every entry of a skew-Hermitian matrix satisfies $a_{jk} = -\overline{a_{kj}}$. In particular, its diagonal entries satisfy $a_{jj} = -\overline{a_{jj}}$, which means

$$\begin{aligned} a_{jj} + \overline{a_{jj}} &= 2\,\mathrm{Re}(a_{jj}) \\ &= 0. \end{aligned}$$

Therefore, every diagonal entry of a skew-Hermitian matrix must be purely imaginary (or zero). For example, $a_{jj} = 3\,i$ works, since

$$\begin{aligned} -\overline{3\,i} &= -(-3\,i) \\ &= 3\,i, \end{aligned}$$

but $a_{jj} = 3$ does not, since $-\overline{3} = -3$.

Example 0.7 (A skew-Hermitian matrix)**.** The matrix

$$\mathbf{A} = \begin{bmatrix} 2\,i & 1+i \\ -1+i & 0 \end{bmatrix}$$

is skew-Hermitian: transposing and conjugating gives

$$\begin{aligned} \mathbf{A}^{\mathsf{H}} &= \begin{bmatrix} \overline{2\,i} & \overline{-1+i} \\ \overline{1+i} & \overline{0} \end{bmatrix} \\ &= \begin{bmatrix} -2\,i & -1-i \\ 1-i & 0 \end{bmatrix} \\ &= -\mathbf{A}. \end{aligned}$$

Its diagonal entries $2\,i$ and 0 are purely imaginary.

Theorem 0.4 (Hermitian and skew-Hermitian decomposition)**.** *Every square complex matrix* $\mathbf{A} \in \mathbb{C}^{n\times n}$ *can be uniquely expressed as the sum of a Hermitian matrix* $\mathbf{H}$ *(Definition 0.5) and a skew-Hermitian matrix* $\mathbf{S}$ *(Definition 0.6):*

$$\mathbf{A} = \mathbf{H} + \mathbf{S},$$

where

$$\mathbf{H} = \frac{\mathbf{A} + \mathbf{A}^{\mathsf{H}}}{2} \quad \textit{and} \quad \mathbf{S} = \frac{\mathbf{A} - \mathbf{A}^{\mathsf{H}}}{2}.$$

Proof

Proof. **Existence**: Define $\mathbf{H} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^{\mathsf{H}})$ and $\mathbf{S} = \frac{1}{2}(\mathbf{A} - \mathbf{A}^{\mathsf{H}})$. Their sum is:

$$\begin{aligned}\mathbf{H} + \mathbf{S} &= \frac{\mathbf{A} + \mathbf{A}^{\mathsf{H}}}{2} + \frac{\mathbf{A} - \mathbf{A}^{\mathsf{H}}}{2} \\ &= \frac{2\mathbf{A}}{2} \\ &= \mathbf{A}.\end{aligned}$$

Applying the properties of the Hermitian adjoint (Theorem 0.1):

$$\begin{aligned}\mathbf{H}^{\mathsf{H}} &= \left(\frac{\mathbf{A} + \mathbf{A}^{\mathsf{H}}}{2}\right)^{\mathsf{H}} \\ &= \frac{\mathbf{A}^{\mathsf{H}} + (\mathbf{A}^{\mathsf{H}})^{\mathsf{H}}}{2} \\ &= \frac{\mathbf{A}^{\mathsf{H}} + \mathbf{A}}{2} \\ &= \mathbf{H}, \\ \mathbf{S}^{\mathsf{H}} &= \left(\frac{\mathbf{A} - \mathbf{A}^{\mathsf{H}}}{2}\right)^{\mathsf{H}} \\ &= \frac{\mathbf{A}^{\mathsf{H}} - (\mathbf{A}^{\mathsf{H}})^{\mathsf{H}}}{2} \\ &= \frac{\mathbf{A}^{\mathsf{H}} - \mathbf{A}}{2} \\ &= -\mathbf{S}.\end{aligned}$$

Thus $\mathbf{H}$ is Hermitian and $\mathbf{S}$ is skew-Hermitian.
Uniqueness: Suppose $\mathbf{A} = \mathbf{H}' + \mathbf{S}'$ with $(\mathbf{H}')^{\mathsf{H}} = \mathbf{H}'$ and $(\mathbf{S}')^{\mathsf{H}} = -\mathbf{S}'$. Taking the Hermitian adjoint of both sides yields:

$$\begin{aligned}\mathbf{A}^{\mathsf{H}} &= (\mathbf{H}' + \mathbf{S}')^{\mathsf{H}} \\ &= (\mathbf{H}')^{\mathsf{H}} + (\mathbf{S}')^{\mathsf{H}} \\ &= \mathbf{H}' - \mathbf{S}'.\end{aligned}$$

Adding $\mathbf{A} = \mathbf{H}' + \mathbf{S}'$ and $\mathbf{A}^{\mathsf{H}} = \mathbf{H}' - \mathbf{S}'$ gives $\mathbf{A} + \mathbf{A}^{\mathsf{H}} = 2\mathbf{H}'$, so

$$\begin{aligned}\mathbf{H}' &= \frac{1}{2}(\mathbf{A} + \mathbf{A}^{\mathsf{H}}) \\ &= \mathbf{H}.\end{aligned}$$

Subtracting the two equations gives $\mathbf{A} - \mathbf{A}^{\mathsf{H}} = 2\mathbf{S}'$, so

$$\begin{aligned}\mathbf{S}' &= \frac{1}{2}(\mathbf{A} - \mathbf{A}^{\mathsf{H}}) \\ &= \mathbf{S}.\end{aligned}$$

□

Definition 0.7 (Matrix analysis)**. Matrix analysis** is the study of matrices and their algebraic properties, with operations on matrices, functions of matrices and the eigenvalues of matrices as example topics [104].

Example 0.8 (Eigenvalues of a small matrix)**.** The matrix $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ has the eigenvalues 3 and 1. Finding these eigenvalues is a question about $\mathbf{A}$ alone, so it is a topic of matrix analysis.

Remark (Linear algebra and matrix analysis)

Remark 0.6 (Linear algebra and matrix analysis)*.* Matrix analysis is usually placed within linear algebra and its applications [104], and these notes do not treat it as a separate subject: they use the name *linear algebra* for the material in the pages above. Banerjee and Roy [48] pairs the two names in its title. Its preface says that its last two chapters cover convergence of matrices and Hilbert spaces [48, p. xvi].

Further reading

- Strang [105] is a widely used first course in linear algebra. It covers vectors, elimination, vector spaces, orthogonality, determinants, eigenvalues, and the singular value decomposition.
- Nield [106] is a short, code-driven introduction to the linear algebra used in data science (Chapter 4).
- Axler [52] is a proof-based treatment centered on vector spaces and linear maps. A free electronic version is available from the publisher.
- Lange [107] covers computing eigenvalues and eigenvectors, including Jacobi's method (Chapter 8), and the singular value decomposition, its properties, and its applications (Chapter 9).
- Fieller [108]
- Banerjee and Roy [48]
- Searle and Khuri [109]
- Petersen and Pedersen [110] is a free desk reference that collects matrix identities: it

lists results on determinants, inverses, decompositions, and derivatives of vectors and matrices, so it is a quick place to look up a formula these pages prove.

- Stanford University [111] (Stanford EE 278, taught by Tsachy Weissman) applies this page's tools to probability and inference. Its "MMSE Geometry" lecture treats the best estimate as an orthogonal projection, and its lectures on spectral decomposition, the Karhunen-Loeve expansion and principal component analysis treat diagonalization of a covariance matrix.
- Essence of Linear Algebra is a 16-video YouTube playlist by Grant Sanderson (3Blue1Brown) that develops visual and geometric intuition for vectors, linear combinations, spans, linear transformations, matrix multiplication, determinants, inverses, dot and cross products, change of basis, and eigenvectors and eigenvalues.

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